# Theory of Mathematics

## Volume VIII: Algebraic Topology

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# Part I

# Homotopy

# Chapter 1

# Homotopies of Maps

Algebraic topology begins by weakening the notion of equality of maps. Two continuous maps need not agree pointwise to carry the same large-scale topological information. The basic relation is *homotopy*: one map may be continuously deformed into the other while the domain and target remain fixed. This chapter develops the definition, the elementary calculus of homotopies, relative and based variants, and the first explicit deformation formulas.

## 1.1 The unit interval as a deformation parameter

Write

$$I = [0, 1].$$

A point of $X \times I$ is written $(x, t)$, with $t$ interpreted as time. At $t = 0$ we see the initial map and at $t = 1$ the final map.

**Definition 1.1** (Homotopy)**.** Let $f, g : X \to Y$ be continuous maps. A *homotopy from $f$ to $g$* is a continuous map

$$H : X \times I \longrightarrow Y$$

such that

$$H(x, 0) = f(x), \qquad H(x, 1) = g(x)$$

for every $x \in X$. We write

$$f \simeq g.$$

For each fixed $t$, the map

$$H_t : X \to Y, \qquad H_t(x) = H(x, t),$$

is a time slice of the deformation.

*Warning* 1.2 (Pointwise paths are not enough). For every $x$, the curve $t \mapsto H(x, t)$ is a path in $Y$, but choosing an arbitrary path from $f(x)$ to $g(x)$ for each $x$ does not automatically produce a homotopy. The dependence on $x$ and $t$ must be jointly continuous.

## 1.2 First explicit homotopies

**Example 1.3** (Straight-line homotopy)**.** If $Y$ is a convex subset of a real topological vector space and $f, g : X \to Y$ are continuous, then

$$H(x, t) = (1 - t)f(x) + tg(x)$$

is a homotopy from $f$ to $g$.

**Example 1.4** (Maps into an interval)**.** Any two continuous maps $f, g : X \to [a, b]$ are homotopic by the straight-line formula. Thus the interval has no obstruction to deforming one incoming map into another.

**Example 1.5** (Rotation of the circle)**.** For a fixed angle $\alpha$, the identity map on $S^1 \subset \mathbb{C}$ is homotopic to the rotation $R_\alpha(z) = e^{i\alpha}z$ via

$$H(z, t) = e^{it\alpha}z.$$

## 1.3 Homotopy is an equivalence relation

**Theorem 1.6** (Equivalence relation)**.** *Homotopy is reflexive, symmetric, and transitive on the set of continuous maps $X \to Y$.*

*Proof.* Reflexivity is given by the constant homotopy $H(x, t) = f(x)$. If $H$ runs from $f$ to $g$, then

$$\overline{H}(x, t) = H(x, 1 - t)$$

runs from $g$ to $f$. For transitivity, suppose $H : f \simeq g$ and $K : g \simeq h$. Define

$$L(x, t) = \begin{cases} H(x, 2t), & 0 \le t \le \frac{1}{2}, \\ K(x, 2t - 1), & \frac{1}{2} \le t \le 1. \end{cases}$$

The two formulas agree at $t = \frac{1}{2}$, where both equal $g(x)$. The gluing lemma gives continuity. $\square$

*Remark* 1.7 (Homotopy classes)*.* The equivalence class of $f : X \to Y$ is denoted $[f]$. The set of homotopy classes is often written

$$[X, Y].$$

At this stage it is only a set; later chapters supply algebraic structure in important cases.

## 1.4 Compatibility with composition

**Proposition 1.8** (Postcomposition)**.** *If $f \simeq g : X \to Y$ and $u : Y \to Z$ is continuous, then*

$$u \circ f \simeq u \circ g.$$

*Proof.* If $H$ is a homotopy from $f$ to $g$, use

$$(x, t) \longmapsto u(H(x, t)).$$

$\square$

**Proposition 1.9** (Precomposition)**.** *If $f \simeq g : X \to Y$ and $v : W \to X$ is continuous, then*

$$f \circ v \simeq g \circ v.$$

*Proof.* Use

$$(w, t) \longmapsto H(v(w), t).$$

$\square$

**Corollary 1.10** (Two-sided compatibility)**.** *If $f_0 \simeq f_1 : X \to Y$ and $g_0 \simeq g_1 : Y \to Z$, then*

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

## 1.5 Null-homotopic maps

**Definition 1.11** (Null-homotopic map)**.** A map $f : X \to Y$ is *null-homotopic* if it is homotopic to a constant map.

**Example 1.12** (Maps into a convex set)**.** Every map into a convex subset $C \subset \mathbb{R}^n$ is null-homotopic: choose $c_0 \in C$ and use

$$H(x,t) = (1-t)f(x) + tc_0.$$

**Proposition 1.13** (Factorization through a point)**.** *A map is null-homotopic precisely when it is homotopic to a map factoring as*

$$X \longrightarrow \{*\} \longrightarrow Y.$$

## 1.6 Homotopies relative to a subspace

Often a deformation must leave part of the domain fixed.

**Definition 1.14** (Homotopy relative to $A$)**.** Let $A \subset X$. A homotopy $H : f \simeq g$ is a *homotopy relative to* $A$, written

$$f \simeq g \pmod{A},$$

if

$$H(a,t) = f(a) = g(a)$$

for all $a \in A$ and $t \in I$.

*Remark* 1.15. A relative homotopy can exist only if $f|_A = g|_A$. The condition says more: the common restriction is fixed throughout the deformation.

**Example 1.16** (Fixing endpoints)**.** For paths $\alpha, \beta : I \to Y$ with common endpoints, a homotopy relative to $\{0,1\}$ is exactly an endpoint-fixing deformation of one path into the other. This is the relation used later in the fundamental group.

## 1.7 Based homotopy

Let $(X, x_0)$ and $(Y, y_0)$ be pointed spaces.

**Definition 1.17** (Based map and based homotopy)**.** A map $f : (X, x_0) \to (Y, y_0)$ is based if $f(x_0) = y_0$. A homotopy through based maps is a homotopy relative to $\{x_0\}$:

$$H(x_0, t) = y_0 \qquad \text{for all } t.$$

*Warning* 1.18 (Free versus based homotopy)*.* Two based maps can be freely homotopic without being based-homotopic. The distinction becomes essential when loops and fundamental groups enter.

## 1.8 Reparameterizing a homotopy

**Lemma 1.19** (Monotone reparameterization)**.** *If $H : f \simeq g$ and $\rho : I \to I$ is continuous with $\rho(0) = 0$ and $\rho(1) = 1$, then*

$$K(x,t) = H(x, \rho(t))$$

*is another homotopy from $f$ to $g$.*

This permits slowing, speeding, or pausing a deformation without changing its homotopy class.

## 1.9 Homotopy through a subspace

**Proposition 1.20** (Image control)**.** *Suppose $A \subset Y$, and a homotopy $H : X \times I \to Y$ satisfies $H(X \times I) \subset A$. Then $H$, regarded as a map into $A$, is a homotopy in the subspace $A$.*

The proposition is elementary, but important: whether a deformation is allowed depends on the target in which the deformation is required to live.

## 1.10 A punctured-plane obstruction preview

The straight-line formula can fail when the target is not convex.

**Example 1.21** (Why the ambient target matters)**.** In $\mathbb{R}^2 \setminus \{0\}$, the formula

$$H(x,t) = (1-t)x + t(-x)$$

passes through 0 at $t = \frac{1}{2}$. Therefore it is not a homotopy in the punctured plane. Later invariants will explain why certain maps cannot be deformed while avoiding the missing point.

## 1.11 Cylinder viewpoint

A homotopy $H : X \times I \to Y$ is simply a map out of the cylinder on $X$, with prescribed values on the two ends

$$X \times \{0\}, \qquad X \times \{1\}.$$

This point of view anticipates mapping cylinders, cofibrations, and relative homotopy constructions.

**Proposition 1.22** (Restriction to time intervals)**.** *If $H : f \simeq g$ and $0 \leq a < b \leq 1$, then after affine reparameterization the restriction of $H$ to $X \times [a,b]$ is a homotopy from $H_a$ to $H_b$.*

## 1.12 Products of homotopies

**Proposition 1.23** (Product homotopy)**.** *If $f_0 \simeq f_1 : X \to Y$ and $g_0 \simeq g_1 : W \to Z$, then*

$$f_0 \times g_0 \simeq f_1 \times g_1 : X \times W \to Y \times Z.$$

*Proof.* For homotopies $H$ and $K$, set

$$L((x,w),t) = \big(H(x,t), K(w,t)\big).$$

□

## 1.13 What a homotopy does not claim

A homotopy does not assert that any $H_t$ is injective, surjective, an embedding, a homeomorphism, or a diffeomorphism. It is a deformation in the space of continuous maps. If every time slice is required to be an embedding or homeomorphism, one obtains stronger notions such as isotopy.

*Warning* 1.24 (Homotopy is weaker than isotopy)*.* Do not infer geometric rigidity from a homotopy. Intermediate maps may fold, collapse, or identify points unless additional hypotheses prohibit this.

## 1.14 Exercises

**Exercise 1.25.** Write the defining endpoint equations for a homotopy $H : f \simeq g$.

*Hint.* Evaluate at $t = 0$ and $t = 1$. Regard the two restrictions as maps on the entire copies of $X \times \{0\}$ and $X \times \{1\}$, not as equations at one chosen point. □

*Solution.* They are $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for every $x \in X$. □

**Exercise 1.26.** Show that every map is homotopic to itself.

*Hint.* Do not move the map in time. Factor the proposed cylinder map through the projection $X \times I \to X$; this proves joint continuity as well as the endpoint conditions. □

*Solution.* Use $H(x, t) = f(x)$. □

**Exercise 1.27.** Reverse a given homotopy $H : f \simeq g$.

*Hint.* Reverse the interval parameter. Use an endpoint-exchanging continuous self-map of $I$, and check which of $f$ and $g$ appears at each new endpoint. □

*Solution.* Set $\overline{H}(x, t) = H(x, 1 - t)$. □

**Exercise 1.28.** Concatenate homotopies $H : f \simeq g$ and $K : g \simeq h$.

*Hint.* Give each homotopy half the time interval. Write the formulas on the closed halves of $X \times I$. Besides the endpoints, check their common value on $X \times \{1/2\}$. □

*Solution.* Use $H(x, 2t)$ for $t \leq 1/2$ and $K(x, 2t - 1)$ for $t \geq 1/2$. □

**Exercise 1.29.** Why is the concatenated homotopy continuous at $t = 1/2$?

*Hint.* Compare the two values there. Apply the pasting lemma to the two closed cylinder halves; separate continuity on each half alone does not settle agreement at their intersection. □

*Solution.* Both pieces equal $g(x)$ at the joining time, so the gluing lemma applies. □

**Exercise 1.30.** Give a homotopy from $f(x) = x$ to $g(x) = 0$ on $\mathbb{R}^n$.

*Hint.* Use scalar multiplication. Choose a continuous scalar coefficient that changes from 1 to 0; check the resulting vector-valued map on all of $\mathbb{R}^n \times I$. □

*Solution.* $H(x, t) = (1 - t)x$. □

**Exercise 1.31.** Show that two maps $X \to [a, b]$ are homotopic.

*Hint.* The interval is convex. Interpolate the two values with nonnegative weights summing to one. Verify both the inequalities $a \leq H(x, t) \leq b$ and continuity. □

*Solution.* Use $H(x, t) = (1 - t)f(x) + tg(x)$. □

**Exercise 1.32.** Construct a homotopy from the identity of $S^1$ to rotation by angle $\pi/3$.

*Hint.* Rotate gradually. Multiply $z$ by a unit complex number whose argument changes continuously from 0 to $\pi/3$; verify the modulus throughout the deformation. □

*Solution.* $H(z, t) = e^{it\pi/3}z$. □

**Exercise 1.33.** If $f \simeq g$, prove $u \circ f \simeq u \circ g$.

*Hint.* Postcompose the homotopy. Write the cylinder map as a composite $X \times I \to Y \to Z$. Its two restrictions should be the composites named in the question. □

*Solution.* For $H : f \simeq g$, use $K(x,t) = u(H(x,t))$. □

**Exercise 1.34.** If $f \simeq g$, prove $f \circ v \simeq g \circ v$.

*Hint.* Precompose the homotopy. Use $v \times \mathrm{id}_I : W \times I \to X \times I$ before the given homotopy; check that the time coordinate is unchanged. □

*Solution.* Use $K(w,t) = H(v(w),t)$. □

**Exercise 1.35.** Define a null-homotopic map.

*Hint.* Compare with a constant map. Specify a single target point independent of $x$; the terminal time slice, rather than every intermediate slice, must take that constant value. □

*Solution.* It is a map homotopic to a constant map. □

**Exercise 1.36.** Show every map into a convex subset of $\mathbb{R}^n$ is null-homotopic.

*Hint.* Choose a point in the convex set. Assume the convex target is nonempty so a contraction point can be chosen. Explain why every segment to that point stays inside the target. □

*Solution.* For $c_0$ in the target, $H(x,t) = (1-t)f(x) + tc_0$ remains in the convex set. □

**Exercise 1.37.** State what $f \simeq g \pmod{A}$ means.

*Hint.* The subspace must not move. Write the restriction of the cylinder map to $A \times I$; constancy in time is stronger than agreement of only the initial and final restrictions. □

*Solution.* It means there is a homotopy $H : f \simeq g$ with $H(a,t) = f(a) = g(a)$ for all $a \in A$. □

**Exercise 1.38.** What preliminary condition on $f$ and $g$ is necessary for a homotopy rel $A$?

*Hint.* Set $t = 0, 1$ on $A$. Apply the fixed-subspace condition at both ends of the interval. Distinguish this necessary agreement from a sufficient criterion for constructing a relative homotopy. □

*Solution.* They must agree on $A$: $f|_A = g|_A$. □

**Exercise 1.39.** Interpret homotopy rel $\{0,1\}$ for paths.

*Hint.* The endpoints are the chosen subspace. For a path parameter $s$ and deformation parameter $t$, write separately what happens on the edges $s = 0$ and $s = 1$ of the square. □

*Solution.* It is a deformation of paths that keeps both endpoints fixed at all times. □

**Exercise 1.40.** What is a based homotopy?

*Hint.* Fix the basepoint throughout time. Use the relative-homotopy definition with $A = \{x_0\}$, keeping the chosen target basepoint the same for every time slice. □

*Solution.* It is a homotopy $H$ satisfying $H(x_0,t) = y_0$ for every $t$. □

**Exercise 1.41.** Why can an arbitrary collection of paths from $f(x)$ to $g(x)$ fail to be a homotopy?

*Hint.* Continuity has two variables. A family continuous in the time variable for each fixed $x$ need not define a continuous map on the product. Identify the missing dependence on $x$. $\square$

*Solution.* The chosen paths may fail to depend continuously on $x$, so the resulting map $X \times I \to Y$ need not be continuous. $\square$

**Exercise 1.42.** Reparameterize a homotopy so that it pauses at $t = 1/2$.

*Hint.* Choose a continuous $\rho : I \to I$ that is constant on a small interval. Construct a piecewise-linear $\rho$ with a plateau on $[1/3, 2/3]$ and values $0, 1$ at the endpoints; compose it with the time variable of $H$. $\square$

*Solution.* Choose any continuous endpoint-preserving $\rho$ that is constant near $1/2$, and use $K(x,t) = H(x, \rho(t))$. $\square$

**Exercise 1.43.** Show that the product of two null-homotopic maps is null-homotopic.

*Hint.* Take the product of their homotopies. Use the same time parameter in both coordinates. The universal property of the product checks continuity, and the terminal pair must be independent of both input points. $\square$

*Solution.* If $H$ and $K$ contract the two maps to constants, then $(H, K)$ contracts their product to the product constant. $\square$

**Exercise 1.44.** Why does the straight-line deformation from $x$ to $-x$ fail in $\mathbb{R}^2 \setminus \{0\}$?

*Hint.* Evaluate at $t = 1/2$. Compute the scalar factor in $(1-t)x + t(-x)$. A continuous ambient formula is not a homotopy into the prescribed target when that factor vanishes. $\square$

*Solution.* It equals 0 halfway through, which is not in the target. $\square$

**Exercise 1.45.** If $H : X \times I \to A \subset Y$, in which two targets can the same formula be viewed?

*Hint.* Use the inclusion $A \hookrightarrow Y$. Distinguish the map with codomain $A$ from its composite with $A \hookrightarrow Y$. Neither operation asserts that a homotopy in $Y$ stays in $A$. $\square$

*Solution.* It is a homotopy into $A$, and after composing with the inclusion it is also a homotopy into $Y$. $\square$

**Exercise 1.46.** Does a homotopy require every time slice to be injective?

*Hint.* Return to the definition. Consider a contraction of an interval as a test case: its final slice collapses distinct points while the cylinder map remains continuous. $\square$

*Solution.* No. Only continuity and the endpoint conditions are required. $\square$

**Exercise 1.47.** Does a homotopy between homeomorphisms have to pass through homeomorphisms?

*Hint.* Compare homotopy with isotopy. On $\mathbb{R}$, interpolate between $x \mapsto x$ and $x \mapsto -x$. Examine the middle slice, despite both endpoint maps being homeomorphisms. $\square$

*Solution.* No. That stronger requirement belongs to isotopy. $\square$

**Exercise 1.48.** Explain why composition descends to homotopy classes.

*Hint.* Use pre- and postcomposition compatibility. Replace one factor at a time: first hold the outer map fixed, then the inner map. Concatenate the resulting homotopies to check independence of representatives. □

*Solution.* Replacing either factor by a homotopic map changes the composite only by a homotopy, so the homotopy class of a composite depends only on the classes of its factors. □

## 1.15 Solved problem dossiers

**Problem 1.49** (Piecewise concatenation)**.** Verify directly that concatenating two relative homotopies rel $A$ again gives a homotopy rel $A$.

*Solution.* Use the usual two-piece concatenation. On $A$, both original homotopies are constant at the common restriction $f|_A = g|_A = h|_A$, so both pieces of the concatenation are constant there as well. □

**Problem 1.50** (Radial contraction)**.** Let $D^n = \{x : \|x\| \leq 1\}$. Construct a homotopy from the identity of $D^n$ to the constant map at 0.

*Solution.* Set $H(x,t) = (1-t)x$. Since $\|H(x,t)\| = (1-t)\|x\| \leq 1$, the image remains in $D^n$. □

**Problem 1.51** (Affine homotopy in a convex target)**.** Prove that the straight-line formula is jointly continuous.

*Solution.* The maps $(x,t) \mapsto (1-t)f(x)$ and $(x,t) \mapsto tg(x)$ are continuous by continuity of scalar multiplication and of $f, g$; their sum is continuous. Convexity ensures the value remains in the target. □

**Problem 1.52** (Homotopy of constant maps)**.** If $Y$ is path connected, prove that any two constant maps $X \to Y$ are homotopic.

*Solution.* If the constants are $y_0, y_1$, choose a path $\gamma : I \to Y$ from $y_0$ to $y_1$ and define $H(x,t) = \gamma(t)$. □

**Problem 1.53** (Path components detected by maps from a point)**.** Describe $[\{*\}, Y]$.

*Solution.* A map from a point chooses a point of $Y$. Two such maps are homotopic exactly when the chosen points are joined by a path. Hence $[\{*\}, Y]$ is the set of path components of $Y$. □

**Problem 1.54** (Restriction of a relative homotopy)**.** If $H : f \simeq g \pmod{A}$ and $B \subset A$, show $H$ is a homotopy rel $B$.

*Solution.* The defining fixed-point equation on all of $A$ in particular holds on its subset $B$. □

**Problem 1.55** (Composition of based homotopies)**.** Show that composing based homotopies produces a based homotopy.

*Solution.* If $H(x_0,t) = y_0$ and $K(y_0,t) = z_0$, the standard two-sided compatibility construction keeps the distinguished point at $z_0$ throughout. □

**Problem 1.56** (A stationary first half)**.** Given $H : f \simeq g$, construct a homotopy that stays equal to $f$ until time $1/2$ and then performs $H$.

*Solution.* Use

$$K(x,t) = \begin{cases} f(x), & 0 \le t \le 1/2, \\ H(x, 2t-1), & 1/2 \le t \le 1. \end{cases}$$

The formulas agree at $t = 1/2$. □

**Problem 1.57** (Product deformation)**.** If $X$ and $W$ each admit a contraction to a chosen point, write the contraction of $X \times W$.

*Solution.* For contractions $H$ and $K$, use

$$L((x,w),t) = (H(x,t), K(w,t)).$$

□

**Problem 1.58** (Target inclusion)**.** Let $A \subset Y$ and suppose $f, g : X \to A$ are homotopic as maps into $A$. Prove they are homotopic as maps into $Y$.

*Solution.* Compose the homotopy $X \times I \to A$ with the inclusion $A \hookrightarrow Y$. □

**Problem 1.59** (Two-stage deformation)**.** Suppose $f \simeq c$ and $c \simeq g$, where $c$ is constant. Prove $f \simeq g$.

*Solution.* This is transitivity: concatenate the two homotopies. □

**Problem 1.60** (Bridge to VIII/02)**.** Explain why a homotopy $f \simeq g$ alone does not say that the spaces $X$ and $Y$ have the same homotopy type.

*Solution.* The relation compares two maps with the same domain and target. To compare spaces, one needs maps in both directions whose composites are homotopic to the appropriate identity maps. That is the homotopy-equivalence notion developed in VIII/02. □

## 1.16 Supplementary worked examples

**Example 1.61** (Straight-line homotopy in a convex target)**.** For maps $f, g : X \to C$ into a convex subset $C \subset \mathbb{R}^m$, the formula $H(x,t) = (1-t)f(x) + tg(x)$ stays in $C$ and gives a homotopy. The key point is joint continuity on $X \times I$, not merely a path for each fixed $x$.

**Example 1.62** (A rotation of the circle)**.** For $R_\alpha(z) = e^{i\alpha}z$ on $S^1$, $H(z,t) = e^{it\alpha}z$ deforms the identity to $R_\alpha$. Every time slice is even a homeomorphism, although homotopy theory does not require this stronger property.

**Example 1.63** (Relative homotopy of paths)**.** If paths $\alpha, \beta : I \to Y$ have the same endpoints, a homotopy relative to $\{0,1\}$ keeps both endpoints fixed for every time. This is exactly the equivalence relation later used to define $\pi_1(Y, y_0)$.

## 1.17 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 1.64** (Endpoint equations)**.** State the two endpoint equations for $H : f \simeq g$.

*Hint.* Restrict $H$ to $X \times \{0\}$ and $X \times \{1\}$. □

*Solution.* They are $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for every $x \in X$. □

**Exercise 1.65** (Reverse a homotopy)**.** Given $H : f \simeq g$, write a homotopy from $g$ to $f$.

*Hint.* Reverse the interval parameter. □

*Solution.* Use $\overline{H}(x,t) = H(x, 1-t)$. □

**Exercise 1.66** (Concatenate homotopies)**.** Write the standard concatenation of $H : f \simeq g$ and $K : g \simeq h$.

*Hint.* Give each deformation half of $I$. □

*Solution.* Use $H(x, 2t)$ for $t \leq 1/2$ and $K(x, 2t-1)$ for $t \geq 1/2$; the gluing lemma gives continuity. □

**Exercise 1.67** (Convex null-homotopy)**.** Construct a null-homotopy of $f : X \to \mathbb{R}^m$.

*Hint.* Choose a constant target point $c$. □

*Solution.* $H(x,t) = (1-t)f(x) + tc$ contracts $f$ to the constant map $c$. □

**Exercise 1.68** (Based condition)**.** What additional equation makes a homotopy based at $x_0 \mapsto y_0$?

*Hint.* Keep the base point fixed for all time. □

*Solution.* Require $H(x_0, t) = y_0$ for every $t \in I$. □

### Proofs

**Exercise 1.69** (Equivalence relation)**.** Prove homotopy is an equivalence relation on maps $X \to Y$.

*Hint.* Use constant, reversed, and concatenated homotopies. □

*Solution.* The constant homotopy gives reflexivity, $t \mapsto 1 - t$ gives symmetry, and piecewise concatenation plus the gluing lemma gives transitivity. □

**Exercise 1.70** (Postcomposition)**.** Prove $f \simeq g$ implies $u \circ f \simeq u \circ g$ for continuous $u : Y \to Z$.

*Hint.* Compose the homotopy with $u$. □

*Solution.* If $H$ joins $f$ to $g$, then $(x, t) \mapsto u(H(x, t))$ joins the composites. □

**Exercise 1.71** (Precomposition)**.** Prove $f \simeq g$ implies $f \circ v \simeq g \circ v$ for continuous $v : W \to X$.

*Hint.* Insert $v$ into the first variable. □

*Solution.* Use $(w, t) \mapsto H(v(w), t)$. □

**Exercise 1.72** (Product homotopy)**.** Prove products of homotopic maps are homotopic.

*Hint.* Pair the two homotopies at the same time. □

*Solution.* If $H$ and $K$ are the homotopies, $((x, w), t) \mapsto (H(x, t), K(w, t))$ is continuous and has the required endpoints. □

### Counterexamples and hypothesis tests

**Exercise 1.73** (Pointwise paths suffice)**.** Test: choosing a path from $f(x)$ to $g(x)$ for every $x$ always defines a homotopy.

*Hint.* Joint continuity is the missing condition. □

*Solution.* False. The paths can vary discontinuously with $x$. □

**Exercise 1.74** (Free equals based)**.** Test: freely homotopic based maps are always based-homotopic.

*Hint.* The base point may move during a free homotopy. □

*Solution.* False in general; based homotopy imposes the additional fixed-basepoint condition. □

**Exercise 1.75** (Homotopy implies isotopy)**.** Test: every homotopy between embeddings is an isotopy.

*Hint.* Intermediate time slices need not remain embeddings. □

*Solution.* False. Homotopy only asks for continuous time slices, which may fold or identify points. □

## Applications and investigations

**Exercise 1.76** (Deformation planning)**.** Explain why convexity is an algorithmically useful certificate for constructing homotopies.

*Hint.* It makes line segments stay in the target. □

*Solution.* Convexity guarantees the straight-line interpolation remains admissible, giving an explicit continuous deformation with no obstruction search. □

**Exercise 1.77** (Cylinder viewpoint)**.** Interpret a homotopy as an extension problem from the ends of a cylinder.

*Hint.* The prescribed data live on $X \times \{0, 1\}$. □

*Solution.* A homotopy is a map $X \times I \to Y$ extending the two endpoint maps on the boundary copies of $X$. □

## Challenge problems

**Exercise 1.78** (Punctured-plane warning)**.** Why can the straight-line homotopy from $x$ to $-x$ fail in $\mathbb{R}^2 \setminus \{0\}$?

*Hint.* Evaluate at $t = 1/2$. □

*Solution.* It passes through the excluded origin, so it is not a homotopy in the punctured target. □

**Exercise 1.79** (Relative synthesis)**.** Construct a homotopy of a map on a disk while fixing its boundary, and state exactly what must be checked.

*Hint.* Use a formula that agrees with the original map on $S^1$ for all $t$. □

*Solution.* A valid construction must be jointly continuous on $D^2 \times I$, have the prescribed endpoint maps, and satisfy $H(a, t) = f(a)$ for every $a \in S^1$ and all $t$. □

# Chapter 2

# Homotopy Equivalence and Contractibility

Homotopy of maps becomes a relation between spaces when maps are available in both directions. The resulting notion, homotopy equivalence, is much weaker than homeomorphism but strong enough to preserve the invariants that algebraic topology is designed to compute. Contractible spaces, deformation retracts, and homotopy type are the central themes of this chapter.

## 2.1 Homotopy equivalence

**Definition 2.1** (Homotopy equivalence)**.** Spaces $X$ and $Y$ are *homotopy equivalent*, written

$$X \simeq Y,$$

if there exist continuous maps

$$f : X \to Y, \qquad g : Y \to X$$

such that

$$g \circ f \simeq \mathrm{id}_X, \qquad f \circ g \simeq \mathrm{id}_Y .$$

The maps $f$ and $g$ are called homotopy inverses.

*Warning* 2.2 (Not an actual inverse). A homotopy inverse need not satisfy $g \circ f = \mathrm{id}_X$ or $f \circ g = \mathrm{id}_Y$ pointwise. Equality is replaced by homotopy.

## 2.2 Homotopy type

**Definition 2.3** (Homotopy type)**.** Two spaces have the same *homotopy type* if they are homotopy equivalent.

**Proposition 2.4** (Homotopy equivalence is an equivalence relation)**.** *Homotopy equivalence is reflexive, symmetric, and transitive.*

*Proof.* Reflexivity uses the identity. Symmetry is built into the definition. For transitivity, if $X \simeq Y$ via $f, g$ and $Y \simeq Z$ via $u, v$, then $u \circ f : X \to Z$ and $g \circ v : Z \to X$ are homotopy inverses. The verification uses compatibility of homotopy with composition from VIII/01. $\square$

## 2.3 Contractible spaces

**Definition 2.5** (Contractible space)**.** A space $X$ is *contractible* if

$$\mathrm{id}_X$$

is homotopic to a constant map.

**Theorem 2.6** (Contractible means homotopy equivalent to a point)**.** *A nonempty space $X$ is contractible if and only if*

$$X \simeq \{*\}.$$

*Proof.* If $X$ is contractible to $x_0$, let $p : X \to \{*\}$ be the unique map and $i : \{*\} \to X$ send the point to $x_0$. Then $p \circ i$ is the identity on the point, while $i \circ p$ is the constant map at $x_0$, homotopic to $\mathrm{id}_X$. The converse is the same argument read backwards. □

## 2.4 Convex and star-shaped examples

**Proposition 2.7** (Convex sets are contractible)**.** *Every nonempty convex subset $C$ of a real topological vector space is contractible.*

*Proof.* Fix $c_0 \in C$ and use

$$H(x,t) = (1-t)x + tc_0.$$

□

**Definition 2.8** (Star-shaped set)**.** A subset $A \subset \mathbb{R}^n$ is star-shaped with center $a \in A$ if the line segment from $a$ to every $x \in A$ lies in $A$.

**Proposition 2.9** (Star-shaped sets are contractible)**.** *Every nonempty star-shaped subset of $\mathbb{R}^n$ is contractible.*

**Example 2.10.** Intervals, disks, balls, cubes, and $\mathbb{R}^n$ are contractible.

## 2.5 Retractions

**Definition 2.11** (Retraction)**.** If $A \subset X$, a *retraction* is a map

$$r : X \to A$$

such that

$$r|_A = \mathrm{id}_A .$$

Then $A$ is called a retract of $X$.

Let $i : A \hookrightarrow X$ denote inclusion. The retraction condition is

$$r \circ i = \mathrm{id}_A .$$

*Warning* 2.12 (Retraction alone is not a deformation retract). A retraction provides an exact one-sided inverse to inclusion, but does not require $i \circ r$ to be homotopic to $\mathrm{id}_X$.

## 2.6 Deformation retracts

**Definition 2.13** (Deformation retract)**.** A subspace $A \subset X$ is a *deformation retract* of $X$ if there is a retraction $r : X \to A$ such that

$$i \circ r \simeq \mathrm{id}_X .$$

Equivalently, there is a homotopy from $\mathrm{id}_X$ to a map with image in $A$, ending at the retraction followed by inclusion.

**Theorem 2.14** (Deformation retracts preserve homotopy type)**.** *If $A$ is a deformation retract of $X$, then*

$$A \simeq X.$$

*Proof.* The inclusion $i : A \to X$ and retraction $r : X \to A$ satisfy $r \circ i = \mathrm{id}_A$ and $i \circ r \simeq \mathrm{id}_X$. □

## 2.7 Strong deformation retracts

**Definition 2.15** (Strong deformation retract)**.** A deformation retract $A \subset X$ is *strong* if the deformation

$$H : X \times I \to X$$

from $\mathrm{id}_X$ to $i \circ r$ fixes $A$ pointwise:

$$H(a, t) = a \qquad (a \in A,\ t \in I).$$

Thus a strong deformation retract is a relative homotopy from VIII/01.

## 2.8 Radial examples

**Example 2.16** (Punctured Euclidean space)**.** The sphere $S^{n-1}$ is a strong deformation retract of $\mathbb{R}^n \setminus \{0\}$. Put

$$r(x) = \frac{x}{\|x\|}$$

and

$$H(x, t) = \left((1 - t) + \frac{t}{\|x\|}\right) x.$$

The scalar coefficient is positive, so the homotopy never reaches 0. For $x \in S^{n-1}$, $H(x, t) = x$.

**Corollary 2.17.**

$$\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}.$$

**Example 2.18** (Annulus)**.** For

$$A = \{x \in \mathbb{R}^2 : 1 \le \|x\| \le 2\},$$

the unit circle is a strong deformation retract by radial normalization.

## 2.9 Cylinder and circle

**Example 2.19** (Cylinder)**.** The cylinder $S^1 \times I$ strongly deformation retracts onto $S^1 \times \{0\}$ by

$$H((z, s), t) = (z, (1 - t)s).$$

Hence

$$S^1 \times I \simeq S^1.$$

## 2.10 Graphs with dangling trees

A contractible branch attached to a graph often contributes no new homotopy type if it can be collapsed continuously while the remaining graph is fixed.

**Example 2.20** (Circle with a whisker)**.** Attach an interval to $S^1$ at one endpoint. Collapsing the interval linearly to its attachment point gives a strong deformation retract onto $S^1$.

This elementary picture foreshadows cell collapses and CW complexes.

## 2.11 Contractibility versus path connectedness

**Proposition 2.21** (Contractible implies path connected)**.** *Every nonempty contractible space is path connected.*

*Proof.* Let $H$ contract $X$ to $x_0$. For each $x$, the path $t \mapsto H(x,t)$ joins $x$ to $x_0$. Concatenating such paths joins any two points. □

*Warning* 2.22*.* The converse fails. The circle $S^1$ is path connected but is not contractible. Proving noncontractibility requires an invariant; degree in VIII/03 gives one route, and later fundamental-group or homology calculations give others.

## 2.12 Homotopy invariants

**Definition 2.23** (Homotopy invariant)**.** A construction $F$ on spaces is a *homotopy invariant* if

$$X \simeq Y \quad \Longrightarrow \quad F(X) \cong F(Y)$$

in the appropriate category.

The central strategy of algebraic topology is:

$$\text{space} \longmapsto \text{algebraic invariant},$$

then use an invariant mismatch to prove that two spaces are not homotopy equivalent.

## 2.13 Homotopy equivalent need not mean homeomorphic

**Example 2.24** (Interval and point)**.** The interval $I$ is contractible, so $I \simeq \{*\}$, but $I$ is not homeomorphic to a point.

**Example 2.25** (Disk and point)**.** The closed disk $D^n$ has the homotopy type of a point while having many more points and nontrivial local geometry.

*Warning* 2.26 (Do not transfer non-homotopy properties)*.* Homotopy equivalence does not preserve dimension, cardinality, smooth structure, metric data, or local geometric shape in general.

## 2.14 A useful criterion

**Proposition 2.27** (One-sided strict inverse plus homotopy)**.** *If maps $f : X \to Y$ and $g : Y \to X$ satisfy*

$$g \circ f = \mathrm{id}_X \qquad \textit{and} \qquad f \circ g \simeq \mathrm{id}_Y,$$

*then $f$ and $g$ are homotopy inverses.*

Deformation retracts are the main source of this pattern.

## 2.15 Exercises

**Exercise 2.28.** State the two equations defining homotopy inverse maps $f : X \to Y$ and $g : Y \to X$.

*Hint.* Compose in both orders. Check domains before writing the identities: one composite is an endomorphism of $X$, the other of $Y$. Equality is not required in either condition. □

*Solution.* $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. □

**Exercise 2.29.** Why is every space homotopy equivalent to itself?

*Hint.* Use the identity map twice. Check both composite conditions separately, even though this choice makes them identical strict equalities. □

*Solution.* The identity is its own strict inverse, hence also its own homotopy inverse. □

**Exercise 2.30.** Prove symmetry of homotopy equivalence.

*Hint.* The same pair of maps works in the opposite order. Exchange the domain and codomain names in the two homotopy-inverse conditions; no new deformation needs to be constructed. □

*Solution.* If $f$ and $g$ are homotopy inverses, then $g$ and $f$ are homotopy inverses after swapping the roles of $X$ and $Y$. □

**Exercise 2.31.** Define a contractible space.

*Hint.* Contract the identity map. Specify a terminal point in the space and a cylinder map beginning at its identity; contracting just one map into the space is not enough. □

*Solution.* $X$ is contractible when $\mathrm{id}_X$ is homotopic to a constant map. □

**Exercise 2.32.** Show $\mathbb{R}^n$ is contractible.

*Hint.* Contract to the origin. Look for a scalar multiple of $x$ with coefficients 1 and 0 at the two times; the ambient vector-space operations give joint continuity. □

*Solution.* Use $H(x, t) = (1 - t)x$. □

**Exercise 2.33.** Show a convex set is contractible.

*Hint.* Choose any point $c_0$ in it. The statement requires a nonempty convex set. Once $c_0$ is chosen, use the segment from each $x$ to $c_0$, rather than between arbitrary pairs of points. □

*Solution.* The straight-line homotopy $(1 - t)x + tc_0$ remains in the convex set. □

**Exercise 2.34.** Show a star-shaped set is contractible.

*Hint.* Use its star center. Only segments ending at the star center are guaranteed to lie in the set. Do not assume the stronger hypothesis of convexity. □

*Solution.* The segment from each $x$ to the center lies in the set, giving the usual linear contraction. □

**Exercise 2.35.** Define a retraction $r : X \to A$.

*Hint.* What must happen on $A$? Restrict $r$ to the subspace with its inherited topology. A map merely taking values in $A$ need not leave its points unchanged. □

*Solution.* It must satisfy $r(a) = a$ for all $a \in A$. ☐

**Exercise 2.36.** Translate the retraction condition using the inclusion $i : A \hookrightarrow X$.

*Hint.* Compose $r$ after $i$. Track an element $a \in A$ through the inclusion and back. The other composite has domain $X$, so it expresses a different condition. ☐

*Solution.* $r \circ i = \mathrm{id}_A$. ☐

**Exercise 2.37.** What additional condition turns a retract into a deformation retract?

*Hint.* Compare $i \circ r$ with the identity of $X$. The proposed homotopy must live in $X$; requiring its image to stay in $A$ would generally exclude the initial identity slice. ☐

*Solution.* One requires $i \circ r \simeq \mathrm{id}_X$. ☐

**Exercise 2.38.** What makes a deformation retract strong?

*Hint.* The subspace should remain fixed during the homotopy. Write an equation on $A \times I$, not just at the two times. This is the extra relative condition beyond the chapter's deformation-retract convention. ☐

*Solution.* The deformation from $\mathrm{id}_X$ to $i \circ r$ must be relative to $A$. ☐

**Exercise 2.39.** Prove a deformation retract has the same homotopy type as the ambient space.

*Hint.* Use inclusion and retraction as the two maps. Identify which composite is already an identity and which is supplied by the deformation. These are exactly the two homotopy-inverse tests. ☐

*Solution.* They satisfy $ri = \mathrm{id}_A$ and $ir \simeq \mathrm{id}_X$, so they are homotopy inverses. ☐

**Exercise 2.40.** Write the radial retraction $\mathbb{R}^n \setminus \{0\} \to S^{n-1}$.

*Hint.* Normalize the vector. Require both unit norm and agreement on already-unit vectors; also locate the excluded point where the normalization formula is undefined. ☐

*Solution.* $r(x) = x/\|x\|$. ☐

**Exercise 2.41.** Why does the radial deformation avoid the origin?

*Hint.* Inspect the scalar multiplying $x$. Write the coefficient as a convex combination of the positive numbers 1 and $1/\|x\|$, and check the endpoint cases as well. ☐

*Solution.* The coefficient $(1 - t) + t/\|x\|$ is positive, so a nonzero $x$ remains nonzero. ☐

**Exercise 2.42.** Show the radial deformation fixes $S^{n-1}$.

*Hint.* Set $\|x\| = 1$. Substitute the unit-norm condition into the entire time-dependent coefficient, not only into the terminal retraction. ☐

*Solution.* The scalar coefficient becomes $(1 - t) + t = 1$, so $H(x, t) = x$. ☐

**Exercise 2.43.** Show $S^1 \times I \simeq S^1$.

*Hint.* Collapse the interval coordinate. Keep the $S^1$ coordinate unchanged, shrink the interval coordinate, and check that the bottom circle is stationary throughout. ☐

*Solution.* The homotopy $H((z,s),t) = (z,(1-t)s)$ strongly deformation retracts onto $S^1 \times \{0\} \cong S^1$. □

**Exercise 2.44.** Why is a contractible space path connected?

*Hint.* Follow each point along its contraction path. Join a point to the contraction center, then traverse another such path backwards. Check the common value when the two paths are pasted. □

*Solution.* Every point is joined by the contraction to the same terminal point, so any two points can be joined through that point. □

**Exercise 2.45.** Give a path-connected space that is not contractible.

*Hint.* The basic example is one-dimensional and closed. Compare the degree of the identity on a circle with that of a constant map; this gives the obstruction missing from path connectedness alone. □

*Solution.* $S^1$. □

**Exercise 2.46.** Give homotopy-equivalent spaces with different cardinalities.

*Hint.* Use a contractible continuum and a point. Use the contraction to construct maps in both directions, then compare the numbers of points; a homotopy equivalence need not be bijective. □

*Solution.* The interval $I$ and a one-point space are homotopy equivalent but have different cardinalities. □

**Exercise 2.47.** Does homotopy equivalence preserve dimension?

*Hint.* Compare $D^n$ with a point. Choose $n > 0$, exhibit the contraction, and distinguish the dimension of a disk from that of the space to which it contracts. □

*Solution.* No. Every disk $D^n$ is homotopy equivalent to a point. □

**Exercise 2.48.** If $X \simeq Y$ and $Y \simeq Z$, what maps exhibit $X \simeq Z$?

*Hint.* Compose the forward maps and compose the backward maps. For the return composite, write $gvuf$ and replace $vu$ by its homotopic identity. Treat $ufgv$ in the other order similarly. □

*Solution.* If $f : X \to Y$, $g : Y \to X$, $u : Y \to Z$, $v : Z \to Y$, use $uf : X \to Z$ and $gv : Z \to X$. □

**Exercise 2.49.** Why is a circle with a contractible whisker homotopy equivalent to the circle?

*Hint.* Collapse the whisker to its attachment point. Parametrize the attached interval with its attachment point at 0. Check that linear shrinking agrees with the stationary circle at the identified endpoint. □

*Solution.* That collapse is a strong deformation retraction onto the circle. □

**Exercise 2.50.** What is a homotopy invariant?

*Hint.* It cannot distinguish homotopy-equivalent spaces. Specify the algebraic object and the equivalence notion used to compare its values, for example isomorphism of homology groups. □

*Solution.* It is a construction whose value is isomorphic for homotopy-equivalent spaces. □

**Exercise 2.51.** How can a homotopy invariant prove $X \not\simeq Y$?

*Hint.* Compare the invariant values. Argue contrapositively: assume a homotopy equivalence and apply the invariant. The resulting forced isomorphism must contradict the computed values. □

*Solution.* If the invariant values are not isomorphic, homotopy invariance implies the spaces cannot be homotopy equivalent. □

## 2.16 Solved problem dossiers

**Problem 2.52** (Interval versus point)**.** Write explicit homotopy-inverse maps between $I$ and a point.

*Solution.* Let $p : I \to \{*\}$ be unique and $i(*) = 0$. Then $pi$ is the identity of the point and $ip$ is the constant map at 0, homotopic to $\mathrm{id}_I$ via $H(s,t) = (1-t)s$. □

**Problem 2.53** (Annulus collapse)**.** Construct a strong deformation retraction of $\{x : 1 \leq \|x\| \leq 2\}$ onto $S^1$.

*Solution.* Use
$$H(x,t) = \left((1-t) + \frac{t}{\|x\|}\right) x.$$
Its radius is $(1-t)\|x\| + t$, which remains in $[1,2]$, and unit vectors are fixed. □

**Problem 2.54** (Punctured plane)**.** Determine the homotopy type of $\mathbb{R}^2 \setminus \{0\}$.

*Solution.* Radial normalization gives a strong deformation retraction onto $S^1$, so $\mathbb{R}^2 \setminus \{0\} \simeq S^1$. □

**Problem 2.55** (Punctured three-space)**.** Determine the homotopy type of $\mathbb{R}^3 \setminus \{0\}$.

*Solution.* The same radial construction retracts it strongly onto $S^2$. □

**Problem 2.56** (Cylinder core)**.** Show $S^1 \times [-1,1]$ has the homotopy type of $S^1$.

*Solution.* Collapse the second coordinate linearly: $H((z,s),t) = (z,(1-t)s)$. □

**Problem 2.57** (Whisker graph)**.** Let $X = S^1 \cup_p I$ attach $0 \in I$ to $p \in S^1$. Give a strong deformation retraction $X \to S^1$.

*Solution.* Fix every point of $S^1$, and send a whisker point $s \in I$ to $(1-t)s$ along the whisker. At $t = 1$ the entire whisker is at $p$. □

**Problem 2.58** (Product of contractible spaces)**.** Prove $X \times Y$ is contractible if $X$ and $Y$ are contractible.

*Solution.* Take contractions $H$ of $X$ and $K$ of $Y$, and define $L((x,y),t) = (H(x,t), K(y,t))$. □

**Problem 2.59** (Homotopy type of a strip)**.** Determine the homotopy type of $\mathbb{R} \times [0, 1]$.

*Solution.* The strip is convex, hence contractible and therefore homotopy equivalent to a point. □

**Problem 2.60** (Retract versus deformation retract)**.** Explain logically why the equation $ri = \mathrm{id}_A$ alone does not establish $A \simeq X$.

*Solution.* Homotopy equivalence requires information about both composites. The retraction equation controls $ri$, but without $ir \simeq \mathrm{id}_X$ there is no homotopy inverse condition in the other direction. □

**Problem 2.61** (Transporting an invariant)**.** Suppose $F$ is a homotopy invariant and $A$ deformation retracts from $X$. What relation holds between $F(A)$ and $F(X)$?

*Solution.* Since $A \simeq X$, homotopy invariance gives $F(A) \cong F(X)$. □

**Problem 2.62** (Noncontractibility strategy)**.** What would be enough to prove that $S^1$ is not contractible?

*Solution.* It is enough to find any homotopy invariant that takes different values on $S^1$ and on a point. VIII/03 introduces degree as an early obstruction, while later chapters supply fundamental group and homology. □

**Problem 2.63** (Bridge to VIII/03)**.** Why does the notion of homotopy invariant naturally lead to degree?

*Solution.* To distinguish maps and spaces up to homotopy, one seeks quantities unchanged by deformation. Degree assigns an integer to maps between oriented spheres (or, more generally, oriented closed manifolds) and is invariant under homotopy, so it supplies the first quantitative obstruction in this volume. □

## 2.17 Supplementary worked examples

**Example 2.64** (Contracting a star-shaped set)**.** If $X \subset \mathbb{R}^n$ is star-shaped about $x_0$, $H(x,t) = (1-t)x + tx_0$ contracts $X$ to $x_0$. Thus star-shaped spaces are contractible even when they are not convex.

**Example 2.65** (A deformation retract)**.** The punctured plane deformation retracts onto $S^1$ by $H(x,t) = ((1-t) + t/\|x\|)x$. Hence $\mathbb{R}^2 \setminus \{0\}$ and $S^1$ have the same homotopy type without being homeomorphic.

**Example 2.66** (Projection from a cylinder)**.** The projection $S^1 \times I \to S^1$ and inclusion $S^1 \hookrightarrow S^1 \times I$ are homotopy inverses. One composite is the identity and the other is deformed to the identity by sliding the interval coordinate.

## 2.18 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 2.67** (Homotopy inverse equations)**.** State the equations defining a homotopy equivalence $f : X \to Y$ with inverse candidate $g$.

*Hint.* The composites need only be homotopic to identities. ☐

*Solution.* Require $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. ☐

**Exercise 2.68** (Contract a disk)**.** Give an explicit contraction of $D^n$ to the origin.

*Hint.* Use scalar multiplication. ☐

*Solution.* $H(x,t) = (1-t)x$ stays in $D^n$ and ends at 0. ☐

**Exercise 2.69** (Cylinder retraction)**.** Write a deformation retraction of $X \times I$ onto $X \times \{0\}$.

*Hint.* Move only the interval coordinate. ☐

*Solution.* $H((x,s),t) = (x,(1-t)s)$ fixes $X \times \{0\}$ pointwise. ☐

**Exercise 2.70** (Retract maps)**.** For a retraction $r : X \to A$ with inclusion $i : A \hookrightarrow X$, compute $r \circ i$.

*Hint.* A retraction is the identity on $A$. ☐

*Solution.* $r \circ i = \mathrm{id}_A$. ☐

**Exercise 2.71** (Homotopy type transfer)**.** If $X \simeq Y$ and $Y \simeq Z$, what should be the homotopy type relation between $X$ and $Z$?

*Hint.* Compose homotopy equivalences. ☐

*Solution.* $X \simeq Z$; composition of homotopy equivalences is a homotopy equivalence. ☐

## Proofs

**Exercise 2.72** (Contractible implies identity null)**.** Prove $X$ is contractible iff $\mathrm{id}_X$ is null-homotopic.

*Hint.* A contraction is exactly such a homotopy. □

*Solution.* A contraction joins $\mathrm{id}_X$ to a constant; conversely a null-homotopy of the identity is a contraction. □

**Exercise 2.73** (Deformation retract gives equivalence)**.** Prove a deformation retract $A \subset X$ is homotopy equivalent to $X$.

*Hint.* Use inclusion and retraction. □

*Solution.* $r \circ i = \mathrm{id}_A$, while $i \circ r \simeq \mathrm{id}_X$ by the deformation retraction. □

**Exercise 2.74** (Equivalence transitivity)**.** Prove the composite of two homotopy equivalences is a homotopy equivalence.

*Hint.* Compose chosen homotopy inverses in reverse order. □

*Solution.* If $g$ and $k$ are inverse candidates for $f$ and $h$, then $g \circ k$ is an inverse candidate for $h \circ f$; composition compatibility transports the homotopies. □

**Exercise 2.75** (Maps into contractible target)**.** Prove any two maps $X \to Y$ are homotopic when $Y$ is contractible.

*Hint.* First deform each map to the contraction point. □

*Solution.* Postcompose the contraction of $Y$ with each map, obtaining homotopies to the same constant map; concatenate one with the reverse of the other. □

## Counterexamples and hypothesis tests

**Exercise 2.76** (Equivalent means homeomorphic)**.** Test: homotopy equivalent spaces must be homeomorphic.

*Hint.* Compare $S^1$ and $S^1 \times I$. □

*Solution.* False. They are homotopy equivalent but have different dimensions/boundary behavior. □

**Exercise 2.77** (Retract means deformation retract)**.** Test: every retract is automatically a deformation retract.

*Hint.* A retraction alone supplies no homotopy from $i \circ r$ to the identity. □

*Solution.* False; deformation retraction is strictly stronger. □

**Exercise 2.78** (Contractible means one point)**.** Test: every contractible space is a singleton.

*Hint.* Use $D^n$ or an interval. □

*Solution.* False. Contractibility is a homotopy-type statement, not equality of underlying sets. □

## Applications and investigations

**Exercise 2.79** (Model reduction). Why is replacing a space by a deformation retract useful before computing invariants?

*Hint.* Homotopy invariants are unchanged. ☐

*Solution.* The retract often has fewer cells or simpler geometry while preserving homotopy type, so later algebraic invariants can be computed on the simpler model. ☐

**Exercise 2.80** (Robot configuration intuition). Interpret contractibility as absence of homotopy-level obstacles in a configuration region.

*Hint.* Every configuration can be continuously moved to one reference configuration. ☐

*Solution.* A contraction supplies a coherent motion of all points toward one reference point; this does not encode metric feasibility, but it certifies trivial homotopy type. ☐

## Challenge problems

**Exercise 2.81** (Punctured plane). Use a deformation retraction to reduce the homotopy type of $\mathbb{R}^2 \setminus \{0\}$.

*Hint.* Normalize the radius. ☐

*Solution.* Radial deformation retracts the punctured plane onto $S^1$, so all homotopy invariants can be computed on the circle. ☐

**Exercise 2.82** (Two-sided synthesis). Given explicit maps $f : X \to Y$ and $g : Y \to X$, list the exact data needed to certify a homotopy equivalence.

*Hint.* Maps alone are insufficient. ☐

*Solution.* One must provide continuous homotopies $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$, with any required based or relative conditions checked separately. ☐

# Chapter 3

# Degree of Maps

Homotopy theory becomes effective when a deformation-invariant quantity can be computed. Degree is the first such integer-valued invariant in this volume. For a map between oriented spaces of the same dimension, degree records the signed number of times the domain covers the target. It detects maps that cannot be deformed into one another and provides immediate applications to spheres and antipodal maps.

## 3.1 Orientation and the signed-counting idea

For a smooth map

$$f : M^n \to N^n$$

between oriented $n$-manifolds, a point $x \in M$ is regular if the derivative

$$df_x : T_xM \to T_{f(x)}N$$

is an isomorphism. At such a point, $df_x$ either preserves or reverses orientation.

**Definition 3.1** (Local sign)**.** At a regular point $x$, define

$$\operatorname{sgn}_x(f) = \begin{cases} +1, & df_x \text{ preserves orientation,} \\ -1, & df_x \text{ reverses orientation.} \end{cases}$$

## 3.2 Degree by a regular value

**Definition 3.2** (Degree for a smooth map)**.** Let $M, N$ be connected compact oriented $n$-manifolds without boundary, and let $f : M \to N$ be smooth. If $y \in N$ is a regular value, define

$$\deg(f) = \sum_{x \in f^{-1}(y)} \operatorname{sgn}_x(f).$$

Compactness makes the regular fiber finite.

**Theorem 3.3** (Independence of regular value)**.** *The signed sum above is independent of the chosen regular value $y$.*

This theorem is the key reason degree is global rather than merely local. A full proof can be given by oriented cobordism of inverse images of a path between regular values. Later homology theory will provide a shorter algebraic formulation.

## 3.3 Continuous maps

**Theorem 3.4** (Continuous degree). *Every continuous map $f : M \to N$ between connected closed oriented $n$-manifolds has an integer degree. It agrees with the signed regular-value count for smooth maps and is invariant under homotopy.*

One construction approximates a continuous map by a smooth map and proves that different sufficiently close approximations are homotopic. Later, after homology is available, degree will be characterized by the induced map on the top-dimensional fundamental class.

## 3.4 Normalization

**Proposition 3.5** (Identity and constant map). *For a connected oriented closed $n$-manifold,*

$$\deg(\mathrm{id}) = 1.$$

*If $n > 0$, every constant map has degree $0$.*

*Proof.* The identity has one preimage of a regular value and preserves orientation. A constant map misses every point other than its image, so choosing a regular value outside that image gives an empty signed sum. □

## 3.5 Homotopy invariance

**Theorem 3.6** (Homotopy invariance of degree). *If*

$$f \simeq g : M \to N,$$

*then*

$$\deg(f) = \deg(g).$$

**Corollary 3.7** (Nonzero degree obstructs null-homotopy). *If $n > 0$ and $\deg(f) \neq 0$, then $f$ is not null-homotopic.*

*Proof.* A null-homotopic map is homotopic to a constant map, whose degree is 0. □

## 3.6 Multiplicativity

**Theorem 3.8** (Degree of a composite). *For maps of connected closed oriented $n$-manifolds*

$$M \xrightarrow{f} N \xrightarrow{g} P,$$

*one has*

$$\deg(g \circ f) = \deg(g)\deg(f).$$

**Corollary 3.9** (Degree of a homotopy equivalence). *If $f : M \to N$ is a homotopy equivalence between connected closed oriented $n$-manifolds, then*

$$\deg(f) = \pm 1.$$

*Proof.* For a homotopy inverse $g$,

$$1 = \deg(\mathrm{id}_M) = \deg(g \circ f) = \deg(g)\deg(f).$$

The only units in $\mathbb{Z}$ are $\pm 1$. □

## 3.7 Degree on the circle

Identify

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

**Theorem 3.10** (Power map). *For*

$$p_k : S^1 \to S^1, \qquad p_k(z) = z^k,$$

*with $k \in \mathbb{Z}$,*

$$\deg(p_k) = k.$$

For $k > 0$, a regular value has $k$ preimages, each with positive local sign. For $k < 0$, complex conjugation reverses orientation and contributes the negative sign.

**Example 3.11.**

$$\deg(z \mapsto z^5) = 5, \qquad \deg(z \mapsto z^{-3}) = -3.$$

## 3.8 Degree classifies circle maps up to homotopy

**Theorem 3.12** (Classification of maps $S^1 \to S^1$). *Two continuous maps*

$$f, g : S^1 \to S^1$$

*are homotopic if and only if*

$$\deg(f) = \deg(g).$$

This is an early instance of a complete homotopy classification by an algebraic invariant. Later the fundamental group identifies the same integer as a winding number.

## 3.9 Orientation-reversing maps

**Proposition 3.13** (Reflection). *A reflection of $S^n$ across a hyperplane has degree $-1$.*

**Proposition 3.14** (Orientation-preserving homeomorphism). *An orientation-preserving homeomorphism of an oriented closed manifold has degree $+1$; an orientation-reversing one has degree $-1$.*

## 3.10 The antipodal map

Let

$$a : S^n \to S^n, \qquad a(x) = -x.$$

**Theorem 3.15** (Degree of the antipodal map). *The antipodal map on $S^n$ has degree*

$$\deg(a) = (-1)^{n+1}.$$

*Proof.* The antipodal map is the restriction of $-I : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$. Its determinant is

$$\det(-I) = (-1)^{n+1}.$$

The induced orientation sign on the sphere agrees with this value. □

Thus the antipodal map has degree $+1$ for odd-dimensional spheres and degree $-1$ for even-dimensional spheres.

## 3.11 No-retraction preview

Degree can obstruct retractions.

**Theorem 3.16** (No retraction of a ball onto its boundary)**.** *For $n \geq 1$, there is no retraction*

$$r : D^{n+1} \to S^n.$$

*Proof.* Suppose $r$ existed and let $i : S^n \hookrightarrow D^{n+1}$ be inclusion. Then

$$r \circ i = \mathrm{id}_{S^n}.$$

Because $D^{n+1}$ is contractible, $i$ is null-homotopic as a map into the disk, so $r \circ i$ is null-homotopic by postcomposition. But the identity of $S^n$ has degree 1, while a null-homotopic map has degree 0, a contradiction. ☐

## 3.12 Surjectivity consequence

**Proposition 3.17** (Nonzero degree implies surjective)**.** *If $f : M \to N$ is a continuous map between connected closed oriented $n$-manifolds and*

$$\deg(f) \neq 0,$$

*then $f$ is surjective.*

*Proof.* If $f$ missed a point $y \in N$, then in the smooth regular-value picture a nearby regular value outside the image would have empty preimage and hence degree 0. Equivalently, a map missing a point factors through a punctured target, which has no top-dimensional degree contribution. ☐

## 3.13 Degree and homotopy equivalence of spheres

**Corollary 3.18.** *A self-map $f : S^n \to S^n$ of degree other than $\pm 1$ cannot be a homotopy equivalence.*

For example, $z \mapsto z^k$ on $S^1$ is a homotopy equivalence exactly when $k = \pm 1$.

## 3.14 Regular-value computation protocol

For a smooth map $f : M^n \to N^n$, a practical degree computation is:

1. choose a regular value $y \in N$;
2. solve $f(x) = y$;
3. determine whether each $df_x$ preserves or reverses orientation;
4. sum the local signs.

*Warning* 3.19 (Unsigned counting is not degree). The number $|f^{-1}(y)|$ is generally not homotopy invariant. Oppositely oriented sheets can be created or cancelled in pairs. Degree is the *signed* count.

## 3.15 Exercises

**Exercise 3.20.** Define the local sign of a smooth map at a regular point.

*Hint.* Inspect whether the derivative preserves orientation. Choose positively oriented tangent bases in source and target; regularity makes the derivative an isomorphism, so its determinant has a nonzero sign. ☐

*Solution.* It is $+1$ if $df_x$ preserves orientation and $-1$ if it reverses orientation. ☐

**Exercise 3.21.** Write the regular-value formula for degree.

*Hint.* Sum local signs over the inverse image. Keep the chosen orientations and a regular value fixed. Compactness and regularity make the inverse image finite, while each summand retains its orientation sign. ☐

*Solution.*

$$\deg(f) = \sum_{x \in f^{-1}(y)} \operatorname{sgn}_x(f).$$

☐

**Exercise 3.22.** What is the degree of the identity map?

*Hint.* A regular value has one preimage. Check the derivative at that unique preimage in compatible oriented charts; uniqueness of the preimage alone does not determine its sign. ☐

*Solution.* 1. ☐

**Exercise 3.23.** What is the degree of a constant map $S^n \to S^n$ for $n > 0$?

*Hint.* Choose a point not equal to the constant value. The chosen value has empty inverse image and is vacuously regular. Interpret the corresponding signed sum, using the positive-dimensional hypothesis. ☐

*Solution.* 0. ☐

**Exercise 3.24.** If $f \simeq g$, how are their degrees related?

*Hint.* Degree is a homotopy invariant. For oriented closed manifolds, compare the homomorphisms induced on top homology by the two maps; a homotopy makes these homomorphisms equal. ☐

*Solution.* $\deg(f) = \deg(g)$. ☐

**Exercise 3.25.** Why can a nonzero-degree map not be null-homotopic?

*Hint.* Compare with a constant map. A null-homotopy identifies the induced top-dimensional homology map with that of a map factoring through a point. Use positive dimension here. ☐

*Solution.* A null-homotopic map has the degree of a constant map, namely 0. ☐

**Exercise 3.26.** State the multiplicativity formula.

*Hint.* Take the degree of a composite. Track a chosen fundamental class through $f_*$ and then $g_*$; the two integer multipliers occur successively, not additively. ☐

*Solution.* $\deg(g \circ f) = \deg(g) \deg(f)$. ☐

**Exercise 3.27.** What degrees are possible for a homotopy equivalence of oriented closed manifolds?

*Hint.* Its degree must be a unit in $\mathbb{Z}$. Introduce a homotopy inverse and take the degree of its composite with $f$. Ask which pairs of integers have product 1. $\square$

*Solution.* Only $+1$ and $-1$. $\square$

**Exercise 3.28.** Compute $\deg(z \mapsto z^7)$ on $S^1$.

*Hint.* Use the power-map theorem. Use $1 \in S^1$ as a regular value and count its seventh roots; check the direction of the angle map near each root. $\square$

*Solution.* 7. $\square$

**Exercise 3.29.** Compute $\deg(z \mapsto z^{-4})$ on $S^1$.

*Hint.* Negative exponent reverses winding orientation. Write the map in angle coordinates and track the sign of the angular derivative, rather than counting four preimages without signs. $\square$

*Solution.* $-4$. $\square$

**Exercise 3.30.** When are two maps $S^1 \to S^1$ homotopic?

*Hint.* Degree is complete in this case. Separate necessity, from homotopy invariance, from sufficiency, which uses the classification theorem specifically for circle maps. $\square$

*Solution.* Exactly when they have the same degree. $\square$

**Exercise 3.31.** What is the degree of a reflection of $S^n$?

*Hint.* A reflection reverses orientation. Extend the reflection to an ambient linear map with one negative eigenvalue, and compare its determinant with the induced boundary orientation sign. $\square$

*Solution.* $-1$. $\square$

**Exercise 3.32.** Compute the degree of the antipodal map on $S^1$.

*Hint.* Use $(-1)^{n+1}$ with $n = 1$. Distinguish the circle's dimension from that of its ambient plane when substituting in the antipodal formula. $\square$

*Solution.* $+1$. $\square$

**Exercise 3.33.** Compute the degree of the antipodal map on $S^2$.

*Hint.* Use $(-1)^{n+1}$ with $n = 2$. Count sign reversals in the three-dimensional ambient space. Reversing all ambient coordinates is not the same as reversing two tangent coordinates in isolation. $\square$

*Solution.* $-1$. $\square$

**Exercise 3.34.** Compute the degree of the antipodal map on $S^3$.

*Hint.* Use parity. The relevant determinant is that of $-I$ on $\mathbb{R}^4$. Determine the parity of its number of negative eigenvalues. $\square$

*Solution.* $+1$. $\square$

**Exercise 3.35.** If $\deg f = 2$ and $\deg g = -3$, compute $\deg(g \circ f)$.

*Hint.* Multiply. Apply multiplicativity before doing the integer arithmetic, and keep the order of the named composite clear. ☐

*Solution.* $-6$. ☐

**Exercise 3.36.** Can a degree-2 self-map of $S^n$ be a homotopy equivalence?

*Hint.* A homotopy equivalence has degree $\pm 1$. Assume a homotopy inverse exists and derive an equation $2k = 1$ for an integer $k$; this exposes the obstruction directly. ☐

*Solution.* No. ☐

**Exercise 3.37.** Why does nonzero degree force surjectivity?

*Hint.* A missed regular value has no preimages. In the smooth case, a missed point is a regular value with empty fiber. For a continuous sphere map, factor through the contractible punctured sphere instead. ☐

*Solution.* If the map missed a regular value, the signed inverse-image sum would be empty, forcing degree 0. ☐

**Exercise 3.38.** Why is ordinary preimage cardinality not a homotopy invariant?

*Hint.* Pairs with opposite local signs can appear or disappear. Compare the change in unsigned count with the contribution $(+1) + (-1)$ of a newly created pair; only one of these quantities cancels. ☐

*Solution.* Homotopies can create or cancel pairs of preimages whose local signs sum to zero, changing cardinality while preserving degree. ☐

**Exercise 3.39.** What degree would a retraction argument force on $r \circ i : S^n \to S^n$?

*Hint.* A retraction satisfies $ri = \mathrm{id}$. Apply degree to the strict retraction equation on $S^n$, with the same orientation used for both copies of the sphere. ☐

*Solution.* It would have degree 1. ☐

**Exercise 3.40.** Why is $r \circ i$ null-homotopic if $i : S^n \hookrightarrow D^{n+1}$?

*Hint.* The disk is contractible. Contract the inclusion inside the disk first, then postcompose the entire homotopy by the hypothetical retraction; the intermediate maps need not land on the boundary before postcomposition. ☐

*Solution.* The inclusion $i$ is homotopic in the disk to a constant map; postcomposing that homotopy by $r$ makes $ri$ null-homotopic. ☐

**Exercise 3.41.** What contradiction proves the no-retraction theorem?

*Hint.* Compare degrees 1 and 0. Compute the degree of the same composite in two ways: from its identity property and from its factorization through a contractible space. ☐

*Solution.* A retraction would make $ri = \mathrm{id}_{S^n}$ have degree 1, while the contraction through the disk would make $ri$ null-homotopic and hence degree 0. ☐

**Exercise 3.42.** If $\deg f = 1$, must $f$ be a homeomorphism?

*Hint.* Degree measures global signed covering, not injectivity. Try a circle map $e^{it} \mapsto e^{i(t+2\sin t)}$. Its angle increases by $2\pi$ over one period, but its derivative changes sign. ☐

*Solution.* No. Degree 1 does not imply injectivity or local homeomorphism behavior. ☐

**Exercise 3.43.** What topic does the antipodal degree formula prepare for next?

*Hint.* Look at VIII/04. Compare the parity-dependent degree of antipodes with the identity degree; this specifies a concrete obstruction to test in VIII/04 rather than just a chapter reference. ☐

*Solution.* It prepares the study of spheres and antipodal maps, including parity-sensitive obstructions and Borsuk–Ulam-type phenomena. ☐

## 3.16 Solved problem dossiers

**Problem 3.44** (Power map). Compute the degree of $p_6(z) = z^6$.

*Solution.* A regular value has six preimages, each orientation preserving, so $\deg(p_6) = 6$. ☐

**Problem 3.45** (Negative power). Compute the degree of $q(z) = z^{-5}$.

*Solution.* The map winds five times with reversed orientation, so $\deg(q) = -5$. ☐

**Problem 3.46** (Composition). Let $f(z) = z^3$ and $g(z) = z^{-2}$. Compute the degree of $g \circ f$.

*Solution.*

$$\deg(g \circ f) = (-2)(3) = -6.$$

Indeed $g(f(z)) = z^{-6}$. ☐

**Problem 3.47** (Homotopy obstruction). Can $z \mapsto z^2$ be homotopic to $z \mapsto z^3$?

*Solution.* No. Their degrees are 2 and 3, and homotopic maps have equal degree. ☐

**Problem 3.48** (Null-homotopy test). Can $z \mapsto z^{-1}$ be null-homotopic as a map $S^1 \to S^1$?

*Solution.* No. Its degree is $-1$, whereas a null-homotopic circle map has degree 0. ☐

**Problem 3.49** (Antipodal parity). For which $n$ does the antipodal map $S^n \to S^n$ have degree $+1$?

*Solution.* When $n$ is odd, because $(-1)^{n+1} = +1$. ☐

**Problem 3.50** (Antipodal parity II). For which $n$ does the antipodal map have degree $-1$?

*Solution.* When $n$ is even. ☐

**Problem 3.51** (No retraction). Give the degree contradiction for a hypothetical retraction $D^2 \to S^1$.

*Solution.* For inclusion $i : S^1 \hookrightarrow D^2$, a retraction $r$ gives $ri = \mathrm{id}_{S^1}$, degree 1. But $i$ contracts through $D^2$, so $ri$ is null-homotopic, degree 0. Contradiction. ☐

**Problem 3.52** (Surjectivity). Suppose $f : S^2 \to S^2$ has degree 4. What can be said about its image?

*Solution.* It must be all of $S^2$, since nonzero degree implies surjectivity. ☐

**Problem 3.53** (Homotopy equivalence test)**.** Suppose $f : S^3 \to S^3$ has degree $-1$. Does degree alone prove that $f$ is a homotopy equivalence?

*Solution.* Degree $-1$ is necessary and, for sphere self-maps, it is sufficient because sphere maps of the same degree are homotopic and a degree $-1$ reflection is a homeomorphism. In general manifolds, degree $\pm 1$ alone need not imply homotopy equivalence. □

**Problem 3.54** (Classification on the circle)**.** Classify the homotopy class of $f(z) = \overline{z}^4$.

*Solution.* Since $\overline{z} = z^{-1}$ on $S^1$, $f(z) = z^{-4}$, so $\deg f = -4$. Its homotopy class is the unique circle-map class of degree $-4$. □

**Problem 3.55** (Bridge to VIII/04)**.** Explain what the formula $\deg(a) = (-1)^{n+1}$ says qualitatively about antipodal maps.

*Solution.* Their homotopy behavior depends on the parity of the sphere dimension: odd-dimensional spheres give degree $+1$, even-dimensional spheres give degree $-1$. VIII/04 turns this parity into concrete sphere and antipodal-map phenomena. □

## 3.17 Supplementary worked examples

**Example 3.56** (Power maps on the circle)**.** The map $f_k : S^1 \to S^1$, $f_k(z) = z^k$, has degree $k$. Positive $k$ wraps counterclockwise $k$ times, negative $k$ reverses orientation and wraps $|k|$ times, and $k = 0$ is constant.

**Example 3.57** (Degree of the antipodal map)**.** On $S^n$, the antipodal map $A(x) = -x$ has degree $(-1)^{n+1}$. This follows from viewing $A$ as the restriction of the linear map $-I$ on $\mathbb{R}^{n+1}$, whose determinant is $(-1)^{n+1}$.

**Example 3.58** (Composition multiplies degree)**.** For maps $S^n \xrightarrow{f} S^n \xrightarrow{g} S^n$, induced maps on top homology are multiplication by $\deg f$ and $\deg g$, hence $\deg(g \circ f) = \deg g \deg f$.

## 3.18 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 3.59** (Identity degree)**.** Compute the degree of $\mathrm{id}_{S^n}$.

*Hint.* Look at its action on the fundamental class. □

*Solution.* The identity induces the identity on $H_n(S^n) \cong \mathbb{Z}$, so its degree is 1. □

**Exercise 3.60** (Constant degree)**.** Compute the degree of a constant map $S^n \to S^n$ for $n > 0$.

*Hint.* It factors through a point. □

*Solution.* Its induced map on $H_n$ is zero, so the degree is 0. □

**Exercise 3.61** (Circle power map)**.** Compute $\deg(z \mapsto z^{-3})$ on $S^1$.

*Hint.* Count signed windings. □

*Solution.* The map winds three times with reversed orientation, so the degree is $-3$. □

**Exercise 3.62** (Composite)**.** If $\deg f = -2$ and $\deg g = 5$, compute $\deg(g \circ f)$.

*Hint.* Use multiplicativity. □

*Solution.* The degree is $5(-2) = -10$. □

**Exercise 3.63** (Antipodal parity)**.** Compute the degree of the antipodal map on $S^4$.

*Hint.* Use $(-1)^{n+1}$. □

*Solution.* $(-1)^5 = -1$. □

### Proofs

**Exercise 3.64** (Homotopy invariance)**.** Prove homotopic maps $S^n \to S^n$ have the same degree.

*Hint.* Homotopic maps induce the same map on homology. □

*Solution.* Since degree is the integer describing the induced map on $H_n(S^n)$, homotopy invariance of homology gives equal degrees. □

**Exercise 3.65** (Multiplicativity)**.** Prove $\deg(g \circ f) = \deg g \deg f$.

*Hint.* Write each induced homomorphism as multiplication by an integer. □

*Solution.* Functoriality gives $(g \circ f)_* = g_* f_*$; composing multiplication by the two degrees multiplies the integers. □

**Exercise 3.66** (Nonzero degree not null)**.** Prove a map $S^n \to S^n$ with nonzero degree is not null-homotopic.

*Hint.* A constant map has degree zero. □

*Solution.* If it were null-homotopic, homotopy invariance would force its degree to equal that of a constant map, namely zero. □

**Exercise 3.67** (Homotopy equivalence degree)**.** Prove a homotopy equivalence $S^n \to S^n$ has degree $\pm 1$.

*Hint.* Use a homotopy inverse and multiplicativity. □

*Solution.* If $g$ is a homotopy inverse, $\deg g \deg f = 1$ in $\mathbb{Z}$, so both degrees are units, hence $\pm 1$. □

### Counterexamples and hypothesis tests

**Exercise 3.68** (Degree determines map pointwise)**.** Test: two maps of equal degree are equal as functions.

*Hint.* Degree is a homotopy invariant, much coarser than pointwise data. □

*Solution.* False. Many distinct maps have the same degree. □

**Exercise 3.69** (Degree zero means constant)**.** Test: degree zero forces a map $S^n \to S^n$ to be constant.

*Hint.* It only detects a homotopy class obstruction. □

*Solution.* False pointwise; degree zero maps can be nonconstant, although in the sphere-to-sphere setting they are null-homotopic under the standard classification theorem. □

**Exercise 3.70** (Every integer occurs)**.** Test: only $-1, 0, 1$ can occur as degrees of maps $S^n \to S^n$.

*Hint.* Use wrapping maps or suspension constructions. □

*Solution.* False. Every integer occurs as the degree of a suitable sphere map. □

## Applications and investigations

**Exercise 3.71** (Surjectivity certificate)**.** Explain why nonzero degree forces a map $S^n \to S^n$ to be surjective.

*Hint.* A map missing one point factors through a contractible punctured sphere. □

*Solution.* If the image misses a point, the map factors through $S^n \setminus \{p\} \simeq \mathbb{R}^n$, hence is null-homotopic and has degree zero; contradiction. □

**Exercise 3.72** (Orientation bookkeeping)**.** Why is degree a useful compact summary of orientation and covering multiplicity?

*Hint.* Local sheets contribute signed counts. □

*Solution.* At regular values, the degree is the sum of local orientation signs, combining multiplicity and orientation into one homotopy-invariant integer. □

## Challenge problems

**Exercise 3.73** (Fixed-point style obstruction)**.** Suppose $f : S^n \to S^n$ is homotopic to the antipodal map. What is its degree?

*Hint.* Use homotopy invariance. □

*Solution.* It has degree $(-1)^{n+1}$, the degree of the antipodal map. □

**Exercise 3.74** (Degree synthesis)**.** A map $f : S^n \to S^n$ satisfies $f \circ f \simeq f$ and has nonzero degree. Determine $\deg f$.

*Hint.* Translate the homotopy relation into an integer equation. □

*Solution.* Let $d = \deg f$. Multiplicativity and homotopy invariance give $d^2 = d$. Since $d \neq 0$, $d = 1$. □

# Chapter 4

# Spheres and Antipodal Maps

Spheres are the first family on which homotopy and degree interact systematically. The involution $x \mapsto -x$ exposes a parity phenomenon, while the same geometry leads to Borsuk–Ulam obstructions, projective quotients, and the cell structures used in the next chapters.

## 4.1 Spheres, disks, and hemispheres

Write

$$S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}, \qquad D^{n+1} = \{x \in \mathbb{R}^{n+1} : \|x\| \leq 1\}.$$

Then $\partial D^{n+1} = S^n$. The upper and lower closed hemispheres of $S^n$ are copies of $D^n$ meeting in the equatorial $S^{n-1}$.

**Proposition 4.1** (Two-disk description)**.** *For $n \geq 1$, $S^n$ is obtained by gluing two copies of $D^n$ along their boundary $S^{n-1}$ by the identity.*

## 4.2 The antipodal map

**Definition 4.2** (Antipodal map)**.** The antipodal map is

$$a : S^n \to S^n, \qquad a(x) = -x.$$

It is a fixed-point-free involution: $a^2 = \mathrm{id}$.

**Theorem 4.3** (Antipodal degree)**.** *With the standard orientation,*

$$\deg(a) = (-1)^{n+1}.$$

**Corollary 4.4** (Parity obstruction)**.** *If $n$ is even, $a$ is not homotopic to the identity map of $S^n$.*

## 4.3 Odd and even maps

**Definition 4.5** (Odd and even)**.** A map $f : S^n \to \mathbb{R}^m$ is odd if $f(-x) = -f(x)$, and even if $f(-x) = f(x)$.

**Theorem 4.6** (Odd self-map theorem)**.** *Every continuous odd map $f : S^n \to S^n$ has odd degree.*

**Corollary 4.7.** *An odd self-map of $S^n$ is not null-homotopic.*

## 4.4 Borsuk–Ulam

**Theorem 4.8** (Borsuk–Ulam)**.** *For every continuous $f : S^n \to \mathbb{R}^n$, there exists $x \in S^n$ such that*

$$f(x) = f(-x).$$

**Theorem 4.9** (Equivalent no-odd-map form)**.** *There is no continuous odd map $S^n \to S^{n-1}$.*

If a map $f : S^n \to \mathbb{R}^n$ separated every antipodal pair, then

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

would be an odd map $S^n \to S^{n-1}$. This proves the equivalence of the two formulations.

## 4.5 The one-dimensional case

For $f : S^1 \to \mathbb{R}$, put $h(x) = f(x) - f(-x)$. Then $h(-x) = -h(x)$, so the intermediate value theorem on a semicircle forces a zero of $h$. Thus some antipodal pair has the same value.

## 4.6 No retraction

**Theorem 4.10** (No boundary retraction)**.** *There is no retraction $D^{n+1} \to S^n$.*

For $n \geq 1$, degree proves this: if $r$ were a retraction and $i : S^n \hookrightarrow D^{n+1}$, then $ri = \mathrm{id}$ has degree 1, while $i$ is null-homotopic in the disk and hence $ri$ would have degree 0.

## 4.7 Real projective space

**Definition 4.11** (Real projective space)**.**

$$\mathbb{R}P^n = S^n/(x \sim -x).$$

Equivalently, $\mathbb{R}P^n$ is the set of one-dimensional linear subspaces of $\mathbb{R}^{n+1}$.

Let $q : S^n \to \mathbb{R}P^n$ be the quotient map.

**Proposition 4.12** (Even-map factorization)**.** *If $f : S^n \to Y$ is continuous and even, then it factors uniquely as*

$$S^n \xrightarrow{q} \mathbb{R}P^n \xrightarrow{\bar{f}} Y.$$

## 4.8 Low-dimensional projective spaces

$$\mathbb{R}P^0 \cong \{*\}, \qquad \mathbb{R}P^1 \cong S^1.$$

Under a standard identification $\mathbb{R}P^1 \cong S^1$, the quotient map is the degree-two circle map $z \mapsto z^2$.

## 4.9 Projective filtration

There are natural inclusions

$$\mathbb{R}P^0 \subset \mathbb{R}P^1 \subset \cdots \subset \mathbb{R}P^n.$$

Moreover,

$$\mathbb{R}P^k \setminus \mathbb{R}P^{k-1} \cong \mathbb{R}^k,$$

so $\mathbb{R}P^n$ has one open cell in each dimension $0, \ldots, n$.

## 4.10 One-point compactification

Stereographic projection gives

$$S^n \setminus \{N\} \cong \mathbb{R}^n,$$

so $S^n$ is the one-point compactification of $\mathbb{R}^n$.

## 4.11 Suspension

The sphere also satisfies

$$S^{n+1} \cong \Sigma S^n.$$

This viewpoint connects sphere geometry to cones and mapping cones.

## 4.12 Bridge to cell attachment

The projective filtration can be written schematically as

$$\mathbb{R}P^n = \mathbb{R}P^{n-1} \cup e^n.$$

VIII/05 formalizes this operation as attachment of an $n$-disk along its boundary.

*Warning* 4.13 (Quotient versus subspace). The antipodal quotient identifies points; it is not an inclusion. Retraction, subspace, and quotient arguments must therefore be kept distinct.

## 4.13 Exercises

**Exercise 4.14.** Compute $a^2$ for $a(x) = -x$.

*Hint.* Apply the map twice.

Compute the two successive ambient sign changes, and check that both intermediate and final vectors still lie on the unit sphere. □

*Solution.* $a(a(x)) = x$. □

**Exercise 4.15.** Does $a$ have a fixed point on $S^n$?

*Hint.* Solve $x = -x$.

Solve the fixed-point equation in the ambient real vector space, then impose the sphere's unit-norm condition. □

*Solution.* No; that would force $x = 0$. □

**Exercise 4.16.** Find $\deg a$ on $S^4$.

*Hint.* Use $(-1)^{n+1}$.

Use the determinant of $-I$ in $\mathbb{R}^5$; the exponent counts ambient coordinates, not the sphere dimension alone. □

*Solution.* $-1$. □

**Exercise 4.17.** Find $\deg a$ on $S^5$.

*Hint.* Use parity.

There are six ambient coordinate sign reversals for $S^5$. Relate their parity to the orientation sign. □

*Solution.* 1. □

**Exercise 4.18.** When does degree rule out $a \simeq \mathrm{id}$?

*Hint.* Compare their degrees.

Use homotopy invariance to obtain a necessary equality of degrees. Equality by itself is not being used here as a general classification theorem. □

*Solution.* For even $n$. □

**Exercise 4.19.** State Borsuk–Ulam on $S^2$.

*Hint.* The target is $\mathbb{R}^2$.

Specify that the assertion concerns every continuous map and at least one pair $x, -x$, with equality in both real coordinates simultaneously. □

*Solution.* Every continuous $S^2 \to \mathbb{R}^2$ identifies an antipodal pair. □

**Exercise 4.20.** Why would an odd $S^n \to S^{n-1}$ violate Borsuk–Ulam?

*Hint.* Embed $S^{n-1}$ in $\mathbb{R}^n$.

Equal antipodal images combined with oddness would force the common image to be zero; test whether zero belongs to the unit-sphere target. □

*Solution.* Oddness prevents equal antipodal images. □

**Exercise 4.21.** Construct an odd map from an antipode-separating $f : S^n \to \mathbb{R}^n$.

*Hint.* Normalize $f(x) - f(-x)$.

First prove the denominator is nowhere zero from antipode separation. Then check continuity and that the normalized vector has unit norm. □

*Solution.* $g(x) = \frac{f(x)-f(-x)}{\|f(x)-f(-x)\|}$. □

**Exercise 4.22.** Show that this $g$ is odd.

*Hint.* Replace $x$ by $-x$.

Compute the numerator and its norm separately after replacing $x$ by $-x$; the norm is unchanged by sign reversal. □

*Solution.* Its numerator changes sign and denominator does not. □

**Exercise 4.23.** State Borsuk–Ulam for $S^1 \to \mathbb{R}$.

*Hint.* Set $n = 1$.

Set $h(x) = f(x) - f(-x)$ and restrict to a semicircle joining antipodes. Its endpoint values allow an intermediate-value argument. □

*Solution.* Some antipodal pair has equal value. □

**Exercise 4.24.** Define $\mathbb{R}P^n$ as a quotient.

*Hint.* Identify antipodes.

Describe both the equivalence relation and the quotient topology; naming the orbit set alone does not specify the topological space. □

*Solution.* $\mathbb{R}P^n = S^n/(x \sim -x)$. □

**Exercise 4.25.** Give the linear-algebra description of $\mathbb{R}P^n$.

*Hint.* Use lines through the origin.

Intersect a line through the origin with $S^n$; determine the pair of unit representatives and compare it with one antipodal equivalence class. ☐

*Solution.* One-dimensional linear subspaces of $\mathbb{R}^{n+1}$. ☐

**Exercise 4.26.** When does $f : S^n \to Y$ factor through $q$?

*Hint.* Be constant on quotient fibers.

Define a candidate map on classes by $[x] \mapsto f(x)$. Separate well-definedness on each two-point fiber from continuity supplied by the quotient property. ☐

*Solution.* When $f(x) = f(-x)$. ☐

**Exercise 4.27.** Identify $\mathbb{R}P^0$.

*Hint.* There is one line in $\mathbb{R}$.

Compare the two points of $S^0$ with their single orbit under the antipodal involution. ☐

*Solution.* A point. ☐

**Exercise 4.28.** Identify $\mathbb{R}P^1$.

*Hint.* Antipodal circle quotient.

Represent a line by an angle modulo $\pi$ and double the angle to obtain a well-defined point on the circle. ☐

*Solution.* It is homeomorphic to $S^1$. ☐

**Exercise 4.29.** What circle map models $q : S^1 \to \mathbb{R}P^1 \cong S^1$?

*Hint.* Two-to-one covering.

Seek a circle map unchanged when $z$ is replaced by $-z$; then check that its fibers contain exactly those two points. ☐

*Solution.* $z \mapsto z^2$. ☐

**Exercise 4.30.** What is the degree of that map?

*Hint.* Use the power-map theorem.

Use the angle-doubling identification from the preceding exercise with its chosen orientation; an arbitrary orientation change of the target changes the degree's sign. ☐

*Solution.* 2. ☐

**Exercise 4.31.** Why is $S^n$ a one-point compactification of $\mathbb{R}^n$?

*Hint.* Use stereographic projection.

After stereographic projection, verify that neighborhoods of the omitted pole correspond to complements of compact subsets of $\mathbb{R}^n$. ☐

*Solution.* Removing one point gives $\mathbb{R}^n$. ☐

**Exercise 4.32.** Express $S^{n+1}$ as a suspension.

*Hint.* Suspend $S^n$.

Use height $t \in [-1, 1]$ and the map $(x, t) \mapsto (\sqrt{1 - t^2}x, t)$; examine what happens at the two extreme heights. □

*Solution.* $S^{n+1} \cong \Sigma S^n$. □

**Exercise 4.33.** How many standard cells does $\mathbb{R}P^n$ have?

*Hint.* One in each dimension $0, \ldots, n$.

Count the dimensions in the filtration $\mathbb{R}P^0 \subset \cdots \subset \mathbb{R}P^n$, remembering the zero-dimensional cell. □

*Solution.* $n + 1$. □

**Exercise 4.34.** What is the equator of $S^n$?

*Hint.* Set the last coordinate to zero.

With the last coordinate fixed to zero, the remaining coordinates still satisfy one unit-sphere equation in $\mathbb{R}^n$. □

*Solution.* A copy of $S^{n-1}$. □

**Exercise 4.35.** Why is a closed hemisphere a disk?

*Hint.* Project to the first $n$ coordinates.

On the upper hemisphere recover the last coordinate as $\sqrt{1 - \|u\|^2}$. This constructs the inverse to projection, including the boundary. □

*Solution.* It is homeomorphic to $D^n$. □

**Exercise 4.36.** Why is $S^n$ noncontractible for $n \geq 1$?

*Hint.* Use degree of the identity.

A contraction would make the identity homotopic to a constant sphere map. Compare their degrees, keeping the condition $n \geq 1$. □

*Solution.* A contraction would force degree $1 = 0$, impossible. □

**Exercise 4.37.** What is the bridge to VIII/05?

*Hint.* Use projective filtration.

Normalize projective coordinates with nonzero last coordinate to 1. Identify this open $\mathbb{R}^n$ chart as the new cell and the complementary projective subspace as the old skeleton. □

*Solution.* $\mathbb{R}P^n$ is built from $\mathbb{R}P^{n-1}$ by adding one $n$-cell. □

## 4.14 Solved problem dossiers

**Problem 4.38** (Antipodal parity)**.** Use degree to compare the antipodal map with the identity.

*Solution.* For even $n$, the degrees are $-1$ and $1$, so they are not homotopic. For odd $n$, both degrees are 1, so degree gives no obstruction. □

**Problem 4.39** (Circle antipodes)**.** Prove directly that every continuous $f : S^1 \to \mathbb{R}$ has equal values at some antipodal pair.

*Solution.* Set $h(x) = f(x) - f(-x)$. On a semicircle from $x$ to $-x$, the endpoint values have opposite signs, so the intermediate value theorem gives $h = 0$. □

**Problem 4.40** (No odd map to $S^0$). Prove there is no odd continuous map $S^1 \to S^0$.

*Solution.* A continuous image of connected $S^1$ in discrete $S^0$ is a point, but a constant map cannot be odd. □

**Problem 4.41** (Equivalent Borsuk–Ulam form). Derive the no-odd-map formulation and its converse.

*Solution.* An odd $S^n \to S^{n-1} \subset \mathbb{R}^n$ contradicts equal antipodal images. Conversely, an antipode-separating $f$ yields the normalized odd difference map. □

**Problem 4.42** (Odd self-map). Explain why an odd self-map of $S^n$ cannot be null-homotopic.

*Solution.* Its degree is odd and therefore nonzero, while a null-homotopic sphere map has degree zero. □

**Problem 4.43** (Antipodal $S^2$). Compare the antipodal map on $S^2$ with the identity and a constant map.

*Solution.* Its degree is $-1$, versus 1 for the identity and 0 for a constant; hence it is homotopic to neither. □

**Problem 4.44** (Antipodal $S^3$). What does degree say about the antipodal map on $S^3$?

*Solution.* Its degree is 1, equal to the identity degree. Thus degree gives no obstruction to a homotopy with the identity; this chapter does not infer the homotopy from degree alone. □

**Problem 4.45** (No retraction). Recover the no-retraction theorem from degree.

*Solution.* If $ri = \mathrm{id}_{S^n}$, then degree is 1; but $i$ contracts through the disk, making $ri$ null-homotopic of degree 0. □

**Problem 4.46** (Even-map quotient). Prove the factorization of an even map through $\mathbb{R}P^n$.

*Solution.* Define $\bar{f}([x]) = f(x)$. Evenness gives well-definedness, and the quotient property of $q$ gives continuity and uniqueness. □

**Problem 4.47** (Projective line). Construct $\mathbb{R}P^1 \cong S^1$.

*Solution.* Represent a line by angle $\theta$ modulo $\pi$ and send it to $e^{2i\theta}$. This descends to a continuous bijection from compact $\mathbb{R}P^1$ to Hausdorff $S^1$. □

**Problem 4.48** (Quotient degree). Compute the degree of $S^1 \to \mathbb{R}P^1 \cong S^1$.

*Solution.* It is $z \mapsto z^2$, hence degree 2. □

**Problem 4.49** (Sphere from hemispheres). Describe $S^n$ as two disks glued along the boundary.

*Solution.* Glue $D^n_+$ and $D^n_-$ by the identity on $S^{n-1}$; the quotient is $S^n$. □

**Problem 4.50** (One-point compactification). Explain the stereographic model of $S^n$.

*Solution.* Stereographic projection identifies $S^n \setminus \{N\}$ with $\mathbb{R}^n$; the north pole becomes the point at infinity. □

**Problem 4.51** (Suspension sphere)**.** Explain geometrically why $\Sigma S^n \cong S^{n+1}$.

*Solution.* Use height in $[-1, 1]$; at the two extreme heights all directions collapse to the suspension points. □

**Problem 4.52** (Antipodal orbit size)**.** Why is $q : S^n \to \mathbb{R}P^n$ two-to-one?

*Solution.* Each orbit is exactly $\{x, -x\}$, and $xe - x$ on a unit sphere. □

**Problem 4.53** (Projective inclusion)**.** Explain $\mathbb{R}P^{n-1} \subset \mathbb{R}P^n$.

*Solution.* Lines lying in the coordinate hyperplane $\mathbb{R}^n \subset \mathbb{R}^{n+1}$ form the embedded lower projective space. □

**Problem 4.54** (Projective open cell)**.** Why is $\mathbb{R}P^n \setminus \mathbb{R}P^{n-1} \cong \mathbb{R}^n$?

*Solution.* Normalize a line not in the coordinate hyperplane so its last coordinate is 1; the remaining coordinates are arbitrary in $\mathbb{R}^n$. □

**Problem 4.55** (Borsuk–Ulam application)**.** Apply Borsuk–Ulam to two continuous measurements on $S^2$.

*Solution.* For $f = (f_1, f_2) : S^2 \to \mathbb{R}^2$, some antipodal pair has both measurements equal. □

**Problem 4.56** (Odd-vector obstruction)**.** Why can there be no continuous rule $v : S^n \to S^{n-1}$ with $v(-x) = -v(x)$?

*Solution.* That is exactly an odd map $S^n \to S^{n-1}$, forbidden by Borsuk–Ulam. □

**Problem 4.57** (Bridge to cells)**.** Explain how projective geometry motivates cell attachment.

*Solution.* The filtration $\mathbb{R}P^{n-1} \subset \mathbb{R}P^n$ adds an open $n$-cell; VIII/05 abstracts the gluing of its closed disk along the boundary. □

## 4.15 Supplementary worked examples

**Example 4.58** (Antipodal quotient)**.** Identifying $x \sim -x$ on $S^n$ produces real projective space $\mathbb{RP}^n$. The quotient map $S^n \to \mathbb{RP}^n$ is a two-sheeted covering for $n \geq 1$.

**Example 4.59** (Antipodal degree parity)**.** The antipodal map on $S^n$ has degree $(-1)^{n+1}$: it preserves orientation when $n$ is odd and reverses orientation when $n$ is even.

**Example 4.60** (Odd maps and antipodal symmetry)**.** Maps satisfying $f(-x) = -f(x)$ respect the antipodal involution and therefore descend to maps on projective space. Equivariance converts a symmetry condition upstairs into a well-defined quotient map downstairs.

## 4.16 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 4.61** (Antipodal square)**.** Compute $A \circ A$ for $A(x) = -x$.

*Hint.* Apply the sign twice. □

*Solution.* $A^2 = \mathrm{id}_{S^n}$. □

**Exercise 4.62** (Fixed points)**.** Does the antipodal map on $S^n$ have fixed points?

*Hint.* Solve $x = -x$ on the unit sphere. □

*Solution.* No; $x = -x$ implies $x = 0$, which is not on $S^n$. □

**Exercise 4.63** (Degree on $S^3$)**.** Compute the degree of the antipodal map on $S^3$.

*Hint.* Use $(-1)^{n+1}$. □

*Solution.* $(-1)^4 = 1$. □

**Exercise 4.64** (Degree on $S^2$)**.** Compute the degree of the antipodal map on $S^2$.

*Hint.* Use the parity formula. □

*Solution.* $(-1)^3 = -1$. □

**Exercise 4.65** (Quotient fiber)**.** How many points are in a fiber of $S^n \to \mathbb{RP}^n$?

*Hint.* Each projective point is a line through the origin. □

*Solution.* Exactly two: $x$ and $-x$. □

## Proofs

**Exercise 4.66** (Free action)**.** Prove the antipodal $\mathbb{Z}/2$-action on $S^n$ is free.

*Hint.* A nontrivial stabilizer would give a fixed point. ☐

*Solution.* The nonidentity element sends $x$ to $-x$, and $x = -x$ has no unit-vector solution, so all stabilizers are trivial. ☐

**Exercise 4.67** (Quotient map is covering)**.** Explain why the antipodal quotient is locally a two-sheeted covering.

*Hint.* Choose a small neighborhood disjoint from its antipode. ☐

*Solution.* For sufficiently small $U$ around $x$, $U \cap (-U) = \varnothing$; the quotient evenly covers its image by $U$ and $-U$. ☐

**Exercise 4.68** (Equivariant descent)**.** Prove an odd map $f : S^n \to S^m$ descends to $\mathbb{RP}^n \to \mathbb{RP}^m$.

*Hint.* Check independence of the representative $x$ versus $-x$. ☐

*Solution.* Since $f(-x) = -f(x)$, the two representatives have the same projective image, so the quotient universal property gives a continuous descended map. ☐

**Exercise 4.69** (Orientation parity)**.** Derive the orientation behavior of the antipodal map from its degree.

*Hint.* Orientation preserving means degree $+1$. ☐

*Solution.* $(-1)^{n+1} = 1$ exactly for odd $n$, and equals $-1$ for even $n$. ☐

## Counterexamples and hypothesis tests

**Exercise 4.70** (Antipodal has fixed points)**.** Test: the antipodal map always has two fixed points.

*Hint.* Do not confuse it with a reflection. ☐

*Solution.* False. It has no fixed points on any sphere. ☐

**Exercise 4.71** (Quotient is sphere)**.** Test: $S^n/(x \sim -x)$ is always homeomorphic to $S^n$.

*Hint.* The quotient is projective space. ☐

*Solution.* False in general; it is $\mathbb{RP}^n$. ☐

**Exercise 4.72** (Degree always minus one)**.** Test: the antipodal map always reverses orientation.

*Hint.* Check the parity of $n$. ☐

*Solution.* False. It preserves orientation for odd-dimensional spheres. ☐

## Applications and investigations

**Exercise 4.73** (Projective modeling)**.** Why is antipodal identification a natural model for unoriented directions?

*Hint.* A line does not distinguish $x$ from $-x$. □

*Solution.* Unit vectors encode directions with an orientation; quotienting by the antipodal relation forgets the sign and yields the space of unoriented lines. □

**Exercise 4.74** (Symmetry reduction)**.** How does equivariance simplify maps between quotient spaces?

*Hint.* Work upstairs where spheres are simpler. □

*Solution.* An equivariant map between covering spaces descends automatically, allowing calculations on spheres before passing to projective quotients. □

## Challenge problems

**Exercise 4.75** (Odd-map obstruction)**.** Suppose an odd map $S^n \to S^n$ were null-homotopic through odd maps. What extra structure must such a homotopy preserve?

*Hint.* Track antipodal equivariance at every time. □

*Solution.* It must satisfy $H(-x,t) = -H(x,t)$ for all $(x,t)$, so it descends to a homotopy on $\mathbb{RP}^n$; ordinary null-homotopy alone does not guarantee this. □

**Exercise 4.76** (Parity synthesis)**.** For which $n$ can the antipodal map be homotopic to the identity based solely on degree obstruction?

*Hint.* Compare degrees. □

*Solution.* Degree does not obstruct a homotopy when $(-1)^{n+1} = 1$, i.e. odd $n$; for even $n$ the degrees differ, so no homotopy exists. □

# Part II

# CW Complexes

# Chapter 5

# Cell Attachments

Cell attachment turns local disks into global spaces. Starting from $X$, an $n$-disk is glued along its boundary by a map

$$\varphi : S^{n-1} \to X.$$

The quotient remembers the old space, the new cell dimension, and the attaching map. This chapter develops the full adjunction-space language and the major problem families mapped to cell attachments, while reserving the theorem about homotopic attaching maps for VIII/08.

## 5.1 Adjunction spaces

**Definition 5.1** (Attaching an $n$-cell)**.**

$$X \cup_\varphi D^n = (X \sqcup D^n)/\!\sim,$$

where $u \sim \varphi(u)$ for $u \in S^{n-1} = \partial D^n$.

## 5.2 Characteristic map and open cell

Let $q : X \sqcup D^n \to X \cup_\varphi D^n$ be the quotient map.

**Definition 5.2** (Characteristic map)**.** The characteristic map is

$$\chi = q|_{D^n} : D^n \to X \cup_\varphi D^n.$$

Its restriction to $\mathring{D}^n$ is a homeomorphism onto the open $n$-cell $e^n$.

*Warning* 5.3 (Boundary identifications)*.* The characteristic map need not embed the closed disk. Distinct boundary points can be identified by the attaching map.

## 5.3 Universal property

**Proposition 5.4** (Gluing criterion)**.** *To define $F : X \cup_\varphi D^n \to Y$ is equivalent to giving continuous maps $f : X \to Y$ and $g : D^n \to Y$ satisfying*

$$g|_{S^{n-1}} = f \circ \varphi.$$

This is the quotient, equivalently pushout, universal property.

## 5.4 Zero- and one-cells

A 0-cell is a point and has empty boundary, so attaching it adds a new disjoint point. A 1-cell has boundary

$$S^0 = \{-1, +1\},$$

so its attaching map chooses two points of the old space, possibly equal.

**Example 5.5** (Loop)**.** Attaching both endpoints of $D^1$ to one vertex produces $S^1$.

**Example 5.6** (Graphs)**.** Finite graphs arise from finitely many zero-cells followed by finitely many one-cell attachments.

## 5.5 Constant attachments

**Proposition 5.7** (Constant attachment)**.** *If $S^{n-1} \to X$ is constant at a basepoint $x_0$, the whole boundary of the new disk collapses at $x_0$. In the standard pointed setting the result is*

$$X \vee S^n.$$

*In particular,*

$$D^n/S^{n-1} \cong S^n.$$

## 5.6 Degree-$m$ two-cell attachments

For

$$\varphi_m : S^1 \to S^1, \qquad \varphi_m(z) = z^m,$$

define

$$X_m = S^1 \cup_{\varphi_m} D^2.$$

**Example 5.8** (Degree one)**.** $X_1 \cong D^2$.

**Example 5.9** (Degree zero)**.** For a constant attaching map,

$$X_0 \cong S^1 \vee S^2.$$

**Example 5.10** (Projective plane)**.**

$$\mathbb{R}P^2 \cong S^1 \cup_{\varphi_2} D^2.$$

The identical cell counts can therefore carry very different topology because the attaching maps differ.

## 5.7 Sphere with two cells

**Proposition 5.11** (Minimal sphere structure)**.** *For $n \geq 1$, $S^n$ is obtained from one zero-cell by attaching one $n$-cell with constant attaching map.*

## 5.8 Torus polygon

The standard torus has one 0-cell, two 1-cells $a, b$, and one 2-cell. Its one-skeleton is

$$S^1 \vee S^1,$$

and the two-cell attaches by the commutator word

$$aba^{-1}b^{-1}.$$

## 5.9 Closed orientable surfaces

A genus-$g$ orientable surface has one 0-cell, $2g$ one-cells, and one two-cell, with attaching word

$$[a_1, b_1] \cdots [a_g, b_g].$$

## 5.10 Projective filtration

**Proposition 5.12** (Projective cells). *$\mathbb{R}P^n$ is obtained from $\mathbb{R}P^{n-1}$ by attaching one $n$-cell. Hence it has one cell in every dimension $0, \ldots, n$.*

## 5.11 Attaching several cells

For a family $\varphi_\alpha : S^{n-1} \to X$, attach all cells simultaneously:

$$\left( X \sqcup \coprod_\alpha D^n_\alpha \right) / \sim, \qquad u \sim \varphi_\alpha(u).$$

Distinct disk interiors remain disjoint.

## 5.12 Successive attachment

Dimension-by-dimension construction yields

$$X^0 \subset X^1 \subset X^2 \subset \cdots,$$

where $X^n$ is obtained from $X^{n-1}$ by attaching $n$-cells. VIII/06 formalizes these $X^n$ as skeleta.

## 5.13 Attaching versus characteristic maps

The attaching map

$$\varphi : S^{n-1} \to X$$

describes how the boundary meets the old space. The characteristic map

$$\chi : D^n \to X \cup_\varphi D^n$$

parametrizes the closure of the new cell.

## 5.14 Pushout formulation

Cell attachment is the pushout

$$\begin{array}{ccc} S^{n-1} & \xrightarrow{\varphi} & X \\ \downarrow & & \downarrow \\ D^n & \longrightarrow & X \cup_\varphi D^n. \end{array}$$

## 5.15 Dependence on the attaching map

*Warning* 5.13 (The map matters). The number and dimensions of cells do not generally determine the resulting space. The attaching maps are essential data.

Homotopic attaching maps often lead to homotopy-equivalent adjunction spaces, but the rigorous theorem and its homotopy-extension argument belong to VIII/08. Blocks explicitly mapped there are therefore not absorbed into this chapter.

## 5.16 Mapping-cone preview

For a map $f : A \to X$, the mapping cone is

$$C_f = X \cup_f CA.$$

Since $CS^{n-1} \cong D^n$, ordinary cell attachment is the sphere-domain special case. VIII/07 develops mapping cones in full.

## 5.17 Why attachments compute topology

Cellular homology later converts $n$-cell attaching maps into boundary homomorphisms. The degree-$m$ family already predicts the integer $m$ that will appear in a cellular boundary map.

*Warning* 5.14 (Use the quotient topology). The construction is not merely a set-theoretic union. Continuity is controlled by the quotient topology induced by the boundary identifications.

## 5.18 Exercises

**Exercise 5.15.** Write the quotient defining $X \cup_\varphi D^n$.

*Hint.* Begin with the disjoint union.

Identify only boundary points with their prescribed images; explain which equivalence classes contain points of the disk interior. ☐

*Solution.* $(X \sqcup D^n)/\sim$, with $u \sim \varphi(u)$ on $S^{n-1}$. ☐

**Exercise 5.16.** What is the domain of an $n$-cell attaching map?

*Hint.* Use the disk boundary.

For $n > 0$, compare the dimensions of the disk, its interior, and its boundary; treat the zero-cell boundary as empty. ☐

*Solution.* $S^{n-1}$. ☐

**Exercise 5.17.** Define the characteristic map.

*Hint.* Restrict the quotient map.

Start with the quotient map on $X \sqcup D^n$, then restrict its domain to the new disk, including its boundary. ☐

*Solution.* $\chi = q|_{D^n}$. ☐

**Exercise 5.18.** Where is $\chi$ a homeomorphism?

*Hint.* Use the disk interior.

Check that no interior point is identified with another point; do not extend this injectivity claim to the disk boundary. ☐

*Solution.* On $\mathring{D}^n$, onto the open cell. ☐

**Exercise 5.19.** State the compatibility for maps $f : X \to Y$, $g : D^n \to Y$.

*Hint.* Compare on the boundary.

A point $u$ on the boundary represents both a disk point and $\varphi(u)$ in the old space; their proposed images must agree. ☐

*Solution.* $g|_{S^{n-1}} = f \circ \varphi$. □

**Exercise 5.20.** What does a zero-cell attachment do?

*Hint.* Its boundary is empty.

Since there are no boundary points to identify, use the disjoint-union topology to decide how the new point meets $X$. □

*Solution.* It adds a new disjoint point. □

**Exercise 5.21.** What data specifies a one-cell attachment?

*Hint.* $S^0$ has two points.

The two endpoint images may coincide or differ; compare the loop-forming and edge-joining possibilities. □

*Solution.* The images of the two endpoints. □

**Exercise 5.22.** Build $S^1$ from one zero-cell and one one-cell.

*Hint.* Attach both endpoints to one vertex.

Parametrize the interval by $t \mapsto e^{2\pi i t}$, and check precisely which endpoint identification makes this descend. □

*Solution.* The quotient $D^1/\partial D^1$ is $S^1$. □

**Exercise 5.23.** How are finite graphs built cellularly?

*Hint.* Use dimensions 0 and 1.

Assign each edge an interval and record its two endpoint vertices; loops and multiple edges are allowed in this CW description. □

*Solution.* Attach finitely many one-cells to finitely many vertices. □

**Exercise 5.24.** What is $D^n/S^{n-1}$?

*Hint.* Collapse the boundary.

Use a disk as a compactification of its interior after collapsing the boundary; check that the collapsed class is the single point at infinity. □

*Solution.* $S^n$. □

**Exercise 5.25.** What does a constant $n$-attachment give in a pointed space?

*Hint.* The new sphere meets at the basepoint.

The entire boundary is identified to the chosen basepoint, while the disk interior stays separate from the old space. □

*Solution.* $X \vee S^n$. □

**Exercise 5.26.** Define the degree-$m$ attaching map on $S^1$.

*Hint.* Use a power map.

Write $z = e^{i\theta}$ and let the output angle advance by $m\theta$; negative integers must reverse winding direction. □

*Solution.* $\varphi_m(z) = z^m$. □

**Exercise 5.27.** Identify $S^1 \cup_{\varphi_1} D^2$.

*Hint.* Boundary goes once around.

The old circle is identified with the new disk's actual boundary, not collapsed to a point as in the sphere construction. □

*Solution.* $D^2$. □

**Exercise 5.28.** Identify a constant two-cell attachment to $S^1$.

*Hint.* Use a wedge.

Keep the old circle and replace the disk with its boundary quotient; identify the single point where these two pieces meet. □

*Solution.* $S^1 \vee S^2$. □

**Exercise 5.29.** Give the cell attachment for $\mathbb{R}P^2$.

*Hint.* Use degree two.

In the disk model of $\mathbb{R}P^2$, opposite boundary points represent the same line; track the boundary map after identifying $\mathbb{R}P^1$ with a circle. □

*Solution.* $S^1 \cup_{\varphi_2} D^2$. □

**Exercise 5.30.** Give minimal cell counts for $S^n$.

*Hint.* One old point, one new cell.

For $n > 0$, collapse the boundary of one $n$-disk to the old vertex; for $n = 0$, regard $S^0$ as two vertices. □

*Solution.* One 0-cell and one $n$-cell. □

**Exercise 5.31.** Give the torus cell counts.

*Hint.* Use the square model.

Track the equivalence classes of the square's four vertices and its pairs of opposite edges before counting the open interior. □

*Solution.* One 0-cell, two 1-cells, one 2-cell. □

**Exercise 5.32.** What word attaches the torus two-cell?

*Hint.* Use the commutator.

Choose orientations on the two identified edges and read the oriented square boundary once, using inverses when traversal opposes the chosen edge direction. □

*Solution.* $aba^{-1}b^{-1}$. □

**Exercise 5.33.** Give genus-$g$ surface cell counts.

*Hint.* Generalize the torus.

Use a $4g$-gon with paired sides; count the resulting vertex class, distinct edge pairs, and open polygon interior. □

*Solution.* One 0-cell, $2g$ one-cells, one two-cell. □

**Exercise 5.34.** Give its attaching word.

*Hint.* Multiply commutators.

Group the oriented boundary traversal into one four-letter block for each handle, retaining the order of the blocks. □

*Solution.* $[a_1, b_1] \cdots [a_g, b_g]$. □

**Exercise 5.35.** How is $\mathbb{R}P^n$ built from $\mathbb{R}P^{n-1}$?

*Hint.* Use projective filtration.

The projective chart with last homogeneous coordinate nonzero is an open $n$-cell; its complement is the old projective subspace. □

*Solution.* Attach one $n$-cell. □

**Exercise 5.36.** What is an $n$-skeleton informally?

*Hint.* Stop at dimension $n$.

Check that the boundary of every included cell lands in a lower skeleton, so the stopped construction is closed under attachments. □

*Solution.* The union of all cells of dimension at most $n$. □

**Exercise 5.37.** Why is attachment a mapping-cone construction?

*Hint.* A cone on $S^{n-1}$ is $D^n$.

Use a radial parameter to identify the cone on the attaching sphere with a disk, with cone base corresponding to disk boundary. □

*Solution.* $X \cup_\varphi D^n = X \cup_\varphi CS^{n-1}$. □

**Exercise 5.38.** Why do cell dimensions alone not determine a space?

*Hint.* Compare degree-$m$ attachments.

Compare the constant and identity attachments of a two-cell to a circle: the cell dimensions agree, but one retains a circle homotopy class and the other fills it. □

*Solution.* Different attaching maps with identical cell counts can yield different topology. □

## 5.19 Solved problem dossiers

**Problem 5.39** (Adjunction universal property)**.** Prove the gluing criterion for maps out of $X \cup_\varphi D^n$.

*Solution.* A map out restricts to compatible maps on $X$ and $D^n$. Conversely, compatible maps are constant on the quotient equivalence classes and therefore descend uniquely by the quotient property. □

**Problem 5.40** (One-cell loop)**.** Attach one interval to one vertex at both endpoints and identify the result.

*Solution.* The quotient $D^1/\partial D^1$ is a circle. □

**Problem 5.41** (One-cell edge)**.** Attach an interval between two distinct vertices.

*Solution.* The result is a graph with two vertices joined by one edge, homeomorphic to a closed interval. □

**Problem 5.42** (Parallel edges)**.** Attach two one-cells between the same two vertices. Determine the homotopy type.

*Solution.* The two edges form a circle: go out along one and return along the other. The space is homeomorphic to $S^1$. □

**Problem 5.43** (Bouquet of circles)**.** Attach $r$ one-cells with both endpoints at one vertex.

*Solution.* Each interval becomes a circle and all circles meet only at the common vertex, giving $\bigvee_{j=1}^{r} S^1$. □

**Problem 5.44** (Constant attachment)**.** Prove $D^n/S^{n-1} \cong S^n$.

*Solution.* The quotient is the one-point compactification of the open disk $\mathring{D}^n \cong \mathbb{R}^n$, hence $S^n$. □

**Problem 5.45** (Minimal sphere)**.** Give the two-cell structure of $S^n$.

*Solution.* Start from one vertex and attach one $n$-disk by the constant boundary map. The quotient is $D^n/S^{n-1} \cong S^n$. □

**Problem 5.46** (Degree-one two-cell)**.** Identify $S^1 \cup_{\mathrm{id}} D^2$.

*Solution.* The existing circle becomes exactly the boundary of the new disk, so the result is $D^2$. □

**Problem 5.47** (Constant two-cell)**.** Identify $S^1 \cup_c D^2$.

*Solution.* The new disk boundary collapses at the chosen point, turning the disk into $S^2$ wedged with the old $S^1$. □

**Problem 5.48** (Degree-two two-cell)**.** Explain the $\mathbb{R}P^2$ disk model.

*Solution.* Identify antipodal points of $\partial D^2$. The quotient boundary is $\mathbb{R}P^1 \cong S^1$, and its attaching map is the two-to-one map of degree 2. □

**Problem 5.49** (Degree-$m$ family)**.** Why can $X_m = S^1 \cup_{\varphi_m} D^2$ vary with $m$ despite fixed cell counts?

*Solution.* The boundary winds $m$ times around the one-skeleton. Cellular homology later records this as multiplication by $m$, so the attaching data are topologically significant. □

**Problem 5.50** (Torus polygon)**.** Derive the torus attaching word.

*Solution.* Label the square boundary in order $a, b, a^{-1}, b^{-1}$; after opposite edge identifications this becomes the loop $aba^{-1}b^{-1}$. □

**Problem 5.51** (Genus-two surface)**.** Give a cell attachment for the closed genus-two surface.

*Solution.* Use one vertex, four one-cells $a_1, b_1, a_2, b_2$, and attach one two-cell along $[a_1, b_1][a_2, b_2]$. □

**Problem 5.52** (Projective filtration)**.** Why does $\mathbb{R}P^n$ have one cell in every dimension?

*Solution.* Each difference $\mathbb{R}P^k \setminus \mathbb{R}P^{k-1}$ is $\mathbb{R}^k$, so stage $k$ adds exactly one open $k$-cell. □

**Problem 5.53** (Simultaneous cells)**.** Describe the quotient for a family of $n$-cell attachments.

*Solution.* Use $X \sqcup \coprod_\alpha D^n_\alpha$ and identify each boundary point $u$ with $\varphi_\alpha(u)$; disk interiors remain mutually disjoint. ☐

**Problem 5.54** (Characteristic boundary)**.** Why can a characteristic map fail to be injective on the boundary?

*Solution.* If distinct $u, v \in S^{n-1}$ satisfy $\varphi(u) = \varphi(v)$, the quotient identifies them. ☐

**Problem 5.55** (Open-cell embedding)**.** Why is the interior embedded?

*Solution.* The attachment relation only touches boundary points. No two disk interior points are identified, and none is identified with the old space. ☐

**Problem 5.56** (Pushout uniqueness)**.** State the uniqueness clause for the cell-attachment pushout.

*Solution.* Compatible maps on $X$ and $D^n$ determine exactly one map on the quotient because their images cover the adjunction space. ☐

**Problem 5.57** (Successive attachments)**.** What data determine a two-dimensional complex built on a graph?

*Solution.* The one-skeleton graph plus one attaching loop $S^1 \to X^1$ for every two-cell. ☐

**Problem 5.58** (Attaching word)**.** Why can a word in graph edges specify a two-cell attachment?

*Solution.* Reading the oriented edges in sequence defines a loop in the graph, hence a map $S^1 \to X^1$ used as the attaching map. ☐

**Problem 5.59** (Cell as cone)**.** Show an $n$-cell attachment is a mapping cone of a sphere map.

*Solution.* $CS^{n-1} \cong D^n$, with the cone base equal to the disk boundary, so $X \cup_\varphi CS^{n-1}$ is the same adjunction space. ☐

**Problem 5.60** (Homotopic attaching maps boundary)**.** What conclusion is expected for homotopic attaching maps, and why is it reserved for VIII/08?

*Solution.* Under standard hypotheses their adjunction spaces are homotopy equivalent, but the proof requires homotopy extension/pushout control; VIII/08 supplies that theorem rather than treating the quotients as automatically homeomorphic. ☐

**Problem 5.61** (Quotient topology)**.** Why is the quotient topology essential?

*Solution.* It is what makes the gluing universal property valid and determines continuity of maps into and out of the adjunction space. ☐

**Problem 5.62** (Bridge to CW complexes)**.** How does repeated cell attachment lead to VIII/06?

*Solution.* Build $X^0, X^1, X^2, \ldots$ dimension by dimension; a CW complex formalizes this filtration together with closure-finiteness and weak-topology conditions. ☐

## 5.20 Supplementary worked examples

**Example 5.63** (Attaching a 1-cell)**.** Attaching a 1-cell to two points by its boundary $S^0$ creates an interval joining them; attaching both endpoints to one point creates a circle. The attaching map records the topology created at the boundary.

**Example 5.64** (Attaching a 2-cell to a circle)**.** Attach $D^2$ to $S^1$ by the identity map on $S^1$. The result is homeomorphic to $D^2 \cup_{S^1} D^2 \cong S^2$, exhibiting the sphere as a 0-cell plus a 2-cell.

**Example 5.65** (Degree-$m$ 2-cell attachment)**.** Attaching a 2-cell to $S^1$ by a map of degree $m$ produces a standard two-dimensional CW complex whose first homology becomes $\mathbb{Z}/m$ for $m \neq 0$. The attaching degree becomes a cellular boundary coefficient.

## 5.21 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 5.66** (Boundary of a cell)**.** What is the boundary sphere used to attach an $n$-cell?

*Hint.* An $n$-cell is a copy of $D^n$. ☐

*Solution.* Its boundary is $S^{n-1} = \partial D^n$. ☐

**Exercise 5.67** (Attach a 0-cell)**.** What data are needed to attach a 0-cell?

*Hint.* The boundary of $D^0$ is empty. ☐

*Solution.* No attaching map data are needed; a 0-cell simply adds a point. ☐

**Exercise 5.68** (Circle from one cell)**.** Describe $S^1$ as a CW complex with one 0-cell and one 1-cell.

*Hint.* Identify both endpoints of $D^1$ with the 0-cell. ☐

*Solution.* Attach the boundary $S^0 = \{-1, 1\}$ of the 1-cell to the same 0-cell; the quotient is a circle. ☐

**Exercise 5.69** (Sphere cells)**.** Give the two-cell CW structure on $S^n$.

*Hint.* Use one 0-cell and one $n$-cell. ☐

*Solution.* Attach $D^n$ to a point by the unique map $S^{n-1} \to \{*\}$; the quotient $D^n/S^{n-1}$ is $S^n$. ☐

**Exercise 5.70** (Degree attachment)**.** If a 2-cell is attached to $S^1$ by degree $m$, what integer will appear in the cellular boundary?

*Hint.* The boundary coefficient is the attaching degree. ☐

*Solution.* The differential $C_2 \to C_1$ is multiplication by $m$ in the one-cell model. ☐

### Proofs

**Exercise 5.71** (Quotient description)**.** Prove cell attachment $X \cup_f D^n$ is a pushout/quotient construction.

*Hint.* Identify each $u \in S^{n-1}$ with $f(u) \in X$. ☐

*Solution.* Take the disjoint union $X \sqcup D^n$ and impose precisely the relations $u \sim f(u)$ for $u \in \partial D^n$; this realizes the pushout of $S^{n-1} \hookrightarrow D^n$ and $f$. ☐

**Exercise 5.72** (Sphere quotient)**.** Prove $D^n/S^{n-1} \cong S^n$.

*Hint.* Collapse the boundary to the point at infinity. ☐

*Solution.* Identifying the boundary of a closed ball to one point gives the one-point compactification of its interior $\mathbb{R}^n$, which is $S^n$. ☐

**Exercise 5.73** (Attachment functoriality)**.** Explain why a homeomorphism of attaching data induces a homeomorphism of attached spaces.

*Hint.* Use the universal property of the quotient/pushout. ☐

*Solution.* Compatible homeomorphisms on the old space and disk descend to mutually inverse maps on the quotients. ☐

**Exercise 5.74** (Homotopic maps preview)**.** Why should homotopic attaching maps often lead to the same homotopy type?

*Hint.* A homotopy provides a cylinder between the boundary maps. ☐

*Solution.* The homotopy allows one to interpolate the gluing in a mapping-cylinder construction; under standard CW/cofibration hypotheses this yields a homotopy equivalence of the adjunction spaces. ☐

### Counterexamples and hypothesis tests

**Exercise 5.75** (Dimension always increases)**.** Test: attaching an $n$-cell always changes homotopy type.

*Hint.* An attachment can kill or create topology, but may also be redundant. ☐

*Solution.* False; depending on the attaching map and existing space, the resulting inclusion can sometimes be a homotopy equivalence or add contractible structure. ☐

**Exercise 5.76** (Attaching map interior)**.** Test: an attaching map is defined on all of $D^n$.

*Hint.* Only the boundary is prescribed. ☐

*Solution.* False. The attaching map has domain $S^{n-1} = \partial D^n$. ☐

**Exercise 5.77** (All attachments are wedges)**.** Test: attaching a cell always gives a wedge with $S^n$.

*Hint.* That occurs for the constant attaching map under suitable based conditions. ☐

*Solution.* False; nontrivial attaching maps encode essential relations. ☐

## Applications and investigations

**Exercise 5.78** (Topological modeling)**.** Why are cell attachments a useful finite description of spaces?

*Hint.* They separate local pieces from gluing data. ☐

*Solution.* A complicated space can be encoded by simple disks plus attaching maps, turning topology into finite combinatorial/algebraic data when only finitely many cells are used. ☐

**Exercise 5.79** (Relation encoding)**.** Interpret a 2-cell attachment to a graph as imposing a relation on loops.

*Hint.* The attaching circle traces a word in the graph. ☐

*Solution.* The boundary loop becomes null-homotopic after filling it by the new disk, so its word is imposed as a relation in the fundamental group. ☐

## Challenge problems

**Exercise 5.80** (Moore-space preview)**.** Analyze the cellular chain groups for one 0-cell, one 1-cell, and one 2-cell attached by degree $m$.

*Hint.* There is one generator in each dimension $0, 1, 2$. ☐

*Solution.* $C_2 \cong C_1 \cong C_0 \cong \mathbb{Z}$, with $d_2$ multiplication by $m$ and $d_1 = 0$; hence $H_1 \cong \mathbb{Z}/m$ for $m \neq 0$ and $H_2 = 0$. ☐

**Exercise 5.81** (Attachment synthesis)**.** Explain how the same attaching map participates in both homotopy and homology calculations.

*Hint.* It is a boundary map geometrically and induces algebraic boundary coefficients. ☐

*Solution.* Geometrically it determines which boundary sphere is glued to the lower skeleton; algebraically its degrees against lower cells become entries of the cellular differential, connecting homotopy data to homology. ☐

# Chapter 6

# CW Complexes

Cell attachments become a general language when organized into skeleta. CW complexes are built from disks in increasing dimension, together with closure finiteness and the weak topology. They include graphs, spheres, surfaces, projective spaces, and most spaces used in classical algebraic topology.

## 6.1 Cells and characteristic maps

An $n$-cell is homeomorphic to $\mathring{D}^n \cong \mathbb{R}^n$. A characteristic map

$$\chi_\alpha : D^n \to X$$

restricts to a homeomorphism from $\mathring{D}^n$ onto its open cell and sends the boundary into lower-dimensional cells.

## 6.2 Skeleta

**Definition 6.1** (Skeleton)**.** The $n$-skeleton $X^n$ is the union of all cells of dimension at most $n$, with $X^{-1} = \varnothing$.

Inductively, $X^n$ is obtained from $X^{n-1}$ by attaching $n$-cells.

## 6.3 CW complex

**Definition 6.2** (CW complex)**.** A CW complex is a cell decomposition satisfying:

1. every cell has a characteristic map with boundary in the lower skeleton;
2. closure finiteness: the closure of each cell meets only finitely many cells;
3. weak topology: $A \subset X$ is closed exactly when $A \cap \overline{e}$ is closed in $\overline{e}$ for every cell $e$.

The letters C and W refer to closure finiteness and weak topology.

## 6.4 Skeletal pushouts

At stage $n$,

$$\begin{array}{ccc} \coprod_\alpha S^{n-1}_\alpha & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_\alpha D^n_\alpha & \longrightarrow & X^n \end{array}$$

is a pushout. This is VIII/05 repeated dimension by dimension.

## 6.5 Dimension

**Definition 6.3** (Dimension)**.**

$$\dim X = \sup\{n : X \text{ has an } n\text{-cell}\}.$$

A zero-dimensional CW complex is discrete; a one-dimensional CW complex is a topological graph.

## 6.6 Finite and locally finite complexes

**Definition 6.4** (Finite CW complex)**.** A CW complex is finite if it has only finitely many cells.

**Proposition 6.5** (Finite implies compact)**.** *Every finite CW complex is compact.*

**Definition 6.6** (Locally finite)**.** A CW complex is locally finite if every point has a neighborhood meeting only finitely many cells.

*Warning* 6.7 (Different finiteness notions)*.* Closure finiteness and local finiteness are distinct conditions.

## 6.7 Subcomplexes

**Definition 6.8** (Subcomplex)**.** A subspace $A \subset X$ is a CW subcomplex if it is a union of cells and, with every cell it contains, it contains the closure of that cell.

Every skeleton is a subcomplex.

**Proposition 6.9** (Subcomplex closedness)**.** *Every subcomplex of a CW complex is closed.*

## 6.8 Cellular maps

**Definition 6.10** (Cellular map)**.** A map $f : X \to Y$ of CW complexes is cellular if

$$f(X^n) \subset Y^n$$

for every $n$.

## 6.9 Standard examples

**Example 6.11** (Sphere)**.** $S^n$ has one 0-cell and one $n$-cell.

**Example 6.12** (Graph)**.** Graphs are one-dimensional CW complexes with vertices as zero-cells and edge interiors as one-cells.

**Example 6.13** (Torus)**.** The torus has one 0-cell, two 1-cells, and one 2-cell.

**Example 6.14** (Projective space)**.** $\mathbb{R}P^n$ has one cell in each dimension $0, \dots, n$.

## 6.10 CW structure is extra data

The same underlying space may carry different CW structures. For example, $S^1$ can have one zero-cell and one one-cell, or two zero-cells and two one-cells.

*Warning* 6.15 (Raw cell counts)*.* Individual cell counts depend on the CW structure. Invariant combinations, such as Euler characteristic for finite CW complexes, require proof of independence from the decomposition.

## 6.11 Infinite examples

The ray $[0, \infty)$ has a locally finite CW structure with zero-cells at nonnegative integers and one-cells $(k, k+1)$. Also,

$$\mathbb{R}P^\infty = \bigcup_{n\geq 0} \mathbb{R}P^n$$

has one cell in every nonnegative dimension.

## 6.12 Weak topology and cellwise continuity

**Proposition 6.16** (Cellwise criterion)**.** *A function $f : X \to Y$ from a CW complex is continuous if each composite*

$$f \circ \chi_\alpha : D^{n_\alpha} \to Y$$

*is continuous.*

## 6.13 Compact subsets

**Theorem 6.17** (Compact-cell finiteness)**.** *Every compact subset of a CW complex meets only finitely many open cells.*

Consequently a compact image lies in a finite subcomplex after adjoining the finitely many cell closures that it meets.

## 6.14 Connectedness and the one-skeleton

**Proposition 6.18** (One-skeleton detects components)**.** *For a nonempty CW complex, path components are already detected by the one-skeleton. In particular, $X$ is path connected exactly when $X^1$ is path connected.*

## 6.15 CW pairs

**Definition 6.19** (CW pair)**.** A CW pair $(X, A)$ consists of a CW complex $X$ and a subcomplex $A \subset X$.

## 6.16 Quotients by subcomplexes

**Proposition 6.20** (CW quotient)**.** *If $A \subset X$ is a subcomplex, then $X/A$ has a natural CW structure: $A$ becomes a zero-cell and cells not in $A$ survive with their dimensions.*

## 6.17 Euler characteristic preview

For a finite CW complex with $c_n$ $n$-cells, define

$$\chi(X) = \sum_n (-1)^n c_n.$$

For the standard torus structure,

$$\chi(T^2) = 1 - 2 + 1 = 0,$$

while

$$\chi(S^n) = 1 + (-1)^n.$$

Later homology proves this alternating sum is independent of the finite CW structure.

## 6.18 Why CW complexes are effective

CW complexes combine constructive flexibility with strong control: compact data live in finite subcomplexes, maps can often be made cellular, subcomplex inclusions support homotopy extension, and homology can be computed from the skeletal filtration and attaching maps. VIII/07 and VIII/08 exploit exactly these features.

## 6.19 Exercises

**Exercise 6.21.** What do C and W stand for?

*Hint.* Recall the defining global conditions.

Separate the finiteness assertion about each closed cell from the topological test using characteristic maps of all closed disks. ☐

*Solution.* Closure finite and weak topology. ☐

**Exercise 6.22.** Define $X^n$.

*Hint.* Collect cells up to dimension $n$.

Include every dimension 0 through $n$, not merely the cells of exactly dimension $n$. ☐

*Solution.* The union of all cells of dimension at most $n$. ☐

**Exercise 6.23.** What is $X^{-1}$?

*Hint.* It is the initial skeleton.

Identify the starting object before any zero-cells have been attached; no points have yet been introduced. ☐

*Solution.* $\varnothing$. ☐

**Exercise 6.24.** How is $X^n$ built from $X^{n-1}$?

*Hint.* Use VIII/05.

Write one attaching map $S^{n-1}_\alpha \to X^{n-1}$ per new cell and form the pushout of the disjoint union of boundary inclusions. ☐

*Solution.* Attach the $n$-cells along maps $S^{n-1} \to X^{n-1}$. ☐

**Exercise 6.25.** Define CW dimension.

*Hint.* Take the largest cell dimension or supremum.

An infinite number of cells need not imply unbounded dimension; compare an infinite graph with projective space of unbounded dimension. ☐

*Solution.* $\dim X = \sup\{n : X \text{ has an } n\text{-cell}\}$. ☐

**Exercise 6.26.** Describe a zero-dimensional CW complex.

*Hint.* It has only points.

Apply the weak-topology test to an arbitrary subset: its inverse image in each characteristic zero-disk is automatically open and closed. ☐

*Solution.* It is discrete. ☐

**Exercise 6.27.** Describe a one-dimensional CW complex.

*Hint.* Use vertices and edges.

Record each one-cell's two endpoints and allow them to coincide; this captures loops as well as ordinary edges. ☐

*Solution.* It is a topological graph. ☐

**Exercise 6.28.** Define a finite CW complex.

*Hint.* Count all cells.

Distinguish finitely many cells in total from finitely many cells in each dimension and from finite dimension. ☐

*Solution.* It has finitely many cells. ☐

**Exercise 6.29.** Why is a finite CW complex compact?

*Hint.* Use characteristic disks.

Map the finite disjoint union of closed characteristic disks onto the complex; use compactness of that union and continuity of the quotient map. ☐

*Solution.* It is a continuous image of a finite union of compact closed disks. ☐

**Exercise 6.30.** Define a subcomplex.

*Hint.* Require closure under cell closures.

Including an open cell obliges you to include all boundary cells in its closure, rather than only selected interior points. ☐

*Solution.* A union of cells containing the closure of every cell it contains. ☐

**Exercise 6.31.** Are skeleta subcomplexes?

*Hint.* Apply the definition.

The attaching boundary of a cell of dimension at most $n$ lies in a lower skeleton, so check the subcomplex condition cell by cell. ☐

*Solution.* Yes. ☐

**Exercise 6.32.** Define a cellular map.

*Hint.* Respect dimensions.

The requirement concerns every skeleton simultaneously; it does not require individual cells to map bijectively to cells. ☐

*Solution.* $f(X^n) \subset Y^n$ for all $n$. ☐

**Exercise 6.33.** Give minimal cell counts for $S^n$.

*Hint.* Use VIII/05.

Use $D^n/\partial D^n$ for $n > 0$; check separately that $S^0$ has two zero-cells. ☐

*Solution.* One 0-cell and one $n$-cell. □

**Exercise 6.34.** Give torus cell counts.

*Hint.* Use the polygon model.

In the square quotient, count vertex equivalence classes and edge equivalence classes before the single open face. □

*Solution.* One 0-cell, two 1-cells, one 2-cell. □

**Exercise 6.35.** Give $\mathbb{R}P^n$ cell counts.

*Hint.* Use projective filtration.

At each inclusion $\mathbb{R}P^{k-1} \subset \mathbb{R}P^k$, identify the complement as $\mathbb{R}^k$. □

*Solution.* One cell in every dimension $0, \ldots, n$. □

**Exercise 6.36.** Can one space have different CW structures?

*Hint.* Think of $S^1$.

Subdivide a circle edge by adding a vertex; compare its old and new vertex and edge counts without changing the underlying space. □

*Solution.* Yes. □

**Exercise 6.37.** Define local finiteness.

*Hint.* Look at neighborhoods.

State the quantifiers in the order: for each point, there exists a neighborhood, meeting only finitely many cells. □

*Solution.* Every point has a neighborhood meeting finitely many cells. □

**Exercise 6.38.** Is closure finiteness the same as local finiteness?

*Hint.* They quantify different things.

Use an infinite wedge of circles: each individual closed edge meets finitely many cells, while every neighborhood of the wedge point meets infinitely many edges. □

*Solution.* No. □

**Exercise 6.39.** State the cellwise continuity test.

*Hint.* Use characteristic maps.

Test the composite on each closed characteristic disk, including its boundary; continuity on open cells alone is insufficient. □

*Solution.* Continuity follows if the composite with every characteristic map is continuous. □

**Exercise 6.40.** What is true of compact subsets and cells?

*Hint.* Use compact-cell finiteness.

After identifying finitely many cells met by the compact set, use closure finiteness to place them inside a finite subcomplex. □

*Solution.* A compact subset meets only finitely many open cells. □

**Exercise 6.41.** Define a CW pair.

*Hint.* Choose a subcomplex.

The subspace must be a union of whole cells together with their boundary cells, not an arbitrary subset of a CW complex. □

*Solution.* A pair $(X, A)$ with $A$ a subcomplex of CW complex $X$. □

**Exercise 6.42.** Describe $X/A$ cellularly.

*Hint.* Collapse the subcomplex.

For nonempty $A$, first record the new collapsed vertex, then inspect the characteristic maps of cells outside $A$. □

*Solution.* $A$ becomes a zero-cell; other cells survive with their dimensions. □

**Exercise 6.43.** Compute $\chi(T^2)$.

*Hint.* Use $1 - 2 + 1$.

Use the alternating cell count with dimensions attached to each number; this prevents adding both edge contributions with the wrong sign. □

*Solution.* 0. □

**Exercise 6.44.** Compute $\chi(S^n)$.

*Hint.* Use one 0-cell and one $n$-cell.

The top cell contributes according to the parity of $n$; distinguish the even- and odd-dimensional cases, including $S^0$. □

*Solution.* $1 + (-1)^n$. □

## 6.20 Solved problem dossiers

**Problem 6.45** (Skeletal pushout)**.** Write the construction of $X^n$ from $X^{n-1}$.

*Solution.* Take the pushout of $\coprod S^{n-1}_\alpha \to X^{n-1}$ and the boundary inclusions $\coprod S^{n-1}_\alpha \hookrightarrow \coprod D^n_\alpha$. □

**Problem 6.46** (Two circle structures)**.** Compare the one-vertex/one-edge and two-vertex/two-edge structures on $S^1$.

*Solution.* They have the same underlying space and different raw cell counts. Both have Euler alternating sum zero. □

**Problem 6.47** (Sphere skeletons)**.** Describe the minimal CW filtration of $S^n$.

*Solution.* There is one zero-cell, no cells in dimensions $1, \dots, n-1$, and one $n$-cell attached constantly. □

**Problem 6.48** (Torus skeleta)**.** Describe the standard torus skeleta.

*Solution.* $X^0$ is a point, $X^1 = S^1 \vee S^1$, and $X^2$ attaches one disk along $aba^{-1}b^{-1}$. □

**Problem 6.49** (Projective skeleta)**.** Identify the $k$-skeleton of standard $\mathbb{R}P^n$.

*Solution.* It is $\mathbb{R}P^k$, because the projective filtration adds exactly one cell at each dimension. □

**Problem 6.50** (Finite implies compact)**.** Give the compactness proof for a finite CW complex.

*Solution.* Take the finite disjoint union of all compact characteristic disks and map it continuously and surjectively to the complex. □

**Problem 6.51** (Subcomplex closure)**.** Why does a subcomplex contain cell boundaries?

*Solution.* By definition it contains the closure of each cell it contains, so all cells appearing in those boundaries are included. □

**Problem 6.52** (Skeleton subcomplex)**.** Prove $X^n$ is a subcomplex.

*Solution.* The closure of a $k$-cell with $k \leq n$ lies in cells of dimension at most $k$, hence remains in $X^n$. □

**Problem 6.53** (Cellular map test)**.** For minimal sphere CW structures, when is $f : S^n \to S^n$ cellular?

*Solution.* It suffices that the unique zero-cell maps to the unique zero-cell; the full $n$-skeleton condition is automatic. □

**Problem 6.54** (Graph realization)**.** Explain how a multigraph becomes a CW complex.

*Solution.* Use vertices as zero-cells and attach one interval per edge to its endpoint vertices, allowing loops and parallel edges. □

**Problem 6.55** (Infinite ray)**.** Verify the integer subdivision of $[0, \infty)$ is locally finite.

*Solution.* A small neighborhood of a nonvertex meets one edge; near an integer it meets at most that vertex and its adjacent edges. □

**Problem 6.56** (Finiteness distinction)**.** Contrast closure finiteness and local finiteness.

*Solution.* Closure finiteness starts from one cell and studies its closure; local finiteness starts from one point and studies a neighborhood. □

**Problem 6.57** (Weak topology)**.** Why do compatible cellwise continuous maps assemble globally?

*Solution.* Preimages of closed sets intersect every closed cell in closed sets; the weak topology then makes the full preimage closed. □

**Problem 6.58** (Compact image)**.** What does a continuous map from compact $K$ into a CW complex reduce to?

*Solution.* Its image meets finitely many cells and is contained in a finite subcomplex after adjoining their closures. □

**Problem 6.59** (CW quotient)**.** Describe cells of $X/A$ for subcomplex $A$.

*Solution.* Collapse $A$ to one zero-cell and retain each open cell outside $A$, with the attaching map followed by the quotient. □

**Problem 6.60** (CW pair)**.** Give a natural example of a CW pair.

*Solution.* With compatible cell structures, $(D^n, S^{n-1})$ is a CW pair because the boundary is a subcomplex. □

**Problem 6.61** (Euler sphere)**.** Compute Euler characteristics of even and odd spheres.

*Solution.* $\chi(S^{2k}) = 2$ and $\chi(S^{2k+1}) = 0$. □

**Problem 6.62** (Euler projective)**.** Compute the cellular Euler sum for $\mathbb{R}P^n$.

*Solution.* $\sum_{k=0}^{n}(-1)^k$, equal to 1 for even $n$ and 0 for odd $n$. □

**Problem 6.63** (One-skeleton connectedness)**.** Why can higher-dimensional cells not join two distinct components of the lower skeleton?

*Solution.* For $n \geq 2$, the connected boundary sphere of an attached $n$-cell maps into a single component, so the disk is added to that component only. □

**Problem 6.64** (Bridge to mapping cones)**.** Why are CW attachments naturally related to mapping cones?

*Solution.* Each disk is $CS^{n-1}$; attaching it to $X^{n-1}$ is the mapping cone of its sphere attaching map, the subject of VIII/07. □

## 6.21 Supplementary worked examples

**Example 6.65** (CW structure on the torus)**.** A torus admits one 0-cell, two 1-cells $a, b$, and one 2-cell attached along the commutator word $aba^{-1}b^{-1}$. This compact cell structure exposes both its fundamental group relation and its cellular chains.

**Example 6.66** (Projective space skeletons)**.** Real projective space has one cell in each dimension $0, 1, \dots, n$. Its skeletal filtration $\mathbb{RP}^0 \subset \cdots \subset \mathbb{RP}^n$ makes induction and cellular homology particularly efficient.

**Example 6.67** (Spheres as minimal CW complexes)**.** $S^n$ has a CW structure with exactly one 0-cell and one $n$-cell. This illustrates that a CW decomposition need not resemble a triangulation and can be dramatically smaller.

## 6.22 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 6.68** (Skeleton definition)**.** What is the $n$-skeleton $X^n$ of a CW complex?

*Hint.* Collect cells of dimension at most $n$. ☐

*Solution.* $X^n$ is the subspace obtained from all cells of dimensions $\leq n$. ☐

**Exercise 6.69** (Sphere cells)**.** How many cells occur in the standard two-cell CW structure on $S^n$?

*Hint.* Use dimensions 0 and $n$. ☐

*Solution.* Two cells: one 0-cell and one $n$-cell. ☐

**Exercise 6.70** (Torus chain ranks)**.** For the standard torus CW structure, state the ranks of $C_0, C_1, C_2$.

*Hint.* Count cells in each dimension. ☐

*Solution.* The ranks are $1, 2, 1$ respectively. ☐

**Exercise 6.71** (Wedge of circles)**.** Give a CW structure on a wedge of $r$ circles.

*Hint.* Use a common 0-cell. ☐

*Solution.* Use one 0-cell and $r$ 1-cells, each with both endpoints attached to the common vertex. ☐

**Exercise 6.72** (Finite CW Euler count)**.** For a finite CW complex with $c_n$ cells in dimension $n$, write the cell-count expression for Euler characteristic.

*Hint.* Use alternating signs. ☐

*Solution.* $\chi(X) = \sum_n (-1)^n c_n$. ☐

## Proofs

**Exercise 6.73** (Skeleta are subcomplexes)**.** Prove each skeleton $X^n$ is a CW subcomplex of $X$.

*Hint.* Its cells are closed under taking attaching boundaries into lower dimensions. □

*Solution.* By construction every $k$-cell with $k \leq n$ attaches to $X^{k-1} \subset X^n$, so their union carries the induced CW structure. □

**Exercise 6.74** (Closure finite principle)**.** Explain why the closure of a cell in a CW complex meets only finitely many cells under the closure-finite axiom.

*Hint.* This is one of the defining finiteness conditions. □

*Solution.* The characteristic map of a closed disk meets the lower skeleton through a compact boundary and the CW closure-finite condition requires only finitely many cell intersections. □

**Exercise 6.75** (Weak topology)**.** State the weak topology test for a CW complex and explain its role.

*Hint.* Test closedness on every closed cell/finite skeleton piece. □

*Solution.* A set is closed when its intersection with each closed cell (equivalently each skeleton in the standard formulation) is closed; this ensures the topology is determined by the attaching process. □

**Exercise 6.76** (Cellular induction)**.** Why can many maps/properties on CW complexes be constructed inductively over skeleta?

*Hint.* Pass from $X^{n-1}$ to $X^n$ by attaching $n$-cells. □

*Solution.* At each step the extension problem reduces to extending data over disks from their boundary spheres, allowing dimension-by-dimension induction. □

## Counterexamples and hypothesis tests

**Exercise 6.77** (CW equals simplicial)**.** Test: every CW decomposition is a simplicial complex decomposition.

*Hint.* Cells need not be simplices and attaching maps need not be injective. □

*Solution.* False. CW complexes allow general disks and boundary identifications. □

**Exercise 6.78** (One CW structure)**.** Test: a space has at most one CW structure.

*Hint.* Compare a circle with different subdivisions. □

*Solution.* False. The same space can admit many CW structures with very different cell counts. □

**Exercise 6.79** (Finite dimensional means finite cells)**.** Test: every finite-dimensional CW complex has finitely many cells.

*Hint.* Dimension bounds cell dimensions, not their number. □

*Solution.* False; there may be infinitely many cells in a fixed dimension. □

## Applications and investigations

**Exercise 6.80** (Computation compression)**.** Why can a coarse CW structure outperform a triangulation for algebraic computations?

*Hint.* Chain ranks equal cell counts. □

*Solution.* Fewer cells produce much smaller cellular chain groups and boundary matrices while preserving the topology. □

**Exercise 6.81** (Incremental topology)**.** Explain why the skeletal filtration is natural for persistent or staged calculations.

*Hint.* Each stage adds one dimension of cells. □

*Solution.* The filtration isolates how topology changes as higher-dimensional cells are attached, making inductive exact sequences and cellular differentials natural. □

## Challenge problems

**Exercise 6.82** (Torus relation)**.** Explain how the torus 2-cell attaching word encodes commutativity of the two fundamental loops.

*Hint.* The attached disk kills its boundary loop. □

*Solution.* The word $aba^{-1}b^{-1}$ becomes null-homotopic, imposing $ab = ba$ in the fundamental group. □

**Exercise 6.83** (CW synthesis)**.** For a finite CW complex, explain the bridge from geometric cell counts to Euler characteristic and then to homology ranks.

*Hint.* Use cellular chains and the Euler-Poincare identity. □

*Solution.* Cell counts give alternating ranks of cellular chain groups; cancellation of boundary ranks shows this equals the alternating sum of homology ranks, so $\chi$ is both combinatorial and homological. □

# Chapter 7

# Mapping Cones

Cell attachment is a special case of a more flexible construction. Given a map $f : A \to X$, instead of attaching only a disk along a sphere, one can attach the whole cone on $A$. The resulting mapping cone records the map itself as part of a new space. Mapping cones organize quotient spaces, cofibers, suspensions, and many exact-sequence constructions that appear later in homology.

## 7.1 The cone on a space

**Definition 7.1** (Cone)**.** For a space $A$, the cone on $A$ is

$$CA = (A \times I)/(A \times \{1\}),$$

where all points of $A \times \{1\}$ are identified to one cone vertex.

The copy $A \times \{0\}$ is the base of the cone.

**Proposition 7.2** (Cones are contractible)**.** *For every nonempty space $A$, the cone $CA$ is contractible.*

*Proof.* Push every point linearly toward the cone vertex in the $I$-coordinate. The quotient description makes the terminal slice a single point. ☐

## 7.2 The suspension

**Definition 7.3** (Suspension)**.** The suspension of $A$ is

$$\Sigma A = (A \times I)/(A \times \{0\}, A \times \{1\}),$$

where each end is collapsed to a point.

Equivalently, $\Sigma A$ is obtained by gluing two cones on $A$ along their common base. In particular,

$$\Sigma S^n \cong S^{n+1}.$$

## 7.3 Definition of the mapping cone

**Definition 7.4** (Mapping cone)**.** For a continuous map $f : A \to X$, the mapping cone is

$$C_f = X \cup_f CA,$$

where the base $A \subset CA$ is identified with $X$ using $f$.

Thus $C_f$ is the pushout

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ CA & \longrightarrow & C_f. \end{array}$$

## 7.4 Cell attachment as a mapping cone

If $A = S^{n-1}$, then

$$CA \cong D^n.$$

Therefore

$$X \cup_\varphi D^n \cong C_\varphi$$

for every attaching map $\varphi : S^{n-1} \to X$.

**Proposition 7.5** (Cell-attachment specialization)**.** *Ordinary $n$-cell attachment is the mapping-cone construction for a map whose domain is $S^{n-1}$.*

## 7.5 Mapping cone of the identity

**Proposition 7.6** (Cone of the identity)**.** *For every space $A$,*

$$C_{\mathrm{id}_A} \cong CA.$$

*Hence it is contractible.*

The copy of $A$ already present as $X = A$ is exactly the base to which the cone is attached.

## 7.6 Mapping cone of a constant map

**Proposition 7.7** (Constant map cone)**.** *If $f : A \to X$ is constant at a chosen basepoint $x_0$, then under standard pointed hypotheses,*

$$C_f \simeq X \vee \Sigma A.$$

*In the literal quotient model, the cone base is attached at one point of $X$.*

For $X = \{*\}$,

$$C_f \cong \Sigma A.$$

## 7.7 Mapping cone of an inclusion

Let $i : A \hookrightarrow X$ be an inclusion.

**Proposition 7.8** (Cone of a cofibration-like inclusion)**.** *When $A \subset X$ is a CW subcomplex, the mapping cone $C_i$ is naturally homotopy equivalent to the quotient*

$$X/A.$$

For CW pairs this is one of the main reasons mapping cones model relative topology.

## 7.8 The cofiber viewpoint

The sequence

$$A \xrightarrow{f} X \longrightarrow C_f$$

is called a cofiber sequence at its first stage. After suspending, one obtains the extended pattern

$$A \xrightarrow{f} X \longrightarrow C_f \longrightarrow \Sigma A \longrightarrow \Sigma X \longrightarrow \cdots .$$

Later homology turns this geometric sequence into exact algebraic sequences.

## 7.9 Examples from spheres

For the degree-$m$ map

$$f_m : S^1 \to S^1, \qquad f_m(z) = z^m,$$

the mapping cone is

$$C_{f_m} = S^1 \cup_{f_m} D^2.$$

Thus:

$$C_{f_0} \simeq S^1 \vee S^2, \qquad C_{f_1} \cong D^2, \qquad C_{f_2} \cong \mathbb{R}P^2.$$

## 7.10 Moore-space preview

A basic Moore space can be obtained by attaching an $(n+1)$-cell to $S^n$ along a degree-$m$ map:

$$M(\mathbb{Z}/m, n) = S^n \cup_m D^{n+1}.$$

Its cellular homology later realizes a single $\mathbb{Z}/m$ group in the desired degree.

## 7.11 Functorial behavior

Suppose there is a commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow u & & \downarrow v \\ B & \xrightarrow{g} & Y. \end{array}$$

The maps $u$ and $v$ induce a natural map

$$C_f \to C_g$$

by applying $v$ on the old space and the cone on $u$ on the cone part.

**Proposition 7.9** (Map of mapping cones)**.** *A strictly commutative square of maps induces a continuous map between the corresponding mapping cones.*

## 7.12 Quotient description

One can write

$$C_f = (X \sqcup A \times I)/\sim,$$

where

$$(a, 0) \sim f(a)$$

and all $(a, 1)$ are identified to the cone vertex.

This description is useful for checking maps explicitly.

## 7.13 Mapping cylinders versus mapping cones

**Definition 7.10** (Mapping cylinder)**.** The mapping cylinder of $f : A \to X$ is

$$M_f = (A \times I) \sqcup X/((a, 0) \sim f(a)).$$

The mapping cone is obtained from $M_f$ by collapsing the free end $A \times \{1\}$ to a point.

**Proposition 7.11** (Cylinder retract)**.** *The mapping cylinder $M_f$ deformation retracts onto the copy of $X$.*

## 7.14 Null-homotopic maps

If $f : A \to X$ is null-homotopic, the mapping cone is closely related to $X \vee \Sigma A$. For CW-type spaces this relationship is a homotopy equivalence.

**Theorem 7.12** (Null-map splitting)**.** *For a based null-homotopic map between suitable CW spaces,*

$$C_f \simeq X \vee \Sigma A.$$

## 7.15 Homotopy invariance preview

If $f_0 \simeq f_1 : A \to X$, one expects

$$C_{f_0} \simeq C_{f_1}.$$

This is the mapping-cone version of the homotopic-attaching-map theorem. VIII/08 proves the relevant principle in the cell-attachment setting.

*Warning* 7.13 (Not automatically homeomorphic)*.* Homotopic maps need not produce literally identical quotient spaces. The natural invariant conclusion is homotopy equivalence, not homeomorphism.

## 7.16 Exercises

**Exercise 7.14.** Define the cone $CA$.

*Hint.* Collapse one end of $A \times I$.

Specify which end becomes the vertex and which end remains the cone base; this convention determines the later attaching identifications. □

*Solution.* $CA = (A \times I)/(A \times \{1\})$. □

**Exercise 7.15.** Why is $CA$ contractible?

*Hint.* Move every point toward the cone vertex.

For a representative $[a, s]$, vary the height toward 1 and verify that representatives of the existing vertex always have the same image. □

*Solution.* The homotopy that increases the $I$-coordinate to 1 contracts the cone to its vertex. □

**Exercise 7.16.** Define $\Sigma A$.

*Hint.* Collapse both ends.

Collapse each end separately, obtaining two suspension points; identifying both ends to the same point would be a different quotient. □

*Solution.* $\Sigma A = (A \times I)/(A \times \{0\}, A \times \{1\})$. □

**Exercise 7.17.** Express $S^{n+1}$ as a suspension.

*Hint.* Suspend $S^n$.

Use latitude as suspension height and shrink the $S^n$ factor to a point at each of the two poles. □

*Solution.* $S^{n+1} \cong \Sigma S^n$. □

**Exercise 7.18.** Define $C_f$ for $f : A \to X$.

*Hint.* Attach the cone base using $f$.

Write the boundary relation $(a, 0) \sim f(a)$ as well as the common cone-vertex relation at height 1. □

*Solution.* $C_f = X \cup_f CA$. □

**Exercise 7.19.** How is an $n$-cell attachment a mapping cone?

*Hint.* Use $CS^{n-1} \cong D^n$.

Identify the cone base with the boundary of the new disk, keeping the original map into $X$ as the attaching map. □

*Solution.* $X \cup_\varphi D^n \cong C_\varphi$. □

**Exercise 7.20.** What is $C_{\mathrm{id}_A}$?

*Hint.* The cone is attached to its own base copy.

The copy of $A$ in the target is already the cone base after gluing by the identity; no additional boundary collapse occurs. □

*Solution.* It is homeomorphic to $CA$, hence contractible. □

**Exercise 7.21.** What is the mapping cone of a constant map to a point?

*Hint.* The target is one point.

Collapsing the cone base to the target point creates the second suspension pole, distinct from the existing cone vertex. □

*Solution.* It is $\Sigma A$. □

**Exercise 7.22.** For a CW subcomplex inclusion $A \hookrightarrow X$, what quotient models the cone?

*Hint.* Collapse $A$.

Retain the CW-subcomplex hypothesis and use the cofiber-to-quotient comparison; its conclusion is a homotopy equivalence, not an automatic homeomorphism. □

*Solution.* $C_i \simeq X/A$. □

**Exercise 7.23.** Write the first three terms of a cofiber sequence.

*Hint.* Start with $f$.

The second arrow includes the old target space into the cone attachment; check its composite with $f$ contracts along the cone direction. □

*Solution.* $A \xrightarrow{f} X \to C_f$. □

**Exercise 7.24.** What follows $C_f$ in the extended cofiber sequence?

*Hint.* Suspension of the domain.

Collapse the entire copy of $X$ in $C_f$; the cone base and cone vertex then become the two suspension poles. □

*Solution.* $\Sigma A$. □

**Exercise 7.25.** Identify $C_{z \mapsto z}$ for $S^1 \to S^1$.

*Hint.* Degree one attachment.

Gluing the disk boundary bijectively to the existing circle simply makes that circle the boundary of the filled disk. □

*Solution.* It is $D^2$. □

**Exercise 7.26.** Identify $C_{z \mapsto z^2}$.

*Hint.* Recall the projective-plane attachment.

Track the degree-two boundary identification in the disk model of the projective plane, rather than treating the cone as a suspension. □

*Solution.* It is $\mathbb{R}P^2$. □

**Exercise 7.27.** Identify the constant-map cone $S^1 \to S^1$ up to homotopy.

*Hint.* Use the null-map splitting.

The constant attachment collapses only the new disk's boundary to one point of the old circle; identify the resulting sphere summand. □

*Solution.* $S^1 \vee S^2$. □

**Exercise 7.28.** What is a Moore-space attachment?

*Hint.* Attach by a degree-$m$ map.

For $m \neq 0$, the two relevant cellular groups are linked by multiplication by $m$; locate the cokernel degree and the kernel degree separately. □

*Solution.* $S^n \cup_m D^{n+1}$. □

**Exercise 7.29.** Define the mapping cylinder $M_f$.

*Hint.* Do not collapse the free end.

Identify the base end using $f$ but leave all free-end points distinct; this is precisely the distinction from $C_f$. □

*Solution.* $(A \times I) \sqcup X/((a, 0) \sim f(a))$. □

**Exercise 7.30.** How do you get $C_f$ from $M_f$?

*Hint.* Collapse the free end.

The collapsed subspace is the free copy of the domain, not the old target $X$; those two collapses yield different constructions. □

*Solution.* Collapse $A \times \{1\}$ to one point. □

**Exercise 7.31.** To what does $M_f$ deformation retract?

*Hint.* Push the cylinder into the glued end.

Decrease the cylinder height to zero and keep every point of $X$ fixed; check agreement at each glued point $(a, 0)$. □

*Solution.* To the copy of $X$. □

**Exercise 7.32.** What does a commutative square induce on cones?

*Hint.* Map the target part and cone part separately.

On a cone representative use $[a, t] \mapsto [u(a), t]$; on $X$ use $v$. Commutativity checks their agreement on the glued base. □

*Solution.* A natural map $C_f \to C_g$. □

**Exercise 7.33.** What splitting holds for a suitable null-homotopic map?

*Hint.* Use suspension.

Choose a null-homotopy to replace $f$ by a constant map, then identify the attached cone with a suspension joined at its basepoint. □

*Solution.* $C_f \simeq X \vee \Sigma A$. □

**Exercise 7.34.** Are mapping cones of homotopic maps necessarily homeomorphic?

*Hint.* The expected invariant is weaker.

A homotopy changes attaching data through continuous deformations; the comparison provides homotopy inverses, not generally a pointwise bijection of quotients. □

*Solution.* No; the natural conclusion is homotopy equivalence. □

**Exercise 7.35.** Write the quotient model of $C_f$.

*Hint.* Use $X \sqcup A \times I$.

List the relations at heights 0 and 1 separately, and check that no extra identifications are imposed at interior heights. □

*Solution.* Identify $(a, 0)$ with $f(a)$ and collapse all $(a, 1)$ to the cone vertex. □

**Exercise 7.36.** Why are mapping cones useful for relative topology?

*Hint.* Consider a subcomplex inclusion.

For a CW pair, combine the cone-to-quotient equivalence with the interpretation of reduced homology of $X/A$ as relative homology. □

*Solution.* For $A \subset X$, $C_i \simeq X/A$, so relative information becomes an ordinary quotient space. □

**Exercise 7.37.** What is the bridge from VIII/07 to VIII/08?

*Hint.* Vary the attaching map through a homotopy.

Locate the sphere-boundary inclusion that permits homotopy extension; it controls how a deformation of attaching data changes the whole attachment. □

*Solution.* VIII/08 proves that homotopic attaching maps give the same homotopy type under the standard CW hypotheses. □

## 7.17 Solved problem dossiers

**Problem 7.38** (Cone contraction)**.** Construct an explicit contraction of $CA$ to its vertex.

*Solution.* Represent a cone point by $[a, s]$. Define $H([a, s], t) = [a, (1-t)s + t]$. At $t = 1$ every point has coordinate 1, hence is the cone vertex. □

**Problem 7.39** (Suspension of a point)**.** Identify $\Sigma\{*\}$.

*Solution.* The interval has each endpoint separately collapsed, which changes nothing; the result is an interval, hence contractible. □

**Problem 7.40** (Suspension of $S^0$)**.** Show $\Sigma S^0 \cong S^1$.

*Solution.* Two intervals, one for each point of $S^0$, share both suspension endpoints. Their union is a circle. □

**Problem 7.41** (Identity cone)**.** Prove $C_{\mathrm{id}_A} \cong CA$.

*Solution.* The target copy $A$ is glued pointwise to the cone base. It contributes no additional points beyond that base, leaving precisely the cone. □

**Problem 7.42** (Constant cone)**.** Compute the mapping cone of the constant map $S^{n-1} \to \{*\}$.

*Solution.* It is the suspension $\Sigma S^{n-1} \cong S^n$. □

**Problem 7.43** (Degree-one circle cone)**.** Compute $C_f$ for $f : S^1 \to S^1$ of degree 1 represented by the identity.

*Solution.* The cone is a disk attached along its full boundary by the identity, so the result is $D^2$, hence contractible. □

**Problem 7.44** (Degree-two circle cone)**.** Explain why the cone of $z \mapsto z^2$ is $\mathbb{R}P^2$.

*Solution.* The cone on $S^1$ is $D^2$; attaching its boundary to $S^1$ by the degree-two quotient map is the standard cell model of $\mathbb{R}P^2$. □

**Problem 7.45** (Null circle cone)**.** Compute the cone of a constant $S^1 \to S^1$.

*Solution.* The disk boundary collapses at one point of the old circle, producing $S^1 \vee S^2$. □

**Problem 7.46** (Inclusion cone)**.** For the inclusion $S^{n-1} \hookrightarrow D^n$, identify the quotient model $D^n/S^{n-1}$.

*Solution.* It is $S^n$. Thus the mapping cone of the boundary inclusion has homotopy type $S^n$. □

**Problem 7.47** (Pair quotient)**.** Let $A$ be a CW subcomplex of $X$. Explain why $C_i$ models $X/A$.

*Solution.* The cone on $A$ contracts the entire attached copy of $A$ to the cone vertex while the base is identified with the original $A \subset X$. For a cofibration such as a CW-subcomplex inclusion, this yields the quotient homotopy type. ☐

**Problem 7.48** (Mapping-cylinder retraction)**.** Write an explicit deformation retraction $M_f \to X$.

*Solution.* On the cylinder send $[a, s]$ to $[a, (1-t)s]$ during the homotopy, keeping $X$ fixed. At $t = 1$ every cylinder point lies at $s = 0$, identified with $f(a) \in X$. ☐

**Problem 7.49** (Cone from cylinder)**.** Describe $C_f$ as a quotient of $M_f$.

*Solution.* Collapse the free copy $A \times \{1\} \subset M_f$ to a single point. The result is exactly the mapping cone. ☐

**Problem 7.50** (Map of cones)**.** Given $vf = gu$, construct $C_f \to C_g$.

*Solution.* Use $v$ on $X$ and send $[a, t] \in CA$ to $[u(a), t] \in CB$. Commutativity ensures agreement on the glued base. ☐

**Problem 7.51** (Moore-space preview)**.** For $M = S^n \cup_m D^{n+1}$, predict the nontrivial reduced homology group.

*Solution.* Cellular homology later gives a boundary map multiplication by $m$, so the reduced homology in degree $n$ is $\mathbb{Z}/m$. ☐

**Problem 7.52** (Cofiber continuation)**.** Explain geometrically why a map $C_f \to \Sigma A$ exists.

*Solution.* Collapse the copy of $X \subset C_f$ to a point. The cone part then has its base collapsed as well as its vertex, producing the suspension $\Sigma A$. ☐

**Problem 7.53** (Null-map splitting)**.** Why should a null-homotopy of $f : A \to X$ lead to $C_f \simeq X \vee \Sigma A$?

*Solution.* The null-homotopy lets the cone base attachment deform to a constant attachment. A constant attachment wedges the suspended domain onto $X$. ☐

**Problem 7.54** (Two homotopic maps)**.** Suppose $f_0 \simeq f_1 : A \to X$. What mapping-cone relation should be expected?

*Solution.* Under the standard CW/cofibration hypotheses, $C_{f_0} \simeq C_{f_1}$. VIII/08 develops the corresponding attaching-map theorem. ☐

**Problem 7.55** (Quotient map check)**.** Verify that the natural $X \to C_f$ is continuous.

*Solution.* It is the composite of the inclusion $X \hookrightarrow X \sqcup CA$ with the quotient map defining $C_f$, hence continuous. ☐

**Problem 7.56** (Suspension as double cone)**.** Show that $\Sigma A$ is obtained from two copies of $CA$ glued along $A$.

*Solution.* Split $A \times I$ at $t = 1/2$. Each half becomes a cone after collapsing its outer end, and their common middle slice is a copy of $A$. ☐

**Problem 7.57** (Bridge to exact sequences)**.** Why does the cofiber sequence foreshadow homology exact sequences?

*Solution.* Homology converts the geometric passage $A \to X \to C_f \to \Sigma A$ into groups linked by induced maps; suspension shifts degree, yielding the connecting morphisms of the long exact sequence. ☐

## 7.18 Supplementary worked examples

**Example 7.58** (Cone of the identity)**.** The mapping cone of $\mathrm{id}_X$ is contractible: one glues the cone $CX$ to $X$ along the identity, and the resulting space is naturally homeomorphic to the cone with its base retained as the attached copy.

**Example 7.59** (Cone of a constant map)**.** For a based constant map $f : X \to Y$, the mapping cone is homotopy equivalent to $Y \vee \Sigma X$. The cone base is collapsed at one point of $Y$, leaving the suspension contribution.

**Example 7.60** (Degree-$m$ cone)**.** The mapping cone of a degree-$m$ map $S^1 \to S^1$ is a two-dimensional CW complex with one cell in dimensions $0, 1, 2$ and cellular differential $d_2 = m$, producing $H_1 \cong \mathbb{Z}/m$ for $m \neq 0$.

## 7.19 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 7.61** (Cone definition)**.** Describe the cone $CX$ as a quotient of $X \times I$.

*Hint.* Collapse one end of the cylinder. □

*Solution.* $CX = X \times I/(X \times \{1\}$ collapsed to a point) (up to the chosen endpoint convention). □

**Exercise 7.62** (Mapping cone gluing)**.** What is glued to $Y$ in the mapping cone of $f : X \to Y$?

*Hint.* Attach the base of $CX$ using $f$. □

*Solution.* One forms $C_f = Y \cup_f CX$, identifying $(x, 0)$ in the cone base with $f(x) \in Y$. □

**Exercise 7.63** (Identity cone)**.** What is the homotopy type of the mapping cone of $\mathrm{id}_X$?

*Hint.* The identity attachment fills the copy of $X$ by its cone. □

*Solution.* It is contractible. □

**Exercise 7.64** (Point target)**.** What is the mapping cone of the map $X \to *$?

*Hint.* The cone base is collapsed to the target point. □

*Solution.* It is the reduced suspension $\Sigma X$ in the based setting. □

**Exercise 7.65** (Degree-$m$ homology)**.** For the cone of degree $m : S^1 \to S^1$, compute $H_1$ when $m \neq 0$.

*Hint.* Use the cellular differential $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$. □

*Solution.* $H_1 \cong \operatorname{coker}(m) = \mathbb{Z}/m$. □

## Proofs

**Exercise 7.66** (Identity cone contraction)**.** Prove the cone of the identity is contractible.

*Hint.* Use the cone coordinate to slide all points to the cone vertex. ☐

*Solution.* The attached copy of $X$ is exactly the cone base; the ordinary cone contraction descends to the mapping cone. ☐

**Exercise 7.67** (Homotopy functoriality)**.** Explain why homotopic maps have homotopy-equivalent mapping cones under CW/cofibration hypotheses.

*Hint.* Use the homotopy as gluing data over $X \times I$. ☐

*Solution.* A mapping-cylinder comparison interpolates the attaching maps and gives maps between the two cones that are homotopy inverses. ☐

**Exercise 7.68** (Cofiber sequence intuition)**.** Why does $X \xrightarrow{f} Y \to C_f$ behave like a quotient that kills the image of $f$ homotopically?

*Hint.* The cone on $X$ is contractible and fills each $f(x)$ along a cone line. ☐

*Solution.* After adjoining $CX$, the map $X \to C_f$ through $Y$ becomes null-homotopic; $C_f$ is the homotopy-theoretic cofiber of $f$. ☐

**Exercise 7.69** (Suspension case)**.** Prove $C_{X\to *} \cong \Sigma X$ in the reduced based model.

*Hint.* Both constructions collapse the two cylinder ends appropriately. ☐

*Solution.* The target point collapses the base $X \times \{0\}$ while the cone collapses $X \times \{1\}$, exactly yielding the suspension quotient. ☐

## Counterexamples and hypothesis tests

**Exercise 7.70** (Cone always contractible)**.** Test: every mapping cone is contractible.

*Hint.* Only the ordinary cone is automatically contractible. ☐

*Solution.* False; degree-$m$ mapping cones can have nontrivial homology. ☐

**Exercise 7.71** (Cone remembers only image)**.** Test: mapping cones depend only on the set-theoretic image of $f$.

*Hint.* The attaching homotopy class matters. ☐

*Solution.* False. Different attaching degrees with the same image can give different homology. ☐

**Exercise 7.72** (Cone of zero is just $Y$)**.** Test: the cone of a constant map $X \to Y$ is homotopy equivalent to $Y$.

*Hint.* There is generally a suspension summand. ☐

*Solution.* False; in the based setting it is typically $Y \vee \Sigma X$. ☐

## Applications and investigations

**Exercise 7.73** (Relation-to-homology)**.** Why are mapping cones useful for turning a map into an object whose homology measures failure of isomorphism?

*Hint.* Cone homology fits into a long exact sequence with the source and target. □

*Solution.* The cofiber sequence yields an exact sequence in homology, so kernels and cokernels of $f_*$ appear as pieces of the cone's homology. □

**Exercise 7.74** (Data structure analogy)**.** Interpret a mapping cone as recording both a target and a certificate that the source map becomes null.

*Hint.* Each source point receives a cone path to one vertex. □

*Solution.* The attached cone supplies explicit null-homotopy paths for the composite $X \to Y \to C_f$, while retaining the rest of $Y$; it packages a map together with its homotopy cofiber. □

## Challenge problems

**Exercise 7.75** (Moore space)**.** Use a mapping cone to construct a space with reduced homology $\mathbb{Z}/m$ in degree one.

*Hint.* Take the degree-$m$ map on $S^1$. □

*Solution.* The mapping cone of $S^1 \xrightarrow{m} S^1$ has cellular chain differential $d_2 = m$, so $\widetilde{H}_1 \cong \mathbb{Z}/m$ and the adjacent reduced groups vanish. □

**Exercise 7.76** (Cone synthesis)**.** For $f : S^n \to S^n$ of degree $m$, predict the key cellular differential in $C_f$.

*Hint.* There is an $n$-cell from the target and an $(n+1)$-cell from the cone. □

*Solution.* The cellular differential $C_{n+1} \to C_n$ is multiplication by $m$, so the cone converts degree into torsion homology. □

# Chapter 8

# Homotopic Attaching Maps

The topology of an adjunction space depends on its attaching map, but it does not depend on that map pointwise. Under the standard CW/cofibration hypotheses, deforming the attaching map through a homotopy preserves the homotopy type of the resulting space. This chapter makes that principle precise and develops the homotopy-extension mechanism behind it.

## 8.1 The basic problem

Let

$$\varphi_0, \varphi_1 : S^{n-1} \to X$$

be homotopic maps. Form

$$Y_i = X \cup_{\varphi_i} D^n.$$

The two quotient spaces need not be literally identical, but one expects

$$Y_0 \simeq Y_1.$$

## 8.2 Homotopy extension property

**Definition 8.1** (Homotopy extension property)**.** A pair $(X, A)$ has the homotopy extension property if, whenever

$$f : X \to Y$$

and

$$H : A \times I \to Y$$

satisfy $H(a, 0) = f(a)$, there exists

$$\widetilde{H} : X \times I \to Y$$

with

$$\widetilde{H}(x, 0) = f(x), \qquad \widetilde{H}|_{A \times I} = H.$$

**Definition 8.2** (Cofibration)**.** An inclusion $i : A \hookrightarrow X$ is a cofibration if the pair $(X, A)$ has the homotopy extension property.

## 8.3 CW pairs are cofibrations

**Theorem 8.3** (CW subcomplex inclusion)**.** *If $A$ is a CW subcomplex of a CW complex $X$, then*

$$A \hookrightarrow X$$

*is a cofibration.*

This theorem is one of the main reasons CW complexes form a convenient category for homotopy theory.

## 8.4 Homotopic attachment theorem

**Theorem 8.4** (Homotopic attaching maps)**.** *Let $X$ be a CW complex and let*

$$\varphi_0, \varphi_1 : S^{n-1} \to X$$

*be homotopic. Then*

$$X \cup_{\varphi_0} D^n \simeq X \cup_{\varphi_1} D^n.$$

The conclusion is homotopy equivalence, not generally homeomorphism.

## 8.5 Idea of the proof

Let

$$H : S^{n-1} \times I \to X$$

be a homotopy from $\varphi_0$ to $\varphi_1$. One can insert a collar between the old space and the disk boundary and use $H$ along this collar. The resulting intermediate adjunction space contains both attachments as deformation retracts.

Equivalently, one can use a mapping-cylinder construction to replace the attaching map by a homotopic representative without changing homotopy type.

## 8.6 A cylinder model

Attach

$$S^{n-1} \times I$$

to $X$ using $H$, and place $D^n$ at the far end. At one end the boundary is attached by $\varphi_0$; at the other by $\varphi_1$. Sliding along the cylinder provides the comparison homotopies.

**Proposition 8.5** (Interpolation space)**.** *There is a space $Z_H$ constructed from the homotopy $H$ that deformation retracts onto each of*

$$X \cup_{\varphi_0} D^n \quad \textit{and} \quad X \cup_{\varphi_1} D^n.$$

## 8.7 Constant versus null-homotopic attachment

A null-homotopic attaching map need not be literally constant, but it produces the same homotopy type as a constant attachment.

**Corollary 8.6** (Null attachment splitting)**.** *If*

$$\varphi : S^{n-1} \to X$$

*is null-homotopic, then*

$$X \cup_\varphi D^n \simeq X \vee S^n$$

*under the usual pointed CW hypotheses.*

## 8.8 Degree classifies circle attachments up to homotopy

For maps

$$S^1 \to S^1,$$

homotopy classes are classified by degree.

**Corollary 8.7** (Degree determines two-cell attachment type)**.** *If*

$$\varphi, \psi : S^1 \to S^1$$

*have the same degree, then*

$$S^1 \cup_\varphi D^2 \simeq S^1 \cup_\psi D^2.$$

This does not say the two quotient spaces are homeomorphic for arbitrary representatives; it says the homotopy type is determined by the homotopy class of the attachment.

## 8.9 Several cells

Suppose a family of $n$-cells is attached by maps

$$\varphi_\alpha : S^{n-1} \to X.$$

If each $\varphi_\alpha$ is replaced by a homotopic map $\psi_\alpha$, then under the standard CW hypotheses the resulting spaces are homotopy equivalent.

**Theorem 8.8** (Simultaneous homotopic attachments)**.** *Homotopic replacement of a family of attaching maps preserves the homotopy type of the corresponding CW attachment stage.*

## 8.10 Relative version

If the attaching-map homotopy is fixed on a specified subspace of the boundary, the comparison can be performed relative to the corresponding attached subspace. Relative homotopies therefore fit naturally into the attachment theory developed in VIII/01.

## 8.11 Mapping-cone formulation

For maps

$$f_0, f_1 : A \to X$$

with $f_0 \simeq f_1$, mapping cones satisfy

$$C_{f_0} \simeq C_{f_1}$$

under the standard cofibration/CW assumptions.

**Theorem 8.9** (Homotopy invariance of mapping cones)**.** *Homotopic maps between suitable CW spaces have homotopy-equivalent mapping cones.*

This is the general form of the cell-attachment theorem.

## 8.12 Why cofibrations matter

A naive quotient argument can fail if the inclusion being pushed out behaves badly. Cofibrations guarantee that a homotopy on the attaching subspace can be extended and controlled inside the ambient construction.

*Warning* 8.10 (Do not replace cofibration hypotheses silently). The theorem is robust for CW attachments because sphere-boundary inclusions and CW-subcomplex inclusions are cofibrations. For arbitrary quotient constructions, one should not assume homotopy invariance without checking the relevant hypotheses.

## 8.13 Application: different formulas, same cell complex homotopy type

Two polygonal attaching loops can look quite different as parametrized maps but represent the same element of

$$[S^1, X].$$

When they are homotopic, attaching a two-cell along either loop yields the same homotopy type.

## 8.14 Application to null loops

If a loop

$$\varphi : S^1 \to X$$

is null-homotopic, adding a two-cell along $\varphi$ has homotopy type

$$X \vee S^2.$$

This is often easier to recognize than the original quotient.

## 8.15 Dependence only on homotopy class

The correct conceptual object attached to a cell is not the raw parametrized formula

$$\varphi : S^{n-1} \to X,$$

but its homotopy class

$$[\varphi] \in [S^{n-1}, X],$$

as far as the resulting homotopy type is concerned.

## 8.16 Limits of the conclusion

Homotopy-equivalent attachment spaces may differ in geometric or quotient-level details. The theorem does not preserve a chosen cell parametrization, metric, smooth structure, or literal point-set presentation.

*Warning* 8.11 (Homotopy equivalence is the invariant). Never strengthen the conclusion

$$X \cup_{\varphi_0} D^n \simeq X \cup_{\varphi_1} D^n$$

to homeomorphism without additional proof.

## 8.17 Bridge to the fundamental group

For two-cell attachments to a graph, the homotopy class of the attaching loop is encoded by an element of the fundamental group. Thus the next chapter's group law on loops gives an algebraic language for the attachment data of two-dimensional CW complexes.

## 8.18 Exercises

**Exercise 8.12.** Define the homotopy extension property.

*Hint.* Extend a homotopy from the subspace while preserving the initial map.

The prescribed maps must agree on $A \times \{0\}$; ask for an extension from $(X \times \{0\}) \cup (A \times I)$ to the full cylinder. □

*Solution.* A pair $(X, A)$ has HEP when every compatible $f : X \to Y$ and $H : A \times I \to Y$ extend to $\widetilde{H} : X \times I \to Y$. □

**Exercise 8.13.** Define a cofibration.

*Hint.* Use HEP.

Include the quantifier over all target spaces and all compatible initial maps and subspace homotopies for the specified inclusion. □

*Solution.* An inclusion $A \hookrightarrow X$ is a cofibration when $(X, A)$ has the homotopy extension property. □

**Exercise 8.14.** What important inclusions in CW complexes are cofibrations?

*Hint.* Think subcomplexes.

Check that the subspace consists of cells together with their attaching boundaries, so the standard CW-pair homotopy-extension theorem applies. □

*Solution.* Every CW-subcomplex inclusion $A \hookrightarrow X$. □

**Exercise 8.15.** State the homotopic-attaching-map theorem.

*Hint.* Compare two maps $S^{n-1} \to X$.

State the common source, common target, and homotopy between the two attaching maps before identifying the two adjunction spaces to compare. □

*Solution.* If $\varphi_0 \simeq \varphi_1$, then $X \cup_{\varphi_0} D^n \simeq X \cup_{\varphi_1} D^n$ under the standard CW hypotheses. □

**Exercise 8.16.** What is the conclusion: homeomorphism or homotopy equivalence?

*Hint.* Use the weaker invariant.

Ask whether the construction supplies a continuous bijection or maps inverse only up to homotopy; the theorem guarantees only the latter. □

*Solution.* Homotopy equivalence. □

**Exercise 8.17.** What geometric device interpolates between two attaching maps?

*Hint.* Use the homotopy parameter as a collar.

Use the two ends of $S^{n-1} \times I$ for the two boundary maps; the side carries their joint continuous interpolation. □

*Solution.* A cylinder $S^{n-1} \times I$ carrying the homotopy. □

**Exercise 8.18.** What happens if the attaching map is null-homotopic?

*Hint.* Replace it by a constant attachment.

Replace $\varphi$ by its constant representative and identify $D^n/\partial D^n$, while retaining the old space $X$. □

*Solution.* $X \cup_\varphi D^n \simeq X \vee S^n$. □

**Exercise 8.19.** When are two maps $S^1 \to S^1$ homotopic?

*Hint.* Use degree.

Use the classification theorem for circle maps, not just the fact that degree is invariant; sufficiency is special information. □

*Solution.* Exactly when they have the same degree. □

**Exercise 8.20.** What does equal degree imply for two two-cell attachments to $S^1$?

*Hint.* Apply the attachment theorem.

First convert equal degrees into a homotopy of circle maps, then invoke homotopy invariance of the cell attachment. □

*Solution.* The resulting adjunction spaces are homotopy equivalent. □

**Exercise 8.21.** Can several attaching maps be changed simultaneously by homotopy?

*Hint.* Use the family version.

Form the disjoint union of the homotopies on the separate boundary spheres; check the associated CW pushout rather than assuming arbitrary quotients behave well. □

*Solution.* Yes, under the standard CW hypotheses the resulting stage has the same homotopy type. □

**Exercise 8.22.** What relative condition can be imposed on an attaching homotopy?

*Hint.* Fix a subspace during the deformation.

For the selected boundary subspace $B$, require $H(b,t)$ to be independent of $t$; endpoint agreement alone is not a relative homotopy. □

*Solution.* The homotopy may be taken relative to a specified boundary subspace, leading to a relative comparison. □

**Exercise 8.23.** State the mapping-cone version of the theorem.

*Hint.* Replace $S^{n-1}$ by a general domain.

Use the cone-cylinder model to replace a sphere domain by $A$, and retain the hypotheses supporting the homotopy pushout comparison. □

*Solution.* If $f_0 \simeq f_1 : A \to X$, then $C_{f_0} \simeq C_{f_1}$ under suitable CW/cofibration hypotheses. □

**Exercise 8.24.** Why are cofibrations used?

*Hint.* They control extension of homotopies.

Identify the extension problem on the old space and attaching collar; HEP supplies the compatible ambient deformation missing from a set-theoretic quotient argument. □

*Solution.* They ensure a homotopy on the attaching subspace extends compatibly to the ambient construction. ☐

**Exercise 8.25.** What point-set warning accompanies the theorem?

*Hint.* Do not confuse equivalence notions.

Do not transfer local point-set properties merely from homotopy equivalence; the comparison theorem does not claim the quotients have the same local neighborhoods. ☐

*Solution.* Homotopy-equivalent quotients need not be homeomorphic. ☐

**Exercise 8.26.** On what datum does the homotopy type of a cell attachment depend?

*Hint.* Pass from map to class.

Keep the target $X$ and cell dimension fixed when replacing the attaching map by a representative of its homotopy class. ☐

*Solution.* On the homotopy class $[\varphi] \in [S^{n-1}, X]$, under the standard hypotheses. ☐

**Exercise 8.27.** If $\varphi : S^1 \to X$ is null-homotopic, what is the two-cell attachment type?

*Hint.* Set $n = 2$.

The new disk has dimension two while its boundary loop has dimension one; keep this shift when identifying the wedge summand. ☐

*Solution.* $X \vee S^2$ up to homotopy. ☐

**Exercise 8.28.** What is an interpolation space $Z_H$?

*Hint.* Use a cylinder carrying $H$.

Use the collar carrying $H$ to compare the endpoint attachments, then identify the retractions or comparison maps proving the endpoint homotopy types agree. ☐

*Solution.* A space built from the homotopy between attachments that deformation retracts onto both endpoint adjunction spaces. ☐

**Exercise 8.29.** Why can different formulas represent the same attachment homotopy type?

*Hint.* Parametrized maps may be homotopic.

A reparameterization or deformation may alter every pointwise formula while keeping its class in $[S^{n-1}, X]$ unchanged. ☐

*Solution.* The theorem depends on the homotopy class, not the literal formula. ☐

**Exercise 8.30.** Does HEP say every homotopy on every subset extends?

*Hint.* The pair matters.

Distinguish a fixed pair's HEP from a claim about all subsets, and include compatibility with the chosen initial map. ☐

*Solution.* No. HEP is a property of a specified inclusion $A \hookrightarrow X$. ☐

**Exercise 8.31.** Why is $S^{n-1} \hookrightarrow D^n$ a good attaching inclusion?

*Hint.* It is a CW-subcomplex inclusion.

Choose a CW structure on the disk containing its boundary as a subcomplex; apply the corresponding subcomplex-cofibration result. ☐

*Solution.* It is a cofibration. ☐

**Exercise 8.32.** How does VIII/08 refine VIII/05?

*Hint.* VIII/05 said the map matters.

Separate exact quotient data from homotopy-type data: raw attaching formulas specify the quotient, whereas homotopy classes suffice for the theorem's weaker conclusion. ☐

*Solution.* VIII/08 says the homotopy class matters for homotopy type, so pointwise changes within the same class do not matter. ☐

**Exercise 8.33.** How does VIII/08 refine VIII/07?

*Hint.* Apply the same principle to cones.

Replace an attached disk by the cone on its boundary; this identifies the attachment comparison as a particular mapping-cone comparison. ☐

*Solution.* It proves homotopy invariance of mapping cones under suitable hypotheses. ☐

**Exercise 8.34.** What algebraic object will encode attaching loops in dimension two?

*Hint.* Go to VIII/09.

Choose a basepoint and a path to the attaching loop when necessary; free loop classes correspond to conjugacy classes rather than a canonical based element. ☐

*Solution.* The fundamental group. ☐

**Exercise 8.35.** Why is this theorem especially useful for CW complexes?

*Hint.* Attaching maps define skeleta.

At one skeleton, replace a complicated boundary map by a homotopic representative with simpler combinatorics; check the CW hypotheses before proceeding to later attachments. ☐

*Solution.* It permits replacement of complicated attaching maps by simpler homotopic representatives without changing the homotopy type. ☐

## 8.19 Solved problem dossiers

**Problem 8.36** (Null attachment)**.** Let $\varphi : S^{n-1} \to X$ be null-homotopic. Derive the splitting of the attachment.

*Solution.* Choose a homotopy from $\varphi$ to a constant map $c$. The homotopic-attaching-map theorem gives $X \cup_\varphi D^n \simeq X \cup_c D^n$, and the constant attachment has homotopy type $X \vee S^n$. ☐

**Problem 8.37** (Degree-three representatives)**.** Let $\varphi, \psi : S^1 \to S^1$ both have degree 3. Compare the two attached spaces.

*Solution.* Circle maps of equal degree are homotopic. Therefore $S^1 \cup_\varphi D^2 \simeq S^1 \cup_\psi D^2$. ☐

**Problem 8.38** (Degree mismatch)**.** Suppose $\deg \varphi = 2$ and $\deg \psi = 3$. Can VIII/08 identify their attachments?

*Solution.* No. The maps are not homotopic, so the theorem does not apply. Later cellular homology in fact distinguishes the corresponding degree attachments. ☐

**Problem 8.39** (Constant versus wiggly null loop)**.** A loop in $X$ is visibly complicated but null-homotopic. What two-cell attachment homotopy type results?

*Solution.* Replace the loop by the constant loop through a null-homotopy. The result is homotopy equivalent to $X \vee S^2$. □

**Problem 8.40** (Cylinder interpolation)**.** Explain how $S^{n-1} \times I$ records a homotopy between two attachments.

*Solution.* At time 0 its base is glued to $X$ by $\varphi_0$, at time 1 by $\varphi_1$, and intermediate slices follow $H(-, t)$. Sliding along this collar produces comparison deformation retracts. □

**Problem 8.41** (Why not homeomorphic?)**.** Why does the theorem not automatically produce a homeomorphism between the endpoint quotient spaces?

*Solution.* The construction controls deformations and deformation retracts, not pointwise quotient identifications. Homotopy equivalence is invariant under the interpolation, while homeomorphism would require stronger point-set control. □

**Problem 8.42** (Mapping-cone corollary)**.** Deduce the homotopy invariance of $C_f$ from the attachment principle.

*Solution.* A mapping cone is obtained by attaching the cone $CA$ along its base $A$ using $f$. Replacing $f$ by a homotopic map changes the attachment through a homotopy, hence preserves the cone homotopy type under the cofibration hypotheses. □

**Problem 8.43** (CW-pair role)**.** Why does the theorem apply cleanly to a cell attachment in a CW complex?

*Solution.* The boundary inclusion $S^{n-1} \hookrightarrow D^n$ and CW-subcomplex inclusions are cofibrations, so the required homotopies can be extended in the pushout construction. □

**Problem 8.44** (Two-cell graph attachment)**.** Let $X$ be a graph and let two loops $\alpha, \beta : S^1 \to X$ be homotopic rel basepoint. Compare the attached two-complexes.

*Solution.* Attaching one two-cell along $\alpha$ or $\beta$ produces homotopy-equivalent spaces. The based homotopy is stronger than needed for the unbased conclusion. □

**Problem 8.45** (Relative attachment)**.** Suppose a homotopy of attaching maps is fixed on a subset $B \subset S^{n-1}$. What extra control should the comparison preserve?

*Solution.* The interpolation can be performed relative to the image of $B$, so the resulting homotopy equivalence can preserve the corresponding attached subspace. □

**Problem 8.46** (Family of null attachments)**.** Attach $r$ $n$-cells to $X$ by null-homotopic maps. Predict the homotopy type.

*Solution.* Replace all attaching maps simultaneously by constants. Under the standard pointed hypotheses the result is homotopy equivalent to $X \vee \bigvee_{j=1}^{r} S^n$. □

**Problem 8.47** (HEP extension)**.** Given $f : X \to Y$ and a homotopy $H : A \times I \to Y$ starting at $f|_A$, what does HEP produce?

*Solution.* It produces $\widetilde{H} : X \times I \to Y$ extending $H$ and satisfying $\widetilde{H}(-, 0) = f$. □

**Problem 8.48** (Subcomplex example)**.** Give a concrete cofibration from the previous chapters.

*Solution.* The inclusion $S^{n-1} \hookrightarrow D^n$ is an inclusion of a CW subcomplex and therefore a cofibration. □

**Problem 8.49** (Attachment simplification)**.** Why is it legitimate to replace an attaching loop by a simpler loop in the same homotopy class before computing invariants?

*Solution.* VIII/08 ensures the replacement preserves the homotopy type of the resulting CW complex, and homotopy invariants therefore have the same values. □

**Problem 8.50** (Same word, different parametrization)**.** Two polygon boundary maps traverse the same loop word but with different speeds and pauses. Compare their attachments.

*Solution.* Reparameterization gives a homotopy of the loops, so their two-cell attachments are homotopy equivalent. □

**Problem 8.51** (Reverse parametrization warning)**.** Is a loop necessarily homotopic to the same geometric loop traversed backward?

*Solution.* No. Reversal represents the inverse fundamental-group element and need not be homotopic to the original loop. VIII/09 makes this distinction algebraic. □

**Problem 8.52** (Degree-zero circle attachment)**.** Let $\varphi : S^1 \to S^1$ have degree 0. Identify the attachment homotopy type.

*Solution.* Every degree-zero circle map is null-homotopic, so $S^1 \cup_\varphi D^2 \simeq S^1 \vee S^2$. □

**Problem 8.53** (Degree-one circle attachment)**.** Let $\varphi : S^1 \to S^1$ have degree 1. Compare its attachment with the identity attachment.

*Solution.* $\varphi$ is homotopic to the identity, so the attachment is homotopy equivalent to $S^1 \cup_{\mathrm{id}} D^2 \cong D^2$, hence contractible. □

**Problem 8.54** (Homotopy-class principle)**.** State precisely what attachment data VIII/08 says can be simplified.

*Solution.* For homotopy-type questions under the standard hypotheses, the attaching map may be replaced by any representative of its homotopy class in $[S^{n-1}, X]$. □

**Problem 8.55** (Bridge to $\pi_1$)**.** Why is the fundamental group the natural next tool for two-cell attachments?

*Solution.* An attaching map for a two-cell is a loop $S^1 \to X^1$. Based homotopy classes of loops form $\pi_1(X^1)$, converting the attaching data into group elements and words. □

## 8.20 Supplementary worked examples

**Example 8.56** (Two degree-one attachments)**.** Any two degree-one maps $S^{n-1} \to S^{n-1}$ are homotopic, so attaching an $n$-cell along either gives homotopy-equivalent adjunction spaces under the standard CW hypotheses.

**Example 8.57** (Changing a loop representative)**.** If two loops in a graph are homotopic through based loops, attaching a 2-cell along either loop imposes the same homotopy-level relation. The geometric routes can differ while the resulting CW homotopy type agrees.

**Example 8.58** (Constant attachments)**.** Any two based constant attaching maps $S^{n-1} \to X$ with the same base point coincide; attaching along the constant map gives a wedge $X \vee S^n$ in the standard based CW setting.

## 8.21 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 8.59** (Attachment data)**.** Which part of an $n$-cell attachment may be replaced by a homotopic map?

*Hint.* The map on $S^{n-1}$. ☐

*Solution.* The attaching map $f : S^{n-1} \to X$ can be replaced by a homotopic attaching map $g$. ☐

**Exercise 8.60** (Constant attachment)**.** What is the homotopy type of attaching an $n$-cell by a constant map to a based space $X$?

*Hint.* The boundary collapses to the base point. ☐

*Solution.* It gives $X \vee S^n$ (indeed the quotient has this standard form). ☐

**Exercise 8.61** (Degree classes)**.** When $X = S^{n-1}$, what invariant classifies attaching maps $S^{n-1} \to S^{n-1}$ up to homotopy?

*Hint.* Use degree. ☐

*Solution.* Their degree in $\mathbb{Z}$ classifies the homotopy class for sphere self-maps. ☐

**Exercise 8.62** (Same degree)**.** If two sphere self-maps have the same degree, what can be concluded about the corresponding one-cell attachments?

*Hint.* First identify their homotopy classes. ☐

*Solution.* They are homotopic, hence their adjunction spaces are homotopy equivalent under the standard attachment theorem. ☐

**Exercise 8.63** (Zero degree)**.** What is special about a degree-zero attaching map $S^{n-1} \to S^{n-1}$?

*Hint.* It is null-homotopic in the sphere range. ☐

*Solution.* It is homotopic to a constant map, so the attachment has the homotopy type of $S^{n-1} \vee S^n$. ☐

## Proofs

**Exercise 8.64** (Homotopic attachment theorem)**.** Outline why homotopic attaching maps give homotopy-equivalent adjunction spaces.

*Hint.* Use the attaching homotopy on a collar/cylinder. ☐

*Solution.* Insert the homotopy between the two boundary identifications into a mapping-cylinder comparison; the inclusion of the boundary sphere is a cofibration, allowing the homotopy to extend to homotopy equivalences of pushouts. ☐

**Exercise 8.65** (Degree consequence)**.** Prove equal-degree sphere attachments yield homotopy-equivalent spaces.

*Hint.* Equal degree gives homotopic maps. ☐

*Solution.* Sphere self-maps with equal degree are homotopic; apply the homotopic attaching-map theorem. ☐

**Exercise 8.66** (Null attachment wedge)**.** Prove a null-homotopic attaching map yields the homotopy type $X \vee S^n$.

*Hint.* Replace it by a constant attaching map. ☐

*Solution.* Homotopic attachments have the same homotopy type, and the constant attachment collapses $S^{n-1}$ to the base point, producing the wedge with $S^n$. ☐

**Exercise 8.67** (Iterated attachments)**.** Explain how the theorem supports inductive comparisons of CW complexes.

*Hint.* Compare one skeleton at a time. ☐

*Solution.* If corresponding attaching maps are homotopic after identifying the lower skeleta, each attachment step preserves a homotopy equivalence, enabling cellular induction. ☐

## Counterexamples and hypothesis tests

**Exercise 8.68** (Homotopic gives homeomorphic)**.** Test: homotopic attaching maps always give homeomorphic adjunction spaces.

*Hint.* The theorem guarantees homotopy type, not necessarily homeomorphism. ☐

*Solution.* False in that strength; homotopy equivalence is the robust conclusion. ☐

**Exercise 8.69** (Same image suffices)**.** Test: attaching maps with the same image subset always yield the same homotopy type.

*Hint.* Winding/degree can differ despite identical image. ☐

*Solution.* False. The homotopy class, not merely image, controls the attachment. ☐

**Exercise 8.70** (Different degree harmless)**.** Test: changing the degree of a sphere attaching map never changes homology.

*Hint.* Degree becomes a cellular boundary coefficient. ☐

*Solution.* False. Mapping-cone examples show different degrees produce different torsion homology. ☐

## Applications and investigations

**Exercise 8.71** (Model simplification)**.** How can one simplify an attaching map before computation without changing homotopy type?

*Hint.* Replace it by a convenient representative of its homotopy class. ☐

*Solution.* Choose a homotopic map with simpler geometry or combinatorics; the resulting attached space retains the same homotopy type while calculations become easier. ☐

**Exercise 8.72** (Presentation robustness)**.** Why does this theorem make CW descriptions robust under geometric perturbation?

*Hint.* Small/controlled deformations often preserve homotopy class. ☐

*Solution.* The exact geometric route of an attaching sphere can change while its homotopy class stays fixed, so homotopy-level invariants of the CW complex remain stable. ☐

## Challenge problems

**Exercise 8.73** (Degree-$m$ family)**.** Compare the spaces obtained by attaching a 2-cell to $S^1$ with degrees $m$ and $-m$.

*Hint.* Their cellular homology depends on $|m|$; consider an orientation reversal of the cell or circle. ☐

*Solution.* They have isomorphic cellular homology and are in fact related by pre/postcomposition with degree $-1$ homeomorphisms, so the sign can be absorbed by orientation reversal; the magnitude records the essential torsion. ☐

**Exercise 8.74** (Attachment synthesis)**.** Explain the hierarchy: equal maps, homotopic maps, equal degree maps on spheres, and resulting attached spaces.

*Hint.* Each implication weakens geometric data but preserves enough homotopy information. ☐

*Solution.* Equal maps are homotopic; on spheres equal degree implies homotopic; homotopic attaching maps yield homotopy-equivalent adjunction spaces. Thus degree can control an entire family of CW homotopy types. ☐

# Part III

# Fundamental Groups and Coverings

# Chapter 9

# Paths and Fundamental Groups

The first genuinely algebraic invariant of a space is built from loops. Paths record how points are connected, while loops based at a chosen point can be concatenated. Passing to homotopy classes converts this concatenation into a group law. The resulting fundamental group detects holes, organizes covering spaces, and encodes two-cell attaching loops in low-dimensional CW complexes.

## 9.1 Paths

**Definition 9.1** (Path)**.** A path in $X$ from $x_0$ to $x_1$ is a continuous map

$$\alpha : I \to X$$

with

$$\alpha(0) = x_0, \qquad \alpha(1) = x_1.$$

A loop based at $x_0$ is a path satisfying

$$\alpha(0) = \alpha(1) = x_0.$$

## 9.2 Path reversal

For a path $\alpha$, define

$$\overline{\alpha}(t) = \alpha(1 - t).$$

Then $\overline{\alpha}$ runs from the endpoint of $\alpha$ back to its starting point.

## 9.3 Concatenation

If

$$\alpha(1) = \beta(0),$$

define

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t), & 0 \le t \le \frac{1}{2}, \\ \beta(2t - 1), & \frac{1}{2} \le t \le 1. \end{cases}$$

*Warning* 9.2 (Convention)*.* This chapter reads $\alpha * \beta$ as first traverse $\alpha$, then $\beta$. Some texts adopt the opposite multiplication convention.

## 9.4 Path homotopy

**Definition 9.3** (Homotopy rel endpoints)**.** Two paths $\alpha, \beta : I \to X$ with the same endpoints are path-homotopic if there exists

$$H : I \times I \to X$$

such that

$$H(s,0) = \alpha(s), \qquad H(s,1) = \beta(s),$$

and

$$H(0,t) = x_0, \qquad H(1,t) = x_1.$$

This is exactly homotopy relative to $\{0,1\}$ from VIII/01.

## 9.5 The fundamental group

**Definition 9.4** (Fundamental group)**.** The fundamental group of $X$ at $x_0$ is

$$\pi_1(X, x_0) = \{\text{loops at } x_0\}/\text{path homotopy},$$

with multiplication

$$[\alpha][\beta] = [\alpha * \beta].$$

**Theorem 9.5** (Group structure)**.** *Path concatenation induces a well-defined group structure on $\pi_1(X, x_0)$.*

## 9.6 Identity and inverse

The identity element is represented by the constant loop

$$c_{x_0}(t) = x_0.$$

The inverse of $[\alpha]$ is

$$[\overline{\alpha}].$$

Associativity of literal path concatenation holds only up to reparameterization, but that is enough because reparameterized paths are homotopic rel endpoints.

## 9.7 Well-defined multiplication

**Proposition 9.6** (Compatibility with path homotopy)**.** *If*

$$\alpha \simeq \alpha' \quad \textit{and} \quad \beta \simeq \beta'$$

*rel endpoints, then*

$$\alpha * \beta \simeq \alpha' * \beta'$$

*rel endpoints.*

Hence multiplication depends only on homotopy classes.

## 9.8 Simply connected spaces

**Definition 9.7** (Simply connected)**.** A space $X$ is simply connected if it is path connected and

$$\pi_1(X, x_0) = 0$$

for one, equivalently every, basepoint.

Contractible spaces are simply connected.

**Example 9.8.** For $n \geq 2$,

$$\pi_1(S^n) = 0.$$

## 9.9 The circle

**Theorem 9.9** (Fundamental group of the circle)**.**

$$\pi_1(S^1, 1) \cong \mathbb{Z}.$$

The integer records winding number. The loop

$$\omega_n(t) = e^{2\pi i n t}$$

represents the integer $n$.

## 9.10 Winding and degree

A loop in $S^1$ is equivalently a based map

$$S^1 \to S^1.$$

Its fundamental-group class is measured by the same integer that appeared as degree in VIII/03.

**Proposition 9.10** (Degree and $\pi_1(S^1)$)**.** *Under the identification $\pi_1(S^1) \cong \mathbb{Z}$,*

$$[\omega_n] \longleftrightarrow n.$$

## 9.11 Basepoint change

Let $\gamma$ be a path from $x_0$ to $x_1$.

**Theorem 9.11** (Change of basepoint)**.** *There is an isomorphism*

$$\gamma_\# : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

*given by*

$$[\alpha] \longmapsto [\overline{\gamma} * \alpha * \gamma]$$

*with the concatenation convention used in this chapter, after interpreting the order consistently.*

Different choices of connecting path may change the isomorphism by an inner automorphism.

## 9.12 Induced homomorphisms

For a based map

$$f : (X, x_0) \to (Y, y_0),$$

define

$$f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

by

$$f_*([\alpha]) = [f \circ \alpha].$$

**Theorem 9.12** (Functoriality)**.**

$$(\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X)}, \qquad (g \circ f)_* = g_* \circ f_*.$$

## 9.13 Homotopy invariance

**Theorem 9.13** (Based homotopy invariance)**.** *Based-homotopic maps induce the same homomorphism on fundamental groups.*

**Corollary 9.14** (Homotopy equivalence)**.** *A homotopy equivalence of path-connected spaces induces an isomorphism of fundamental groups, after the usual basepoint identifications.*

## 9.14 Products

**Theorem 9.15** (Product formula)**.** *For path-connected based spaces,*

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0).$$

Hence

$$\pi_1(T^2) \cong \mathbb{Z}^2.$$

## 9.15 Wedges and free groups: preview

For a wedge of $r$ circles,

$$\bigvee_{j=1}^{r} S^1,$$

the fundamental group is the free group

$$F_r.$$

A complete computation uses van Kampen's theorem, developed later in the fundamental-group arc.

## 9.16 Attaching loops and relations

A two-cell attaching map

$$\varphi : S^1 \to X^1$$

represents a loop class in

$$\pi_1(X^1).$$

Attaching the two-cell makes that loop null-homotopic in the enlarged space. Thus two-cell attachments impose relations on the fundamental group.

**Example 9.16** (Torus relation)**.** The torus one-skeleton has two loop generators $a, b$. Its two-cell attaches along

$$aba^{-1}b^{-1},$$

forcing the commutator to become trivial. This is the geometric source of

$$\pi_1(T^2) \cong \mathbb{Z}^2.$$

## 9.17 Fundamental group as a homotopy invariant

If two path-connected spaces have nonisomorphic fundamental groups, then they cannot be homotopy equivalent. In particular,

$$S^1 \not\simeq \{*\}$$

because

$$\pi_1(S^1) \cong \mathbb{Z}$$

whereas the point has trivial fundamental group.

## 9.18 Bridge to covering spaces

The circle computation is most naturally understood using the covering map

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t}.$$

Loops lift to paths in $\mathbb{R}$, and the endpoint difference is an integer. VIII/10 develops covering spaces systematically and turns this observation into a general lifting theory.

## 9.19 Exercises

**Exercise 9.17.** Define a path from $x_0$ to $x_1$.

*Hint.* Use a continuous map on $I$.

Specify both endpoint equations and the target space; continuity is required on the whole closed interval. □

*Solution.* A continuous $\alpha : I \to X$ with $\alpha(0) = x_0$ and $\alpha(1) = x_1$. □

**Exercise 9.18.** Define a loop at $x_0$.

*Hint.* Make the endpoints equal.

The two equal endpoint values must be the designated basepoint, not an unspecified point of the space. □

*Solution.* A path $\alpha$ with $\alpha(0) = \alpha(1) = x_0$. □

**Exercise 9.19.** Define the reverse path $\overline{\alpha}$.

*Hint.* Reverse the parameter.

Check how the endpoint values change under $t \mapsto 1 - t$, and apply the operation twice as a consistency test. □

*Solution.* $\overline{\alpha}(t) = \alpha(1 - t)$. □

**Exercise 9.20.** When is $\alpha * \beta$ defined?

*Hint.* Match the endpoint of the first to the start of the second.

Evaluate the proposed two pieces at the joining time; mismatched endpoints prevent the pasting lemma from applying. □

*Solution.* When $\alpha(1) = \beta(0)$. □

**Exercise 9.21.** Write the concatenation formula.

*Hint.* Give each path half the interval.

Rescale each half-interval to $I$, retaining this chapter's convention that $\alpha$ is traversed before $\beta$. □

*Solution.* $(\alpha * \beta)(t) = \alpha(2t)$ for $t \leq 1/2$, and $\beta(2t-1)$ for $t \geq 1/2$. □

**Exercise 9.22.** What must a path homotopy keep fixed?

*Hint.* Use homotopy rel endpoints.

Use a square with path coordinate $s$ and homotopy coordinate $t$; the two vertical edges must remain at the fixed endpoints. □

*Solution.* The initial and terminal endpoints. □

**Exercise 9.23.** Define $\pi_1(X, x_0)$.

*Hint.* Take based loops modulo endpoint-fixing homotopy.

Distinguish individual loops from their relative-homotopy classes, then define multiplication on those classes by representative concatenation. □

*Solution.* The set of based loop homotopy classes with concatenation. □

**Exercise 9.24.** What is the identity in $\pi_1(X, x_0)$?

*Hint.* Use the constant loop.

A constant segment changes the timing of a concatenated loop; use endpoint-preserving reparameterization to verify the identity law on classes. □

*Solution.* $[c_{x_0}]$. □

**Exercise 9.25.** What is the inverse of $[\alpha]$?

*Hint.* Reverse the loop.

Contract the path followed by its reverse by shortening the distance traveled along it, keeping the basepoint fixed throughout. □

*Solution.* $[\overline{\alpha}]$. □

**Exercise 9.26.** Why is multiplication well-defined?

*Hint.* Concatenate representatives of homotopies.

Paste the two path homotopies along the rescaled path coordinate; their stationary common endpoint ensures continuity on the joining edge. □

*Solution.* Path-homotopic representatives have path-homotopic concatenations. □

**Exercise 9.27.** Is literal path concatenation strictly associative?

*Hint.* Compare parametrizations.

Locate the three traversal intervals in each parenthesization; a piecewise-linear change of timing compares them relative to the endpoints. □

*Solution.* No; it is associative up to endpoint-fixing reparameterization homotopy. □

**Exercise 9.28.** Define simply connected.

*Hint.* Require connectivity and trivial $\pi_1$.

Path connectedness is part of the definition; trivial groups in separate components alone do not make their disjoint union simply connected. □

*Solution.* Path connected with trivial fundamental group. □

**Exercise 9.29.** What is $\pi_1(S^n)$ for $n \geq 2$?

*Hint.* Higher spheres are simply connected.

Use two enlarged hemispheres and van Kampen: their intersection is path connected for $n \geq 2$, while each hemisphere neighborhood is contractible. □

*Solution.* Trivial. □

**Exercise 9.30.** What is $\pi_1(S^1)$?

*Hint.* Use winding number.

Lift a loop to $\mathbb{R}$ through $t \mapsto e^{2\pi i t}$ with fixed starting point; inspect its integer endpoint displacement. □

*Solution.* $\mathbb{Z}$. □

**Exercise 9.31.** What integer does $\omega_n(t) = e^{2\pi i n t}$ represent?

*Hint.* Count windings.

Use the lift $t \mapsto nt$ starting at zero and distinguish positive from negative displacement. □

*Solution.* $n$. □

**Exercise 9.32.** How is degree related to $\pi_1(S^1)$?

*Hint.* Both classify circle maps.

Apply the circle map to the positively oriented generator; its induced homomorphism on $\mathbb{Z}$ multiplies by the degree. □

*Solution.* The degree equals the winding-number integer under $\pi_1(S^1) \cong \mathbb{Z}$. □

**Exercise 9.33.** What does a based map $f : X \to Y$ induce?

*Hint.* Compose loops with $f$.

Postcompose both a loop and any endpoint-fixing homotopy with $f$; this proves the assignment is independent of the representative. □

*Solution.* A homomorphism $f_* : \pi_1(X) \to \pi_1(Y)$. □

**Exercise 9.34.** State functoriality for $g \circ f$.

*Hint.* Induced maps compose.

Evaluate both proposed homomorphisms on an arbitrary loop class, following the order of the two map compositions. □

*Solution.* $(g \circ f)_* = g_* \circ f_*$. ☐

**Exercise 9.35.** What do based-homotopic maps induce?

*Hint.* Use homotopy invariance.

For a loop $\alpha$, substitute $\alpha(s)$ into the given based homotopy; the resulting square keeps both path endpoints fixed. ☐

*Solution.* The same homomorphism on $\pi_1$. ☐

**Exercise 9.36.** What does a homotopy equivalence induce on $\pi_1$?

*Hint.* Use a homotopy inverse.

Track the path traced by the basepoint under a nonbased homotopy; the induced isomorphism requires that basepoint change, not an unsupported equality of based maps. ☐

*Solution.* An isomorphism, after appropriate basepoint identification. ☐

**Exercise 9.37.** State the product formula.

*Hint.* Project loops to both factors.

The inverse to projection pairs two loops with the same time parameter; pair their homotopies to check well-definedness. ☐

*Solution.* $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$. ☐

**Exercise 9.38.** Compute $\pi_1(T^2)$.

*Hint.* Use $T^2 = S^1 \times S^1$.

Choose generators by traversing each circle coordinate with the other fixed; use the product formula, not the free-group formula for a wedge. ☐

*Solution.* $\mathbb{Z}^2$. ☐

**Exercise 9.39.** What does attaching a two-cell along a loop do to that loop class?

*Hint.* The disk fills the loop.

The attached disk supplies a null-homotopy of its boundary loop; at the group level the relation is imposed through its normal closure. ☐

*Solution.* It makes the attaching loop null-homotopic in the enlarged space. ☐

**Exercise 9.40.** What is the bridge to VIII/10?

*Hint.* Use the universal cover of the circle.

Fix the initial lift of a circle loop and compare its endpoints under $\mathbb{R} \to S^1$; this motivates the general path-lifting theorem. ☐

*Solution.* Covering-space lifting turns loop classes into endpoint data of lifted paths. ☐

## 9.20 Solved problem dossiers

**Problem 9.41** (Reverse twice)**.** Show $\overline{\overline{\alpha}} = \alpha$.

*Solution.* By definition, $\overline{\overline{\alpha}}(t) = \overline{\alpha}(1-t) = \alpha(1-(1-t)) = \alpha(t)$. □

**Problem 9.42** (Concatenation endpoints)**.** Verify the endpoints of $\alpha * \beta$.

*Solution.* At $t = 0$ the value is $\alpha(0)$; at $t = 1$ it is $\beta(1)$. At $t = 1/2$, both formulas equal the common point $\alpha(1) = \beta(0)$. □

**Problem 9.43** (Identity loop)**.** Why is $\alpha * c_{x_1}$ homotopic rel endpoints to $\alpha$?

*Solution.* Reparameterize the traversal of $\alpha$ so that it reaches the endpoint later and removes the terminal pause. Endpoint-preserving reparameterization gives the required homotopy. □

**Problem 9.44** (Inverse loop)**.** Why is $\alpha * \overline{\alpha}$ null-homotopic rel basepoint?

*Solution.* Shrink the out-and-back traversal symmetrically toward the basepoint; at each stage traverse only an initial segment of $\alpha$ and immediately return along it. □

**Problem 9.45** (Associativity)**.** Explain why $(\alpha * \beta) * \gamma$ and $\alpha * (\beta * \gamma)$ represent the same fundamental-group element.

*Solution.* They traverse the same three paths in the same order with different time allocations. A continuous reparameterization homotopy rel endpoints changes one allocation into the other. □

**Problem 9.46** (Circle generator)**.** Show that $\omega_1(t) = e^{2\pi i t}$ generates $\pi_1(S^1) \cong \mathbb{Z}$.

*Solution.* Every loop class has an integer winding number $n$, and $\omega_n$ is homotopic to the $n$-fold concatenation of $\omega_1$ for $n > 0$, with reversed copies for $n < 0$. □

**Problem 9.47** (Degree and winding)**.** For $f(z) = z^m$, compute the induced map $f_* : \pi_1(S^1) \to \pi_1(S^1)$.

*Solution.* The generator loop $\omega_1$ is sent to $t \mapsto e^{2\pi i m t} = \omega_m$. Thus $f_*$ is multiplication by $m$. □

**Problem 9.48** (Contractible target)**.** Show that a contractible path-connected space has trivial fundamental group.

*Solution.* The identity is homotopic to a constant map. Homotopy invariance makes the identity homomorphism on $\pi_1$ equal to the zero homomorphism induced by the constant map, so the group is trivial. □

**Problem 9.49** (Product loop)**.** Describe a loop in $X \times Y$ in terms of loops in the factors.

*Solution.* A loop $\alpha(t) = (\alpha_X(t), \alpha_Y(t))$ is exactly a pair of loops. Homotopies and concatenation occur componentwise, yielding the direct-product formula. □

**Problem 9.50** (Torus group)**.** Compute $\pi_1(T^2)$ from the product formula.

*Solution.* $\pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$. □

**Problem 9.51** (Basepoint path)**.** Explain why path-connected spaces have isomorphic fundamental groups at different basepoints.

*Solution.* Choose a path joining the basepoints and conjugate loops by that path and its reverse. This gives an isomorphism, with different path choices differing by inner automorphisms. □

**Problem 9.52** (Homotopy-equivalence obstruction)**.** Why can $S^1$ not be homotopy equivalent to a point?

*Solution.* Homotopy equivalence would induce an isomorphism of fundamental groups, but $\pi_1(S^1) \cong \mathbb{Z}$ and the point has trivial fundamental group. □

**Problem 9.53** (Higher sphere comparison)**.** Use $\pi_1$ to distinguish $S^1$ from $S^2$.

*Solution.* $\pi_1(S^1) \cong \mathbb{Z}$ while $\pi_1(S^2) = 0$, so the spaces are not homotopy equivalent. □

**Problem 9.54** (Wedge preview)**.** What group should be expected for $S^1 \vee S^1$?

*Solution.* The expected group is the free group $F_2$ on the two circle loops. Van Kampen's theorem gives the formal computation. □

**Problem 9.55** (Torus attaching relation)**.** Explain geometrically why the torus two-cell kills the commutator.

*Solution.* The boundary of the two-cell is the loop $aba^{-1}b^{-1}$. Once the disk is attached, that boundary loop bounds the disk and is therefore null-homotopic, imposing $[a, b] = 1$. □

**Problem 9.56** (Null two-cell attachment)**.** If a two-cell attaches along a null-homotopic loop in $X$, what does VIII/08 predict?

*Solution.* The result is homotopy equivalent to $X \vee S^2$, because the attaching loop can be replaced by the constant loop. □

**Problem 9.57** (Functoriality check)**.** Prove $(g \circ f)_* = g_* \circ f_*$ on a loop class.

*Solution.* For $[\alpha]$, $(g \circ f)_*[\alpha] = [g \circ f \circ \alpha] = g_*([f \circ \alpha]) = (g_* \circ f_*)[\alpha]$. □

**Problem 9.58** (Constant map)**.** What homomorphism on fundamental groups is induced by a constant based map?

*Solution.* Every loop is sent to the constant loop, so the induced homomorphism is trivial. □

**Problem 9.59** (Circle covering preview)**.** Lift $\omega_n(t) = e^{2\pi int}$ to a path in $\mathbb{R}$ starting at 0.

*Solution.* The lift is $\widetilde{\omega}_n(t) = nt$, because $e^{2\pi i(nt)} = \omega_n(t)$. Its endpoint is the integer $n$. □

**Problem 9.60** (Why coverings next?)**.** Explain how the circle cover suggests a general method for studying $\pi_1$.

*Solution.* Loop classes can be detected by how their lifts move between points in a fiber. VIII/10 generalizes path lifting and endpoint behavior to arbitrary covering spaces. □

**Problem 9.61** (Loop reparameterization)**.** Show that changing the speed of a loop by an endpoint-preserving map $\rho : I \to I$ does not change its fundamental-group class.

*Solution.* The homotopy $H(s, t) = \alpha((1 - t)s + t\rho(s))$ works when the interpolation remains endpoint preserving; more generally VIII/01's reparameterization principle provides the endpoint-fixing homotopy. □

**Problem 9.62** (Path-connected basepoint independence)**.** Why is the statement '$\pi_1(X)$' usually acceptable without naming a basepoint for a path-connected space?

*Solution.* All basepoint fundamental groups are isomorphic. The isomorphism is not completely canonical in nonabelian groups, but the group isomorphism type is independent of basepoint. ☐

**Problem 9.63** (Two-cell relation principle)**.** State the conceptual relation between VIII/08 and VIII/09 for two-dimensional CW complexes.

*Solution.* VIII/08 says only the homotopy class of the attaching loop matters. VIII/09 packages based loop homotopy classes into $\pi_1$, so attaching data become group elements and relations. ☐

**Problem 9.64** (Bridge dossier)**.** Describe the role of $p : \mathbb{R} \to S^1$, $p(t) = e^{2\pi i t}$, in the next chapter.

*Solution.* It is the model covering map. Paths and loops lift uniquely after choosing a starting point, and a loop's winding number is recovered from the endpoint of its lift. ☐

## 9.21 Supplementary worked examples

**Example 9.65** (Fundamental group of the circle)**.** Loops $\gamma_k(t) = e^{2\pi ikt}$ represent all classes in $\pi_1(S^1, 1)$, and concatenation adds winding numbers. Thus $\pi_1(S^1) \cong \mathbb{Z}$.

**Example 9.66** (Product loops)**.** A loop in $X \times Y$ is a pair of loops, and endpoint-fixing homotopies act coordinatewise. Consequently $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.

**Example 9.67** (Simply connected disk)**.** Every loop in $D^n$ contracts by the straight-line contraction of the disk. Hence $\pi_1(D^n) = 0$, illustrating how contractibility forces trivial fundamental group.

## 9.22 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 9.68** (Loop endpoints)**.** What endpoint conditions define a loop based at $x_0$?

*Hint.* Both ends of $I$ map to the base point. □

*Solution.* $\gamma(0) = \gamma(1) = x_0$. □

**Exercise 9.69** (Constant loop)**.** Write the identity element of $\pi_1(X, x_0)$.

*Hint.* Use the constant path. □

*Solution.* It is the class of $c_{x_0}(t) = x_0$. □

**Exercise 9.70** (Inverse loop)**.** Given $\gamma$, write a representative of its inverse class.

*Hint.* Reverse time. □

*Solution.* $\overline{\gamma}(t) = \gamma(1 - t)$ represents $[\gamma]^{-1}$. □

**Exercise 9.71** (Circle product)**.** In $\pi_1(S^1) \cong \mathbb{Z}$, what class is $[\gamma_2] * [\gamma_{-5}]$?

*Hint.* Concatenation adds winding numbers. □

*Solution.* It corresponds to $2 + (-5) = -3$. □

**Exercise 9.72** (Torus group)**.** Compute $\pi_1(S^1 \times S^1)$.

*Hint.* Use the product theorem. □

*Solution.* It is $\mathbb{Z} \times \mathbb{Z}$. □

## Proofs

**Exercise 9.73** (Well-defined product)**.** Prove loop concatenation descends to endpoint-homotopy classes.

*Hint.* Concatenate homotopies in the path variable. ☐

*Solution.* If $\alpha \simeq \alpha'$ and $\beta \simeq \beta'$ rel endpoints, piece the homotopies on the two halves of the interval; this gives $\alpha * \beta \simeq \alpha' * \beta'$ rel endpoints. ☐

**Exercise 9.74** (Group inverse)**.** Prove $[\gamma][\overline{\gamma}] = 1$ in $\pi_1$.

*Hint.* Shrink the out-and-back path to the base point. ☐

*Solution.* There is an endpoint-fixed homotopy that progressively shortens the traversal of $\gamma$ followed by its reverse, yielding the constant loop. ☐

**Exercise 9.75** (Functoriality)**.** Prove a based map $f : (X, x_0) \to (Y, y_0)$ induces a homomorphism $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$.

*Hint.* Postcompose loops and homotopies with $f$. ☐

*Solution.* Define $f_*[\gamma] = [f \circ \gamma]$. It is well defined by homotopy compatibility and respects concatenation because $f \circ (\alpha * \beta) = (f \circ \alpha) * (f \circ \beta)$ up to the same parameterization. ☐

**Exercise 9.76** (Contractible group)**.** Prove a contractible path-connected space has trivial fundamental group.

*Hint.* Use the contraction to homotope every loop to the constant loop. ☐

*Solution.* Composing a loop with the contraction gives a free contraction; after arranging the based condition (or using a contraction fixing the chosen point when available and change-of-basepoint), every class is trivial. ☐

## Counterexamples and hypothesis tests

**Exercise 9.77** (Fundamental group always abelian)**.** Test: $\pi_1(X)$ is always abelian.

*Hint.* Wedges of circles give free groups. ☐

*Solution.* False. Fundamental groups can be highly nonabelian. ☐

**Exercise 9.78** (Free homotopy defines group)**.** Test: the fundamental group uses free homotopy classes of loops.

*Hint.* The base point must remain fixed. ☐

*Solution.* False. It uses homotopies relative to the endpoints/base point. ☐

**Exercise 9.79** (Homeomorphism required)**.** Test: homotopy equivalent path-connected spaces can have different fundamental groups.

*Hint.* $\pi_1$ is a homotopy invariant up to basepoint transport. ☐

*Solution.* False in the based/isomorphism sense: a homotopy equivalence induces an isomorphism of fundamental groups. ☐

## Applications and investigations

**Exercise 9.80** (Obstacle detection)**.** Why does $\pi_1$ detect holes relevant to path planning?

*Hint.* Nonhomotopic loops cannot be continuously deformed through admissible paths. □

*Solution.* Distinct loop classes encode qualitatively different ways of winding around obstacles, so the group organizes path-homotopy constraints ignored by metric distance alone. □

**Exercise 9.81** (Presentation from CW structure)**.** How does a 2-dimensional CW complex lead toward a presentation of $\pi_1$?

*Hint.* 1-cells generate loops and 2-cell boundaries impose relations. □

*Solution.* After choosing a spanning tree/basepoint, remaining 1-cells supply generators; each 2-cell attaching loop becomes a relation via van Kampen. □

## Challenge problems

**Exercise 9.82** (Figure-eight preview)**.** Predict the fundamental group of a wedge of two circles.

*Hint.* No 2-cell imposes a relation between the two basic loops. □

*Solution.* It is the free group $F_2$ on the two circle loops. □

**Exercise 9.83** (Idempotent endomorphism)**.** If a based self-map $f : S^1 \to S^1$ satisfies $f \circ f \simeq f$ based, what winding numbers are possible?

*Hint.* The induced map on $\pi_1(S^1) = \mathbb{Z}$ is multiplication by an integer $d$. □

*Solution.* The relation gives $d^2 = d$, hence $d = 0$ or $1$. □

# Chapter 10

# Covering Spaces

Covering spaces turn global topology into locally trivial sheets. Around every point of the base, a covering map looks like a disjoint union of copies of one open neighborhood. This simple local model gives strong global information: fibers are discrete, connected covers have a constant number of sheets, the circle admits the exponential cover by the real line, and coverings interact directly with fundamental groups.

## 10.1 Covering maps

**Definition 10.1** (Covering map)**.** A continuous surjection

$$p : \widetilde{X} \to X$$

is a covering map if every $x \in X$ has an open neighborhood $U$ such that

$$p^{-1}(U) = \coprod_{\alpha \in A} V_\alpha$$

with each $V_\alpha$ open in $\widetilde{X}$ and

$$p|_{V_\alpha} : V_\alpha \to U$$

a homeomorphism.

Such a neighborhood $U$ is called evenly covered.

## 10.2 Sheets and fibers

For $x \in X$, the fiber is

$$p^{-1}(x).$$

**Proposition 10.2** (Fibers are discrete)**.** *Every fiber of a covering map is a discrete subspace of $\widetilde{X}$.*

If $U$ is evenly covered, each sheet above $U$ contains exactly one point of the fiber over each $x \in U$.

## 10.3 Number of sheets

**Proposition 10.3** (Constant sheet number on connected bases)**.** *If $X$ is connected, then the cardinality*

$$|p^{-1}(x)|$$

*is independent of $x \in X$.*

A cover with $n$ points in each fiber is called an $n$-sheeted cover.

## 10.4 The universal example

**Example 10.4** (The exponential cover)**.** The map

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t},$$

is a covering map. Every sufficiently short open arc in $S^1$ is evenly covered by countably many translated intervals.

The fiber over $1 \in S^1$ is

$$p^{-1}(1) = \mathbb{Z}.$$

## 10.5 Finite circle covers

For $n \geq 1$,

$$p_n : S^1 \to S^1, \qquad p_n(z) = z^n$$

is an $n$-sheeted covering map.

**Example 10.5.** The map $z \mapsto z^2$ is a two-sheeted covering of the circle by itself.

## 10.6 Trivial covers

**Definition 10.6** (Trivial covering)**.** For a discrete set $F$, the projection

$$X \times F \to X$$

is a trivial covering.

Every covering is locally of this form over an evenly covered neighborhood, but need not be globally a product.

## 10.7 Local homeomorphisms

**Proposition 10.7** (Coverings are local homeomorphisms)**.** *Every covering map is a local homeomorphism.*

*Warning* 10.8 (The converse fails)*.* A surjective local homeomorphism need not be a covering map without additional global control on the sheets.

## 10.8 Restriction of a covering

**Proposition 10.9** (Restriction)**.** *If $p : \widetilde{X} \to X$ is a covering and $A \subset X$, then the restricted map*

$$p^{-1}(A) \to A$$

*is a covering with the subspace topologies.*

## 10.9 Pullback coverings

Given $p : \widetilde{X} \to X$ and a map $f : Y \to X$, define

$$f^*\widetilde{X} = \{(y, \tilde{x}) \in Y \times \widetilde{X} : f(y) = p(\tilde{x})\}.$$

**Theorem 10.10** (Pullback covering)**.** *The projection*

$$f^*\widetilde{X} \to Y$$

*is a covering map.*

This gives a systematic way to transport coverings along maps.

## 10.10 Connected coverings

A covering space $\widetilde{X}$ may be disconnected even when $X$ is connected. Its connected components themselves cover the base when the base is connected and locally path connected.

**Proposition 10.11** (Components of a covering)**.** *For a connected locally path-connected base, every connected component of a covering space maps onto the base and is itself a covering space.*

## 10.11 Universal covers

**Definition 10.12** (Universal covering space)**.** A covering

$$p : \widetilde{X} \to X$$

is universal if $\widetilde{X}$ is simply connected.

The real line is the universal cover of $S^1$.

**Example 10.13.** For $n \geq 2$,

$$S^n \to \mathbb{R}P^n$$

is a two-sheeted cover, and $S^n$ is simply connected. Thus it is the universal cover of $\mathbb{R}P^n$.

## 10.12 Evenly covered neighborhoods

If $U \subset X$ is evenly covered, choosing a point

$$\tilde{x} \in p^{-1}(x), \qquad x \in U,$$

selects a unique sheet $V$ above $U$. The inverse

$$(p|_V)^{-1} : U \to V$$

is a local section.

## 10.13 Sections

**Definition 10.14** (Section)**.** A section of $p : \widetilde{X} \to X$ is a map

$$s : X \to \widetilde{X}$$

such that

$$p \circ s = \mathrm{id}_X .$$

Every covering has local sections over evenly covered neighborhoods. A global section is much stronger and can force a connected cover to be trivial in many situations.

## 10.14 Covering isomorphisms

**Definition 10.15** (Isomorphism of coverings)**.** Two coverings $p_i : \widetilde{X}_i \to X$ are isomorphic if there is a homeomorphism

$$h : \widetilde{X}_1 \to \widetilde{X}_2$$

such that

$$p_2 \circ h = p_1.$$

The base is held fixed.

## 10.15 Coverings and fundamental groups

A connected covering determines a subgroup of the base fundamental group. If

$$p : (\widetilde{X}, \tilde{x}_0) \to (X, x_0),$$

then

$$p_* : \pi_1(\widetilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$$

is injective.

**Theorem 10.16** (Fundamental-group injection)**.** *For a covering map, the induced homomorphism on fundamental groups is injective.*

The proof uses lifting uniqueness, developed systematically in VIII/11.

## 10.16 Index and number of sheets

Under the standard hypotheses used in covering-space classification, a connected covering corresponds to a subgroup

$$H \leq \pi_1(X, x_0),$$

and the number of sheets is the index

$$[\pi_1(X, x_0) : H].$$

## 10.17 Universal cover and the trivial subgroup

For a simply connected universal cover,

$$\pi_1(\widetilde{X}) = 0,$$

so the corresponding subgroup of $\pi_1(X)$ is trivial.

## 10.18 Classification preview

For path-connected, locally path-connected, semilocally simply connected spaces, connected coverings are classified by conjugacy classes of subgroups of $\pi_1(X)$. Pointed connected coverings correspond to actual subgroups.

This theorem relies on the lifting machinery of VIII/11 and the deck-group structure of VIII/12.

## 10.19 Why coverings matter

Covering spaces convert loops into sheet changes. A loop in the base lifts after a starting point is chosen; whether its lift closes or ends at another point of the fiber detects its fundamental-group behavior. The next chapter develops this lifting principle precisely.

## 10.20 Exercises

**Exercise 10.17.** Define an evenly covered neighborhood.

*Hint.* Write the preimage as disjoint sheets.

Require the sheets to be open and pairwise disjoint, and check that each restriction maps onto the same entire neighborhood $U$. ☐

*Solution.* An open $U \subset X$ is evenly covered if $p^{-1}(U) = \coprod V_\alpha$ and every $p|_{V_\alpha} : V_\alpha \to U$ is a homeomorphism. ☐

**Exercise 10.18.** Define a covering map.

*Hint.* Every base point needs an evenly covered neighborhood.

A local inverse near one point upstairs is not enough: one neighborhood downstairs must simultaneously account for its whole preimage. ☐

*Solution.* A continuous surjection locally decomposing into disjoint homeomorphic sheets. ☐

**Exercise 10.19.** What is a fiber?

*Hint.* Take the inverse image of one point.

Distinguish the fiber $p^{-1}(x)$ from the preimage of a neighborhood, which is a union of open sheets. ☐

*Solution.* $p^{-1}(x)$. ☐

**Exercise 10.20.** Why are covering fibers discrete?

*Hint.* Use one evenly covered neighborhood.

Intersect one open sheet with $p^{-1}(x)$; the homeomorphism onto an evenly covered neighborhood makes this intersection a singleton. ☐

*Solution.* Each sheet contains one fiber point, isolated inside that sheet. ☐

**Exercise 10.21.** What is an $n$-sheeted cover?

*Hint.* Count fiber points.

Count points over a fixed basepoint, not connected components of the total space; a connected cover can have several sheets. ☐

*Solution.* A covering whose fibers have exactly $n$ points. ☐

**Exercise 10.22.** When is sheet number constant?

*Hint.* Use connectedness of the base.

An evenly covered neighborhood identifies its fibers with the same sheet-index set. Thus fiber cardinality is locally constant before connectedness is used. ☐

*Solution.* For a connected base, fiber cardinality is constant. ☐

**Exercise 10.23.** Give the universal cover of $S^1$.

*Hint.* Use the exponential map.

Choose an arc shorter than a full turn and lift its angle to disjoint intervals differing by integers; also check that the total space is simply connected. $\square$

*Solution.* $p : \mathbb{R} \to S^1,\ p(t) = e^{2\pi i t}$. $\square$

**Exercise 10.24.** What is the fiber over 1 in the exponential cover?

*Hint.* Solve $e^{2\pi i t} = 1$.

Divide the angle by $2\pi$ and use the periodicity of the complex exponential to identify every solution, including negative ones. $\square$

*Solution.* $\mathbb{Z}$. $\square$

**Exercise 10.25.** Give an $n$-sheeted self-cover of $S^1$.

*Hint.* Use the power map.

For $n \geq 1$, solve $z^n = w$ on a small arc and separate its $n$ continuous root branches. $\square$

*Solution.* $z \mapsto z^n$. $\square$

**Exercise 10.26.** Define a trivial covering.

*Hint.* Use a discrete fiber.

The discreteness of $F$ makes every $X \times \{a\}$ an open sheet mapping homeomorphically onto $X$. $\square$

*Solution.* The projection $X \times F \to X$ for discrete $F$. $\square$

**Exercise 10.27.** What local property does every covering have?

*Hint.* Restrict to one sheet.

Restrict $p$ to the unique sheet containing a chosen point upstairs; that restriction supplies the required local homeomorphism. $\square$

*Solution.* It is a local homeomorphism. $\square$

**Exercise 10.28.** Does every surjective local homeomorphism automatically give a covering?

*Hint.* Recall the warning.

Test the projection $\mathbb{R} \sqcup (0, \infty) \to \mathbb{R}$ given by inclusion on each summand: compare fiber cardinalities near zero. $\square$

*Solution.* Not in complete generality; global sheet control can fail. $\square$

**Exercise 10.29.** What happens when a covering is restricted to a subspace $A \subset X$?

*Hint.* Restrict domain to the inverse image.

Intersect an evenly covered neighborhood with $A$ and its sheets with $p^{-1}(A)$; use the subspace topologies on both sides. $\square$

*Solution.* $p^{-1}(A) \to A$ is again a covering. $\square$

**Exercise 10.30.** Define the pullback covering set.

*Hint.* Impose $f(y) = p(\tilde{x})$.

Form a subspace of $Y \times \widetilde{X}$, not an unrestricted product: both coordinates must represent the same point of $X$. ☐

*Solution.* $f^*\widetilde{X} = \{(y, \tilde{x}) : f(y) = p(\tilde{x})\}$. ☐

**Exercise 10.31.** What map is the pullback covering projection?

*Hint.* Project to $Y$.

Over $f^{-1}(U)$, each sheet of $p$ supplies a graph $y \mapsto (y, s(f(y)))$; use these graphs to verify the local covering model. ☐

*Solution.* $(y, \tilde{x}) \mapsto y$. ☐

**Exercise 10.32.** Define a universal cover.

*Hint.* Require the total space to be simply connected.

Include path connectedness in simple connectivity and keep the covering condition separate from the condition on its total space. ☐

*Solution.* A covering $p : \widetilde{X} \to X$ with simply connected $\widetilde{X}$. ☐

**Exercise 10.33.** What universal cover does $\mathbb{R}P^n$ have for $n \geq 2$?

*Hint.* Use the antipodal quotient.

Check both ingredients: the antipodal quotient is a double cover, and $S^n$ is simply connected in the stated range $n \geq 2$. ☐

*Solution.* $S^n \to \mathbb{R}P^n$. ☐

**Exercise 10.34.** Define a section of a covering.

*Hint.* It is a right inverse.

Type the maps carefully: $s : X \to \widetilde{X}$, so the right-inverse equation is an equality of maps from $X$ to itself. ☐

*Solution.* A map $s : X \to \widetilde{X}$ with $p \circ s = \mathrm{id}_X$. ☐

**Exercise 10.35.** Do coverings always have local sections?

*Hint.* Use an evenly covered neighborhood.

Choose a sheet $V$ above $U$; the inverse of $p|_V$ is continuous and its composite with $p$ is $\mathrm{id}_U$. ☐

*Solution.* Yes. ☐

**Exercise 10.36.** Define an isomorphism of coverings over $X$.

*Hint.* Use a homeomorphism commuting with projections.

The commuting equation holds over the fixed base, and $h$ must be a homeomorphism, not merely a bijection between fibers. ☐

*Solution.* A homeomorphism $h : \widetilde{X}_1 \to \widetilde{X}_2$ with $p_2 h = p_1$. ☐

**Exercise 10.37.** What does a covering induce on $\pi_1$?

*Hint.* Use VIII/11 lifting uniqueness.

Start with an upstairs loop whose projection is null-homotopic; lift the based null-homotopy to show the original upstairs class vanishes. ☐

*Solution.* An injective homomorphism $p_* : \pi_1(\widetilde{X}) \hookrightarrow \pi_1(X)$. ☐

**Exercise 10.38.** What subgroup corresponds to a universal cover?

*Hint.* The total fundamental group is trivial.

Compute the image of the upstairs fundamental group under $p_*$; simple connectivity makes the source group trivial. ☐

*Solution.* The trivial subgroup. ☐

**Exercise 10.39.** How is sheet number related to subgroup index in the standard classification setting?

*Hint.* Count cosets.

Identify the fiber with cosets of the covering subgroup after choosing a basepoint and a left/right action convention; count those cosets. ☐

*Solution.* It equals $[\pi_1(X) : H]$. ☐

**Exercise 10.40.** What is the bridge to VIII/11?

*Hint.* Loops and paths must be lifted.

For a path downstairs, specify one starting point upstairs, then ask separately whether a lift exists, is unique, and returns to its starting fiber point. ☐

*Solution.* VIII/11 proves path lifting, homotopy lifting, uniqueness, and the general lifting criterion. ☐

## 10.21 Solved problem dossiers

**Problem 10.41** (Exponential evenly covered arc)**.** Show a sufficiently short open arc $U \subset S^1$ is evenly covered by $p(t) = e^{2\pi i t}$.

*Solution.* Choose $U$ avoiding one point so that it admits a continuous argument interval $(a, b)$ of length less than 1. Then $p^{-1}(U) = \coprod_{k\in\mathbb{Z}}(a + k, b + k)$, and $p$ is a homeomorphism from each interval to $U$. ☐

**Problem 10.42** (Two-sheeted circle cover)**.** Show $p(z) = z^2$ is a two-sheeted cover of $S^1$.

*Solution.* Every point $w \in S^1$ has two square roots. On a sufficiently short arc choose a continuous square-root branch; the two branches give disjoint sheets each mapped homeomorphically onto the arc. ☐

**Problem 10.43** (Degree-$n$ circle cover)**.** Show $z \mapsto z^n$ is $n$-sheeted.

*Solution.* Each $w = e^{i\theta}$ has the $n$ roots $e^{i(\theta+2\pi k)/n}$, and short arcs admit $n$ disjoint continuous root branches. ☐

**Problem 10.44** (Trivial finite cover)**.** Describe the $n$-sheeted trivial covering of $X$.

*Solution.* Take $X \times \{1, \ldots, n\} \to X$. Its $n$ components are sheets globally homeomorphic to $X$. ☐

**Problem 10.45** (Discrete fiber proof)**.** Prove directly that $p^{-1}(x)$ is discrete.

*Solution.* Choose evenly covered $U \ni x$. Each fiber point lies in a different sheet $V_\alpha$, and $V_\alpha \cap p^{-1}(x)$ is that single point. ☐

**Problem 10.46** (Sheet-number constancy)**.** Why is fiber cardinality locally constant on the base?

*Solution.* Over an evenly covered neighborhood, every fiber has one point in each sheet. Hence all fibers over that neighborhood have the same cardinality. On a connected base a locally constant cardinal-valued function is constant. ☐

**Problem 10.47** (Restriction cover)**.** Prove $p^{-1}(A) \to A$ is a covering for arbitrary $A \subset X$.

*Solution.* If $U$ evenly covers $x \in A$, then $U \cap A$ is open in $A$, and the sheets $V_\alpha \cap p^{-1}(A)$ map homeomorphically onto $U \cap A$. ☐

**Problem 10.48** (Pullback cover)**.** Prove the pullback of a covering is a covering.

*Solution.* Choose an evenly covered $U$ around $f(y)$. Over $f^{-1}(U)$, the pullback splits into copies indexed by the sheets above $U$, with each copy projected homeomorphically to $f^{-1}(U)$. ☐

**Problem 10.49** (Projective cover)**.** Why is $S^n \to \mathbb{R}P^n$ two-sheeted?

*Solution.* Each projective point is a line with exactly two unit representatives $x, -x$. Small neighborhoods of a line lift to disjoint neighborhoods around these two representatives. ☐

**Problem 10.50** (Universal projective cover)**.** For $n \geq 2$, why is $S^n \to \mathbb{R}P^n$ universal?

*Solution.* The antipodal quotient is a covering, and $S^n$ is simply connected for $n \geq 2$. ☐

**Problem 10.51** (Local sections)**.** Construct a local section over an evenly covered neighborhood.

*Solution.* Choose one sheet $V$ over $U$ and set $s = (p|_V)^{-1} : U \to V \subset \widetilde{X}$. ☐

**Problem 10.52** (Global section warning)**.** Why can the exponential cover $\mathbb{R} \to S^1$ have no global section?

*Solution.* A global section would induce a homomorphism splitting $p_* : 0 = \pi_1(\mathbb{R}) \to \pi_1(S^1) \cong \mathbb{Z}$, impossible because $p \circ s = \mathrm{id}$ would imply $p_* s_* = \mathrm{id}_{\mathbb{Z}}$. ☐

**Problem 10.53** (Covering isomorphism)**.** Show an isomorphism of coverings maps each fiber bijectively to the corresponding fiber.

*Solution.* If $p_2 h = p_1$, then $p_1(\tilde{x}) = x$ implies $p_2(h(\tilde{x})) = x$. Since $h$ is a homeomorphism, the induced fiber map is bijective. ☐

**Problem 10.54** (Connected component cover)**.** Why does a component of a cover over a connected locally path-connected base map onto the base?

*Solution.* The image of a component is open by local sheets and closed under local continuation; equivalently path lifting from one image point reaches points along base paths. Connectedness forces the image to be all of the base. ☐

**Problem 10.55** (Universal cover subgroup)**.** What subgroup of $\pi_1(S^1)$ corresponds to $\mathbb{R} \to S^1$?

*Solution.* The total space is simply connected, so the subgroup $p_*\pi_1(\mathbb{R})$ is $0 \leq \mathbb{Z}$. □

**Problem 10.56** (Finite circle subgroup)**.** What subgroup of $\pi_1(S^1) \cong \mathbb{Z}$ corresponds to $z \mapsto z^n$?

*Solution.* The induced map on $\pi_1$ is multiplication by $n$, so the subgroup is $n\mathbb{Z}$, of index $n$. □

**Problem 10.57** (Product pullback)**.** Pull back the trivial cover $X \times F \to X$ along $f : Y \to X$. Identify the result.

*Solution.* The pullback consists of triples effectively determined by $(y, a) \in Y \times F$, so it is naturally the trivial cover $Y \times F \to Y$. □

**Problem 10.58** (Cover of a simply connected base)**.** What can be said about a connected covering of a simply connected, suitably nice base?

*Solution.* The corresponding subgroup of the trivial fundamental group is trivial, so the connected covering is isomorphic to the identity cover under the standard classification hypotheses. □

**Problem 10.59** (Cover and local topology)**.** Why do base and total space have the same local topological type under a covering?

*Solution.* Near every total point, choose the sheet over an evenly covered neighborhood; the covering restriction is a homeomorphism onto the base neighborhood. □

**Problem 10.60** (Fiber cardinality of disconnected cover)**.** Can a covering over a connected base be disconnected while having constant fiber size?

*Solution.* Yes. A trivial $n$-sheeted covering $X \times \{1, \dots, n\} \to X$ has $n$ disconnected components when $X$ is connected. □

**Problem 10.61** (Identity cover)**.** Show $\mathrm{id}_X : X \to X$ is a one-sheeted covering.

*Solution.* Every open neighborhood is evenly covered by itself, with one sheet. □

**Problem 10.62** (Composition preview)**.** Under suitable hypotheses, why is a composition of covering maps again locally a covering?

*Solution.* Choose evenly covered neighborhoods successively so the upper sheets map homeomorphically through the intermediate cover to the base neighborhood. The local sheet decomposition composes. □

**Problem 10.63** (Classification preview)**.** State the subgroup object associated with a pointed connected cover.

*Solution.* Choose $\tilde{x}_0$ above $x_0$; the associated subgroup is $p_*\pi_1(\widetilde{X}, \tilde{x}_0) \leq \pi_1(X, x_0)$. □

**Problem 10.64** (Bridge to lifting)**.** Why is lifting the mechanism behind covering classification?

*Solution.* Loops lift uniquely from a chosen sheet. Whether a lifted loop closes detects membership in the associated subgroup; maps lift precisely when their fundamental-group images land in that subgroup. □

## 10.22 Supplementary worked examples

**Example 10.65** (Exponential covering of the circle)**.** The map $p : \mathbb{R} \to S^1$, $p(t) = e^{2\pi it}$, evenly covers every sufficiently short arc. Each fiber is $t + \mathbb{Z}$, making $\mathbb{R}$ the universal cover of $S^1$.

**Example 10.66** (Power covering)**.** The map $p_n : S^1 \to S^1$, $p_n(z) = z^n$, is an $n$-sheeted covering. A small arc avoiding wraparound has $n$ disjoint inverse arcs, each mapped homeomorphically onto it.

**Example 10.67** (Product covering)**.** If $p : E \to B$ is a covering and $X$ is any space, then $p \times \mathrm{id}_X : E \times X \to B \times X$ is a covering; evenly covered neighborhoods are products of evenly covered opens with opens in $X$.

## 10.23 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 10.68** (Evenly covered)**.** State what it means for an open set $U \subset B$ to be evenly covered by $p : E \to B$.

*Hint.* Describe $p^{-1}(U)$ as disjoint sheets. ☐

*Solution.* $p^{-1}(U)$ is a disjoint union of open sets $V_\alpha$, each mapped homeomorphically onto $U$ by $p$. ☐

**Exercise 10.69** (Fiber of exponential cover)**.** Compute $p^{-1}(1)$ for $p(t) = e^{2\pi it}$.

*Hint.* Solve $e^{2\pi it} = 1$. ☐

*Solution.* The fiber is $\mathbb{Z}$. ☐

**Exercise 10.70** (Fiber of power map)**.** How many points lie over 1 under $z \mapsto z^n$?

*Hint.* Solve $z^n = 1$. ☐

*Solution.* Exactly $n$, the $n$th roots of unity. ☐

**Exercise 10.71** (Local homeomorphism)**.** Why is every covering map a local homeomorphism?

*Hint.* Use one sheet of an evenly covered neighborhood. ☐

*Solution.* For $e \in E$, choose an evenly covered $U$ around $p(e)$; the sheet containing $e$ maps homeomorphically to $U$. ☐

**Exercise 10.72** (Product sheets)**.** If $U$ is evenly covered by $p$ and $W \subset X$ is open, describe the sheets over $U \times W$ for $p \times \mathrm{id}_X$.

*Hint.* Take products with each sheet. ☐

*Solution.* They are $V_\alpha \times W$, each homeomorphic to $U \times W$. ☐

### Proofs

**Exercise 10.73** (Discrete fibers)**.** Prove fibers of a covering map are discrete subspaces of $E$.

*Hint.* Use an evenly covered neighborhood around the image point. □

*Solution.* Each sheet contains exactly one point of a given fiber, and the sheet is an open neighborhood separating that point from the other fiber points. □

**Exercise 10.74** (Constant sheet number)**.** Prove a covering over a connected base has constant fiber cardinality.

*Hint.* The number of sheets is locally constant. □

*Solution.* An evenly covered neighborhood identifies fibers over all its points with the same set of sheets; hence the cardinality is locally constant and therefore constant on a connected base. □

**Exercise 10.75** (Composition)**.** Under standard hypotheses, prove a composition of covering maps is a covering.

*Hint.* Refine evenly covered neighborhoods successively. □

*Solution.* Choose an evenly covered neighborhood for the lower cover and then neighborhoods whose sheets are evenly covered by the upper cover; a common refinement produces disjoint composite sheets. □

**Exercise 10.76** (Pullback preview)**.** Explain why pulling back a covering along a map again yields a covering.

*Hint.* Pull back each local trivialization. □

*Solution.* Over an evenly covered $U$, the cover is a disjoint union of copies of $U$; its fiber product with $X \to B$ is a disjoint union of copies of the preimage of $U$, giving local covering charts. □

### Counterexamples and hypothesis tests

**Exercise 10.77** (Local homeomorphism is covering)**.** Test: every local homeomorphism is a covering map.

*Hint.* Covering requires simultaneous evenly covered neighborhoods for all points of a fiber. □

*Solution.* False without additional hypotheses; local homeomorphism is a pointwise condition. □

**Exercise 10.78** (Fibers connected)**.** Test: fibers of a nontrivial covering are connected.

*Hint.* Fibers are discrete. □

*Solution.* False; a multi-sheeted fiber is a discrete set with several points. □

**Exercise 10.79** (Sheet count can jump)**.** Test: an $n$-sheeted covering over a connected base may have different fiber sizes at different points.

*Hint.* Use local constancy. □

*Solution.* False. The sheet number is constant on each connected component of the base. □

## Applications and investigations

**Exercise 10.80** (Unwrapping periodicity)**.** Explain how the universal cover $\mathbb{R} \to S^1$ converts angular periodicity into translation symmetry.

*Hint.* Angles differing by integers project to the same point. □

*Solution.* Upstairs, the circle coordinate becomes a real coordinate; the lost periodicity reappears as the deck translations $t \mapsto t + k$. □

**Exercise 10.81** (Covering as local copies)**.** Why are covering spaces useful for transferring local calculations?

*Hint.* Each sheet is locally identical to the base. □

*Solution.* Any local construction on an evenly covered neighborhood can be lifted independently to each sheet, while global topology is encoded by how the sheets connect and permute. □

## Challenge problems

**Exercise 10.82** (Universal-cover criterion preview)**.** What extra global property distinguishes a universal cover from an arbitrary connected cover?

*Hint.* Its total space has trivial fundamental group. □

*Solution.* A universal cover is connected and simply connected, so every other connected cover is obtained as a quotient under the standard classification hypotheses. □

**Exercise 10.83** (Circle-cover synthesis)**.** Relate the $n$-fold cover $S^1 \to S^1$ to the universal cover $\mathbb{R} \to S^1$.

*Hint.* Quotient the universal cover by a subgroup of integer translations. □

*Solution.* The subgroup $n\mathbb{Z}$ acts on $\mathbb{R}$ by translations; $\mathbb{R}/n\mathbb{Z} \cong S^1$, and the induced map to $\mathbb{R}/\mathbb{Z} \cong S^1$ is the $n$-fold cover. □

# Chapter 11

# Lifting Properties

Covering maps are powerful because maps into the base can often be lifted to the covering space once an initial point is chosen. Paths always lift, homotopies of paths lift compatibly, and more general maps lift exactly when their fundamental-group image lands inside the subgroup represented by the cover. This chapter develops those principles carefully.

## 11.1 Lifts

**Definition 11.1** (Lift). Let

$$p : \widetilde{X} \to X$$

be a covering and $f : Y \to X$ a map. A lift of $f$ is a map

$$\widetilde{f} : Y \to \widetilde{X}$$

such that

$$p \circ \widetilde{f} = f.$$

A lift therefore makes the triangle commute.

## 11.2 Path lifting

**Theorem 11.2** (Path lifting). *Let*

$$\alpha : I \to X$$

*be a path and choose*

$$\tilde{x}_0 \in p^{-1}(\alpha(0)).$$

*Then there exists a unique path*

$$\widetilde{\alpha} : I \to \widetilde{X}$$

*such that*

$$\widetilde{\alpha}(0) = \tilde{x}_0 \qquad \textit{and} \qquad p \circ \widetilde{\alpha} = \alpha.$$

The proof subdivides $I$ into finitely many intervals whose images lie in evenly covered neighborhoods and continues sheet by sheet.

## 11.3 Uniqueness of lifts

**Theorem 11.3** (Uniqueness on connected domains)**.** *Let $Y$ be connected. If*

$$\widetilde{f}_0, \widetilde{f}_1 : Y \to \widetilde{X}$$

*are lifts of the same map $f : Y \to X$ and agree at one point of $Y$, then*

$$\widetilde{f}_0 = \widetilde{f}_1.$$

For locally path-connected applications, this is often proved by showing the agreement set is both open and closed.

## 11.4 Homotopy lifting

**Theorem 11.4** (Homotopy lifting)**.** *Let*

$$H : Y \times I \to X$$

*be a homotopy and suppose*

$$\widetilde{H}_0 : Y \to \widetilde{X}$$

*lifts the initial map $H(-, 0)$. Then there is a unique lift*

$$\widetilde{H} : Y \times I \to \widetilde{X}$$

*with*

$$\widetilde{H}(-, 0) = \widetilde{H}_0.$$

Path lifting is the case $Y = \{*\}$.

## 11.5 Endpoint behavior of lifted homotopies

**Corollary 11.5** (Homotopic paths have matching lifted endpoints)**.** *If paths $\alpha, \beta : I \to X$ are homotopic rel endpoints and their lifts begin at the same point, then their lifts end at the same point.*

This makes the endpoint of a lifted loop depend only on its fundamental-group class.

## 11.6 Loop lifting criterion

Let

$$p : (\widetilde{X}, \tilde{x}_0) \to (X, x_0)$$

be a pointed covering.

**Proposition 11.6** (Closed-lift criterion)**.** *A loop $\alpha$ at $x_0$ lifts to a loop at $\tilde{x}_0$ if and only if*

$$[\alpha] \in p_* \pi_1(\widetilde{X}, \tilde{x}_0).$$

## 11.7 Injection on fundamental groups

**Theorem 11.7** (Coverings inject on $\pi_1$)**.** *The homomorphism*

$$p_* : \pi_1(\widetilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$$

*induced by a covering map is injective.*

*Proof.* If a loop upstairs projects to a null-homotopic loop downstairs, lift the null-homotopy beginning with the original loop. The lifted final path is the constant loop at $\tilde{x}_0$, so the original loop is null-homotopic. □

## 11.8 General lifting criterion

**Theorem 11.8** (Lifting criterion)**.** *Let $Y$ be path connected and locally path connected, choose $y_0 \in Y$, and let*

$$f : (Y, y_0) \to (X, x_0)$$

*be a based map. For a covering*

$$p : (\widetilde{X}, \tilde{x}_0) \to (X, x_0),$$

*a based lift*

$$\widetilde{f} : (Y, y_0) \to (\widetilde{X}, \tilde{x}_0)$$

*exists if and only if*

$$f_*\pi_1(Y, y_0) \subset p_*\pi_1(\widetilde{X}, \tilde{x}_0).$$

This is one of the central structural results of covering-space theory.

## 11.9 Construction of the lifted map

Assume the subgroup inclusion in the lifting criterion. For each $y \in Y$, choose a path $\gamma$ from $y_0$ to $y$. Lift

$$f \circ \gamma$$

starting at $\tilde{x}_0$, and define

$$\widetilde{f}(y)$$

to be its endpoint. The subgroup condition guarantees independence of the chosen path.

## 11.10 Why path independence works

If $\gamma_0, \gamma_1$ both run from $y_0$ to $y$, then

$$\gamma_0 * \overline{\gamma_1}$$

is a loop. Its image under $f$ lies in the covering subgroup, so its lift from $\tilde{x}_0$ is closed. Therefore the lifts of $f\gamma_0$ and $f\gamma_1$ end at the same point.

## 11.11 Lifting maps to the universal cover

If $p : \widetilde{X} \to X$ is universal, then

$$p_* \pi_1(\widetilde{X}) = 0.$$

Hence a based map $f : Y \to X$ lifts to $\widetilde{X}$ exactly when

$$f_* \pi_1(Y) = 0.$$

**Corollary 11.9** (Simply connected domains lift)**.** *If $Y$ is simply connected and satisfies the standard local hypotheses, then every based map $Y \to X$ lifts to every covering of $X$ after a lift of the basepoint is chosen.*

## 11.12 The logarithm as a lift

For the exponential cover

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t},$$

a lift of a map

$$f : Y \to S^1$$

is a continuous real-valued argument function

$$\widetilde{f} : Y \to \mathbb{R}.$$

Thus the lifting criterion becomes a precise obstruction to the existence of a global continuous argument.

## 11.13 Roots as lifting problems

The map

$$p_n : S^1 \to S^1, \qquad p_n(z) = z^n,$$

is a covering. A continuous $n$th root of

$$f : Y \to S^1$$

is exactly a lift of $f$ through $p_n$.

## 11.14 Winding obstruction

For a loop

$$f : S^1 \to S^1$$

of degree $m$, a lift through

$$z \mapsto z^n$$

exists if and only if

$$m \in n\mathbb{Z}.$$

Equivalently,

$$n \mid m.$$

## 11.15 Monodromy on a fiber

Fix $x_0 \in X$. A loop $\alpha$ at $x_0$ lifts uniquely from every

$$\tilde{x} \in p^{-1}(x_0).$$

Taking the endpoint defines a permutation of the fiber. Homotopic loops rel basepoint give the same permutation.

This produces an action

$$\pi_1(X, x_0) \curvearrowright p^{-1}(x_0),$$

called monodromy.

## 11.16 Connectedness and transitivity

For a connected covering of a path-connected locally path-connected base, the monodromy action on the fiber is transitive. Any two points in the fiber can be joined upstairs by a path; its projection is a loop downstairs whose lift moves the first fiber point to the second.

## 11.17 Lifting and covering classification

The subgroup

$$H = p_* \pi_1(\widetilde{X}, \tilde{x}_0)$$

controls which loops close upstairs and which maps lift. This subgroup is the algebraic fingerprint of the pointed connected covering.

## 11.18 Bridge to deck transformations

Deck transformations are self-homeomorphisms of the covering space that lie over the identity of the base. Lifting uniqueness makes them rigid: on a connected cover, a deck transformation is completely determined by the image of one point. VIII/12 develops the resulting group action.

## 11.19 Exercises

**Exercise 11.10.** Define a lift of $f : Y \to X$.

*Hint.* Make the triangle commute.

Check the types $\widetilde{f} : Y \to \widetilde{X}$ and $p : \widetilde{X} \to X$; their composite must equal the given map on all of $Y$. □

*Solution.* A map $\widetilde{f} : Y \to \widetilde{X}$ with $p\widetilde{f} = f$. □

**Exercise 11.11.** What extra datum is needed to lift a path uniquely?

*Hint.* Choose a starting point upstairs.

The chosen point must lie over $\alpha(0)$. Different choices in that fiber generally produce different lifts of the same path. □

*Solution.* A point $\tilde{x}_0 \in p^{-1}(\alpha(0))$. □

**Exercise 11.12.** State path-lift uniqueness.

*Hint.* Same path and same initial lift point.

Inside an evenly covered neighborhood, the starting sheet forces the local inverse; propagate this agreement along the interval. □

*Solution.* There is exactly one lifted path with the chosen starting point. □

**Exercise 11.13.** How is path lifting proved locally?

*Hint.* Use evenly covered neighborhoods.

Use compactness of $I$ to choose a finite subdivision subordinate to evenly covered neighborhoods, then match adjacent lifted endpoints. □

*Solution.* Subdivide the parameter interval and continue the lift sheet by sheet. □

**Exercise 11.14.** When are two lifts of one map equal on a connected domain?

*Hint.* They need agree once.

Show that both the agreement set and its complement are open using local sheets; connectedness and one agreement point finish the argument. □

*Solution.* If they agree at one point, uniqueness implies they agree everywhere. □

**Exercise 11.15.** State homotopy lifting.

*Hint.* Lift the initial time slice first.

Prescribe a continuous lift of the entire initial map $H(-,0)$, not just one initial point when the parameter space is an arbitrary $Y$. □

*Solution.* A homotopy $H : Y \times I \to X$ has a unique lift once a lift of $H(-,0)$ is fixed. □

**Exercise 11.16.** What happens to lifted endpoints of endpoint-homotopic paths?

*Hint.* Lift the path homotopy.

The lifted endpoint varies continuously with the homotopy parameter inside a discrete fiber. A continuous path in that fiber must be constant. □

*Solution.* Starting at the same point, the lifted paths end at the same point. □

**Exercise 11.17.** When does a loop lift to a loop at $\tilde{x}_0$?

*Hint.* Use the covering subgroup.

For one direction project the closed lift; for the other lift a representative of an upstairs loop and use invariance of lifted endpoints. □

*Solution.* When its class lies in $p_*\pi_1(\widetilde{X}, \tilde{x}_0)$. □

**Exercise 11.18.** What does a covering induce on fundamental groups?

*Hint.* Use lifted null-homotopies.

Lift a based null-homotopy of the projected loop, keeping its boundary in the appropriate fiber; compare the resulting lift with the original loop. □

*Solution.* An injective homomorphism. □

**Exercise 11.19.** State the general lifting criterion.

*Hint.* Compare subgroup images.

Choose compatible basepoints and assume the usual path-connected, locally path-connected domain. Compare images inside the same based group $\pi_1(X, x_0)$. □

*Solution.* A based lift exists iff $f_*\pi_1(Y) \subset p_*\pi_1(\widetilde{X})$, under the standard path/local hypotheses. □

**Exercise 11.20.** How is a lift constructed from the criterion?

*Hint.* Choose a path from the basepoint.

For each $y$, lift $f \circ \gamma$ from the chosen point over $f(y_0)$; the next task is to prove its endpoint is independent of $\gamma$. □

*Solution.* Lift $f \circ \gamma$ from $\tilde{x}_0$ and define $\widetilde{f}(y)$ as its endpoint. □

**Exercise 11.21.** Why is the constructed value independent of path?

*Hint.* Compare two paths via a loop.

Join one candidate path to the reverse of the other. The subgroup hypothesis forces the image loop to have a closed lift. □

*Solution.* Their difference loop maps into the covering subgroup, so its lift closes. □

**Exercise 11.22.** What is the covering subgroup of a universal cover?

*Hint.* The total space is simply connected.

Apply $p_*$ to the trivial group of the simply connected total space; do not identify this subgroup with all of $\pi_1(X)$. □

*Solution.* The trivial subgroup. □

**Exercise 11.23.** When does a map lift to a universal cover?

*Hint.* Use the trivial subgroup criterion.

Substitute the trivial covering subgroup into the based lifting criterion, retaining its local hypotheses on the domain. □

*Solution.* Exactly when $f_*\pi_1(Y) = 0$. □

**Exercise 11.24.** Why does a simply connected domain lift?

*Hint.* Its $\pi_1$ is trivial.

Every homomorphism from $\pi_1(Y) = 0$ has trivial image, which lies in any covering subgroup; choose the lifted basepoint before applying the criterion. □

*Solution.* The subgroup criterion is automatically satisfied. □

**Exercise 11.25.** What is a lift to $\mathbb{R}$ of a map into $S^1$?

*Hint.* Use the exponential cover.

With $p(t) = e^{2\pi i t}$, the equation is $e^{2\pi i \widetilde{f}(y)} = f(y)$; the usual angle in radians is $2\pi\widetilde{f}$. □

*Solution.* A continuous real-valued argument. □

**Exercise 11.26.** How is a continuous $n$th root a lifting problem?

*Hint.* Use $z \mapsto z^n$.

Write the desired equation $g(y)^n = f(y)$ and view $g$ as a map to the total circle of the power covering. □

*Solution.* It is a lift through the $n$-sheeted power covering. □

**Exercise 11.27.** When does a degree-$m$ circle map have an $n$th-root lift?

*Hint.* Compare $m\mathbb{Z}$ with $n\mathbb{Z}$.

The induced image of the degree-$m$ map is $m\mathbb{Z}$; determine when this subgroup lies in the image $n\mathbb{Z}$ of the covering map. □

*Solution.* Exactly when $n \mid m$. □

**Exercise 11.28.** What is monodromy?

*Hint.* Lift loops from points of the fiber.

Lift each loop from each point of the fiber and retain its endpoint; state the concatenation convention before deciding whether this is a left or right action. □

*Solution.* The permutation action of $\pi_1(X, x_0)$ on $p^{-1}(x_0)$ by lifted endpoints. □

**Exercise 11.29.** Why does monodromy depend only on loop homotopy class?

*Hint.* Use homotopy lifting rel endpoints.

Apply homotopy lifting to an endpoint-fixed loop homotopy; the lifted boundary tracks lie in discrete fibers and cannot move. □

*Solution.* Homotopic loops have lifts with the same endpoints. □

**Exercise 11.30.** When is monodromy transitive?

*Hint.* Use connectedness upstairs.

Local path connectedness makes a connected covering space path connected. Project a path joining two points of the same fiber to obtain the required loop downstairs. □

*Solution.* For a connected cover over a path-connected locally path-connected base. □

**Exercise 11.31.** What subgroup records a pointed connected covering?

*Hint.* Image of upstairs $\pi_1$.

Keep the chosen point $\tilde{x}_0$ in the notation: changing it can change the subgroup by conjugation rather than equality. □

*Solution.* $p_*\pi_1(\widetilde{X}, \tilde{x}_0) \leq \pi_1(X, x_0)$. □

**Exercise 11.32.** What does this subgroup tell about loops?

*Hint.* Use the closed-lift criterion.

Compare the subgroup definition with the closed-lift criterion in both directions, always lifting from the designated $\tilde{x}_0$. □

*Solution.* It is exactly the set of loop classes whose lifts from $\tilde{x}_0$ close. □

**Exercise 11.33.** What is the bridge to VIII/12?

*Hint.* Use lifting uniqueness for self-lifts.

Type the self-lift correctly: a deck map $h : \widetilde{X} \to \widetilde{X}$ satisfies $p \circ h = p$. It lifts $p$ through itself; a lift of $\mathrm{id}_X$ would instead be a section. ☐

*Solution.* Deck transformations are lifts of the identity and are determined by one point on a connected cover. ☐

## 11.20 Solved problem dossiers

**Problem 11.34** (Lift a circle loop)**.** Lift $\alpha(t) = e^{2\pi int}$ through $\mathbb{R} \to S^1$ starting at 0.

*Solution.* The unique lift is $\widetilde{\alpha}(t) = nt$. ☐

**Problem 11.35** (Lift from another sheet)**.** Lift the same loop starting at $k \in \mathbb{Z}$.

*Solution.* The unique lift is $\widetilde{\alpha}(t) = k + nt$, ending at $k + n$. ☐

**Problem 11.36** (Closed lift test on the universal circle cover)**.** Which loops in $S^1$ lift to loops in $\mathbb{R}$?

*Solution.* Only degree-zero loops. The covering subgroup is trivial, so the lift closes exactly for the zero class in $\mathbb{Z}$. ☐

**Problem 11.37** (Finite circle cover criterion)**.** For $p_n(z) = z^n$, which loop classes in $\pi_1(S^1) = \mathbb{Z}$ lift to loops?

*Solution.* The subgroup is $n\mathbb{Z}$, so exactly multiples of $n$ lift closed. ☐

**Problem 11.38** (Injection proof)**.** Prove $p_*$ is injective.

*Solution.* If an upstairs loop projects to a null-homotopic loop, lift a null-homotopy starting with the upstairs loop. The opposite edge is the constant lift, showing the original loop is null-homotopic. ☐

**Problem 11.39** (Uniqueness of map lifts)**.** Why do two lifts agreeing at one point agree everywhere on a connected domain?

*Solution.* Locally, both lifts must lie in the same sheet and equal the inverse sheet map applied to $f$, so the agreement set is open; its complement is also open. Connectedness forces global agreement. ☐

**Problem 11.40** (Homotopy endpoint invariant)**.** Why do homotopic based loops induce the same endpoint permutation of a fiber?

*Solution.* Lift the based homotopy from a chosen fiber point. The boundary of the lifted square forces the two lifted loops to have the same endpoint. ☐

**Problem 11.41** (Continuous argument on an interval)**.** Why does every map $f : I \to S^1$ admit a continuous real argument after choosing the initial value?

*Solution.* The interval is simply connected, so the lifting criterion to $\mathbb{R} \to S^1$ is automatic. ☐

**Problem 11.42** (No global argument on $S^1$)**.** Why does the identity map $S^1 \to S^1$ not lift to $\mathbb{R}$?

*Solution.* Its induced map on $\pi_1$ is the identity $\mathbb{Z} \to \mathbb{Z}$, not contained in the trivial covering subgroup. ☐

**Problem 11.43** (Square root of a circle map)**.** When does $f : S^1 \to S^1$ admit a continuous square root?

*Solution.* A lift through $z \mapsto z^2$ exists iff $\deg f \in 2\mathbb{Z}$, so exactly when the degree is even. □

**Problem 11.44** (Cube root)**.** When does a degree-8 circle map admit a continuous cube root?

*Solution.* Never, because $3 \nmid 8$. □

**Problem 11.45** (Fourth root)**.** When does a degree-12 circle map admit a continuous fourth root?

*Solution.* Yes, because $4 \mid 12$. □

**Problem 11.46** (Lift from simply connected domain)**.** Let $Y = D^2$ and $f : D^2 \to X$. Why does $f$ lift to any cover after choosing the initial lift point?

*Solution.* $\pi_1(D^2) = 0$, so the lifting subgroup criterion is automatically satisfied. □

**Problem 11.47** (Universal-cover lift)**.** Let $f : Y \to X$ induce the trivial map on $\pi_1$. What can be said about a lift to a universal cover?

*Solution.* Under the standard path-connected/local hypotheses and a chosen basepoint lift, a unique based lift exists. □

**Problem 11.48** (Path-independence calculation)**.** Explain explicitly how two paths $\gamma_0, \gamma_1$ from $y_0$ to $y$ are compared in the lifting criterion proof.

*Solution.* Form the loop $\gamma_0 * \overline{\gamma_1}$. Its image class lies in the covering subgroup, so its lift closes. Hence the lifted endpoints of $f\gamma_0$ and $f\gamma_1$ coincide. □

**Problem 11.49** (Monodromy of $z \mapsto z^n$)**.** Describe the action of the generator of $\pi_1(S^1)$ on the $n$-point fiber over 1.

*Solution.* Label the roots by $0, \ldots, n-1$. Lifting one positive turn sends label $k$ to $k+1$ $(\mathrm{mod}\ n)$, an $n$-cycle. □

**Problem 11.50** (Transitivity)**.** Why is that monodromy action transitive?

*Solution.* The generator acts by one $n$-cycle, so any fiber point can be reached from any other. □

**Problem 11.51** (Disconnected trivial cover monodromy)**.** Describe monodromy for the trivial cover $X \times F \to X$.

*Solution.* Every lifted loop stays in the same global sheet, so the action on $F$ is trivial and therefore not transitive when $|F| > 1$. □

**Problem 11.52** (Connectedness test preview)**.** What does transitivity of monodromy suggest about a cover?

*Solution.* For standard path-connected locally path-connected bases, transitive monodromy corresponds to connectedness of the covering space. □

**Problem 11.53** (Lift of identity)**.** What are lifts of $\mathrm{id}_X : X \to X$ through $p : \widetilde{X} \to X$?

*Solution.* They are sections $s : X \to \widetilde{X}$ satisfying $ps = \mathrm{id}$. □

**Problem 11.54** (Deck transformation as a lift)**.** Why is a deck transformation a lift of $p : \widetilde{X} \to X$ through itself?

*Solution.* If $h$ is a deck transformation, $ph = p$. Thus $h : \widetilde{X} \to \widetilde{X}$ is a lift of the map $p$ to the covering $p$. □

**Problem 11.55** (Bridge to deck rigidity)**.** Why is a deck transformation on a connected cover determined by one point?

*Solution.* Two deck transformations are lifts of the same map $p$; if they agree at one point, uniqueness of lifts on the connected domain forces them to agree everywhere. □

## 11.21 Supplementary worked examples

**Example 11.56** (Lifting a circle path)**.** For $p : \mathbb{R} \to S^1$, a path $\gamma(t) = e^{2\pi i\theta(t)}$ with chosen initial lift $\widetilde{\gamma}(0) = a$ has a unique continuous lift whose real coordinate tracks the angle continuously from $a$.

**Example 11.57** (Lift of a loop need not close)**.** A loop on $S^1$ of winding number $k$ lifts to a path in $\mathbb{R}$ whose endpoints differ by $k$. Closing upstairs is therefore equivalent to zero winding for the universal circle cover.

**Example 11.58** (Homotopy lifting)**.** Once a lift of the initial time slice is chosen, a homotopy $H : X \times I \to B$ lifts uniquely through a covering under the standard hypotheses. The lifted homotopy coherently lifts every path $t \mapsto H(x, t)$.

## 11.22 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 11.59** (Initial lift)**.** What extra datum is needed to lift a path $\gamma : I \to B$?

*Hint.* Choose a point in the fiber over $\gamma(0)$. ☐

*Solution.* Choose $\widetilde{x}_0 \in E$ with $p(\widetilde{x}_0) = \gamma(0)$. ☐

**Exercise 11.60** (Endpoint of winding one)**.** For $p(t) = e^{2\pi it}$, lift the loop $e^{2\pi it}$ starting at 0. What is the endpoint?

*Hint.* Use the obvious real lift. ☐

*Solution.* The lift is $\widetilde{\gamma}(t) = t$, so it ends at 1. ☐

**Exercise 11.61** (Endpoint of winding $k$)**.** For the same cover, where does the lift of $e^{2\pi ikt}$ starting at 0 end?

*Hint.* Lift the angle $kt$. ☐

*Solution.* It ends at $k$. ☐

**Exercise 11.62** (Constant path)**.** Lift a constant path at $b$ starting at $e \in p^{-1}(b)$.

*Hint.* Uniqueness prevents movement between discrete sheets. ☐

*Solution.* The unique lift is the constant path at $e$. ☐

**Exercise 11.63** (Project lifted homotopy)**.** What equation must a lifted homotopy $\widetilde{H}$ satisfy?

*Hint.* It lies over $H$. ☐

*Solution.* $p \circ \widetilde{H} = H$, together with the prescribed lift at the initial time (or other specified boundary data). ☐

## Proofs

**Exercise 11.64** (Path-lift uniqueness)**.** Prove two lifts of the same path with the same initial point agree.

*Hint.* The agreement set is open and closed in $I$ using evenly covered neighborhoods. □

*Solution.* Where the lifts agree, an evenly covered neighborhood forces them into the same sheet and hence to agree locally; the agreement set is nonempty, open, and closed, so connectedness of $I$ gives equality everywhere. □

**Exercise 11.65** (Homotopic loops endpoint)**.** Prove endpoint-fixed homotopic loops have lifts with the same endpoint when started at the same lift.

*Hint.* Lift the homotopy and inspect the endpoint edge. □

*Solution.* Homotopy lifting gives a continuous path in the discrete fiber traced by lifted endpoints; it must be constant, so endpoints agree. □

**Exercise 11.66** (Lift criterion intuition)**.** Explain why a global lift $f : X \to B$ is constrained by loops in $X$.

*Hint.* Every loop upstairs must project to a loop whose lift closes. □

*Solution.* If $f$ lifts, then $f_*$ sends $\pi_1(X)$ into the subgroup $p_*\pi_1(E)$; otherwise some loop would lift with mismatched endpoints. □

**Exercise 11.67** (Uniqueness of map lift)**.** Prove two lifts of a map from a connected space that agree at one point agree everywhere under the covering hypotheses.

*Hint.* Use local sheet uniqueness and connectedness. □

*Solution.* The equalizer is nonempty, open, and closed because local lifts into an evenly covered neighborhood are determined by the sheet; connectedness forces global equality. □

## Counterexamples and hypothesis tests

**Exercise 11.68** (Any initial point works)**.** Test: a path can be lifted from an arbitrary point of $E$, regardless of its projection.

*Hint.* The initial point must lie over the path start. □

*Solution.* False. Necessarily $p(\widetilde{\gamma}(0)) = \gamma(0)$. □

**Exercise 11.69** (Loop lift always loop)**.** Test: the lift of every loop is a loop.

*Hint.* Use a nonzero winding loop under $\mathbb{R} \to S^1$. □

*Solution.* False. Its endpoint can be a different point in the same fiber. □

**Exercise 11.70** (Two lifts can cross)**.** Test: distinct lifts of the same path with the same starting point can later separate.

*Hint.* Use uniqueness. □

*Solution.* False. Path-lift uniqueness makes them identical. □

## Applications and investigations

**Exercise 11.71** (Winding detector)**.** How does path lifting numerically reveal winding number for loops on $S^1$?

*Hint.* Track the endpoint displacement in the universal cover. ☐

*Solution.* Lift the loop to $\mathbb{R}$ from 0; the integer endpoint is exactly its winding number. ☐

**Exercise 11.72** (Continuation algorithm)**.** Why can a lift be constructed incrementally along a compact interval?

*Hint.* Cover the base path by finitely many evenly covered neighborhoods. ☐

*Solution.* Compactness gives a finite subdivision on which the path stays in trivializing neighborhoods; choose the unique continuing sheet successively, producing the global lift. ☐

## Challenge problems

**Exercise 11.73** (Subgroup criterion)**.** State the standard fundamental-group criterion for lifting a based map $f : (X, x_0) \to (B, b_0)$ to $(E, e_0)$.

*Hint.* Compare induced subgroups. ☐

*Solution.* For path-connected, locally path-connected spaces in the usual covering setting, a lift with $\widetilde{f}(x_0) = e_0$ exists iff $f_*\pi_1(X, x_0) \subset p_*\pi_1(E, e_0)$. ☐

**Exercise 11.74** (Homotopy-lift synthesis)**.** Explain how path-lift uniqueness and homotopy lifting together prove homotopy invariance of the endpoint permutation of a loop.

*Hint.* Lift the entire endpoint-fixed homotopy. ☐

*Solution.* The endpoint of each lifted loop lies in a discrete fiber and varies continuously with homotopy time; it is therefore constant, so endpoint action depends only on the loop homotopy class. ☐

# Chapter 12

# Deck Transformations and Group Actions

A covering space has internal symmetries that preserve every point of the base. These symmetries are deck transformations. On connected covers they are extremely rigid: one point determines the whole transformation. Their group acts on fibers, regular coverings are characterized by transitivity of this action, and universal covers recover the base fundamental group as a deck group.

## 12.1 Deck transformations

**Definition 12.1** (Deck transformation)**.** Let

$$p : \widetilde{X} \to X$$

be a covering. A deck transformation is a homeomorphism

$$h : \widetilde{X} \to \widetilde{X}$$

such that

$$p \circ h = p.$$

The deck transformations form a group under composition:

$$\mathrm{Deck}(p).$$

## 12.2 Deck maps preserve fibers

**Proposition 12.2** (Fiber preservation)**.** *For every $x \in X$,*

$$h(p^{-1}(x)) = p^{-1}(x)$$

*for each deck transformation $h$.*

Thus the deck group acts by permutations on every fiber.

## 12.3 Rigidity on connected covers

**Theorem 12.3** (One point determines a deck transformation)**.** *If $\widetilde{X}$ is connected, two deck transformations that agree at one point are equal everywhere.*

*Proof.* Both maps are lifts of the same covering map

$$p : \widetilde{X} \to X$$

through the covering $p$. Apply uniqueness of lifts from VIII/11. □

**Corollary 12.4** (Free action)**.** *If a deck transformation of a connected covering fixes one point, it is the identity. Hence the deck group acts freely on* $\widetilde{X}$*.*

## 12.4 The universal circle cover

For

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t},$$

the translations

$$T_n(t) = t + n, \qquad n \in \mathbb{Z},$$

are deck transformations.

**Theorem 12.5** (Deck group of the circle cover)**.**

$$\mathrm{Deck}(\mathbb{R} \to S^1) \cong \mathbb{Z}.$$

## 12.5 Finite circle covers

For

$$p_n : S^1 \to S^1, \qquad p_n(z) = z^n,$$

the rotations

$$R_k(z) = e^{2\pi i k/n} z$$

are deck transformations.

**Theorem 12.6** (Deck group of $z^n$)**.**

$$\mathrm{Deck}(p_n) \cong \mathbb{Z}/n.$$

## 12.6 Antipodal covering

For $n \geq 2$,

$$q : S^n \to \mathbb{R}P^n$$

has the nontrivial deck transformation

$$a(x) = -x.$$

**Proposition 12.7** (Projective deck group)**.**

$$\mathrm{Deck}(S^n \to \mathbb{R}P^n) \cong \mathbb{Z}/2.$$

## 12.7 Deck action on a fiber

Fix $x_0 \in X$. Restriction gives an action

$$\mathrm{Deck}(p) \curvearrowright p^{-1}(x_0).$$

Because the action is free for connected covers, no nonidentity deck transformation fixes a fiber point.

## 12.8 Regular coverings

**Definition 12.8** (Regular covering)**.** A connected covering is regular, normal, or Galois if the deck group acts transitively on one, equivalently every, fiber.

**Theorem 12.9** (Normal-subgroup criterion)**.** *Under the standard covering-classification hypotheses, a connected pointed cover corresponding to*

$$H \leq \pi_1(X, x_0)$$

*is regular if and only if*

$$H \trianglelefteq \pi_1(X, x_0).$$

## 12.9 Deck group as a quotient

**Theorem 12.10** (Deck group of a regular cover)**.** *If a connected covering corresponds to a normal subgroup*

$$H \trianglelefteq G = \pi_1(X, x_0),$$

*then*

$$\mathrm{Deck}(p) \cong G/H.$$

For a universal cover $H = 1$, so

$$\mathrm{Deck}(\widetilde{X} \to X) \cong \pi_1(X, x_0)$$

under the standard hypotheses.

## 12.10 Universal covers and group actions

If $\widetilde{X}$ is a universal cover, the fundamental group acts on $\widetilde{X}$ by deck transformations. The quotient by this action recovers the base:

$$X \cong \widetilde{X}/\pi_1(X).$$

This is the geometric realization of the fundamental group as a symmetry group of the universal cover.

## 12.11 Properly discontinuous actions

A group $G$ acting on a space $\widetilde{X}$ is properly discontinuous in the covering-space sense when every point has a neighborhood whose distinct translates do not overlap in a way that destroys the local quotient model.

**Theorem 12.11** (Quotient construction)**.** *If $G$ acts freely and properly discontinuously on a suitable space $\widetilde{X}$, then*

$$\widetilde{X} \to \widetilde{X}/G$$

*is a covering map, and $G$ embeds into its deck group.*

Under standard connectedness hypotheses and an effective action realizing all deck symmetries, the deck group is exactly $G$.

## 12.12 Translations on the line

The action

$$\mathbb{Z} \curvearrowright \mathbb{R}, \qquad n \cdot t = t + n,$$

is free and properly discontinuous. Its quotient is

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

## 12.13 Lattice actions

For a lattice

$$\Lambda \cong \mathbb{Z}^n$$

acting by translations on $\mathbb{R}^n$,

$$\mathbb{R}^n/\Lambda$$

is an $n$-torus, and

$$\mathbb{R}^n \to \mathbb{R}^n/\Lambda$$

is a universal covering map with deck group $\Lambda$.

In particular,

$$\mathrm{Deck}(\mathbb{R}^2 \to T^2) \cong \mathbb{Z}^2.$$

## 12.14 Monodromy versus deck action

Monodromy is the action

$$\pi_1(X, x_0) \curvearrowright p^{-1}(x_0)$$

obtained by lifting loops. The deck group also acts on the same fiber. These actions are related but not identical in arbitrary covers.

For a regular cover, the deck action is transitive and can be identified with a quotient of the fundamental group.

## 12.15 The normalizer formula

Let a connected pointed cover correspond to

$$H \leq G = \pi_1(X, x_0).$$

**Theorem 12.12** (Normalizer formula)**.** *Under the standard classification hypotheses,*

$$\mathrm{Deck}(p) \cong N_G(H)/H,$$

*where*

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

If $H$ is normal, then $N_G(H) = G$, recovering the regular-cover quotient formula.

## 12.16 Deck transformations and basepoint choices

A deck transformation can move the chosen point

$$\tilde{x}_0 \in p^{-1}(x_0)$$

to another point of the same fiber. Changing the pointed cover in this way conjugates the associated subgroup. This explains why unpointed connected coverings correspond to conjugacy classes of subgroups.

## 12.17 Automorphisms of coverings

Deck transformations are precisely automorphisms of the covering object over the fixed base $X$. Thus

$$\mathrm{Deck}(p) = \mathrm{Aut}_X(\widetilde{X}).$$

## 12.18 Regularity examples

The universal circle cover is regular because the trivial subgroup of $\mathbb{Z}$ is normal. The $n$-fold circle cover corresponds to $n\mathbb{Z} \trianglelefteq \mathbb{Z}$, so it is regular with deck group

$$\mathbb{Z}/n.$$

The universal projective cover corresponds to the trivial subgroup of $\mathbb{Z}/2$, hence is regular.

## 12.19 Bridge to $SU(2) \to SO(3)$

The next chapter studies one of the most important geometric two-sheeted covers:

$$SU(2) \to SO(3).$$

Its kernel is

$$\{\pm I\},$$

and the nontrivial deck transformation is multiplication by $-I$. This provides a Lie-group realization of the same covering and quotient principles developed here.

## 12.20 Exercises

**Exercise 12.13.** Define a deck transformation.

*Hint.* Use a homeomorphism over the identity of the base.

The map acts on the total space and obeys $p \circ h = p$; distinguish this from a section, whose domain is the base. □

*Solution.* A homeomorphism $h : \widetilde{X} \to \widetilde{X}$ satisfying $ph = p$. □

**Exercise 12.14.** Why does a deck transformation preserve each fiber?

*Hint.* Apply $p$ after $h$.

Evaluate the commuting equation at an arbitrary $\tilde{x} \in p^{-1}(x)$, then identify the fiber containing $h(\tilde{x})$. □

*Solution.* If $p(\tilde{x}) = x$, then $p(h(\tilde{x})) = p(\tilde{x}) = x$. □

**Exercise 12.15.** What group do deck transformations form?

*Hint.* Use composition.

Check that the identity, composition, and inverse of homeomorphisms satisfying $ph = p$ again satisfy that same equation. □

*Solution.* $\mathrm{Deck}(p)$. □

**Exercise 12.16.** On a connected cover, how much data determines a deck transformation?

*Hint.* Use lift uniqueness.

Two deck maps are lifts of the same map $p$; apply uniqueness for lifts on a connected domain when their values agree once. □

*Solution.* Its value at one point. □

**Exercise 12.17.** What happens if a deck transformation fixes one point of a connected cover?

*Hint.* Compare with the identity deck map.

Compare the given map with $\mathrm{id}_{\widetilde{X}}$ as two lifts of $p$, using the fixed point as their common value. □

*Solution.* It is the identity. □

**Exercise 12.18.** What deck transformations act on $\mathbb{R} \to S^1$?

*Hint.* Use integer translations.

The equation $e^{2\pi i h(t)} = e^{2\pi i t}$ makes $h(t) - t$ an integer-valued continuous function on a connected line. □

*Solution.* $T_n(t) = t + n$. □

**Exercise 12.19.** What is the deck group of $\mathbb{R} \to S^1$?

*Hint.* Translations are indexed by integers.

Compose translations and track the resulting integer parameter; this identifies the group operation, not just the set of deck maps. □

*Solution.* $\mathbb{Z}$. □

**Exercise 12.20.** What deck transformations act on $z \mapsto z^n$?

*Hint.* Rotate by $n$th roots of unity.

For any deck map, $h(z)/z$ lies among the $n$th roots of unity. Its continuity on the connected circle forces it to be constant. □

*Solution.* $R_k(z) = e^{2\pi i k/n} z$. □

**Exercise 12.21.** What is the deck group of $z \mapsto z^n$?

*Hint.* There are $n$ rotations.

Choose a primitive $n$th root of unity and find the least positive power of its rotation that is the identity. □

*Solution.* $\mathbb{Z}/n$. □

**Exercise 12.22.** What is the nontrivial deck map of $S^n \to \mathbb{R}P^n$?

*Hint.* Use the antipodal involution.

Verify that the antipodal map preserves every quotient fiber and is not the identity; connected-cover uniqueness rules out further possibilities once one image is chosen. □

*Solution.* $x \mapsto -x$. □

**Exercise 12.23.** What is that deck group?

*Hint.* Two elements.

Compute the square of the nonidentity fiber-preserving map and use the two-point fiber to bound the number of deck transformations. ☐

*Solution.* $\mathbb{Z}/2$. ☐

**Exercise 12.24.** Define a regular covering.

*Hint.* Use the deck action on a fiber.

Transitivity means any chosen point of a fiber can be moved to any other by a deck map; connectedness of the cover remains a separate hypothesis. ☐

*Solution.* A connected cover whose deck group acts transitively on a fiber. ☐

**Exercise 12.25.** What subgroup condition characterizes regular covers?

*Hint.* Use normality.

Use the covering classification to compare conjugates of the subgroup when the chosen point in a fiber changes. ☐

*Solution.* The corresponding subgroup $H$ is normal in $\pi_1(X)$. ☐

**Exercise 12.26.** For a regular cover corresponding to $H \trianglelefteq G$, what is the deck group?

*Hint.* Take the quotient.

Normality is what makes multiplication of cosets define a quotient group; it cannot be omitted from this formula. ☐

*Solution.* $G/H$. ☐

**Exercise 12.27.** What is the deck group of a universal cover?

*Hint.* Set $H = 1$.

Set the covering subgroup to the identity subgroup, and remember that a concrete identification with $\pi_1(X, x_0)$ uses basepoint and action conventions. ☐

*Solution.* It is isomorphic to $\pi_1(X)$ under the standard hypotheses. ☐

**Exercise 12.28.** How can a group action produce a covering?

*Hint.* Require free proper discontinuity.

Choose a neighborhood whose distinct group translates are disjoint. Its quotient image should be evenly covered by those translates. ☐

*Solution.* A free properly discontinuous action gives $\widetilde{X} \to \widetilde{X}/G$ as a covering under standard hypotheses. ☐

**Exercise 12.29.** What quotient does $\mathbb{Z} \curvearrowright \mathbb{R}$ produce?

*Hint.* Translate by integers.

Use $t \mapsto e^{2\pi i t}$ to identify exactly the integer-translation orbits and give the quotient a circle model. ☐

*Solution.* $\mathbb{R}/\mathbb{Z} \cong S^1$. ☐

**Exercise 12.30.** What quotient does $\mathbb{Z}^2 \curvearrowright \mathbb{R}^2$ produce?

*Hint.* Use a lattice.

Choose a fundamental parallelogram and match its opposite sides; its two lattice generators determine the two circle directions. □

*Solution.* A two-torus. □

**Exercise 12.31.** What is the deck group of $\mathbb{R}^2 \to T^2$?

*Hint.* Use lattice translations.

A deck map differs from the identity by a continuous lattice-valued displacement; connectedness makes that displacement constant. □

*Solution.* $\mathbb{Z}^2$. □

**Exercise 12.32.** State the normalizer formula.

*Hint.* Use $N_G(H)$.

First determine which elements conjugate $H$ to itself, then divide by $H$; using $G/H$ before checking normality gives the wrong general formula. □

*Solution.* $\mathrm{Deck}(p) \cong N_G(H)/H$. □

**Exercise 12.33.** What happens to the normalizer formula when $H$ is normal?

*Hint.* Then $N_G(H) = G$.

Expand the definition $gHg^{-1} = H$ for a normal subgroup and substitute the resulting normalizer into the general deck-group formula. □

*Solution.* It reduces to $G/H$. □

**Exercise 12.34.** Why do unpointed connected covers correspond to conjugacy classes?

*Hint.* Change the chosen lift basepoint.

Transport the lifted basepoint along an upstairs path and project it; conjugation by the resulting downstairs path explains the change in subgroup. □

*Solution.* Moving the basepoint in a fiber conjugates the associated subgroup. □

**Exercise 12.35.** How can deck transformations be described categorically?

*Hint.* Automorphisms over the fixed base.

In the category of spaces over $X$, a morphism must commute with projection, and an automorphism must also have an inverse over $X$. □

*Solution.* $\mathrm{Deck}(p) = \mathrm{Aut}_X(\widetilde{X})$. □

**Exercise 12.36.** What is the bridge to VIII/13?

*Hint.* Use the two-sheeted Lie-group cover.

For $SU(2) \to SO(3)$, compare multiplication by the two kernel elements $\pm I$; verify directly that both leave the projected rotation unchanged. □

*Solution.* VIII/13 studies $SU(2) \to SO(3)$ with deck group $\{\pm I\} \cong \mathbb{Z}/2$. □

## 12.21 Solved problem dossiers

**Problem 12.37** (Circle translations)**.** Prove $T_n(t) = t + n$ is a deck transformation of $\mathbb{R} \to S^1$.

*Solution.* $p(T_n(t)) = e^{2\pi i(t+n)} = e^{2\pi it} = p(t)$, and $T_n$ is a homeomorphism. ☐

**Problem 12.38** (All circle deck maps)**.** Show every deck transformation of $\mathbb{R} \to S^1$ is an integer translation.

*Solution.* A deck map sends 0 to a point in the fiber $\mathbb{Z}$, say $n$. The translation $T_n$ is another deck map with the same value at 0; connected-cover rigidity gives equality. ☐

**Problem 12.39** (Finite circle rotations)**.** Verify $R_k(z) = e^{2\pi ik/n}z$ is a deck map of $p_n(z) = z^n$.

*Solution.* $p_n(R_k(z)) = e^{2\pi ik}z^n = z^n = p_n(z)$. ☐

**Problem 12.40** (All finite circle deck maps)**.** Why are there exactly $n$ deck transformations of $z \mapsto z^n$?

*Solution.* A deck transformation is determined by the image of 1, which can be any of the $n$ points in the fiber over 1. Each choice is realized by one root-of-unity rotation. ☐

**Problem 12.41** (Antipodal deck map)**.** Verify $a(x) = -x$ is a deck transformation of $S^n \to \mathbb{R}P^n$.

*Solution.* The quotient map sends $x$ and $-x$ to the same projective line, so $qa = q$. ☐

**Problem 12.42** (Projective deck group)**.** Why are identity and antipodal map the only deck transformations of the connected cover $S^n \to \mathbb{R}P^n$?

*Solution.* A deck transformation is determined by the image of one point, and the fiber contains only $x$ and $-x$. ☐

**Problem 12.43** (Free action proof)**.** Prove the deck action on a connected covering is free.

*Solution.* If $h(\tilde{x}) = \tilde{x}$, then $h$ and the identity deck map agree at one point. Lift uniqueness makes them equal. ☐

**Problem 12.44** (Universal-cover quotient)**.** Explain why a universal cover can be viewed as a quotient by its deck group.

*Solution.* The deck group acts freely and properly discontinuously, and its orbits are fibers for a regular universal cover. The quotient therefore recovers the base. ☐

**Problem 12.45** (Torus lattice cover)**.** Show $\mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2$ has deck group $\mathbb{Z}^2$.

*Solution.* Each lattice translation is a deck map. Any deck map is determined by the image of 0, which must be another point of the fiber $\mathbb{Z}^2$, so it is the corresponding translation. ☐

**Problem 12.46** (Regularity of $z^n$)**.** Why is the $n$-fold circle cover regular?

*Solution.* It corresponds to $n\mathbb{Z} \trianglelefteq \mathbb{Z}$. Equivalently, the root-of-unity deck rotations act transitively on every fiber. ☐

**Problem 12.47** (Quotient deck group)**.** Use subgroup theory to compute the deck group of the cover corresponding to $n\mathbb{Z} \leq \mathbb{Z}$.

*Solution.* Since $\mathbb{Z}$ is abelian, $n\mathbb{Z}$ is normal and the deck group is $\mathbb{Z}/n\mathbb{Z}$. □

**Problem 12.48** (Universal circle quotient)**.** Apply the regular-cover formula to the trivial subgroup of $\mathbb{Z}$.

*Solution.* The deck group is $\mathbb{Z}/0 \cong \mathbb{Z}$, matching integer translations of $\mathbb{R}$. □

**Problem 12.49** (Normalizer example)**.** If $H \leq G$ is self-normalizing, what is the deck group of the corresponding connected cover?

*Solution.* The normalizer formula gives $N_G(H)/H = H/H$, so the deck group is trivial. □

**Problem 12.50** (Normal subgroup example)**.** If $H \trianglelefteq G$ has index 5, what can be said about the regular cover?

*Solution.* It is five-sheeted and its deck group $G/H$ has five elements. □

**Problem 12.51** (Deck versus monodromy)**.** Explain one difference between deck action and monodromy action.

*Solution.* Monodromy is an action of the base fundamental group on a fiber obtained by lifting loops. Deck action is an action of automorphisms of the entire covering. They coincide only through additional regularity identifications. □

**Problem 12.52** (Connected-cover rigidity)**.** Why can a deck transformation not permute one fiber arbitrarily?

*Solution.* The image of one point determines the whole map, but not every fiber permutation need extend continuously to a deck transformation. Only permutations compatible with the global covering structure do. □

**Problem 12.53** (Trivial cover automorphisms)**.** Describe deck transformations of $X \times F \to X$ when $X$ is connected and $F$ is discrete.

*Solution.* Any fixed permutation $\sigma \in \mathrm{Sym}(F)$ defines $h(x, a) = (x, \sigma(a))$. Conversely continuity over connected $X$ forces the permutation to be constant, so the deck group is $\mathrm{Sym}(F)$. □

**Problem 12.54** (Why connectedness mattered)**.** Why does one-point rigidity fail for a disconnected covering space?

*Solution.* A deck map can agree with another on one component but differ on another component, so agreement at one point need not control all components. □

**Problem 12.55** (Conjugate subgroup viewpoint)**.** How does moving $\tilde{x}_0$ in the fiber change the associated subgroup?

*Solution.* A path/monodromy element relating the two fiber points conjugates $H = p_*\pi_1(\widetilde{X}, \tilde{x}_0)$ inside $G = \pi_1(X, x_0)$. □

**Problem 12.56** (Regular cover fiber action)**.** Why does transitivity of the deck group on a fiber imply every fiber point is equivalent as a pointed cover?

*Solution.* For any two fiber points, a deck transformation carries one to the other while commuting with the projection, giving an isomorphism of the corresponding pointed covering structures. □

**Problem 12.57** (Deck group as quotient symmetry)**.** Interpret $G/H$ geometrically for a regular cover.

*Solution.* Cosets label sheets/fiber positions, and quotient elements act by translating those positions; this algebraic action is realized by deck transformations. ☐

**Problem 12.58** (Proper discontinuity of translations)**.** Why is the integer-translation action on $\mathbb{R}$ properly discontinuous?

*Solution.* For any $t$, choose an interval of length less than 1. Its nonzero integer translates are disjoint from it, giving a local quotient sheet. ☐

**Problem 12.59** (Free action necessity)**.** What goes wrong for covering quotients if a group action has a fixed point?

*Solution.* At a fixed point the quotient map cannot look locally like disjoint homeomorphic sheets indexed by group elements; distinct orbit representatives collapse at the fixed point. ☐

**Problem 12.60** (Bridge to $SU(2)$)**.** What deck symmetry should appear in the cover $SU(2) \to SO(3)$?

*Solution.* The kernel is $\{\pm I\}$. Multiplication by $-I$ is the nontrivial deck transformation, so the deck group is $\mathbb{Z}/2$. ☐

## 12.22 Supplementary worked examples

**Example 12.61** (Deck group of the circle universal cover)**.** For $p : \mathbb{R} \to S^1$, every deck transformation is $T_k(t) = t + k$ with $k \in \mathbb{Z}$. Hence $\mathrm{Deck}(p) \cong \mathbb{Z}$.

**Example 12.62** (Deck group of $z \mapsto z^n$)**.** For $p_n : S^1 \to S^1$, multiplication by an $n$th root of unity preserves fibers and commutes with $p_n$. These rotations form a cyclic deck group of order $n$.

**Example 12.63** (Free properly discontinuous action)**.** If a discrete group $G$ acts freely and properly discontinuously on a suitable space $E$, the quotient map $E \to E/G$ is a covering and $G$ acts by deck transformations. Regular coverings arise naturally this way.

## 12.23 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 12.64** (Deck equation)**.** State the defining equation for a deck transformation $h : E \to E$ over $B$.

*Hint.* It must preserve the projection. ☐

*Solution.* $p \circ h = p$, with $h$ a homeomorphism of $E$. ☐

**Exercise 12.65** (Integer translation)**.** Check $T_k(t) = t + k$ is a deck transformation of $\mathbb{R} \to S^1$.

*Hint.* Exponentials are periodic by integers. ☐

*Solution.* $p(t + k) = e^{2\pi i(t+k)} = e^{2\pi it} = p(t)$, and $T_k$ is a homeomorphism. ☐

**Exercise 12.66** (Root rotation)**.** For $p_n(z) = z^n$, check $h_\zeta(z) = \zeta z$ is a deck transformation when $\zeta^n = 1$.

*Hint.* Compute $(\zeta z)^n$. ☐

*Solution.* $p_n(\zeta z) = \zeta^n z^n = z^n = p_n(z)$. ☐

**Exercise 12.67** (Composition law)**.** What is the composition law for deck transformations of $\mathbb{R} \to S^1$?

*Hint.* Compose translations. ☐

*Solution.* $T_k T_\ell = T_{k+\ell}$. ☐

**Exercise 12.68** (Fiber action)**.** Where can a deck transformation send a point $e \in E$?

*Hint.* Use $p(h(e)) = p(e)$. ☐

*Solution.* It must remain in the same fiber $p^{-1}(p(e))$. ☐

## Proofs

**Exercise 12.69** (Deck determined by one point)**.** Prove a deck transformation of a connected cover is determined by its value at one point.

*Hint.* Two deck maps are lifts of the same map $p : E \to B$. ☐

*Solution.* If two deck transformations agree at one point, map-lift uniqueness on connected $E$ forces them to agree everywhere. ☐

**Exercise 12.70** (Free action on total space)**.** Prove the deck group of a connected cover acts freely on $E$.

*Hint.* A deck map fixing one point agrees there with the identity. ☐

*Solution.* By the previous uniqueness principle, a deck transformation with a fixed point equals the identity. ☐

**Exercise 12.71** (Quotient construction)**.** Explain why a free properly discontinuous group action yields evenly covered quotient neighborhoods.

*Hint.* Choose $U$ disjoint from all nontrivial translates. ☐

*Solution.* The quotient restricts to a homeomorphism on each translate $gU$, and the inverse image of its quotient image is the disjoint union of these translates. ☐

**Exercise 12.72** (Regular cover transitivity)**.** Show that if deck transformations act transitively on one fiber of a connected cover, they act transitively on every fiber.

*Hint.* Transport points along lifted paths. ☐

*Solution.* Path lifting conjugates the fiber actions along any base path; a deck transformation matching two points over one basepoint propagates to the corresponding endpoints over another. ☐

## Counterexamples and hypothesis tests

**Exercise 12.73** (Deck maps can fix points)**.** Test: a nonidentity deck transformation of a connected cover may fix isolated points.

*Hint.* Use lift uniqueness. ☐

*Solution.* False. The deck action is free. ☐

**Exercise 12.74** (Every cover regular)**.** Test: the deck group always acts transitively on fibers.

*Hint.* That is the definition/characterization of regular coverings, not all coverings. ☐

*Solution.* False. Nonregular connected covers can have smaller deck groups. ☐

**Exercise 12.75** (Deck group equals fiber)**.** Test: for every connected cover, deck-group cardinality equals fiber cardinality.

*Hint.* Only in regular covers does the action become transitive. ☐

*Solution.* False in general; a deck transformation is determined by a point image, so its size is at most the fiber size, with equality in regular cases. ☐

## Applications and investigations

**Exercise 12.76** (Symmetry extraction)**.** Why is the deck group a useful algebraic summary of a covering?

*Hint.* It records global symmetries invisible in one local sheet. ☐

*Solution.* Deck transformations permute sheets compatibly over every base point, encoding how the covering repeats globally and connecting the cover to subgroup normalizers. ☐

**Exercise 12.77** (Quotient reconstruction)**.** When can a covering be recovered as a quotient by its deck group?

*Hint.* Think regular connected coverings. ☐

*Solution.* For a regular connected cover, the deck group acts freely and transitively on fibers, and the base is naturally the quotient $E/\operatorname{Deck}(p)$ under standard hypotheses. ☐

## Challenge problems

**Exercise 12.78** (Normalizer formula preview)**.** For a connected cover corresponding to $H \leq \pi_1(B)$, what group controls deck transformations?

*Hint.* Use the normalizer of $H$. ☐

*Solution.* Under the classification hypotheses, $\operatorname{Deck}(p) \cong N(H)/H$; if $H$ is normal, this becomes $\pi_1(B)/H$. ☐

**Exercise 12.79** (Monodromy versus deck symmetry)**.** Distinguish the action of $\pi_1(B)$ on a fiber by path lifting from the deck-group action.

*Hint.* One comes from loops downstairs; the other from homeomorphisms upstairs. ☐

*Solution.* Monodromy permutes a chosen fiber via lifted loop endpoints and exists for every cover; deck transformations are global symmetries of the total space. For regular covers the two structures align through the quotient group. ☐

# Chapter 13

# $SU(2)$ to $SO(3)$

The double cover $SU(2) \to SO(3)$ is a central example where topology, group structure, and geometry reinforce one another. The group $SU(2)$ is a 3-sphere, while $SO(3)$ is obtained by identifying antipodal points. Unit quaternions provide the most transparent formula for the covering map.

## 13.1 The group $SU(2)$

**Definition 13.1** (Special unitary group)**.**

$$SU(2) = \{A \in M_2(\mathbb{C}) : A^*A = I,\ \det A = 1\}.$$

Every element has the form

$$A = \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

**Proposition 13.2** (Three-sphere model)**.** *As a topological space,*

$$SU(2) \cong S^3.$$

Indeed $(\alpha, \beta) \in \mathbb{C}^2 \cong \mathbb{R}^4$ lies on the unit 3-sphere.

## 13.2 Unit quaternions

Write

$$\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}.$$

The unit quaternions

$$S^3 = \{q \in \mathbb{H} : |q| = 1\}$$

form a group under quaternion multiplication and identify with $SU(2)$.

The purely imaginary quaternions

$$\operatorname{Im} \mathbb{H} = \{bi + cj + dk\}$$

are naturally $\mathbb{R}^3$.

## 13.3 Quaternion conjugation

For a unit quaternion $q$, define

$$R_q(v) = qvq^{-1}, \qquad v \in \operatorname{Im} \mathbb{H}.$$

**Theorem 13.3** (Quaternion rotation). *$R_q$ is an orientation-preserving orthogonal transformation of $\mathbb{R}^3$. Hence*

$$R_q \in SO(3).$$

Thus

$$\rho : S^3 \cong SU(2) \longrightarrow SO(3), \qquad q \longmapsto R_q$$

is a group homomorphism.

## 13.4 Kernel

**Theorem 13.4** (Kernel of the rotation map).

$$\ker \rho = \{\pm 1\}.$$

*Proof.* Both 1 and $-1$ commute with every imaginary quaternion and therefore act trivially. Conversely, a unit quaternion acting trivially by conjugation commutes with all imaginary quaternions, hence lies in the real center of $\mathbb{H}$, so it is $\pm 1$. $\square$

## 13.5 Axis-angle formula

Let $u \in \operatorname{Im} \mathbb{H}$ be a unit vector. Put

$$q = \cos \frac{\theta}{2} + u \sin \frac{\theta}{2}.$$

**Theorem 13.5** (Half-angle formula). *Conjugation by $q$ rotates $\mathbb{R}^3$ through angle $\theta$ about the axis $u$.*

The half-angle is the geometric source of the double cover.

## 13.6 Surjectivity

Every rotation in $SO(3)$ has an axis and angle. The half-angle formula therefore gives:

**Theorem 13.6** (Surjectivity). *The map*

$$\rho : SU(2) \to SO(3)$$

*is surjective.*

Together with the kernel computation,

$$SO(3) \cong SU(2)/\{\pm I\}.$$

## 13.7 Double covering

**Theorem 13.7** (Double cover)**.**

$$\rho : SU(2) \to SO(3)$$

*is a two-sheeted covering map with deck group*

$$\{\pm I\} \cong \mathbb{Z}/2.$$

The nontrivial deck transformation is multiplication by $-I$.

## 13.8 $SO(3)$ as projective 3-space

Since

$$SU(2) \cong S^3$$

and the kernel identifies $q$ with $-q$,

$$SO(3) \cong S^3/\{\pm 1\} \cong \mathbb{R}P^3.$$

**Corollary 13.8.**

$$SO(3) \cong \mathbb{R}P^3$$

*as topological manifolds.*

## 13.9 Fundamental group of $SO(3)$

Because $S^3$ is simply connected, the double cover is universal.

**Theorem 13.9** (Fundamental group)**.**

$$\pi_1(SO(3)) \cong \mathbb{Z}/2.$$

## 13.10 A $2\pi$ rotation loop

A path in $SU(2)$ from 1 to $-1$ projects to a loop in $SO(3)$. For example

$$q(t) = \cos(\pi t) + u \sin(\pi t)$$

projects to rotation through angle $2\pi t$ about $u$.

Its lift is not closed, so the projected loop represents the nontrivial element of $\pi_1(SO(3))$.

## 13.11 A $4\pi$ rotation

Traversing twice gives a lift from 1 to 1. Hence the $4\pi$-rotation loop is null-homotopic in $SO(3)$.

This is the topological content of the familiar distinction between $2\pi$ and $4\pi$ rotation paths.

## 13.12 Deck symmetry and lifting

A loop in $SO(3)$ lifts from $I \in SU(2)$ either back to $I$ or to $-I$. These two possible endpoints detect the two elements of

$$\pi_1(SO(3)) \cong \mathbb{Z}/2.$$

## 13.13 Matrix picture

The quaternion formula is equivalent to the adjoint action of $SU(2)$ on its three-dimensional Lie algebra. After identifying the Lie algebra with $\mathbb{R}^3$, the adjoint representation is precisely

$$\mathrm{Ad} : SU(2) \to SO(3).$$

## 13.14 Central quotient principle

The covering map is a group quotient by a discrete central subgroup:

$$1 \longrightarrow \{\pm I\} \longrightarrow SU(2) \longrightarrow SO(3) \longrightarrow 1.$$

This is a model for many Lie-group covering constructions.

## 13.15 Bridge to covering graphs

The $SU(2) \to SO(3)$ example shows how a simply connected total space carries a deck action realizing the base fundamental group. VIII/14 develops the same principle combinatorially for graphs, where universal covers become trees and fundamental groups become free groups.

## 13.16 Exercises

**Exercise 13.10.** Define $SU(2)$.

*Hint.* Unitary matrices with determinant one.

Specify $2 \times 2$ complex matrices, then separate the orthonormal-column condition $A^*A = I$ from the determinant-one condition. ☐

*Solution.* $SU(2) = \{A : A^*A = I, \det A = 1\}$. ☐

**Exercise 13.11.** Write the standard form of an $SU(2)$ matrix.

*Hint.* Use two complex parameters.

Choose the first unit column $(\alpha, \beta)^T$; find a unit orthogonal second column, and use the determinant to fix its otherwise free phase. ☐

*Solution.* $\begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix}$ with $|\alpha|^2 + |\beta|^2 = 1$. ☐

**Exercise 13.12.** Why is $SU(2) \cong S^3$?

*Hint.* Count real coordinates.

Write the real and imaginary parts of $\alpha, \beta$. The norm equation gives a sphere, and the matrix parametrization and its inverse are continuous. ☐

*Solution.* $(\alpha, \beta) \in \mathbb{C}^2 \cong \mathbb{R}^4$ lies on the unit sphere. ☐

**Exercise 13.13.** What group is another model of $SU(2)$?

*Hint.* Use quaternions.

Compare quaternion multiplication with the corresponding matrix multiplication, not merely their common underlying three-sphere. ☐

*Solution.* The unit quaternions. □

**Exercise 13.14.** Which quaternion subspace is $\mathbb{R}^3$?

*Hint.* Discard the real part.

Use the basis $i, j, k$ and the condition $\overline{v} = -v$; this distinguishes the three-dimensional imaginary subspace from all of $\mathbb{H}$. □

*Solution.* $\operatorname{Im}\mathbb{H}$. □

**Exercise 13.15.** Define $R_q$.

*Hint.* Conjugate imaginary quaternions.

Use a unit quaternion so that $q^{-1} = \overline{q}$; check that conjugating an imaginary quaternion leaves it imaginary. □

*Solution.* $R_q(v) = qvq^{-1}$. □

**Exercise 13.16.** In which group does $R_q$ lie?

*Hint.* It preserves orientation and norm.

Multiplicativity of quaternion norm gives orthogonality. To determine the determinant sign, use continuity on connected $S^3$ and evaluate at $q = 1$. □

*Solution.* $SO(3)$. □

**Exercise 13.17.** What is $\ker\rho$?

*Hint.* Find unit quaternions commuting with all imaginary ones.

A kernel element commutes with $i, j, k$, hence lies in the real center of $\mathbb{H}$; impose the unit-norm condition last. □

*Solution.* $\{\pm 1\}$. □

**Exercise 13.18.** Why do $q$ and $-q$ give the same rotation?

*Hint.* The signs cancel in conjugation.

Compute $(-q)^{-1}$ before simplifying $(-q)v(-q)^{-1}$; both minus signs must be accounted for. □

*Solution.* $(-q)v(-q)^{-1} = qvq^{-1}$. □

**Exercise 13.19.** Write the quaternion for rotation by $\theta$ about unit axis $u$.

*Hint.* Use half-angle.

Use $u^2 = -1$ and test conjugation on a vector perpendicular to $u$; its rotation angle is twice the angle in the quaternion coefficients. □

*Solution.* $q = \cos(\theta/2) + u\sin(\theta/2)$. □

**Exercise 13.20.** Why is $\rho$ surjective?

*Hint.* Every 3D rotation has axis-angle form.

Choose an axis and angle for the desired element of $SO(3)$, then substitute them into the unit-quaternion half-angle formula. □

*Solution.* Choose the half-angle quaternion for that axis and angle. □

**Exercise 13.21.** How many points are in each fiber of $\rho$?

*Hint.* Use the kernel.

Once one preimage $q$ is known, its entire fiber is $q \ker \rho$; count the elements of that coset. □

*Solution.* Two. □

**Exercise 13.22.** What is the deck group?

*Hint.* Use multiplication by kernel elements.

Multiply by the central kernel elements and check the projection equation; a two-point fiber and connected-cover uniqueness bound the deck group. □

*Solution.* $\mathbb{Z}/2$. □

**Exercise 13.23.** Identify $SO(3)$ as a quotient of $S^3$.

*Hint.* Identify antipodes.

Verify that two unit quaternions have the same image precisely when they differ by a kernel element, then apply the quotient description. □

*Solution.* $S^3/\{\pm 1\}$. □

**Exercise 13.24.** Identify that quotient as a familiar space.

*Hint.* Use real projective space.

Each line through zero in $\mathbb{R}^4$ meets $S^3$ in an antipodal pair; this identifies the equivalence classes in the quotient. □

*Solution.* $\mathbb{R}P^3$. □

**Exercise 13.25.** What is $\pi_1(SO(3))$?

*Hint.* The cover is universal.

Use that $S^3$ is simply connected before identifying the fundamental group downstairs with the deck group of this universal cover. □

*Solution.* $\mathbb{Z}/2$. □

**Exercise 13.26.** What does a $2\pi$-rotation loop lift to?

*Hint.* Start at 1.

Insert $\theta(t) = 2\pi t$ into the half-angle formula and evaluate the lifted path at both endpoints. □

*Solution.* A path ending at $-1$. □

**Exercise 13.27.** Is the $2\pi$-rotation loop null-homotopic?

*Hint.* Its lift is not closed.

A null-homotopy would lift to a based null-homotopy upstairs; in a universal cover its lifted boundary loop must close. □

*Solution.* No. □

**Exercise 13.28.** What does a $4\pi$-rotation loop lift to?

*Hint.* Traverse the nontrivial loop twice.

For the twice-traversed rotation, insert $\theta(t) = 4\pi t$ and compare the endpoint quaternion with its initial value. ☐

*Solution.* A loop from 1 back to 1. ☐

**Exercise 13.29.** What is the homotopy class of the $4\pi$-rotation loop?

*Hint.* Use the order-two fundamental group.

The closed lifted path contracts because the total space $S^3$ is simply connected. Closedness alone in an arbitrary cover would not prove null-homotopy downstairs. ☐

*Solution.* It is trivial. ☐

**Exercise 13.30.** What representation gives the matrix version of $\rho$?

*Hint.* Use conjugation on the Lie algebra.

Conjugate elements of the real three-dimensional Lie algebra $\mathfrak{su}(2)$; an invariant inner product identifies the induced transformations as orthogonal. ☐

*Solution.* The adjoint representation $\mathrm{Ad} : SU(2) \to SO(3)$. ☐

**Exercise 13.31.** Write the short exact sequence.

*Hint.* Kernel, source, quotient.

Check injectivity of the kernel inclusion, equality of its image with the kernel of the next map, and surjectivity onto $SO(3)$. ☐

*Solution.* $1 \to \{\pm I\} \to SU(2) \to SO(3) \to 1$. ☐

**Exercise 13.32.** Why is the kernel central?

*Hint.* $\pm I$ commute with every matrix.

Scalar multiplication commutes with every $2 \times 2$ matrix; verify separately that these scalar matrices satisfy the defining conditions of $SU(2)$. ☐

*Solution.* Both scalar matrices lie in the center of $SU(2)$. ☐

**Exercise 13.33.** What is the bridge to VIII/14?

*Hint.* Compare universal covers.

Compare two universal-cover models: the simply connected three-sphere here and a tree for a connected graph, with their respective deck groups. ☐

*Solution.* Graphs have simply connected universal covers that are trees, with free groups acting as deck groups. ☐

## 13.17 Solved problem dossiers

**Problem 13.34** (Three-sphere parametrization)**.** Verify the usual two-complex-parameter form of $SU(2)$.

*Solution.* Unitarity forces the second column to be a unit orthogonal vector to the first, hence $(-\overline{\beta}, \overline{\alpha})^T$ up to phase; determinant one fixes the phase. The norm condition is $|\alpha|^2 + |\beta|^2 = 1$. ☐

**Problem 13.35** (Quaternion norm preservation)**.** Why does $v \mapsto qvq^{-1}$ preserve Euclidean norm on imaginary quaternions?

*Solution.* Quaternion norm is multiplicative, so $|qvq^{-1}| = |q||v||q|^{-1} = |v|$. Conjugation also preserves the imaginary subspace. □

**Problem 13.36** (Orientation)**.** Why does quaternion conjugation land in $SO(3)$, not merely $O(3)$?

*Solution.* The map depends continuously on $q \in S^3$, and at $q = 1$ it has determinant 1. Since determinant takes values $\pm 1$ and $S^3$ is connected, the determinant is always 1. □

**Problem 13.37** (Kernel computation)**.** Prove the kernel is exactly $\{\pm 1\}$.

*Solution.* An element in the kernel commutes with $i, j, k$, hence with all quaternions. The center of $\mathbb{H}$ is $\mathbb{R}$; a real unit quaternion is $\pm 1$. □

**Problem 13.38** (Half-angle check)**.** Why does a $2\pi$ rotation correspond to the quaternion $-1$?

*Solution.* Set $\theta = 2\pi$: $q = \cos \pi + u \sin \pi = -1$. □

**Problem 13.39** (Double cover)**.** Why does each rotation have exactly two quaternion lifts?

*Solution.* Surjectivity gives at least one lift $q$. The fiber is the coset $q \ker \rho = \{q, -q\}$. □

**Problem 13.40** (Projective identification)**.** Derive $SO(3) \cong \mathbb{R}P^3$.

*Solution.* Use $SU(2) \cong S^3$ and quotient by the central kernel $\{\pm 1\}$, which is exactly the antipodal equivalence relation defining $\mathbb{R}P^3$. □

**Problem 13.41** (Universal cover)**.** Why is $SU(2) \to SO(3)$ universal?

*Solution.* $SU(2) \cong S^3$ and $S^3$ is simply connected. A covering with simply connected total space is universal. □

**Problem 13.42** (Fundamental group)**.** Recover $\pi_1(SO(3))$ from deck transformations.

*Solution.* For a universal cover, the deck group is isomorphic to the base fundamental group. The deck group here is $\{\pm I\} \cong \mathbb{Z}/2$. □

**Problem 13.43** (Two-pi loop)**.** Construct a loop in $SO(3)$ representing the nontrivial element.

*Solution.* Fix an axis $u$ and project $q(t) = \cos(\pi t) + u \sin(\pi t)$. The endpoints $1, -1$ project to the same rotation, so the projection is a loop whose lift is not closed. □

**Problem 13.44** (Four-pi loop)**.** Explain why a $4\pi$ rotation path is null-homotopic.

*Solution.* It represents twice the generator of $\mathbb{Z}/2$, hence the identity. Equivalently its lift to $S^3$ is closed, and $S^3$ is simply connected. □

**Problem 13.45** (Fiber endpoint test)**.** How can one decide which element of $\pi_1(SO(3))$ a loop represents?

*Solution.* Lift it from $I \in SU(2)$. If the lift ends at $I$, the class is trivial; if it ends at $-I$, it is the nontrivial class. □

**Problem 13.46** (Axis reversal)**.** Compare axis-angle data $(u, \theta)$ and $(-u, -\theta)$.

*Solution.* They give the same quaternion $\cos(\theta/2) + u\sin(\theta/2)$, hence the same rotation. ☐

**Problem 13.47** (Angle $2\pi - \theta$)**.** Explain why distinct quaternion signs can encode the same rotation.

*Solution.* Changing $q$ to $-q$ alters the quaternion representative but not conjugation. Axis-angle parametrizations reflect this double-valuedness at the group level. ☐

**Problem 13.48** (Central quotient)**.** Why is $SU(2)/\{\pm I\}$ naturally a group?

*Solution.* $\{\pm I\}$ is a normal central subgroup, so the quotient group is defined. The homomorphism theorem identifies it with $SO(3)$. ☐

**Problem 13.49** (Deck action)**.** Describe the nontrivial deck transformation explicitly.

*Solution.* It is $q \mapsto -q$, or matrix multiplication $A \mapsto -A$. It preserves every $SO(3)$ image. ☐

**Problem 13.50** (Quaternion example)**.** What rotation is represented by $q = \cos(\pi/4) + k\sin(\pi/4)$?

*Solution.* The half-angle is $\pi/4$, so the rotation angle is $\pi/2$ about the $k$-axis, i.e. the $z$-axis. ☐

**Problem 13.51** (Quaternion $i$)**.** What rotation does $q = i$ represent?

*Solution.* Write $i = \cos(\pi/2) + i\sin(\pi/2)$; it represents rotation by $\pi$ about the $x$-axis. ☐

**Problem 13.52** (Adjoint viewpoint)**.** Why does the adjoint action have three real dimensions?

*Solution.* The Lie algebra $\mathfrak{su}(2)$ is three-dimensional over $\mathbb{R}$, and its invariant inner product identifies the adjoint action with orthogonal rotations of $\mathbb{R}^3$. ☐

**Problem 13.53** (Bridge to graph covers)**.** State the common pattern between $SU(2) \to SO(3)$ and a graph universal cover.

*Solution.* A simply connected total space carries a free deck action by the base fundamental group, and the quotient by that action recovers the base. ☐

## 13.18 Supplementary worked examples

**Example 13.54** (Unit quaternions as $S^3$)**.** $SU(2)$ is diffeomorphic to the unit quaternions $S^3$. A unit quaternion $q$ acts on the imaginary quaternions $\operatorname{Im}\mathbb{H} \cong \mathbb{R}^3$ by $v \mapsto qvq^{-1}$, giving a rotation.

**Example 13.55** (Kernel $\{\pm 1\}$)**.** The conjugation action of $q$ and $-q$ on imaginary quaternions is identical. These are exactly the elements in the kernel, so the map $SU(2) \to SO(3)$ is two-to-one.

**Example 13.56** (Double cover and fundamental group)**.** Because $SU(2) \cong S^3$ is simply connected and covers $SO(3)$ with deck group $\mathbb{Z}/2$, it is the universal cover. Consequently $\pi_1(SO(3)) \cong \mathbb{Z}/2$.

## 13.19 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 13.57** (Dimension)**.** What is the real dimension of $SU(2)$?

*Hint.* Identify it with unit quaternions. □

*Solution.* It is 3, since the unit quaternions form $S^3 \subset \mathbb{R}^4$. □

**Exercise 13.58** (Kernel elements)**.** Which two elements of $SU(2)$ map to the identity rotation in $SO(3)$?

*Hint.* Central unit quaternions act trivially by conjugation. □

*Solution.* $q = 1$ and $q = -1$. □

**Exercise 13.59** (Fiber size)**.** How many points are in each fiber of $SU(2) \to SO(3)$?

*Hint.* It is a double cover. □

*Solution.* Exactly two. □

**Exercise 13.60** (Deck involution)**.** Write the nontrivial deck transformation of the double cover.

*Hint.* Swap $q$ with the other point in its fiber. □

*Solution.* $q \mapsto -q$. □

**Exercise 13.61** (Fundamental group)**.** Compute $\pi_1(SO(3))$ using the universal cover.

*Hint.* The deck group of a universal cover identifies with the fundamental group. □

*Solution.* $\pi_1(SO(3)) \cong \mathbb{Z}/2$. □

### Proofs

**Exercise 13.62** (Conjugation preserves norm)**.** Prove $v \mapsto qvq^{-1}$ preserves the Euclidean norm on imaginary quaternions.

*Hint.* Quaternion norm is multiplicative. □

*Solution.* $|qvq^{-1}| = |q||v||q|^{-1} = |v|$ for unit $q$; conjugation also preserves the imaginary subspace, hence gives an orthogonal transformation. □

**Exercise 13.63** (Orientation)**.** Explain why the induced orthogonal transformation lies in $SO(3)$ rather than merely $O(3)$.

*Hint.* Use continuity and the value at $q = 1$. □

*Solution.* The determinant is a continuous map from connected $SU(2)$ to $\{\pm 1\}$ and equals $+1$ at the identity, so it is always $+1$. □

**Exercise 13.64** (Kernel)**.** Prove the kernel of the conjugation homomorphism is $\{\pm 1\}$.

*Hint.* An element commuting with every imaginary quaternion is central in $\mathbb{H}$. □

*Solution.* Such a unit quaternion must be real, hence equals $\pm 1$. □

**Exercise 13.65** (Universal cover)**.** Prove the double cover is universal.

*Hint.* Use $SU(2) \cong S^3$. □

*Solution.* $S^3$ is simply connected, so the connected covering $SU(2) \to SO(3)$ has simply connected total space and is universal. □

### Counterexamples and hypothesis tests

**Exercise 13.66** ($SU(2) = SO(3)$)**.** Test: $SU(2)$ and $SO(3)$ are isomorphic as Lie groups.

*Hint.* The map has a two-element kernel. □

*Solution.* False. $SU(2)$ is a double cover of $SO(3)$, not isomorphic to it. □

**Exercise 13.67** (Rotation has unique lift)**.** Test: every rotation has a unique representing unit quaternion.

*Hint.* Remember the kernel. □

*Solution.* False. Each rotation has exactly two lifts $q$ and $-q$. □

**Exercise 13.68** ($SO(3)$ simply connected)**.** Test: $SO(3)$ is simply connected because it is connected.

*Hint.* Connectedness is much weaker. □

*Solution.* False. Its fundamental group is $\mathbb{Z}/2$. □

## Applications and investigations

**Exercise 13.69** (Quaternion rotation). Why are unit quaternions numerically attractive for representing rotations?

*Hint.* They lift $SO(3)$ to a sphere without Euler-angle singularities. ☐

*Solution.* A unit quaternion provides a smooth normalized four-coordinate representation; the only redundancy is the harmless sign pair $q \sim -q$, avoiding gimbal-lock coordinate singularities. ☐

**Exercise 13.70** (Spinorial sign). Interpret the fact that a $2\pi$ rotation represents the nontrivial loop in $SO(3)$ while a $4\pi$ rotation becomes null-homotopic.

*Hint.* Lift the loop to $SU(2)$. ☐

*Solution.* A $2\pi$ rotation lifts from 1 to $-1$ and therefore does not close; traversing twice reaches 1 again, reflecting the order-two fundamental group. ☐

## Challenge problems

**Exercise 13.71** (Loop lifting). Describe the lift of a generator of $\pi_1(SO(3))$ to $SU(2)$.

*Hint.* A generator should switch the two points over the identity rotation. ☐

*Solution.* Starting at 1, its lift ends at $-1$. Squaring the loop lifts to a closed path, consistent with order two. ☐

**Exercise 13.72** (Cover synthesis). Connect the algebraic kernel, deck group, and fundamental group in the $SU(2) \to SO(3)$ example.

*Hint.* All three involve the same $\{\pm 1\}$ symmetry. ☐

*Solution.* The kernel is $\{\pm 1\}$; multiplication by $-1$ is the nontrivial deck transformation; since the cover is universal, this deck group is canonically isomorphic to $\pi_1(SO(3)) \cong \mathbb{Z}/2$. ☐

# Chapter 14

# Free Groups and Covering Graphs

Graphs are the combinatorial laboratory of covering-space theory. Their fundamental groups are free, their universal covers are trees, and connected covering graphs correspond to subgroups of free groups. This chapter develops the algebra of reduced words together with the geometry of graph covers.

## 14.1 Free groups

**Definition 14.1** (Free group)**.** Let $S$ be a set. The free group $F(S)$ is the group of reduced finite words in symbols

$$s^{\pm 1}, \qquad s \in S,$$

with multiplication given by concatenation followed by cancellation of adjacent inverse pairs.

If $S = \{x_1, \dots, x_r\}$, write

$$F_r = F(x_1, \dots, x_r).$$

## 14.2 Reduced words

A word is reduced if it contains no adjacent subword

$$xx^{-1} \quad \text{or} \quad x^{-1}x.$$

**Proposition 14.2** (Unique reduced form)**.** *Every element of a free group has a unique reduced-word representative.*

This normal form is the basis for concrete computations.

## 14.3 Universal property

**Theorem 14.3** (Universal property of a free group)**.** *Given a set map*

$$f : S \to G$$

*into a group $G$, there exists a unique homomorphism*

$$\overline{f} : F(S) \to G$$

*extending $f$.*

Thus a homomorphism out of a free group is completely determined by the images of its free generators.

## 14.4 Graphs as one-dimensional CW complexes

A connected graph $X$ is a one-dimensional CW complex. Choose a base vertex $v_0$. Edge paths determine loops and homotopies can be simplified combinatorially by canceling immediate backtracking.

## 14.5 Spanning trees

**Definition 14.4** (Spanning tree)**.** A spanning tree $T \subset X$ of a connected graph is a connected subgraph containing every vertex and having no cycles.

A finite connected graph always has a spanning tree.

## 14.6 Collapsing a spanning tree

**Theorem 14.5** (Graph collapse)**.** *If $T$ is a spanning tree of a connected graph $X$, then the quotient*

$$X/T$$

*is a wedge of circles, one for each edge of $X \setminus T$, and*

$$X \simeq X/T.$$

Hence graph fundamental groups are free.

## 14.7 Rank formula

For a finite connected graph with $V$ vertices and $E$ edges, a spanning tree has $V - 1$ edges. Therefore the number of edges outside the tree is

$$E - (V - 1) = E - V + 1.$$

**Theorem 14.6** (Fundamental group of a finite connected graph)**.**

$$\pi_1(X) \cong F_{E-V+1}.$$

## 14.8 Wedges of circles

For the bouquet

$$\bigvee_{j=1}^{r} S^1,$$

one obtains

$$\pi_1 \left( \bigvee_{j=1}^{r} S^1 \right) \cong F_r.$$

Each circle contributes one free generator.

## 14.9 Trees

**Definition 14.7** (Tree)**.** A tree is a connected graph containing no nontrivial cycles.

**Theorem 14.8** (Trees are simply connected)**.** *Every tree is contractible and therefore simply connected.*

A finite tree collapses edge by edge to a vertex. Infinite trees admit the same path-uniqueness intuition and are simply connected as graphs.

## 14.10 Universal covers of graphs

**Theorem 14.9** (Universal covering graph)**.** *The universal cover of a connected graph is a tree.*

Conversely, a tree with a free properly discontinuous graph automorphism action gives a graph quotient whose universal cover is that tree.

## 14.11 Cayley graphs

For a group $G$ generated by a set $S$, the Cayley graph has vertex set $G$ and directed edges

$$g \longrightarrow gs$$

for $s \in S$.

**Example 14.10** (Cayley tree of a free group)**.** The Cayley graph of $F_r$ with respect to its free generators is a regular tree of valence $2r$.

## 14.12 Deck action of a free group

The universal cover of the wedge of $r$ circles is the Cayley tree of $F_r$. The group acts by left multiplication on vertices and edges.

**Theorem 14.11** (Free-group deck action)**.**

$$F_r \cong \mathrm{Deck}(\widetilde{X} \to X)$$

*for the universal cover of the $r$-circle wedge.*

## 14.13 Covering graphs

A graph map

$$p : \widetilde{X} \to X$$

is a covering graph when the incident edge pattern around every vertex maps bijectively to the incident edge pattern around its image. This is the combinatorial form of an evenly covered neighborhood.

## 14.14 Subgroups and connected graph covers

For a connected based graph $X$, pointed connected covers correspond to subgroups

$$H \leq \pi_1(X).$$

Since $\pi_1(X)$ is free, graph coverings give a geometric realization of subgroups of free groups.

## 14.15 Schreier graphs

Let $F_r$ act on the right cosets of a subgroup $H$. The associated Schreier graph has vertices

$$Hg, \qquad g \in F_r,$$

and edges labeled by free generators.

This graph realizes the covering corresponding to $H$.

## 14.16 Nielsen–Schreier theorem

**Theorem 14.12** (Nielsen–Schreier)**.** *Every subgroup of a free group is free.*

A covering-space proof is immediate in spirit: a subgroup determines a connected covering graph, and the fundamental group of that graph is free.

## 14.17 Finite-index rank formula

**Theorem 14.13** (Schreier index formula)**.** *If*

$$H \leq F_r$$

*has finite index $d$, then*

$$\operatorname{rank}(H) = 1 + d(r-1).$$

This can be seen by counting vertices and edges in the finite covering graph.

## 14.18 Regular graph covers

A covering graph corresponding to

$$H \leq F_r$$

is regular exactly when $H$ is normal. Then the deck group is

$$F_r/H.$$

## 14.19 Core graphs

For finitely generated subgroups of free groups, one can prune hanging trees from a covering graph and retain a finite core containing every reduced loop. This gives a compact combinatorial model of the subgroup.

## 14.20 Edge paths and reduced words

In a rose with labeled oriented edges

$$x_1, \dots, x_r,$$

a based edge loop reads a word in the symbols $x_i^{\pm 1}$. Removing immediate backtracking corresponds exactly to free reduction.

Thus graph homotopy and free-group reduction are two versions of the same operation.

## 14.21 Bridge to simplicial complexes

Graphs are one-dimensional simplicial complexes after subdividing loops and multiple edges if necessary. VIII/15 generalizes the vertex-edge language to higher-dimensional simplices and prepares the chain complexes used in homology.

## 14.22 Exercises

**Exercise 14.14.** Define $F(S)$.

*Hint.* Use reduced words.

Use the empty word as identity and define multiplication by concatenation followed by complete free reduction; inverses reverse order and invert every letter. ☐

*Solution.* The group of reduced finite words in $s^{\pm 1}$ for $s \in S$. ☐

**Exercise 14.15.** What is a reduced word?

*Hint.* No adjacent inverse pair.

After cancelling an inverse pair, check the newly adjacent letters again. A reduction may expose further cancellations that were not adjacent initially. ☐

*Solution.* A word containing neither $xx^{-1}$ nor $x^{-1}x$ adjacently. ☐

**Exercise 14.16.** What is $F_r$?

*Hint.* Choose $r$ free generators.

Choose an alphabet of $r$ generators and their formal inverses; do not impose commutativity when constructing its reduced words. ☐

*Solution.* The free group on $r$ generators. ☐

**Exercise 14.17.** State the universal property of $F(S)$.

*Hint.* Extend any set map to a homomorphism.

The image of every inverse letter is forced by the image of its generator. Evaluate reduced words in $G$ to obtain existence and uniqueness. ☐

*Solution.* Every map $S \to G$ extends uniquely to $F(S) \to G$. ☐

**Exercise 14.18.** Define a spanning tree.

*Hint.* All vertices, connected, no cycles.

A spanning tree must contain every vertex, but not every edge; check connectedness and the absence of cycles independently. ☐

*Solution.* A connected cycle-free subgraph containing every vertex. ☐

**Exercise 14.19.** How many edges does a finite spanning tree on $V$ vertices have?

*Hint.* Use the tree formula.

Remove a leaf and its incident edge and use induction on the number of vertices; identify the single-vertex starting case. ☐

*Solution.* $V - 1$. ☐

**Exercise 14.20.** What is $X/T$ for a spanning tree $T$?

*Hint.* Collapse the tree.

All tree vertices become one point after collapse. Each edge not in the tree then has both endpoints at that point and supplies a circle. ☐

*Solution.* A wedge of circles indexed by edges outside $T$. ☐

**Exercise 14.21.** Give the rank of $\pi_1(X)$ for a finite connected graph.

*Hint.* Count non-tree edges.

A spanning tree uses $V-1$ of the $E$ edges; count the remaining generators rather than counting all edges as independent loops. ☐

*Solution.* $E-V+1$. ☐

**Exercise 14.22.** What is $\pi_1(\bigvee^r S^1)$?

*Hint.* Each circle gives one generator.

Read a based loop as an ordered word in oriented circles, cancelling immediate backtracking; for $r>1$, order cannot be discarded as in $\mathbb{Z}^r$. ☐

*Solution.* $F_r$. ☐

**Exercise 14.23.** Define a tree.

*Hint.* Connected graph with no cycles.

Use the equivalent unique-simple-path characterization: two distinct simple paths between the same vertices would form a cycle. ☐

*Solution.* A connected cycle-free graph. ☐

**Exercise 14.24.** What is $\pi_1$ of a tree?

*Hint.* Trees are contractible.

The image of a loop lies in a finite subtree of the CW tree; contract that finite subtree to show the loop class vanishes. ☐

*Solution.* Trivial. ☐

**Exercise 14.25.** What is the universal cover of a connected graph?

*Hint.* Use simple connectivity.

A universal cover is a simply connected graph. A reduced cycle would survive as a nontrivial fundamental-group element, so no such cycle can occur. ☐

*Solution.* A tree. ☐

**Exercise 14.26.** What is the Cayley graph vertex set of $G$?

*Hint.* One vertex per group element.

For a generator $s$, draw an oriented edge from the vertex $g$ to $gs$; this records both the vertex set and its group structure. ☐

*Solution.* $G$. ☐

**Exercise 14.27.** What valence has the standard Cayley graph of $F_r$?

*Hint.* Each generator and inverse gives an edge direction.

Count outgoing edge germs for generators and inverse generators; keep the two directions distinct even when discussing one unoriented edge. ☐

*Solution.* $2r$. ☐

**Exercise 14.28.** What group acts by deck transformations on the universal cover of the $r$-rose?

*Hint.* Use the fundamental group.

Use that the rose has fundamental group $F_r$, then apply the deck-group identification for its simply connected covering tree. ☐

*Solution.* $F_r$. ☐

**Exercise 14.29.** What local condition defines a covering graph?

*Hint.* Incident edges must match bijectively.

Count incident edge germs, not just distinct adjacent vertices; a loop edge contributes two germs at the same vertex. ☐

*Solution.* At every vertex, the star maps bijectively to the star of its image. ☐

**Exercise 14.30.** What algebraic objects classify pointed connected graph covers?

*Hint.* Use covering classification.

Fix a basepoint upstairs to obtain an actual subgroup of $\pi_1(X)$; forgetting that choice retains only its conjugacy class. ☐

*Solution.* Subgroups of the graph fundamental group. ☐

**Exercise 14.31.** What is a Schreier graph?

*Hint.* Use cosets of a subgroup.

From the right coset $Hg$, draw the generator-labeled edge to $Hgs$. Check that changing the representative of $Hg$ does not change this endpoint. ☐

*Solution.* A labeled graph whose vertices are cosets $Hg$ and whose edges encode multiplication by free generators. ☐

**Exercise 14.32.** State Nielsen–Schreier.

*Hint.* Subgroups preserve freeness.

Realize the subgroup by a connected covering graph and compute that graph's fundamental group by a spanning tree, allowing infinite rank. ☐

*Solution.* Every subgroup of a free group is free. ☐

**Exercise 14.33.** State Schreier's finite-index rank formula.

*Hint.* Use index $d$.

A $d$-sheeted cover of the $r$-rose has $d$ vertices and $dr$ edges; substitute these counts into the connected-graph rank formula. ☐

*Solution.* $\operatorname{rank}(H) = 1 + d(r-1)$. ☐

**Exercise 14.34.** When is a connected graph cover regular?

*Hint.* Use subgroup normality.

Apply the normal-subgroup criterion for connected covers with compatible basepoints; transitivity of the deck action is the geometric counterpart. □

*Solution.* When the corresponding subgroup is normal. □

**Exercise 14.35.** What is its deck group in the normal case?

*Hint.* Take quotient.

For a normal subgroup the normalizer is all of $F_r$; substitute this into the general normalizer-quotient formula. □

*Solution.* $F_r/H$. □

**Exercise 14.36.** What does free reduction correspond to geometrically?

*Hint.* Cancel immediate edge backtracking.

Match a letter and its inverse with traversing one edge and immediately retracing it; repeat until newly adjacent backtracks have also disappeared. □

*Solution.* Removing an edge immediately followed by the same edge in reverse. □

**Exercise 14.37.** What is the bridge to VIII/15?

*Hint.* Graphs are 1-dimensional simplicial objects.

Subdivide loops and multiple edges before treating a graph as an abstract simplicial complex, then replace oriented edges by higher-dimensional oriented simplices. □

*Solution.* VIII/15 generalizes vertices and edges to higher-dimensional simplices. □

## 14.23 Solved problem dossiers

**Problem 14.38** (Reduce a word)**.** Reduce $aba^{-1}ab^{-1}b$ in the free group $F(a,b)$.

*Solution.* The terminal $b^{-1}b$ cancels, leaving $aba^{-1}a$. No further adjacent inverse pair occurs. □

**Problem 14.39** (Free universal property)**.** Define a homomorphism $F(a,b) \to \mathbb{Z}$ by $a \mapsto 2$, $b \mapsto -1$. What is the image of $ab^{-1}a$?

*Solution.* By the universal property, extend additively: $2 - (-1) + 2 = 5$. □

**Problem 14.40** (Triangle graph rank)**.** Compute the fundamental-group rank of a triangle graph.

*Solution.* It has $V = 3$, $E = 3$, so the rank is $3 - 3 + 1 = 1$, hence $\pi_1 \cong \mathbb{Z}$. □

**Problem 14.41** (Square with diagonal)**.** Compute the rank for a square with one diagonal.

*Solution.* Here $V = 4$, $E = 5$, so rank $= 5 - 4 + 1 = 2$. □

**Problem 14.42** (Complete graph $K_4$)**.** Compute the rank of $\pi_1(K_4)$.

*Solution.* $K_4$ has $V = 4$, $E = 6$, so the rank is 3, hence $\pi_1(K_4) \cong F_3$. □

**Problem 14.43** (Tree criterion)**.** Show a finite connected graph is a tree iff $E = V - 1$.

*Solution.* A spanning tree already has $V-1$ edges. If the connected graph has no extra edges it is the tree; any extra edge creates a cycle. Conversely every tree has $V-1$ edges. $\square$

**Problem 14.44** (Rose cover)**.** Describe the universal cover of $S^1 \vee S^1$.

*Solution.* It is the 4-regular Cayley tree of $F_2$, with edges labeled by the two generators and their inverse orientations. $\square$

**Problem 14.45** (Deck action)**.** How does $F_2$ act on that tree?

*Solution.* By left multiplication on vertex labels $g \mapsto hg$, carrying each labeled edge to the corresponding labeled edge. $\square$

**Problem 14.46** (Index-two subgroup rank)**.** If $H \leq F_3$ has index 2, find its rank.

*Solution.* Schreier gives $1 + 2(3-1) = 5$. $\square$

**Problem 14.47** (Index-five subgroup of $F_2$)**.** Find the rank.

*Solution.* $1 + 5(2-1) = 6$. $\square$

**Problem 14.48** (Subgroup of $\mathbb{Z}$)**.** Interpret $n\mathbb{Z} \leq F_1 \cong \mathbb{Z}$ as a graph cover.

*Solution.* It corresponds to the $n$-sheeted circle cover $S^1 \to S^1$, with covering graph an $n$-cycle. $\square$

**Problem 14.49** (Nielsen–Schreier proof idea)**.** Explain the covering-space proof that every subgroup $H \leq F_r$ is free.

*Solution.* Realize $F_r$ as $\pi_1$ of the $r$-rose. The subgroup gives a connected covering graph. Fundamental groups of graphs are free, and that group identifies with $H$. $\square$

**Problem 14.50** (Normal subgroup cover)**.** What extra symmetry appears when $H \trianglelefteq F_r$?

*Solution.* The associated cover is regular, and deck transformations act transitively on fibers with group $F_r/H$. $\square$

**Problem 14.51** (Non-normal subgroup)**.** What does the deck group become for a general subgroup?

*Solution.* Under the covering classification, it is $N_{F_r}(H)/H$, which can be smaller than the full quotient of $F_r$. $\square$

**Problem 14.52** (Schreier graph vertices)**.** If $H \leq F_r$ has index $d$, how many vertices does its Schreier graph have?

*Solution.* Exactly $d$, one for each right coset. $\square$

**Problem 14.53** (Schreier edge count)**.** For a degree-$d$ cover of the $r$-rose, how many unoriented edges are there?

*Solution.* Each of the $r$ base loops lifts with one outgoing edge from each of $d$ vertices, giving $rd$ unoriented labeled edges. $\square$

**Problem 14.54** (Derive Schreier rank)**.** Use the previous count to recover the index formula.

*Solution.* The connected covering graph has $V = d$, $E = rd$, hence rank $E - V + 1 = rd - d + 1 = 1 + d(r-1)$. $\square$

**Problem 14.55** (Core graph)**.** Why can hanging trees be removed without changing the subgroup represented by a covering graph?

*Solution.* Hanging trees contain no reduced closed loops. Collapsing or pruning them does not alter the fundamental group, leaving the loop-carrying core. ☐

**Problem 14.56** (Backtracking cancellation)**.** Why does an edge followed immediately by its reverse represent a null segment in a graph path?

*Solution.* That out-and-back segment contracts through the edge, so deleting it is an endpoint-fixing path homotopy. ☐

**Problem 14.57** (Bridge to simplices)**.** How does a graph become a simplicial complex?

*Solution.* Use vertices as 0-simplices and non-loop edges as 1-simplices; subdivide loops or multiple-edge configurations as needed to fit the abstract simplicial-complex rules. ☐

## 14.24 Supplementary worked examples

**Example 14.58** (Rose graph)**.** The wedge $R_r = \bigvee_{i=1}^{r} S^1$ has one vertex and $r$ oriented loop edges. Reduced edge words give its loop classes, yielding $\pi_1(R_r) \cong F_r$.

**Example 14.59** (Universal covering tree)**.** The universal cover of a connected graph is a tree. For the rose $R_r$, vertices can be labeled by reduced words in $F_r$, and an edge appends one generator before free reduction.

**Example 14.60** (Subgroups as covering graphs)**.** Under covering-space classification, a subgroup $H \leq F_r$ determines a connected covering graph of the rose. Cosets describe the fiber, and loops based at a chosen lift encode exactly the subgroup $H$.

## 14.25 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 14.61** (Rank-one rose)**.** What is $\pi_1(R_1)$?

*Hint.* $R_1 = S^1$. □

*Solution.* $\pi_1(R_1) \cong \mathbb{Z} = F_1$. □

**Exercise 14.62** (Rank-two rose)**.** Name the group $\pi_1(R_2)$.

*Hint.* There are two loop generators and no 2-cell relations. □

*Solution.* It is the free group $F_2 = \langle a, b\rangle$. □

**Exercise 14.63** (Tree group)**.** Compute the fundamental group of a tree.

*Hint.* A tree contracts to a point. □

*Solution.* It is trivial. □

**Exercise 14.64** (Index and sheets)**.** If $H \leq F_r$ has finite index $d$, how many sheets does the associated connected cover have?

*Hint.* Fibers correspond to cosets. □

*Solution.* It is a $d$-sheeted covering. □

**Exercise 14.65** (Reduced word)**.** Reduce the word $abb^{-1}a^{-1}c$ in a free group.

*Hint.* Cancel adjacent inverse pairs. □

*Solution.* It reduces to $c$. □

### Proofs

**Exercise 14.66** (Graph universal cover is tree)**.** Prove a simply connected connected graph has no cycles and hence is a tree.

*Hint.* A cycle would define a nontrivial loop after removing backtracking. ☐

*Solution.* Any embedded cycle contributes a nontrivial free-group element; simply connectedness forbids cycles, so a connected simply connected graph is a tree. ☐

**Exercise 14.67** (Rose free group)**.** Prove $\pi_1(R_r) \cong F_r$.

*Hint.* Use van Kampen or reduced edge loops. ☐

*Solution.* Each oriented loop edge gives a generator; endpoint homotopy cancels only immediate backtracking, producing precisely reduced words with no further relations. ☐

**Exercise 14.68** (Subgroup from cover)**.** Show $p_*\pi_1(E, e_0)$ is a subgroup of $\pi_1(B, b_0)$ for a based cover.

*Hint.* $p_*$ is a group homomorphism and injective for coverings. ☐

*Solution.* The image of a homomorphism is a subgroup; covering path lifting also shows $p_*$ is injective by lifting a null-homotopy. ☐

**Exercise 14.69** (Coset fiber)**.** Explain why fiber points correspond to cosets of the covering subgroup.

*Hint.* Transport the base lift along paths representing group elements. ☐

*Solution.* Two loop classes send the base lift to the same endpoint exactly when their quotient lies in the subgroup of loops that lift closed; thus endpoints are indexed by the appropriate cosets. ☐

### Counterexamples and hypothesis tests

**Exercise 14.70** (All graph groups abelian)**.** Test: fundamental groups of graphs are always abelian.

*Hint.* $F_2$ is not abelian. ☐

*Solution.* False. Connected graphs have free fundamental groups, usually nonabelian. ☐

**Exercise 14.71** (Universal cover finite)**.** Test: the universal cover of a finite graph is always finite.

*Hint.* Consider a circle. ☐

*Solution.* False. The universal cover of $S^1$ is an infinite line; nontrivial free groups generally give infinite trees. ☐

**Exercise 14.72** (Subgroups of free groups need not free)**.** Test: a subgroup of a free group can have torsion and need not be free.

*Hint.* Use covering graphs. ☐

*Solution.* False. Nielsen–Schreier says every subgroup of a free group is free; covering graphs provide a topological proof. ☐

## Applications and investigations

**Exercise 14.73** (Automata viewpoint)**.** How can a finite covering graph encode a finite-index subgroup of a free group?

*Hint.* Edges labeled by generators define transitions between sheets/cosets. □

*Solution.* Choose a base vertex; reading a word follows labeled edges, and the accepted words returning to the base vertex are precisely the subgroup. The finite cover acts like a deterministic finite-state machine for coset action. □

**Exercise 14.74** (Spanning tree computation)**.** Explain how a spanning tree gives a basis for the fundamental group of a finite connected graph.

*Hint.* Tree edges create no cycles. □

*Solution.* Collapse the spanning tree to a point; each remaining edge becomes one loop, so the graph is homotopy equivalent to a wedge of $E-V+1$ circles and has free rank $E-V+1$. □

## Challenge problems

**Exercise 14.75** (Nielsen–Schreier rank)**.** If $H$ has index $d$ in $F_r$, compute the rank predicted by a $d$-sheeted covering of the rose.

*Hint.* Count vertices and edges of the covering graph. □

*Solution.* The cover has $V = d$ and $E = dr$, so its free rank is $E - V + 1 = d(r-1) + 1$. □

**Exercise 14.76** (Covering graph synthesis)**.** Explain how lifting, deck actions, and subgroup classification meet in covering graphs.

*Hint.* Loops act on the fiber, closed lifts form a subgroup, symmetries normalize that subgroup. □

*Solution.* Path lifting gives the monodromy action of $F_r$ on fiber vertices; the stabilizer of the base vertex is the subgroup defining the cover; global graph automorphisms over the base form the deck group, governed by the subgroup normalizer. □

# Part IV

# Homology

# Chapter 15

# Simplicial Complexes

Simplicial complexes replace a topological space by vertices, edges, triangles, tetrahedra, and higher-dimensional simplices glued face to face. They are the first combinatorial models used systematically in homology. This chapter develops abstract and geometric simplicial complexes, realizations, links and stars, subdivisions, orientations, and simplicial maps, while leaving the full chain-complex formalism to VIII/16.

## 15.1 Geometric simplices

Let

$$v_0, \dots, v_k \in \mathbb{R}^N$$

be affinely independent.

**Definition 15.1** (Geometric simplex)**.** The $k$-simplex spanned by these vertices is

$$[v_0, \dots, v_k] = \left\{ \sum_{i=0}^{k} t_i v_i : t_i \geq 0,\ \sum_{i=0}^{k} t_i = 1 \right\}.$$

The coefficients $t_i$ are barycentric coordinates.

## 15.2 Faces

A face of

$$[v_0, \dots, v_k]$$

is obtained by retaining a subset of its vertices.

The $i$th codimension-one face is

$$[v_0, \dots, \widehat{v}_i, \dots, v_k].$$

**Definition 15.2** (Dimension)**.** A simplex with $k+1$ vertices has dimension $k$.

Thus vertices are 0-simplices, edges are 1-simplices, triangles are 2-simplices, and tetrahedra are 3-simplices.

## 15.3 Geometric simplicial complexes

**Definition 15.3** (Geometric simplicial complex)**.** A geometric simplicial complex $K$ is a collection of geometric simplices such that:

1. every face of a simplex in $K$ also lies in $K$;
2. the intersection of two simplices of $K$ is either empty or a common face of both.

The second condition rules out uncontrolled crossings.

## 15.4 Abstract simplicial complexes

**Definition 15.4** (Abstract simplicial complex)**.** An abstract simplicial complex $K$ on a vertex set $V$ is a collection of finite nonempty subsets of $V$ such that

$$\sigma \in K, \quad \varnothing \neq \tau \subset \sigma \quad \Longrightarrow \quad \tau \in K.$$

Members $\sigma \in K$ are simplices, and

$$\dim \sigma = |\sigma| - 1.$$

## 15.5 Dimension of a complex

**Definition 15.5** (Complex dimension)**.**

$$\dim K = \sup_{\sigma \in K} \dim \sigma.$$

The dimension may be finite or infinite.

## 15.6 Subcomplexes

**Definition 15.6** (Subcomplex)**.** A subcollection $L \subset K$ is a subcomplex if it is itself a simplicial complex: whenever $\sigma \in L$, every nonempty face of $\sigma$ also lies in $L$.

## 15.7 Geometric realization

An abstract complex becomes a topological space by assigning basis vectors to vertices and realizing each abstract simplex as their convex hull.

**Definition 15.7** (Geometric realization)**.** The geometric realization of $K$, denoted

$$|K|,$$

is the union of its realized simplices with the natural weak topology.

Different geometric embeddings of the same abstract complex yield homeomorphic realizations.

## 15.8 Basic examples

**Example 15.8** (Simplex and boundary)**.** The full simplex on $n+1$ vertices realizes as $D^n$, while its proper-face subcomplex realizes as

$$S^{n-1}.$$

**Example 15.9** (Cycle)**.** A polygonal cycle with $m \geq 3$ vertices realizes as $S^1$.

**Example 15.10** (Triangulated disk)**.** Two triangles sharing one edge can realize a quadrilateral disk.

## 15.9 Triangulations

**Definition 15.11** (Triangulation)**.** A triangulation of a topological space $X$ is a simplicial complex $K$ together with a homeomorphism

$$|K| \cong X.$$

Not every arbitrary decomposition is a triangulation; the simplicial intersection conditions must hold.

## 15.10 Stars

**Definition 15.12** (Open star)**.** For a vertex $v \in K$, the open star $\mathrm{st}^\circ(v)$ is the union of interiors of all simplices containing $v$.

**Definition 15.13** (Closed star)**.** The closed star $\mathrm{st}(v)$ is the subcomplex generated by all simplices containing $v$.

The closed star is contractible: it is a cone with vertex $v$.

## 15.11 Links

**Definition 15.14** (Link)**.** The link of a vertex $v$ is

$$\mathrm{lk}(v) = \{\sigma \in K : v \notin \sigma,\ \sigma \cup \{v\} \in K\}.$$

Informally, the link records the directions surrounding $v$.

**Proposition 15.15** (Star as a cone)**.**

$$\mathrm{st}(v) = v * \mathrm{lk}(v),$$

*the simplicial cone on the link.*

## 15.12 Manifold links

In a triangulated $n$-manifold, an interior vertex should have link homeomorphic to

$$S^{n-1},$$

while a boundary vertex has a link resembling

$$D^{n-1}.$$

This local criterion motivates later combinatorial manifold theory.

## 15.13 Barycentric subdivision

**Definition 15.16** (Barycentric subdivision)**.** The barycentric subdivision $\mathrm{sd}\, K$ has one vertex for each nonempty simplex of $K$. A collection

$$\sigma_0, \ldots, \sigma_r$$

forms a simplex in $\mathrm{sd}\, K$ exactly when, after reordering,

$$\sigma_0 \subsetneq \sigma_1 \subsetneq \cdots \subsetneq \sigma_r.$$

**Theorem 15.17** (Subdivision preserves realization)**.**

$$|\operatorname{sd} K| \cong |K|.$$

Subdivision refines the combinatorics without changing the underlying topological space.

## 15.14 Orientation of simplices

An orientation of a $k$-simplex is an ordering

$$[v_0, \dots, v_k]$$

of its vertices, modulo even permutations.

Reversing orientation changes the sign:

$$[v_{\pi(0)}, \dots, v_{\pi(k)}] = \operatorname{sgn}(\pi)[v_0, \dots, v_k].$$

This sign convention prepares the simplicial boundary operator in VIII/16.

## 15.15 Boundary preview

For an oriented simplex, the formal alternating boundary is

$$\partial[v_0, \dots, v_k] = \sum_{i=0}^{k} (-1)^i [v_0, \dots, \widehat{v}_i, \dots, v_k].$$

The central identity

$$\partial^2 = 0$$

will become the defining algebraic condition for chain complexes in VIII/16.

## 15.16 Simplicial maps

**Definition 15.18** (Simplicial map)**.** Let $K, L$ be abstract simplicial complexes. A map on vertices

$$f : V(K) \to V(L)$$

is simplicial if for every simplex

$$\sigma = \{v_0, \dots, v_k\} \in K,$$

the image set

$$f(\sigma) = \{f(v_0), \dots, f(v_k)\}$$

is a simplex of $L$.

Distinct source vertices are allowed to have the same image.

## 15.17 Realization of a simplicial map

A simplicial map extends linearly over each simplex:

$$|f| \left( \sum_i t_i v_i \right) = \sum_i t_i f(v_i).$$

**Theorem 15.19** (Continuity of realization)**.** *A simplicial map induces a continuous map*

$$|f| : |K| \to |L|.$$

## 15.18 Composition

**Proposition 15.20** (Functoriality)**.** *The composite of simplicial maps is simplicial, and*

$$|g \circ f| = |g| \circ |f|.$$

## 15.19 Simplicial isomorphisms

**Definition 15.21** (Simplicial isomorphism)**.** A simplicial map is a simplicial isomorphism if it is bijective on vertices and its inverse vertex map is also simplicial.

A simplicial isomorphism induces a homeomorphism of realizations.

## 15.20 Why subdivision helps maps

A continuous map between polyhedra need not be simplicial with respect to a given triangulation. After sufficiently fine subdivisions, it can often be approximated by a simplicial map. This is the content of the simplicial approximation theorem, developed later as needed.

## 15.21 Bridge to chain complexes

Once each oriented $k$-simplex is treated as a generator of an abelian group, the alternating face formula becomes a homomorphism

$$\partial_k : C_k \to C_{k-1}.$$

The cancellation

$$\partial_{k-1}\partial_k = 0$$

creates the chain-complex structure of VIII/16.

## 15.22 Exercises

**Exercise 15.22.** Define a geometric $k$-simplex.

*Hint.* Use convex combinations of $k + 1$ affinely independent vertices.

Require nonnegative barycentric coordinates summing to one; affine independence ensures the vertices span dimension $k$ and coordinates are unique. □

*Solution.* $[v_0, \ldots, v_k] = \{\sum t_i v_i : t_i \geq 0, \sum t_i = 1\}$. □

**Exercise 15.23.** How many vertices does a $k$-simplex have?

*Hint.* Dimension is one less than vertex count.

Check the cases of a vertex, edge, and triangle before expressing the general relation between affine dimension and vertex count. □

*Solution.* $k + 1$. □

**Exercise 15.24.** What is a face of a simplex?

*Hint.* Retain a subset of vertices.

Set the coordinates of omitted vertices to zero; use the chapter's convention that only nonempty faces are included. □

*Solution.* The simplex spanned by a nonempty subset of the original vertices. □

**Exercise 15.25.** State the intersection condition in a geometric simplicial complex.

*Hint.* Two simplices meet cleanly.

An arbitrary overlap, such as crossing edges without a shared vertex, fails the condition; identify the common face in each allowed intersection. □

*Solution.* Their intersection is empty or a common face of both. □

**Exercise 15.26.** Define an abstract simplicial complex.

*Hint.* Use finite subsets closed under nonempty subsets.

Test downward closure: including a triangle forces its edges and vertices to be present, while including its edges does not force the triangle. □

*Solution.* A family $K$ of finite nonempty vertex sets such that every nonempty subset of a simplex is again a simplex. □

**Exercise 15.27.** What is $\dim \sigma$ for an abstract simplex?

*Hint.* Count vertices.

Count elements of the vertex set, not its proper subsets; a singleton must have dimension zero. □

*Solution.* $|\sigma| - 1$. □

**Exercise 15.28.** Define a subcomplex.

*Hint.* Close under faces.

For each selected simplex, verify that every nonempty subset of its vertices is also selected and belongs to the original complex. □

*Solution.* A subcollection that contains every face of each of its simplices. □

**Exercise 15.29.** What is $|K|$?

*Hint.* Pass from combinatorics to topology.

Glue standard simplices along the common faces prescribed by the abstract complex and specify the resulting realization topology. □

*Solution.* The geometric realization of the abstract simplicial complex. □

**Exercise 15.30.** What space does a full $n$-simplex realize?

*Hint.* It is convex.

Convexity alone gives contractibility, not a ball homeomorphism; use the standard simplex's radial parametrization from an interior point. □

*Solution.* $D^n$. □

**Exercise 15.31.** What does its boundary subcomplex realize?

*Hint.* Use the simplex boundary.

For $n \geq 1$, project radially from the barycenter to identify the simplex boundary with a sphere of dimension one less. □

*Solution.* $S^{n-1}$. □

**Exercise 15.32.** Define a triangulation of $X$.

*Hint.* Use a simplicial complex plus homeomorphism.

A homotopy equivalence is not enough: a triangulation includes a homeomorphism from the realization onto the given space. □

*Solution.* A complex $K$ with a homeomorphism $|K| \cong X$. □

**Exercise 15.33.** Define the closed star of a vertex.

*Hint.* Collect simplices containing the vertex and all their faces.

Include faces opposite the chosen vertex as well as simplices containing it; otherwise the proposed collection need not be a subcomplex. □

*Solution.* The subcomplex generated by all simplices containing the vertex. □

**Exercise 15.34.** Why is a closed star contractible?

*Hint.* It is a cone.

Join every point of the closed star to $v$ within a containing simplex; barycentric interpolation gives a contraction that agrees on shared faces. □

*Solution.* $\mathrm{st}(v) = v * \mathrm{lk}(v)$. □

**Exercise 15.35.** Define the link of $v$.

*Hint.* Take faces opposite $v$.

Impose both conditions: the simplex excludes $v$, but adjoining $v$ produces a simplex of $K$. □

*Solution.* $\mathrm{lk}(v) = \{\sigma : v \notin \sigma, \sigma \cup \{v\} \in K\}$. □

**Exercise 15.36.** What is the expected link of an interior vertex in an $n$-manifold triangulation?

*Hint.* One dimension lower.

In a combinatorial or PL manifold triangulation, examine a small vertex neighborhood as the cone on its link; do not silently extend this sphere-link criterion to all topological triangulations. □

*Solution.* $S^{n-1}$. □

**Exercise 15.37.** What are vertices of $\mathrm{sd}\, K$?

*Hint.* Use simplices of $K$.

Each nonempty original simplex contributes one new vertex at its barycenter, including original vertices viewed as zero-dimensional simplices. □

*Solution.* Nonempty simplices of $K$, represented by their barycenters. □

**Exercise 15.38.** When do vertices form a simplex in $\mathrm{sd}\, K$?

*Hint.* Require a strict inclusion chain.

Check comparability under inclusion, not merely nonempty intersection; a simplex in the subdivision encodes a strict flag of original faces. □

*Solution.* When the corresponding simplices of $K$ form a chain under inclusion. □

**Exercise 15.39.** What happens to the realization under barycentric subdivision?

*Hint.* Topology is unchanged.

Map a new vertex to the barycenter of its original simplex and extend affinely on flags; the pieces triangulate each original simplex without changing it. ☐

*Solution.* $|\operatorname{sd} K| \cong |K|$. ☐

**Exercise 15.40.** Define an orientation of a simplex.

*Hint.* Order vertices modulo even permutations.

Compare the parity of two vertex orderings; one transposition reverses orientation while an even permutation preserves it. ☐

*Solution.* An equivalence class of vertex orderings under even permutations. ☐

**Exercise 15.41.** Write the alternating boundary of $[v_0, \ldots, v_k]$.

*Hint.* Delete one vertex at a time.

Keep the surviving vertices in their inherited order and assign the sign from the position of the deleted vertex, starting at index zero. ☐

*Solution.* $\partial[v_0, \ldots, v_k] = \sum_i (-1)^i [v_0, \ldots, \widehat{v}_i, \ldots, v_k]$. ☐

**Exercise 15.42.** Define a simplicial map.

*Hint.* Images of vertex sets must remain simplices.

The map is specified on vertices, but its defining condition must hold for every simplex, not only for its edges. ☐

*Solution.* A vertex map $f : V(K) \to V(L)$ sending every simplex of $K$ to a simplex of $L$. ☐

**Exercise 15.43.** May a simplicial map identify two vertices?

*Hint.* Injectivity is not required.

Test the collapse of one edge to a vertex; the image has lower dimension but remains a simplex, so simpliciality need not imply injectivity. ☐

*Solution.* Yes, provided the resulting image set is a simplex. ☐

**Exercise 15.44.** How is $|f|$ defined on barycentric coordinates?

*Hint.* Extend linearly.

When several vertices have the same image, add their barycentric coefficients; nonnegativity and the total sum of one are preserved. ☐

*Solution.* $|f|(\sum t_i v_i) = \sum t_i f(v_i)$. ☐

**Exercise 15.45.** What is the bridge to VIII/16?

*Hint.* Turn oriented simplices into generators.

Form a free abelian group in each dimension, impose the orientation sign relation, and check cancellation of each codimension-two face in the double boundary. ☐

*Solution.* The alternating face maps become boundary homomorphisms in a chain complex. ☐

## 15.23 Solved problem dossiers

**Problem 15.46** (Triangle faces)**.** List all nonempty faces of the abstract simplex $\{0, 1, 2\}$.

*Solution.* The three vertices, the three edges $\{0, 1\}, \{0, 2\}, \{1, 2\}$, and the triangle $\{0, 1, 2\}$. ☐

**Problem 15.47** (Tetrahedron count)**.** Count the 0-, 1-, 2-, and 3-faces of a tetrahedron.

*Solution.* There are 4 vertices, $\binom{4}{2} = 6$ edges, $\binom{4}{3} = 4$ triangular faces, and one tetrahedron. ☐

**Problem 15.48** (Boundary of a triangle)**.** Identify the realization of the proper-face subcomplex of a 2-simplex.

*Solution.* It is the three-edge cycle, homeomorphic to $S^1$. ☐

**Problem 15.49** (Boundary of a tetrahedron)**.** What space is realized by the boundary complex of a 3-simplex?

*Solution.* The four triangular faces glue to a 2-sphere $S^2$. ☐

**Problem 15.50** (Invalid geometric complex)**.** Why do two triangles crossing through their interiors fail to form a geometric simplicial complex?

*Solution.* Their intersection is not empty and is not a common face of the two triangles, violating the intersection axiom. ☐

**Problem 15.51** (Cycle complex)**.** Build an abstract simplicial complex realizing a square circle.

*Solution.* Use vertices $1, 2, 3, 4$, all singleton vertices, and edges $\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}$. Its realization is $S^1$. ☐

**Problem 15.52** (Filled versus unfilled triangle)**.** Compare the complexes with and without the 2-simplex $\{0, 1, 2\}$.

*Solution.* With the 2-simplex the realization is a disk; with only its proper faces the realization is a circle. ☐

**Problem 15.53** (Star and link in a triangle)**.** For vertex 0 in a filled triangle, identify its closed star and link.

*Solution.* The star is the entire triangle and all faces incident/generated by it. The link consists of vertices $1, 2$ and edge $\{1, 2\}$, hence realizes as an interval. ☐

**Problem 15.54** (Link in a cycle)**.** What is the link of a vertex in a triangulated circle?

*Solution.* It consists of the two neighboring vertices as two isolated 0-simplices, realizing $S^0$. ☐

**Problem 15.55** (Link in a triangulated surface)**.** What local link should an interior vertex have?

*Solution.* A circle $S^1$, matching the $n = 2$ manifold link rule. ☐

**Problem 15.56** (First barycentric subdivision of an edge)**.** Describe sd of a single edge.

*Solution.* The original two vertices and the edge barycenter become subdivision vertices; the edge is replaced by two edges meeting at the barycenter. ☐

**Problem 15.57** (Subdivision of a triangle)**.** How many small triangles appear in the first barycentric subdivision of one 2-simplex?

*Solution.* There are $3! = 6$ maximal chains vertex $\subset$ edge $\subset$ triangle, hence six small triangles. □

**Problem 15.58** (Orientation reversal)**.** Compare $[v_0, v_1, v_2]$ and $[v_1, v_0, v_2]$.

*Solution.* The transposition is odd, so the orientations differ by a minus sign. □

**Problem 15.59** (Triangle boundary)**.** Compute the formal oriented boundary of $[v_0, v_1, v_2]$.

*Solution.* $\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$. □

**Problem 15.60** (Boundary-squared preview)**.** Check formally that $\partial^2[v_0, v_1, v_2] = 0$.

*Solution.* Apply $\partial[v_i, v_j] = [v_j] - [v_i]$. Every vertex occurs twice with opposite signs, so the sum is zero. □

**Problem 15.61** (Constant simplicial map)**.** When is a constant vertex map $K \to L$ simplicial?

*Solution.* If its common image vertex belongs to $L$, every simplex image is that singleton vertex, hence a simplex. Thus any constant map to a vertex is simplicial. □

**Problem 15.62** (Vertex identification)**.** Map a triangle's vertices $0, 1, 2$ to $a, a, b$. When is the map simplicial?

*Solution.* It is simplicial provided $\{a, b\}$ is a simplex of the target; the triangle image set is then the edge $\{a, b\}$. □

**Problem 15.63** (Composition)**.** Prove a composite of simplicial maps is simplicial.

*Solution.* If $\sigma \in K$, then $f(\sigma)$ is a simplex of $L$, and $g(f(\sigma))$ is a simplex of $M$. Hence $g \circ f$ is simplicial. □

**Problem 15.64** (Realization continuity)**.** Why is the realization of a simplicial map continuous?

*Solution.* On each simplex it is affine and hence continuous; the formulas agree on common faces, and the weak topology of the realization glues them into a global continuous map. □

**Problem 15.65** (Simplicial isomorphism)**.** Why does a simplicial isomorphism induce a homeomorphism?

*Solution.* Both the map and its inverse are simplicial, so their realizations are continuous and inverse to one another. □

**Problem 15.66** (Graph as a simplicial complex)**.** What restriction must be handled when viewing an arbitrary graph as an abstract simplicial complex?

*Solution.* Abstract simplicial complexes do not allow loop edges with repeated vertices or multiple distinct edges with the same endpoint set. Subdivision can remove these obstacles. □

**Problem 15.67** (Triangulated disk Euler preview)**.** For a square cut by one diagonal, count vertices, edges, and triangles.

*Solution.* There are 4 vertices, 5 edges, and 2 triangles. The alternating count is $4 - 5 + 2 = 1$, foreshadowing disk Euler characteristic. □

**Problem 15.68** (Triangulated sphere Euler preview)**.** For the tetrahedral sphere, compute $V - E + F$.

*Solution.* $4 - 6 + 4 = 2$, foreshadowing $\chi(S^2) = 2$. □

**Problem 15.69** (Bridge to chains)**.** Explain why orientations and alternating faces are exactly the data needed for simplicial homology.

*Solution.* Oriented $k$-simplices become generators of free abelian groups, and the signed face sum defines the boundary map. The identity $\partial^2 = 0$ makes these groups into the chain complex developed next. □

## 15.24 Supplementary worked examples

**Example 15.70** (Boundary of a triangle)**.** An oriented 2-simplex $[v_0v_1v_2]$ has boundary $[v_1v_2] - [v_0v_2] + [v_0v_1]$. The alternating signs ensure shared edges cancel when adjacent triangles are summed with compatible orientations.

**Example 15.71** (Triangulated circle)**.** A polygon with cyclically arranged edges is a simplicial model of $S^1$. Its 1-cycle is the sum of oriented edges; every vertex appears once with sign $+1$ and once with sign $-1$ in the boundary.

**Example 15.72** (Triangulated sphere)**.** The boundary of a tetrahedron is a simplicial complex homeomorphic to $S^2$. The signed sum of its four oriented triangular faces is a 2-cycle because every edge appears twice with opposite induced orientations.

## 15.25 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 15.73** (0-simplex faces)**.** How many proper faces does a 0-simplex have in the ordinary nonempty-face convention?

*Hint.* A vertex has no nonempty lower-dimensional face. ☐

*Solution.* None. ☐

**Exercise 15.74** (1-simplex boundary)**.** Compute $\partial[v_0v_1]$.

*Hint.* Delete one vertex at a time with alternating signs. ☐

*Solution.* $\partial[v_0v_1] = [v_1] - [v_0]$. ☐

**Exercise 15.75** (2-simplex boundary)**.** Compute $\partial[v_0v_1v_2]$.

*Hint.* Use the alternating face formula. ☐

*Solution.* $[v_1v_2] - [v_0v_2] + [v_0v_1]$. ☐

**Exercise 15.76** (Dimension)**.** What is the dimension of a simplex with $k + 1$ vertices?

*Hint.* A $k$-simplex has $k + 1$ vertices. ☐

*Solution.* Its dimension is $k$. ☐

**Exercise 15.77** (Euler count tetrahedron)**.** Compute $V - E + F$ for the boundary of a tetrahedron.

*Hint.* Count 4 vertices, 6 edges, 4 faces. ☐

*Solution.* $4 - 6 + 4 = 2$. ☐

### Proofs

**Exercise 15.78** (Boundary squared)**.** Prove $\partial^2 = 0$ on an oriented simplex.

*Hint.* Each codimension-two face is obtained by deleting two vertices in two orders. ☐

*Solution.* The two deletion orders contribute opposite signs, so every codimension-two face cancels pairwise. ☐

**Exercise 15.79** (Shared-edge cancellation)**.** Explain why internal edges cancel in the boundary of an oriented triangulated surface chain.

*Hint.* Adjacent faces induce opposite orientations on their common edge. ☐

*Solution.* Each internal edge appears in two face boundaries with opposite signs, leaving only boundary edges of the surface. ☐

**Exercise 15.80** (Subcomplex closure)**.** Prove a collection of simplices closed under faces forms a simplicial subcomplex.

*Hint.* Check exactly the defining face condition. ☐

*Solution.* Every face of every selected simplex remains selected, so the collection inherits all required incidence data and is a subcomplex. ☐

**Exercise 15.81** (Simplicial map determined by vertices)**.** Why is a simplicial map determined by its action on vertices?

*Hint.* A simplex is the set/span of its vertices. ☐

*Solution.* Once vertex images are fixed, each simplex must map to the simplex spanned by those images; affine extension on geometric realizations is then unique. ☐

### Counterexamples and hypothesis tests

**Exercise 15.82** (Any vertex map simplicial)**.** Test: every function between vertex sets defines a simplicial map.

*Hint.* Images of simplices must again span simplices. ☐

*Solution.* False; the simplex condition can fail. ☐

**Exercise 15.83** (Orientation changes simplex)**.** Test: reversing orientation creates a different underlying simplex.

*Hint.* Orientation is extra algebraic data. ☐

*Solution.* False. The geometric simplex is unchanged; its oriented chain generator changes sign. ☐

**Exercise 15.84** (Triangulation unique)**.** Test: a topological space has at most one triangulation.

*Hint.* Subdivision already gives alternatives. ☐

*Solution.* False. Spaces can admit many distinct triangulations. ☐

## Applications and investigations

**Exercise 15.85** (Mesh topology)**.** Why is a simplicial complex a natural data model for a triangle mesh?

*Hint.* It separates combinatorial adjacency from geometric coordinates. ☐

*Solution.* Vertices, edges, and faces encode incidence combinatorially; geometric realization assigns positions. Topological chains can therefore be computed from connectivity independent of embedding. ☐

**Exercise 15.86** (Boundary extraction)**.** How can the simplicial boundary operator identify boundary edges of an oriented triangle mesh?

*Hint.* Sum all oriented faces. ☐

*Solution.* Internal edges cancel in pairs, while edges incident to only one face remain in the total boundary chain, directly extracting the mesh boundary. ☐

## Challenge problems

**Exercise 15.87** (Sphere cycle)**.** Write the conceptual fundamental 2-cycle of the tetrahedral sphere.

*Hint.* Orient all four faces coherently. ☐

*Solution.* Take the signed sum of the four oriented triangular faces induced by an orientation of the tetrahedron boundary; shared edges cancel, so the chain has zero boundary. ☐

**Exercise 15.88** (Subdivision synthesis)**.** Why should subdividing a simplicial complex preserve the homology one intends to compute?

*Hint.* Subdivision changes combinatorics but not the underlying polyhedron. ☐

*Solution.* A subdivision has homeomorphic geometric realization; the induced chain maps are chain-homotopy equivalences, so homology remains unchanged while mesh resolution changes. ☐

# Chapter 16

# Chain Complexes

Homology begins with a simple algebraic pattern: a sequence of abelian groups linked by homomorphisms whose consecutive composites vanish. The groups encode objects in each dimension, while the homomorphisms encode boundaries. This chapter develops the algebra of chain complexes before any particular topological chain model is chosen.

## 16.1 Chain complexes

**Definition 16.1** (Chain complex)**.** A chain complex $C_\bullet$ of abelian groups is a sequence

$$\cdots \xrightarrow{\partial_{n+2}} C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots$$

such that

$$\partial_n \partial_{n+1} = 0$$

for every $n$.

The homomorphisms $\partial_n$ are boundary operators.

## 16.2 Cycles and boundaries

**Definition 16.2** (Cycles and boundaries)**.** Define

$$Z_n(C) = \ker \partial_n, \qquad B_n(C) = \operatorname{im} \partial_{n+1}.$$

Elements of $Z_n(C)$ are $n$-cycles and elements of $B_n(C)$ are $n$-boundaries.

The identity $\partial^2 = 0$ gives

$$B_n(C) \subset Z_n(C).$$

## 16.3 Homology of a chain complex

**Definition 16.3** (Homology group)**.** The $n$th homology group of $C_\bullet$ is

$$H_n(C) = Z_n(C)/B_n(C).$$

At this stage homology is purely algebraic. VIII/17 constructs chain complexes from spaces and simplicial complexes.

## 16.4 Concentrated complexes

A group $A$ may be viewed as a chain complex concentrated in degree $k$:

$$C_k = A, \qquad C_n = 0 \text{ for } n \neq k.$$

**Example 16.4.** For this complex,

$$H_k(C) \cong A$$

and every other homology group vanishes.

## 16.5 Two-term complexes

Consider

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow 0,$$

with $A$ in degree 1 and $B$ in degree 0.

**Proposition 16.5** (Two-term homology)**.**

$$H_1 \cong \ker f, \qquad H_0 \cong B/\operatorname{im} f.$$

Thus kernels and cokernels are already homology groups.

## 16.6 Exact chain complexes

**Definition 16.6** (Exactness)**.** A chain complex is exact at $C_n$ if

$$\operatorname{im} \partial_{n+1} = \ker \partial_n.$$

It is exact if this holds in every degree.

**Proposition 16.7** (Exact iff acyclic)**.** *A chain complex is exact if and only if*

$$H_n(C) = 0$$

*for all $n$.*

## 16.7 Chain maps

**Definition 16.8** (Chain map)**.** A chain map

$$f_\bullet : C_\bullet \to D_\bullet$$

is a family of homomorphisms

$$f_n : C_n \to D_n$$

such that

$$\partial_n^D f_n = f_{n-1} \partial_n^C$$

for every $n$.

Equivalently, every boundary square commutes.

## 16.8 Induced maps on homology

**Theorem 16.9** (Homology is functorial on chain maps). *A chain map sends cycles to cycles and boundaries to boundaries, and therefore induces*

$$H_n(f) : H_n(C) \to H_n(D)$$

*by*

$$[z] \longmapsto [f_n(z)].$$

## 16.9 Identity and composition

If $f : C \to D$ and $g : D \to E$ are chain maps, then $g \circ f$ is a chain map and

$$H_n(g \circ f) = H_n(g) \circ H_n(f).$$

Also

$$H_n(\mathrm{id}_C) = \mathrm{id}_{H_n(C)}\,.$$

## 16.10 Chain isomorphisms

**Definition 16.10** (Chain isomorphism). A chain map $f : C \to D$ is a chain isomorphism if every

$$f_n : C_n \to D_n$$

is an isomorphism.

A chain isomorphism induces isomorphisms on every homology group.

## 16.11 Subcomplexes

**Definition 16.11** (Chain subcomplex). A chain subcomplex $A_\bullet \subset C_\bullet$ consists of subgroups

$$A_n \subset C_n$$

with

$$\partial_C(A_n) \subset A_{n-1}.$$

The restricted maps form a chain complex.

## 16.12 Quotient complexes

If $A_\bullet \subset C_\bullet$ is a chain subcomplex, define

$$(C/A)_n = C_n/A_n.$$

**Proposition 16.12** (Quotient boundary). *The formula*

$$\partial[c] = [\partial c]$$

*is well defined and makes $C_\bullet/A_\bullet$ a chain complex.*

## 16.13 Short exact sequences of complexes

**Definition 16.13** (Short exact sequence of chain complexes)**.** A sequence

$$0 \to A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{p} C_\bullet \to 0$$

is short exact if

$$0 \to A_n \xrightarrow{i_n} B_n \xrightarrow{p_n} C_n \to 0$$

is exact for every $n$, and all maps are chain maps.

Such a sequence produces a long exact sequence in homology. The construction of the connecting homomorphism is developed in VIII/19.

## 16.14 Direct sums

Given chain complexes $C_\bullet^\alpha$, define

$$\left(\bigoplus_\alpha C^\alpha\right)_n = \bigoplus_\alpha C_n^\alpha$$

with boundary acting componentwise.

**Proposition 16.14** (Homology and direct sums)**.** *For ordinary direct sums of abelian-group chain complexes,*

$$H_n\left(\bigoplus_\alpha C^\alpha\right) \cong \bigoplus_\alpha H_n(C^\alpha).$$

## 16.15 Augmented complexes

An augmented chain complex continues one step beyond degree zero:

$$C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\epsilon} A \to 0,$$

with

$$\epsilon\partial_1 = 0.$$

For simplicial complexes, the augmentation

$$\epsilon : C_0 \to \mathbb{Z}$$

sends every vertex to 1. This leads to reduced homology in VIII/17.

## 16.16 Reduced chain complexes

The reduced degree-zero information is obtained by replacing the ordinary degree-zero homology with the kernel of the augmentation. For a connected space this removes the universal $\mathbb{Z}$ summand of $H_0$.

## 16.17 Graded viewpoint

A chain complex may be written compactly as a graded group

$$C_* = \bigoplus_n C_n$$

equipped with a degree-$-1$ map

$$\partial : C_* \to C_*$$

satisfying

$$\partial^2 = 0.$$

This notation becomes especially convenient for chain homotopies in VIII/22.

## 16.18 Matrices for finite free complexes

If every $C_n$ is finitely generated free abelian, choose bases and represent

$$\partial_n$$

by an integer matrix $D_n$. The chain condition becomes

$$D_{n-1}D_n = 0.$$

Homology can then be computed using kernels, images, and Smith normal form.

## 16.19 Smith normal form preview

For an integer matrix $D$, unimodular row and column operations can put $D$ into diagonal form

$$\mathrm{diag}(d_1, \dots, d_r, 0, \dots, 0), \qquad d_i \mid d_{i+1}.$$

This is the standard computational tool for finitely generated homology groups.

## 16.20 Euler characteristic preview

For a finite chain complex of finitely generated free abelian groups, let

$$c_n = \operatorname{rank} C_n.$$

The alternating chain rank

$$\sum_n (-1)^n c_n$$

will later agree with the alternating sum of homology ranks.

## 16.21 Bridge to topological chains

VIII/15 provided oriented simplices and the alternating face formula. In VIII/17 those simplices become generators of chain groups and the face formula becomes a boundary operator satisfying

$$\partial^2 = 0.$$

The present chapter supplies the algebra into which those constructions fit.

## 16.22 Exercises

**Exercise 16.15.** State the chain condition.

*Hint.* Compose consecutive boundaries.

Write the domains $C_{n+1} \to C_n \to C_{n-1}$; the composite must be the zero homomorphism, not merely have a nontrivial kernel. □

*Solution.* $\partial_n \partial_{n+1} = 0$. □

**Exercise 16.16.** Define $Z_n(C)$.

*Hint.* Take a kernel.

A cycle belongs to degree $n$ and has zero outgoing boundary; identify the relevant kernel inside $C_n$. □

*Solution.* $Z_n(C) = \ker \partial_n$. □

**Exercise 16.17.** Define $B_n(C)$.

*Hint.* Take the preceding image.

A degree-$n$ boundary comes from degree $n + 1$, so use the incoming differential rather than the outgoing one. □

*Solution.* $B_n(C) = \operatorname{im} \partial_{n+1}$. □

**Exercise 16.18.** Why is $B_n \subset Z_n$?

*Hint.* Use $\partial^2 = 0$.

Write an arbitrary boundary as $b = \partial_{n+1} c$, then apply $\partial_n$ and use the chain condition. □

*Solution.* If $b = \partial_{n+1} c$, then $\partial_n b = \partial_n \partial_{n+1} c = 0$. □

**Exercise 16.19.** Define $H_n(C)$.

*Hint.* Cycles modulo boundaries.

First justify that boundaries form a subgroup of cycles; two cycle representatives define the same class exactly when their difference is a boundary. □

*Solution.* $H_n(C) = Z_n(C)/B_n(C)$. □

**Exercise 16.20.** What is the homology of a complex concentrated in degree $k$ at $A$?

*Hint.* All boundaries vanish.

At the sole nonzero degree, the outgoing differential has full kernel and the incoming differential has zero image; inspect the other degrees separately. □

*Solution.* $H_k \cong A$, all other $H_n = 0$. □

**Exercise 16.21.** For $0 \to A \xrightarrow{f} B \to 0$, what is $H_1$?

*Hint.* Take the kernel of $f$.

Place $A$ in degree one and $B$ in degree zero; there is no incoming degree-two boundary to quotient out. □

*Solution.* $\ker f$. □

**Exercise 16.22.** For the same complex, what is $H_0$?

*Hint.* Take the cokernel.

The outgoing map from $B$ is zero, so all of $B$ consists of cycles; divide by the image arriving from $A$. □

*Solution.* $B/\operatorname{im} f$. □

**Exercise 16.23.** Define exactness at $C_n$.

*Hint.* Image equals kernel.

The chain condition gives only image contained in kernel; exactness requires the reverse containment as well. □

*Solution.* $\operatorname{im} \partial_{n+1} = \ker \partial_n$. □

**Exercise 16.24.** When is a chain complex acyclic?

*Hint.* All homology vanishes.

Apply the cycles-modulo-boundaries definition in every degree, distinguishing ordinary acyclicity from reduced acyclicity of a topological space. □

*Solution.* When $H_n(C) = 0$ for all $n$. □

**Exercise 16.25.** Define a chain map.

*Hint.* Boundary squares commute.

For each $n$, compare the two maps $C_n \to D_{n-1}$ obtained by applying the differential before or after $f$. □

*Solution.* $\partial^D f_n = f_{n-1}\partial^C$. □

**Exercise 16.26.** Why does a chain map send cycles to cycles?

*Hint.* Apply the commuting square.

Substitute a cycle $z$ into $\partial^D f_n = f_{n-1}\partial^C$; its zero boundary should remain zero after applying $f$. □

*Solution.* If $\partial z = 0$, then $\partial f(z) = f(\partial z) = 0$. □

**Exercise 16.27.** Why does a chain map send boundaries to boundaries?

*Hint.* Move the boundary through $f$.

Write the input as $\partial^C_{n+1} c$ and use the chain-map equation in degree $n+1$ to exhibit an explicit boundary preimage. □

*Solution.* $f(\partial c) = \partial f(c)$. □

**Exercise 16.28.** What map does a chain map induce?

*Hint.* Pass to quotient classes.

Define the image of a cycle class by $[z] \mapsto [f_n(z)]$, then check that changing $z$ by a boundary does not change the output class. □

*Solution.* $H_n(f) : H_n(C) \to H_n(D)$. □

**Exercise 16.29.** What is a chain isomorphism?

*Hint.* Degreewise isomorphism.

Invert the degreewise isomorphisms in the commuting differential square to verify that their inverses also form a chain map. □

*Solution.* A chain map whose every component $f_n$ is an isomorphism. □

**Exercise 16.30.** Define a chain subcomplex.

*Hint.* Boundary preserves the subgroups.

Containment $A_n \subset C_n$ alone is insufficient: apply the ambient differential and require its image to lie in $A_{n-1}$. □

*Solution.* Subgroups $A_n \subset C_n$ with $\partial(A_n) \subset A_{n-1}$. □

**Exercise 16.31.** How is the quotient boundary defined?

*Hint.* Apply boundary before taking the coset.

Replace a representative $c$ by $c + a$ with $a \in A_n$; boundary stability of the subcomplex makes both images give the same quotient class. □

*Solution.* $\partial[c] = [\partial c]$. □

**Exercise 16.32.** Define short exactness for chain complexes.

*Hint.* Require exactness degreewise.

Require chain maps as well as exactness of the underlying groups in each degree; these commuting differentials enable the homology construction. □

*Solution.* Every $0 \to A_n \to B_n \to C_n \to 0$ is exact. □

**Exercise 16.33.** What does a short exact sequence of complexes later produce?

*Hint.* Recall VIII/19.

Start with a cycle in the quotient complex, lift it to the middle complex, and take its boundary to locate the degree-lowering connecting map. □

*Solution.* A long exact sequence in homology. □

**Exercise 16.34.** How is the boundary on a direct sum defined?

*Hint.* Componentwise.

Elements of a direct sum have finite support; applying each differential separately preserves that finite-support condition. □

*Solution.* $\partial((c_\alpha)) = (\partial c_\alpha)$. □

**Exercise 16.35.** What is an augmentation?

*Hint.* Continue beyond $C_0$.

Treat the augmentation as an extra differential leaving degree zero and verify its composite with $\partial_1$ vanishes. □

*Solution.* A map $\epsilon : C_0 \to A$ satisfying $\epsilon\partial_1 = 0$. □

**Exercise 16.36.** What does the standard simplicial augmentation send a vertex to?

*Hint.* Every vertex contributes one component.

Evaluate augmentation on the edge boundary $[v_1] - [v_0]$; sending both vertices to one makes the result zero. □

*Solution.* $1 \in \mathbb{Z}$. □

**Exercise 16.37.** What matrix identity expresses the chain condition?

*Hint.* Choose bases.

Check matrix sizes against $C_n \to C_{n-1} \to C_{n-2}$, then multiply in that order using a consistent column-vector convention. □

*Solution.* $D_{n-1}D_n = 0$. □

**Exercise 16.38.** What is the bridge to VIII/17?

*Hint.* Use oriented simplices.

Apply the alternating face operator twice to one generator; pairing the two deletion orders for each pair of vertices supplies the chain condition. □

*Solution.* VIII/17 turns simplices into chain generators and the alternating face sum into the boundary operator. □

## 16.23 Solved problem dossiers

**Problem 16.39** (Zero differential)**.** Let $C_n = A_n$ with every boundary zero. Compute $H_n(C)$.

*Solution.* Every element is a cycle and no nonzero element is a boundary, so $H_n(C) \cong A_n$ for every $n$. □

**Problem 16.40** (Identity differential warning)**.** Can a nonzero chain complex have every differential equal to the identity?

*Solution.* Not in consecutive nonzero degrees, because the chain condition would give $\mathrm{id} \circ \mathrm{id} = 0$, impossible on a nonzero group. □

**Problem 16.41** (Multiplication-by-$m$ complex)**.** Compute the homology of $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to 0$ for $m \neq 0$.

*Solution.* The map is injective, so $H_1 = 0$. Its image is $m\mathbb{Z}$, so $H_0 \cong \mathbb{Z}/m$. □

**Problem 16.42** (Zero multiplication)**.** Repeat the previous computation for $m = 0$.

*Solution.* The boundary is zero, hence $H_1 \cong \mathbb{Z}$ and $H_0 \cong \mathbb{Z}$. □

**Problem 16.43** (Exactness test)**.** Show $0 \to \mathbb{Z} \xrightarrow{\mathrm{id}} \mathbb{Z} \to 0$ is exact.

*Solution.* The identity has zero kernel and full image. Thus the degree-one and degree-zero homology groups both vanish. □

**Problem 16.44** (Cycle-to-cycle proof)**.** Prove a chain map preserves cycles.

*Solution.* For $z \in Z_n(C)$, $\partial^D f_n(z) = f_{n-1}\partial^C(z) = 0$, so $f_n(z) \in Z_n(D)$. □

**Problem 16.45** (Boundary-to-boundary proof)**.** Prove a chain map preserves boundaries.

*Solution.* If $b = \partial^C_{n+1}c$, then $f_n(b) = f_n\partial^C c = \partial^D f_{n+1}(c)$, so it is a boundary. $\square$

**Problem 16.46** (Well-defined induced map)**.** Why does $[z] \mapsto [f(z)]$ not depend on the chosen cycle representative?

*Solution.* If $z - z' = \partial c$, then $f(z) - f(z') = f(\partial c) = \partial f(c)$, so the images differ by a boundary. $\square$

**Problem 16.47** (Composition on homology)**.** Verify $H_n(gf) = H_n(g)H_n(f)$.

*Solution.* On a class $[z]$, both sides equal $[g_n(f_n(z))]$. $\square$

**Problem 16.48** (Quotient differential)**.** Prove $\partial[c] = [\partial c]$ is well defined on $C/A$.

*Solution.* If $c - c' \in A_n$, then $\partial(c - c') \in A_{n-1}$, so $[\partial c] = [\partial c']$. $\square$

**Problem 16.49** (Quotient chain condition)**.** Verify the quotient differential squares to zero.

*Solution.* $\partial^2[c] = [\partial^2 c] = [0]$. $\square$

**Problem 16.50** (Direct sum homology)**.** Why do cycles and boundaries of a direct sum split componentwise?

*Solution.* The direct-sum differential acts componentwise, so kernels and images are direct sums of the component kernels and images, giving the homology direct-sum formula. $\square$

**Problem 16.51** (Augmentation of a connected graph)**.** Why does sending each vertex generator to 1 annihilate every edge boundary?

*Solution.* An oriented edge has boundary $v_1 - v_0$, and the augmentation sends this to $1 - 1 = 0$. $\square$

**Problem 16.52** (Matrix chain condition)**.** Let $D_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $D_1 = \begin{pmatrix} 1 & -1 \end{pmatrix}$. Verify the chain condition.

*Solution.* $D_1D_2 = 1 - 1 = 0$. $\square$

**Problem 16.53** (Matrix homology)**.** For the preceding complex $0 \to \mathbb{Z} \xrightarrow{D_2} \mathbb{Z}^2 \xrightarrow{D_1} \mathbb{Z} \to 0$, compute the homology.

*Solution.* $D_2$ is injective, so $H_2 = 0$. The kernel of $D_1$ is generated by $(1, 1)$, exactly the image of $D_2$, so $H_1 = 0$. $D_1$ is surjective, so $H_0 = 0$. $\square$

**Problem 16.54** (Nonexact matrix example)**.** Replace $D_2$ above by multiplication of the generator by $(2, 2)$. Compute $H_1$.

*Solution.* $\ker D_1 = \mathbb{Z}(1, 1)$ and $\operatorname{im} D_2 = 2\mathbb{Z}(1, 1)$, so $H_1 \cong \mathbb{Z}/2$. $\square$

**Problem 16.55** (Chain isomorphism)**.** Why must a chain isomorphism induce homology isomorphisms?

*Solution.* Its inverse is also a chain map, and functoriality gives mutually inverse induced maps on homology. $\square$

**Problem 16.56** (Exact iff zero homology)**.** Prove the equivalence.

*Solution.* Exactness says $Z_n = B_n$ in every degree, which is equivalent to the quotient $Z_n/B_n$ being zero. $\square$

**Problem 16.57** (Short exact quotient example)**.** Given a subcomplex $A \subset C$, write the canonical short exact sequence.

*Solution.* $0 \to A_\bullet \to C_\bullet \to (C/A)_\bullet \to 0$, degreewise the standard subgroup-quotient sequence. □

**Problem 16.58** (Smith-form preview)**.** What does the complex $0 \to \mathbb{Z} \xrightarrow{\times 12} \mathbb{Z} \to 0$ predict for torsion?

*Solution.* The single Smith invariant is 12, and the degree-zero homology is $\mathbb{Z}/12$. □

**Problem 16.59** (Euler rank check)**.** For a finite exact complex of finite-rank free groups, what alternating rank should one expect?

*Solution.* Zero. Exactness lets ranks telescope across images and kernels; later the Euler–Poincare formula formalizes this. □

**Problem 16.60** (Reduced degree zero)**.** Why is reduced homology useful for connected spaces?

*Solution.* Ordinary $H_0$ always contains the universal $\mathbb{Z}$ component recording connectedness. The augmentation removes that component, making reduced $H_0$ vanish for connected spaces. □

## 16.24 Supplementary worked examples

**Example 16.61** (A two-term complex)**.** For $C_1 \cong \mathbb{Z} \xrightarrow{m} C_0 \cong \mathbb{Z}$, with no other nonzero groups, $H_1 = \ker(m)$ and $H_0 = \mathbb{Z}/m$. Thus a single integer differential creates torsion as a cokernel.

**Example 16.62** (Circle chain complex)**.** The cellular chain complex of a circle with one 0-cell and one 1-cell is $0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$, yielding $H_1 \cong \mathbb{Z}$ and $H_0 \cong \mathbb{Z}$.

**Example 16.63** (Why $d^2 = 0$ matters)**.** The condition $d_n d_{n+1} = 0$ says every boundary is automatically a cycle, so $\operatorname{im} d_{n+1} \subseteq \ker d_n$ and the quotient defining homology is meaningful.

## 16.25 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 16.64** (Cycle group)**.** Define $Z_n(C)$ for a chain complex.

*Hint.* Cycles are killed by the outgoing differential. □

*Solution.* $Z_n(C) = \ker(d_n : C_n \to C_{n-1})$. □

**Exercise 16.65** (Boundary group)**.** Define $B_n(C)$.

*Hint.* Boundaries come from one degree higher. □

*Solution.* $B_n(C) = \operatorname{im}(d_{n+1} : C_{n+1} \to C_n)$. □

**Exercise 16.66** (Homology)**.** Define $H_n(C)$.

*Hint.* Quotient cycles by boundaries. □

*Solution.* $H_n(C) = Z_n(C)/B_n(C)$. □

**Exercise 16.67** (Zero differential)**.** If all differentials vanish, what is $H_n(C)$?

*Hint.* Then every chain is a cycle and only zero is a boundary. □

*Solution.* $H_n(C) \cong C_n$. □

**Exercise 16.68** (Multiplication complex)**.** For $0 \to \mathbb{Z} \xrightarrow{4} \mathbb{Z} \to 0$, compute the homology in the two nonzero degrees.

*Hint.* Multiplication by 4 is injective with cokernel $\mathbb{Z}/4$. □

*Solution.* The upper homology is 0 and the lower is $\mathbb{Z}/4$. □

## Proofs

**Exercise 16.69** (Boundaries are cycles)**.** Prove $B_n \subseteq Z_n$.

*Hint.* Use $d_n d_{n+1} = 0$. □

*Solution.* If $x = d_{n+1}y$, then $d_n x = d_n d_{n+1} y = 0$, so $x \in \ker d_n$. □

**Exercise 16.70** (Homology functor on chain maps)**.** Show a chain map $f : C \to D$ sends cycles to cycles and boundaries to boundaries.

*Hint.* Use $d_D f = f d_C$. □

*Solution.* For a cycle $z$, $d_D f(z) = f(d_C z) = 0$; for a boundary $d_C c$, $f(d_C c) = d_D f(c)$ is a boundary. Hence $f$ induces $H_n(f)$. □

**Exercise 16.71** (Composition)**.** Prove induced homology maps respect composition.

*Hint.* Apply the maps to representatives. □

*Solution.* $H(g \circ f)[z] = [(g \circ f)(z)] = H(g)(H(f)[z])$, so homology is functorial. □

**Exercise 16.72** (Isomorphism of complexes)**.** Prove a chain isomorphism induces isomorphisms on all homology groups.

*Hint.* Its inverse is also a chain map. □

*Solution.* Functoriality makes the induced maps inverse to each other on homology. □

## Counterexamples and hypothesis tests

**Exercise 16.73** (Any graded map is chain map)**.** Test: any family $f_n : C_n \to D_n$ defines a chain map.

*Hint.* It must commute with differentials. □

*Solution.* False; the chain condition $d_D f = f d_C$ is essential. □

**Exercise 16.74** (Cycles equal homology)**.** Test: $H_n(C) = Z_n(C)$ always.

*Hint.* Boundaries may be nonzero. □

*Solution.* False. Homology is the quotient $Z_n/B_n$. □

**Exercise 16.75** ($d^2 = 0$ optional)**.** Test: the equation $d^2 = 0$ is merely a convenient convention.

*Hint.* Without it, boundaries need not be cycles. □

*Solution.* False. It is necessary for the homology quotient to be defined in the usual way. □

## Applications and investigations

**Exercise 16.76** (Matrix computation)**.** How is homology computed from finite free chain complexes in practice?

*Hint.* Represent differentials by integer matrices. ☐

*Solution.* Compute kernels and images, often via Smith normal form; free ranks and invariant factors then give the homology groups. ☐

**Exercise 16.77** (Error checking)**.** Why is verifying $D_n D_{n+1} = 0$ a useful implementation test for boundary matrices?

*Hint.* It is the chain-complex identity. ☐

*Solution.* A nonzero product signals inconsistent orientations/incidence or assembly errors before any homology computation is trusted. ☐

## Challenge problems

**Exercise 16.78** (Smith form preview)**.** For a differential represented by diagonal Smith form $\mathrm{diag}(d_1, \ldots, d_r, 0, \ldots)$, what information becomes visible?

*Hint.* Nonzero invariant factors control cokernel torsion; zeros control kernel rank. ☐

*Solution.* The diagonal immediately separates rank, free directions, and torsion factors $\mathbb{Z}/d_i$, enabling explicit kernel/cokernel and hence homology calculations. ☐

**Exercise 16.79** (Exactness synthesis)**.** At one position $C_{n+1} \to C_n \to C_{n-1}$, relate exactness to homology.

*Hint.* Exactness means image equals kernel. ☐

*Solution.* The complex is exact at $C_n$ iff $B_n = Z_n$, which is equivalent to $H_n(C) = 0$. ☐

# Chapter 17

# Simplicial and Singular Homology

Chain complexes become topological invariants once their generators are chosen from a space. Simplicial homology uses oriented simplices in a triangulation. Singular homology replaces a triangulation by all continuous maps from standard simplices into the space. The two theories agree on triangulable spaces, but singular homology is defined for arbitrary topological spaces and is therefore the canonical general theory.

## 17.1 Simplicial chain groups

Let $K$ be a simplicial complex.

**Definition 17.1** (Simplicial chain group)**.** The group $C_n(K)$ is the free abelian group generated by oriented $n$-simplices of $K$, with the relation that reversing orientation changes sign.

## 17.2 Simplicial boundary

For an oriented simplex,

$$[v_0, \dots, v_n],$$

define

$$\partial_n[v_0, \dots, v_n] = \sum_{i=0}^{n} (-1)^i [v_0, \dots, \widehat{v}_i, \dots, v_n].$$

**Theorem 17.2** (Boundary squares to zero)**.**

$$\partial_{n-1}\partial_n = 0.$$

Every codimension-two face appears twice with opposite signs.

## 17.3 Simplicial homology

**Definition 17.3** (Simplicial homology)**.**

$$H_n(K) = \ker \partial_n / \operatorname{im} \partial_{n+1}.$$

If $X \cong |K|$, one often writes these groups as homology groups of the triangulated space.

## 17.4 Degree zero

For a simplicial complex,

$$H_0(K)$$

records path components.

**Theorem 17.4** (Zeroth homology)**.** *If $K$ has $c$ path components, then*

$$H_0(K) \cong \mathbb{Z}^c.$$

## 17.5 The circle

Triangulate $S^1$ by a polygon. There is one independent one-cycle going around the loop and no two-simplex filling it.

**Theorem 17.5** (Homology of the circle)**.**

$$H_0(S^1) \cong \mathbb{Z}, \qquad H_1(S^1) \cong \mathbb{Z}, \qquad H_n(S^1) = 0 \quad (n \geq 2).$$

## 17.6 The sphere

For $n \geq 1$,

$$H_k(S^n) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

The top-dimensional class records the fundamental cycle of the oriented sphere.

## 17.7 The disk

Because a disk is contractible,

$$H_0(D^n) \cong \mathbb{Z}, \qquad H_k(D^n) = 0 \quad (k > 0).$$

A direct simplicial calculation sees every positive-dimensional cycle as a boundary after filling the triangulation.

## 17.8 Reduced homology

**Definition 17.6** (Reduced homology)**.** The reduced homology groups $\widetilde{H}_n(X)$ are defined using the augmented chain complex. For nonempty $X$,

$$H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z}.$$

If $X$ is path connected,

$$\widetilde{H}_0(X) = 0.$$

For a point,

$$\widetilde{H}_n(*) = 0$$

for every $n$.

## 17.9 Singular simplices

Let

$$\Delta^n = \left\{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0,\ \sum t_i = 1\right\}.$$

**Definition 17.7** (Singular simplex)**.** A singular $n$-simplex in $X$ is a continuous map

$$\sigma : \Delta^n \to X.$$

## 17.10 Singular chain groups

**Definition 17.8** (Singular chains)**.** The singular chain group

$$C_n^{\text{sing}}(X)$$

is the free abelian group generated by all singular $n$-simplices in $X$.

Unlike simplicial chains, there are generally infinitely many generators even for very simple spaces.

## 17.11 Face maps and singular boundary

Let

$$\delta_i : \Delta^{n-1} \to \Delta^n$$

include the $i$th face. Define

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \sigma \circ \delta_i.$$

The same cancellation proves

$$\partial^2 = 0.$$

## 17.12 Singular homology

**Definition 17.9** (Singular homology)**.**

$$H_n^{\text{sing}}(X) = H_n(C_*^{\text{sing}}(X)).$$

This construction requires no triangulation.

## 17.13 Functoriality

A continuous map

$$f : X \to Y$$

acts on a singular simplex by composition:

$$\sigma \longmapsto f \circ \sigma.$$

**Theorem 17.10** (Induced chain map)**.** *The maps*

$$f_\# : C_n^{\text{sing}}(X) \to C_n^{\text{sing}}(Y)$$

*defined by*

$$f_\#(\sigma) = f \circ \sigma$$

*form a chain map.*

Hence every continuous map induces

$$f_* : H_n(X) \to H_n(Y).$$

## 17.14 Functoriality laws

$$(\mathrm{id}_X)_* = \mathrm{id}_{H_n(X)}, \qquad (g \circ f)_* = g_* \circ f_*.$$

Thus homology is a functor from spaces and continuous maps to graded abelian groups.

## 17.15 Simplicial maps and homology

A simplicial map

$$f : K \to L$$

induces a simplicial chain map by sending an oriented simplex to its oriented image when the dimension is preserved, and to zero when vertices collapse and the image has smaller dimension.

This agrees with the induced map on singular homology after geometric realization.

## 17.16 Simplicial versus singular homology

**Theorem 17.11** (Comparison theorem)**.** *For a simplicial complex $K$,*

$$H_n^{\mathrm{simp}}(K) \cong H_n^{\mathrm{sing}}(|K|).$$

The proof uses subdivision and approximation machinery. The theorem allows combinatorial calculations to compute the general topological invariant.

## 17.17 Homology of disjoint unions

For a disjoint union

$$X = \coprod_\alpha X_\alpha,$$

singular simplices lie in individual components, so

$$H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha).$$

## 17.18 Wedges in positive degree

For well-pointed spaces and positive degrees, reduced homology satisfies the wedge formula

$$\widetilde{H}_n(X \vee Y) \cong \widetilde{H}_n(X) \oplus \widetilde{H}_n(Y).$$

This predicts, for example,

$$H_1(S^1 \vee S^1) \cong \mathbb{Z}^2.$$

## 17.19 Homology and holes

Homology does not record all homotopy information. Instead it linearizes cycles:

$$\text{cycles that do not bound} \quad \rightsquigarrow \quad \text{homology classes.}$$

The groups are abelian even when the fundamental group is not.

## 17.20 First homology and abelianization

**Theorem 17.12** (First homology)**.** *For a path-connected space,*

$$H_1(X;\mathbb{Z}) \cong \pi_1(X)_{\mathrm{ab}},$$

*the abelianization of the fundamental group.*

This result provides a direct bridge between the covering/fundamental-group arc and homology.

## 17.21 Homotopy invariance preview

Homotopic maps induce the same maps in singular homology. The full chain-level proof uses prism operators and chain homotopies and is deferred to VIII/20, with chain-homotopy machinery developed further in VIII/22.

## 17.22 Bridge to cellular homology

Simplicial and singular chain groups can be very large. A CW complex offers a smaller chain complex with one generator per cell. VIII/18 derives that cellular chain complex from the skeletal filtration and computes its boundary maps from attaching maps.

## 17.23 Exercises

**Exercise 17.13.** Define $C_n(K)$.

*Hint.* Use oriented simplices as free generators.

Choose one orientation for each $n$-simplex as a basis; reversing that orientation represents its negative, not an additional independent generator. □

*Solution.* The free abelian group on oriented $n$-simplices modulo orientation reversal. □

**Exercise 17.14.** Write the simplicial boundary formula.

*Hint.* Delete one vertex with alternating signs.

Delete each vertex once, retain the order of the remaining vertices, and use its original position to determine the sign. □

*Solution.* $\partial[v_0,\ldots,v_n] = \sum_i(-1)^i[v_0,\ldots,\widehat{v}_i,\ldots,v_n]$. □

**Exercise 17.15.** What key identity makes simplicial chains a chain complex?

*Hint.* Apply boundary twice.

For two distinct deleted vertices, compare the two possible deletion orders; the resulting codimension-two face occurs with opposite signs. □

*Solution.* $\partial^2 = 0$. □

**Exercise 17.16.** Define simplicial $H_n(K)$.

*Hint.* Cycles modulo boundaries.

Use the outgoing differential for cycles and the incoming differential for boundaries; both are subgroups of the same degree-$n$ chain group. □

*Solution.* $\ker \partial_n / \operatorname{im} \partial_{n+1}$. ☐

**Exercise 17.17.** What does $H_0$ count?

*Hint.* Think components.

The boundary of a path identifies its endpoints in $H_0$. This detects path components, which need not equal connected components for an arbitrary space. ☐

*Solution.* Free abelian rank equal to the number of path components. ☐

**Exercise 17.18.** Compute $H_1(S^1)$.

*Hint.* The fundamental loop does not bound.

Triangulate the circle as a polygon; the consistently oriented sum of its edges is a cycle and there are no two-simplices in this model. ☐

*Solution.* $\mathbb{Z}$. ☐

**Exercise 17.19.** Compute $H_k(S^n)$.

*Hint.* Only degrees zero and $n$.

For $n \geq 1$, separate the connected degree-zero class from the top-dimensional fundamental class; treat $S^0$, which has two components, separately. ☐

*Solution.* $\mathbb{Z}$ for $k = 0, n$, otherwise 0. ☐

**Exercise 17.20.** Compute positive-dimensional homology of $D^n$.

*Hint.* The disk is contractible.

Use the contraction to a point and homotopy invariance; do not infer vanishing from the disk having no singular simplices in positive degrees. ☐

*Solution.* Zero. ☐

**Exercise 17.21.** Define reduced homology conceptually.

*Hint.* Use the augmented complex.

Augment degree-zero chains by total coefficient sum; reduced degree-zero homology retains the kernel rather than the single universal component class. ☐

*Solution.* It removes the universal degree-zero $\mathbb{Z}$ summand. ☐

**Exercise 17.22.** What is $\widetilde{H}_0(X)$ for path-connected nonempty $X$?

*Hint.* Only one component.

For a nonempty path-connected space, augmentation induces an isomorphism $H_0(X) \to \mathbb{Z}$; inspect its kernel. ☐

*Solution.* Zero. ☐

**Exercise 17.23.** Define a singular $n$-simplex.

*Hint.* Map the standard simplex into $X$.

A singular simplex is a continuous map, not necessarily an embedding; even a constant map from $\Delta^n$ is allowed. ☐

*Solution.* A continuous $\sigma : \Delta^n \to X$. ☐

**Exercise 17.24.** Define $C_n^{\text{sing}}(X)$.

*Hint.* Free group on all singular simplices.

Use finite integer linear combinations of all such maps. Different maps with the same image are still different generators. □

*Solution.* The free abelian group generated by continuous maps $\Delta^n \to X$. □

**Exercise 17.25.** How is the singular boundary defined?

*Hint.* Restrict to faces with signs.

Precompose $\sigma$ with the standard face inclusions $\delta_i : \Delta^{n-1} \to \Delta^n$, preserving their alternating signs. □

*Solution.* $\partial\sigma = \sum_i (-1)^i \sigma\delta_i$. □

**Exercise 17.26.** Define singular homology.

*Hint.* Take homology of the singular chain complex.

Use the kernel of the singular boundary in degree $n$ modulo the image of the singular boundary from degree $n+1$. □

*Solution.* $H_n^{\text{sing}}(X) = H_n(C_*^{\text{sing}}(X))$. □

**Exercise 17.27.** How does $f : X \to Y$ act on a singular simplex?

*Hint.* Compose.

The target changes from $X$ to $Y$, while the domain remains $\Delta^n$; composition therefore sends a generator to another singular simplex. □

*Solution.* $\sigma \mapsto f \circ \sigma$. □

**Exercise 17.28.** Why is $f_\#$ a chain map?

*Hint.* Composition commutes with face restriction.

For every face inclusion $\delta_i$, compare $(f \circ \sigma) \circ \delta_i$ with $f \circ (\sigma \circ \delta_i)$, then sum with signs. □

*Solution.* $\partial(f\sigma) = f_\#(\partial\sigma)$. □

**Exercise 17.29.** What map does $f$ induce on homology?

*Hint.* Pass from the chain map.

Apply the chain-map result that preserves cycles and boundaries; this justifies passing to homology classes rather than only defining a map on chains. □

*Solution.* $f_* : H_n(X) \to H_n(Y)$. □

**Exercise 17.30.** State composition functoriality.

*Hint.* Induced maps compose.

Evaluate the two proposed induced maps on a representative cycle and use associativity of composition on each singular simplex. □

*Solution.* $(g \circ f)_* = g_* \circ f_*$. □

**Exercise 17.31.** Compare simplicial and singular homology on $|K|$.

*Hint.* Use the comparison theorem.

The comparison map sends an oriented geometric simplex to its characteristic singular simplex; its induced isomorphism concerns homology, not equality of chain groups. □

*Solution.* They are naturally isomorphic. □

**Exercise 17.32.** State the disjoint-union formula.

*Hint.* Chains split by components.

A singular simplex has connected domain and hence lands in one summand; finite chain support gives a direct sum rather than an unrestricted product. □

*Solution.* $H_n(\coprod X_\alpha) \cong \bigoplus H_n(X_\alpha)$. □

**Exercise 17.33.** Compute $H_1(S^1 \vee S^1)$.

*Hint.* Use reduced wedge additivity.

Each circle contributes one independent one-dimensional class. Contrast the abelian result with the nonabelian free fundamental group of the wedge. □

*Solution.* $\mathbb{Z}^2$. □

**Exercise 17.34.** How is $H_1$ related to $\pi_1$?

*Hint.* Abelianize.

Pass from $\pi_1(X)$ to its quotient by the commutator subgroup; retain the path-connectedness hypothesis when using this identification. □

*Solution.* $H_1(X) \cong \pi_1(X)_{\mathrm{ab}}$ for path-connected $X$. □

**Exercise 17.35.** Do homotopic maps induce the same homology map?

*Hint.* Preview VIII/20.

Seek a prism operator whose boundary formula is $g_\# - f_\# = \partial P + P\partial$; applying it to cycles makes the two image cycles differ by a boundary. □

*Solution.* Yes; VIII/20 proves this at chain level. □

**Exercise 17.36.** What is the bridge to VIII/18?

*Hint.* Replace huge chain groups by cells.

Use the skeletal groups $H_n(X^n, X^{n-1})$; their cell generators lead to a smaller complex without making singular chain groups themselves smaller. □

*Solution.* Cellular homology uses one generator per CW cell. □

## 17.24 Solved problem dossiers

**Problem 17.37** (Triangle boundary squared)**.** Verify $\partial^2[v_0, v_1, v_2] = 0$.

*Solution.* First $\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$. Applying $\partial[v_i, v_j] = [v_j] - [v_i]$, every vertex cancels. □

**Problem 17.38** (Polygonal circle)**.** Compute simplicial $H_1$ of an $m$-gon with no 2-simplices.

*Solution.* The sum of all coherently oriented edges is a cycle. The kernel of $\partial_1$ has rank one and there are no 2-boundaries, so $H_1 \cong \mathbb{Z}$. □

**Problem 17.39** (Filled triangle)**.** Compute $H_1$ of a filled 2-simplex.

*Solution.* The boundary cycle of the triangle is exactly the boundary of the 2-simplex, so it dies. The disk is contractible, hence $H_1 = 0$. ☐

**Problem 17.40** (Two components)**.** Compute $H_0$ of two disjoint circles.

*Solution.* There are two path components, so $H_0 \cong \mathbb{Z}^2$. ☐

**Problem 17.41** (Reduced zero homology)**.** Compute $\widetilde{H}_0$ of a space with $c$ nonempty path components.

*Solution.* Ordinary $H_0 \cong \mathbb{Z}^c$; removing the diagonal augmentation summand leaves $\mathbb{Z}^{c-1}$. ☐

**Problem 17.42** (Sphere top cycle)**.** Why does an oriented triangulated sphere define a top-dimensional cycle?

*Solution.* Every codimension-one face belongs to two top simplices with opposite induced orientations, so those face contributions cancel in the total boundary. ☐

**Problem 17.43** (Disk boundary class)**.** Why is the boundary sphere class zero in the homology of the disk?

*Solution.* The boundary fundamental cycle is the boundary of the sum of oriented top-dimensional simplices filling the disk. ☐

**Problem 17.44** (Singular 0-simplices)**.** What are singular 0-simplices in $X$?

*Solution.* $\Delta^0$ is a point, so singular 0-simplices are exactly points of $X$. ☐

**Problem 17.45** (Singular 1-simplex)**.** What familiar object is a singular 1-simplex?

*Solution.* A continuous map $I \cong \Delta^1 \to X$, i.e. a path with an orientation. ☐

**Problem 17.46** (Boundary of a singular path)**.** For a singular 1-simplex $\sigma$, what is $\partial\sigma$?

*Solution.* It is the endpoint minus the starting point: $\sigma(1) - \sigma(0)$. ☐

**Problem 17.47** (Functorial boundary)**.** Verify $f_\#\partial = \partial f_\#$ on a singular simplex.

*Solution.* Both expressions are $\sum_i (-1)^i f \circ \sigma \circ \delta_i$. ☐

**Problem 17.48** (Constant map on positive homology)**.** What does a constant map induce on reduced positive-degree homology?

*Solution.* It factors through a point, whose reduced homology vanishes, so the induced map is zero. ☐

**Problem 17.49** (Identity map)**.** What map does $\mathrm{id}_X$ induce on homology?

*Solution.* The identity on every $H_n(X)$. ☐

**Problem 17.50** (Circle degree map)**.** For $f : S^1 \to S^1$ of degree $m$, what is $f_* : H_1(S^1) \to H_1(S^1)$?

*Solution.* Under $H_1(S^1) \cong \mathbb{Z}$, it is multiplication by $m$. ☐

**Problem 17.51** (Two-circle wedge)**.** Interpret $H_1(S^1 \vee S^1)$ using abelianization.

*Solution.* $\pi_1 \cong F_2$, whose abelianization is $\mathbb{Z}^2$, so $H_1 \cong \mathbb{Z}^2$. ☐

**Problem 17.52** (Torus first homology). Use $\pi_1(T^2) \cong \mathbb{Z}^2$ to compute $H_1(T^2)$.

*Solution.* The group is already abelian, so its abelianization is $\mathbb{Z}^2$. ☐

**Problem 17.53** (Projective-plane first homology). Use $\pi_1(\mathbb{R}P^2) \cong \mathbb{Z}/2$ to compute $H_1$.

*Solution.* The fundamental group is abelian already, so $H_1(\mathbb{R}P^2) \cong \mathbb{Z}/2$. ☐

**Problem 17.54** (Simply connected space). What is $H_1(X)$ for path-connected simply connected $X$?

*Solution.* The fundamental group is trivial, so its abelianization and therefore $H_1$ are zero. ☐

**Problem 17.55** (Disjoint-union chains). Why do singular chain groups split over components?

*Solution.* The connected simplex $\Delta^n$ has image in a single component, so every singular generator belongs to one component and the chain group is the direct sum of component chain groups. ☐

**Problem 17.56** (Comparison theorem use). Why is simplicial homology computationally useful if singular homology is the general invariant?

*Solution.* For a triangulated space, the comparison theorem identifies the small combinatorial simplicial chain complex with singular homology, avoiding infinitely many singular generators. ☐

**Problem 17.57** (Simplicial collapse map). If a simplicial map sends an edge's two vertices to the same vertex, what happens to that oriented edge on chains?

*Solution.* Its image has dimension zero, so the induced simplicial 1-chain map sends the edge to zero. ☐

**Problem 17.58** (Orientation sign). Why must simplicial chains impose sign under odd vertex permutations?

*Solution.* Without the sign convention, adjacent top simplices would not cancel correctly along shared faces and the alternating boundary formula would fail to encode orientation. ☐

**Problem 17.59** (Homotopy-invariance boundary). Why is the proof that homotopic maps induce the same map postponed?

*Solution.* It requires a chain homotopy, concretely a prism operator on singular chains. VIII/20 develops this proof after the basic homology constructions are established. ☐

**Problem 17.60** (Bridge to cellular chains). Why can cellular homology be much smaller than singular homology?

*Solution.* A CW complex with finitely many cells has one cellular generator per cell, while singular chains contain every continuous simplex map. The cellular boundary encodes attaching-map degrees. ☐

## 17.25 Supplementary worked examples

**Example 17.61** (Singular homology of a point)**.** Although the singular chain groups of a point contain many degenerate simplices, the resulting homology is $H_0(*) \cong \mathbb{Z}$ and $H_n(*) = 0$ for $n > 0$. Homology discards most chain-level redundancy.

**Example 17.62** (Simplicial circle)**.** For a triangulated circle, one independent 1-cycle survives modulo boundaries, giving $H_1(S^1) \cong \mathbb{Z}$ and $H_0(S^1) \cong \mathbb{Z}$. Singular homology gives the same groups.

**Example 17.63** (Comparison principle)**.** For a simplicial complex $K$, simplicial chains include into singular chains of $|K|$. The comparison theorem shows this inclusion induces isomorphisms on homology, so combinatorial and continuous models agree.

## 17.26 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 17.64** (Point homology)**.** State $H_n(*)$ with integer coefficients.

*Hint.* Only connected-component information survives in degree zero. □

*Solution.* $H_0(*) \cong \mathbb{Z}$ and $H_n(*) = 0$ for $n > 0$. □

**Exercise 17.65** (Circle homology)**.** State the integral homology of $S^1$.

*Hint.* There is one component and one essential loop. □

*Solution.* $H_0(S^1) \cong \mathbb{Z}$, $H_1(S^1) \cong \mathbb{Z}$, and $H_n = 0$ for $n \geq 2$. □

**Exercise 17.66** (Sphere homology)**.** State the reduced integral homology of $S^n$.

*Hint.* Only the top-dimensional class survives. □

*Solution.* $\widetilde{H}_k(S^n) \cong \mathbb{Z}$ for $k = n$ and is zero otherwise. □

**Exercise 17.67** (Disconnected $H_0$)**.** If a space has $c$ path components under the standard niceness assumptions, what is $H_0$?

*Hint.* One generator survives per component. □

*Solution.* $H_0 \cong \mathbb{Z}^c$. □

**Exercise 17.68** (Triangulated disk)**.** What is the homology of a triangulated disk?

*Hint.* A disk is contractible. □

*Solution.* $H_0 \cong \mathbb{Z}$ and $H_n = 0$ for $n > 0$. □

### Proofs

**Exercise 17.69** (Simplicial cycles map to singular cycles)**.** Show the inclusion of simplicial chains into singular chains commutes with boundary.

*Hint.* Each oriented affine simplex has the same alternating face restrictions. ☐

*Solution.* The singular boundary of the affine characteristic simplex is exactly the signed sum of its simplicial faces, so the inclusion is a chain map. ☐

**Exercise 17.70** (Triangulation invariance)**.** Explain why two triangulations of the same polyhedron yield isomorphic homology.

*Hint.* Both compare to singular homology of the same space. ☐

*Solution.* Each simplicial homology is naturally isomorphic to singular homology of the common underlying polyhedron, hence the groups are isomorphic independently of triangulation. ☐

**Exercise 17.71** (Homeomorphism invariance)**.** Prove a homeomorphism induces isomorphisms on singular homology.

*Hint.* Use the induced chain maps of the map and its inverse. ☐

*Solution.* Functoriality gives induced homology maps whose compositions are identities, so they are inverse isomorphisms. ☐

**Exercise 17.72** (Contractible spaces)**.** Prove a contractible space has the homology of a point.

*Hint.* Homotopy invariance is the key theorem. ☐

*Solution.* A contraction gives a homotopy equivalence with a point, hence isomorphisms on all homology groups. ☐

### Counterexamples and hypothesis tests

**Exercise 17.73** (Singular depends on triangulation)**.** Test: singular homology requires choosing a triangulation and changes if the triangulation changes.

*Hint.* Singular simplices are all continuous maps from standard simplices. ☐

*Solution.* False. Singular homology is defined directly for topological spaces and is triangulation-independent. ☐

**Exercise 17.74** (Simplicial always available)**.** Test: every topological space is already a simplicial complex.

*Hint.* Triangulability is an extra property. ☐

*Solution.* False; singular homology is designed to work without a triangulation. ☐

**Exercise 17.75** (Same homology implies homeomorphic)**.** Test: spaces with isomorphic homology groups are homeomorphic.

*Hint.* Homology is not a complete invariant. ☐

*Solution.* False. Many nonhomeomorphic and even non-homotopy-equivalent spaces share the same homology groups. ☐

## Applications and investigations

**Exercise 17.76** (Mesh verification)**.** Why compare a mesh's simplicial homology with a known singular-homology target?

*Hint.* It checks that the discretization has the intended topology. ☐

*Solution.* For a mesh approximating a known space, matching Betti numbers/torsion provides a strong topological sanity check independent of metric fidelity. ☐

**Exercise 17.77** (Continuous-to-discrete bridge)**.** What is gained by the simplicial-singular comparison theorem computationally?

*Hint.* Use finite combinatorics without changing the invariant. ☐

*Solution.* One may compute with sparse finite boundary matrices of a triangulation while knowing the result equals the intrinsic singular homology of the underlying space. ☐

## Challenge problems

**Exercise 17.78** (Subdivision chain equivalence)**.** Explain how repeated subdivision can refine geometry without changing homology.

*Hint.* Subdivision maps are chain-homotopy equivalences. ☐

*Solution.* The subdivision operator and a suitable simplicial approximation induce inverse maps on homology up to chain homotopy, so refinement preserves the groups. ☐

**Exercise 17.79** (Comparison synthesis)**.** Describe the logical route from a triangulated surface to a topological invariant independent of that mesh.

*Hint.* Build simplicial chains, compare to singular chains, then use topological invariance. ☐

*Solution.* Compute simplicial homology from the mesh; the comparison theorem identifies it with singular homology of the realization; singular homology is invariant under homeomorphism/homotopy, so the answer reflects the space rather than the chosen triangulation. ☐

# Chapter 18

# Cellular Homology

For a CW complex, singular chains are usually far larger than necessary. Cellular homology compresses the calculation to one generator for each cell. The cellular boundary map is determined by how each $n$-cell attaches to the $(n-1)$-skeleton. This makes the topology of CW constructions directly computable from degrees of attaching maps.

## 18.1 CW filtration and relative groups

Let

$$X^0 \subset X^1 \subset X^2 \subset \cdots$$

be the skeletal filtration of a CW complex.

The cellular chain group in degree $n$ is built from

$$H_n(X^n, X^{n-1}).$$

**Theorem 18.1** (Relative cell decomposition). *If $X$ has $c_n$ $n$-cells, then*

$$H_k(X^n, X^{n-1}) \cong \begin{cases} \mathbb{Z}^{c_n}, & k = n, \\ 0, & k \neq n. \end{cases}$$

Thus every oriented $n$-cell contributes one free generator.

## 18.2 Cellular chain groups

**Definition 18.2** (Cellular chains). Define

$$C_n^{\mathrm{cell}}(X) = H_n(X^n, X^{n-1}).$$

After choosing orientations of the $n$-cells,

$$C_n^{\mathrm{cell}}(X) \cong \mathbb{Z}^{c_n}.$$

## 18.3 Cellular boundary

The cellular boundary is the composite

$$d_n : H_n(X^n, X^{n-1}) \longrightarrow H_{n-1}(X^{n-1}, X^{n-2})$$

obtained from the connecting map in the long exact sequence of the triple

$$(X^n, X^{n-1}, X^{n-2}).$$

**Theorem 18.3** (Cellular chain complex)**.**

$$\cdots \to C_n^{\mathrm{cell}}(X) \xrightarrow{d_n} C_{n-1}^{\mathrm{cell}}(X) \to \cdots$$

*is a chain complex, and its homology is naturally isomorphic to singular homology:*

$$H_n(C_*^{\mathrm{cell}}(X)) \cong H_n(X).$$

## 18.4 Boundary coefficients from attaching maps

Let $e_\alpha^n$ be an $n$-cell with attaching map

$$\varphi_\alpha : S^{n-1} \to X^{n-1}.$$

For an $(n-1)$-cell $e_\beta^{n-1}$, collapse

$$X^{n-1} \setminus e_\beta^{n-1}$$

to a point. The resulting quotient is a sphere

$$X^{n-1}/(X^{n-1} \setminus e_\beta^{n-1}) \cong S^{n-1}.$$

The composite

$$S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \longrightarrow S^{n-1}$$

has an integer degree.

**Theorem 18.4** (Cellular boundary formula)**.** *The coefficient of $e_\beta^{n-1}$ in*

$$d_n(e_\alpha^n)$$

*is the degree of this composite map.*

This is the computational heart of cellular homology.

## 18.5 A CW complex with no adjacent dimensions

If there are no $(n-1)$-cells, then

$$C_{n-1}^{\mathrm{cell}}(X) = 0$$

and therefore

$$d_n = 0.$$

This makes the minimal CW structure of a sphere immediate.

## 18.6 Spheres

For the CW structure on $S^n$ with one 0-cell and one $n$-cell,

$$C_k = \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

All differentials vanish, giving

$$H_k(S^n) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

## 18.7 The torus

The standard torus CW structure has one 0-cell, two 1-cells, and one 2-cell:

$$C_2 \cong \mathbb{Z}, \qquad C_1 \cong \mathbb{Z}^2, \qquad C_0 \cong \mathbb{Z}.$$

The two-cell attaching word is

$$aba^{-1}b^{-1}.$$

Its total degree on each one-cell is zero, so

$$d_2 = 0.$$

Since the one-cells are loops at one vertex,

$$d_1 = 0.$$

Hence

$$H_2(T^2) \cong \mathbb{Z}, \qquad H_1(T^2) \cong \mathbb{Z}^2, \qquad H_0(T^2) \cong \mathbb{Z}.$$

## 18.8 Real projective spaces

The standard CW structure on

$$\mathbb{R}P^n$$

has one cell in each dimension $0, \dots, n$.

**Theorem 18.5** (Projective cellular differential)**.** *With suitable orientations,*

$$d_k = \begin{cases} 0, & k \text{ odd}, \\ \times 2, & k \text{ even}. \end{cases}$$

Thus the cellular chain complex alternates:

$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}.$$

## 18.9 Homology of real projective space

For $0 < k < n$,

$$H_k(\mathbb{R}P^n) \cong \begin{cases} \mathbb{Z}/2, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases}$$

At the top dimension,

$$H_n(\mathbb{R}P^n) \cong \begin{cases} \mathbb{Z}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

## 18.10 Degree-$m$ two-cell attachments

Consider

$$X_m = S^1 \cup_m D^2.$$

Its cellular chain complex is

$$0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0.$$

For $m \neq 0$,

$$H_2(X_m) = 0, \qquad H_1(X_m) \cong \mathbb{Z}/m, \qquad H_0(X_m) \cong \mathbb{Z}.$$

For $m = 0$,

$$X_0 \simeq S^1 \vee S^2$$

and both $H_1$ and $H_2$ are $\mathbb{Z}$.

## 18.11 Moore spaces

For $n \geq 1$, attach an $(n+1)$-cell to $S^n$ by a degree-$m$ map:

$$M(\mathbb{Z}/m, n) = S^n \cup_m D^{n+1}.$$

The relevant cellular boundary is

$$\mathbb{Z} \xrightarrow{\times m} \mathbb{Z}.$$

Hence, for $m \neq 0$,

$$\widetilde{H}_k(M(\mathbb{Z}/m, n)) \cong \begin{cases} \mathbb{Z}/m, & k = n, \\ 0, & k \neq n. \end{cases}$$

## 18.12 Lens-space pattern

A lens space admits a CW structure with one cell in each dimension

$$0, 1, 2, 3.$$

The cellular boundary maps encode the finite cyclic fundamental-group data, and a standard three-dimensional lens space $L(p, q)$ has homology

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}/p, \qquad H_2 = 0, \qquad H_3 \cong \mathbb{Z}.$$

The parameter $q$ affects finer topology but not these integral homology groups.

## 18.13 Closed orientable surfaces

A genus-$g$ orientable surface has one 0-cell, $2g$ one-cells, and one 2-cell. The two-cell attaches by a product of commutators.

The total exponent of every one-cell in that attaching word is zero, so

$$d_2 = 0.$$

Also $d_1 = 0$.

Therefore

$$H_2(\Sigma_g) \cong \mathbb{Z}, \qquad H_1(\Sigma_g) \cong \mathbb{Z}^{2g}, \qquad H_0(\Sigma_g) \cong \mathbb{Z}.$$

## 18.14 Nonorientable surface pattern

For a closed nonorientable surface built from one vertex, $g$ one-cells, and one two-cell, the two-cell boundary contains squared generators. Cellular homology therefore produces a nontrivial multiplication-by-2 component and detects 2-torsion in first homology.

## 18.15 Euler characteristic

For a finite CW complex,

$$\chi(X) = \sum_n (-1)^n c_n.$$

Since

$$C_n^{\mathrm{cell}}(X) \cong \mathbb{Z}^{c_n},$$

the Euler–Poincare principle later gives

$$\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X).$$

## 18.16 Cellular maps and induced maps

A cellular map

$$f : X \to Y$$

induces maps between the skeletal pairs and hence a chain map

$$C_*^{\mathrm{cell}}(X) \to C_*^{\mathrm{cell}}(Y).$$

The induced homology map agrees with the usual singular-homology map.

## 18.17 Why cellular homology is efficient

For a finite CW complex, the cellular chain groups have ranks equal to the finite cell counts. By contrast, singular chain groups are typically infinitely generated in every positive degree.

The price for this compression is that the cellular differential requires careful analysis of attaching maps.

## 18.18 Support-diagram provenance

The legacy corpus contains support figures for quotient, lens-space, and Moore space calculations mapped explicitly into this chapter. These belong with the cellular computations rather than as standalone chapters. Their mathematical content is represented here, while final one-to-one figure/dossier reconciliation is reserved for the post-VIII/35 corpus audit.

## 18.19 Bridge to relative homology

The definition

$$C_n^{\mathrm{cell}}(X) = H_n(X^n, X^{n-1})$$

already uses relative homology. VIII/19 develops relative groups and the long exact sequence systematically, explaining the connecting maps that underlie the cellular boundary operator.

## 18.20 Exercises

**Exercise 18.6.** Define $C_n^{\mathrm{cell}}(X)$.

*Hint.* Use adjacent skeleta.

Take relative homology of the pair $(X^n, X^{n-1})$ in degree $n$; the cell interiors supply relative generators after the lower skeleton is collapsed. □

*Solution.* $H_n(X^n, X^{n-1})$. □

**Exercise 18.7.** If $X$ has $c_n$ $n$-cells, what is $C_n^{\text{cell}}(X)$?

*Hint.* Choose orientations.

Orient each characteristic disk to choose a generator. For infinitely many cells the group is a direct sum of copies of $\mathbb{Z}$, not a product. □

*Solution.* $\mathbb{Z}^{c_n}$. □

**Exercise 18.8.** What determines a cellular boundary coefficient?

*Hint.* Collapse all but one lower cell.

For $n \geq 2$, compose the attaching sphere with the quotient onto a chosen $(n-1)$-cell sphere and compute its signed degree; in degree one use terminal minus initial vertex. □

*Solution.* The degree of the attaching-map composite $S^{n-1} \to S^{n-1}$. □

**Exercise 18.9.** What happens to $d_n$ if there are no $(n-1)$-cells?

*Hint.* The target chain group vanishes.

The cellular differential lands in $C_{n-1}^{\text{cell}}(X)$; a zero target forces this map to vanish regardless of the attaching formula. □

*Solution.* $d_n = 0$. □

**Exercise 18.10.** Give the cellular groups for minimal $S^n$.

*Hint.* One 0-cell and one $n$-cell.

For $n \geq 1$, distinguish the base zero-cell from the top cell. When $n = 1$, the edge boundary vanishes because its endpoints coincide, not because its target is zero. □

*Solution.* $C_0 = C_n = \mathbb{Z}$, all others zero. □

**Exercise 18.11.** Compute $H_n(S^n)$ cellularly.

*Hint.* All differentials vanish.

Compute both the outgoing kernel and incoming image at the top cell; there are no cells in degree $n+1$ in the minimal sphere model. □

*Solution.* $\mathbb{Z}$. □

**Exercise 18.12.** Give the torus cellular ranks.

*Hint.* Use 1,2,1 cells.

Read the standard square identifications: all corners give one vertex, the opposite edge pairs give two edges, and the square interior gives one face. □

*Solution.* $C_2 = \mathbb{Z},\ C_1 = \mathbb{Z}^2,\ C_0 = \mathbb{Z}$. □

**Exercise 18.13.** What is $d_2$ for the standard torus?

*Hint.* The commutator has zero exponent sum in each generator.

For each oriented edge, count its appearances with sign in $aba^{-1}b^{-1}$. This abelian boundary computation does not make the attaching loop itself trivial in the rose. □

*Solution.* Zero. □

**Exercise 18.14.** Compute $H_1(T^2)$.

*Hint.* Both differentials around $C_1$ vanish.

Use $H_1 = \ker d_1 / \operatorname{im} d_2$, checking the endpoint boundary of each loop edge and the exponent sums of the face word. ☐

*Solution.* $\mathbb{Z}^2$. ☐

**Exercise 18.15.** How many standard cells does $\mathbb{R}P^n$ have in each dimension?

*Hint.* One per dimension.

Use the filtration $\mathbb{R}P^{k-1} \subset \mathbb{R}P^k$; the complement contributes one open $k$-cell at each stage. ☐

*Solution.* Exactly one in each dimension $0, \ldots, n$. ☐

**Exercise 18.16.** What are the projective cellular differentials?

*Hint.* Alternate by parity.

Check the low-dimensional case $d_2 = 2$, then use the two hemispherical contributions to distinguish even and odd indices consistently. ☐

*Solution.* $d_k = 0$ for odd $k$, and $d_k = \times 2$ for even $k$. ☐

**Exercise 18.17.** Compute $H_1(\mathbb{R}P^2)$.

*Hint.* Use $d_2 = \times 2$ and $d_1 = 0$.

All degree-one chains are cycles, but the attached two-cell contributes the subgroup $2\mathbb{Z}$ of boundaries. ☐

*Solution.* $\mathbb{Z}/2$. ☐

**Exercise 18.18.** Compute $H_2(\mathbb{R}P^2)$.

*Hint.* Take the kernel of multiplication by 2.

Over $\mathbb{Z}$, multiplication by two is injective. There are no three-cells, so compute the top kernel without introducing mod-two coefficients. ☐

*Solution.* Zero. ☐

**Exercise 18.19.** Compute $H_3(\mathbb{R}P^3)$.

*Hint.* The top differential is zero.

At degree three the outgoing differential is zero and there is no incoming degree-four map; both facts are needed for the top homology group. ☐

*Solution.* $\mathbb{Z}$. ☐

**Exercise 18.20.** What chain complex computes $S^1 \cup_m D^2$?

*Hint.* One cell in dimensions 0,1,2.

Place the three generators in degrees two, one, and zero; the attaching degree belongs to $d_2$, while the loop edge has zero vertex boundary. ☐

*Solution.* $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$. ☐

**Exercise 18.21.** For $m \neq 0$, compute $H_1(S^1 \cup_m D^2)$.

*Hint.* Take cokernel of $\times m$.

Quotient the degree-one cycle group by $m\mathbb{Z}$; for negative $m$, the subgroup and cyclic group depend only on $|m|$. ☐

*Solution.* $\mathbb{Z}/m$. ☐

**Exercise 18.22.** What is a Moore space $M(\mathbb{Z}/m, n)$?

*Hint.* Attach one higher cell by degree $m$.

Use $n \geq 1$ and a nonzero attaching degree $m$ for the stated Moore-space homology pattern; identify the only two positive-dimensional chain groups. ☐

*Solution.* $S^n \cup_m D^{n+1}$. ☐

**Exercise 18.23.** What reduced homology does that Moore space realize?

*Hint.* Use the multiplication-by-$m$ boundary.

Compute the kernel and cokernel of $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$. The assertion that other reduced groups vanish uses $m \neq 0$. ☐

*Solution.* $\widetilde{H}_n \cong \mathbb{Z}/m$, all other reduced groups zero. ☐

**Exercise 18.24.** State the homology of a 3D lens space $L(p, q)$.

*Hint.* Only $H_1$ sees $p$.

Use the standard cellular differentials in degrees three, two, and one: zero, multiplication by $p$, and zero; keep the integral coefficients fixed. ☐

*Solution.* $H_0 = H_3 = \mathbb{Z},\ H_1 = \mathbb{Z}/p,\ H_2 = 0$. ☐

**Exercise 18.25.** Give $H_1$ of an orientable genus-$g$ surface.

*Hint.* There are $2g$ one-cells and zero cellular differentials there.

The surface relator is a product of commutators, each with zero exponent sum in every generator; combine this with the common vertex of all one-cells. ☐

*Solution.* $\mathbb{Z}^{2g}$. ☐

**Exercise 18.26.** Give $H_2$ of an orientable closed surface.

*Hint.* The top cellular differential is zero.

For a connected closed orientable surface, the top cellular cycle has no incoming three-dimensional boundaries; retain connectedness when counting its generators. ☐

*Solution.* $\mathbb{Z}$. ☐

**Exercise 18.27.** How is Euler characteristic read from cells?

*Hint.* Take alternating cell counts.

Count cells with sign $(-1)^n$ in each dimension and require a finite CW complex for the unqualified finite alternating sum. ☐

*Solution.* $\chi(X) = \sum(-1)^n c_n$. ☐

**Exercise 18.28.** What does a cellular map induce?

*Hint.* Use skeletal pairs.

Use the induced maps on $(X^n, X^{n-1}) \to (Y^n, Y^{n-1})$; naturality of the connecting maps gives the commuting differential squares. □

*Solution.* A cellular chain map and hence maps on homology. □

**Exercise 18.29.** What is the bridge to VIII/19?

*Hint.* Cellular chains are relative groups.

Write the triple $X^{n-2} \subset X^{n-1} \subset X^n$ and identify the degree-lowering connecting map between its two consecutive cellular chain groups. □

*Solution.* VIII/19 develops relative homology and long exact sequences underlying the cellular boundary. □

## 18.21 Solved problem dossiers

**Problem 18.30** (Minimal sphere)**.** Compute cellular homology of the two-cell CW structure on $S^n$.

*Solution.* The chain groups are $\mathbb{Z}$ in degrees $n$ and 0 and zero elsewhere. Since there is no adjacent nonzero target, every differential is zero, giving $H_0 = H_n = \mathbb{Z}$. □

**Problem 18.31** (Circle)**.** Compute cellular homology of the one-vertex one-edge circle.

*Solution.* The chain complex is $0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$, so $H_1 \cong H_0 \cong \mathbb{Z}$. □

**Problem 18.32** (Two-circle wedge)**.** Compute cellular homology of $S^1 \vee S^1$.

*Solution.* One vertex and two loop one-cells give $0 \to \mathbb{Z}^2 \xrightarrow{0} \mathbb{Z} \to 0$, so $H_1 \cong \mathbb{Z}^2$. □

**Problem 18.33** (Torus)**.** Compute all cellular homology groups of the standard torus.

*Solution.* The chain complex is $0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z}^2 \xrightarrow{0} \mathbb{Z} \to 0$, hence $H_2 = \mathbb{Z},\ H_1 = \mathbb{Z}^2,\ H_0 = \mathbb{Z}$. □

**Problem 18.34** (Genus two surface)**.** Compute $H_*(\Sigma_2)$.

*Solution.* One 0-cell, four 1-cells, and one 2-cell with commutator-product attaching word give zero differentials, so $H_2 = \mathbb{Z},\ H_1 = \mathbb{Z}^4,\ H_0 = \mathbb{Z}$. □

**Problem 18.35** (Degree-three attachment)**.** Compute $H_*(S^1 \cup_3 D^2)$.

*Solution.* The chain complex has $d_2 = \times 3$ and $d_1 = 0$. Thus $H_2 = 0,\ H_1 = \mathbb{Z}/3,\ H_0 = \mathbb{Z}$. □

**Problem 18.36** (Degree-one attachment)**.** Compute the homology of $S^1 \cup_1 D^2$.

*Solution.* Multiplication by one is an isomorphism, so positive homology vanishes and $H_0 = \mathbb{Z}$, agreeing with the disk model. □

**Problem 18.37** (Degree-zero attachment)**.** Compute the homology of $S^1 \cup_0 D^2$.

*Solution.* All differentials vanish, giving $H_2 = \mathbb{Z},\ H_1 = \mathbb{Z},\ H_0 = \mathbb{Z}$, as for $S^1 \vee S^2$. □

**Problem 18.38** (Projective plane)**.** Compute cellular homology of $\mathbb{R}P^2$.

*Solution.* The complex is $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$. Hence $H_2 = 0,\ H_1 = \mathbb{Z}/2,\ H_0 = \mathbb{Z}$. □

**Problem 18.39** (Projective 3-space)**.** Compute cellular homology of $\mathbb{R}P^3$.

*Solution.* The complex is $0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$. Thus $H_3 = \mathbb{Z},\ H_2 = 0,\ H_1 = \mathbb{Z}/2,\ H_0 = \mathbb{Z}$. □

**Problem 18.40** (Projective parity)**.** Why is top homology of $\mathbb{R}P^n$ nonzero exactly for odd $n$?

*Solution.* The top differential is zero for odd $n$, giving kernel $\mathbb{Z}$; for even $n$ it is multiplication by 2, whose kernel is zero. □

**Problem 18.41** (Moore $M(\mathbb{Z}/5, n)$)**.** Compute its reduced homology.

*Solution.* The only relevant differential is $\mathbb{Z} \xrightarrow{5} \mathbb{Z}$, so $\widetilde{H}_n \cong \mathbb{Z}/5$ and all other reduced groups vanish. □

**Problem 18.42** (Moore $m = 1$)**.** What homology does $S^n \cup_1 D^{n+1}$ have?

*Solution.* The multiplication-by-one differential is an isomorphism, so all reduced homology vanishes; the space is homologically contractible. □

**Problem 18.43** (Lens space)**.** Explain why $H_1(L(p, q))$ is $\mathbb{Z}/p$ in the standard cell model.

*Solution.* The one-cell and two-cell portion contains a cellular differential whose essential coefficient is $p$, so the relevant cokernel is $\mathbb{Z}/p$. □

**Problem 18.44** (Lens parameter $q$)**.** Why can cellular integral homology fail to distinguish $L(p, q)$ for different $q$?

*Solution.* The standard cellular boundary ranks and the torsion coefficient $p$ do not depend on $q$. The parameter affects finer invariants beyond these homology groups. □

**Problem 18.45** (Euler torus)**.** Compute the torus Euler characteristic from its cells.

*Solution.* $\chi = 1 - 2 + 1 = 0$. □

**Problem 18.46** (Euler genus $g$)**.** Compute $\chi(\Sigma_g)$ from the standard cells.

*Solution.* $\chi = 1 - 2g + 1 = 2 - 2g$. □

**Problem 18.47** (Euler projective plane)**.** Compute $\chi(\mathbb{R}P^2)$.

*Solution.* There is one cell in dimensions 0,1,2, so $\chi = 1 - 1 + 1 = 1$. □

**Problem 18.48** (Boundary coefficient meaning)**.** Explain how an attaching map contributes an integer to a cellular differential.

*Solution.* Project the attaching sphere onto the quotient sphere corresponding to one lower-dimensional cell; the degree of that sphere map is the incidence coefficient. □

**Problem 18.49** (Orientation change)**.** What happens to a cellular boundary matrix if the orientation of one cell is reversed?

*Solution.* The corresponding basis generator changes sign, so one row or column changes sign. Homology is unchanged because this is a basis change by a unit. □

**Problem 18.50** (Cell subdivision)**.** Why should subdividing a CW or simplicial model not change the final homology?

*Solution.* Subdivision changes the chain complex but not the underlying topological space; cellular/simplicial homology agrees with singular homology, which is an invariant of the space. ☐

**Problem 18.51** (Support-figure role)**.** Where do the mapped quotient/lens/Moore support diagrams belong conceptually?

*Solution.* They illustrate cellular attaching maps and the resulting boundary matrices, so their canonical mathematical home is VIII/18 rather than an independent chapter. ☐

**Problem 18.52** (Cellular versus singular size)**.** Compare chain sizes for the minimal CW sphere.

*Solution.* Cellular chains have only two generators total, while singular chains have infinitely many singular simplices in every relevant degree. Both yield the same homology. ☐

**Problem 18.53** (Bridge to relative exactness)**.** Why does the cellular construction naturally lead to VIII/19?

*Solution.* Each cellular group is a relative group $H_n(X^n, X^{n-1})$, and the cellular differential is built from connecting homomorphisms in exact sequences of pairs and triples. ☐

## 18.22 Supplementary worked examples

**Example 18.54** (Cellular homology of the sphere)**.** With one 0-cell and one $n$-cell, the cellular chain complex of $S^n$ has $C_n = C_0 = \mathbb{Z}$ and no adjacent nonzero groups, so all differentials vanish and $H_n \cong H_0 \cong \mathbb{Z}$.

**Example 18.55** (Cellular homology of the torus)**.** For the one-vertex, two-edge, one-face CW structure of $T^2$, both cellular differentials vanish: $d_1 = 0$, and the commutator attaching word has zero total degree on each 1-cell. Thus $H_2 \cong \mathbb{Z}$, $H_1 \cong \mathbb{Z}^2$, $H_0 \cong \mathbb{Z}$.

**Example 18.56** (Projective-space differential)**.** In the one-cell-per-dimension CW structure of $\mathbb{RP}^n$, cellular differentials alternate between multiplication by 2 and 0 (with convention depending on dimension indexing), producing the characteristic $\mathbb{Z}/2$ torsion.

## 18.23 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 18.57** (Cellular chain rank)**.** What is the rank of $C_n^{\mathrm{cell}}(X)$ for a CW complex with $c_n$ $n$-cells?

*Hint.* One generator per cell. ☐

*Solution.* $C_n^{\mathrm{cell}}(X) \cong \mathbb{Z}^{c_n}$ for integral coefficients. ☐

**Exercise 18.58** (Sphere differential)**.** Why are cellular differentials zero in the two-cell CW structure of $S^n$?

*Hint.* There are no cells in dimensions $n-1$ or 1 when $n > 1$. ☐

*Solution.* The target/source adjacent chain groups vanish, so the differentials are zero (with the circle case also having $d_1 = 0$ because the single edge has the same initial and terminal vertex). ☐

**Exercise 18.59** (Torus ranks)**.** State the cellular chain ranks for the standard torus CW structure.

*Hint.* Count cells $1, 2, 1$. ☐

*Solution.* $\operatorname{rank} C_0 = 1$, $\operatorname{rank} C_1 = 2$, $\operatorname{rank} C_2 = 1$. ☐

**Exercise 18.60** (Degree boundary)**.** If one 2-cell attaches to one 1-cell by degree $m$, what is $d_2$?

*Hint.* Cellular incidence is the attaching degree. ☐

*Solution.* $d_2 : \mathbb{Z} \to \mathbb{Z}$ is multiplication by $m$. ☐

**Exercise 18.61** (Euler from cells)**.** For a finite CW complex, express Euler characteristic using cellular chain ranks.

*Hint.* Alternate the cell counts. ☐

*Solution.* $\chi(X) = \sum_n (-1)^n \operatorname{rank} C_n^{\mathrm{cell}} = \sum_n (-1)^n c_n$. ☐

### Proofs

**Exercise 18.62** (Cellular chains from relative homology)**.** Explain why $C_n^{\text{cell}}(X) \cong H_n(X^n, X^{n-1})$ is free on the $n$-cells.

*Hint.* Collapse $X^{n-1}$ and obtain a wedge of $n$-spheres. □

*Solution.* $X^n/X^{n-1} \cong \bigvee S^n$ over the $n$-cells; reduced homology in degree $n$ is one copy of $\mathbb{Z}$ per sphere/cell. □

**Exercise 18.63** ($d^2 = 0$)**.** Why do cellular boundary maps satisfy $d_{n-1}d_n = 0$?

*Hint.* They arise as connecting maps in exact sequences of skeletal triples/pairs. □

*Solution.* Naturality and exactness of the long exact sequences of the filtration make consecutive connecting homomorphisms compose to zero, producing a chain complex. □

**Exercise 18.64** (Cellular equals singular)**.** State why cellular homology computes ordinary singular homology of a CW complex.

*Hint.* Use the skeletal filtration spectral/exact-sequence argument. □

*Solution.* The filtration yields a cellular chain complex whose homology is naturally isomorphic to $H_*(X)$; only relative groups in the matching cell dimension survive. □

**Exercise 18.65** (Euler-Poincare)**.** Prove the alternating sum of cell counts equals the alternating sum of homology ranks for a finite CW complex.

*Hint.* Use a finite free chain complex and cancellation of boundary ranks. □

*Solution.* Writing $\operatorname{rank} C_n = \operatorname{rank} B_n + \operatorname{rank} H_n + \operatorname{rank} B_{n-1}$, the boundary-rank terms cancel in the alternating sum. □

### Counterexamples and hypothesis tests

**Exercise 18.66** (Need triangulation)**.** Test: cellular homology requires subdividing every CW cell into simplices.

*Hint.* It works directly from CW skeleta. □

*Solution.* False. The point is to compute from cells without triangulating them. □

**Exercise 18.67** (Boundary is cell count only)**.** Test: cellular boundary matrices are determined solely by the number of cells.

*Hint.* Attaching maps determine incidence degrees. □

*Solution.* False. Equal cell counts can have different attaching maps and different homology. □

**Exercise 18.68** (All CW differentials zero)**.** Test: cellular differentials vanish because cells are open disks.

*Hint.* The attaching maps carry the topology. □

*Solution.* False. Nonzero attaching degrees give nonzero differentials and torsion. □

## Applications and investigations

**Exercise 18.69** (Sparse topology)**.** Why is cellular homology often much smaller than simplicial homology for the same space?

*Hint.* A CW model can use very few cells. ☐

*Solution.* Each cell contributes one chain generator, so a minimal CW decomposition can replace a huge triangulation by tiny boundary matrices. ☐

**Exercise 18.70** (Attachment diagnostics)**.** How can a cellular boundary matrix help diagnose whether a proposed CW attachment encoded the intended relation?

*Hint.* Its entries are signed degrees/incidences of attaching maps. ☐

*Solution.* Unexpected matrix entries reveal incorrect winding/orientation of attaching maps; homology then provides a global consistency check. ☐

## Challenge problems

**Exercise 18.71** (Projective torsion)**.** Explain qualitatively how a multiplication-by-2 cellular differential creates $\mathbb{Z}/2$ homology in projective space.

*Hint.* Take a cokernel of $2 : \mathbb{Z} \to \mathbb{Z}$ when the adjacent differential is zero. ☐

*Solution.* At the relevant degree, cycles are all of $\mathbb{Z}$ while boundaries are $2\mathbb{Z}$, so the quotient is $\mathbb{Z}/2$. ☐

**Exercise 18.72** (Cellular synthesis)**.** Describe the complete workflow from a finite CW description to homology.

*Hint.* Count cells, determine attaching degrees, build matrices, compute kernels/images. ☐

*Solution.* Choose orientations for cells; form one free generator per cell; compute cellular differentials from attaching-map degrees; verify $d^2 = 0$; reduce the integer matrices (e.g. Smith form) to obtain kernels modulo images and hence all homology groups. ☐

# Chapter 19

# Relative Homology and Exact Sequences

Relative homology measures what remains in a space after a chosen subspace is regarded as already filled or already understood. Algebraically, it is the homology of a quotient chain complex. Geometrically, it is the natural language for pairs, CW skeleta, excision, and connecting maps. This chapter develops the long exact sequence of a pair and the related exact-sequence machinery used throughout the rest of the volume.

## 19.1 Relative chains

Let $A \subset X$. The singular chain groups satisfy

$$C_n(A) \subset C_n(X),$$

since every singular simplex in $A$ is also one in $X$.

**Definition 19.1** (Relative chain group)**.** The relative chain group is

$$C_n(X, A) = C_n(X)/C_n(A).$$

The singular boundary preserves $C_n(A)$, so it descends to a boundary map on the quotient.

**Proposition 19.2** (Relative chain complex)**.** *The groups $C_n(X, A)$ form a chain complex with*

$$\partial[\sigma] = [\partial\sigma].$$

## 19.2 Relative homology

**Definition 19.3** (Relative homology)**.**

$$H_n(X, A) = H_n(C_*(X, A)).$$

A relative cycle may have boundary inside $A$, because that boundary becomes zero in the quotient chain complex.

## 19.3 Basic examples

If $A = \varnothing$, then

$$H_n(X, \varnothing) \cong H_n(X).$$

If $A = X$, then

$$H_n(X, X) = 0.$$

If $X$ is path connected and $A = \{x_0\}$, then positive-degree relative homology agrees with reduced homology:

$$H_n(X, \{x_0\}) \cong \widetilde{H}_n(X) \qquad (n > 0),$$

with the degree-zero relation controlled by the long exact sequence.

## 19.4 Short exact sequence of chain complexes

There is a natural short exact sequence

$$0 \longrightarrow C_*(A) \xrightarrow{i_\#} C_*(X) \xrightarrow{q_\#} C_*(X, A) \longrightarrow 0.$$

This is exact in every degree.

## 19.5 Connecting homomorphism

A short exact sequence of chain complexes

$$0 \to A_* \xrightarrow{i} B_* \xrightarrow{p} C_* \to 0$$

produces connecting homomorphisms

$$\delta_n : H_n(C) \to H_{n-1}(A).$$

To define $\delta_n$, take a cycle $c \in C_n$, lift it to $b \in B_n$, observe that $\partial b$ lies in the image of $A_{n-1}$, and take the corresponding homology class in $H_{n-1}(A)$.

## 19.6 Long exact sequence of a pair

**Theorem 19.4** (Long exact sequence of a pair)**.** *For every pair $A \subset X$, there is a natural long exact sequence*

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{q_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots .$$

In low degrees,

$$H_1(X, A) \xrightarrow{\partial} H_0(A) \to H_0(X) \to H_0(X, A) \to 0.$$

## 19.7 Exactness meaning

Exactness at $H_n(X)$ means

$$\operatorname{im} i_* = \ker q_*.$$

Thus an absolute homology class becomes zero relatively exactly when it comes from the subspace.

Exactness at $H_n(X, A)$ means

$$\operatorname{im} q_* = \ker \partial.$$

Thus a relative class comes from an absolute cycle exactly when its connecting boundary class vanishes.

## 19.8 Naturality

A map of pairs

$$f : (X, A) \to (Y, B), \qquad f(A) \subset B,$$

induces maps on absolute and relative homology.

**Theorem 19.5** (Naturality of the pair sequence)**.** *The induced maps fit into a commutative diagram of long exact sequences for $(X, A)$ and $(Y, B)$.*

## 19.9 The pair $(D^n, S^{n-1})$

Because $D^n$ is contractible and $S^{n-1}$ has reduced homology concentrated in degree $n-1$, the pair sequence gives

$$H_k(D^n, S^{n-1}) \cong \begin{cases} \mathbb{Z}, & k = n, \\ 0, & k \neq n. \end{cases}$$

This is the relative calculation underlying cellular chain groups.

## 19.10 Quotient-space interpretation

For a good pair, such as a CW pair $(X, A)$, collapsing $A$ to a point relates relative homology to reduced homology of the quotient.

**Theorem 19.6** (Relative quotient theorem)**.** *For a CW pair $(X, A)$,*

$$H_n(X, A) \cong \widetilde{H}_n(X/A)$$

*for all $n$.*

This is one of the cleanest geometric interpretations of relative homology.

## 19.11 Excision

**Theorem 19.7** (Excision)**.** *Suppose $Z \subset A \subset X$ and*

$$\overline{Z} \subset \operatorname{int} A.$$

*Then inclusion induces an isomorphism*

$$H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A).$$

Excision says relative homology ignores material lying completely inside the interior of the distinguished subspace.

## 19.12 Why excision matters

Excision allows local decomposition. It is the technical engine behind:

- cellular homology,
- Mayer–Vietoris sequences,
- local homology,
- manifold orientation theory.

## 19.13 Triples

Let

$$B \subset A \subset X.$$

There is a short exact sequence of relative chain complexes

$$0 \to C_*(A, B) \to C_*(X, B) \to C_*(X, A) \to 0.$$

**Theorem 19.8** (Long exact sequence of a triple)**.** *There is a natural long exact sequence*

$$\cdots \to H_n(A, B) \to H_n(X, B) \to H_n(X, A) \to H_{n-1}(A, B) \to \cdots .$$

The cellular differential from VIII/18 can be interpreted through such triples of consecutive skeleta.

## 19.14 Cellular boundary revisited

For

$$X^{n-2} \subset X^{n-1} \subset X^n,$$

the triple sequence contains

$$H_n(X^n, X^{n-1}) \longrightarrow H_{n-1}(X^{n-1}, X^{n-2}),$$

which is exactly the cellular boundary map.

Thus the exact-sequence framework explains the construction used in VIII/18.

## 19.15 Relative homology of attached cells

If

$$X = A \cup \bigcup_\alpha e^n_\alpha$$

is obtained from $A$ by attaching only $n$-cells, then

$$H_k(X, A) \cong \begin{cases} \mathbb{Z}^{\#\{n\text{-cells}\}}, & k = n, \\ 0, & k \neq n. \end{cases}$$

This is the cell-by-cell relative calculation behind CW homology.

## 19.16 Relative $H_0$

The group $H_0(X, A)$ measures components of $X$ not represented by $A$. If $A$ meets every path component of $X$, then

$$H_0(X, A) = 0.$$

## 19.17 Reduced homology from a pair

For nonempty $X$ with basepoint $x_0$,

$$\widetilde{H}_n(X) \cong H_n(X, \{x_0\})$$

for $n > 0$, and the pair long exact sequence organizes the degree-zero adjustment.

## 19.18 Boundary map geometry

A relative class may be represented by a chain whose boundary lies in $A$. The connecting map sends that relative class to the homology class of its boundary inside $A$.

This is often the most intuitive way to understand $\partial$.

## 19.19 Exactness as obstruction theory

The pair sequence can be read as a succession of obstruction statements:

- an $A$-class dies in $X$ exactly when it comes from a relative class one degree higher;
- a relative class comes from $X$ exactly when its boundary in $A$ vanishes;
- an $X$-class dies relatively exactly when it comes from $A$.

## 19.20 Bridge to homotopy invariance

Relative homology is functorial, so homotopies of maps of pairs should induce the same relative maps. VIII/20 proves the stronger chain-level statement for singular chains using prism operators and chain homotopies.

## 19.21 Exercises

**Exercise 19.9.** Define $C_n(X, A)$.

*Hint.* Quotient absolute chains by chains in $A$.

The quotient identifies chains differing by a chain supported in $A$; it does not require every simplex representative to avoid $A$. □

*Solution.* $C_n(X, A) = C_n(X)/C_n(A)$. □

**Exercise 19.10.** Why does the boundary descend to the quotient?

*Hint.* $C_*(A)$ is a subcomplex.

Replace $c$ by $c + a$, with $a \in C_n(A)$, and verify that their boundaries differ by a chain in $C_{n-1}(A)$. □

*Solution.* $\partial C_n(A) \subset C_{n-1}(A)$, so $\partial[c] = [\partial c]$ is well defined. □

**Exercise 19.11.** Define $H_n(X, A)$.

*Hint.* Take homology of the relative chain complex.

A relative cycle may have a nonzero absolute boundary, provided that boundary lies in $C_{n-1}(A)$; apply the quotient differential when testing cycles. □

*Solution.* $H_n(X, A) = H_n(C_*(X, A))$. □

**Exercise 19.12.** What is $H_n(X, \varnothing)$?

*Hint.* The quotient changes nothing.

There are no singular simplices with nonempty standard domain mapping to the empty set, so its chain groups vanish. □

*Solution.* $H_n(X)$. □

**Exercise 19.13.** What is $H_n(X, X)$?

*Hint.* The quotient chain complex is zero.

Every absolute chain is identified with zero in the quotient by the same chain group; inspect the resulting zero complex in each degree. □

*Solution.* Zero. □

**Exercise 19.14.** Write the chain short exact sequence for a pair.

*Hint.* Subspace chains, space chains, quotient chains.

Use the inclusion and quotient maps and verify injectivity, surjectivity, and image equal to kernel degree by degree. □

*Solution.* $0 \to C_*(A) \to C_*(X) \to C_*(X, A) \to 0$. □

**Exercise 19.15.** What does this short exact sequence produce?

*Hint.* Use homological algebra.

To construct the new arrow, lift a quotient cycle to an absolute chain and take its boundary; that boundary lies in the subspace chain complex. □

*Solution.* The long exact sequence of the pair. □

**Exercise 19.16.** Write the central part of the pair LES.

*Hint.* Absolute, relative, then one lower subspace group.

Track degrees on the connecting arrow: it starts at relative degree $n$ and ends at subspace degree $n - 1$. □

*Solution.* $H_n(A) \to H_n(X) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A)$. □

**Exercise 19.17.** What does exactness at $H_n(X)$ say?

*Hint.* Image equals kernel.

Name the inclusion-induced map $i_*$ and quotient-induced map $j_*$; a class is killed by $j_*$ precisely when it lies in $\operatorname{im} i_*$. □

*Solution.* $\operatorname{im} H_n(A) \to H_n(X) = \ker(H_n(X) \to H_n(X, A))$. □

**Exercise 19.18.** What does the connecting map do geometrically?

*Hint.* Take the boundary of a relative cycle in $A$.

Choose $c \in C_n(X)$ with $\partial c \in C_{n-1}(A)$; replacing the relative representative must change $\partial c$ only by an $A$-boundary. □

*Solution.* It sends a relative class to the homology class of its boundary inside $A$. □

**Exercise 19.19.** What is $H_k(D^n, S^{n-1})$?

*Hint.* Use the pair sequence.

Use $D^n/S^{n-1} \cong S^n$ for this CW pair, or use the reduced pair sequence to handle the two endpoints when $n = 1$. □

*Solution.* $\mathbb{Z}$ for $k = n$, zero otherwise. □

**Exercise 19.20.** For a CW pair, how is $H_n(X, A)$ related to $X/A$?

*Hint.* Collapse $A$.

Require a nonempty CW subcomplex $A$ for the usual quotient statement and use reduced homology of the quotient, especially in degree zero. □

*Solution.* $H_n(X, A) \cong \widetilde{H}_n(X/A)$. □

**Exercise 19.21.** State excision.

*Hint.* Remove a subset whose closure lies in the interior of $A$.

Write the hypothesis $\overline{Z} \subset \mathrm{int}_X(A)$ and identify the inclusion of pairs that induces the claimed isomorphism. □

*Solution.* $H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A)$. □

**Exercise 19.22.** What is the purpose of excision?

*Hint.* Ignore interior material of the subspace.

Apply excision to reduce a cell calculation to $(D^n, S^{n-1})$, checking the interior condition before removing any subset. □

*Solution.* It localizes relative calculations and underlies cellular and Mayer–Vietoris arguments. □

**Exercise 19.23.** Write the inclusions for a triple.

*Hint.* Three nested spaces.

Distinguish the three quotient complexes $C_*(A)/C_*(B)$, $C_*(X)/C_*(B)$, and $C_*(X)/C_*(A)$. □

*Solution.* $B \subset A \subset X$. □

**Exercise 19.24.** What LES comes from a triple?

*Hint.* Use relative pairs.

Apply the homology long exact sequence to $0 \to C_*(A, B) \to C_*(X, B) \to C_*(X, A) \to 0$, retaining the degree shift. □

*Solution.* $\cdots \to H_n(A, B) \to H_n(X, B) \to H_n(X, A) \to H_{n-1}(A, B) \to \cdots$. □

**Exercise 19.25.** How does the cellular boundary arise from a triple?

*Hint.* Use consecutive skeleta.

Set $(X, A, B) = (X^n, X^{n-1}, X^{n-2})$; the triple connecting arrow has target $H_{n-1}(X^{n-1}, X^{n-2})$. □

*Solution.* From $X^{n-2} \subset X^{n-1} \subset X^n$. □

**Exercise 19.26.** If $X$ is obtained from $A$ by $r$ $n$-cells only, what is $H_n(X, A)$?

*Hint.* One generator per cell.

Collapse the old subcomplex so that the newly attached disks become a wedge of $n$-spheres; orient the disks to choose the relative generators. □

*Solution.* $\mathbb{Z}^r$. □

**Exercise 19.27.** What is $H_k(X, A)$ in other degrees in that situation?

*Hint.* Only $n$-cells were added.

The relative cellular complex has no generators outside degree $n$; distinguish this relative vanishing from the absolute homology of $X$. ☐

*Solution.* Zero. ☐

**Exercise 19.28.** When is $H_0(X, A) = 0$?

*Hint.* Make $A$ meet every component.

Examine the cokernel of $H_0(A) \to H_0(X)$; the inclusion must hit the generator for every path component of $X$. ☐

*Solution.* When $A$ meets every path component of $X$. ☐

**Exercise 19.29.** How is reduced homology related to a based pair?

*Hint.* Use a point subspace.

Apply the pair sequence with $A = \{x_0\}$; the map on $H_0$ is injective, explaining why the degree-one case also works. ☐

*Solution.* For $n > 0$, $\widetilde{H}_n(X) \cong H_n(X, \{x_0\})$. ☐

**Exercise 19.30.** What maps induce morphisms of pair LESs?

*Hint.* Maps preserving subspaces.

Require $f(A) \subset B$, so the absolute chain map carries the subcomplex being quotiented out into the corresponding target subcomplex. ☐

*Solution.* Maps of pairs $f : (X, A) \to (Y, B)$. ☐

**Exercise 19.31.** Is the pair LES natural?

*Hint.* Yes; induced maps commute with all arrows.

For the connecting square, compare $\partial(f_\# c)$ with $f_\#(\partial c)$ on a lifted relative cycle. ☐

*Solution.* Yes. ☐

**Exercise 19.32.** What is the bridge to VIII/20?

*Hint.* Homotopies should induce equal homology maps.

A homotopy of pairs must carry $A \times I$ into $B$; this is the condition that allows the prism operator to descend to relative chains. ☐

*Solution.* VIII/20 proves this by a chain homotopy on singular chains. ☐

## 19.22 Solved problem dossiers

**Problem 19.33** (Pair of a disk and boundary)**.** Compute $H_*(D^2, S^1)$.

*Solution.* The only nonzero relative group is $H_2(D^2, S^1) \cong \mathbb{Z}$. ☐

**Problem 19.34** (Interval endpoints)**.** Compute $H_1(I, \partial I)$.

*Solution.* Since $I/\partial I \cong S^1$, the quotient theorem gives $H_1(I, \partial I) \cong \widetilde{H}_1(S^1) \cong \mathbb{Z}$. ☐

**Problem 19.35** (Sphere from disk pair)**.** Use the quotient theorem to rederive $H_n(D^n, S^{n-1})$.

*Solution.* Collapse $S^{n-1}$ to a point: $D^n/S^{n-1} \cong S^n$, so relative homology equals $\widetilde{H}_*(S^n)$. ☐

**Problem 19.36** (Relative circle to point)**.** Compute $H_1(S^1, \{*\})$.

*Solution.* By reduced homology, it is $\widetilde{H}_1(S^1) \cong \mathbb{Z}$. □

**Problem 19.37** (Relative $H_0$)**.** Let $X$ have three components and let $A$ meet exactly two. Compute $H_0(X, A)$.

*Solution.* One component remains unrepresented by $A$, so $H_0(X, A) \cong \mathbb{Z}$. □

**Problem 19.38** (Connecting map in disk pair)**.** What does $\partial : H_n(D^n, S^{n-1}) \to H_{n-1}(S^{n-1})$ do?

*Solution.* Both groups are $\mathbb{Z}$, and the connecting map sends the relative fundamental class of the disk to the fundamental class of its boundary sphere; it is an isomorphism up to orientation sign. □

**Problem 19.39** (Exactness at a sphere)**.** Why does $H_n(D^n) = 0$ force the connecting map above to be injective?

*Solution.* In the pair LES, exactness gives $\ker \partial = \operatorname{im}(H_n(D^n) \to H_n(D^n, S^{n-1})) = 0$. □

**Problem 19.40** (Cell attachment pair)**.** If $X = A \cup e^n$, compute $H_*(X, A)$.

*Solution.* The quotient $X/A$ is $S^n$, so $H_n(X, A) \cong \mathbb{Z}$ and all other relative groups vanish. □

**Problem 19.41** (Two attached cells)**.** If two $n$-cells are attached to $A$, compute relative homology.

*Solution.* Collapsing $A$ gives a wedge $S^n \vee S^n$ up to the relevant quotient cell structure, so $H_n(X, A) \cong \mathbb{Z}^2$. □

**Problem 19.42** (Excision on a disk)**.** Remove a small closed disk from the interior of $D^2$ while keeping an annular subspace inside $A$. Why can the relative homology remain unchanged?

*Solution.* If the removed set lies with closure inside the interior of the distinguished subspace, excision applies and the inclusion of complements induces an isomorphism. □

**Problem 19.43** (Triple of skeleta)**.** For a CW complex, identify the three spaces used to define $d_n^{\mathrm{cell}}$.

*Solution.* Use $X^{n-2} \subset X^{n-1} \subset X^n$; the triple connecting map runs from $H_n(X^n, X^{n-1})$ to $H_{n-1}(X^{n-1}, X^{n-2})$. □

**Problem 19.44** (Absolute class dies relatively)**.** Suppose $[z] \in H_n(X)$ maps to zero in $H_n(X, A)$. What does exactness imply?

*Solution.* $[z]$ lies in the image of $H_n(A) \to H_n(X)$; it is represented, up to homology, by a class from the subspace. □

**Problem 19.45** (Relative class lifts absolutely)**.** Suppose $\xi \in H_n(X, A)$ has $\partial\xi = 0$. What follows?

*Solution.* Exactness says $\xi$ lies in the image of $H_n(X) \to H_n(X, A)$. □

**Problem 19.46** (Subspace class dies in space)**.** Suppose $\eta \in H_{n-1}(A)$ maps to zero in $H_{n-1}(X)$. What follows?

*Solution.* Exactness says $\eta$ lies in the image of the connecting map $H_n(X, A) \to H_{n-1}(A)$. $\square$

**Problem 19.47** (Quotient pair)**.** For a CW pair $(X, A)$, why is quotient homology reduced?

*Solution.* Collapsing $A$ creates a distinguished basepoint. Relative chains ignore chains inside $A$, corresponding to reduced chains of the quotient rather than an extra degree-zero component. $\square$

**Problem 19.48** (Relative homology of a point pair)**.** Compute $H_*(X, X)$.

*Solution.* The quotient chain complex $C_*(X)/C_*(X)$ is zero, so every relative homology group is zero. $\square$

**Problem 19.49** (Naturality square)**.** What commutativity condition must hold for connecting maps under a map of pairs?

*Solution.* If $f : (X, A) \to (Y, B)$, then $f_*\partial = \partial f_*$ between $H_n(X, A) \to H_{n-1}(A)$ and $H_n(Y, B) \to H_{n-1}(B)$. $\square$

**Problem 19.50** (Reduced $H_0$ of two components)**.** Let $X$ have two path components and basepoint in one. Compute $H_0(X, \{x_0\})$.

*Solution.* The pair kills the component containing the basepoint but leaves one independent component, so $H_0(X, \{x_0\}) \cong \mathbb{Z}$, matching $\widetilde{H}_0(X)$. $\square$

**Problem 19.51** (Relative torus skeleton)**.** For the torus $T^2$, what is $H_2(T^2, T^1)$ in the standard CW structure?

*Solution.* There is one 2-cell, so $H_2(T^2, T^1) \cong \mathbb{Z}$. $\square$

**Problem 19.52** (Relative projective skeleton)**.** For $\mathbb{R}P^n$, compute $H_n(\mathbb{R}P^n, \mathbb{R}P^{n-1})$.

*Solution.* There is one $n$-cell, so the relative group is $\mathbb{Z}$. $\square$

**Problem 19.53** (Excision intuition)**.** Why should relative homology ignore a region lying deep inside $A$?

*Solution.* Relative chains already regard chains in $A$ as zero. Removing material whose closure lies inside $\operatorname{int} A$ does not affect how chains outside $A$ meet its boundary. $\square$

**Problem 19.54** (Boundary as obstruction)**.** Interpret a nonzero connecting class $\partial\xi$.

*Solution.* It is the obstruction to representing the relative class $\xi$ by an absolute cycle in $X$; exactness says precisely that $\xi$ lifts absolutely iff this boundary class vanishes. $\square$

**Problem 19.55** (Long exact sequence source)**.** What algebraic input produces the LES of a pair?

*Solution.* The short exact sequence of chain complexes $0 \to C_*(A) \to C_*(X) \to C_*(X, A) \to 0$. $\square$

**Problem 19.56** (Bridge to homotopy)**.** Why should a homotopy of maps of pairs preserve the entire long exact sequence map?

*Solution.* If the two maps induce chain-homotopic chain maps, they induce equal maps on absolute and relative homology, and naturality then preserves all exact-sequence arrows. $\square$

## 19.23 Supplementary worked examples

**Example 19.57** (The fundamental relative class of a disk)**.** For the pair $(D^n, S^{n-1})$, the long exact sequence and the contractibility of $D^n$ give

$$H_k(D^n, S^{n-1}) \cong \begin{cases} \mathbb{Z}, & k = n, \\ 0, & k \neq n. \end{cases}$$

The generator is represented by the oriented $n$-cell; its boundary becomes zero only after passing to relative chains.

**Example 19.58** (Relative homology as homology of a quotient)**.** For a good pair $(X, A)$, collapsing $A$ to a point gives

$$H_n(X, A) \cong \widetilde{H}_n(X/A).$$

Thus $(D^n, S^{n-1})$ becomes $S^n$, turning the relative fundamental class into the ordinary reduced fundamental class of the sphere.

**Example 19.59** (Exactness detects what survives an inclusion)**.** In the long exact sequence

$$H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A),$$

exactness says that an absolute class dies in relative homology exactly when it came from $A$, while the boundary of a relative class measures its residual boundary in $A$.

## 19.24 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 19.60** (Relative chain group)**.** Write $C_n(X, A)$ in terms of singular chain groups.

*Hint.* Quotient out chains supported in $A$. ☐

*Solution.* $C_n(X, A) = C_n(X)/C_n(A)$. ☐

**Exercise 19.61** (Empty subspace)**.** Compute $H_n(X, \varnothing)$.

*Hint.* The relative chain quotient changes nothing. ☐

*Solution.* $H_n(X, \varnothing) \cong H_n(X)$. ☐

**Exercise 19.62** (Identical pair)**.** Compute $H_n(X, X)$.

*Hint.* The relative chain complex is zero. ☐

*Solution.* $H_n(X, X) = 0$ in every degree. ☐

**Exercise 19.63** (Disk pair)**.** Compute $H_n(D^n, S^{n-1})$ and $H_k(D^n, S^{n-1})$ for $k \neq n$.

*Hint.* Use the long exact sequence of the pair. ☐

*Solution.* The only nonzero group is $H_n(D^n, S^{n-1}) \cong \mathbb{Z}$. ☐

**Exercise 19.64** (Boundary map of a relative class)**.** What does the connecting map $\partial : H_n(X, A) \to H_{n-1}(A)$ do to a relative cycle represented by $c$?

*Hint.* Take the ordinary boundary before passing to homology. ☐

*Solution.* Since $\partial c$ lies in $C_{n-1}(A)$, the connecting map sends $[c]$ to $[\partial c]$. ☐

### Proofs

**Exercise 19.65** (Exactness at absolute homology)**.** Prove $\ker j_* = \operatorname{im} i_*$ in the long exact sequence of a pair.

*Hint.* Translate $j_*[z] = 0$ into a relative boundary statement. ☐

*Solution.* If $j_*[z] = 0$, then $z = \partial c + a$ with $a \in C_n(A)$, so $[z] = i_*[a]$; the reverse inclusion follows because chains from $A$ vanish relatively. ☐

**Exercise 19.66** (Deformation retract pair)**.** Show that if $A$ is a deformation retract of $X$, then $H_n(X, A) = 0$ for all $n$.

*Hint.* The inclusion induces an isomorphism on homology. ☐

*Solution.* In the long exact sequence, $i_* : H_n(A) \to H_n(X)$ is an isomorphism in every degree, forcing all relative groups to vanish. ☐

**Exercise 19.67** (Naturality)**.** Explain why a map of pairs $f : (X, A) \to (Y, B)$ induces a morphism between the two long exact sequences.

*Hint.* Use the induced chain map on quotients. ☐

*Solution.* $f_\#$ preserves subcomplexes, descends to relative chains, and commutes with boundaries; the connecting maps are therefore natural. ☐

**Exercise 19.68** (Reduced quotient relation)**.** Under the usual good-pair hypothesis, prove the relation $H_n(X, A) \cong \widetilde{H}_n(X/A)$.

*Hint.* Compare the relative chain model with chains after collapsing $A$. ☐

*Solution.* The quotient map induces a chain equivalence between the relative complex and the reduced chain complex of $X/A$, hence an isomorphism on homology. ☐

### Counterexamples and hypothesis tests

**Exercise 19.69** (Naive quotient formula)**.** Test: $H_n(X, A)$ is always $H_n(X)/H_n(A)$.

*Hint.* Look at the connecting homomorphism in the long exact sequence. ☐

*Solution.* False. The relative group also records classes whose boundary lies in $A$; the connecting map can contribute information not captured by a quotient of homology groups. ☐

**Exercise 19.70** (Separate homology determines the pair)**.** Test: knowing $H_*(X)$ and $H_*(A)$ determines $H_*(X, A)$.

*Hint.* The inclusion map matters. ☐

*Solution.* False. The long exact sequence depends on the induced map $H_*(A) \to H_*(X)$, not only on the two groups separately. ☐

**Exercise 19.71** (Relative zero-cycles)**.** Test: $H_0(X, A)$ is always zero when $A$ is nonempty.

*Hint.* Consider components of $X$ disjoint from $A$. ☐

*Solution.* False. $H_0(X, A)$ detects components of $X$ not met by $A$; it vanishes when $A$ meets every path component of $X$. ☐

## Applications and investigations

**Exercise 19.72** (Cell attachment)**.** Explain why relative homology is the natural language for adding an $n$-cell.

*Hint.* Compare $(X^n, X^{n-1})$. $\square$

*Solution.* The relative group isolates the newly added cells: $H_k(X^n, X^{n-1})$ is free on the $n$-cells when $k = n$ and zero otherwise. $\square$

**Exercise 19.73** (Local topology)**.** How can $H_n(X, X - \{x\})$ be used as a local invariant near $x$?

*Hint.* Collapse everything away from a small neighborhood of $x$. $\square$

*Solution.* It records homology supported near $x$; for an $n$-manifold it is locally $\mathbb{Z}$ in degree $n$ for orientable local coefficients and zero in other degrees. $\square$

## Challenge problems

**Exercise 19.74** (Sphere from a pair)**.** Recover $\widetilde{H}_*(S^n)$ from the pair $(D^n, S^{n-1})$.

*Hint.* Use $D^n/S^{n-1} \cong S^n$. $\square$

*Solution.* The relative computation gives $\widetilde{H}_n(S^n) \cong \mathbb{Z}$ and zero in all other degrees. $\square$

**Exercise 19.75** (Exact-sequence synthesis)**.** Given $H_*(A)$, $H_*(X)$, and the inclusion map, describe a systematic workflow for computing $H_*(X, A)$.

*Hint.* Use exactness degree by degree. $\square$

*Solution.* Insert the known groups and maps into the long exact sequence, compute kernels and cokernels of the inclusion maps, and resolve the short exact pieces determined by exactness. $\square$

# Chapter 20

# Homotopy Invariance

Homology is useful only if it ignores deformations that preserve homotopy type. The key theorem is that homotopic maps induce the same homomorphism on homology. At chain level, the difference between the two induced chain maps is a boundary commutator. For singular chains this is implemented by the prism operator.

## 20.1 Chain homotopies

Let

$$f_\bullet, g_\bullet : C_\bullet \to D_\bullet$$

be chain maps.

**Definition 20.1** (Chain homotopy). A chain homotopy from $f$ to $g$ is a family of homomorphisms

$$P_n : C_n \to D_{n+1}$$

such that

$$g_n - f_n = \partial_{n+1} P_n + P_{n-1} \partial_n.$$

The operator $P$ has degree $+1$.

## 20.2 Chain-homotopic maps induce equal homology maps

**Theorem 20.2** (Chain-homotopy invariance). *If chain maps $f, g : C_\bullet \to D_\bullet$ are chain homotopic, then*

$$H_n(f) = H_n(g)$$

*for every $n$.*

*Proof.* If $z$ is a cycle, then

$$g(z) - f(z) = \partial P(z) + P(\partial z) = \partial P(z),$$

so $f(z)$ and $g(z)$ differ by a boundary and represent the same homology class. □

## 20.3 Topological homotopies

Suppose

$$f, g : X \to Y$$

are homotopic through

$$H : X \times I \to Y.$$

The goal is to construct a chain homotopy between the induced singular chain maps

$$f_\#, g_\# : C_*^{\text{sing}}(X) \to C_*^{\text{sing}}(Y).$$

## 20.4 Prisms

For a singular simplex

$$\sigma : \Delta^n \to X,$$

the homotopy gives a map

$$\Delta^n \times I \to Y.$$

The prism $\Delta^n \times I$ can be subdivided into $n + 1$ $(n + 1)$-simplices.

**Definition 20.3** (Prism operator)**.** The prism operator

$$P_n : C_n(X) \to C_{n+1}(Y)$$

is the signed sum of the singular $(n+1)$-simplices obtained by this standard prism subdivision followed by $H \circ (\sigma \times \text{id})$.

## 20.5 Prism identity

**Theorem 20.4** (Prism identity)**.** *The prism operator satisfies*

$$g_\# - f_\# = \partial P + P\partial.$$

Thus $P$ is a chain homotopy between $f_\#$ and $g_\#$.

## 20.6 Homotopy invariance theorem

**Theorem 20.5** (Homotopic maps induce the same map)**.** *If*

$$f \simeq g : X \to Y,$$

*then*

$$f_* = g_* : H_n(X) \to H_n(Y)$$

*for every $n$.*

## 20.7 Homotopy equivalence

**Corollary 20.6** (Homotopy equivalent spaces have isomorphic homology)**.** *If*

$$X \simeq Y,$$

*then*

$$H_n(X) \cong H_n(Y)$$

*for all $n$.*

Indeed, if $f : X \to Y$ and $g : Y \to X$ are homotopy inverses, then

$$g_* f_* = (g \circ f)_* = (\text{id}_X)_*,$$

and similarly $f_* g_* = \text{id}$.

## 20.8 Contractible spaces

**Corollary 20.7** (Homology of a contractible space). *If $X$ is nonempty and contractible, then*

$$H_0(X) \cong \mathbb{Z}, \qquad H_n(X) = 0 \quad (n > 0).$$

*Equivalently,*

$$\widetilde{H}_n(X) = 0$$

*for every $n$.*

## 20.9 Deformation retracts

If $A \subset X$ is a deformation retract, then

$$A \simeq X.$$

**Corollary 20.8** (Deformation retract). *The inclusion*

$$A \hookrightarrow X$$

*induces homology isomorphisms in every degree.*

## 20.10 Strong deformation retracts

A strong deformation retract gives the same conclusion, with the homotopy fixed on $A$. The additional relative control is important in pairs and CW constructions but does not alter the absolute homology conclusion.

## 20.11 Retracts and split maps

If

$$r : X \to A, \qquad i : A \hookrightarrow X, \qquad ri = \mathrm{id}_A,$$

then

$$r_* i_* = \mathrm{id}_{H_*(A)} .$$

Thus $i_*$ is injective and $r_*$ is surjective even without assuming a deformation retraction.

## 20.12 Null-homotopic maps

If

$$f : X \to Y$$

is null-homotopic, then on reduced homology

$$\widetilde{f}_* = 0.$$

For positive degrees,

$$f_* : H_n(X) \to H_n(Y)$$

is zero.

## 20.13 Degree and homotopy

For maps

$$f, g : S^n \to S^n,$$

homotopy invariance of top homology gives

$$f \simeq g \implies \deg f = \deg g.$$

This recovers the degree obstruction from VIII/03 inside homology.

## 20.14 No-retraction theorem via homology

Suppose

$$r : D^{n+1} \to S^n$$

were a retraction. Then inclusion

$$i : S^n \hookrightarrow D^{n+1}$$

would satisfy

$$r_* i_* = \mathrm{id}_{H_n(S^n)}.$$

But

$$H_n(D^{n+1}) = 0,$$

so $i_* = 0$, contradiction.

**Theorem 20.9** (Homological no-retraction)**.** *There is no retraction*

$$D^{n+1} \to S^n.$$

## 20.15 Homology detects noncontractibility

If a connected space has

$$\widetilde{H}_n(X) \neq 0$$

for some $n$, then $X$ is not contractible.

Thus:

$$S^n, \quad T^n, \quad \mathbb{R}P^{2k+1}$$

and many other spaces are immediately seen to be noncontractible.

## 20.16 Homology is not complete

The converse is false: spaces can have the same homology groups without being homotopy equivalent.

Homology is a powerful but coarse invariant. It is abelian and loses much noncommutative information visible to the fundamental group.

## 20.17 Acyclic does not mean contractible

A space is acyclic if its reduced homology vanishes. Acyclic spaces need not be contractible.

Thus

$$\widetilde{H}_*(X) = 0$$

is necessary but not sufficient for contractibility.

## 20.18 Homotopy invariance for pairs

If maps of pairs
$$f, g : (X, A) \to (Y, B)$$
are homotopic through maps of pairs, then
$$f_* = g_* : H_n(X, A) \to H_n(Y, B).$$

The prism construction descends to relative quotient chain complexes.

## 20.19 Naturality with exact sequences

Homotopy invariance is compatible with the long exact sequence of a pair. Homotopic maps of pairs induce the same morphism of long exact sequences.

## 20.20 Cellular consequence

If two CW complexes are homotopy equivalent, cellular homology calculations for either must give isomorphic homology groups. This justifies simplifying cell structures through homotopy-preserving operations before computation.

## 20.21 Chain-homotopy viewpoint

The topological statement
$$f \simeq g$$
is converted into the algebraic statement
$$f_\# \simeq_{\text{ch}} g_\#.$$

This is the first major instance of a recurring principle:
$$\text{topological deformation} \quad \rightsquigarrow \quad \text{algebraic chain homotopy.}$$

VIII/22 develops chain homotopies as algebraic objects in their own right.

## 20.22 Bridge to Euler characteristic

For finite CW complexes, homology ranks determine Euler characteristic through the Euler–Poincare formula. Since homology is homotopy invariant, Euler characteristic becomes a homotopy invariant as well. VIII/21 develops this systematically.

## 20.23 Exercises

**Exercise 20.10.** Define a chain homotopy $P$ between $f$ and $g$.

*Hint.* Use a degree +1 operator.

Write $P_n : C_n \to D_{n+1}$, then verify that both $\partial^D_{n+1} P_n$ and $P_{n-1}\partial^C_n$ have the same domain and target as $g_n - f_n$. ☐

*Solution.* $g - f = \partial P + P\partial$. ☐

**Exercise 20.11.** What degree does a chain homotopy have?

*Hint.* It raises chain degree.

A differential lowers degree by one, so its composite with the desired operator must have degree zero to compare two chain maps. □

*Solution.* $+1$. □

**Exercise 20.12.** Why do chain-homotopic maps agree on homology?

*Hint.* Evaluate on a cycle.

Substitute $\partial z = 0$ into the chain-homotopy equation and identify the explicit chain whose boundary is the difference of image cycles. □

*Solution.* For $\partial z = 0$, $g(z) - f(z) = \partial P(z)$, a boundary. □

**Exercise 20.13.** What topological object produces the singular-chain homotopy?

*Hint.* A homotopy $H : X \times I \to Y$.

Apply the homotopy to the prism $\sigma \times \mathrm{id}_I$ over each singular simplex, using an oriented triangulation of that prism. □

*Solution.* The prism operator. □

**Exercise 20.14.** What is the prism built from?

*Hint.* Use $\Delta^n \times I$.

A prism over an $n$-simplex has dimension $n+1$; choose compatible orientations so internal faces cancel when the pieces are added. □

*Solution.* The product of a singular simplex with the interval, subdivided into $(n+1)$-simplices. □

**Exercise 20.15.** State the prism identity.

*Hint.* Compare endpoint chain maps.

Separate the top and bottom faces from the side faces of the prism; their signs determine the endpoint difference and the $P\partial$ term. □

*Solution.* $g_\# - f_\# = \partial P + P\partial$. □

**Exercise 20.16.** What do homotopic maps induce on homology?

*Hint.* Apply the prism identity.

Combine the prism identity with the cycle argument; the conclusion is equality of induced maps, not necessarily equality of chain maps. □

*Solution.* The same homomorphism. □

**Exercise 20.17.** What does a homotopy equivalence induce?

*Hint.* Use homotopy inverses.

Apply homology to both homotopy-inverse composites and use functoriality to obtain two actual inverse homomorphisms. □

*Solution.* Isomorphisms on all homology groups. □

**Exercise 20.18.** What is the reduced homology of a contractible space?

*Hint.* Compare to a point.

Use reduced rather than ordinary homology: the point has zero reduced groups but has an ordinary degree-zero copy of $\mathbb{Z}$. ☐

*Solution.* Zero in every degree. ☐

**Exercise 20.19.** What is $H_0$ of a nonempty contractible space?

*Hint.* It is path connected.

A contraction supplies paths from all points to its terminal point; use the path-component interpretation of $H_0$. ☐

*Solution.* $\mathbb{Z}$. ☐

**Exercise 20.20.** What does a deformation retract induce?

*Hint.* It is a homotopy equivalence.

Let $i$ be the inclusion and $r$ the deformation retraction; apply homology to $ri = \mathrm{id}$ and $ir \simeq \mathrm{id}$. ☐

*Solution.* Homology isomorphisms. ☐

**Exercise 20.21.** If $ri = \mathrm{id}$, what follows for $i_*$?

*Hint.* Apply homology.

From $r_* i_* = \mathrm{id}$, apply $r_*$ to an equation $i_*(a) = 0$; this identifies the kernel of $i_*$. ☐

*Solution.* $i_*$ is injective. ☐

**Exercise 20.22.** If $ri = \mathrm{id}$, what follows for $r_*$?

*Hint.* Same splitting equation.

For a target class $a$, the element $i_*(a)$ provides an explicit preimage under $r_*$. ☐

*Solution.* $r_*$ is surjective. ☐

**Exercise 20.23.** What does a null-homotopic map induce on reduced homology?

*Hint.* Compare to a constant map.

A constant map factors through a point, whose reduced homology vanishes. Ordinary $H_0$ would require a different statement. ☐

*Solution.* The zero map. ☐

**Exercise 20.24.** What does homotopy imply about degree of sphere maps?

*Hint.* Top homology maps agree.

Choose oriented generators of the top homology groups for $n \geq 1$; the multiplication integer is the degree and is unchanged when the induced homomorphism is unchanged. ☐

*Solution.* Homotopic sphere maps have equal degree. ☐

**Exercise 20.25.** Why is $S^n$ not contractible?

*Hint.* Use nonzero reduced homology.

Use the nonzero reduced top class of the sphere; for $S^0$, reduced degree-zero homology also detects the failure of contractibility. ☐

*Solution.* $\widetilde{H}_n(S^n) \cong \mathbb{Z} \neq 0$. ☐

**Exercise 20.26.** Can equal homology groups imply homotopy equivalence?

*Hint.* Homology is incomplete.

Compare $T^2$ with $S^1 \vee S^1 \vee S^2$: their integral homology groups agree, but their fundamental groups are respectively abelian and nonabelian. ☐

*Solution.* No. ☐

**Exercise 20.27.** Does acyclic imply contractible?

*Hint.* Recall the warning.

Vanishing reduced homology does not test nonabelian fundamental-group information; a contractibility argument would need more than all homology classes vanishing. ☐

*Solution.* No. ☐

**Exercise 20.28.** State homotopy invariance for relative homology.

*Hint.* Use maps of pairs.

State the condition $H(A \times I) \subset B$ throughout the homotopy, not merely that the endpoint maps preserve the subspaces. ☐

*Solution.* Homotopic maps of pairs induce the same maps on $H_n(X, A)$. ☐

**Exercise 20.29.** How does the prism operator act on quotient chains?

*Hint.* It respects subspace chains under a homotopy of pairs.

For a simplex lying in $A$, its entire homotopy prism maps into $B$; hence the operator preserves the subcomplexes being quotiented out. ☐

*Solution.* It descends to relative chain complexes. ☐

**Exercise 20.30.** Why does homotopy invariance help cellular computations?

*Hint.* Simplify by homotopy equivalence.

Replace the CW model by a homotopy-equivalent simpler one and compute there; do not assume a smaller complex is valid solely from matching cell counts. ☐

*Solution.* Homotopy-equivalent CW models have isomorphic homology. ☐

**Exercise 20.31.** What algebraic relation corresponds to topological homotopy?

*Hint.* Pass to singular chains.

Distinguish a homotopy of continuous maps from a degree-raising algebraic operator on singular chains satisfying the boundary equation. ☐

*Solution.* Chain homotopy. ☐

**Exercise 20.32.** What chapter develops chain homotopy abstractly later?

*Hint.* Look ahead.

Follow the algebraic equation $g - f = \partial P + P\partial$ to VIII/22, where its symmetry, composition, and equivalence properties are studied independently of spaces. ☐

*Solution.* VIII/22. ☐

**Exercise 20.33.** What is the bridge to VIII/21?

*Hint.* Euler characteristic is computed from homology ranks.

For finite CW complexes, apply Euler–Poincare to the homology ranks; the finiteness condition makes the alternating sum an ordinary finite sum. ☐

*Solution.* Homotopy invariance of homology implies homotopy invariance of Euler characteristic for finite CW complexes. ☐

## 20.24 Solved problem dossiers

**Problem 20.34** (Cycle proof)**.** Prove chain-homotopic maps induce equal maps on homology.

*Solution.* If $z$ is a cycle, $g(z) - f(z) = \partial P(z) + P\partial z = \partial P(z)$, so the two images differ by a boundary. ☐

**Problem 20.35** (Homotopy inverse)**.** Let $f : X \to Y$, $g : Y \to X$ be homotopy inverses. Prove $f_*$ is an isomorphism.

*Solution.* Homotopy invariance gives $g_* f_* = (gf)_* = (\mathrm{id}_X)_*$ and $f_* g_* = \mathrm{id}_{H_*(Y)}$, so $g_*$ is the inverse. ☐

**Problem 20.36** (Contractible disk)**.** Compute the homology of $D^n$ from homotopy invariance.

*Solution.* $D^n \simeq *$, so $H_0(D^n) = \mathbb{Z}$ and all positive homology vanishes. ☐

**Problem 20.37** (Deformation retract annulus)**.** Compute the homology of an annulus by deformation retracting to its core circle.

*Solution.* The annulus is homotopy equivalent to $S^1$, so $H_0 = \mathbb{Z},\ H_1 = \mathbb{Z}$, and higher groups vanish. ☐

**Problem 20.38** (Punctured plane)**.** Compute $H_*(\mathbb{R}^2 \setminus \{0\})$.

*Solution.* Radial deformation retract gives $S^1$, hence $H_0 = \mathbb{Z},\ H_1 = \mathbb{Z}$, higher groups zero. ☐

**Problem 20.39** (Punctured Euclidean space)**.** Compute $H_*(\mathbb{R}^{n+1} \setminus \{0\})$.

*Solution.* It deformation retracts onto $S^n$, so homology is $\mathbb{Z}$ in degrees $0, n$ and zero otherwise. ☐

**Problem 20.40** (No retraction)**.** Give the homological proof that $D^{n+1}$ does not retract onto $S^n$.

*Solution.* A retraction would give $r_* i_* = \mathrm{id}$ on $H_n(S^n) = \mathbb{Z}$, but $i_*$ factors through $H_n(D^{n+1}) = 0$, contradiction. ☐

**Problem 20.41** (Null map)**.** If $f : S^n \to Y$ is null-homotopic, what is $f_*$ on $H_n$ for $n > 0$?

*Solution.* Zero, because $f$ is homotopic to a constant map and positive homology of a point vanishes. ☐

**Problem 20.42** (Degree obstruction)**.** Why can maps $S^n \to S^n$ of different degree not be homotopic?

*Solution.* The induced map on $H_n(S^n) \cong \mathbb{Z}$ is multiplication by degree. Homotopic maps induce the same homology map. ☐

**Problem 20.43** (Antipodal even sphere)**.** Use homology to distinguish the antipodal map from the identity on even $S^n$.

*Solution.* The identity induces multiplication by 1 on $H_n$; the antipodal map has degree $-1$ for even $n$, so the induced maps differ. □

**Problem 20.44** (Retract splitting)**.** Suppose $A$ is a retract of $X$. What algebraic information follows on homology?

*Solution.* The inclusion-induced map is split injective and the retraction-induced map is split surjective; $H_*(A)$ occurs as a direct summand in the appropriate algebraic sense. □

**Problem 20.45** (Noncontractible torus)**.** Why is $T^2$ not contractible?

*Solution.* $H_1(T^2) \cong \mathbb{Z}^2 \neq 0$, while a contractible space has zero positive homology. □

**Problem 20.46** (Projective noncontractibility)**.** Why is $\mathbb{R}P^2$ not contractible?

*Solution.* $H_1(\mathbb{R}P^2) \cong \mathbb{Z}/2 \neq 0$. □

**Problem 20.47** (Acyclic warning)**.** Why can vanishing reduced homology not prove contractibility?

*Solution.* Homology forgets nonabelian and higher homotopy information; there exist noncontractible acyclic spaces. □

**Problem 20.48** (Relative deformation)**.** If $(X, A)$ deformation retracts as a pair onto $(Y, B)$, what happens to relative homology?

*Solution.* The pair maps are homotopy equivalences through maps of pairs, so they induce isomorphisms $H_n(X, A) \cong H_n(Y, B)$. □

**Problem 20.49** (Prism in dimension zero)**.** What does the prism operator do to a singular 0-simplex $x$?

*Solution.* The homotopy path $t \mapsto H(x, t)$ is a singular 1-simplex whose boundary is $g(x) - f(x)$, exactly the degree-zero prism identity. □

**Problem 20.50** (Prism boundary meaning)**.** Geometrically, what are the endpoint terms in the boundary of a prism?

*Solution.* The top and bottom faces are the singular simplices obtained from $g \circ \sigma$ and $f \circ \sigma$; the side faces combine into $P(\partial\sigma)$. □

**Problem 20.51** (Identity homotopy)**.** What chain homotopy is needed between $\mathrm{id}_\#$ and itself?

*Solution.* The zero operator $P = 0$ satisfies $\mathrm{id} - \mathrm{id} = 0 = \partial P + P\partial$. □

**Problem 20.52** (Composition with homotopy)**.** If $f_0 \simeq f_1 : X \to Y$ and $h : Y \to Z$, compare $hf_0$ and $hf_1$ on homology.

*Solution.* They are homotopic, so $(hf_0)_* = (hf_1)_*$; equivalently $h_* f_{0*} = h_* f_{1*}$. □

**Problem 20.53** (Precomposition)**.** If $f_0 \simeq f_1 : X \to Y$ and $k : W \to X$, compare $f_0 k$ and $f_1 k$.

*Solution.* They are homotopic via $H(k(w), t)$, hence induce the same homology map. □

**Problem 20.54** (Euler consequence)**.** Why must finite CW complexes of different Euler characteristic fail to be homotopy equivalent?

*Solution.* Euler characteristic equals the alternating sum of homology ranks and is therefore preserved by homotopy equivalence. □

**Problem 20.55** (Bridge to algebraic chain homotopy)**.** What additional structure will VIII/22 study beyond this chapter?

*Solution.* It treats chain homotopies abstractly, including chain contractions and algebraic equivalences, rather than only the prism operator arising from topological homotopies. □

## 20.25 Supplementary worked examples

**Example 20.56** (A deformation retract does not change homology)**.** If $A \subset X$ is a deformation retract, the inclusion $i : A \hookrightarrow X$ and retraction $r : X \to A$ satisfy $ri = \mathrm{id}_A$ and $ir \simeq \mathrm{id}_X$. Homotopy invariance gives $i_* r_* = \mathrm{id}$ and $r_* i_* = \mathrm{id}$, so $H_*(A) \cong H_*(X)$.

**Example 20.57** (Contractible spaces have point homology)**.** If $X$ is contractible, $\mathrm{id}_X$ is homotopic to a constant map. Hence $X$ has the homology of a point:

$$H_0(X) \cong \mathbb{Z}, \qquad H_n(X) = 0 \quad (n > 0)$$

when $X$ is nonempty and path connected.

**Example 20.58** (Homotopic maps act identically on homology)**.** For $f, g : X \to Y$ with $f \simeq g$, the induced chain maps are chain homotopic. Therefore $f_* = g_* : H_n(X) \to H_n(Y)$ in every degree. This is the algebraic mechanism behind homotopy invariance.

## 20.26 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 20.59** (Constant map in positive degree)**.** What map does a constant map induce on $H_n$ for $n > 0$?

*Hint.* It factors through a point. □

*Solution.* It induces the zero map because $H_n(\{*\}) = 0$ for $n > 0$. □

**Exercise 20.60** (Interval homology)**.** Compute $H_*(I)$ using homotopy invariance.

*Hint.* $I$ contracts to a point. □

*Solution.* $H_0(I) \cong \mathbb{Z}$ and $H_n(I) = 0$ for $n > 0$. □

**Exercise 20.61** (Cylinder homology)**.** Compute $H_*(X \times I)$ in terms of $H_*(X)$.

*Hint.* Project to $X$. □

*Solution.* $X \times I$ deformation retracts onto $X \times \{0\}$, so $H_*(X \times I) \cong H_*(X)$. □

**Exercise 20.62** (Punctured plane)**.** Compute the homology of $\mathbb{R}^2 - \{0\}$.

*Hint.* Radially deformation retract onto $S^1$. □

*Solution.* $H_0 \cong \mathbb{Z}$, $H_1 \cong \mathbb{Z}$, and all higher groups vanish. □

**Exercise 20.63** (Annulus)**.** Compute the homology of an annulus.

*Hint.* Deformation retract onto its core circle. □

*Solution.* It has the homology of $S^1$: $\mathbb{Z}$ in degrees 0 and 1, zero otherwise. □

## Proofs

**Exercise 20.64** (Homotopic maps)**.** Prove that if $f \simeq g$, then $f_* = g_*$ on homology.

*Hint.* Use the prism operator or the chain-homotopy theorem. □

*Solution.* The homotopy produces a chain homotopy $P$ with $g_\# - f_\# = \partial P + P\partial$; boundaries differ by boundaries on cycles, so induced homology maps agree. □

**Exercise 20.65** (Homotopy equivalence)**.** Prove a homotopy equivalence induces homology isomorphisms.

*Hint.* Apply homotopy invariance to the two homotopy-inverse composites. □

*Solution.* If $g$ is a homotopy inverse to $f$, then $g_*f_* = \mathrm{id}$ and $f_*g_* = \mathrm{id}$, so $f_*$ is an isomorphism. □

**Exercise 20.66** (Retract injection)**.** If $A$ is a retract of $X$, prove $H_n(A) \to H_n(X)$ is injective.

*Hint.* Use $r \circ i = \mathrm{id}_A$. □

*Solution.* On homology, $r_*i_* = \mathrm{id}$, so $i_*$ has a left inverse and is injective. □

**Exercise 20.67** (Contractible reduced homology)**.** Prove a nonempty contractible space has zero reduced homology.

*Hint.* Compare it with a point. □

*Solution.* Homotopy equivalence with a point gives $\widetilde{H}_n(X) = 0$ in every degree. □

## Counterexamples and hypothesis tests

**Exercise 20.68** (Converse failure)**.** Test: spaces with isomorphic homology groups are homotopy equivalent.

*Hint.* Homology is not a complete invariant. □

*Solution.* False. Distinct homotopy types can share all homology groups; homology forgets information such as much of the fundamental group and higher operations. □

**Exercise 20.69** (Acyclic means contractible)**.** Test: if $\widetilde{H}_*(X) = 0$, then $X$ is contractible.

*Hint.* Acyclicity is weaker than contractibility. □

*Solution.* False in general; there are noncontractible acyclic spaces. □

**Exercise 20.70** (Homology detects all homotopies)**.** Test: if $f_* = g_*$ on all homology groups, then $f \simeq g$.

*Hint.* Induced homology maps are coarse invariants. □

*Solution.* False. Different homotopy classes can induce the same homology maps. □

## Applications and investigations

**Exercise 20.71** (Simplifying a space)**.** Why is deformation retraction a computational tool rather than only a qualitative statement?

*Hint.* It replaces a space by a smaller homotopy-equivalent model. □

*Solution.* Any homology computation may be carried out on the retract, often reducing a complicated geometric space to a graph, sphere, or CW skeleton. □

**Exercise 20.72** (Model validation)**.** How can homology invariance help check a numerical or combinatorial simplification of a shape?

*Hint.* A claimed homotopy-preserving simplification should preserve homology. □

*Solution.* If the simplification is intended to be a deformation retract or homotopy equivalence, differing Betti numbers expose a topological error immediately. □

## Challenge problems

**Exercise 20.73** (Wedge computation)**.** Use deformation and known homology to determine $H_1$ of a figure-eight graph.

*Hint.* It is already a wedge $S^1 \vee S^1$. □

*Solution.* $H_1 \cong \mathbb{Z}^2$; the two loop classes are independent. □

**Exercise 20.74** (Invariant hierarchy)**.** Explain why homotopy invariance is essential when defining any topological invariant from a chain model.

*Hint.* Different models of the same homotopy type should agree. □

*Solution.* It guarantees the chain-level construction descends to the homotopy category: homotopy-equivalent spaces receive canonically isomorphic homology groups. □

# Chapter 21

# Euler Characteristic

Euler characteristic compresses a finite cell decomposition into a single integer. Remarkably, the same integer can be computed from homology ranks, so it is independent of the chosen finite CW or simplicial structure and is preserved by homotopy equivalence. This chapter develops the combinatorial, homological, and structural forms of the invariant.

## 21.1 Cell-count definition

Let $X$ be a finite CW complex with $c_n$ cells of dimension $n$.

**Definition 21.1** (Euler characteristic)**.**

$$\chi(X) = \sum_{n \geq 0} (-1)^n c_n.$$

Only finitely many terms are nonzero.

## 21.2 Simplicial form

If a finite simplicial complex has $f_n$ $n$-simplices, then

$$\chi(K) = \sum_n (-1)^n f_n.$$

For a triangulated space $X \cong |K|$, this agrees with the CW definition.

## 21.3 Basic examples

For a finite discrete set with $m$ points,

$$\chi = m.$$

For an interval,

$$\chi = 2 - 1 = 1.$$

For a polygonal circle,

$$\chi = V - E = 0.$$

For the standard two-cell sphere structure,

$$\chi(S^n) = 1 + (-1)^n.$$

## 21.4 Graphs

For a finite graph,

$$\chi(X) = V - E.$$

If the graph is connected and has fundamental-group rank $r$, then

$$r = E - V + 1,$$

so

$$\chi(X) = 1 - r.$$

Thus a tree has

$$\chi = 1.$$

## 21.5 Closed orientable surfaces

The standard CW structure on a genus-$g$ orientable surface has one 0-cell, $2g$ 1-cells, and one 2-cell. Hence

$$\chi(\Sigma_g) = 2 - 2g.$$

In particular,

$$\chi(S^2) = 2, \qquad \chi(T^2) = 0.$$

## 21.6 Real projective spaces

The standard CW structure on $\mathbb{R}P^n$ has one cell in every dimension $0, \ldots, n$, so

$$\chi(\mathbb{R}P^n) = \sum_{k=0}^{n} (-1)^k.$$

Therefore

$$\chi(\mathbb{R}P^n) = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

## 21.7 Euler–Poincare formula

Let $X$ be a finite CW complex. Write

$$b_n = \operatorname{rank} H_n(X; \mathbb{Z}).$$

**Theorem 21.2** (Euler–Poincare)**.**

$$\chi(X) = \sum_n (-1)^n b_n.$$

Torsion in homology does not contribute to the rank sum.

## 21.8 Algebraic proof

For a finite chain complex of finite-rank free abelian groups,

$$C_n,$$

write

$$Z_n = \ker \partial_n, \qquad B_n = \operatorname{im} \partial_{n+1}.$$

From

$$0 \to Z_n \to C_n \to B_{n-1} \to 0$$

we obtain

$$\operatorname{rank} C_n = \operatorname{rank} Z_n + \operatorname{rank} B_{n-1}.$$

From

$$0 \to B_n \to Z_n \to H_n \to 0$$

we obtain

$$\operatorname{rank} Z_n = \operatorname{rank} B_n + \operatorname{rank} H_n.$$

Alternating sums telescope, leaving the homology-rank formula.

## 21.9 Independence of cell structure

**Corollary 21.3** (Well-definedness)**.** *The alternating cell count is independent of the chosen finite CW structure.*

Indeed both cell decompositions compute the same homology ranks.

## 21.10 Homotopy invariance

**Corollary 21.4** (Homotopy invariance)**.** *If finite CW complexes $X$ and $Y$ are homotopy equivalent, then*

$$\chi(X) = \chi(Y).$$

This follows immediately from VIII/20.

## 21.11 Contractible finite complexes

If $X$ is a nonempty finite contractible CW complex, then

$$H_0(X) \cong \mathbb{Z}, \qquad H_n(X) = 0 \ (n > 0),$$

so

$$\chi(X) = 1.$$

Therefore a finite CW complex with

$$\chi(X) \neq 1$$

cannot be contractible.

The converse is false.

## 21.12 Disjoint unions

For finite CW complexes,

$$\chi(X \sqcup Y) = \chi(X) + \chi(Y).$$

This is immediate from either cell counts or homology direct sums.

## 21.13 Wedges

For connected finite CW complexes with chosen basepoints,

$$\chi(X \vee Y) = \chi(X) + \chi(Y) - 1.$$

The subtraction corrects for identifying two zero-cells into one.
More generally,

$$\chi\left(\bigvee_{j=1}^{r} X_j\right) = \sum_j \chi(X_j) - (r-1).$$

## 21.14 Product formula

**Theorem 21.5** (Product formula)**.** *For finite CW complexes,*

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

One proof uses product cells: an $i$-cell times a $j$-cell contributes an $(i+j)$-cell, and the alternating double sum factors.

## 21.15 Examples of products

Since

$$\chi(S^1) = 0,$$

every finite CW product

$$S^1 \times X$$

has

$$\chi = 0.$$

For example,

$$\chi(T^n) = 0$$

for every $n \geq 1$.
Also,

$$\chi(S^2 \times S^2) = 4.$$

## 21.16 Inclusion–exclusion

Suppose finite subcomplexes $A, B$ cover a finite CW complex $X$:

$$X = A \cup B.$$

**Theorem 21.6** (Euler inclusion–exclusion)**.**

$$\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B).$$

This follows from cell counts when the decompositions are compatible, and more generally from Mayer–Vietoris/homological arguments.

## 21.17 Attaching one cell

If $Y$ is obtained from $X$ by attaching one $n$-cell, then

$$\chi(Y) = \chi(X) + (-1)^n.$$

Thus:

- adding a 0-cell increases $\chi$ by 1;
- adding a 1-cell decreases $\chi$ by 1;
- adding a 2-cell increases $\chi$ by 1.

The attaching map does not affect this raw Euler change.

## 21.18 Relative Euler characteristic

For a finite CW pair $(X, A)$, define

$$\chi(X, A) = \chi(X) - \chi(A).$$

Equivalently, if the relative homology groups have finite rank,

$$\chi(X, A) = \sum_n (-1)^n \operatorname{rank} H_n(X, A).$$

## 21.19 Exact sequences and Euler sums

For a finite exact sequence of finite-rank abelian groups

$$0 \to A_0 \to A_1 \to \cdots \to A_m \to 0,$$

the alternating sum of ranks is zero.

This elementary fact is the algebraic reason Euler characteristic behaves well with long exact sequences.

## 21.20 Euler characteristic and coverings

If

$$p : \widetilde{X} \to X$$

is a finite $d$-sheeted covering of finite CW complexes, then

$$\chi(\widetilde{X}) = d\,\chi(X).$$

Each cell downstairs lifts to $d$ cells upstairs.

## 21.21 Covering examples

For the $d$-fold circle cover,

$$0 = \chi(S^1) = d \cdot 0.$$

For a finite-sheeted covering

$$\widetilde{\Sigma} \to \Sigma_g$$

of closed orientable surfaces,

$$\chi(\widetilde{\Sigma}) = d(2 - 2g).$$

This constrains the genus upstairs.

## 21.22 Odd-dimensional closed manifolds preview

For closed orientable odd-dimensional manifolds, Poincare duality later implies

$$\chi(M) = 0.$$

That theorem belongs to VIII/33; here it is only a preview of how deeper homological symmetry controls Euler characteristic.

## 21.23 What Euler characteristic cannot see

Euler characteristic is extremely coarse. Spaces with the same $\chi$ can have very different topology.

For example,

$$\chi(S^1) = 0 \qquad \text{and} \qquad \chi(T^2) = 0,$$

yet their homology and fundamental groups are different.

## 21.24 Bridge to homological machinery

The Euler–Poincare argument is a rank shadow of chain-level algebra. The next part develops that algebra more deeply through chain homotopies, contractions, mapping cones, coefficient changes, universal coefficient formulas, and products.

## 21.25 Exercises

**Exercise 21.7.** Define $\chi(X)$ from finite CW cells.

*Hint.* Take an alternating count.

Specify that $c_n$ counts $n$-cells and that only finitely many cells occur; signs depend on cell dimension, not on orientations. □

*Solution.* $\chi(X) = \sum_n (-1)^n c_n$. □

**Exercise 21.8.** What is the simplicial formula?

*Hint.* Replace cell counts by simplex counts.

Count simplices of every dimension, including vertices, exactly once; do not count each simplex again every time it is a face of a larger one. □

*Solution.* $\chi(K) = \sum_n (-1)^n f_n$. □

**Exercise 21.9.** Compute $\chi(*)$.

*Hint.* One 0-cell.

Use the cell decomposition with one degree-zero generator and no positive-dimensional cells to fix the normalization. □

*Solution.* 1. □

**Exercise 21.10.** Compute $\chi(I)$.

*Hint.* Two vertices and one edge.

Compare the endpoint-and-edge count with the homology of a point; these two calculations should agree although the cell counts differ. □

*Solution.* $2 - 1 = 1$. □

**Exercise 21.11.** Compute $\chi(S^1)$.

*Hint.* One 0-cell and one 1-cell.

Track the contribution of the common vertex and the loop edge separately; the endpoint identification does not remove the vertex cell. □

*Solution.* 0. □

**Exercise 21.12.** Compute $\chi(S^n)$.

*Hint.* One 0-cell and one $n$-cell.

The top cell contributes with sign $(-1)^n$; check both parity cases and treat $S^0$ as two zero-cells. □

*Solution.* $1 + (-1)^n$. □

**Exercise 21.13.** For a finite graph, what is $\chi$?

*Hint.* Only dimensions 0 and 1.

All terms above dimension one vanish. The formula does not require graph connectedness, although rank formulas may. □

*Solution.* $V - E$. □

**Exercise 21.14.** For a connected graph of rank $r$, what is $\chi$?

*Hint.* Use $r = E - V + 1$.

Solve the connected-graph rank equation for $V - E$; the connectedness assumption supplies the single degree-zero homology generator. □

*Solution.* $1 - r$. □

**Exercise 21.15.** What is $\chi$ of a tree?

*Hint.* Rank zero.

For a finite tree, use $E = V - 1$, or compare its homology with a point; avoid an undefined infinite alternating count. □

*Solution.* 1. □

**Exercise 21.16.** Compute $\chi(\Sigma_g)$.

*Hint.* Use 1, $2g$, 1 cells.

Insert the surface's one vertex, $2g$ edges, and one face with their respective dimension signs. □

*Solution.* $2 - 2g$. □

**Exercise 21.17.** Compute $\chi(\mathbb{R}P^n)$.

*Hint.* One cell per dimension.

Pair consecutive terms in $1 - 1 + 1 - \cdots + (-1)^n$; determine whether an unpaired term remains. □

*Solution.* 1 for even $n$, 0 for odd $n$. □

**Exercise 21.18.** State Euler–Poincare.

*Hint.* Use homology ranks.

Use ranks of integral homology, equivalently dimensions after tensoring with $\mathbb{Q}$, and require finite CW type for this finite formula. □

*Solution.* $\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X)$. □

**Exercise 21.19.** Does torsion contribute to Euler–Poincare?

*Hint.* Ranks ignore finite torsion.

A finite cyclic group becomes zero after tensoring with $\mathbb{Q}$; its order is not a Betti number. □

*Solution.* No. □

**Exercise 21.20.** Why is Euler characteristic independent of finite CW structure?

*Hint.* Both compute the same homology rank sum.

Compute the alternating homology rank sum for either CW decomposition; singular homology depends on the space rather than its particular cells. □

*Solution.* Euler–Poincare makes the cell count equal a homological invariant. □

**Exercise 21.21.** Is Euler characteristic a homotopy invariant for finite CW complexes?

*Hint.* Use homology invariance.

A homotopy equivalence induces isomorphisms in each homology degree, hence preserves every rank appearing in the finite Euler sum. □

*Solution.* Yes. □

**Exercise 21.22.** What is $\chi$ of a nonempty finite contractible CW complex?

*Hint.* Same homology as a point.

Ordinary homology has one degree-zero generator and no positive-degree classes; reduced acyclicity alone is not a proof of contractibility. □

*Solution.* 1. □

**Exercise 21.23.** State disjoint-union additivity.

*Hint.* Cell counts add.

The cell sets of a finite disjoint union are disjoint in every dimension; add their signed contributions before simplifying. □

*Solution.* $\chi(X \sqcup Y) = \chi(X) + \chi(Y)$. □

**Exercise 21.24.** State wedge additivity.

*Hint.* One basepoint is identified.

Choose basepoints that are zero-cells; gluing identifies two such cells as one, so compare the count with disjoint-union additivity. □

*Solution.* $\chi(X \vee Y) = \chi(X) + \chi(Y) - 1$. □

**Exercise 21.25.** State the product formula.

*Hint.* Product cell dimensions add.

A product of cells of dimensions $i$ and $j$ contributes sign $(-1)^{i+j} = (-1)^i(-1)^j$; factor the finite double sum. ☐

*Solution.* $\chi(X \times Y) = \chi(X)\chi(Y)$. ☐

**Exercise 21.26.** Compute $\chi(T^n)$ for $n \geq 1$.

*Hint.* It contains an $S^1$ factor.

Write the torus as a finite product of circles and apply multiplicativity, noting why the restriction $n \geq 1$ excludes the point. ☐

*Solution.* 0. ☐

**Exercise 21.27.** State inclusion–exclusion.

*Hint.* Subtract the overlap.

For compatible finite subcomplexes, a cell in $A \cap B$ has been counted twice; subtract its contribution exactly once. ☐

*Solution.* $\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B)$. ☐

**Exercise 21.28.** What happens to $\chi$ when one $n$-cell is attached?

*Hint.* Add its alternating contribution.

Only the count in dimension $n$ changes. The attaching degree can affect homology without changing this Euler increment. ☐

*Solution.* $\chi$ changes by $(-1)^n$. ☐

**Exercise 21.29.** How does a $d$-sheeted finite cover affect $\chi$?

*Hint.* Every cell lifts $d$ times.

Lift the CW structure from the base and count $d$ copies of every open cell, then take the same alternating sum upstairs. ☐

*Solution.* $\chi(\widetilde{X}) = d\chi(X)$. ☐

**Exercise 21.30.** What is the bridge beyond VIII/21?

*Hint.* Move from ranks to chain-level algebra.

Rank calculations discard torsion and maps; the next machinery retains chain homotopies, cone differentials, and coefficient-change information that Euler characteristic cannot see. ☐

*Solution.* Part V develops chain homotopies, mapping cones, coefficients, UCT, and products. ☐

## 21.26 Solved problem dossiers

**Problem 21.31** (Triangle disk)**.** Compute Euler characteristic of a filled triangle.

*Solution.* There are 3 vertices, 3 edges, and 1 face, so $\chi = 3 - 3 + 1 = 1$. ☐

**Problem 21.32** (Tetrahedral sphere)**.** Compute $\chi$ of the boundary of a tetrahedron.

*Solution.* $V = 4, E = 6, F = 4$, so $\chi = 4 - 6 + 4 = 2$. ☐

**Problem 21.33** (Square with diagonal). Compute $\chi$ of a square triangulated by one diagonal.

*Solution.* $V = 4, E = 5, F = 2$, so $\chi = 4 - 5 + 2 = 1$. ☐

**Problem 21.34** (Graph rank). A connected finite graph has $V = 10, E = 13$. Compute its fundamental-group rank and Euler characteristic.

*Solution.* The rank is $E - V + 1 = 4$, and $\chi = V - E = -3 = 1 - 4$. ☐

**Problem 21.35** (Rose). Compute $\chi$ of the wedge of $r$ circles.

*Solution.* One vertex and $r$ one-cells give $\chi = 1 - r$. ☐

**Problem 21.36** (Genus three). Compute $\chi(\Sigma_3)$.

*Solution.* $2 - 2 \cdot 3 = -4$. ☐

**Problem 21.37** (Projective plane). Compute $\chi(\mathbb{R}P^2)$ from cells and homology.

*Solution.* Cell count gives $1 - 1 + 1 = 1$. Homology ranks are $1, 0, 0$ since $H_1$ is torsion, so Euler–Poincare also gives 1. ☐

**Problem 21.38** (Projective 3-space). Compute $\chi(\mathbb{R}P^3)$.

*Solution.* Cell count $1 - 1 + 1 - 1 = 0$. Homology ranks $b_0 = b_3 = 1$ give $1 - 1 = 0$. ☐

**Problem 21.39** (Torus). Compute $\chi(T^2)$ from homology.

*Solution.* The Betti numbers are $1, 2, 1$, so $\chi = 1 - 2 + 1 = 0$. ☐

**Problem 21.40** (Sphere product). Compute $\chi(S^2 \times S^2)$.

*Solution.* $\chi(S^2) = 2$, so the product formula gives 4. ☐

**Problem 21.41** (Circle product). Why does $S^1 \times X$ have zero Euler characteristic for finite CW $X$?

*Solution.* $\chi(S^1) = 0$, so the product formula gives $0 \cdot \chi(X) = 0$. ☐

**Problem 21.42** (Wedge of sphere and circle). Compute $\chi(S^2 \vee S^1)$.

*Solution.* $\chi(S^2) + \chi(S^1) - 1 = 2 + 0 - 1 = 1$. ☐

**Problem 21.43** (Three-sphere wedge). Compute $\chi(S^2 \vee S^2 \vee S^2)$.

*Solution.* The sum is 6, and identifying three basepoints costs 2, so $\chi = 4$. ☐

**Problem 21.44** (Attach a one-cell). If $X$ has $\chi = 5$, what is the Euler characteristic after attaching one 1-cell?

*Solution.* It decreases by 1, giving 4. ☐

**Problem 21.45** (Attach a two-cell). Continue from the previous space by attaching one 2-cell.

*Solution.* It increases by 1, returning to 5. ☐

**Problem 21.46** (Contractibility obstruction). A finite CW complex has $\chi = 0$. What can you conclude?

*Solution.* It cannot be contractible, since every nonempty finite contractible CW complex has Euler characteristic 1. ☐

**Problem 21.47** (Same Euler warning)**.** Why can $\chi = 0$ not distinguish $S^1$ from $T^2$?

*Solution.* Both have Euler characteristic zero, but their homology and fundamental groups differ. Euler characteristic is only one alternating integer. ☐

**Problem 21.48** (Covering surface genus)**.** Let $\widetilde{\Sigma} \to \Sigma_2$ be a 3-sheeted cover of closed orientable surfaces. Find the genus upstairs.

*Solution.* $\chi(\Sigma_2) = -2$, so upstairs $\chi = -6$. Solve $2 - 2\tilde{g} = -6$, giving $\tilde{g} = 4$. ☐

**Problem 21.49** (Cover of a torus)**.** What Euler characteristic must every finite-sheeted cover of $T^2$ have?

*Solution.* Since $\chi(T^2) = 0$, the covering formula gives 0. ☐

**Problem 21.50** (Relative Euler disk pair)**.** Compute $\chi(D^n, S^{n-1})$.

*Solution.* $\chi(D^n) - \chi(S^{n-1}) = 1 - [1 + (-1)^{n-1}] = (-1)^n$, matching the single relative $n$-cell. ☐

**Problem 21.51** (Exact sequence rank cancellation)**.** Why does a finite exact sequence have zero alternating rank sum?

*Solution.* Each image rank appears once with each sign when the groups are decomposed into successive kernels/images, so the contributions telescope. ☐

**Problem 21.52** (Euler–Poincare torsion)**.** Why does $\mathbb{Z}/5$ in homology contribute nothing to Euler characteristic?

*Solution.* Euler–Poincare uses free ranks (Betti numbers); a finite torsion group has rank zero. ☐

**Problem 21.53** (Subdivision)**.** Why does barycentric subdivision preserve Euler characteristic?

*Solution.* It preserves the homeomorphism type, hence homology and Euler characteristic. Combinatorially the larger simplex counts have the same alternating sum. ☐

**Problem 21.54** (Bridge to duality)**.** What later theorem forces $\chi = 0$ for closed orientable odd-dimensional manifolds?

*Solution.* Poincare duality in VIII/33 pairs Betti numbers symmetrically across complementary dimensions, causing alternating cancellation. ☐

## 21.27 Supplementary worked examples

**Example 21.55** (Euler characteristic of a sphere)**.** The CW structure on $S^n$ with one 0-cell and one $n$-cell gives

$$\chi(S^n) = 1 + (-1)^n.$$

The same number is obtained from Betti numbers, illustrating the Euler–Poincare identity.

**Example 21.56** (Euler characteristic of a closed orientable surface)**.** A genus-$g$ surface has a CW structure with one vertex, $2g$ one-cells, and one two-cell. Therefore

$$\chi(\Sigma_g) = 1 - 2g + 1 = 2 - 2g.$$

**Example 21.57** (Inclusion–exclusion)**.** For finite CW subcomplexes $A, B$ with $X = A \cup B$,

$$\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B).$$

This is the Euler-characteristic shadow of the Mayer–Vietoris exact sequence.

## 21.28 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 21.58** (Circle)**.** Compute $\chi(S^1)$.

*Hint.* Use one 0-cell and one 1-cell. □

*Solution.* $\chi(S^1) = 1 - 1 = 0$. □

**Exercise 21.59** (Two-sphere)**.** Compute $\chi(S^2)$.

*Hint.* Use one 0-cell and one 2-cell. □

*Solution.* $\chi(S^2) = 2$. □

**Exercise 21.60** (Torus)**.** Compute $\chi(T^2)$ from its standard CW structure.

*Hint.* Use cell counts $1, 2, 1$. □

*Solution.* $\chi(T^2) = 1 - 2 + 1 = 0$. □

**Exercise 21.61** (Wedge of circles)**.** Compute $\chi(\bigvee_{i=1}^r S^1)$.

*Hint.* Use one vertex and $r$ edges. □

*Solution.* $\chi = 1 - r$. □

**Exercise 21.62** (Projective plane)**.** Compute $\chi(\mathbb{RP}^2)$ from its one-cell-per-dimension CW structure.

*Hint.* Count dimensions $0, 1, 2$. □

*Solution.* $\chi(\mathbb{RP}^2) = 1 - 1 + 1 = 1$. □

## Proofs

**Exercise 21.63** (Euler–Poincare)**.** Prove for a finite free chain complex that $\sum(-1)^n \operatorname{rank} C_n = \sum(-1)^n \operatorname{rank} H_n$.

*Hint.* Split ranks into boundaries, homology, and shifted boundaries. ☐

*Solution.* Since $\operatorname{rank} C_n = \operatorname{rank} B_n + \operatorname{rank} H_n + \operatorname{rank} B_{n-1}$, the boundary ranks cancel pairwise in the alternating sum. ☐

**Exercise 21.64** (Homotopy invariance of chi)**.** Prove $\chi$ is a homotopy invariant for finite CW complexes.

*Hint.* Use homology ranks. ☐

*Solution.* Homotopy-equivalent finite CW complexes have isomorphic homology groups, so the alternating sum of Betti numbers agrees. ☐

**Exercise 21.65** (Wedge formula)**.** Prove $\chi(X \vee Y) = \chi(X) + \chi(Y) - 1$ for connected finite CW complexes.

*Hint.* Apply inclusion–exclusion at the wedge point. ☐

*Solution.* The intersection is a point of Euler characteristic 1, yielding the formula. ☐

**Exercise 21.66** (Product formula)**.** Prove $\chi(X \times Y) = \chi(X)\chi(Y)$ for finite CW complexes.

*Hint.* Product cells have dimensions that add. ☐

*Solution.* Summing $(-1)^{p+q} c_p(X) c_q(Y)$ factors into the product of the two alternating sums. ☐

## Counterexamples and hypothesis tests

**Exercise 21.67** (Cell decomposition dependence)**.** Test: $\chi(X)$ depends on the chosen finite CW decomposition.

*Hint.* Compare the homology formula. ☐

*Solution.* False. Different finite CW structures give the same alternating sum because both equal the homological Euler characteristic. ☐

**Exercise 21.68** (Zero chi means torus)**.** Test: a connected closed surface with $\chi = 0$ must be a torus.

*Hint.* Remember nonorientable surfaces. ☐

*Solution.* False without orientability; the Klein bottle also has Euler characteristic zero. ☐

**Exercise 21.69** (Chi determines homology)**.** Test: equal Euler characteristics imply isomorphic homology groups.

*Hint.* An alternating sum loses information. ☐

*Solution.* False. Many different Betti-number patterns can have the same alternating sum. ☐

## Applications and investigations

**Exercise 21.70** (Mesh sanity check)**.** Explain the formula $\chi = V - E + F$ for a finite triangulated surface.

*Hint.* Use simplices as cells. □

*Solution.* Vertices, edges, and faces are the 0-, 1-, and 2-cells, so the cellular alternating sum is $V - E + F$. □

**Exercise 21.71** (Genus recovery)**.** For a connected closed orientable surface, recover genus from $\chi$.

*Hint.* Use $\chi = 2 - 2g$. □

*Solution.* $g = 1 - \chi/2$. □

## Challenge problems

**Exercise 21.72** (Connected sum)**.** Derive $\chi(M\#N) = \chi(M) + \chi(N) - 2$ for closed surfaces.

*Hint.* Remove a disk from each and glue along $S^1$. □

*Solution.* Removing each open disk lowers $\chi$ by 1; gluing along a circle contributes no correction since $\chi(S^1) = 0$, giving the formula. □

**Exercise 21.73** (Euler synthesis)**.** Explain three independent routes to $\chi$ and when each is useful.

*Hint.* Think cells, homology, and inclusion–exclusion. □

*Solution.* One may alternate cell counts for a CW model, alternate Betti numbers after homology computation, or decompose the space and use additivity; agreement is a strong consistency check. □

# Part V

# Homological Machinery

# Chapter 22

# Chain Homotopies

Chain homotopy is the algebraic shadow of topological homotopy. It compares chain maps without requiring equality and is exactly the equivalence relation needed for homology: chain-homotopic maps induce the same maps on homology.

## 22.1 Definition

**Definition 22.1** (Chain homotopy). Let $f, g : C_\bullet \to D_\bullet$ be chain maps. A chain homotopy $s : f \simeq g$ is a family

$$s_n : C_n \to D_{n+1}$$

such that

$$g - f = \partial s + s\partial.$$

## 22.2 Degree convention

The chain differential has degree $-1$, while a chain homotopy has degree $+1$. Thus both terms

$$\partial s, \qquad s\partial$$

have degree 0.

## 22.3 Homology consequence

**Theorem 22.2** (Homology invariance). *If $f \simeq g$ as chain maps, then*

$$H_n(f) = H_n(g)$$

*for every $n$.*

*Proof.* For a cycle $z$,

$$g(z) - f(z) = \partial s(z) + s(\partial z) = \partial s(z),$$

so the two images differ by a boundary. □

## 22.4 Reflexivity

The zero operator gives

$$f \simeq f.$$

## 22.5 Symmetry

If $s : f \simeq g$, then

$$-s : g \simeq f.$$

## 22.6 Transitivity

If $s : f \simeq g$ and $t : g \simeq h$, then

$$s + t : f \simeq h.$$

Hence chain homotopy is an equivalence relation on chain maps.

## 22.7 Compatibility with composition

If $f \simeq g : C \to D$ and $u : D \to E$ is a chain map, then

$$uf \simeq ug.$$

Similarly, if $v : B \to C$, then

$$fv \simeq gv.$$

## 22.8 Chain-homotopy equivalence

**Definition 22.3** (Chain-homotopy equivalence)**.** Chain complexes $C$ and $D$ are chain-homotopy equivalent if there are chain maps

$$f : C \to D, \qquad g : D \to C$$

with

$$gf \simeq \mathrm{id}_C, \qquad fg \simeq \mathrm{id}_D .$$

**Theorem 22.4.** *Chain-homotopy-equivalent complexes have naturally isomorphic homology groups.*

## 22.9 Null-homotopic chain maps

A chain map $f : C \to D$ is null-homotopic if

$$f \simeq 0.$$

Equivalently there are maps $s_n : C_n \to D_{n+1}$ satisfying

$$f = \partial s + s\partial.$$

Every null-homotopic chain map induces the zero map on homology.

## 22.10 The prism operator revisited

VIII/20 constructed, from a homotopy $F : X \times I \to Y$, a prism operator

$$P : C_n^{\mathrm{sing}}(X) \to C_{n+1}^{\mathrm{sing}}(Y)$$

with

$$g_\# - f_\# = \partial P + P\partial.$$

Thus topological homotopy becomes chain homotopy.

## 22.11 Homotopy of simplicial maps

After subdivision or through appropriate simplicial homotopy data, simplicial maps can likewise give chain-homotopic simplicial chain maps.

## 22.12 Chain homotopy and exact complexes

An exact complex need not automatically come equipped with a contraction. Chain homotopy provides the language for stating the stronger property that the identity itself is null-homotopic.

## 22.13 Hom-complex viewpoint

For graded homomorphisms $T : C \to D$, define

$$\delta(T) = \partial_D T - (-1)^{|T|} T \partial_C.$$

For $|s| = 1$,

$$\delta(s) = \partial s + s\partial.$$

Thus $g - f = \delta(s)$: chain-homotopic maps differ by a coboundary in the Hom-complex.

## 22.14 Matrices

For finite free complexes, a chain homotopy is represented by matrices $S_n$ satisfying

$$G_n - F_n = D^D_{n+1} S_n + S_{n-1} D^C_n.$$

This converts chain-homotopy questions into linear Diophantine equations.

## 22.15 Direct sums

Chain homotopies add componentwise under direct sums.

## 22.16 Tensor-product preview

For tensor products, chain homotopies interact with Koszul signs. The full product machinery is deferred to VIII/27.

## 22.17 Relative chains

A homotopy of maps of pairs produces a chain homotopy on relative chain complexes because the prism operator preserves the distinguished subcomplex.

## 22.18 Augmented complexes

For augmented complexes, chain homotopies may be required to respect the augmentation. This is useful when reduced homology is the natural invariant.

## 22.19 Uniqueness up to chain homotopy

Many chain-level constructions are not canonical on the nose but are canonical up to chain homotopy. Since homology kills this ambiguity, homology maps remain canonical.

## 22.20 Why this notion matters

Chain homotopy is the correct algebraic equivalence because it is:

- weaker than equality of chain maps;
- strong enough to force equality on homology;
- stable under composition;
- produced naturally by topological homotopies.

## 22.21 Bridge to contractions

A chain contraction is precisely a chain homotopy

$$\mathrm{id}_C \simeq 0.$$

VIII/23 studies this special but powerful case and its relation to acyclic complexes.

## 22.22 Exercises

**Exercise 22.5.** Exercise 1

*Hint.* For the identity discussed in the solution, write $s_n : C_n \to D_{n+1}$ and check the types of both terms comparing $g_n$ and $f_n$. □

*Solution.* If $s : f \simeq g$, write the identity $g - f = \partial s + s\partial$. □

**Exercise 22.6.** Exercise 2

*Hint.* A differential has degree $-1$; choose the degree of $s$ so that $\partial s + s\partial$ has degree zero. □

*Solution.* The homotopy operator has degree $+1$. □

**Exercise 22.7.** Exercise 3

*Hint.* To compare a chain map with itself, substitute $g = f$ and test the zero operator in every degree. □

*Solution.* The zero homotopy proves reflexivity. □

**Exercise 22.8.** Exercise 4

*Hint.* Negate the entire equation $g - f = \partial s + s\partial$; this reverses the endpoint order without changing the degrees. □

*Solution.* If $s : f \simeq g$, then $-s : g \simeq f$. □

**Exercise 22.9.** Exercise 5

*Hint.* Add the equations for $g - f$ and $h - g$; the middle chain map cancels and the operators add. ☐

*Solution.* If $s : f \simeq g$ and $t : g \simeq h$, then $s + t : f \simeq h$. ☐

**Exercise 22.10.** Exercise 6

*Hint.* For a cycle $z$, the summand $s\partial z$ vanishes; identify the chain whose boundary is $g(z) - f(z)$. ☐

*Solution.* On cycles, $g(z) - f(z) = \partial s(z)$. ☐

**Exercise 22.11.** Exercise 7

*Hint.* Two cycles define the same homology class precisely when their difference is a boundary; apply this to the images in the preceding identity. ☐

*Solution.* Therefore chain-homotopic maps induce the same homology map. ☐

**Exercise 22.12.** Exercise 8

*Hint.* To write $f = \partial s + s\partial$, take a homotopy from zero to $f$; reversing its direction negates $s$ under this chapter's sign convention. ☐

*Solution.* A null-homotopic map satisfies $f = \partial s + s\partial$. ☐

**Exercise 22.13.** Exercise 9

*Hint.* Apply the null-homotopy equation to an arbitrary cycle and exhibit an explicit boundary representing its image under $f$. ☐

*Solution.* A null-homotopic chain map induces zero on homology. ☐

**Exercise 22.14.** Exercise 10

*Hint.* For a chain map $u : D \to E$, test $us$; use $\partial_E u = u\partial_D$ to move the differential past $u$. ☐

*Solution.* Postcomposition by a chain map preserves chain homotopy. ☐

**Exercise 22.15.** Exercise 11

*Hint.* For a chain map $v : B \to C$, test $sv$; use the commuting differential equation for $v$ in the second summand. ☐

*Solution.* Precomposition by a chain map preserves chain homotopy. ☐

**Exercise 22.16.** Exercise 12

*Hint.* Specify maps in both directions and homotopies of both composites to the relevant identities; do not require the composites to be literally equal. ☐

*Solution.* Chain-homotopy equivalence requires homotopy inverses, not strict inverses. ☐

**Exercise 22.17.** Exercise 13

*Hint.* Apply homology to each composite of the proposed homotopy inverses; chain homotopy turns those composites into identity homomorphisms. ☐

*Solution.* Chain-homotopy-equivalent complexes have isomorphic homology. □

**Exercise 22.18.** Exercise 14

*Hint.* Triangulate $\Delta^n \times I$ and apply the topological homotopy; internal face cancellation produces the degree-raising prism operator. □

*Solution.* The prism operator is a topological source of chain homotopies. □

**Exercise 22.19.** Exercise 15

*Hint.* Choose bases and check the dimensions of $D^D_{n+1}S_n$ and $S_{n-1}D^C_n$; both must match the matrix of $g_n - f_n$. □

*Solution.* For finite free complexes the identity becomes a matrix equation. □

**Exercise 22.20.** Exercise 16

*Hint.* Apply each homotopy to its own summand; an input with finite support still has finite support after this componentwise operation. □

*Solution.* Direct sums of chain homotopies are defined componentwise. □

**Exercise 22.21.** Exercise 17

*Hint.* Use the graded Hom differential on degree-zero chain maps and degree-one operators; compare $g - f$ with the differential image of $s$. □

*Solution.* In the Hom-complex, $g - f$ is a coboundary. □

**Exercise 22.22.** Exercise 18

*Hint.* Substitute $|s| = 1$ into $\delta(T) = \partial_D T - (-1)^{|T|} T \partial_C$ before simplifying the sign. □

*Solution.* For a degree-one operator $s$, $\delta(s) = \partial s + s\partial$. □

**Exercise 22.23.** Exercise 19

*Hint.* Require the homotopy operator to carry the distinguished source subcomplex into the distinguished target subcomplex, so it is well defined on quotient chains. □

*Solution.* A homotopy of maps of pairs descends to relative chains. □

**Exercise 22.24.** Exercise 20

*Hint.* View the augmentation as an extra differential in degree zero; include the bottom-degree compatibility rather than checking only positive degrees. □

*Solution.* Augmented chain homotopies can be required to preserve augmentation. □

**Exercise 22.25.** Exercise 21

*Hint.* Compare two choices by an explicit operator $s$; equality of the resulting homology maps follows from the boundary of $s(z)$ for each cycle. □

*Solution.* Homology ignores chain-level choices differing by chain homotopy. □

**Exercise 22.26.** Exercise 22

*Hint.* When $g - f = 0$, a degree-one zero operator satisfies the homotopy equation even when the chain groups themselves are nonzero. □

*Solution.* Equality of chain maps implies chain homotopy via $s = 0$. □

**Exercise 22.27.** Exercise 23

*Hint.* For a comparison between complexes, distinguish chain-homotopy equivalence from chain isomorphism; the elementary identity two-term complex is equivalent to zero but not isomorphic to it. □

*Solution.* Chain homotopy is weaker than chain isomorphism. □

**Exercise 22.28.** Exercise 24

*Hint.* Set one endpoint map to zero and the other to the identity, keeping track of the sign convention used when naming the contraction operator. □

*Solution.* A contraction is a homotopy from the identity map to zero. □

## 22.23 Solved problem dossiers

**Problem 22.29** (Cycle argument)**.** Prove directly that chain-homotopic maps induce equal homology maps.

*Solution.* For a cycle $z$, $g(z) - f(z) = \partial s(z)$, so their classes agree. □

**Problem 22.30** (Equivalence relation)**.** Verify reflexivity, symmetry and transitivity.

*Solution.* Use $0$, $-s$, and $s + t$, respectively. □

**Problem 22.31** (Postcomposition)**.** If $f \simeq g$ and $u$ is a chain map, find a homotopy from $uf$ to $ug$.

*Solution.* Use $us$; the chain-map identity moves $u$ past the differential. □

**Problem 22.32** (Precomposition)**.** If $f \simeq g$ and $v$ is a chain map, find a homotopy from $fv$ to $gv$.

*Solution.* Use $sv$. □

**Problem 22.33** (Null map)**.** Show a null-homotopic map induces zero in homology.

*Solution.* It is chain-homotopic to the zero chain map, so the induced maps agree. □

**Problem 22.34** (Homotopy equivalence)**.** Show chain-homotopy equivalence implies homology isomorphism.

*Solution.* Apply homology to $gf \simeq \mathrm{id}$ and $fg \simeq \mathrm{id}$. □

**Problem 22.35** (Strict isomorphism)**.** Why is every chain isomorphism a chain-homotopy equivalence?

*Solution.* Use its strict inverse; the composites equal the identities, hence are homotopic via zero. □

**Problem 22.36** (Prism)**.** Interpret the prism identity algebraically.

*Solution.* It says the singular chain maps induced by homotopic topological maps differ by $\partial P + P\partial$. $\square$

**Problem 22.37** (Relative prism)**.** Why does the prism operator descend to relative chains for a homotopy of pairs?

*Solution.* Prisms on chains lying in the subspace remain in the target subspace, so the operator respects the quotient. $\square$

**Problem 22.38** (Matrix criterion)**.** State the matrix equation for a chain homotopy.

*Solution.* $G_n - F_n = D^D_{n+1} S_n + S_{n-1} D^C_n$. $\square$

**Problem 22.39** (Zero differential)**.** If both complexes have zero differentials, when are two chain maps chain homotopic?

*Solution.* The homotopy equation becomes $g - f = 0$, so exactly when the maps are equal. $\square$

**Problem 22.40** (Concentrated complex)**.** Let $C$ be concentrated in one degree. What does chain homotopy between endomorphisms reduce to?

*Solution.* Again all differentials vanish, so homotopy is equality. $\square$

**Problem 22.41** (Two-term example)**.** Explain why nontrivial chain homotopies can occur only when adjacent target degrees exist.

*Solution.* The operator $s_n : C_n \to D_{n+1}$ must have somewhere to land; if all $D_{n+1} = 0$, that component vanishes. $\square$

**Problem 22.42** (Hom-complex)**.** Explain $g - f = \delta(s)$.

*Solution.* The differential in the Hom-complex is $\delta(s) = \partial_D s + s\partial_C$ for degree one. $\square$

**Problem 22.43** (Canonical up to homotopy)**.** Why can a chain-level construction still define a canonical homology map if choices are chain homotopic?

*Solution.* All choices induce the same map on homology. $\square$

**Problem 22.44** (Direct sums)**.** Combine homotopies $s^\alpha : f^\alpha \simeq g^\alpha$.

*Solution.* The direct sum $\bigoplus s^\alpha$ satisfies the homotopy identity componentwise. $\square$

**Problem 22.45** (Topological composition)**.** Relate postcomposition of a homotopy $F$ to chain homotopy.

*Solution.* Postcomposing $F$ by a continuous map corresponds to postcomposing its prism chain homotopy by the induced chain map. $\square$

**Problem 22.46** (Augmentation)**.** What extra condition is natural for augmented chain homotopies?

*Solution.* They should be compatible with the augmentation so the homotopy descends correctly to reduced complexes. $\square$

**Problem 22.47** (Homology cannot detect)**.** Can homology distinguish chain maps that are chain homotopic?

*Solution.* No; by definition of chain-homotopy invariance, their induced maps are identical. $\square$

**Problem 22.48** (Bridge)**.** Why is a chain contraction a special case?

*Solution.* It is a chain homotopy from $\mathrm{id}_C$ to 0, i.e. $\mathrm{id}_C = \partial s + s\partial$. ☐

**Problem 22.49** (Two streams)**.** Why is this chapter broader than VIII/20's prism proof?

*Solution.* It treats chain homotopy abstractly for arbitrary chain complexes, not only singular chains arising from topological homotopies. ☐

**Problem 22.50** (Coefficient stability)**.** Why can chain-homotopy identities be transported after coefficient change?

*Solution.* Applying an additive coefficient functor preserves sums and compositions in the defining identity, subject to the usual tensor exactness caveats for later homology calculations. ☐

**Problem 22.51** (Naturality)**.** Explain why chain homotopy is compatible with functorial constructions.

*Solution.* Because the identity $g-f = \partial s+s\partial$ is preserved by additive functors that commute with the relevant differentials. ☐

**Problem 22.52** (Final principle)**.** State the main invariant content.

*Solution.* Chain-homotopy classes of chain maps, not raw chain maps, are what homology sees. ☐

## 22.24 Supplementary worked examples

**Example 22.53** (Chain homotopy kills the difference on homology)**.** If chain maps $f, g : C_* \to D_*$ satisfy

$$g - f = d_D K + K d_C,$$

then for a cycle $z$ the difference $g(z) - f(z) = d_D K(z)$ is a boundary. Hence chain-homotopic maps induce the same homology map.

**Example 22.54** (The prism operator)**.** A topological homotopy $H : X \times I \to Y$ produces a prism operator $P : C_n(X) \to C_{n+1}(Y)$ satisfying

$$g_\# - f_\# = \partial P + P\partial.$$

This chain identity is the concrete bridge from continuous deformation to homology invariance.

**Example 22.55** (Homotopy inverse at chain level)**.** When chain maps $f : C \to D$ and $g : D \to C$ have composites chain homotopic to the identities, $f$ and $g$ induce mutually inverse homology maps even if the composites are not literally equal to the identity on chains.

## 22.25 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 22.56** (Degree of K)**.** What degree does a chain homotopy $K$ have?

*Hint.* Compare degrees in $dK + Kd$. □

*Solution.* $K : C_n \to D_{n+1}$ has degree $+1$. □

**Exercise 22.57** (Cycle difference)**.** If $z$ is a cycle and $g - f = dK + Kd$, simplify $(g - f)(z)$.

*Hint.* $dz = 0$. □

*Solution.* $(g - f)(z) = dK(z)$, a boundary. □

**Exercise 22.58** (Boundary compatibility)**.** Why are $f$ and $g$ required to be chain maps?

*Hint.* Homology classes must map cycles to cycles and boundaries to boundaries. □

*Solution.* Chain-map identities $df = fd$ and $dg = gd$ ensure induced homology maps are defined. □

**Exercise 22.59** (Zero-homotopic map)**.** If $f = dK + Kd$, what map does $f$ induce on homology?

*Hint.* Compare $f$ to the zero chain map. □

*Solution.* $f_* = 0$. □

**Exercise 22.60** (Identity contraction preview)**.** What equation makes $\mathrm{id}_C$ chain homotopic to zero?

*Hint.* Set $f = 0$ and $g = \mathrm{id}$. □

*Solution.* $\mathrm{id}_C = dK + Kd$. □

## Proofs

**Exercise 22.61** (Equality on homology)**.** Prove chain-homotopic chain maps induce equal maps on homology.

*Hint.* Evaluate the chain-homotopy equation on cycles. ☐

*Solution.* The difference on a cycle is a boundary, so the resulting homology classes agree. ☐

**Exercise 22.62** (Composition stability)**.** Show postcomposing chain-homotopic maps with a chain map preserves chain homotopy.

*Hint.* Compose the homotopy operator. ☐

*Solution.* If $g - f = dK + Kd$ and $u$ is a chain map, then $ug - uf = d(uK) + (uK)d$. ☐

**Exercise 22.63** (Precomposition stability)**.** Show precomposing chain-homotopic maps with a chain map preserves chain homotopy.

*Hint.* Use $Kv$ as the new homotopy. ☐

*Solution.* For a chain map $v$, $(g - f)v = d(Kv) + (Kv)d$. ☐

**Exercise 22.64** (Transitivity)**.** Prove chain homotopy is transitive.

*Hint.* Add homotopy operators. ☐

*Solution.* If $g - f = dK + Kd$ and $h - g = dL + Ld$, then $h - f = d(K + L) + (K + L)d$. ☐

## Counterexamples and hypothesis tests

**Exercise 22.65** (Equal homology maps imply chain homotopy)**.** Test: if $f_* = g_*$ on homology, then $f$ and $g$ are chain homotopic.

*Hint.* Chain homotopy is stronger than equality after passing to homology. ☐

*Solution.* False in general; homology forgets extension-level chain information. ☐

**Exercise 22.66** (Homotopy operator degree zero)**.** Test: a chain homotopy has degree 0.

*Hint.* Check the degrees of $dK$. ☐

*Solution.* False. It has degree $+1$ so that $dK + Kd$ has degree 0. ☐

**Exercise 22.67** (Any graded maps)**.** Test: the equation $g - f = dK + Kd$ makes sense without $f, g$ being chain maps.

*Hint.* The equation can be written, but induced homology maps may not exist. ☐

*Solution.* As an algebraic equality it may be written, but the standard conclusion about induced homology requires $f$ and $g$ to be chain maps. ☐

## Applications and investigations

**Exercise 22.68** (Homotopy invariance pipeline)**.** Summarize how a geometric homotopy proves equality of induced homology maps.

*Hint.* Insert the singular-chain functor between geometry and homology. □

*Solution.* A geometric homotopy yields the prism chain homotopy; the chain identity makes the two induced maps differ by boundaries on cycles, hence they agree on homology. □

**Exercise 22.69** (Algorithmic certificate)**.** Why can an explicit matrix $K$ be a certificate that two finite chain maps induce the same homology map?

*Hint.* Check one linear identity. □

*Solution.* Verifying $g - f = dK + Kd$ is a finite matrix calculation and directly certifies equality on homology. □

## Challenge problems

**Exercise 22.70** (Chain-homotopy equivalence)**.** Show a chain-homotopy equivalence is a quasi-isomorphism.

*Hint.* Use homotopy inverses. □

*Solution.* The induced maps of the two chain maps compose to identities on homology, so each is an isomorphism. □

**Exercise 22.71** (Derived viewpoint preview)**.** Why is it useful to identify chain-homotopic maps before taking homology?

*Hint.* They are indistinguishable by homology and behave well under composition. □

*Solution.* Quotienting by chain homotopy forms the homotopy category of complexes, where homology factors naturally and homotopy-equivalent complexes become isomorphic. □

# Chapter 23

# Chain Contractions

A chain contraction is a chain homotopy from the identity to zero. It gives an explicit algebraic certificate that every cycle bounds. Thus contractible chain complexes are acyclic, while the converse requires additional splitting hypotheses.

## 23.1 Definition

**Definition 23.1** (Chain contraction)**.** A chain contraction on $C_\bullet$ is a family

$$s_n : C_n \to C_{n+1}$$

such that

$$\mathrm{id}_C = \partial s + s\partial.$$

A complex admitting such an $s$ is contractible as a chain complex.

## 23.2 Contractible implies acyclic

**Theorem 23.2.** *If $C_\bullet$ admits a chain contraction, then*

$$H_n(C) = 0$$

*for all $n$.*

*Proof.* For a cycle $z$,

$$z = \partial s(z) + s(\partial z) = \partial s(z),$$

so every cycle is a boundary. □

## 23.3 Acyclic versus contractible

Acyclic means only $H_*(C) = 0$. Contractible means the stronger identity map is null-homotopic. In general:

$$\text{contractible} \Longrightarrow \text{acyclic},$$

but not conversely without hypotheses.

## 23.4 Split exact complexes

**Theorem 23.3** (Split exact complexes contract)**.** *An exact chain complex of modules whose short exact sequences*

$$0 \to Z_n \to C_n \to B_{n-1} \to 0$$

*split admits a chain contraction.*

## 23.5 Free complexes

Over a field, every short exact sequence of vector spaces splits. Hence every acyclic chain complex of vector spaces is contractible.

For free abelian complexes, splitting must be checked more carefully.

## 23.6 Elementary contractible pair

Consider

$$0 \to A \xrightarrow{\mathrm{id}} A \to 0$$

in adjacent degrees. It is contractible.

## 23.7 Disk-like complexes

The augmented simplicial chain complex of a simplex is contractible. An explicit contraction can be obtained by coning every simplex to a fixed vertex.

## 23.8 Cone-to-a-vertex operator

Fix a vertex $v$. On a simplex $[v_0, \ldots, v_n]$, define informally

$$s([v_0, \ldots, v_n]) = [v, v_0, \ldots, v_n]$$

when this simplex is valid, with the standard conventions when $v$ is already present. Then

$$\partial s + s\partial = \mathrm{id}$$

on the reduced chain complex of a simplex.

## 23.9 Reduced versus unreduced contraction

A nonempty contractible space has ordinary $H_0 \cong \mathbb{Z}$, so its ordinary chain complex is not acyclic. The reduced or augmented chain complex is the correct object to contract.

## 23.10 Strong deformation retract analogy

Topologically, a contraction deforms a space to a point. Algebraically, a chain contraction deforms the identity chain map to zero. For reduced chains of a contractible space, these ideas align.

## 23.11 Chain contraction and null maps

If $C$ is contractible, every chain map

$$f : C \to D$$

is null-homotopic:

$$f = f\,\mathrm{id}_C = f(\partial s + s\partial) = \partial(fs) + (fs)\partial.$$

Similarly, every chain map $D \to C$ is null-homotopic.

## 23.12 Direct sums

A direct sum of contractible complexes is contractible by taking the direct sum of their contractions.

## 23.13 Tensor preview

Tensoring a contraction with the identity often produces a contraction on a tensor product, with the usual grading signs. This becomes useful in VIII/27.

## 23.14 Filtration-compatible contractions

If a contraction respects a filtration, it can simplify spectral or filtered computations by collapsing contractible pieces without altering homology.

## 23.15 Cancellation of a basis pair

In a finite free chain complex, if a boundary matrix contains a unit entry that pairs one basis generator with another, an elementary chain contraction can often cancel that pair.

This is the algebraic basis of many chain-complex reduction algorithms.

## 23.16 Gaussian elimination over a field

Over a field, any nonzero pivot is a unit. Acyclic finite-dimensional chain complexes can therefore be reduced to direct sums of elementary contractible two-term complexes.

## 23.17 Smith normal form over $\mathbb{Z}$

Over $\mathbb{Z}$, diagonal entries equal to $\pm 1$ correspond to contractible summands, while entries of absolute value greater than 1 produce torsion homology and therefore cannot be contracted away.

## 23.18 Acyclic finite free example

The complex

$$0 \to \mathbb{Z} \xrightarrow{\binom{1}{0}} \mathbb{Z}^2 \xrightarrow{(0\ \ 1)} \mathbb{Z} \to 0$$

is exact and splits, hence contractible.

## 23.19 Noncontractible torsion example

The two-term complex

$$0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to 0$$

with $|m| > 1$ has

$$H_0 \cong \mathbb{Z}/m,$$

so it is not contractible.

## 23.20 Acyclicity certificate

A chain contraction is stronger than merely knowing homology vanishes: it gives a formula that explicitly fills each cycle.

## 23.21 Homological perturbation preview

If a differential is modified slightly, one may try to correct an existing contraction rather than recompute homology from scratch. This idea belongs to more advanced homological perturbation theory.

## 23.22 Bridge to mapping cones

Mapping cones detect chain-homotopy equivalences. In particular, the cone of a chain map is acyclic when the map is a quasi-isomorphism, and is contractible when the map is a chain-homotopy equivalence under the standard algebraic hypotheses. VIII/24 develops this construction.

## 23.23 Exercises

**Exercise 23.4.** Exercise 1

*Hint.* Check $s_n : C_n \to C_{n+1}$ and verify $\partial_{n+1}s_n + s_{n-1}\partial_n = \mathrm{id}_{C_n}$ in each degree. ☐

*Solution.* A chain contraction satisfies $\mathrm{id} = \partial s + s\partial$. ☐

**Exercise 23.5.** Exercise 2

*Hint.* Evaluate the contraction equation on a cycle; the resulting filling shows that every cycle represents the zero homology class. ☐

*Solution.* A contractible chain complex has zero homology. ☐

**Exercise 23.6.** Exercise 3

*Hint.* When $\partial z = 0$, compute $\partial s(z)$; the operator provides an explicit filling rather than just an existence argument. ☐

*Solution.* For a cycle $z$, the formula gives $z = \partial s(z)$. ☐

**Exercise 23.7.** Exercise 4

*Hint.* Test the exact complex $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$; a contraction would force a splitting that does not exist. ☐

*Solution.* Acyclicity alone does not always provide a contraction. ☐

**Exercise 23.8.** Exercise 5

*Hint.* Write $C_n = B_n \oplus L_n$ with $\partial : L_n \to B_{n-1}$ an isomorphism, and use its inverse on the boundary summand to define $s$. ☐

*Solution.* Split exact complexes admit contractions. ☐

**Exercise 23.9.** Exercise 6

*Hint.* Choose a complement to each cycle subspace. Exactness identifies the differential on that complement with the boundary subspace one degree lower. ☐

*Solution.* Every acyclic complex of vector spaces is contractible. ☐

**Exercise 23.10.** Exercise 7

*Hint.* Send the lower copy of $A$ back to the upper copy by the identity, set other components to zero, and verify the equation on both copies. ☐

*Solution.* The elementary complex $0 \to A \xrightarrow{\mathrm{id}} A \to 0$ is contractible. ☐

**Exercise 23.11.** Exercise 8

*Hint.* Choose a cone vertex and prepend it to an oriented simplex, with zero assigned to repeated-vertex simplices; include the augmented degree in the boundary check. ☐

*Solution.* The reduced chain complex of a simplex is contractible. ☐

**Exercise 23.12.** Exercise 9

*Hint.* Ordinary degree-zero homology of a point is nonzero, so its identity cannot be null-homotopic; augmentation removes this obstruction. ☐

*Solution.* Ordinary chains of a point retain $H_0$, so reduced chains are used. ☐

**Exercise 23.13.** Exercise 10

*Hint.* For a chain map $f : C \to D$, test $fs$ as the null-homotopy operator and use that $f$ commutes with differentials. ☐

*Solution.* If $C$ is contractible, every map $C \to D$ is null-homotopic. ☐

**Exercise 23.14.** Exercise 11

*Hint.* For a chain map $f : D \to C$, test $sf$, using the contraction on the target rather than on the source. ☐

*Solution.* If $C$ is contractible, every map $D \to C$ is null-homotopic. ☐

**Exercise 23.15.** Exercise 12

*Hint.* Take the direct sum of the degree-raising operators and check the identity separately on every summand, retaining finite support. ☐

*Solution.* Direct sums of contractions remain contractions. ☐

**Exercise 23.16.** Exercise 13

*Hint.* First isolate a unit boundary coefficient by compatible basis operations; then invert that unit to contract the paired generators and adjust neighboring maps. ☐

*Solution.* A unit pivot in a boundary matrix often determines a cancellable pair. ☐

**Exercise 23.17.** Exercise 14

*Hint.* A nonzero field element has an inverse, whereas an integer pivot such as two cannot be inverted over $\mathbb{Z}$. ☐

*Solution.* Over a field every nonzero pivot is a unit. ☐

**Exercise 23.18.** Exercise 15

*Hint.* On an isolated two-term block with differential $\pm 1$, its inverse supplies the degree-raising contraction of that block. ☐

*Solution.* Smith entries $\pm 1$ correspond to contractible summands. ☐

**Exercise 23.19.** Exercise 16

*Hint.* For an isolated block $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$, compute the cokernel; a nonzero torsion class rules out a contraction. ☐

*Solution.* Smith entries $|m| > 1$ can produce torsion homology. ☐

**Exercise 23.20.** Exercise 17

*Hint.* Given a cycle $z$, the certificate must produce the chain $s(z)$ and verify its boundary is exactly $z$. ☐

*Solution.* A contraction is an explicit certificate that cycles bound. ☐

**Exercise 23.21.** Exercise 18

*Hint.* Since $\partial$ lowers degree, $s$ must raise it for the two composites in the contraction equation to act in the original degree. ☐

*Solution.* A contraction has degree $+1$. ☐

**Exercise 23.22.** Exercise 19

*Hint.* For a chain isomorphism $u : C \to D$, conjugate the contraction to $usu^{-1}$ and verify the identity on $D$. ☐

*Solution.* Contractibility is preserved by chain isomorphism. ☐

**Exercise 23.23.** Exercise 20

*Hint.* The only maps between a complex and zero compose to zero; the homotopy-inverse condition therefore says its identity is null-homotopic. ☐

*Solution.* A chain-homotopy-equivalent complex to zero is contractible. ☐

**Exercise 23.24.** Exercise 21

*Hint.* A topological contraction first compares the identity chain map with a map through a point; pass to the reduced or augmented model to eliminate the point's degree-zero class. ☐

*Solution.* Reduced chains align chain contraction with topological contraction. □

**Exercise 23.25.** Exercise 22

*Hint.* Check that $s$ preserves each filtration level; an unfiltered contraction need not induce a valid contraction of the filtered objects. □

*Solution.* Filtration-compatible contractions can simplify filtered complexes. □

**Exercise 23.26.** Exercise 23

*Hint.* For a contraction on the second tensor factor, test $S(c \otimes d) = (-1)^{|c|} c \otimes s(d)$ and verify cancellation with the tensor differential. □

*Solution.* Tensor contractions require grading signs. □

**Exercise 23.27.** Exercise 24

*Hint.* Compare the two tests: an acyclic cone detects a quasi-isomorphism, while an explicit cone contraction detects the stronger chain-homotopy-equivalence condition. □

*Solution.* Mapping cones provide the next major test for chain equivalences. □

## 23.24 Solved problem dossiers

**Problem 23.28** (Cycle filling)**.** Show a contraction fills every cycle.

*Solution.* If $\partial z = 0$, then $z = \partial s(z) + s\partial z = \partial s(z)$. □

**Problem 23.29** (Elementary pair)**.** Construct a contraction on $0 \to A \xrightarrow{\mathrm{id}} A \to 0$.

*Solution.* Take $s$ from the lower copy of $A$ back to the upper copy as the identity and all other components zero. □

**Problem 23.30** (Split exact)**.** Explain how splittings yield a contraction.

*Solution.* Decompose each $C_n$ into a boundary part and a chosen complement mapping isomorphically onto $B_{n-1}$; invert that isomorphism to define $s$. □

**Problem 23.31** (Vector spaces)**.** Why does acyclic imply contractible over a field?

*Solution.* All short exact sequences of vector spaces split, so the split-exact theorem applies. □

**Problem 23.32** (Torsion obstruction)**.** Why is $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to 0$ not contractible?

*Solution.* Its $H_0$ is $\mathbb{Z}/2$, but contractible complexes are acyclic. □

**Problem 23.33** (Simplex contraction)**.** Describe the vertex-coning contraction on reduced simplicial chains.

*Solution.* Choose a vertex $v$ and prepend it to each oriented simplex, with the usual sign/convention; the boundary identity gives $\partial s + s\partial = \mathrm{id}$. □

**Problem 23.34** (Reduced chains)**.** Why can the unreduced chain complex of a point not be contracted?

*Solution.* It has $H_0 \cong \mathbb{Z} \neq 0$, contradicting contractible-implies-acyclic. □

**Problem 23.35** (Map out)**.** If $C$ contracts, prove every chain map $f : C \to D$ is null-homotopic.

*Solution.* Compose the contraction with $f$: $f = f\partial s + fs\partial = \partial(fs) + (fs)\partial$. ☐

**Problem 23.36** (Map in)**.** If $C$ contracts, prove every chain map $g : D \to C$ is null-homotopic.

*Solution.* Use $sg$: $g = \partial(sg) + (sg)\partial$. ☐

**Problem 23.37** (Direct sum)**.** Construct a contraction on $C \oplus D$ from contractions on each.

*Solution.* Use $s_C \oplus s_D$. ☐

**Problem 23.38** (Unit cancellation)**.** Why does a unit incidence coefficient permit cancellation?

*Solution.* The corresponding differential component is an isomorphism on rank-one summands, producing an elementary contractible two-term summand. ☐

**Problem 23.39** (Field elimination)**.** Explain finite-dimensional acyclic complexes over a field as sums of elementary pairs.

*Solution.* Choose bases adapted to images and complements; exactness pairs each complement with the next boundary space by an isomorphism. ☐

**Problem 23.40** (Smith form)**.** What does a Smith diagonal entry 1 mean homologically?

*Solution.* It yields an isomorphism between rank-one summands, hence no homology and a contractible pair. ☐

**Problem 23.41** (Smith torsion)**.** What does a Smith diagonal entry $m > 1$ mean?

*Solution.* It contributes a cokernel $\mathbb{Z}/m$, so the summand is not contractible. ☐

**Problem 23.42** (Explicit exact example)**.** Verify the displayed $\mathbb{Z} \to \mathbb{Z}^2 \to \mathbb{Z}$ complex is exact.

*Solution.* The first image is $\mathbb{Z}(1, 0)$, exactly the kernel of $(0\ 1)$, and the second map is surjective. ☐

**Problem 23.43** (Contraction strength)**.** Why is a contraction stronger data than vanishing homology?

*Solution.* It gives an explicit degree-one operator solving $\mathrm{id} = \partial s + s\partial$, not just an existence statement about boundaries. ☐

**Problem 23.44** (Homotopy to zero)**.** Restate chain contractibility in VIII/22 language.

*Solution.* The identity chain map is chain homotopic to the zero chain map. ☐

**Problem 23.45** (Invariance)**.** If $C \cong D$ as chain complexes and $C$ contracts, why does $D$ contract?

*Solution.* Transport the contraction through the chain isomorphism and its inverse. ☐

**Problem 23.46** (Topological analogy)**.** What topological object suggests the chain contraction of a simplex?

*Solution.* A simplex is contractible to a vertex; the coning operator is the chain-level analogue. □

**Problem 23.47** (Filtration)**.** Why might a filtration-compatible contraction be useful?

*Solution.* It removes contractible pieces without mixing filtration levels, preserving the structure needed for later filtered calculations. □

**Problem 23.48** (Bridge to cones)**.** What should happen to the mapping cone of a chain-homotopy equivalence?

*Solution.* It should be contractible; VIII/24 proves the cone packages failure of invertibility up to chain homotopy. □

**Problem 23.49** (Acyclic cone preview)**.** What should happen to the cone of a quasi-isomorphism?

*Solution.* Its homology vanishes, so the cone is acyclic. □

**Problem 23.50** (Contraction versus quasi-isomorphism)**.** Why is contractible stronger than quasi-isomorphic to zero?

*Solution.* Quasi-isomorphic to zero only says homology vanishes; contractible supplies a homotopy-level inverse to zero. □

**Problem 23.51** (Final principle)**.** State the practical use of a contraction.

*Solution.* It gives explicit algebraic cancellation and a formula for filling cycles. □

## 23.25 Supplementary worked examples

**Example 23.52** (Contracting a cone complex)**.** For a cone-like chain complex, a degree-+1 operator $s$ with

$$ds + sd = \mathrm{id}$$

is a chain contraction. Every cycle $z$ then satisfies $z = d\,s(z)$, so all positive-degree homology vanishes.

**Example 23.53** (Augmented simplex)**.** The augmented simplicial chain complex of a simplex admits an explicit contraction by coning every simplex to a fixed vertex. This algebraically expresses the geometric contractibility of the simplex.

**Example 23.54** (A split acyclic complex)**.** For the two-term complex $0 \to A \xrightarrow{\mathrm{id}} A \to 0$, define $s$ to be the identity from degree 0 back to degree 1. Then $ds + sd = \mathrm{id}$, giving an explicit contraction rather than merely observing that homology vanishes.

## 23.26 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 23.55** (Contraction equation)**.** State the chain-contraction identity.

*Hint.* The identity map is chain homotopic to zero. ☐

*Solution.* $ds + sd = \mathrm{id}_C$. ☐

**Exercise 23.56** (Acyclic consequence)**.** If $C_*$ has a chain contraction, what is $H_*(C)$?

*Hint.* Apply the identity to a cycle. ☐

*Solution.* All homology groups vanish (or reduced homology in an augmented topological model). ☐

**Exercise 23.57** (Cycle primitive)**.** For a cycle $z$, express it as a boundary using a contraction $s$.

*Hint.* $dz = 0$. ☐

*Solution.* $z = ds(z)$. ☐

**Exercise 23.58** (Two-term identity complex)**.** Contract $0 \to A \xrightarrow{\mathrm{id}} A \to 0$.

*Hint.* Send degree 0 back to degree 1 by the identity. ☐

*Solution.* Take $s_0 = \mathrm{id}_A$ and all other $s_n = 0$; then $ds + sd = \mathrm{id}$. ☐

**Exercise 23.59** (Direct sum)**.** If $C$ and $D$ are contractible chain complexes, contract $C \oplus D$.

*Hint.* Use the direct sum of the two contractions. ☐

*Solution.* $s_C \oplus s_D$ satisfies $d(s_C \oplus s_D) + (s_C \oplus s_D)d = \mathrm{id}$. ☐

## Proofs

**Exercise 23.60** (Contraction implies null identity)**.** Prove a chain contraction is exactly a chain homotopy from id to 0.

*Hint.* Compare definitions. □

*Solution.* The defining equation for a chain homotopy between id and 0 is precisely $\mathrm{id} = ds + sd$. □

**Exercise 23.61** (Homology vanishing)**.** Prove contractible complexes are acyclic.

*Hint.* Apply the contraction to cycles. □

*Solution.* If $dz = 0$, then $z = ds(z)$, hence every cycle is a boundary. □

**Exercise 23.62** (Transport contraction)**.** Show an isomorphic chain complex to a contractible one is contractible.

*Hint.* Conjugate the contraction by the chain isomorphism. □

*Solution.* If $\phi : C \to D$ is an isomorphism and $s$ contracts $C$, then $\phi s \phi^{-1}$ contracts $D$. □

**Exercise 23.63** (Tensor contraction)**.** Under standard sign conventions, explain why tensoring a contractible complex with a free complex often remains contractible.

*Hint.* Use $s \otimes \mathrm{id}$ with the total differential. □

*Solution.* The graded Leibniz signs make the cross terms cancel, leaving the identity on the tensor product. □

## Counterexamples and hypothesis tests

**Exercise 23.64** (Acyclic implies contractible)**.** Test: every acyclic chain complex is chain contractible.

*Hint.* Splittings need not exist. □

*Solution.* False over general rings; acyclicity does not guarantee the exact sequences of cycles and boundaries split. □

**Exercise 23.65** (Topological contractibility automatic)**.** Test: a topologically contractible space gives an unaugmented singular chain complex with zero $H_0$.

*Hint.* Remember ordinary $H_0$ of a point. □

*Solution.* False. Ordinary singular homology has $H_0 \cong \mathbb{Z}$; the reduced or augmented complex is the acyclic object. □

**Exercise 23.66** (Contraction unique)**.** Test: if a contraction exists, it is unique.

*Hint.* Different homotopy operators can satisfy the same identity. □

*Solution.* False; contractions are generally highly nonunique. □

## Applications and investigations

**Exercise 23.67** (Eliminating cells)**.** How does an explicit contraction justify removing an algebraically canceling pair of generators?

*Hint.* The pair forms a contractible summand. □

*Solution.* A contractible direct summand contributes no homology and may be eliminated while preserving the homology of the remaining complex. □

**Exercise 23.68** (Numerical simplification)**.** Why are contractions useful in computational homology?

*Hint.* They produce explicit reductions rather than only isomorphism statements. □

*Solution.* A contraction supplies chain-level maps and homotopies that compress a large complex to a smaller one while certifying that homology is preserved. □

## Challenge problems

**Exercise 23.69** (Cone contraction)**.** Sketch the contraction of the augmented chain complex of a simplex using a chosen vertex $v$.

*Hint.* Cone each simplex to $v$. □

*Solution.* Define $s(\sigma) = v * \sigma$ with the usual orientation convention; the boundary formula for a cone gives $ds + sd = \mathrm{id}$. □

**Exercise 23.70** (Strong reduction)**.** Explain the difference between knowing $H_*(C) = 0$ and possessing a chain contraction.

*Hint.* One is a homology statement; the other is explicit chain data. □

*Solution.* A contraction is a constructive null-homotopy of the identity and can be composed, transported, and used algorithmically; vanishing homology alone provides none of that extra structure. □

# Chapter 24

# Mapping Cones of Chain Maps

The mapping cone of a chain map packages the map into a new chain complex. Its homology measures the failure of the map to be a quasi-isomorphism, while its contractibility detects chain-homotopy equivalences.

## 24.1 Definition

Let $f : C_\bullet \to D_\bullet$ be a chain map.

**Definition 24.1** (Mapping cone)**.** Define

$$\mathrm{Cone}(f)_n = D_n \oplus C_{n-1}$$

with differential

$$d(d, c) = (\partial_D d + f(c), -\partial_C c).$$

## 24.2 Chain condition

**Proposition 24.2.** *The cone differential satisfies*

$$d^2 = 0.$$

The verification uses exactly the chain-map identity

$$\partial_D f = f \partial_C.$$

## 24.3 Shifted complex

Define the suspension/shift $C[1]$ by

$$C[1]_n = C_{n-1}, \qquad \partial_{C[1]} = -\partial_C.$$

Then there is a short exact sequence

$$0 \to D \to \mathrm{Cone}(f) \to C[1] \to 0.$$

## 24.4 Cone exact sequence

The short exact sequence yields

$$\cdots \to H_n(D) \to H_n(\mathrm{Cone}(f)) \to H_{n-1}(C) \xrightarrow{H_{n-1}(f)} H_{n-1}(D) \to \cdots .$$

Equivalently one can arrange it as

$$\cdots \to H_n(C) \xrightarrow{H_n(f)} H_n(D) \to H_n(\mathrm{Cone}(f)) \to H_{n-1}(C) \to \cdots .$$

## 24.5 Cone of the identity

**Theorem 24.3** (Identity cone contracts)**.**

$$\mathrm{Cone}(\mathrm{id}_C)$$

*is contractible.*

## 24.6 Cone of the zero map

If $f = 0$, then

$$\mathrm{Cone}(0) \cong D \oplus C[1]$$

as chain complexes.

Thus

$$H_n(\mathrm{Cone}(0)) \cong H_n(D) \oplus H_{n-1}(C).$$

## 24.7 Quasi-isomorphisms

**Definition 24.4** (Quasi-isomorphism)**.** A chain map $f : C \to D$ is a quasi-isomorphism if

$$H_n(f) : H_n(C) \to H_n(D)$$

is an isomorphism for every $n$.

**Theorem 24.5** (Cone criterion)**.** *A chain map $f$ is a quasi-isomorphism if and only if*

$$H_n(\mathrm{Cone}(f)) = 0$$

*for all $n$.*

## 24.8 Chain-homotopy equivalences

**Theorem 24.6** (Homotopy-equivalence cone)**.** *If $f$ is a chain-homotopy equivalence, then*

$$\mathrm{Cone}(f)$$

*is contractible.*

Thus chain-homotopy equivalence is stronger than quasi-isomorphism, matching contractible versus merely acyclic cones.

## 24.9 Cone of an inclusion

If $i : A \hookrightarrow B$ is an inclusion of chain complexes, the cone provides a chain model closely related to the quotient $B/A$.

Under suitable splitting/cofibration-style algebraic hypotheses, the cone and quotient are quasi-isomorphic.

## 24.10 Relative chains

For a subspace $A \subset X$, the inclusion

$$C_*(A) \to C_*(X)$$

has a cone whose homology is naturally related to relative homology

$$H_*(X, A).$$

This mirrors the topological mapping-cone picture from VIII/07.

## 24.11 Topological versus algebraic cones

A map of spaces

$$f : A \to X$$

has a topological mapping cone $C_f$. Applying singular chains gives a chain complex naturally compared with

$$\mathrm{Cone}(C_*(f)).$$

The two constructions encode the same cofiber geometry up to the expected chain equivalences.

## 24.12 Functoriality

A commutative square of chain maps induces a chain map between mapping cones.

## 24.13 Homotopy-commutative squares

If a square commutes only up to chain homotopy, a corrected cone map can still be built by including the homotopy term. This is one reason mapping cones are central in derived constructions.

## 24.14 Connecting morphism

The cone exact sequence realizes the map

$$H_{n-1}(C) \to H_{n-1}(D)$$

as the homology map induced by $f$. Thus the cone is a universal algebraic device for measuring this map.

## 24.15 Two-term example

For

$$f : \mathbb{Z} \to \mathbb{Z}, \qquad f = \times m$$

with both complexes concentrated in degree 0, the cone is the two-term complex

$$0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to 0.$$

Hence for $m \neq 0$,

$$H_0(\mathrm{Cone}(f)) \cong \mathbb{Z}/m.$$

## 24.16 Moore-space connection

The multiplication-by-$m$ cone is the algebraic analogue of the degree-$m$ cell attachment producing a Moore space. Both produce $\mathbb{Z}/m$ torsion.

## 24.17 Cone of an isomorphism

If $f$ is a strict chain isomorphism, its cone is contractible. This is the strict special case of the homotopy-equivalence theorem.

## 24.18 Cone and exactness

If

$$0 \to C \xrightarrow{f} D \to Q \to 0$$

is a short exact sequence, the cone of $f$ is frequently quasi-isomorphic to the quotient complex $Q$.

## 24.19 Distinguished-triangle preview

The pattern

$$C \xrightarrow{f} D \to \mathrm{Cone}(f) \to C[1]$$

is the prototype of a distinguished triangle. Derived categories formalize this pattern, but no derived-category machinery is needed in this volume.

## 24.20 Why signs matter

The minus sign in the shifted differential is essential. Without it, the cone differential would generally fail to square to zero.

## 24.21 Bridge to coefficients

The next chapter changes coefficient groups. Mapping-cone calculations remain valuable there because tensoring a cone produces the expected cone of the tensored map, subject to standard exactness considerations.

## 24.22 Exercises

**Exercise 24.7.** Exercise 1

*Hint.* The second summand is shifted: an element of $C_{n-1}$ joins an element of $D_n$ to make a cone chain of degree $n$. ☐

*Solution.* $\mathrm{Cone}(f)_n = D_n \oplus C_{n-1}$. ☐

**Exercise 24.8.** Exercise 2

*Hint.* Check that $f(c)$ and $\partial_D d$ both lie in $D_{n-1}$, while the second output lies in $C_{n-2}$. ☐

*Solution.* The cone differential is $d(d, c) = (\partial_D d + f(c), -\partial_C c)$. ☐

**Exercise 24.9.** Exercise 3

*Hint.* The shifted differential contributes $-\partial_C$; trace this sign when applying the cone differential a second time. ☐

*Solution.* The minus sign belongs to the shifted $C$-summand. ☐

**Exercise 24.10.** Exercise 4

*Hint.* Expand both components of $d^2(d, c)$. The mixed first-component terms cancel precisely because $\partial_D f = f\partial_C$. ☐

*Solution.* The chain-map identity is needed to prove $d^2 = 0$. ☐

**Exercise 24.11.** Exercise 5

*Hint.* An element originally in degree $k$ of $C$ occurs in degree $k + 1$ of $C[1]$; use this to check the index direction. ☐

*Solution.* The shifted complex satisfies $C[1]_n = C_{n-1}$. ☐

**Exercise 24.12.** Exercise 6

*Hint.* Compare the projection $\mathrm{Cone}(f) \to C[1]$ with the cone differential; the projected differential must have the same minus sign. ☐

*Solution.* Its differential is $-\partial_C$. ☐

**Exercise 24.13.** Exercise 7

*Hint.* Use $d \mapsto (d, 0)$ for the inclusion and $(d, c) \mapsto c$ for the projection; verify exactness and the chain-map conditions. ☐

*Solution.* There is a short exact sequence $0 \to D \to \mathrm{Cone}(f) \to C[1] \to 0$. ☐

**Exercise 24.14.** Exercise 8

*Hint.* Identify $H_n(C[1])$ with $H_{n-1}(C)$; lifting $c$ as $(0, c)$ identifies the connecting arrow with the map induced by $f$. ☐

*Solution.* The cone therefore has a long exact homology sequence. ☐

**Exercise 24.15.** Exercise 9

*Hint.* For the identity cone, test the degree-raising operator $s(d, c) = (0, d)$ and compute $ds + sd$ explicitly. ☐

*Solution.* The cone of the identity is contractible. ☐

**Exercise 24.16.** Exercise 10

*Hint.* Set $f = 0$ in the differential; the two summands then have independent differentials, yielding a direct sum of complexes. ☐

*Solution.* The cone of zero is $D \oplus C[1]$. ☐

**Exercise 24.17.** Exercise 11

*Hint.* A quasi-isomorphism requires an isomorphism on every homology group, not a degree-wise isomorphism of the chain groups. ☐

*Solution.* A quasi-isomorphism induces homology isomorphisms. ☐

**Exercise 24.18.** Exercise 12

*Hint.* Use exactness on both sides of $H_n(\mathrm{Cone}(f))$ to relate its vanishing to injectivity and surjectivity of the neighboring $H_*(f)$. ☐

*Solution.* A map is a quasi-isomorphism iff its cone is acyclic. ☐

**Exercise 24.19.** Exercise 13

*Hint.* The long exact sequence proves only acyclicity. To obtain contractibility, use a chain-homotopy inverse and the homotopies as additional chain-level data. ☐

*Solution.* A chain-homotopy equivalence has contractible cone. ☐

**Exercise 24.20.** Exercise 14

*Hint.* An acyclic complex can be nonsplit over a general ring; a contraction supplies splittings and explicit fillings, not merely vanishing homology. ☐

*Solution.* Contractible cone is stronger than acyclic cone. ☐

**Exercise 24.21.** Exercise 15

*Hint.* A strict chain inverse is also a chain-homotopy inverse with zero comparison homotopies; apply the stronger cone criterion. ☐

*Solution.* The cone of a strict isomorphism is contractible. ☐

**Exercise 24.22.** Exercise 16

*Hint.* For an inclusion $i : A \hookrightarrow B$, use $(b, a) \mapsto [b]$ from the cone to $B/A$ and examine its kernel to compare their homology. ☐

*Solution.* The cone of a subcomplex inclusion models relative information. ☐

**Exercise 24.23.** Exercise 17

*Hint.* Keep track of augmentation: ordinary chains of a nonempty contractible topological cone retain $H_0$, whereas the algebraic cone of an identity is acyclic. ☐

*Solution.* Topological and algebraic mapping cones are parallel constructions. ☐

**Exercise 24.24.** Exercise 18

*Hint.* For a strictly commuting square $vf = f'u$, apply $v$ on the target summand and $u$ on the shifted source summand; check the mixed term. ☐

*Solution.* A commutative square induces a map of cones. ☐

**Exercise 24.25.** Exercise 19

*Hint.* If $vf - f'u = \partial h + h\partial$, test the corrected map $(d, c) \mapsto (vd + h(c), u(c))$ and verify its differential equation. ☐

*Solution.* Homotopy-commutative squares require a correction term. ☐

**Exercise 24.26.** Exercise 20

*Hint.* Both original complexes occupy degree zero, so the shifted source becomes degree one and the unshifted target remains degree zero. ☐

*Solution.* For $f = \times m$ in degree zero, the cone is a two-term complex. ☐

**Exercise 24.27.** Exercise 21

*Hint.* The degree-zero outgoing differential is zero, while the incoming image is $m\mathbb{Z}$; take that quotient over $\mathbb{Z}$. ☐

*Solution.* Its degree-zero homology is $\mathbb{Z}/m$ for $m \neq 0$. ☐

**Exercise 24.28.** Exercise 22

*Hint.* Compare the cokernel of multiplication by $m$ with the cellular boundary of a degree-$m$ attachment, accounting for the change in degree. ☐

*Solution.* Multiplication-by-$m$ cones parallel Moore-space torsion. ☐

**Exercise 24.29.** Exercise 23

*Hint.* Write the cone inclusion and shifted projection explicitly; their associated long exact sequence explains the triangle pattern without treating it as a short exact sequence of four objects. ☐

*Solution.* The pattern $C \to D \to \mathrm{Cone}(f) \to C[1]$ is a triangle. ☐

**Exercise 24.30.** Exercise 24

*Hint.* Tensor the displayed cone differential with the coefficient group. The cone formula persists, but tensoring need not preserve a quasi-isomorphism without suitable hypotheses. ☐

*Solution.* VIII/25 will revisit cones after changing coefficients. ☐

## 24.23 Solved problem dossiers

**Problem 24.31** (Square-zero check)**.** Verify the cone differential squares to zero.

*Solution.* Compute $d^2(d,c)$: the first component is $\partial_D^2 d+\partial_D f(c)-f\partial_C c=0$ by the chain-map identity; the second is $\partial_C^2 c=0$. ☐

**Problem 24.32** (Identity cone)**.** Explain why Cone(id) contracts.

*Solution.* An explicit degree-one operator swaps the shifted and unshifted components appropriately; equivalently the cone models an algebraic cylinder pair and is chain-homotopy equivalent to zero. ☐

**Problem 24.33** (Zero cone)**.** Compute the homology of Cone(0).

*Solution.* The differential is block diagonal, so the cone is $D \oplus C[1]$ and $H_n \cong H_n(D) \oplus H_{n-1}(C)$. ☐

**Problem 24.34** (Quasi-isomorphism criterion)**.** Prove the cone criterion using the long exact sequence.

*Solution.* Exactness shows all cone homology vanishes exactly when every $H_n(C) \to H_n(D)$ is both injective and surjective. ☐

**Problem 24.35** (Homotopy equivalence)**.** Why does chain-homotopy equivalence imply contractible cone?

*Solution.* A homotopy inverse and the two chain homotopies can be assembled into an explicit cone contraction. ☐

**Problem 24.36** (Strict isomorphism)**.** Deduce the cone of a chain isomorphism contracts.

*Solution.* A strict isomorphism is a chain-homotopy equivalence with zero homotopies. ☐

**Problem 24.37** (Multiplication by $m$)**.** Compute the cone of $\times m : \mathbb{Z} \to \mathbb{Z}$ concentrated in degree zero.

*Solution.* It is $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to 0$, with $H_0 = \mathbb{Z}/m$ and $H_1 = 0$ for $m \neq 0$. ☐

**Problem 24.38** (Case $m = 0$)**.** Compute the same cone for $m = 0$.

*Solution.* The differential is zero, so $H_1 \cong \mathbb{Z}$ and $H_0 \cong \mathbb{Z}$. ☐

**Problem 24.39** (Case $m = 1$)**.** What happens for $m = 1$?

*Solution.* The differential is an isomorphism, so the cone is acyclic and in fact contractible. ☐

**Problem 24.40** (Relative model)**.** Relate the cone of $C_*(A) \to C_*(X)$ to $H_*(X, A)$.

*Solution.* The cone is a standard chain model quasi-isomorphic to the quotient relative complex, up to the conventional degree arrangement. ☐

**Problem 24.41** (Topological cone)**.** Why should singular chains on a topological mapping cone compare with the algebraic cone?

*Solution.* Both encode the same cofiber construction: one geometrically before applying chains, the other algebraically after applying the chain functor. ☐

**Problem 24.42** (Map of cones)**.** Given a strictly commutative square, define the obvious map on cones.

*Solution.* Apply the target map on the $D$-summand and the source map on the shifted $C$-summand; commutativity makes it a chain map. ☐

**Problem 24.43** (Homotopy square)**.** Why is an extra term needed if the square commutes only up to chain homotopy?

*Solution.* The failure of strict commutation appears in the cone differential; the chosen homotopy supplies exactly the correction needed to cancel it. ☐

**Problem 24.44** (Quotient comparison)**.** When does the cone of an inclusion resemble the quotient complex?

*Solution.* For a suitably split inclusion, the cone contains a contractible copy of the subcomplex and reduces up to chain homotopy/quasi-isomorphism to the quotient. ☐

**Problem 24.45** (Acyclic versus contractible)**.** Why can an acyclic cone fail to be contractible?

*Solution.* Acyclicity detects only quasi-isomorphism of $f$; contractibility of the cone requires the stronger chain-homotopy-equivalence property. ☐

**Problem 24.46** (Moore analogy)**.** Explain the algebraic/topological parallel for multiplication by $m$.

*Solution.* The chain cone has $\mathbb{Z}/m$ homology, matching the cellular chain complex of a Moore space built by degree-$m$ attachment. ☐

**Problem 24.47** (Triangle)**.** Why is the cone sequence naturally written $C \to D \to \mathrm{Cone}(f) \to C[1]$?

*Solution.* The inclusion of $D$ and projection to the shifted $C$-summand form the short exact cone sequence and its associated cofiber pattern. ☐

**Problem 24.48** (Sign)**.** Show what fails if the shifted differential had no minus sign.

*Solution.* The cross terms in $d^2$ would add instead of canceling, leaving generally $2f\partial_C$ or equivalent nonzero terms depending on convention. ☐

**Problem 24.49** (Homology map failure)**.** What does nonzero $H_n(\mathrm{Cone}(f))$ tell you?

*Solution.* It witnesses failure of $H_*(f)$ to be an isomorphism in adjacent degrees, as organized by the cone long exact sequence. ☐

**Problem 24.50** (Coefficient bridge)**.** Why are cones useful with changed coefficients?

*Solution.* They convert the question whether a coefficient-changed map remains a quasi-isomorphism into acyclicity of the corresponding coefficient-changed cone. ☐

**Problem 24.51** (Derived preview)**.** What later formalism abstracts mapping-cone triangles?

*Solution.* Derived and triangulated categories, though this volume only uses the concrete chain-complex construction. ☐

**Problem 24.52** (Final principle)**.** State the main diagnostic role of a cone.

*Solution.* The cone measures the failure of a chain map to be an equivalence: acyclic cone means quasi-isomorphism; contractible cone means chain-homotopy equivalence. ☐

## 24.24 Supplementary worked examples

**Example 24.53** (Cone of the identity is acyclic)**.** For a chain map $f : C_* \to D_*$, the mapping cone has groups

$$\mathrm{Cone}(f)_n = D_n \oplus C_{n-1}.$$

When $f = \mathrm{id}_C$, the cone is chain contractible, reflecting that the topological cone on an identity attachment has no residual relative homology.

**Example 24.54** (Cone of the zero map)**.** For $f = 0 : C \to D$, the differential does not couple the two summands, so

$$\mathrm{Cone}(0) \cong D \oplus \Sigma C$$

as chain complexes. Consequently $H_n(\mathrm{Cone}(0)) \cong H_n(D) \oplus H_{n-1}(C)$.

**Example 24.55** (Quasi-isomorphism criterion)**.** The cone fits into a short exact sequence

$$0 \to D \to \mathrm{Cone}(f) \to \Sigma C \to 0.$$

Its long exact homology sequence shows that $f$ is a quasi-isomorphism exactly when $\mathrm{Cone}(f)$ is acyclic.

## 24.25 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 24.56** (Cone group)**.** Write $\mathrm{Cone}(f)_n$.

*Hint.* Shift the source down by one degree. □

*Solution.* $\mathrm{Cone}(f)_n = D_n \oplus C_{n-1}$. □

**Exercise 24.57** (Cone differential)**.** With homological grading, write a standard cone differential.

*Hint.* Include $f(c)$ in the $D$ component and a minus sign on $d_C$. □

*Solution.* $d(d, c) = (d_D d + f(c), -d_C c)$. □

**Exercise 24.58** (Zero map cone)**.** Identify Cone(0).

*Hint.* The coupling term disappears. □

*Solution.* $\mathrm{Cone}(0) \cong D \oplus \Sigma C$. □

**Exercise 24.59** (Identity cone)**.** What is the homology of $\mathrm{Cone}(\mathrm{id}_C)$?

*Hint.* The identity is a quasi-isomorphism. □

*Solution.* It is zero in every degree; the cone is in fact contractible. □

**Exercise 24.60** (Shifted homology)**.** What is $H_n(\Sigma C)$ in terms of $H_*(C)$?

*Hint.* The shift moves degree by one. □

*Solution.* $H_n(\Sigma C) \cong H_{n-1}(C)$. □

## Proofs

**Exercise 24.61** (Check $d^2 = 0$)**.** Verify the cone differential squares to zero.

*Hint.* Use $d_D f = f d_C$. ☐

*Solution.* Applying $d$ twice cancels the two $f(d_C c)$ terms and leaves $d_D^2 d$ and $d_C^2 c$, both zero. ☐

**Exercise 24.62** (Short exact cone sequence)**.** Prove $0 \to D \to \mathrm{Cone}(f) \to \Sigma C \to 0$ is a short exact sequence of complexes.

*Hint.* Use inclusion into the first summand and projection to the second. ☐

*Solution.* The maps are degreewise exact, and the cone differential is compatible with inclusion and projection under the standard shifted differential. ☐

**Exercise 24.63** (Quasi-isomorphism criterion)**.** Prove $f$ is a quasi-isomorphism iff $\mathrm{Cone}(f)$ is acyclic.

*Hint.* Use the long exact homology sequence of the cone short exact sequence. ☐

*Solution.* Exactness identifies the connecting morphism with $f_*$ up to sign; all cone homology vanishes exactly when every $f_*$ is an isomorphism. ☐

**Exercise 24.64** (Homotopic maps and cones)**.** Explain why chain-homotopic maps have chain-homotopy-equivalent cones.

*Hint.* Use the homotopy operator to shear one cone summand. ☐

*Solution.* A chain homotopy gives an explicit triangular chain isomorphism/homotopy equivalence between the two cone constructions, so their homology agrees. ☐

## Counterexamples and hypothesis tests

**Exercise 24.65** (Cone acyclic means map zero)**.** Test: if $\mathrm{Cone}(f)$ is acyclic, then $f$ is the zero map.

*Hint.* Acyclic cone characterizes quasi-isomorphisms. ☐

*Solution.* False. The identity map has an acyclic cone and is usually far from zero. ☐

**Exercise 24.66** (Cone of zero always acyclic)**.** Test: $\mathrm{Cone}(0)$ is always acyclic.

*Hint.* It contains $D$ and a shifted copy of $C$. ☐

*Solution.* False; its homology is $H_n(D) \oplus H_{n-1}(C)$. ☐

**Exercise 24.67** (Cone forgets f)**.** Test: the cone differential depends only on $C$ and $D$, not on $f$.

*Hint.* Inspect the coupling term. ☐

*Solution.* False. The term $f(c)$ is precisely where the map is encoded. ☐

## Applications and investigations

**Exercise 24.68** (Relative model)**.** Why do mapping cones model relative phenomena?

*Hint.* A cone kills the source by attaching a shifted copy. □

*Solution.* The cone packages a map together with a formal null-homotopy of its source, paralleling the topological mapping cone whose reduced homology enters relative exact sequences. □

**Exercise 24.69** (Testing quasi-isomorphisms)**.** How can the cone reduce a many-degree isomorphism test to one homology computation?

*Hint.* Compute whether the cone is acyclic. □

*Solution.* Instead of checking each induced map directly, build the cone and verify that all its homology groups vanish. □

## Challenge problems

**Exercise 24.70** (Cone of multiplication)**.** For $f : \mathbb{Z} \to \mathbb{Z}$ given by multiplication by $m$ between complexes concentrated in degree 0, compute cone homology.

*Hint.* The cone is the two-term complex $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$. □

*Solution.* For $m \neq 0$, $H_0 \cong \mathbb{Z}/m$ and $H_1 = 0$; for $m = 0$, both expected shifted/source terms appear. □

**Exercise 24.71** (Cone synthesis)**.** Describe how mapping cones unify cokernels, shifts, exact sequences, and quasi-isomorphisms.

*Hint.* Use the cone short exact sequence. □

*Solution.* The cone joins $D$ to a shifted $C$, yields a long exact sequence whose connecting map is $f_*$, reduces to a direct sum for $f = 0$, and is acyclic precisely for quasi-isomorphisms. □

# Chapter 25

# Homology with Coefficients

Integral homology is only one member of a family of invariants. Replacing the coefficient group $\mathbb{Z}$ by an abelian group $G$ can reveal torsion, simplify calculations, and produce operations such as Bockstein connecting homomorphisms.

## 25.1 Chains with coefficients

For a chain complex $C_\bullet$ of free abelian groups and an abelian group $G$, define

$$C_n(C;G) = C_n \otimes G$$

with differential

$$\partial \otimes \mathrm{id}_G\,.$$

**Definition 25.1** (Homology with coefficients)**.**

$$H_n(C;G) = H_n(C_\bullet \otimes G).$$

For a space $X$,

$$H_n(X;G) = H_n(C_*^{\mathrm{sing}}(X) \otimes G).$$

## 25.2 Integer coefficients

Taking $G = \mathbb{Z}$ recovers ordinary integral homology.

## 25.3 Field coefficients

If $G = \mathbb{F}_p = \mathbb{Z}/p$, every chain group becomes a vector space over $\mathbb{F}_p$. Boundary matrices may be reduced modulo $p$, and homology can be computed by ordinary linear algebra.

## 25.4 Cellular chains with coefficients

For a CW complex,

$$C_n^{\mathrm{cell}}(X;G) \cong C_n^{\mathrm{cell}}(X;\mathbb{Z}) \otimes G.$$

Thus the same cellular boundary matrices are interpreted in $G$.

## 25.5 Degree-$m$ attachment

For

$$X_m = S^1 \cup_m D^2,$$

the coefficient chain complex is

$$0 \to G \xrightarrow{\times m} G \to G \to 0.$$

Hence the result depends strongly on whether multiplication by $m$ is injective or zero in $G$.

## 25.6 Projective plane mod 2

The integral cellular differential of $\mathbb{R}P^2$ is multiplication by 2:

$$0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0.$$

With $\mathbb{F}_2$-coefficients, this differential becomes zero.

Therefore

$$H_k(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2 \qquad (k = 0, 1, 2).$$

## 25.7 Projective spaces mod 2

More generally, every cellular differential in $\mathbb{R}P^n$ becomes zero modulo 2, so

$$H_k(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2 \qquad 0 \le k \le n.$$

## 25.8 Moore spaces

For

$$M(\mathbb{Z}/m, n) = S^n \cup_m D^{n+1},$$

the coefficient chain complex contains

$$G \xrightarrow{\times m} G.$$

Thus

$$H_{n+1}(M; G) = \ker(\times m : G \to G),$$

and

$$\widetilde{H}_n(M; G) = G/mG.$$

## 25.9 Example with mod $p$

For $G = \mathbb{F}_p$:

- if $p \nmid m$, multiplication by $m$ is an isomorphism;
- if $p \mid m$, multiplication by $m$ is zero.

So mod-$p$ homology detects exactly the $p$-primary part of this Moore attachment.

## 25.10 Tensor products of abelian groups

Basic identities include

$$\mathbb{Z} \otimes G \cong G, \qquad \mathbb{Z}/m \otimes G \cong G/mG,$$

and

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m,n).$$

These formulas foreshadow the universal coefficient theorem.

## 25.11 Tor preview

Tensoring is right exact but not exact in general. The failure of left exactness is measured by

$$\mathrm{Tor}_1^{\mathbb{Z}}(-,G).$$

For cyclic groups,

$$\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n).$$

VIII/26 makes this correction precise.

## 25.12 Short exact coefficient sequences

A short exact sequence of coefficient groups

$$0 \to G' \to G \to G'' \to 0$$

produces a short exact sequence of singular chain complexes because singular chain groups are free abelian.

Therefore it yields a long exact sequence in homology:

$$\cdots \to H_n(X;G') \to H_n(X;G) \to H_n(X;G'') \xrightarrow{\beta} H_{n-1}(X;G') \to \cdots.$$

## 25.13 Bockstein homomorphism

The connecting homomorphism

$$\beta : H_n(X;G'') \to H_{n-1}(X;G')$$

is called a Bockstein homomorphism for the chosen coefficient sequence.

A central example is

$$0 \to \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \to \mathbb{Z}/p \to 0.$$

## 25.14 Bockstein and torsion

The Bockstein can detect when a mod-$p$ class comes from $p$-torsion in integral homology.

For projective spaces this relates the rich mod-2 homology to integral 2-torsion.

## 25.15 Naturality

Continuous maps induce homomorphisms on homology with arbitrary coefficients, and coefficient homomorphisms

$$G \to H$$

also induce maps

$$H_n(X; G) \to H_n(X; H).$$

Thus homology is functorial in both the space and the coefficient group.

## 25.16 Reduced homology

Reduced homology with coefficients is defined using the augmented coefficient chain complex. For a connected space,

$$\widetilde{H}_0(X; G) = 0.$$

## 25.17 Relative coefficients

For a pair $(X, A)$,

$$H_n(X, A; G) = H_n(C_*(X, A) \otimes G).$$

Long exact sequences of pairs remain valid with coefficients.

## 25.18 Orientation with coefficients

A manifold that is not orientable over $\mathbb{Z}$ may become orientable over $\mathbb{F}_2$, because sign reversal is invisible modulo 2.

This phenomenon is developed systematically in the manifold chapters.

## 25.19 Characteristic dependence

Different coefficient fields can produce different Betti numbers. One should therefore always record the coefficient ring when reporting homology:

$$H_n(X; \mathbb{Z}), \quad H_n(X; \mathbb{Q}), \quad H_n(X; \mathbb{F}_p)$$

need not contain equivalent information.

## 25.20 Rational homology

Tensoring with $\mathbb{Q}$ kills finite torsion. Thus rational homology records the free part of integral homology.

## 25.21 Support-diagram provenance

The legacy support corpus contains explicit Moore-space, Bockstein, and projective-space coefficient diagrams mapped to this chapter. Their mathematical roles are integrated here; exact visual-instance reconciliation is reserved for the post-VIII/35 audit.

## 25.22 Bridge to UCT

Computing directly after tensoring chains works, but the universal coefficient theorem gives a far more efficient answer in terms of integral homology: a tensor term plus a Tor correction. VIII/26 develops that theorem.

## 25.23 Exercises

**Exercise 25.2.** Exercise 1

*Hint.* Tensor the chain groups and differential first, then take kernels and images; tensoring the already-computed homology can miss an adjacent Tor contribution. ☐

*Solution.* $H_n(X; G)$ is the homology of $C_*(X) \otimes G$. ☐

**Exercise 25.3.** Exercise 2

*Hint.* Use the chain isomorphism $c \otimes n \mapsto nc$ from $C_*(X) \otimes \mathbb{Z}$ to the integral chain complex. ☐

*Solution.* Taking $G = \mathbb{Z}$ recovers integral homology. ☐

**Exercise 25.4.** Exercise 3

*Hint.* The free integral chain generators become a basis over $\mathbb{F}_p$; compute the differential as a linear map over that field. ☐

*Solution.* With $G = \mathbb{F}_p$, chain groups are vector spaces. ☐

**Exercise 25.5.** Exercise 4

*Hint.* Reduce every integer matrix entry in the same field, then recompute ranks and kernels there rather than reusing rational ranks. ☐

*Solution.* Cellular boundary matrices may be reduced modulo $p$. ☐

**Exercise 25.6.** Exercise 5

*Hint.* The relevant two-term coefficient complex is $G \xrightarrow{m} G$; both its kernel and cokernel may differ from their integral counterparts. ☐

*Solution.* For $S^1 \cup_m D^2$, the key coefficient differential is multiplication by $m$ on $G$. ☐

**Exercise 25.7.** Exercise 6

*Hint.* Replace the integer boundary coefficient two by its residue in $\mathbb{F}_2$; this changes the kernel as well as the image. ☐

*Solution.* For $\mathbb{R}P^2$ mod 2, the multiplication-by-2 boundary becomes zero. ☐

**Exercise 25.8.** Exercise 7

*Hint.* There is one mod-two cellular generator in each degree zero, one, and two, and both differentials vanish in that field. ☐

*Solution.* $H_k(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$ for $k = 0, 1, 2$. ☐

**Exercise 25.9.** Exercise 8

*Hint.* In degree $n$, all chains are cycles and the incoming subgroup is $mG$; compute the quotient in the chosen coefficient group. $\square$

*Solution.* For a Moore space, $\widetilde{H}_n(-;G) \cong G/mG$. $\square$

**Exercise 25.10.** Exercise 9

*Hint.* At degree $n+1$, there is no higher incoming chain group; cycles are exactly the elements of $G$ annihilated by $m$. $\square$

*Solution.* For the same Moore space, $H_{n+1}(-;G) = \ker(\times m)$. $\square$

**Exercise 25.11.** Exercise 10

*Hint.* Use that the residue of $m$ is nonzero in a field; invert it to determine both the kernel and cokernel. $\square$

*Solution.* If $p \nmid m$, multiplication by $m$ is invertible on $\mathbb{F}_p$. $\square$

**Exercise 25.12.** Exercise 11

*Hint.* When $m$ is zero in $\mathbb{F}_p$, the two-term differential is zero, so both adjacent chain groups contribute to homology. $\square$

*Solution.* If $p \mid m$, multiplication by $m$ is zero on $\mathbb{F}_p$. $\square$

**Exercise 25.13.** Exercise 12

*Hint.* Define $n \otimes g \mapsto ng$ and its inverse $g \mapsto 1 \otimes g$; verify they respect the tensor relations. $\square$

*Solution.* $\mathbb{Z} \otimes G \cong G$. $\square$

**Exercise 25.14.** Exercise 13

*Hint.* Tensor the presentation $\mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$ and use right exactness to identify its cokernel. $\square$

*Solution.* $\mathbb{Z}/m \otimes G \cong G/mG$. $\square$

**Exercise 25.15.** Exercise 14

*Hint.* The tensor product is generated by $1 \otimes 1$, annihilated by both $m$ and $n$; use their greatest common divisor to determine its order. $\square$

*Solution.* $\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m,n)$. $\square$

**Exercise 25.16.** Exercise 15

*Hint.* Tensor the injection $\mathbb{Z} \xrightarrow{p} \mathbb{Z}$ with $\mathbb{Z}/p$; its loss of injectivity illustrates the kernel recorded by Tor. $\square$

*Solution.* Tor measures the failure of tensoring to be left exact. $\square$

**Exercise 25.17.** Exercise 16

*Hint.* Singular chain groups are free, hence flat, so tensoring the short exact coefficient sequence remains exact in each chain degree. $\square$

*Solution.* A short exact coefficient sequence yields a long exact homology sequence. $\square$

**Exercise 25.18.** Exercise 17

*Hint.* Lift a quotient-coefficient cycle to a middle-coefficient chain, take its boundary, and identify the resulting class with coefficients in the leftmost group. ☐

*Solution.* The connecting map in a coefficient LES is a Bockstein. ☐

**Exercise 25.19.** Exercise 18

*Hint.* Lift a mod-$p$ cycle to an integral chain $c$; its boundary is divisible by $p$, and $(\partial c)/p$ represents the integral connecting class. ☐

*Solution.* The sequence $0 \to \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \to \mathbb{Z}/p \to 0$ is the standard Bockstein source. ☐

**Exercise 25.20.** Exercise 19

*Hint.* A coefficient map $\phi : G \to H$ gives $\mathrm{id} \otimes \phi$ on chains; verify this commutes with the coefficient differentials. ☐

*Solution.* Homology is functorial in coefficient homomorphisms. ☐

**Exercise 25.21.** Exercise 20

*Hint.* Use the quotient singular chain complex, whose basis consists of simplices not wholly in $A$, then tensor it with $G$. ☐

*Solution.* Relative homology is defined with coefficients by tensoring the relative chain complex. ☐

**Exercise 25.22.** Exercise 21

*Hint.* Use path connectedness for the singular-homology statement: augmentation identifies $H_0(X; G)$ with $G$, leaving a zero reduced kernel. ☐

*Solution.* Reduced $H_0$ of a connected space is zero over any coefficient group. ☐

**Exercise 25.23.** Exercise 22

*Hint.* In characteristic two, $-1 = 1$; oppositely oriented faces have the same coefficient, explaining why orientation obstructions change. ☐

*Solution.* Mod-2 coefficients erase orientation signs. ☐

**Exercise 25.24.** Exercise 23

*Hint.* For a torsion element killed by nonzero $m$, multiplication by $m$ becomes invertible over $\mathbb{Q}$; its tensor contribution vanishes. ☐

*Solution.* Rational coefficients kill finite torsion. ☐

**Exercise 25.25.** Exercise 24

*Hint.* Keep both degrees in view: tensor uses integral $H_n$, while the correction uses integral $H_{n-1}$. ☐

*Solution.* VIII/26 expresses coefficient homology through tensor and Tor terms. ☐

## 25.24 Solved problem dossiers

**Problem 25.26** (RP2 mod 2)**.** Compute $H_*(\mathbb{R}P^2; \mathbb{F}_2)$ cellularly.

*Solution.* The integral differential $d_2 = 2$ becomes zero modulo 2, as does $d_1$; hence one copy of $\mathbb{F}_2$ survives in degrees 0,1,2. ☐

**Problem 25.27** (RP2 mod 3)**.** Compute positive homology of $\mathbb{R}P^2; \mathbb{F}_3$.

*Solution.* Multiplication by 2 is invertible mod 3, so $H_2 = H_1 = 0$. ☐

**Problem 25.28** (Degree m mod p)**.** For $S^1 \cup_m D^2$, compute $H_1(-; \mathbb{F}_p)$.

*Solution.* It is the cokernel of multiplication by $m$ on $\mathbb{F}_p$: zero if $p \nmid m$, and $\mathbb{F}_p$ if $p \mid m$. ☐

**Problem 25.29** (Moore mod p)**.** Compute reduced homology of $M(\mathbb{Z}/m, n)$ over $\mathbb{F}_p$.

*Solution.* If $p \nmid m$, it vanishes; if $p \mid m$, copies of $\mathbb{F}_p$ occur in degrees $n$ and $n + 1$. ☐

**Problem 25.30** (Tensor cyclic)**.** Compute $\mathbb{Z}/12 \otimes \mathbb{Z}/18$.

*Solution.* It is $\mathbb{Z}/\gcd(12, 18) \cong \mathbb{Z}/6$. ☐

**Problem 25.31** (Tensor with Q)**.** Compute $\mathbb{Z}/m \otimes \mathbb{Q}$.

*Solution.* Zero, since multiplication by $m$ is invertible on $\mathbb{Q}$. ☐

**Problem 25.32** (Rational projective)**.** Compute rational homology of $\mathbb{R}P^2$.

*Solution.* Integral $H_1 = \mathbb{Z}/2$ is killed by tensoring with $\mathbb{Q}$; only $H_0 \cong \mathbb{Q}$ remains. ☐

**Problem 25.33** (Rational torus)**.** Compute $H_1(T^2; \mathbb{Q})$.

*Solution.* $\mathbb{Z}^2 \otimes \mathbb{Q} \cong \mathbb{Q}^2$. ☐

**Problem 25.34** (Coefficient map)**.** What map does $\mathbb{Z} \to \mathbb{Z}/p$ induce?

*Solution.* It gives reduction mod $p$: $H_n(X; \mathbb{Z}) \to H_n(X; \mathbb{Z}/p)$, though not every mod-$p$ class need arise this way because of Tor/Bockstein effects. ☐

**Problem 25.35** (Bockstein source)**.** Explain the Bockstein from $0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{Z}/p \to 0$.

*Solution.* It is the connecting map $H_n(X; \mathbb{Z}/p) \to H_{n-1}(X; \mathbb{Z})$. ☐

**Problem 25.36** (Bockstein meaning)**.** What can a nonzero Bockstein indicate?

*Solution.* It indicates an obstruction to lifting a mod-$p$ class to integral coefficients and often detects $p$-torsion one degree lower. ☐

**Problem 25.37** (Field computation)**.** Why are mod-$p$ cellular calculations often easier?

*Solution.* Boundary matrices become matrices over a field, so Gaussian elimination computes kernels and images directly. ☐

**Problem 25.38** (Orientation mod 2)**.** Why can every manifold be locally oriented mod 2?

*Solution.* Changing an orientation sign multiplies by $-1$, which equals 1 in $\mathbb{F}_2$. ☐

**Problem 25.39** (Relative coefficients)**.** State the coefficient LES of a pair.

*Solution.* $\cdots \to H_n(A;G) \to H_n(X;G) \to H_n(X,A;G) \to H_{n-1}(A;G) \to \cdots$. ☐

**Problem 25.40** (Disjoint union)**.** How does coefficient homology behave on disjoint unions?

*Solution.* Chains split by components, so $H_n(\coprod X_\alpha;G) \cong \bigoplus H_n(X_\alpha;G)$. ☐

**Problem 25.41** (Sphere coefficients)**.** Compute $H_*(S^n;G)$.

*Solution.* For nonempty $G$, $H_0 \cong G$, $H_n \cong G$, and other groups vanish. ☐

**Problem 25.42** (Circle coefficients)**.** Compute $H_1(S^1;G)$.

*Solution.* It is $G$, since the cellular differential is zero. ☐

**Problem 25.43** (Moore G/mG)**.** Explain the quotient $G/mG$.

*Solution.* It is the cokernel of the degree-$m$ cellular boundary $G \xrightarrow{\times m} G$. ☐

**Problem 25.44** (Moore kernel)**.** Explain the extra $H_{n+1}$ coefficient group.

*Solution.* It is the kernel of multiplication by $m$ on $G$, which may be nonzero even though integral top homology vanishes. ☐

**Problem 25.45** (Mod 2 projective spaces)**.** Compute $H_k(\mathbb{R}P^n;\mathbb{F}_2)$.

*Solution.* Every cellular differential becomes zero, so it is $\mathbb{F}_2$ in every degree $0 \le k \le n$. ☐

**Problem 25.46** (Coefficient dependence)**.** Give a space whose Betti numbers depend on the field.

*Solution.* $\mathbb{R}P^2$: over $\mathbb{Q}$, positive homology vanishes; over $\mathbb{F}_2$, both $H_1$ and $H_2$ are nonzero. ☐

**Problem 25.47** (Tor preview)**.** Compute $\operatorname{Tor}(\mathbb{Z}/12, \mathbb{Z}/18)$.

*Solution.* It is also $\mathbb{Z}/6$, illustrating that tensor and Tor have the same cyclic gcd size but occur in different UCT degrees. ☐

**Problem 25.48** (Support figures)**.** Where do Bockstein/RP2 and Moore coefficient support diagrams belong?

*Solution.* Their canonical mathematical destination is VIII/25 because they visualize coefficient change and torsion detection. ☐

**Problem 25.49** (Bridge)**.** Why is direct chain tensoring not the end of the story?

*Solution.* Homology does not generally commute with tensor products exactly; VIII/26 adds the Tor correction through the universal coefficient theorem. ☐

## 25.25 Supplementary worked examples

**Example 25.50** (The circle with arbitrary coefficients)**.** For any abelian group $G$,

$$H_0(S^1; G) \cong G, \qquad H_1(S^1; G) \cong G,$$

and higher groups vanish. The cellular chain complex has one copy of $G$ in degrees 0 and 1 with zero differential.

**Example 25.51** (Torsion changes under coefficients)**.** For $\mathbb{RP}^2$, the cellular differential $C_2 \to C_1$ is multiplication by 2. With $\mathbb{F}_2$ coefficients it becomes zero, so mod-2 homology sees a degree-2 class that integral homology does not.

**Example 25.52** (Tensor and Tor effects)**.** Changing coefficients is not merely tensoring homology in all cases. The universal coefficient theorem gives

$$0 \to H_n(X; \mathbb{Z}) \otimes G \to H_n(X; G) \to \operatorname{Tor}(H_{n-1}(X; \mathbb{Z}), G) \to 0,$$

so torsion one degree lower can create new classes.

## 25.26 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 25.53** (Point coefficients)**.** Compute $H_0(\{*\}; G)$.

*Hint.* The degree-zero chain group is $G$. ☐

*Solution.* $H_0(\{*\}; G) \cong G$. ☐

**Exercise 25.54** (Circle coefficients)**.** Compute $H_1(S^1; G)$.

*Hint.* Use the one-cell CW complex. ☐

*Solution.* $H_1(S^1; G) \cong G$. ☐

**Exercise 25.55** (Sphere coefficients)**.** Compute $H_n(S^n; G)$ for $n > 0$.

*Hint.* Use one cell in degrees 0 and $n$. ☐

*Solution.* $H_n(S^n; G) \cong G$, with $H_0 \cong G$ and no other nonzero groups. ☐

**Exercise 25.56** (Tensor with Z/m)**.** Compute $\mathbb{Z}/n \otimes \mathbb{Z}/m$.

*Hint.* The answer has order $\gcd(m, n)$. ☐

*Solution.* $\mathbb{Z}/n \otimes \mathbb{Z}/m \cong \mathbb{Z}/\gcd(m, n)$. ☐

**Exercise 25.57** (Tor with Z/m)**.** Compute $\operatorname{Tor}(\mathbb{Z}/n, \mathbb{Z}/m)$.

*Hint.* Use a two-term free resolution of $\mathbb{Z}/n$. ☐

*Solution.* It is also $\mathbb{Z}/\gcd(m, n)$. ☐

## Proofs

**Exercise 25.58** (Chain-level definition)**.** Explain why $C_*(X;G) = C_*(X;\mathbb{Z}) \otimes G$ is again a chain complex.

*Hint.* Tensor the boundary with the identity. □

*Solution.* The differential is $\partial \otimes \mathrm{id}_G$, whose square is $(\partial^2) \otimes \mathrm{id} = 0$. □

**Exercise 25.59** (Field simplification)**.** Show that over a field, homology can be computed by ordinary linear algebra.

*Hint.* Chain groups become vector spaces. □

*Solution.* Boundary maps are matrices over the field, so $H_n = \ker d_n / \operatorname{im} d_{n+1}$ is obtained from ranks and nullspaces. □

**Exercise 25.60** (Naturality in coefficients)**.** Explain how a homomorphism $G \to G'$ induces $H_n(X;G) \to H_n(X;G')$.

*Hint.* Tensor the chain complex with the coefficient homomorphism. □

*Solution.* The map $\mathrm{id} \otimes \phi$ is a chain map and therefore induces a homology map. □

**Exercise 25.61** (Mod-p detection)**.** Why can mod-$p$ coefficients reveal $p$-torsion?

*Hint.* Both tensor and Tor terms can survive. □

*Solution.* $p$-torsion contributes to tensor products in its own degree and through Tor from the preceding degree, producing nonzero mod-$p$ classes. □

## Counterexamples and hypothesis tests

**Exercise 25.62** (Coefficients never change Betti numbers)**.** Test: changing coefficients never changes homology dimensions.

*Hint.* Compare integral and mod-2 homology of $\mathbb{RP}^2$. □

*Solution.* False. Torsion can become visible over finite fields and change dimensions. □

**Exercise 25.63** (Tensor only)**.** Test: $H_n(X;G)$ is always $H_n(X;\mathbb{Z}) \otimes G$.

*Hint.* Remember the Tor correction. □

*Solution.* False; the universal coefficient theorem includes $\mathrm{Tor}(H_{n-1}(X), G)$. □

**Exercise 25.64** (Field sees all torsion orders)**.** Test: homology over one field determines all integral torsion.

*Hint.* A field only probes its characteristic. □

*Solution.* False. For example, $\mathbb{F}_p$ is sensitive to $p$-primary behavior but not to torsion of unrelated prime order. □

## Applications and investigations

**Exercise 25.65** (Choosing coefficients)**.** Why might one compute first over several finite fields before doing Smith normal form over $\mathbb{Z}$?

*Hint.* Field linear algebra is cheaper. ☐

*Solution.* Finite-field computations quickly reveal Betti numbers and likely torsion primes, guiding or validating the more expensive integral computation. ☐

**Exercise 25.66** (Orientation coefficients)**.** Why do coefficient choices matter for orientability questions?

*Hint.* A sign obstruction disappears in characteristic 2. ☐

*Solution.* Over $\mathbb{F}_2$, $-1 = 1$, so every manifold has a mod-2 fundamental class even when no integral orientation exists. ☐

## Challenge problems

**Exercise 25.67** (RP2 mod 2)**.** Using its cellular differential 2, compute $H_*(\mathbb{RP}^2; \mathbb{F}_2)$.

*Hint.* Multiplication by 2 becomes zero. ☐

*Solution.* There is one $\mathbb{F}_2$ in degrees $0, 1, 2$. ☐

**Exercise 25.68** (Coefficient synthesis)**.** Describe how integral, rational, and mod-$p$ homology complement one another.

*Hint.* Separate free rank from prime torsion information. ☐

*Solution.* Rational homology isolates free ranks; mod-$p$ homology cheaply detects $p$-primary effects; integral homology assembles the full free and torsion structure. ☐

# Chapter 26

# The Universal Coefficient Theorem

The universal coefficient theorem converts integral homology into homology with arbitrary coefficients. The price for tensoring is a Tor correction term.

## 26.1 Statement for homology

**Theorem 26.1** (Universal coefficient theorem for homology)**.** *For a space $X$ and abelian group $G$, there is a natural short exact sequence*

$$0 \to H_n(X;\mathbb{Z}) \otimes G \longrightarrow H_n(X;G) \longrightarrow \mathrm{Tor}(H_{n-1}(X;\mathbb{Z}), G) \to 0.$$

The sequence splits, but not naturally in general.

## 26.2 Meaning

The first term is the expected coefficient change from degree $n$. The Tor term imports torsion information from degree $n-1$.

## 26.3 Free homology

If $H_{n-1}(X;\mathbb{Z})$ is free abelian, then

$$\mathrm{Tor}(H_{n-1}(X), G) = 0,$$

and therefore

$$H_n(X;G) \cong H_n(X;\mathbb{Z}) \otimes G.$$

## 26.4 Field coefficients

For $G = \mathbb{F}_p$,

$$A \otimes \mathbb{F}_p \cong A/pA,$$

while

$$\mathrm{Tor}(A, \mathbb{F}_p)$$

detects $p$-torsion in $A$.

## 26.5 Cyclic formulas

For positive integers $m, n$,

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m,n),$$

and

$$\operatorname{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n).$$

## 26.6 Free summands

$$\operatorname{Tor}(\mathbb{Z}, G) = 0, \qquad \mathbb{Z} \otimes G \cong G.$$

## 26.7 Finite direct sums

Tensor and Tor distribute over finite direct sums, so finitely generated integral homology can be handled one cyclic summand at a time.

## 26.8 Projective plane

For $\mathbb{R}P^2$,

$$H_0 = \mathbb{Z}, \quad H_1 = \mathbb{Z}/2, \quad H_2 = 0.$$

With $G = \mathbb{F}_2$, UCT gives

$$H_1(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$$

from the tensor term and

$$H_2(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$$

from the Tor term.

## 26.9 Why a top mod-2 class appears

The integral top group $H_2(\mathbb{R}P^2; \mathbb{Z})$ vanishes, yet mod-2 $H_2$ is nonzero. UCT explains this as

$$\operatorname{Tor}(H_1(\mathbb{R}P^2), \mathbb{F}_2) \cong \mathbb{F}_2.$$

## 26.10 Moore spaces

For a Moore space with

$$\widetilde{H}_n(M; \mathbb{Z}) = \mathbb{Z}/m,$$

UCT gives coefficient homology in degrees $n$ and $n+1$:

$$\widetilde{H}_n(M; G) \cong (\mathbb{Z}/m) \otimes G,$$

$$\widetilde{H}_{n+1}(M; G) \cong \operatorname{Tor}(\mathbb{Z}/m, G).$$

## 26.11 Rational coefficients

Since $\mathbb{Q}$ is torsion-free and divisible, for finitely generated homology the Tor contribution vanishes and finite torsion tensors to zero. Thus

$$H_n(X; \mathbb{Q}) \cong H_n(X; \mathbb{Z})_{\text{free}} \otimes \mathbb{Q}.$$

## 26.12 Proof idea

Singular chain groups are free abelian. One analyzes

$$0 \to Z_n \to C_n \to B_{n-1} \to 0$$

and the quotient

$$0 \to B_n \to Z_n \to H_n \to 0,$$

then tensors with $G$. The possible failure of exactness is precisely encoded by Tor.

## 26.13 Free resolutions

For an abelian group $A$, choose a free resolution

$$0 \to F_1 \to F_0 \to A \to 0.$$

Then

$$\mathrm{Tor}(A, G) = \ker(F_1 \otimes G \to F_0 \otimes G).$$

## 26.14 Cyclic resolution

For $A = \mathbb{Z}/m$, use

$$0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to \mathbb{Z}/m \to 0.$$

Tensoring with $G$ shows

$$\mathrm{Tor}(\mathbb{Z}/m, G) = \ker(\times m : G \to G).$$

## 26.15 Tensor cyclic quotient

The same resolution gives

$$(\mathbb{Z}/m) \otimes G \cong G/mG.$$

Thus the UCT terms match the direct Moore-space calculation from VIII/25.

## 26.16 Naturality

The UCT short exact sequence is natural in both $X$ and $G$.

A map $X \to Y$ induces a commutative diagram of UCT sequences, as does a homomorphism of coefficient groups.

## 26.17 Nonnatural splitting

Although the UCT short exact sequence splits as a sequence of abelian groups, there is generally no preferred splitting compatible with all maps.

Therefore one should treat

$$H_n(X; G)$$

as an extension unless a canonical identification is available.

## 26.18 Dimension over a field

For finite-type homology and $G = \mathbb{F}_p$,

$$\dim_{\mathbb{F}_p} H_n(X; \mathbb{F}_p)$$

receives contributions from the free rank of $H_n(X;\mathbb{Z})$, from $p$-torsion in $H_n$, and from $p$-torsion in $H_{n-1}$.

## 26.19 Betti numbers

Over $\mathbb{Q}$, the dimension is exactly the integral Betti number:

$$b_n = \dim_{\mathbb{Q}} H_n(X; \mathbb{Q}).$$

## 26.20 Relative UCT

The same theorem applies to relative chain complexes:

$$0 \to H_n(X, A; \mathbb{Z}) \otimes G \to H_n(X, A; G) \to \mathrm{Tor}(H_{n-1}(X, A; \mathbb{Z}), G) \to 0.$$

## 26.21 Reduced UCT

Reduced homology satisfies the same formula in the expected degrees.

## 26.22 Computational workflow

For finitely generated integral homology:

1. decompose each group into free and cyclic torsion summands;
2. tensor the degree-$n$ group with $G$;
3. compute Tor from the degree-$(n-1)$ group;
4. assemble the resulting short exact sequence.

## 26.23 Bridge to Künneth

UCT concerns one chain complex with changed coefficients. The Künneth theorem concerns tensor products of two chain complexes. Again, a tensor term gives the main answer and Tor provides the correction.

## 26.24 Exercises

**Exercise 26.2.** Exercise 1

*Hint.* Identify the left-hand map $H_n(X;\mathbb{Z}) \otimes G \to H_n(X;G)$; the integral group is in the same degree as the requested coefficient homology. □

*Solution.* UCT has a tensor term from $H_n(X;\mathbb{Z})$. □

**Exercise 26.3.** Exercise 2

*Hint.* The Tor correction uses the integral group one degree lower, $H_{n-1}(X;\mathbb{Z})$, rather than another copy of $H_n$. □

*Solution.* Its correction is $\mathrm{Tor}(H_{n-1}(X;\mathbb{Z}), G)$. □

**Exercise 26.4.** Exercise 3

*Hint.* Distinguish a natural short exact sequence from a chosen splitting of that sequence; an abstract direct-sum isomorphism need not commute with induced maps. □

*Solution.* The UCT sequence splits but generally not naturally. □

**Exercise 26.5.** Exercise 4

*Hint.* Free abelian groups have zero first Tor with any coefficient group; apply this to the lower-degree integral homology term only. □

*Solution.* If $H_{n-1}$ is free, the Tor term vanishes. □

**Exercise 26.6.** Exercise 5

*Hint.* Use the map $n \otimes g \mapsto ng$ and inverse $g \mapsto 1 \otimes g$ to identify the free rank-one contribution. □

*Solution.* $\mathbb{Z} \otimes G \cong G$. □

**Exercise 26.7.** Exercise 6

*Hint.* Resolve the free group $\mathbb{Z}$ by itself in degree zero; there is no positive resolution homology from which Tor could arise. □

*Solution.* $\mathrm{Tor}(\mathbb{Z}, G) = 0$. □

**Exercise 26.8.** Exercise 7

*Hint.* Tensor the presentation of $\mathbb{Z}/m$ and take the cokernel of multiplication by $m$ on $G$. □

*Solution.* $\mathbb{Z}/m \otimes G \cong G/mG$. □

**Exercise 26.9.** Exercise 8

*Hint.* Use the same two-term free resolution, but now take the kernel of multiplication by $m$, not its cokernel. □

*Solution.* $\mathrm{Tor}(\mathbb{Z}/m, G) = \ker(\times m : G \to G)$. □

**Exercise 26.10.** Exercise 9

*Hint.* Compute $(\mathbb{Z}/n)/m(\mathbb{Z}/n)$; its order is the index of the image of multiplication by $m$. □

*Solution.* $\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$. □

**Exercise 26.11.** Exercise 10

*Hint.* Solve $mx = 0$ in $\mathbb{Z}/n$ and identify that cyclic subgroup; distinguish this kernel from the quotient used for the tensor term. □

*Solution.* $\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n)$. □

**Exercise 26.12.** Exercise 11

*Hint.* List integral $H_2(\mathbb{R}P^2) = 0$ and $H_1(\mathbb{R}P^2) = \mathbb{Z}/2$; the zero tensor term does not force coefficient homology to vanish. $\square$

*Solution.* UCT explains the top mod-2 class of $\mathbb{R}P^2$. $\square$

**Exercise 26.13.** Exercise 12

*Hint.* Compute $\ker(2 : \mathbb{F}_2 \to \mathbb{F}_2)$ for the lower-degree torsion group and place its contribution in mod-two degree two. $\square$

*Solution.* For $\mathbb{R}P^2$, mod-2 $H_2$ comes from Tor of integral $H_1$. $\square$

**Exercise 26.14.** Exercise 13

*Hint.* For a Moore space, place $G/mG$ in the torsion group's own degree and $\ker(m : G \to G)$ in the next degree. $\square$

*Solution.* Moore-space coefficient homology has tensor and Tor in adjacent degrees. $\square$

**Exercise 26.15.** Exercise 14

*Hint.* Multiplication by any nonzero integer is invertible on $\mathbb{Q}$, so both cyclic torsion tensor terms and their Tor terms vanish. $\square$

*Solution.* Rational coefficients kill finite torsion. $\square$

**Exercise 26.16.** Exercise 15

*Hint.* Decompose finitely generated integral homology into free and torsion summands; rationalization preserves exactly one dimension per free generator. $\square$

*Solution.* Over $\mathbb{Q}$, homology dimensions are Betti numbers. $\square$

**Exercise 26.17.** Exercise 16

*Hint.* For $f : X \to Y$, use $H_n(f) \otimes \mathrm{id}$ on the left and the induced Tor map from $H_{n-1}(f)$ on the right. $\square$

*Solution.* UCT is natural in space maps. $\square$

**Exercise 26.18.** Exercise 17

*Hint.* For $G \to G'$, compare the tensor and Tor functors in their coefficient argument, and keep the same integral homology groups. $\square$

*Solution.* UCT is natural in coefficient homomorphisms. $\square$

**Exercise 26.19.** Exercise 18

*Hint.* Apply $- \otimes G$ to $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$; the two derived outputs are its kernel and cokernel. $\square$

*Solution.* A cyclic free resolution computes both tensor quotient and Tor kernel. $\square$

**Exercise 26.20.** Exercise 19

*Hint.* Relative singular chains are free abelian on simplices not lying wholly in the subspace, so the free-chain UCT applies to that quotient complex. $\square$

*Solution.* Relative homology satisfies UCT. ☐

**Exercise 26.21.** Exercise 20

*Hint.* Use the augmented complex and track the bottom degree separately; the reduced version removes the universal component contribution, not higher torsion. ☐

*Solution.* Reduced homology satisfies UCT. ☐

**Exercise 26.22.** Exercise 21

*Hint.* Distribute tensor and Tor over the finite free-and-cyclic decomposition before combining the contributions to the short exact sequence. ☐

*Solution.* Finite direct sums can be handled summandwise. ☐

**Exercise 26.23.** Exercise 22

*Hint.* Each cyclic summand divisible by $p$ contributes one dimension, from both degree $n$ tensor and degree $n-1$ Tor, in addition to the free rank. ☐

*Solution.* Mod-$p$ dimensions detect $p$-torsion in adjacent integral degrees. ☐

**Exercise 26.24.** Exercise 23

*Hint.* A splitting selects lifts of quotient elements into the middle group; explain why such a selection is extra data rather than part of the natural UCT maps. ☐

*Solution.* The splitting is algebraic, not canonically functorial. ☐

**Exercise 26.25.** Exercise 24

*Hint.* For a product, replace the fixed coefficient group by the graded homology of a second complex and track total degrees in tensor and Tor sums. ☐

*Solution.* VIII/27 applies the same tensor-plus-Tor philosophy to products. ☐

## 26.25 Solved problem dossiers

**Problem 26.26** (RP2 H2 mod2)**.** Use UCT to compute $H_2(\mathbb{R}P^2; \mathbb{F}_2)$.

*Solution.* The tensor term is zero because integral $H_2 = 0$. The Tor term is $\mathrm{Tor}(\mathbb{Z}/2, \mathbb{F}_2) \cong \mathbb{F}_2$. ☐

**Problem 26.27** (RP2 H1 mod2)**.** Compute $H_1(\mathbb{R}P^2; \mathbb{F}_2)$.

*Solution.* $(\mathbb{Z}/2) \otimes \mathbb{F}_2 \cong \mathbb{F}_2$, while Tor from $H_0 = \mathbb{Z}$ vanishes. ☐

**Problem 26.28** (RP2 mod3)**.** Compute positive homology with $\mathbb{F}_3$.

*Solution.* Both tensor and Tor of $\mathbb{Z}/2$ with $\mathbb{F}_3$ vanish, so positive homology is zero. ☐

**Problem 26.29** (Moore mod p)**.** Apply UCT to $M(\mathbb{Z}/m, n)$ with $\mathbb{F}_p$.

*Solution.* If $p \mid m$, tensor gives $\mathbb{F}_p$ in degree $n$ and Tor gives $\mathbb{F}_p$ in degree $n+1$; otherwise both vanish. ☐

**Problem 26.30** (Cyclic G)**.** Compute $(\mathbb{Z}/12) \otimes \mathbb{Z}/18$ and Tor.

*Solution.* Both are $\mathbb{Z}/6$. □

**Problem 26.31** (Free plus torsion)**.** Let $H_n = \mathbb{Z}^2 \oplus \mathbb{Z}/6$. Tensor with $\mathbb{F}_2$.

*Solution.* The free part gives $\mathbb{F}_2^2$ and the torsion summand gives $\mathbb{F}_2$, so the tensor term is $\mathbb{F}_2^3$. □

**Problem 26.32** (Adjacent Tor)**.** If $H_{n-1} = \mathbb{Z}/4$, what Tor contribution appears with $\mathbb{F}_2$?

*Solution.* $\mathrm{Tor}(\mathbb{Z}/4, \mathbb{F}_2) \cong \mathbb{F}_2$. □

**Problem 26.33** (Rationalization)**.** Let $H_n = \mathbb{Z}^3 \oplus \mathbb{Z}/12$. Compute $H_n(-;\mathbb{Q})$ assuming adjacent Tor vanishes.

*Solution.* $\mathbb{Q}^3$. □

**Problem 26.34** (Cyclic resolution)**.** Use the standard resolution to compute $\mathrm{Tor}(\mathbb{Z}/m, G)$.

*Solution.* Tensor $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m \to 0$; Tor is the kernel of multiplication by $m$ on $G$. □

**Problem 26.35** (Tensor quotient)**.** Use the same resolution to compute $\mathbb{Z}/m \otimes G$.

*Solution.* It is the cokernel of multiplication by $m$, namely $G/mG$. □

**Problem 26.36** (Natural map)**.** What happens to UCT under $f : X \to Y$?

*Solution.* Tensor, coefficient homology, and Tor maps form a commutative diagram of short exact UCT sequences. □

**Problem 26.37** (Nonnatural split)**.** Why should one not write a canonical direct sum in UCT?

*Solution.* A splitting exists as groups, but no choice is generally compatible with all maps; only the short exact sequence is canonical. □

**Problem 26.38** (Relative example)**.** Apply UCT to $H_n(D^n, S^{n-1}; G)$.

*Solution.* Integral relative homology is $\mathbb{Z}$ in degree $n$, so coefficient relative homology is $G$ there and zero elsewhere. □

**Problem 26.39** (Sphere)**.** Use UCT to compute $H_n(S^n; G)$.

*Solution.* Integral top homology is free $\mathbb{Z}$ and adjacent lower homology is zero for $n > 1$, so top homology is $G$. □

**Problem 26.40** (Circle)**.** Use UCT to compute $H_1(S^1; G)$.

*Solution.* $\mathbb{Z} \otimes G \cong G$, and Tor from $H_0 = \mathbb{Z}$ vanishes. □

**Problem 26.41** (Betti)**.** Why does rational homology recover Betti numbers?

*Solution.* Finite torsion disappears and Tor vanishes, leaving one $\mathbb{Q}$ for each free integral generator. □

**Problem 26.42** (Mod p dimension)**.** Why can $\dim H_n(X; \mathbb{F}_p)$ exceed $b_n$?

*Solution.* Besides free rank, $p$-torsion in $H_n$ contributes through tensor and $p$-torsion in $H_{n-1}$ contributes through Tor. □

**Problem 26.43** (UCT workflow)**.** Given finitely generated integral homology, what is the first computational step?

*Solution.* Decompose each group into free and cyclic torsion summands, then apply tensor and Tor summandwise. □

**Problem 26.44** (Extension issue)**.** What information can remain after tensor and Tor are known?

*Solution.* One must still know how the two terms fit into the short exact sequence; the theorem says it splits noncanonically for homology over $\mathbb{Z}$. □

**Problem 26.45** (Coefficient consistency)**.** How does UCT confirm direct cellular mod-$p$ calculations?

*Solution.* Both methods produce the same coefficient homology; UCT explains which classes come from tensor and which are new Tor classes. □

**Problem 26.46** (Bridge)**.** How does Künneth resemble UCT?

*Solution.* Both have a primary tensor expression and a Tor correction measuring failure of exactness. □

**Problem 26.47** (Final principle)**.** What does UCT save computationally?

*Solution.* It avoids rebuilding the entire coefficient chain complex once integral homology is known. □

## 26.26 Supplementary worked examples

**Example 26.48** (UCT for homology with a cyclic coefficient group)**.** If $H_n(X;\mathbb{Z})$ and $H_{n-1}(X;\mathbb{Z})$ are known, then

$$0 \to H_n(X) \otimes G \to H_n(X;G) \to \mathrm{Tor}(H_{n-1}(X), G) \to 0.$$

For $G = \mathbb{Z}/m$, both terms reduce to explicit gcd computations on cyclic summands.

**Example 26.49** (UCT over a field)**.** For a field $F$, the homological UCT still records torsion through Tor when $F = \mathbb{F}_p$, whereas with $F = \mathbb{Q}$ all torsion vanishes and

$$H_n(X;\mathbb{Q}) \cong H_n(X;\mathbb{Z}) \otimes \mathbb{Q}.$$

**Example 26.50** (Noncanonical splitting)**.** The UCT short exact sequence splits as a sequence of abelian groups, but generally not naturally. Thus one may identify the isomorphism type of $H_n(X;G)$ without obtaining a preferred decomposition into tensor and Tor summands.

## 26.27 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 26.51** (Free input)**.** If $H_{n-1}(X)$ is free, what happens to the Tor term in homological UCT?

*Hint.* Tor of a free abelian group vanishes. □

*Solution.* The Tor term is zero. □

**Exercise 26.52** (Rational coefficients)**.** What is $\mathrm{Tor}(A,\mathbb{Q})$ for an abelian group $A$?

*Hint.* $\mathbb{Q}$ is flat over $\mathbb{Z}$. □

*Solution.* It vanishes. □

**Exercise 26.53** (Cyclic tensor)**.** Compute $(\mathbb{Z}/6) \otimes \mathbb{Z}/4$.

*Hint.* Use the gcd formula. □

*Solution.* It is $\mathbb{Z}/2$. □

**Exercise 26.54** (Cyclic Tor)**.** Compute $\mathrm{Tor}(\mathbb{Z}/6, \mathbb{Z}/4)$.

*Hint.* Use the same gcd. □

*Solution.* It is $\mathbb{Z}/2$. □

**Exercise 26.55** (Free rank over Q)**.** If $H_n(X) \cong \mathbb{Z}^b \oplus T$, compute $H_n(X;\mathbb{Q})$.

*Hint.* Torsion tensors to zero with $\mathbb{Q}$. □

*Solution.* $H_n(X;\mathbb{Q}) \cong \mathbb{Q}^b$. □

## Proofs

**Exercise 26.56** (Derive cyclic Tor)**.** Derive $\mathrm{Tor}(\mathbb{Z}/n, G)$ from $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n \to 0$.

*Hint.* Tensor the resolution with $G$. ☐

*Solution.* The Tor group is the kernel of multiplication by $n$ on $G$, i.e. $G[n]$. ☐

**Exercise 26.57** (Naturality)**.** Explain what naturality of the UCT means for a map $X \to Y$.

*Hint.* All terms and arrows should commute with induced maps. ☐

*Solution.* The homology maps, tensor maps, Tor maps, and coefficient homology maps form a commutative diagram of short exact sequences. ☐

**Exercise 26.58** (Splitting caveat)**.** Why does existence of a splitting not give a canonical projection onto the Tor summand?

*Hint.* The theorem does not choose a natural splitting. ☐

*Solution.* Different algebraic choices produce different complements, so no functorial projection is determined in general. ☐

**Exercise 26.59** (Torsion shift)**.** Prove $p$-torsion in $H_{n-1}(X)$ can contribute to $H_n(X; \mathbb{F}_p)$.

*Hint.* Inspect the Tor term. ☐

*Solution.* $\mathrm{Tor}(\mathbb{Z}/p, \mathbb{F}_p) \cong \mathbb{F}_p$, so such torsion appears one degree higher in coefficient homology. ☐

## Counterexamples and hypothesis tests

**Exercise 26.60** (UCT sequence canonically splits)**.** Test: the homological UCT has a natural splitting for all spaces and coefficients.

*Hint.* The splitting is generally noncanonical. ☐

*Solution.* False. A splitting exists for abelian groups, but not naturally in the data. ☐

**Exercise 26.61** (No torsion means no coefficient change)**.** Test: if $H_n(X)$ is torsion-free, then $H_n(X; G) = H_n(X) \otimes G$.

*Hint.* Look one degree lower. ☐

*Solution.* False unless the relevant Tor term from $H_{n-1}(X)$ also vanishes. ☐

**Exercise 26.62** (Q determines integral homology)**.** Test: rational homology determines integral homology.

*Hint.* Rationalization loses torsion. ☐

*Solution.* False; spaces with different torsion can have identical rational homology. ☐

### Applications and investigations

**Exercise 26.63** (Predict mod-p ranks)**.** How can UCT predict a jump in mod-$p$ Betti number?

*Hint.* Count both tensor and Tor contributions. □

*Solution.* A $p$-torsion summand in $H_n$ contributes by tensoring, while one in $H_{n-1}$ contributes through Tor, so either can increase the $\mathbb{F}_p$ dimension. □

**Exercise 26.64** (Cross-check computation)**.** How can UCT validate a direct boundary-matrix computation with coefficients?

*Hint.* Compute the expected group independently from integral homology. □

*Solution.* Comparing the direct coefficient result with the tensor-plus-Tor prediction catches rank, torsion, or indexing errors. □

### Challenge problems

**Exercise 26.65** (Moore-space pattern)**.** Suppose $\widetilde{H}_k(X) \cong \mathbb{Z}/m$ and other reduced integral homology vanishes. Describe where mod-$p$ homology can appear when $p \mid m$.

*Hint.* Use tensor in degree $k$ and Tor in degree $k+1$. □

*Solution.* An $\mathbb{F}_p$ class appears in degree $k$ from tensor and another in degree $k+1$ from Tor. □

**Exercise 26.66** (UCT synthesis)**.** Explain the conceptual role of the tensor and Tor terms.

*Hint.* One changes existing classes; the other measures failure of tensoring to preserve exactness. □

*Solution.* The tensor term transports degree-$n$ integral classes to the new coefficients, while Tor records torsion obstructions from degree $n-1$ created by the nonexactness of tensor product. □

# Chapter 27

# Products and the Künneth Theorem

Products of spaces lead to tensor products of chains. The Künneth theorem expresses the homology of $X \times Y$ from the homology of the factors, with Tor terms measuring torsion interactions.

## 27.1 Product cells

If $e^p \subset X$ and $e^q \subset Y$ are cells, then

$$e^p \times e^q$$

is a cell of dimension $p + q$ in the product CW structure.

This immediately gives

$$\chi(X \times Y) = \chi(X)\chi(Y)$$

for finite CW complexes.

## 27.2 Tensor product of chain complexes

For chain complexes $C, D$, define

$$(C \otimes D)_n = \bigoplus_{p+q=n} C_p \otimes D_q.$$

The differential on homogeneous tensors is

$$\partial(c \otimes d) = \partial c \otimes d + (-1)^p c \otimes \partial d, \qquad c \in C_p.$$

## 27.3 Koszul sign

The sign $(-1)^p$ is necessary for

$$\partial^2 = 0.$$

## 27.4 Eilenberg–Zilber principle

For spaces $X, Y$, singular chains on the product are chain-homotopy equivalent to the tensor product:

$$C_*^{\text{sing}}(X \times Y) \simeq C_*^{\text{sing}}(X) \otimes C_*^{\text{sing}}(Y).$$

This is the chain-level bridge from topology to algebra.

## 27.5 Cross product

The Eilenberg–Zilber construction yields the homology cross product

$$\times : H_p(X) \otimes H_q(Y) \to H_{p+q}(X \times Y).$$

## 27.6 Künneth theorem over $\mathbb{Z}$

**Theorem 27.1** (Künneth theorem)**.** *For suitable spaces such as CW complexes, there is a natural short exact sequence*

$$0 \to \bigoplus_{p+q=n} H_p(X) \otimes H_q(Y) \longrightarrow H_n(X \times Y) \longrightarrow \bigoplus_{p+q=n-1} \mathrm{Tor}(H_p(X), H_q(Y)) \to 0.$$

The sequence splits noncanonically.

## 27.7 Field coefficients

Over a field $k$, Tor vanishes and Künneth becomes

$$H_n(X \times Y; k) \cong \bigoplus_{p+q=n} H_p(X; k) \otimes_k H_q(Y; k).$$

## 27.8 Product of circles

Since

$$H_0(S^1) = H_1(S^1) = \mathbb{Z},$$

Künneth gives

$$H_0(T^2) = \mathbb{Z}, \qquad H_1(T^2) = \mathbb{Z}^2, \qquad H_2(T^2) = \mathbb{Z}.$$

## 27.9 Higher tori

For

$$T^n = (S^1)^n,$$

$$H_k(T^n; \mathbb{Z}) \cong \mathbb{Z}^{\binom{n}{k}}.$$

The total homology ranks are the binomial coefficients.

## 27.10 Products of spheres

For $S^m \times S^n$, homology is free and concentrated in degrees

$$0, \quad m, \quad n, \quad m+n,$$

with the middle degrees combining if $m = n$.

## 27.11 $S^2 \times S^2$

$$H_0 = \mathbb{Z}, \quad H_2 = \mathbb{Z}^2, \quad H_4 = \mathbb{Z},$$

and all other groups vanish.

## 27.12 Cylinder

$$S^1 \times I \simeq S^1.$$

Künneth agrees: $I$ has only $H_0 = \mathbb{Z}$, so tensoring introduces no new positive-dimensional classes.

## 27.13 Product with a contractible space

If $Y$ is nonempty and contractible,

$$X \times Y \simeq X,$$

and Künneth reduces correspondingly.

## 27.14 Torsion interaction

If both factors have torsion, Tor can create homology one degree higher than the naive tensor terms predict.

## 27.15 Example with projective spaces

Since

$$H_1(\mathbb{R}P^2) = \mathbb{Z}/2,$$

the product $\mathbb{R}P^2 \times \mathbb{R}P^2$ has tensor torsion and a Tor contribution. In particular,

$$\mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$$

appears in degree 3.

## 27.16 Moore-space products

For Moore spaces, the tensor and Tor terms can be read directly from the cyclic torsion groups:

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m,n).$$

## 27.17 Naturality

The Künneth short exact sequence is natural in maps of both factors.

## 27.18 Nonnatural splitting

As with UCT, the splitting is generally not natural. The short exact sequence, rather than a chosen direct-sum decomposition, is the canonical object.

## 27.19 Reduced Künneth

For pointed connected spaces, reduced homology often gives a cleaner product formula by separating the $H_0$ terms.

## 27.20 Relative Künneth preview

Pairs and products of pairs admit relative forms of Künneth under suitable hypotheses. These are useful in manifold and bundle calculations.

## 27.21 Euler characteristic from Künneth

Over a field, alternating dimensions factor:

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

This recovers the product formula from homology.

## 27.22 Betti-number convolution

Over a field,

$$b_n(X \times Y) = \sum_{p+q=n} b_p(X)b_q(Y).$$

Thus Poincaré polynomials multiply:

$$P_{X\times Y}(t) = P_X(t)P_Y(t).$$

## 27.23 Poincaré polynomial

Define

$$P_X(t) = \sum_n b_n(X)t^n.$$

For the torus,

$$P_{T^n}(t) = (1+t)^n.$$

## 27.24 Chain-homotopy input

The Eilenberg–Zilber and Alexander–Whitney maps are inverse up to chain homotopy, not literally inverse. The machinery of VIII/22 is exactly what allows them to induce inverse homology isomorphisms.

## 27.25 Why the theorem matters

Künneth replaces a potentially large product-space chain calculation with:

- tensor products of known homology groups;
- Tor corrections for torsion.

## 27.26 Bridge to cohomology

Products become even richer in cohomology because the diagonal map and cross product combine to form the cup product. VIII/28 begins cohomology, and VIII/29 develops its multiplicative structure.

## 27.27 Exercises

**Exercise 27.2.** Exercise 1

*Hint.* In total degree $n$, include every bidegree $(p, q)$ with $p + q = n$; distinguish a graded direct sum from a single tensor factor. □

*Solution.* $(C \otimes D)_n = \bigoplus_{p+q=n} C_p \otimes D_q$. □

**Exercise 27.3.** Exercise 2

*Hint.* For homogeneous $c \in C_p$, use $\partial(c \otimes d) = \partial c \otimes d + (-1)^p c \otimes \partial d$. □

*Solution.* The product differential uses the sign $(-1)^p$. □

**Exercise 27.4.** Exercise 3

*Hint.* Expand the double differential; the two mixed terms have signs $(-1)^{p-1}$ and $(-1)^p$ and therefore cancel. □

*Solution.* The sign is needed for $\partial^2 = 0$. □

**Exercise 27.5.** Exercise 4

*Hint.* Compare $C_*(X \times Y)$ with $C_*(X) \otimes C_*(Y)$ by chain maps and chain homotopies, not by claiming their generators coincide. □

*Solution.* Eilenberg–Zilber compares product chains with tensor-product chains. □

**Exercise 27.6.** Exercise 5

*Hint.* The cross product combines representative cycles of dimensions $p$ and $q$; the tensor differential verifies their product has zero boundary. □

*Solution.* The cross product sends $H_p(X) \otimes H_q(Y)$ to $H_{p+q}(X \times Y)$. □

**Exercise 27.7.** Exercise 6

*Hint.* Sum $H_p(X) \otimes H_q(Y)$ over $p + q = n$, including the degree-zero contributions from either factor. □

*Solution.* Künneth has a tensor term in total degree $n$. □

**Exercise 27.8.** Exercise 7

*Hint.* A Tor term from degrees $p, q$ contributes to degree $p + q + 1$; check this shift before computing a product's torsion. □

*Solution.* Its Tor term uses total degree $n - 1$. □

**Exercise 27.9.** Exercise 8

*Hint.* Tensor over the coefficient field itself; vector spaces are flat, so no Tor correction remains in the field-coefficient formula. □

*Solution.* Over a field, Tor vanishes. □

**Exercise 27.10.** Exercise 9

*Hint.* In total degree one, list bidegrees $(1, 0)$ and $(0, 1)$ for the two circles and add their tensor contributions. □

*Solution.* $H_1(T^2) \cong \mathbb{Z}^2$. □

**Exercise 27.11.** Exercise 10

*Hint.* The top class comes from the product of the two degree-one circle generators, in bidegree $(1, 1)$. □

*Solution.* $H_2(T^2) \cong \mathbb{Z}$. □

**Exercise 27.12.** Exercise 11

*Hint.* Choose which $k$ circle factors contribute degree one while the others contribute degree zero; count those choices. □

*Solution.* $H_k(T^n) \cong \mathbb{Z}^{\binom{n}{k}}$. □

**Exercise 27.13.** Exercise 12

*Hint.* List bidegrees $(2, 0)$ and $(0, 2)$; each supplies an independent copy of $\mathbb{Z}$, while the $(2, 2)$ class lies in degree four. □

*Solution.* $H_2(S^2 \times S^2) \cong \mathbb{Z}^2$. □

**Exercise 27.14.** Exercise 13

*Hint.* For a nonempty contractible factor, the projection is a homotopy equivalence; alternatively use its sole ordinary degree-zero homology group in Künneth. □

*Solution.* Product with a contractible factor preserves homology. □

**Exercise 27.15.** Exercise 14

*Hint.* Locate torsion groups in both factors and test their common prime divisors; the resulting Tor class sits one degree above the corresponding tensor total degree. □

*Solution.* Torsion in both factors can create Tor classes. □

**Exercise 27.16.** Exercise 15

*Hint.* Use the resolution of $\mathbb{Z}/2$ and compute the kernel of multiplication by two on the second $\mathbb{Z}/2$. □

*Solution.* $\mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$. □

**Exercise 27.17.** Exercise 16

*Hint.* For torsion in Moore degrees $r, s$, the mixed tensor contribution occurs in $r + s$ and the mixed Tor contribution in $r + s + 1$, both governed by $\gcd(m, n)$. □

*Solution.* Moore-space products produce gcd torsion in tensor and Tor. □

**Exercise 27.18.** Exercise 17

*Hint.* For maps of both spaces, check that the cross product obeys $(f \times g)_*(a \times b) = f_*(a) \times g_*(b)$. □

*Solution.* Künneth is natural in both factors. ☐

**Exercise 27.19.** Exercise 18

*Hint.* The exact sequence has natural maps, but identifying its middle term with the sum of tensor and Tor terms requires a generally nonnatural splitting. ☐

*Solution.* Its splitting is generally nonnatural. ☐

**Exercise 27.20.** Exercise 19

*Hint.* Over a field, take dimensions of each tensor summand and add over $p+q=n$; finite-type assumptions make these ordinary finite Betti-number sums. ☐

*Solution.* Betti numbers convolve under products over a field. ☐

**Exercise 27.21.** Exercise 20

*Hint.* Multiply the two generating functions and identify the coefficient of $t^n$ with the convolution formula for product Betti numbers. ☐

*Solution.* Poincaré polynomials multiply. ☐

**Exercise 27.22.** Exercise 21

*Hint.* The circle contributes $1+t$; multiply one copy for each circle factor and compare coefficients using the binomial theorem. ☐

*Solution.* $P_{T^n}(t) = (1+t)^n$. ☐

**Exercise 27.23.** Exercise 22

*Hint.* For finite CW type, evaluate the product of Poincare polynomials at $t=-1$ to recover the Euler-product formula. ☐

*Solution.* Euler characteristic is multiplicative. ☐

**Exercise 27.24.** Exercise 23

*Hint.* Use the homotopies between each composite and the relevant identity to show the induced homology maps are inverse, even though the chain maps are not strict inverses. ☐

*Solution.* Eilenberg–Zilber inverse maps are only inverse up to chain homotopy. ☐

**Exercise 27.25.** Exercise 24

*Hint.* The diagonal $X \to X \times X$ turns an external cohomology product into an internal one; retain degrees and signs when passing to the cup-product ring. ☐

*Solution.* VIII/29 will turn products into the cup-product ring structure. ☐

## 27.28 Solved problem dossiers

**Problem 27.26** (Torus)**.** Compute $H_*(S^1 \times S^1)$ with Künneth.

*Solution.* All groups are free, so no Tor occurs: tensor terms give $\mathbb{Z}$ in degrees 0 and 2 and $\mathbb{Z}^2$ in degree 1. □

**Problem 27.27** (Three-torus)**.** Compute the ranks of $H_k(T^3)$.

*Solution.* They are $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3} = 1, 3, 3, 1$. □

**Problem 27.28** (Four-torus)**.** Compute $b_2(T^4)$.

*Solution.* $\binom{4}{2} = 6$. □

**Problem 27.29** (Sphere product)**.** Compute $H_*(S^2 \times S^3)$.

*Solution.* Free groups $\mathbb{Z}$ occur in degrees 0,2,3,5 and vanish elsewhere. □

**Problem 27.30** (Equal spheres)**.** Compute $H_2(S^2 \times S^2)$.

*Solution.* There are two tensor contributions $H_2(S^2) \otimes H_0(S^2)$ and $H_0 \otimes H_2$, giving $\mathbb{Z}^2$. □

**Problem 27.31** (Top class)**.** Where does the top class of $S^m \times S^n$ come from?

*Solution.* From $H_m(S^m) \otimes H_n(S^n) \cong \mathbb{Z}$ in total degree $m + n$. □

**Problem 27.32** (Contractible factor)**.** Use Künneth for $X \times I$.

*Solution.* Only $H_0(I) = \mathbb{Z}$ is nonzero, so $H_n(X \times I) \cong H_n(X)$. □

**Problem 27.33** (RP2 product Tor)**.** Locate a Tor contribution in $\mathbb{R}P^2 \times \mathbb{R}P^2$.

*Solution.* Since both factors have $H_1 = \mathbb{Z}/2$, $\mathrm{Tor}(H_1, H_1) = \mathbb{Z}/2$ contributes in total degree $1 + 1 + 1 = 3$. □

**Problem 27.34** (RP2 product H2)**.** Give tensor contributions to $H_2(\mathbb{R}P^2 \times \mathbb{R}P^2)$.

*Solution.* The $H_1 \otimes H_1$ term gives $\mathbb{Z}/2$, while integral $H_2$ of each factor vanishes. □

**Problem 27.35** (Moore gcd)**.** For Moore torsion groups $\mathbb{Z}/12$ and $\mathbb{Z}/18$, compute tensor and Tor.

*Solution.* Both are $\mathbb{Z}/6$. □

**Problem 27.36** (Field Kunneth)**.** Why is Künneth especially simple over $\mathbb{F}_p$?

*Solution.* Vector spaces are flat and Tor vanishes, leaving a direct sum of tensor products. □

**Problem 27.37** (Betti convolution)**.** If $P_X = 1 + 2t + t^2$ and $P_Y = 1 + t$, compute $P_{X \times Y}$.

*Solution.* Multiply to get $1 + 3t + 3t^2 + t^3$. □

**Problem 27.38** (Euler product)**.** Use Poincaré polynomials to recover $\chi(X \times Y)$.

*Solution.* Euler characteristic is $P(-1)$; since product polynomials multiply, $\chi(X \times Y) = \chi(X)\chi(Y)$. □

**Problem 27.39** (Cross product)**.** What homology class does $a \times b$ have degree?

*Solution.* If $a \in H_p(X)$ and $b \in H_q(Y)$, then $a \times b \in H_{p+q}(X \times Y)$. ☐

**Problem 27.40** (Naturality)**.** How does $f \times g$ act on cross products?

*Solution.* $(f \times g)_*(a \times b) = f_*(a) \times g_*(b)$. ☐

**Problem 27.41** (Nonnatural split)**.** Why is the Künneth exact sequence preferable to a chosen direct sum?

*Solution.* The exact sequence is natural; a splitting usually depends on arbitrary choices. ☐

**Problem 27.42** (Reduced product)**.** Why can reduced homology simplify product formulas?

*Solution.* It separates the ubiquitous degree-zero $\mathbb{Z}$ terms, making the genuinely positive-dimensional cross terms more transparent. ☐

**Problem 27.43** (Eilenberg–Zilber)**.** What is the role of chain homotopy in the theorem?

*Solution.* The Alexander–Whitney and shuffle/Eilenberg–Zilber maps compose to maps chain-homotopic to identities, so they induce actual inverse homology isomorphisms. ☐

**Problem 27.44** (Koszul sign)**.** Verify the cross terms in $\partial^2(c \otimes d)$ cancel.

*Solution.* The second application of the differential produces cross terms with opposite signs because the degree of $\partial c$ is $p - 1$. ☐

**Problem 27.45** (Product cells)**.** How does a $p$-cell times a $q$-cell contribute?

*Solution.* It gives a $(p+q)$-cell, explaining both the tensor grading and Euler multiplicativity. ☐

**Problem 27.46** (Two streams)**.** Why is VIII/27 sourced from both topology-13 and topology-17?

*Solution.* One stream carries the main product/Künneth theorem while the other carries complementary product calculations and homological machinery; both have the same canonical destination. ☐

**Problem 27.47** (Bridge)**.** How do products lead toward cup products?

*Solution.* The cohomology cross product composed with the diagonal $X \to X \times X$ produces the cup product. ☐

**Problem 27.48** (Final principle)**.** What does Künneth compute?

*Solution.* It reconstructs homology of a product from factor homologies via tensor products plus Tor corrections. ☐

**Problem 27.49** (Comparison with UCT)**.** State the analogy with VIII/26.

*Solution.* UCT changes coefficients for one complex; Künneth combines two complexes. Both are controlled by tensor products corrected by Tor. ☐

## 27.29 Supplementary worked examples

**Example 27.50** (The torus from two circles)**.** Since $H_*(S^1)$ is free, the Tor terms vanish in Kunneth. Thus

$$H_n(S^1 \times S^1) \cong \bigoplus_{p+q=n} H_p(S^1) \otimes H_q(S^1),$$

giving $\mathbb{Z}$, $\mathbb{Z}^2$, $\mathbb{Z}$ in degrees $0, 1, 2$.

**Example 27.51** (A product of spheres)**.** For $S^m \times S^n$ with $m, n > 0$, free homology gives classes in degrees $0, m, n, m+n$ (with merging when $m = n$). The top class is the cross product of the two sphere fundamental classes.

**Example 27.52** (Where Tor enters)**.** For spaces with torsion homology, Kunneth contains a shifted Tor term. This explains why product homology may contain classes not visible in the simple tensor products of same-total-degree homology groups.

## 27.30 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 27.53** (Product with a point)**.** Compute $H_*(X \times \{*\})$ using Kunneth.

*Hint.* $H_0(\{*\}) = \mathbb{Z}$ is the only nonzero factor. □

*Solution.* One recovers $H_*(X)$. □

**Exercise 27.54** (Torus H1)**.** Compute $H_1(S^1 \times S^1)$.

*Hint.* Use degrees $(1, 0)$ and $(0, 1)$. □

*Solution.* $H_1 \cong \mathbb{Z} \oplus \mathbb{Z}$. □

**Exercise 27.55** (Torus H2)**.** Compute $H_2(T^2)$.

*Hint.* Use the tensor $H_1(S^1) \otimes H_1(S^1)$. □

*Solution.* $H_2(T^2) \cong \mathbb{Z}$. □

**Exercise 27.56** (Sphere product top class)**.** Compute $H_{m+n}(S^m \times S^n)$.

*Hint.* Only the product of top classes contributes. □

*Solution.* It is $\mathbb{Z}$. □

**Exercise 27.57** (Euler product)**.** Use Kunneth ranks to recover $\chi(X \times Y) = \chi(X)\chi(Y)$ when homology is finite and free.

*Hint.* Alternate the tensor-product ranks. □

*Solution.* The double alternating sum factors as the product of the two Euler-characteristic sums. □

### Proofs

**Exercise 27.58** (Free-factor Kunneth)**.** Prove Tor vanishes if one factor has free integral homology.

*Hint.* Tor with a free abelian group is zero. ☐

*Solution.* Every homology group of that factor is free, so all Tor summands in the Kunneth sequence vanish. ☐

**Exercise 27.59** (Cross-product generators)**.** Explain how classes $a \in H_p(X)$ and $b \in H_q(Y)$ produce a class in $H_{p+q}(X \times Y)$.

*Hint.* Use the homology cross product. ☐

*Solution.* The Eilenberg–Zilber map sends the tensor of representative cycles to a cycle on the product, defining $a \times b$. ☐

**Exercise 27.60** (Naturality)**.** State the naturality of cross products under maps $f : X \to X'$ and $g : Y \to Y'$.

*Hint.* Apply the product map. ☐

*Solution.* $(f \times g)_*(a \times b) = f_*(a) \times g_*(b)$. ☐

**Exercise 27.61** (Kunneth short exact structure)**.** Explain why the Tor term is shifted by one total degree.

*Hint.* Recall the standard homological Kunneth exact sequence. ☐

*Solution.* The Tor contribution to $H_n(X \times Y)$ comes from pairs with $p + q = n - 1$, reflecting the first derived functor of tensor product. ☐

### Counterexamples and hypothesis tests

**Exercise 27.62** (Always tensor only)**.** Test: $H_n(X \times Y)$ is always just $\bigoplus_{p+q=n} H_p(X) \otimes H_q(Y)$.

*Hint.* Torsion can create Tor terms. ☐

*Solution.* False over $\mathbb{Z}$ unless suitable freeness hypotheses remove Tor. ☐

**Exercise 27.63** (Product Betti numbers add)**.** Test: $b_n(X \times Y) = b_n(X) + b_n(Y)$.

*Hint.* Over a field, use convolution. ☐

*Solution.* False. Betti numbers satisfy $b_n(X \times Y) = \sum_{p+q=n} b_p(X)b_q(Y)$. ☐

**Exercise 27.64** (Top class without orientability)**.** Test: every product of closed manifolds has an integral top class.

*Hint.* Orientability matters. ☐

*Solution.* False; an integral fundamental class requires orientability of the product, equivalently of both factors in the connected case. ☐

## Applications and investigations

**Exercise 27.65** (High-dimensional construction)**.** How can Kunneth compute a large product without constructing its boundary matrix?

*Hint.* Use already-known homology of the factors. □

*Solution.* It replaces a potentially huge product chain complex by tensor and Tor operations on much smaller homology groups. □

**Exercise 27.66** (Persistent feature intuition)**.** Why is the cross product useful when interpreting independent topological features in two factors?

*Hint.* A feature in each factor combines into a higher-dimensional product feature. □

*Solution.* A $p$-cycle and a $q$-cycle can form a $(p+q)$-cycle, such as two loop classes combining into the torus fundamental class. □

## Challenge problems

**Exercise 27.67** (Three-torus)**.** Compute the integral Betti numbers of $T^3 = (S^1)^3$.

*Hint.* Iterate Kunneth or expand $(1+t)^3$. □

*Solution.* The Betti numbers are $1, 3, 3, 1$. □

**Exercise 27.68** (Kunneth synthesis)**.** Describe a practical Kunneth workflow over $\mathbb{Z}$.

*Hint.* List tensor contributions, then Tor contributions, then resolve the short exact sequence. □

*Solution.* For each total degree, compute all $H_p(X) \otimes H_q(Y)$ with $p+q=n$, all Tor groups with $p+q=n-1$, assemble the Kunneth short exact sequence, and use any available splitting or structural information. □

# Part VI

# Cohomology, Bundles and Manifolds

# Chapter 28

# Cohomology

Homology is built from chains and boundaries. Cohomology reverses the direction: a cochain assigns algebraic data to chains, and the coboundary operator is obtained by precomposing with the chain boundary. The resulting groups are contravariant in spaces and support additional structures that have no direct homological analogue.

## 28.1 Cochains

Let $C_\bullet$ be a chain complex and $G$ an abelian group.

**Definition 28.1** (Cochain group)**.** The group of $n$-cochains with coefficients in $G$ is

$$C^n(C;G) = \mathrm{Hom}(C_n, G).$$

For a space $X$,

$$C^n(X;G) = \mathrm{Hom}(C_n^{\mathrm{sing}}(X), G).$$

## 28.2 Coboundary

**Definition 28.2** (Coboundary operator)**.** For $\varphi \in C^n(C;G)$, define

$$\delta\varphi = \varphi \circ \partial_{n+1}.$$

Thus

$$\delta : C^n \to C^{n+1}.$$

Since $\partial^2 = 0$,

$$\delta^2 = 0.$$

## 28.3 Cocycles and coboundaries

**Definition 28.3.**

$$Z^n(C;G) = \ker\delta, \qquad B^n(C;G) = \mathrm{im}\,\delta.$$

Elements of $Z^n$ are cocycles and elements of $B^n$ are coboundaries.

## 28.4 Cohomology groups

**Definition 28.4** (Cohomology)**.**

$$H^n(C;G) = Z^n(C;G)/B^n(C;G).$$

For a space $X$,

$$H^n(X;G) = H^n(C^*(X;G)).$$

## 28.5 Contravariance

A continuous map

$$f : X \to Y$$

induces

$$f_\# : C_*(X) \to C_*(Y).$$

Precomposition defines

$$f^\# : C^n(Y;G) \to C^n(X;G), \qquad f^\#(\varphi) = \varphi \circ f_\#.$$

**Theorem 28.5** (Functoriality)**.** *The cochain maps $f^\#$ induce*

$$f^* : H^n(Y;G) \to H^n(X;G),$$

*with*

$$(g \circ f)^* = f^* \circ g^*.$$

## 28.6 Degree zero

For a path-connected space and coefficient group $G$,

$$H^0(X;G) \cong G.$$

More generally, $H^0$ is the group of locally constant $G$-valued functions on path components.

## 28.7 Spheres

With integer coefficients,

$$H^k(S^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

## 28.8 Cohomology of a point

$$H^0(*;G) \cong G, \qquad H^n(*;G) = 0 \quad (n > 0).$$

## 28.9 Simplicial cohomology

For a simplicial complex, cochains are homomorphisms from simplicial chain groups to $G$. With chosen oriented simplices, a cochain can be regarded as an assignment of coefficients to oriented simplices, compatible with sign.

Simplicial and singular cohomology agree on simplicial complexes.

## 28.10 Cellular cohomology

For a CW complex, dualizing the cellular chain complex gives the cellular cochain complex

$$C^n_{\text{cell}}(X;G) = \text{Hom}(C_n^{\text{cell}}(X), G).$$

For finite CW complexes, this often gives the fastest computation.

## 28.11 Evaluation pairing

A cochain evaluates on a chain. Passing to homology and cohomology gives the Kronecker pairing

$$\langle\ ,\ \rangle : H^n(X;G) \times H_n(X;\mathbb{Z}) \to G.$$

If $[\varphi] \in H^n$ and $[z] \in H_n$, then

$$\langle [\varphi], [z] \rangle = \varphi(z).$$

This is well defined because cocycles vanish on boundaries and coboundaries vanish on cycles.

## 28.12 Relative cohomology

For a pair $A \subset X$,

$$C^n(X, A; G) = \text{Hom}(C_n(X, A), G).$$

**Definition 28.6** (Relative cohomology)**.**

$$H^n(X, A; G) = H^n(C^*(X, A; G)).$$

## 28.13 Long exact sequence of a pair

A pair has a natural long exact cohomology sequence

$$\cdots \to H^n(X, A; G) \to H^n(X; G) \to H^n(A; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \to \cdots .$$

Notice the connecting map raises degree.

## 28.14 Connecting homomorphism

If a cocycle on $A$ extends to a cochain on $X$, its coboundary vanishes on $A$ and therefore determines a relative cocycle one degree higher. This is the cohomological connecting map.

## 28.15 Annulus example

Let $A$ be one boundary circle of an annulus $X = S^1 \times I$. Since the annulus deformation retracts onto $A$, the relative cohomology

$$H^*(X, A; G)$$

vanishes. The pair exact sequence reflects that the restriction

$$H^1(X; G) \to H^1(A; G)$$

is an isomorphism.

This is the canonical home of the mapped legacy annulus/connecting visual.

## 28.16 Universal coefficient theorem for cohomology

For integral homology and coefficient group $G$, one has a natural short exact sequence

$$0 \to \operatorname{Ext}\big(H_{n-1}(X;\mathbb{Z}),G\big) \to H^n(X;G) \to \operatorname{Hom}\big(H_n(X;\mathbb{Z}),G\big) \to 0.$$

The sequence splits noncanonically in general.

## 28.17 Free homology case

If $H_{n-1}(X;\mathbb{Z})$ is free, then the Ext term vanishes and

$$H^n(X;G) \cong \operatorname{Hom}(H_n(X;\mathbb{Z}),G).$$

## 28.18 Cyclic Ext formula

For positive $m$,

$$\operatorname{Ext}(\mathbb{Z}/m,G) \cong G/mG.$$

For $G=\mathbb{Z}$,

$$\operatorname{Ext}(\mathbb{Z}/m,\mathbb{Z}) \cong \mathbb{Z}/m.$$

Thus torsion in homology may appear one degree higher in integral cohomology.

## 28.19 Projective-plane example

Integral homology of $\mathbb{R}P^2$ has

$$H_1 \cong \mathbb{Z}/2, \qquad H_2 = 0.$$

The cohomology UCT gives

$$H^2(\mathbb{R}P^2;\mathbb{Z}) \cong \mathbb{Z}/2.$$

## 28.20 Homotopy invariance

Homotopic maps induce the same cohomology homomorphisms because their chain maps are chain homotopic and precomposition converts that chain homotopy into a cochain homotopy.

Thus homotopy-equivalent spaces have isomorphic cohomology groups.

## 28.21 Reduced cohomology

Reduced cohomology removes the degree-zero contribution of a basepoint, in parallel with reduced homology. For $n>0$,

$$\widetilde{H}^n(X;G) \cong H^n(X;G).$$

## 28.22 Why cohomology is richer

As graded groups, cohomology often resembles homology. But cohomology admits a natural multiplication

$$H^p(X;R) \times H^q(X;R) \to H^{p+q}(X;R),$$

the cup product. VIII/29 develops this ring structure.

## 28.23 SVG visual laboratory

This chapter ships with editable SVG assets for:

- the chain-to-cochain reversal and evaluation pairing;
- the annulus pair exact-sequence/connecting-map geometry.

The SVG sources are tracked as production assets and do not require an external SVG renderer for the LaTeX build.

## 28.24 Exercises

**Exercise 28.7.** Exercise 1

*Hint.* A degree-$n$ cochain assigns values in $G$ linearly to degree-$n$ chains; for infinitely many free generators, this Hom group is a product of coefficient groups. □

*Solution.* $C^n(C;G) = \mathrm{Hom}(C_n, G)$. □

**Exercise 28.8.** Exercise 2

*Hint.* For $\varphi : C_n \to G$, precompose with $\partial_{n+1} : C_{n+1} \to C_n$; the resulting functional has cochain degree $n+1$. □

*Solution.* The coboundary is $\delta\varphi = \varphi\partial$. □

**Exercise 28.9.** Exercise 3

*Hint.* Evaluate $\delta^2\varphi$ on a chain and use $\partial_n\partial_{n+1} = 0$; keep the order of precomposition explicit. □

*Solution.* $\delta^2 = 0$ follows from $\partial^2 = 0$. □

**Exercise 28.10.** Exercise 4

*Hint.* A cocycle in degree $n$ lies in the kernel of the outgoing map $C^n \to C^{n+1}$. □

*Solution.* Cocycles are $\ker\delta$. □

**Exercise 28.11.** Exercise 5

*Hint.* A degree-$n$ coboundary comes from a degree-$(n-1)$ cochain, so use the incoming coboundary map. □

*Solution.* Coboundaries are $\mathrm{im}\,\delta$. □

**Exercise 28.12.** Exercise 6

*Hint.* Use $\delta^2 = 0$ to place coboundaries inside cocycles; quotient only after these groups have been identified in the same degree. □

*Solution.* $H^n = Z^n/B^n$. □

**Exercise 28.13.** Exercise 7

*Hint.* Pull back a functional by precomposing it with $f_\# : C_n(X) \to C_n(Y)$; this reverses the direction of the induced map. □

*Solution.* A map $f : X \to Y$ induces $f^* : H^n(Y) \to H^n(X)$. ☐

**Exercise 28.14.** Exercise 8

*Hint.* For $X \xrightarrow{f} Y \xrightarrow{g} Z$, check $(g \circ f)^* = f^* \circ g^*$ by evaluating on a chain of $X$. ☐

*Solution.* Cohomology is contravariant. ☐

**Exercise 28.15.** Exercise 9

*Hint.* A zero-cocycle assigns the same value to the endpoints of every path; path connectedness forces a single value throughout $X$. ☐

*Solution.* For path-connected $X$, $H^0(X; G) \cong G$. ☐

**Exercise 28.16.** Exercise 10

*Hint.* Use the point's integral homology in the cohomology UCT, including the vanishing of Ext from its free degree-zero group. ☐

*Solution.* $H^n(*; G) = 0$ for $n > 0$. ☐

**Exercise 28.17.** Exercise 11

*Hint.* A simplicial cochain assigns a coefficient to every oriented simplex with sign reversal under orientation reversal; compute its coboundary on one higher simplex. ☐

*Solution.* Simplicial cochains dualize simplicial chains. ☐

**Exercise 28.18.** Exercise 12

*Hint.* For finite free cellular groups, the cochain matrix in degree $n$ is the transpose of the cellular boundary matrix from degree $n + 1$. ☐

*Solution.* Cellular cochains dualize cellular chains. ☐

**Exercise 28.19.** Exercise 13

*Hint.* Evaluate a cocycle on a cycle; show the value is unchanged when either representative is altered by the appropriate boundary or coboundary. ☐

*Solution.* The Kronecker pairing evaluates cohomology classes on homology classes. ☐

**Exercise 28.20.** Exercise 14

*Hint.* For a cocycle $\varphi$, write a boundary as $\partial c$; its evaluation is $\varphi(\partial c) = (\delta\varphi)(c)$. ☐

*Solution.* A cocycle vanishes on boundaries. ☐

**Exercise 28.21.** Exercise 15

*Hint.* Write a coboundary as $\delta\psi$ and a cycle as $z$; the evaluation becomes $\psi(\partial z)$. ☐

*Solution.* A coboundary vanishes on cycles. ☐

**Exercise 28.22.** Exercise 16

*Hint.* Identify relative cochains with functionals on $C_n(X)$ that vanish on $C_n(A)$, equivalently Hom of the quotient chain group. ☐

*Solution.* Relative cochains dualize relative chains. □

**Exercise 28.23.** Exercise 17

*Hint.* Extend a cocycle from $A$ to a cochain on $X$, then take its coboundary; this vanishes on $A$ and has degree one higher. □

*Solution.* The cohomology connecting map raises degree. □

**Exercise 28.24.** Exercise 18

*Hint.* The inclusion of the chosen boundary circle is a deformation-retract equivalence; use the pair sequence to show all relative cohomology groups vanish. □

*Solution.* The annulus relative to one boundary circle has zero relative cohomology. □

**Exercise 28.25.** Exercise 19

*Hint.* Place $\mathrm{Ext}(H_{n-1}, G)$ on the left and $\mathrm{Hom}(H_n, G)$ on the right of the cohomology UCT sequence. □

*Solution.* The cohomology UCT has Hom and Ext terms. □

**Exercise 28.26.** Exercise 20

*Hint.* A free abelian lower homology group is projective, so its first Ext with any coefficient group is zero. □

*Solution.* If $H_{n-1}$ is free, the Ext term vanishes. □

**Exercise 28.27.** Exercise 21

*Hint.* Apply $\mathrm{Hom}(-, \mathbb{Z})$ to the cyclic free resolution; the Ext group is the cokernel of multiplication by $m$. □

*Solution.* $\mathrm{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m$. □

**Exercise 28.28.** Exercise 22

*Hint.* Integral $H_2(\mathbb{R}P^2)$ is zero, but $H_1$ is $\mathbb{Z}/2$; compute the resulting Ext contribution in cohomological degree two. □

*Solution.* $H^2(\mathbb{R}P^2; \mathbb{Z}) \cong \mathbb{Z}/2$. □

**Exercise 28.29.** Exercise 23

*Hint.* Precompose cochains with the chain-homotopy operator. The resulting cochain homotopy makes the two pullbacks agree on cohomology classes. □

*Solution.* Cohomology is homotopy invariant. □

**Exercise 28.30.** Exercise 24

*Hint.* To define multiplication, specify a coefficient ring rather than only an abelian group; the product combines degrees additively and must descend from cochains. □

*Solution.* VIII/29 equips cohomology with the cup product. □

## 28.25 Solved problem dossiers

**Problem 28.31** (Coboundary squared). Prove $\delta^2 = 0$.

*Solution.* For a cochain $\varphi$, $\delta^2\varphi = \varphi\partial\partial = 0$. □

**Problem 28.32** (Contravariance). Verify $(g \circ f)^* = f^*g^*$.

*Solution.* Precomposition gives $\varphi \mapsto \varphi g_\# f_\#$, which is exactly $f^\#(g^\#\varphi)$. □

**Problem 28.33** (H0). Why is $H^0(X;G) \cong G^c$ for $c$ path components?

*Solution.* A zero-cocycle must take the same value on endpoints of every path, hence is constant on each component; there are no negative-degree coboundaries. □

**Problem 28.34** (Sphere). Compute integral cohomology of $S^n$.

*Solution.* The homology is free in degrees 0 and $n$, so the cohomology UCT gives $\mathbb{Z}$ in those degrees and zero elsewhere. □

**Problem 28.35** (Kronecker). Why is the evaluation pairing well defined?

*Solution.* Changing the cycle by a boundary does not change a cocycle's value, and changing the cocycle by a coboundary does not change its value on a cycle. □

**Problem 28.36** (Circle generator). Interpret a generator of $H^1(S^1;\mathbb{Z})$.

*Solution.* It is a cohomology class evaluating to 1 on the fundamental one-cycle. □

**Problem 28.37** (Torus H1). Compute $H^1(T^2;\mathbb{Z})$.

*Solution.* Since $H_1(T^2) = \mathbb{Z}^2$ is free, $H^1 \cong \mathrm{Hom}(\mathbb{Z}^2,\mathbb{Z}) \cong \mathbb{Z}^2$. □

**Problem 28.38** (RP2 H2). Use cohomology UCT to compute $H^2(\mathbb{R}P^2;\mathbb{Z})$.

*Solution.* The Hom term from $H_2 = 0$ vanishes; the Ext term from $H_1 = \mathbb{Z}/2$ is $\mathbb{Z}/2$. □

**Problem 28.39** (RP2 H1). Compute $H^1(\mathbb{R}P^2;\mathbb{Z})$.

*Solution.* $\mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}) = 0$ and Ext from $H_0 = \mathbb{Z}$ vanishes, so $H^1 = 0$. □

**Problem 28.40** (Relative disk). Compute $H^n(D^n, S^{n-1}; G)$.

*Solution.* The pair has one relative $n$-class; cohomology is $G$ in degree $n$ and zero otherwise. □

**Problem 28.41** (Annulus pair). Why does $H^*(S^1 \times I, S^1 \times \{0\}; G) = 0$?

*Solution.* The chosen boundary circle is a deformation retract of the annulus, so the pair is homotopy equivalent to $(A, A)$. □

**Problem 28.42** (Connecting map). Describe the cohomology connecting homomorphism geometrically.

*Solution.* Extend a cocycle on $A$ to a cochain on $X$; its coboundary vanishes on $A$ and defines a relative cocycle one degree higher. □

**Problem 28.43** (Cellular cochains). For a finite CW complex with $c_n$ $n$-cells, what is $C^n_{\mathrm{cell}}(X;G)$?

*Solution.* It is $\mathrm{Hom}(\mathbb{Z}^{c_n}, G) \cong G^{c_n}$. ☐

**Problem 28.44** (Finite graph)**.** Compute $H^1$ of a connected graph of rank $r$ over $\mathbb{Z}$.

*Solution.* Its $H_1$ is $\mathbb{Z}^r$, so $H^1 \cong \mathbb{Z}^r$. ☐

**Problem 28.45** (Moore cohomology)**.** If $H_n(M) = \mathbb{Z}/m$ and other reduced homology vanishes, where does integral cohomology torsion appear?

*Solution.* $H^{n+1}(M;\mathbb{Z}) \cong \mathrm{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m$. ☐

**Problem 28.46** (Free case)**.** Why does free integral homology make cohomology a dual group?

*Solution.* Ext vanishes, leaving $H^n(X;G) \cong \mathrm{Hom}(H_n(X), G)$. ☐

**Problem 28.47** (Homotopy)**.** Why do homotopic maps induce equal cohomology maps?

*Solution.* The induced chain maps are chain homotopic; dualizing yields cochain-homotopic maps, which agree on cohomology. ☐

**Problem 28.48** (Reduced)**.** For $n > 0$, compare $H^n$ and reduced $H^n$.

*Solution.* They agree; only degree zero is altered. ☐

**Problem 28.49** (Coefficient field)**.** What simplification occurs over a field for finite-dimensional chain groups?

*Solution.* Cochain groups are ordinary vector-space duals and Ext phenomena over the field disappear. ☐

**Problem 28.50** (SVG annulus)**.** What concept should the annulus SVG emphasize?

*Solution.* The deformation-retract geometry together with the restriction/connecting portion of the cohomology long exact sequence. ☐

## 28.26 Supplementary worked examples

**Example 28.51** (Cohomology of a sphere)**.** For $S^n$ with integral coefficients,

$$H^k(S^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

The degree-$n$ generator evaluates to 1 on the oriented fundamental class.

**Example 28.52** (Cocycles as observables on cycles)**.** A cochain $\varphi \in C^n(X;G) = \mathrm{Hom}(C_n(X), G)$ is a cocycle when it vanishes on boundaries. Hence a cohomology class gives a well-defined evaluation on homology classes, turning cochains into algebraic probes of cycles.

**Example 28.53** (Cohomology of the circle)**.** A CW decomposition of $S^1$ with one 0-cell and one 1-cell has zero cellular coboundary. Thus $H^0(S^1;\mathbb{Z}) \cong \mathbb{Z}$ and $H^1(S^1;\mathbb{Z}) \cong \mathbb{Z}$, with the degree-one class measuring winding homologically.

## 28.27 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 28.54** (Cochain group)**.** Define $C^n(X;G)$ from the chain group $C_n(X)$.

*Hint.* Apply $\mathrm{Hom}(-, G)$. □

*Solution.* $C^n(X;G) = \mathrm{Hom}(C_n(X), G)$. □

**Exercise 28.55** (Coboundary)**.** How is the coboundary $\delta : C^n \to C^{n+1}$ defined from the chain boundary?

*Hint.* Precompose with $\partial$. □

*Solution.* $(\delta\varphi)(c) = \varphi(\partial c)$. □

**Exercise 28.56** (Why delta squared vanishes)**.** Show $\delta^2 = 0$.

*Hint.* Use $\partial^2 = 0$. □

*Solution.* $(\delta^2\varphi)(c) = \varphi(\partial^2 c) = 0$. □

**Exercise 28.57** (Point cohomology)**.** Compute $H^*(\{*\};\mathbb{Z})$.

*Hint.* There is one degree-zero cochain generator. □

*Solution.* $H^0 \cong \mathbb{Z}$ and $H^n = 0$ for $n > 0$. □

**Exercise 28.58** (Circle cohomology)**.** Compute $H^0$ and $H^1$ of $S^1$.

*Hint.* Use the one-cell-per-dimension cellular cochain complex. □

*Solution.* Both are $\mathbb{Z}$; higher cohomology vanishes. □

## Proofs

**Exercise 28.59** (Cocycle evaluates on homology)**.** Prove a cocycle has the same value on homologous cycles.

*Hint.* The difference of homologous cycles is a boundary. ☐

*Solution.* If $z - z' = \partial c$ and $\delta\varphi = 0$, then $\varphi(z) - \varphi(z') = \varphi(\partial c) = (\delta\varphi)(c) = 0$. ☐

**Exercise 28.60** (Coboundaries evaluate trivially)**.** Prove a coboundary evaluates to zero on every cycle.

*Hint.* Write $\varphi = \delta\psi$. ☐

*Solution.* For $\partial z = 0$, $\varphi(z) = \psi(\partial z) = 0$. ☐

**Exercise 28.61** (Contravariance)**.** Show a map $f : X \to Y$ induces $f^* : H^n(Y; G) \to H^n(X; G)$.

*Hint.* Precompose cochains with $f_\#$. ☐

*Solution.* $f^*\varphi = \varphi \circ f_\#$ commutes with coboundary, so it descends to cohomology and reverses arrows. ☐

**Exercise 28.62** (Homotopy invariance)**.** Explain why homotopic maps induce the same map on cohomology.

*Hint.* Dualize the chain-homotopy argument. ☐

*Solution.* A chain homotopy between the induced chain maps yields a cochain homotopy after applying $\mathrm{Hom}(-, G)$, so the induced cohomology maps agree. ☐

## Counterexamples and hypothesis tests

**Exercise 28.63** (Cohomology is covariant)**.** Test: cohomology is covariant in the space variable.

*Hint.* Track the direction of pullback. ☐

*Solution.* False. A map $X \to Y$ induces $H^*(Y) \to H^*(X)$, so ordinary cohomology is contravariant. ☐

**Exercise 28.64** (Cocycles are functions on homology only)**.** Test: every cochain defines a function on homology classes.

*Hint.* Boundaries must be annihilated. ☐

*Solution.* False. Only cocycles descend to homology because arbitrary cochains may take nonzero values on boundaries. ☐

**Exercise 28.65** (Same groups mean same ring)**.** Test: knowing only the additive cohomology groups determines all cohomological structure.

*Hint.* Cup products add extra information. ☐

*Solution.* False. Spaces can have isomorphic graded groups but different cup-product rings. ☐

## Applications and investigations

**Exercise 28.66** (Detecting a cycle)**.** How can a cohomology class certify that a cycle is nontrivial?

*Hint.* Pair it with the cycle. ☐

*Solution.* If $[\alpha] \in H^n$ evaluates nontrivially on $[z] \in H_n$, then $[z]$ cannot be zero because cocycles vanish on boundaries. ☐

**Exercise 28.67** (Discrete data interpretation)**.** Why are cochains natural for assigning measurements to simplices, cells, or edges?

*Hint.* They are linear functionals on chains. ☐

*Solution.* A cochain assigns values to generators and extends linearly, so it models quantities such as potentials, circulations, or fluxes while the coboundary encodes consistency. ☐

## Challenge problems

**Exercise 28.68** (Sphere dual generator)**.** Choose generators $u \in H^n(S^n)$ and $[S^n] \in H_n(S^n)$. How may their signs be normalized?

*Hint.* Use the evaluation pairing. ☐

*Solution.* Choose orientations so that $\langle u, [S^n] \rangle = 1$. ☐

**Exercise 28.69** (Cohomology synthesis)**.** Explain the conceptual difference between homology classes and cohomology classes.

*Hint.* One represents cycles; the other probes them. ☐

*Solution.* Homology records cycles modulo boundaries, while cohomology records cocycles modulo coboundaries; the evaluation pairing makes cohomology an algebra of invariants acting on homological geometry. ☐

# Chapter 29

# Cup Products

Cohomology becomes substantially richer when its groups are multiplied. The cup product turns $H^*(X; R)$ into a graded ring and detects topology that the additive groups alone can miss.

## 29.1 Cochain-level cup product

Let $R$ be a commutative coefficient ring. For singular cochains

$$\alpha \in C^p(X; R), \qquad \beta \in C^q(X; R),$$

define their cup product using the Alexander–Whitney front/back face formula.

**Definition 29.1** (Cup product)**.** For a singular $(p+q)$-simplex

$$\sigma : [v_0, \ldots, v_{p+q}] \to X,$$

set

$$(\alpha \smile \beta)(\sigma) = \alpha(\sigma|[v_0, \ldots, v_p])\, \beta(\sigma|[v_p, \ldots, v_{p+q}]).$$

## 29.2 Leibniz rule

**Theorem 29.2** (Coboundary derivation rule)**.**

$$\delta(\alpha \smile \beta) = \delta\alpha \smile \beta + (-1)^p \alpha \smile \delta\beta.$$

Therefore cocycles multiply to cocycles, and changing a factor by a coboundary changes the product by a coboundary.

## 29.3 Product on cohomology

The cochain product descends to

$$\smile : H^p(X; R) \times H^q(X; R) \to H^{p+q}(X; R).$$

## 29.4 Cohomology ring

**Definition 29.3** (Cohomology ring)**.** The graded group

$$H^*(X; R) = \bigoplus_{n \geq 0} H^n(X; R)$$

with cup product is the cohomology ring of $X$.

The unit is the class

$$1 \in H^0(X; R)$$

represented by the constant 1-valued zero-cochain on a connected space.

## 29.5 Associativity

Cup product is associative on cohomology:

$$(a \smile b) \smile c = a \smile (b \smile c).$$

## 29.6 Naturality

**Theorem 29.4** (Naturality)**.** *For $f : X \to Y$,*

$$f^*(a \smile b) = f^*a \smile f^*b.$$

Thus $f^* : H^*(Y; R) \to H^*(X; R)$ is a graded-ring homomorphism.

## 29.7 Graded commutativity

**Theorem 29.5** (Graded commutativity)**.** *For*

$$a \in H^p(X; R), \qquad b \in H^q(X; R),$$

*one has*

$$a \smile b = (-1)^{pq} b \smile a.$$

This equality generally holds only after passing to cohomology, not strictly on the standard cochain-level cup product.

## 29.8 Odd-degree squares

If $a$ has odd degree and 2 is invertible in $R$, then

$$a \smile a = 0.$$

Over characteristic 2, the sign disappears, so odd-degree squares need not vanish for sign reasons.

## 29.9 Sphere ring

For $n > 0$,

$$H^*(S^n; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^2), \qquad |u| = n,$$

where $u^2 = 0$ for dimensional reasons.

## 29.10 Circle ring

$$H^*(S^1; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(a), \qquad |a| = 1.$$

Thus

$$a^2 = 0.$$

## 29.11 Torus ring

Let $a, b \in H^1(T^2;\mathbb{Z})$ be dual to the two fundamental one-cycles. Then

$$H^*(T^2;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(a,b).$$

The top class is

$$a \smile b \in H^2(T^2;\mathbb{Z}) \cong \mathbb{Z},$$

with

$$b \smile a = -a \smile b.$$

## 29.12 Higher tori

For

$$T^n = (S^1)^n,$$

$$H^*(T^n;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(a_1,\ldots,a_n), \qquad |a_i| = 1.$$

This multiplicative structure refines the binomial-rank computation from Künneth.

## 29.13 Real projective space mod 2

Let

$$x \in H^1(\mathbb{R}P^n;\mathbb{F}_2)$$

be the degree-one generator.

A fundamental result is

$$H^*(\mathbb{R}P^n;\mathbb{F}_2) \cong \mathbb{F}_2[x]/(x^{n+1}), \qquad |x| = 1.$$

Thus

$$x^k \neq 0 \quad (0 \le k \le n).$$

## 29.14 Why the ring is stronger

As graded vector spaces, many spaces can have the same Betti numbers. The cup product records which classes multiply nontrivially and can distinguish such spaces.

## 29.15 Example: wedge versus product

Both

$$S^1 \vee S^1 \vee S^2$$

and

$$T^2$$

have additive homology ranks

$$b_0 = 1, \quad b_1 = 2, \quad b_2 = 1.$$

But on the wedge, products of positive-degree classes from different wedge summands vanish, whereas on the torus

$$a \smile b$$

generates $H^2$.

Hence the spaces are not homotopy equivalent.

## 29.16 Cross product and diagonal

The cup product may be described conceptually as

$$a \smile b = \Delta^*(a \times b),$$

where

$$\Delta : X \to X \times X, \qquad \Delta(x) = (x, x),$$

is the diagonal and $\times$ is the cohomology cross product.

## 29.17 Kronecker pairing compatibility

Cup products interact with homology cross products through evaluation: cohomology products detect product cycles and, on manifolds, later encode intersection pairings.

## 29.18 Relative cup products

There are relative forms

$$H^p(X, A; R) \times H^q(X, B; R) \to H^{p+q}(X, A \cup B; R)$$

under suitable hypotheses.

These become important for Thom classes and duality.

## 29.19 Cup length

A basic ring invariant is cup length: the largest number of positive-degree classes whose product is nonzero.

Cup length provides lower bounds for several topological complexity invariants.

## 29.20 Maps and ring obstructions

If $f : X \to Y$ is a homotopy equivalence, then

$$f^* : H^*(Y; R) \to H^*(X; R)$$

is a graded-ring isomorphism.

Therefore a mismatch of cohomology rings obstructs homotopy equivalence.

## 29.21 SVG visual laboratory

This chapter ships with editable SVG assets for:

- the diagonal/cross-product construction of the cup product;
- the contrast between the torus ring and a wedge with the same Betti numbers.

## 29.22 Bridge to bundles

Cohomology rings are the natural receptacle for characteristic classes of bundles. VIII/30 first develops vector bundles and clutching; Thom and Euler classes appear in VIII/31–VIII/32.

## 29.23 Exercises

**Exercise 29.6.** Exercise 1

*Hint.* For classes of degrees $p$ and $q$, identify the target group $H^{p+q}(X;R)$; the coefficient ring supplies multiplication of cochain values. ☐

*Solution.* The cup product raises degree additively. ☐

**Exercise 29.7.** Exercise 2

*Hint.* The sign depends on the degree $p$ of the first cochain; check the degrees of both terms after applying the coboundary. ☐

*Solution.* The Leibniz rule is $\delta(\alpha \smile \beta) = \delta\alpha \smile \beta + (-1)^p \alpha \smile \delta\beta$. ☐

**Exercise 29.8.** Exercise 3

*Hint.* Set both input coboundaries to zero in the Leibniz rule; this makes the product's coboundary vanish. ☐

*Solution.* Cocycles multiply to cocycles. ☐

**Exercise 29.9.** Exercise 4

*Hint.* Replace one cocycle by itself plus $\delta\gamma$; use the Leibniz rule with the other cocycle to exhibit the change as a coboundary. ☐

*Solution.* The product descends to cohomology. ☐

**Exercise 29.10.** Exercise 5

*Hint.* Use the constant degree-zero cochain with value the ring unit; check multiplication on both sides of an arbitrary cohomology class. ☐

*Solution.* The class $1 \in H^0$ is the multiplicative unit. ☐

**Exercise 29.11.** Exercise 6

*Hint.* Track three degrees $p, q, r$ and compare the two bracketings in degree $p + q + r$; associativity does not mean commutativity without signs. ☐

*Solution.* Cup product is associative on cohomology. ☐

**Exercise 29.12.** Exercise 7

*Hint.* Evaluate the cup formula after precomposition by $f$; face restrictions commute with that precomposition, giving the pullback identity. ☐

*Solution.* Pullback preserves cup products. ☐

**Exercise 29.13.** Exercise 8

*Hint.* Write $a \smile b = (-1)^{pq} b \smile a$ for cohomology classes; do not assert this strict identity for arbitrary cochains. ☐

*Solution.* Cup product is graded commutative. ☐

**Exercise 29.14.** Exercise 9

*Hint.* For an odd-degree class, graded commutativity gives $2a^2 = 0$. Conclude $a^2 = 0$ only when multiplication by two is injective, in particular when two is invertible. ☐

*Solution.* Odd-degree squares vanish when 2 is invertible. ☐

**Exercise 29.15.** Exercise 10

*Hint.* In characteristic two, the sign $(-1)^{pq}$ is one, so odd degree alone does not force a square to vanish. ☐

*Solution.* In characteristic two, the sign disappears. ☐

**Exercise 29.16.** Exercise 11

*Hint.* For $n > 0$, the top generator has square in degree $2n$, where sphere cohomology vanishes; record the generator's degree in the ring presentation. ☐

*Solution.* $H^*(S^n; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^2)$ for $n > 0$. ☐

**Exercise 29.17.** Exercise 12

*Hint.* The degree-one generator of the circle squares into $H^2(S^1) = 0$; combine this with its degree-zero unit. ☐

*Solution.* $H^*(S^1; \mathbb{Z})$ is an exterior algebra on one degree-one generator. ☐

**Exercise 29.18.** Exercise 13

*Hint.* Pull back the generators from both circle factors and use the product structure to find their nonzero mixed product and their vanishing squares. ☐

*Solution.* $H^*(T^2; \mathbb{Z})$ is exterior on two degree-one generators. ☐

**Exercise 29.19.** Exercise 14

*Hint.* Choose the product orientation and evaluate $a \smile b$ on the product fundamental class to normalize the top generator. ☐

*Solution.* The torus top class is $a \smile b$. ☐

**Exercise 29.20.** Exercise 15

*Hint.* Both torus generators have degree one, so interchanging them contributes $(-1)^{1\cdot 1}$. ☐

*Solution.* $b \smile a = -a \smile b$ integrally. ☐

**Exercise 29.21.** Exercise 16

*Hint.* Multiply distinct pulled-back degree-one generators in increasing order; count the degree-$k$ basis products by $k$-element subsets. ☐

*Solution.* $H^*(T^n; \mathbb{Z})$ is exterior on $n$ generators. ☐

**Exercise 29.22.** Exercise 17

*Hint.* Use mod-two coefficients and a degree-one generator; distinguish the dimensional relation $x^{n+1} = 0$ from the nonvanishing of lower powers. ☐

*Solution.* $H^*(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2[x]/(x^{n+1})$. ☐

**Exercise 29.23.** Exercise 18

*Hint.* Check the smallest positive nonzero cohomology degree of real projective space with mod-two coefficients to locate $x$. ☐

*Solution.* The projective generator $x$ has degree one. ☐

**Exercise 29.24.** Exercise 19

*Hint.* Use the projective-space cup-product calculation or the mod-two intersection of $n$ generic projective hyperplanes; the group dimensions alone do not prove $x^n \neq 0$. ☐

*Solution.* On $\mathbb{R}P^n$, $x^n \neq 0$. ☐

**Exercise 29.25.** Exercise 20

*Hint.* First form the external class on $X \times X$, then pull it back along $x \mapsto (x, x)$; the two steps have different domains. ☐

*Solution.* The cup product can be written $\Delta^*(a \times b)$. ☐

**Exercise 29.26.** Exercise 21

*Hint.* Compare $T^2$ with $S^1 \vee S^1 \vee S^2$: their graded group ranks agree, but the product of the two degree-one generators differs. ☐

*Solution.* Cohomology rings can distinguish spaces with equal Betti numbers. ☐

**Exercise 29.27.** Exercise 22

*Hint.* For an excisive pair of subspaces, verify that the relative product vanishes on the union in the appropriate cochain model; record $A \cup B$ in the target pair. ☐

*Solution.* Relative cup products land in the union pair. ☐

**Exercise 29.28.** Exercise 23

*Hint.* Apply pullback to a homotopy inverse and use both multiplicativity and homotopy invariance to obtain inverse graded-ring maps. ☐

*Solution.* Homotopy equivalences induce cohomology-ring isomorphisms. ☐

**Exercise 29.29.** Exercise 24

*Hint.* A characteristic class must respect bundle pullback; cohomology-ring naturality supplies the language for comparing these classes and their products. ☐

*Solution.* Characteristic classes will live in these cohomology rings. ☐

## 29.24 Solved problem dossiers

**Problem 29.30** (Leibniz)**.** Why does the cup product descend to cohomology?

*Solution.* The Leibniz rule shows cocycles multiply to cocycles; if one factor changes by a coboundary, the product changes by a coboundary. □

**Problem 29.31** (Naturality)**.** Prove $f^*(a \smile b) = f^*a \smile f^*b$.

*Solution.* The Alexander–Whitney face formula commutes with precomposition by $f$, so the equality already holds on cochains. □

**Problem 29.32** (Odd square)**.** Let $a \in H^1(X;\mathbb{Q})$. Show $a^2 = 0$.

*Solution.* Graded commutativity gives $a^2 = -a^2$, so $2a^2 = 0$; over $\mathbb{Q}$, 2 is invertible. □

**Problem 29.33** (Circle)**.** Compute the integral cohomology ring of $S^1$.

*Solution.* It is $\Lambda(a)$ with $|a| = 1$; dimensional reasons and graded commutativity give $a^2 = 0$. □

**Problem 29.34** (Torus)**.** Describe the ring $H^*(T^2;\mathbb{Z})$.

*Solution.* It is the exterior algebra on $a, b \in H^1$, with top generator $a \smile b$ and $b \smile a = -a \smile b$. □

**Problem 29.35** (Three-torus)**.** How many degree-two ring generators arise as products in $T^3$?

*Solution.* The products $a_1a_2, a_1a_3, a_2a_3$ give three independent degree-two classes. □

**Problem 29.36** (Top torus class)**.** What generates $H^n(T^n;\mathbb{Z})$?

*Solution.* The product $a_1 \smile \cdots \smile a_n$. □

**Problem 29.37** (RP2 mod2)**.** Describe $H^*(\mathbb{R}P^2;\mathbb{F}_2)$.

*Solution.* $\mathbb{F}_2[x]/(x^3)$ with $|x| = 1$; thus $x^2$ is the nonzero top class. □

**Problem 29.38** (RP3 mod2)**.** Describe the nonzero powers of the degree-one generator.

*Solution.* $1, x, x^2, x^3$ are nonzero and $x^4 = 0$. □

**Problem 29.39** (Wedge versus torus)**.** Explain why $S^1 \vee S^1 \vee S^2$ is not homotopy equivalent to $T^2$ despite equal Betti numbers.

*Solution.* The wedge has trivial products between positive-degree classes from different summands, while the torus has $a \smile b \neq 0$. □

**Problem 29.40** (Diagonal)**.** Explain $a \smile b = \Delta^*(a \times b)$.

*Solution.* Take the cohomology cross product on $X \times X$ and pull it back along the diagonal; the Alexander–Whitney formula realizes this construction. □

**Problem 29.41** (Ring map obstruction)**.** If $f : X \to Y$ is a homotopy equivalence, what must happen to cup products?

*Solution.* $f^*$ is an isomorphism preserving multiplication, so the graded cohomology rings must be isomorphic. □

**Problem 29.42** (Cup length)**.** What does a nonzero product $a_1 \cdots a_r$ prove?

*Solution.* The cup length is at least $r$, providing a multiplicative lower bound on topological complexity. □

**Problem 29.43** (Characteristic two)**.** Why need an odd-degree mod-2 class not square to zero?

*Solution.* Graded commutativity contributes the sign $-1$, but $-1 = 1$ over $\mathbb{F}_2$, so no sign-forced cancellation occurs. □

**Problem 29.44** (Sphere)**.** Why is the square of the top sphere class zero?

*Solution.* It would lie in degree $2n$, where $H^{2n}(S^n) = 0$ for $n > 0$. □

**Problem 29.45** (Relative product)**.** State a useful relative cup product.

*Solution.* $H^p(X, A) \times H^q(X, B) \to H^{p+q}(X, A \cup B)$. □

**Problem 29.46** (Pairing)**.** How can a cup product be tested against homology?

*Solution.* Evaluate $a \smile b$ on an appropriate $(p + q)$-cycle using the Kronecker pairing. □

**Problem 29.47** (Product vs additive)**.** What extra information does the ring contain beyond cohomology groups?

*Solution.* It records which additive classes multiply to nonzero higher-degree classes and the relations among these products. □

**Problem 29.48** (SVG torus)**.** What should the torus-vs-wedge SVG show?

*Solution.* Identical Betti-number rows but different multiplication: a nonzero $a \smile b$ cell for the torus and zero mixed products for the wedge. □

**Problem 29.49** (Bridge)**.** Why are cup products relevant to bundles?

*Solution.* Characteristic classes lie in cohomology and multiply; their ring relations encode bundle and manifold structure. □

**Problem 29.50** (Override provenance)**.** Why is cup-product material assigned here rather than to Poincaré duality?

*Solution.* The migration ledger contains a raw-source-verified targeted override for cup-product/cohomology-ring sections into VIII/29. □

**Problem 29.51** (Final principle)**.** State the main improvement from cohomology groups to the cohomology ring.

*Solution.* The ring retains multiplicative interactions between classes, making it strictly more informative than the graded groups alone. □

## 29.25 Supplementary worked examples

**Example 29.52** (The torus cohomology ring)**.** Let $a, b \in H^1(T^2; \mathbb{Z})$ be dual to the two circle factors. Then

$$a \smile a = b \smile b = 0, \qquad a \smile b = -b \smile a,$$

and $a \smile b$ generates $H^2(T^2; \mathbb{Z})$. Thus the ring remembers how the two independent one-dimensional directions combine to form the fundamental class.

**Example 29.53** (Cup products on a sphere)**.** If $u \in H^n(S^n)$ with $n > 0$, then $u \smile u$ lies in $H^{2n}(S^n) = 0$. Hence the positive-degree part of the sphere cohomology ring has trivial multiplication.

**Example 29.54** (Projective-space power structure)**.** With $\mathbb{F}_2$ coefficients, $\mathbb{RP}^m$ has a degree-one class $x$ whose powers survive until dimension $m$:

$$H^*(\mathbb{RP}^m; \mathbb{F}_2) \cong \mathbb{F}_2[x]/(x^{m+1}).$$

This ring structure is much richer than the list of Betti numbers alone.

## 29.26 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 29.55** (Degree of product)**.** If $\alpha \in H^p(X)$ and $\beta \in H^q(X)$, what is the degree of $\alpha \smile \beta$?

*Hint.* Degrees add. □

*Solution.* It lies in $H^{p+q}(X)$. □

**Exercise 29.56** (Unit)**.** What class is the multiplicative unit in the cohomology ring of a connected space?

*Hint.* Look in $H^0(X)$. □

*Solution.* The class of the constant function 1 in $H^0(X)$ is the unit. □

**Exercise 29.57** (Sphere square)**.** For $u \in H^n(S^n)$ with $n > 0$, compute $u^2$.

*Hint.* Check $H^{2n}(S^n)$. □

*Solution.* $u^2 = 0$. □

**Exercise 29.58** (Torus top product)**.** For degree-one generators $a, b$ of $T^2$, what group contains $a \smile b$?

*Hint.* Degrees add to 2. □

*Solution.* $a \smile b \in H^2(T^2) \cong \mathbb{Z}$, and with compatible orientations it is a generator. □

**Exercise 29.59** (Graded sign)**.** State graded commutativity for $\alpha \in H^p$, $\beta \in H^q$.

*Hint.* Interchanging factors contributes a sign. □

*Solution.* $\alpha \smile \beta = (-1)^{pq} \beta \smile \alpha$. □

## Proofs

**Exercise 29.60** (Naturality)**.** Prove pullback preserves cup products.

*Hint.* Use naturality of the cochain-level diagonal approximation. □

*Solution.* For $f : X \to Y$, $f^*(\alpha \smile \beta) = f^*\alpha \smile f^*\beta$, so $f^*$ is a graded-ring homomorphism. □

**Exercise 29.61** (Odd square over Z)**.** Show that if $\alpha$ has odd degree and cohomology has no 2-torsion in degree $2|\alpha|$, then $\alpha^2 = 0$.

*Hint.* Use graded commutativity. □

*Solution.* $\alpha^2 = -\alpha^2$, hence $2\alpha^2 = 0$; without 2-torsion this forces $\alpha^2 = 0$. □

**Exercise 29.62** (Product detects dimension)**.** If $\alpha_1 \smile \cdots \smile \alpha_r \neq 0$, prove the total degree cannot exceed the cohomological dimension.

*Hint.* The product lives in the sum degree. □

*Solution.* A nonzero class must live in a nonzero cohomology group, so the total degree is bounded by the largest degree with nonzero cohomology. □

**Exercise 29.63** (Ring distinguishes spaces)**.** Explain how different products can distinguish spaces with the same additive cohomology groups.

*Hint.* An additive isomorphism need not preserve multiplication. □

*Solution.* If one space has a nonzero product of classes where the corresponding product must vanish in the other, their cohomology rings are not isomorphic and the spaces cannot be homotopy equivalent. □

## Counterexamples and hypothesis tests

**Exercise 29.64** (Cup product strictly commutative)**.** Test: $\alpha \smile \beta = \beta \smile \alpha$ over $\mathbb{Z}$ in all degrees.

*Hint.* Remember the graded sign. □

*Solution.* False. The sign $(-1)^{pq}$ appears. □

**Exercise 29.65** (All positive products vanish)**.** Test: positive-degree cup products always vanish because degree increases.

*Hint.* Consider the torus. □

*Solution.* False. On $T^2$, two degree-one classes have a nonzero degree-two product. □

**Exercise 29.66** (Additive cohomology enough)**.** Test: Betti numbers determine the cohomology ring.

*Hint.* Multiplication is extra structure. □

*Solution.* False; Betti numbers record only graded dimensions. □

## Applications and investigations

**Exercise 29.67** (Cup-length obstruction)**.** How can a nonzero long cup product obstruct a space from being too simple?

*Hint.* Use cup length as a homotopy invariant. □

*Solution.* A nonzero product of many positive-degree classes gives a lower bound on invariants such as Lusternik–Schnirelmann category and rules out homotopy types whose rings have shorter cup length. □

**Exercise 29.68** (Orientation pairing)**.** On a closed oriented surface, how does the cup product encode intersection of one-dimensional classes?

*Hint.* Evaluate the degree-two product on the fundamental class. □

*Solution.* The integer $\langle \alpha \smile \beta, [\Sigma] \rangle$ is the algebraic intersection pairing of the Poincare-dual one-cycles. □

## Challenge problems

**Exercise 29.69** (RP2 mod 2)**.** Let $x \in H^1(\mathbb{RP}^2; \mathbb{F}_2)$. What are $x^2$ and $x^3$?

*Hint.* Use $\mathbb{F}_2[x]/(x^3)$. □

*Solution.* $x^2$ is the nonzero top class and $x^3 = 0$. □

**Exercise 29.70** (Ring synthesis)**.** Why is the cup product a decisive upgrade from cohomology groups to a cohomology ring?

*Hint.* It records interactions among classes. □

*Solution.* The graded groups say which classes exist; the cup product records which classes combine nontrivially, yielding finer homotopy information, intersection pairings, characteristic classes, and duality structures. □

# Chapter 30

# Vector Bundles and Clutching

A vector bundle is a continuously varying family of vector spaces over a base space. Local triviality makes bundles look like products in small neighborhoods, while transition functions record the global twisting. Clutching constructs bundles over spheres by gluing trivial bundles across the equator.

## 30.1 Vector bundles

**Definition 30.1** (Real vector bundle)**.** A rank-$k$ real vector bundle over $X$ is a map

$$\pi : E \to X$$

such that each fiber

$$E_x = \pi^{-1}(x)$$

is a $k$-dimensional real vector space and every $x \in X$ has a neighborhood $U$ with a homeomorphism

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$$

that is linear on fibers.

Complex vector bundles are defined similarly with $\mathbb{C}^k$.

## 30.2 Local trivializations

A local trivialization identifies nearby fibers with one fixed model vector space. The bundle can be globally nontrivial even though every small piece is a product.

## 30.3 Transition functions

If $\Phi_i, \Phi_j$ are trivializations over $U_i, U_j$, then on the overlap

$$U_i \cap U_j$$

their change of coordinates has the form

$$(x, v) \longmapsto (x, g_{ij}(x)v)$$

with

$$g_{ij} : U_i \cap U_j \to GL_k(\mathbb{R}).$$

They satisfy

$$g_{ii} = I, \qquad g_{ij}g_{jk} = g_{ik}.$$

## 30.4 Trivial bundles

The product

$$X \times \mathbb{R}^k \to X$$

is the trivial rank-$k$ real bundle.

A rank-$k$ bundle is trivial if it is isomorphic to this product bundle.

## 30.5 Sections

**Definition 30.2** (Section)**.** A section is a map

$$s : X \to E$$

with

$$\pi \circ s = \mathrm{id}_X\,.$$

The zero vector in every fiber defines the zero section.

A nowhere-zero section is one satisfying

$$s(x) \neq 0$$

for every $x$.

## 30.6 Bundle maps

A bundle map over $f : X \to Y$ is a map of total spaces carrying fibers linearly into fibers and covering $f$.

An isomorphism of bundles over the same base is fiberwise linear and a homeomorphism on total spaces.

## 30.7 Pullback bundles

Given

$$f : Y \to X$$

and a vector bundle $E \to X$, define

$$f^*E = \{(y, e) \in Y \times E : f(y) = \pi(e)\}.$$

**Theorem 30.3** (Pullback)**.** *The projection*

$$f^*E \to Y$$

*is a vector bundle of the same rank.*

## 30.8 Whitney sum

For bundles $E, F \to X$, define

$$(E \oplus F)_x = E_x \oplus F_x.$$

This is the Whitney sum bundle

$$E \oplus F \to X.$$

## 30.9 Subbundles and quotient bundles

A vector subbundle $F \subset E$ has fiberwise vector subspaces of constant rank with local triviality. Under suitable hypotheses one can form the quotient bundle

$$E/F.$$

## 30.10 Line bundles

Rank-one bundles are line bundles.

Real line bundles have transition functions in

$$GL_1(\mathbb{R}) = \mathbb{R}^\times.$$

After choosing a metric, the structure group reduces to

$$O(1) = \{\pm 1\}.$$

## 30.11 The Möbius line bundle

The Möbius strip is the total space of a nontrivial real line bundle over $S^1$. It can be constructed from

$$[0,1] \times \mathbb{R}$$

by the identification

$$(0, v) \sim (1, -v).$$

The sign change is the twisting transition function.

## 30.12 No nowhere-zero section criterion for line bundles

A real line bundle is trivial if and only if it admits a nowhere-zero section.

Hence the Möbius line bundle has no nowhere-zero global section.

## 30.13 Tangent bundles

For a smooth $n$-manifold $M$,

$$TM = \coprod_{x \in M} T_x M$$

is a rank-$n$ vector bundle.

The tangent bundle of $S^1$ is trivial.

The tangent bundle of $S^2$ is nontrivial; the hairy-ball theorem is a manifestation of the absence of a nowhere-zero section.

## 30.14 Bundle metrics

Every vector bundle over a paracompact base admits a fiberwise inner product. This allows the structure group of a real rank-$k$ bundle to be reduced from $GL_k(\mathbb{R})$ to $O(k)$.

For complex bundles one similarly reduces to $U(k)$.

## 30.15 Clutching over a sphere

Write
$$S^n = D^n_+ \cup D^n_-, \qquad D^n_+ \cap D^n_- \simeq S^{n-1}.$$

A bundle is trivial over each hemisphere because each hemisphere is contractible. The global bundle is therefore determined by how the two trivial bundles are glued along the equator.

**Definition 30.4** (Clutching map)**.** A clutching map for a rank-$k$ real bundle over $S^n$ is a map
$$g : S^{n-1} \to GL_k(\mathbb{R})$$
used to identify
$$(x, v)_+ \sim (x, g(x)v)_-$$
on the equator.

## 30.16 Homotopic clutching maps

**Theorem 30.5** (Clutching homotopy invariance)**.** *Homotopic clutching maps determine isomorphic vector bundles over $S^n$.*

After reducing structure group, one can work with
$$g : S^{n-1} \to O(k)$$
for real bundles or
$$g : S^{n-1} \to U(k)$$
for complex bundles.

## 30.17 Classification by clutching

Under the standard rank/base hypotheses, isomorphism classes of rank-$k$ bundles over $S^n$ are classified by homotopy classes
$$[S^{n-1}, GL_k(\mathbb{R})]$$
or equivalently, after metric reduction,
$$[S^{n-1}, O(k)].$$

For complex bundles use $U(k)$.

## 30.18 Real line bundles over the circle

A rank-one real bundle over $S^1$ is clutched along
$$S^0 = \{-1, +1\}.$$

Up to isomorphism there are two possibilities:

- the trivial cylinder bundle;
- the Möbius bundle.

## 30.19 Complex line bundles over $S^2$

A complex line bundle over $S^2$ is clutched by

$$g : S^1 \to U(1) \cong S^1.$$

Its homotopy class is classified by an integer winding number.

This integer will later be identified with the first Chern class after the appropriate characteristic-class machinery is introduced.

## 30.20 Tangent bundle clutching

The tangent bundle of a sphere can be described by a clutching map on the equator. The nontriviality of this clutching map encodes the global twisting of the tangent bundle.

## 30.21 Stable viewpoint

Adding a trivial bundle can simplify classification:

$$E \sim_{\text{st}} F \quad \Longleftrightarrow \quad E \oplus \varepsilon^r \cong F \oplus \varepsilon^s$$

for suitable trivial summands.

This is the entry point to $K$-theory, beyond the scope of this volume.

## 30.22 SVG visual laboratory

This chapter ships with editable SVG assets for:

- local trivializations and transition functions, including the Möbius twist;
- hemisphere clutching over $S^n$, with the equatorial map $S^{n-1} \to O(k)$.

## 30.23 Bridge to Thom classes

A vector bundle has a total space, a zero section, and a disk/sphere bundle. Cohomology of these pairs produces the Thom class. VIII/31 develops that construction and the Thom isomorphism.

## 30.24 Exercises

**Exercise 30.6.** Exercise 1

*Hint.* Specify the scalar field and require local triviality as well as constant fiber dimension; a collection of vector spaces alone is not yet a vector bundle. □

*Solution.* A rank-$k$ bundle has $k$-dimensional vector-space fibers. □

**Exercise 30.7.** Exercise 2

*Hint.* The trivialization must commute with projection to $U$ and restrict to a linear isomorphism on each fiber. □

*Solution.* Local trivializations identify $\pi^{-1}(U)$ with $U \times \mathbb{R}^k$. □

**Exercise 30.8.** Exercise 3

*Hint.* Compose one local trivialization with the inverse of the other; its fiber map is an invertible linear transformation varying continuously over the overlap. ☐

*Solution.* Transition functions take values in $GL_k(\mathbb{R})$. ☐

**Exercise 30.9.** Exercise 4

*Hint.* Fix the convention for $g_{ij}$ before composing across a triple overlap; cancellation of the intermediate trivialization gives the cocycle equation. ☐

*Solution.* They satisfy the cocycle law $g_{ij}g_{jk} = g_{ik}$. ☐

**Exercise 30.10.** Exercise 5

*Hint.* A single trivialization over the entire base identifies the bundle with a product, which is stronger than separate local trivializations. ☐

*Solution.* The product bundle is trivial. ☐

**Exercise 30.11.** Exercise 6

*Hint.* Type a section as $s : B \to E$; the equation $\pi \circ s = \mathrm{id}_B$ says that its value remains in the fiber over its argument. ☐

*Solution.* A section satisfies $\pi s = \mathrm{id}$. ☐

**Exercise 30.12.** Exercise 7

*Hint.* Every linear transition map fixes zero, so local zero sections agree on overlaps and glue without any orientation assumption. ☐

*Solution.* The zero section exists for every vector bundle. ☐

**Exercise 30.13.** Exercise 8

*Hint.* The fiber of $f^*E$ over $y$ is $E_{f(y)}$; use pulled-back trivializations to verify that its dimension stays the same. ☐

*Solution.* Pullback preserves bundle rank. ☐

**Exercise 30.14.** Exercise 9

*Hint.* Over each basepoint take $E_b \oplus F_b$; the transition matrices are block diagonal and their sizes add. ☐

*Solution.* Whitney sum adds fiber dimensions. ☐

**Exercise 30.15.** Exercise 10

*Hint.* For a real line bundle, each fiber is a one-dimensional real vector space, although the total space need not be globally a product. ☐

*Solution.* A real line bundle has rank one. ☐

**Exercise 30.16.** Exercise 11

*Hint.* In an orthonormal fiber coordinate, a real linear isometry is multiplication by one of the two signs; use a bundle metric over the standard paracompact base. ☐

*Solution.* With a metric, a real line bundle has structure group $O(1) = \{\pm 1\}$. ☐

**Exercise 30.17.** Exercise 12

*Hint.* A section of the interval model must satisfy $s(1) = -s(0)$; the intermediate value theorem obstructs a nowhere-zero real section. ☐

*Solution.* The Möbius bundle uses the gluing $(0, v) \sim (1, -v)$. ☐

**Exercise 30.18.** Exercise 13

*Hint.* From a nowhere-zero section construct $(b, t) \mapsto ts(b)$; in rank one this is a fiberwise isomorphism and yields a global trivialization. ☐

*Solution.* A real line bundle is trivial iff it has a nowhere-zero section. ☐

**Exercise 30.19.** Exercise 14

*Hint.* Parametrize the circle and choose its continuous unit tangent vector; one nowhere-zero tangent field gives a global frame for this rank-one bundle. ☐

*Solution.* $TS^1$ is trivial. ☐

**Exercise 30.20.** Exercise 15

*Hint.* A trivial rank-two tangent bundle would have a nowhere-zero section; use the hairy-ball obstruction on the two-sphere to rule this out. ☐

*Solution.* $TS^2$ is nontrivial. ☐

**Exercise 30.21.** Exercise 16

*Hint.* Choose a positive-definite fiber metric and orthonormal local frames; their transition functions lie in $O(k)$. ☐

*Solution.* A bundle metric reduces $GL_k(\mathbb{R})$ to $O(k)$. ☐

**Exercise 30.22.** Exercise 17

*Hint.* Each closed hemisphere is a contractible disk, so its restricted bundle is trivial under the standard bundle hypotheses; choose both trivializations before comparing them. ☐

*Solution.* Each hemisphere of $S^n$ trivializes a bundle. ☐

**Exercise 30.23.** Exercise 18

*Hint.* The two boundary trivializations meet over the equator; their change of coordinates is the clutching map, with values in the appropriate linear group. ☐

*Solution.* A clutching map lives on the equator $S^{n-1}$. ☐

**Exercise 30.24.** Exercise 19

*Hint.* Use metric-compatible trivializations to replace invertible real matrices by orthogonal ones without changing the resulting bundle isomorphism type. ☐

*Solution.* Real rank-$k$ clutching maps may be taken into $O(k)$. ☐

**Exercise 30.25.** Exercise 20

*Hint.* Choose a Hermitian metric in the complex case; unitary transition matrices preserve that metric and supply the corresponding clutching target. ☐

*Solution.* Complex rank-$k$ clutching maps may be taken into $U(k)$. ☐

**Exercise 30.26.** Exercise 21

*Hint.* Glue using the clutching homotopy over $S^n \times I$; the two endpoint restrictions are isomorphic because the interval does not change bundle classification. ☐

*Solution.* Homotopic clutching maps yield isomorphic bundles. ☐

**Exercise 30.27.** Exercise 22

*Hint.* For line bundles on the circle, changing a hemisphere trivialization can flip both endpoint signs simultaneously; only their relative sign distinguishes the two bundle types. ☐

*Solution.* Real line bundles over $S^1$ give the cylinder and Möbius cases. ☐

**Exercise 30.28.** Exercise 23

*Hint.* A clutching map $S^1 \to U(1)$ has an integer winding number; record the equator and fiber orientation conventions before identifying its characteristic-class sign. ☐

*Solution.* Complex line bundles over $S^2$ are classified by winding number. ☐

**Exercise 30.29.** Exercise 24

*Hint.* Choose a bundle metric, form $(D(E), S(E))$, and restrict relative cohomology to the oriented fiber pairs to locate a Thom class. ☐

*Solution.* VIII/31 uses the disk/sphere bundle pair to define Thom classes. ☐

## 30.25 Solved problem dossiers

**Problem 30.30** (Transition cocycle)**.** Derive $g_{ij}g_{jk} = g_{ik}$.

*Solution.* Changing coordinates from chart $k$ to $j$ and then $j$ to $i$ must equal the direct change from $k$ to $i$. ☐

**Problem 30.31** (Pullback)**.** Describe the fiber of $f^*E \to Y$ over $y$.

*Solution.* It is naturally the fiber $E_{f(y)}$, so the pullback has the same rank. ☐

**Problem 30.32** (Whitney rank)**.** If $E$ has rank $r$ and $F$ rank $s$, what is $\mathrm{rank}(E \oplus F)$?

*Solution.* $r + s$. ☐

**Problem 30.33** (Mobius gluing)**.** Why is $(0, v) \sim (1, -v)$ a nontrivial line-bundle gluing?

*Solution.* Going once around the base reverses the fiber coordinate, so no consistent nonzero orientation/vector can return to itself globally. ☐

**Problem 30.34** (Mobius section)**.** Why has the Möbius line bundle no nowhere-zero section?

*Solution.* A nowhere-zero section would trivialize a real line bundle, contradicting the nontrivial sign monodromy. ☐

**Problem 30.35** (Cylinder)**.** What clutching produces the trivial real line bundle over $S^1$?

*Solution.* The constant $+1$ gluing gives the cylinder. ☐

**Problem 30.36** (Two line bundles)**.** Why are there exactly two real line bundles over $S^1$ up to isomorphism?

*Solution.* After metric reduction, the clutching data take values in $O(1) = \{\pm 1\}$, yielding trivial and sign-twisted classes. ☐

**Problem 30.37** (Hemisphere triviality)**.** Why is a bundle over each hemisphere of $S^n$ trivial?

*Solution.* Each hemisphere is contractible; vector bundles over a contractible paracompact base are trivial. ☐

**Problem 30.38** (Clutching homotopy)**.** Why do homotopic clutching maps give isomorphic bundles?

*Solution.* A homotopy of transition data extends across a collar of the equator and gives equivalent gluing data. ☐

**Problem 30.39** (Complex line S2)**.** Classify complex line bundles over $S^2$ by clutching.

*Solution.* Clutching maps are $S^1 \to U(1) \cong S^1$, classified by degree/winding number $\mathbb{Z}$. ☐

**Problem 30.40** (Winding zero)**.** What complex line bundle corresponds to clutching degree zero?

*Solution.* The clutching map is null-homotopic, so the bundle is trivial. ☐

**Problem 30.41** (Bundle metric)**.** Why does a metric reduce the structure group?

*Solution.* Choose orthonormal frames in each trivialization; changes between orthonormal frames lie in $O(k)$ or $U(k)$. ☐

**Problem 30.42** (Tangent circle)**.** Construct a nowhere-zero section of $TS^1$.

*Solution.* The unit tangent vector field $s(e^{it}) = ie^{it}$ is continuous and nowhere zero, trivializing the line bundle. ☐

**Problem 30.43** (Hairy sphere)**.** What bundle fact is expressed by the hairy-ball theorem on $S^2$?

*Solution.* The tangent bundle $TS^2$ has no nowhere-zero section and is therefore nontrivial. ☐

**Problem 30.44** (Bundle isomorphism)**.** What must a bundle isomorphism preserve?

*Solution.* It covers the identity on the base and is a linear isomorphism on every fiber, with a homeomorphic total-space map. ☐

**Problem 30.45** (Stable equivalence)**.** What does adding a trivial summand do conceptually?

*Solution.* It can erase unstable distinctions while retaining stable bundle information, leading toward $K$-theory. ☐

**Problem 30.46** (Clutching tangent sphere)**.** Why can the tangent bundle of $S^n$ be encoded on $S^{n-1}$?

*Solution.* Trivialize over northern and southern hemispheres; their mismatch along the equator is the clutching map. ☐

**Problem 30.47** (SVG transition)**.** What should the transition-function SVG show?

*Solution.* Two overlapping base charts, product-like fibers above each, and a matrix $g_{ij}(x)$ converting fiber coordinates on the overlap. ☐

**Problem 30.48** (SVG clutching)**.** What should the clutching SVG show?

*Solution.* Two trivial hemisphere bundles glued along the equator by $g : S^{n-1} \to O(k)$, making the global twist visually explicit. ☐

**Problem 30.49** (Bridge)**.** Why does the zero section matter for Thom theory?

*Solution.* The Thom class lives in relative cohomology of the bundle total space relative to the complement of the zero section, or equivalently the disk/sphere bundle pair. ☐

**Problem 30.50** (Corpus ownership)**.** Why does this chapter own both vector-bundle and clutching legacy streams?

*Solution.* The migration map sends both explicit sections and all inherited theorem/problem descendants in topology-14 to VIII/30. ☐

**Problem 30.51** (Final principle)**.** What does clutching convert a global bundle problem into?

*Solution.* A homotopy-class problem for a map from the equator into the structure group. ☐

## 30.26 Supplementary worked examples

**Example 30.52** (The Mobius line bundle)**.** Glue $[0,1] \times \mathbb{R}$ by $(0,v) \sim (1,-v)$. The result is a real line bundle over $S^1$ with nontrivial clutching sign. It has no nowhere-zero section compatible with a global choice of orientation, distinguishing it from the trivial cylinder bundle.

**Example 30.53** (Clutching a complex line bundle on S2)**.** Write $S^2 = D^2_+ \cup D^2_-$ and glue trivial complex line bundles along the equator by

$$g_k : S^1 \to U(1), \qquad g_k(z) = z^k.$$

Different winding numbers $k$ yield different complex line bundles; $k$ becomes the first Chern number.

**Example 30.54** (Tangent bundle of the circle)**.** The tangent bundle $TS^1$ is trivial: the vector field $v(z) = iz$ is continuous and nowhere zero, giving a global frame. This contrasts with higher-dimensional sphere tangent bundles where global frames are constrained by Euler classes.

## 30.27 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 30.55** (Bundle chart)**.** What data locally identifies a rank-$r$ vector bundle with a product?

*Hint.* Choose a trivialization over an open set. □

*Solution.* A local trivialization is a homeomorphism or bundle isomorphism $\pi^{-1}(U) \cong U \times \mathbb{F}^r$ respecting projection and linear fiber structure. □

**Exercise 30.56** (Transition function)**.** Where does a transition function take values for a real rank-$r$ vector bundle?

*Hint.* It changes one local frame to another. □

*Solution.* It takes values in $GL_r(\mathbb{R})$. □

**Exercise 30.57** (Trivial line bundle section)**.** Give a nowhere-zero section of $S^1 \times \mathbb{R} \to S^1$.

*Hint.* Use a constant fiber vector. □

*Solution.* $s(x) = (x, 1)$ is a nowhere-zero section. □

**Exercise 30.58** (Clutching domain for Sn)**.** For a bundle over $S^n = D^n_+ \cup D^n_-$, on what space is the clutching map defined?

*Hint.* Use the equatorial overlap. □

*Solution.* On $S^{n-1}$. □

**Exercise 30.59** (Mobius clutching)**.** What is the clutching datum of the Mobius line bundle over $S^1$?

*Hint.* The transition across the overlap reverses sign. ☐

*Solution.* It uses the nontrivial component of $GL_1(\mathbb{R}) = \mathbb{R}^\times$, represented by multiplication by $-1$. ☐

## Proofs

**Exercise 30.60** (Equivalent clutching maps)**.** Explain why homotopic clutching maps yield isomorphic bundles under the standard sphere classification setup.

*Hint.* Extend the homotopy over the collar of the equator. ☐

*Solution.* A homotopy of transition functions gives a bundle over $S^n \times I$ restricting to the two clutched bundles at the ends; standard bundle homotopy invariance identifies them. ☐

**Exercise 30.61** (Trivial bundle criterion)**.** Show a null-homotopic clutching map produces a trivial bundle over $S^n$.

*Hint.* Deform the transition map to the identity. ☐

*Solution.* After the clutching map is deformed to the constant identity transition, the two trivial hemisphere bundles glue to the global product bundle. ☐

**Exercise 30.62** (Frame and triviality)**.** Prove $r$ pointwise linearly independent global sections trivialize a rank-$r$ bundle.

*Hint.* Use the sections as a basis in every fiber. ☐

*Solution.* The map $(x, a_1, \ldots, a_r) \mapsto \sum a_i s_i(x)$ is a continuous fiberwise linear isomorphism from the trivial bundle to the given bundle. ☐

**Exercise 30.63** (Pullback bundle)**.** Explain why pulling a bundle back along $f : X \to B$ again gives a vector bundle.

*Hint.* Pull back each local trivialization. ☐

*Solution.* Over $f^{-1}(U)$, the pullback is identified with $f^{-1}(U) \times \mathbb{F}^r$, with transition functions obtained by composing those of the original bundle with $f$. ☐

## Counterexamples and hypothesis tests

**Exercise 30.64** (Local trivial means global trivial)**.** Test: every vector bundle is globally trivial because it is locally trivial.

*Hint.* The obstruction is in transition functions. ☐

*Solution.* False. Nontrivial gluing, such as the Mobius bundle, prevents a global product trivialization. ☐

**Exercise 30.65** (Nowhere-zero section trivializes every rank)**.** Test: one nowhere-zero section trivializes any vector bundle.

*Hint.* One section supplies only one frame vector. ☐

*Solution.* False for rank $r > 1$; it splits off a trivial line under suitable hypotheses but does not provide a full frame. ☐

**Exercise 30.66** (All clutching maps equivalent)**.** Test: any two clutching maps with the same domain and target define isomorphic bundles.

*Hint.* Homotopy class matters. □

*Solution.* False. Distinct homotopy classes can classify nonisomorphic bundles. □

## Applications and investigations

**Exercise 30.67** (Gauge transition check)**.** How can transition functions be used to verify a computational atlas for a bundle?

*Hint.* Check cocycle identities on overlaps. □

*Solution.* Numerically or symbolically one checks $g_{ij}g_{jk} = g_{ik}$ on triple overlaps and invertibility on pairwise overlaps; failures reveal inconsistent local frames. □

**Exercise 30.68** (Hairy-ball preview)**.** Why would a nowhere-zero section of $TS^2$ be topologically significant?

*Hint.* It would split off a trivial line subbundle. □

*Solution.* Such a section would contradict the nonzero Euler class of $TS^2$; this is the bundle-theoretic form of the hairy-ball theorem. □

## Challenge problems

**Exercise 30.69** (Degree-k clutching)**.** For $g_k(z) = z^k$ and $g_\ell(z) = z^\ell$ from $S^1$ to $U(1)$, when are they homotopic?

*Hint.* Compare winding numbers. □

*Solution.* They are homotopic exactly when $k = \ell$. □

**Exercise 30.70** (Bundle synthesis)**.** Describe the workflow from local trivializations to global bundle topology.

*Hint.* Extract transitions, impose cocycle conditions, and study their homotopy class. □

*Solution.* Local frames determine transition functions; compatibility makes them a cocycle; changes of frame modify the cocycle by gauge transformations; the residual homotopy/cohomology class records the global obstruction to triviality. □

# Chapter 31

# Thom Classes

Thom classes turn the local orientation data of a vector bundle into a global cohomology class. They are the bridge from vector bundles to Euler classes, Gysin maps, Poincaré duality, and intersection theory.

## 31.1 Disk and sphere bundles

Let $\pi : E \to B$ be a rank-$r$ real vector bundle equipped with a bundle metric. Define

$$D(E) = \{v \in E : \|v\| \leq 1\}, \qquad S(E) = \{v \in E : \|v\| = 1\}.$$

The pair $(D(E), S(E))$ is the natural relative space in which the Thom class lives.

## 31.2 Fiber orientation

An orientation of $E$ is a continuous choice of orientation on every fiber. For each $b \in B$,

$$(D(E_b), S(E_b)) \cong (D^r, S^{r-1}), \qquad H^r(D^r, S^{r-1}; R) \cong R.$$

**Definition 31.1** (Thom class)**.** For an oriented rank-$r$ vector bundle $E \to B$, a Thom class is

$$u_E \in H^r(D(E), S(E); R)$$

whose restriction to every oriented fiber pair is the chosen generator.

## 31.3 The Thom isomorphism

**Theorem 31.2** (Thom isomorphism)**.** *For an oriented rank-$r$ vector bundle over a suitable paracompact base,*

$$\Phi_E : H^k(B; R) \longrightarrow H^{k+r}(D(E), S(E); R), \qquad \alpha \longmapsto \pi^*\alpha \smile u_E$$

*is an isomorphism.*

## 31.4 Why relative cohomology appears

The sphere bundle is the fiberwise boundary of the disk bundle. Relative cohomology records the transverse orientation class while forgetting boundary data.

## 31.5 The zero section

The zero section $s : B \to E$ is a homotopy inverse to the projection of the disk bundle, but the relative pair remembers the fiber orientation.

## 31.6 Naturality

For $f : Y \to B$, the pulled-back bundle satisfies

$$\widetilde{f}^*(u_E) = u_{f^*E}.$$

**Theorem 31.3** (Naturality). *The square formed by pullback on base cohomology and pullback on relative cohomology commutes with the two Thom isomorphisms.*

## 31.7 Compactly supported formulation

Under the standard local-compactness hypotheses,

$$H^{k+r}(D(E), S(E); R) \cong H_c^{k+r}(E; R),$$

hence

$$H^k(B; R) \cong H_c^{k+r}(E; R).$$

## 31.8 Trivial bundles and suspension

For $E = B \times \mathbb{R}^r$,

$$\mathrm{Th}(E) = D(E)/S(E) \simeq B_+ \wedge S^r,$$

so the Thom shift becomes an $r$-fold suspension phenomenon.

## 31.9 Orientation reversal

Reversing the orientation negates the integral Thom class. Over $\mathbb{Z}/2$, the sign disappears.

## 31.10 Normal bundles and tubular neighborhoods

If $i : N \hookrightarrow M$ is a closed embedded submanifold with oriented normal bundle $\nu$, a tubular neighborhood identifies a neighborhood of $N$ with $D(\nu)$. The normal Thom class is therefore a cohomological representative concentrated transversely to $N$.

## 31.11 The Thom space and collapse

**Definition 31.4** (Thom space)**.**

$$\mathrm{Th}(E) = D(E)/S(E).$$

Collapsing the complement of a tubular neighborhood gives the Thom collapse

$$M \longrightarrow \mathrm{Th}(\nu).$$

## 31.12 Gysin viewpoint

Cup product with a Thom class, followed by excision and extension, produces wrong-way maps in cohomology. This is the local mechanism behind geometric duality.

## 31.13 From Thom to Euler

Pulling the Thom class back by the zero section gives

$$e(E) = s^*(u_E) \in H^r(B; R).$$

This is the central bridge to VIII/32.

## 31.14 From normal Thom classes to dual cycles

For an oriented embedded submanifold, the normal Thom class extends to ambient cohomology and becomes the class Poincaré dual to that submanifold. This is the bridge to VIII/33 and VIII/34.

## 31.15 Solved dossiers

**Problem 31.5.** Compute the relative cohomology of a fiber pair $(D^r, S^{r-1})$.

*Solution.* It is $\mathbb{Z}$ in degree $r$ and zero otherwise, with the generator fixed by orientation. □

**Problem 31.6.** Why is the Thom class naturally relative?

*Solution.* The sphere bundle is the fiberwise boundary; relative cohomology retains the orientation class of the disk fiber while killing boundary data. □

**Problem 31.7.** Find the Thom class for a trivial oriented line bundle.

*Solution.* It is the pullback of the generator of $H^1(D^1, S^0)$ along the fiber factor. □

**Problem 31.8.** State the rank-five Thom shift.

*Solution.* $H^k(B) \cong H^{k+5}(D(E), S(E))$. □

**Problem 31.9.** What happens under orientation reversal?

*Solution.* The integral Thom class changes sign. □

**Problem 31.10.** Why do $\mathbb{Z}/2$ coefficients remove the orientability requirement?

*Solution.* The two signs agree modulo two. □

**Problem 31.11.** Describe the Thom space of a trivial rank-$r$ bundle.

*Solution.* It is naturally $B_+ \wedge S^r$. □

**Problem 31.12.** What role does the zero section play?

*Solution.* It identifies the base in the total space and pulls the Thom class back to the Euler class. □

**Problem 31.13.** What does a tubular neighborhood provide?

*Solution.* It identifies a neighborhood of an embedded submanifold with the disk bundle of its normal bundle. ☐

**Problem 31.14.** Define the Thom collapse.

*Solution.* Collapse the complement of a tubular neighborhood to the basepoint of the Thom space. ☐

**Problem 31.15.** State naturality under $f : Y \to B$.

*Solution.* For the pulled-back orientation, $\widetilde{f}^* u_E = u_{f^*E}$. ☐

**Problem 31.16.** How is compact support related to the Thom pair?

*Solution.* Under standard hypotheses, $H^*(D(E), S(E)) \cong H_c^*(E)$. ☐

**Problem 31.17.** Where does the bundle rank appear?

*Solution.* As the cohomological degree shift. ☐

**Problem 31.18.** How does the Thom class encode normal orientation?

*Solution.* Its restriction to each normal disk fiber is the chosen orientation generator. ☐

**Problem 31.19.** What is the prototype of a cohomological wrong-way map?

*Solution.* Cup with the Thom class, followed by excision and extension from a tubular neighborhood. ☐

**Problem 31.20.** Relate the Thom space to reduced cohomology.

*Solution.* $\widetilde{H}^{k+r}(\mathrm{Th}(E)) \cong H^k(B)$. ☐

**Problem 31.21.** Why is a bundle metric useful?

*Solution.* It defines disk and sphere subbundles without changing the underlying homotopy-theoretic data. ☐

**Problem 31.22.** What characteristic class comes next?

*Solution.* $e(E) = s^* u_E$. ☐

**Problem 31.23.** How does a submanifold acquire a dual cohomology class?

*Solution.* Transfer the Thom class of its oriented normal bundle into the ambient manifold. ☐

**Problem 31.24.** Why is the Thom theorem a bridge theorem?

*Solution.* It connects bundle orientation, relative cohomology, embeddings, Gysin maps, Euler classes, and duality. ☐

## 31.16 Exercises, Hints, and Solutions

**Exercise 31.25.** Compute $H^*(D^r, S^{r-1}; \mathbb{Z})$.

*Hint.* Use $D^r/S^{r-1} \cong S^r$.

For positive rank, use reduced cohomology of the quotient and identify the single nonzero degree; choose its generator by an orientation of the disk. □

*Solution.* Since $D^r/S^{r-1} \cong S^r$, relative cohomology is reduced cohomology of $S^r$:

$$H^q(D^r, S^{r-1}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = r, \\ 0, & q \neq r. \end{cases}$$

The generator is the orientation class of the disk fiber. □

**Exercise 31.26.** Write the Thom isomorphism for a rank-three bundle over $S^2$.

*Hint.* Shift degree by the rank.

Use integral coefficients and an orientation of the rank-three bundle; list the nonzero base degrees before adding three to each. □

*Solution.* For a rank-three oriented bundle $E \to S^2$,

$$\Phi_E : H^k(S^2) \xrightarrow{\cong} H^{k+3}(D(E), S(E)), \qquad \alpha \mapsto \pi^*\alpha \smile u_E.$$

Thus $1 \in H^0(S^2)$ maps to $u_E \in H^3$, while the degree-two generator maps to a degree-five relative class. □

**Exercise 31.27.** Check the fiber normalization for a trivial oriented line bundle.

*Hint.* Restrict to each fiber pair.

On $(D^1, S^0)$, choose the generator agreeing with the fiber's orientation and check that the same choice is made continuously over the base. □

*Solution.* For $E = B \times \mathbb{R}$, the fiber pair over $b$ is $(D^1, S^0)$. Choose $u_E$ so that

$$u_E|_{(D^1,S^0)} = 1 \in H^1(D^1, S^0; \mathbb{Z}).$$

This fiberwise generator condition is exactly the Thom normalization. □

**Exercise 31.28.** Show that reversing orientation changes the integral Thom class by a sign.

*Hint.* Track the chosen generator.

Restrict both candidate classes to a fiber and compare their generators; uniqueness of the normalized Thom class then determines the global sign. □

*Solution.* Reversing the orientation reverses the chosen generator of every fiber group $H^r(D^r, S^{r-1}; \mathbb{Z})$. By uniqueness of the normalized class, the new Thom class is $-u_E$. □

**Exercise 31.29.** Explain why mod-2 Thom theory does not require orientability.

*Hint.* Reduce signs modulo two.

Compare local generators on overlapping trivializations. The transition can change their signs, but that change is invisible in $\mathbb{F}_2$. □

*Solution.* The only ambiguity in an integral fiber generator is its sign. In $\mathbb{Z}/2$, $1 = -1$, so local generators glue without an orientation choice. Hence every real vector bundle has a mod-two Thom class. $\square$

**Exercise 31.30.** Identify the Thom space of a trivial rank-$r$ bundle.

*Hint.* Use $B_+ \wedge S^r$.

Write the quotient $(B \times D^r)/(B \times S^{r-1})$; the extra basepoint in $B_+$ ensures the entire sphere subbundle is collapsed to one point. $\square$

*Solution.* For $E = B \times \mathbb{R}^r$,

$$\mathrm{Th}(E) = D(E)/S(E) \cong B_+ \wedge (D^r/S^{r-1}) \cong B_+ \wedge S^r.$$

$\square$

**Exercise 31.31.** Compute its reduced cohomology when the base is $S^1$.

*Hint.* Use suspension.

Use the rank shift on ordinary $H^*(S^1)$, retaining its degree-zero class; dropping that class would lose one of the two reduced Thom-space groups. $\square$

*Solution.* With $B = S^1$,

$$\mathrm{Th}(E) \simeq S^1_+ \wedge S^r \simeq S^r \vee S^{r+1}.$$

Therefore its reduced integral cohomology is $\mathbb{Z}$ in degrees $r$ and $r+1$, and zero otherwise. $\square$

**Exercise 31.32.** Describe the pullback Thom class under a map $f : Y \to B$.

*Hint.* Compute the pulled-back bundle first.

Use the pullback metric and orientation so that the bundle map preserves disk/sphere pairs, then check the fiber-generator normalization. $\square$

*Solution.* If $f : Y \to B$, the bundle map $\widetilde{f} : D(f^*E) \to D(E)$ covering $f$ pulls the Thom class back:

$$u_{f^*E} = \widetilde{f}^* u_E.$$

Its restriction to every pulled-back fiber is the same oriented generator. $\square$

**Exercise 31.33.** Verify naturality of the Thom isomorphism.

*Hint.* Apply naturality of cup products.

Compare $\widetilde{f}^*\pi^*\alpha$ with $(\pi')^* f^*\alpha$, and use multiplicativity of pullback on the product with the Thom class. $\square$

*Solution.* For $\alpha \in H^k(B)$,

$$\widetilde{f}^*(\pi^*\alpha \smile u_E) = (\pi')^*(f^*\alpha) \smile u_{f^*E}.$$

Thus pullback commutes with the two Thom isomorphisms. $\square$

**Exercise 31.34.** Describe the normal bundle of an equatorial $S^{n-1} \subset S^n$.

*Hint.* The normal direction is one-dimensional.

Embed the equator as $x_{n+1} = 0$ in the unit sphere; the constant last-coordinate direction is tangent to $S^n$ along the equator and normal to it. $\square$

*Solution.* The equator $S^{n-1} \subset S^n$ has a one-dimensional normal bundle. A continuous unit normal can be chosen pointing toward one hemisphere, so this normal line bundle is trivial:

$$\nu \cong S^{n-1} \times \mathbb{R}.$$

□

**Exercise 31.35.** Construct a tubular neighborhood in that example.

*Hint.* Use a product collar or exponential map.

For small $t$, test $(x, t) \mapsto (\cos t\, x, \sin t)$; check injectivity and identify the equator at $t = 0$. □

*Solution.* Using the normal coordinate, a tubular neighborhood is

$$S^{n-1} \times (-\varepsilon, \varepsilon) \longrightarrow S^n,$$

with $S^{n-1} \times \{0\}$ identified with the equator. Radial/geodesic normal displacement gives the embedding. □

**Exercise 31.36.** Describe the corresponding Thom collapse map.

*Hint.* Collapse the complement.

Choose a closed tubular disk neighborhood and collapse its boundary together with the exterior; the identified boundary is the sphere subbundle in the Thom quotient. □

*Solution.* Collapse the complement of the chosen tubular neighborhood to one point, and identify the neighborhood with $D(\nu)$. This gives

$$S^n \longrightarrow D(\nu)/S(\nu) = \mathrm{Th}(\nu).$$

□

**Exercise 31.37.** Relate relative cohomology of $(D(E), S(E))$ to the Thom quotient.

*Hint.* Use reduced cohomology of a quotient.

Apply the relative-to-reduced quotient theorem to the good disk/sphere pair; the resulting class belongs to reduced rather than ordinary quotient cohomology. □

*Solution.* For any good pair,

$$H^q(D(E), S(E)) \cong \widetilde{H}^q(D(E)/S(E)).$$

Since $D(E)/S(E) = \mathrm{Th}(E)$, relative Thom cohomology is exactly reduced cohomology of the Thom quotient. □

**Exercise 31.38.** Explain the compact-support formulation for $E = \mathbb{R}^r$.

*Hint.* Compactify the fiber direction.

Here the base is a point, so compactifying the fiber is the same as one-point compactification of the total space; identify the added point with infinity. □

*Solution.* For the bundle $\mathbb{R}^r \to \{*\}$, one-point compactification gives $S^r$. Hence

$$H^q_c(\mathbb{R}^r; \mathbb{Z}) \cong \widetilde{H}^q(S^r; \mathbb{Z}),$$

which is $\mathbb{Z}$ in degree $r$ and zero otherwise. □

**Exercise 31.39.** Track the degree shift in a rank-five example.

*Hint.* Write the rank explicitly.

Write a general base degree $k$, add the rank five, and keep the result in relative bundle cohomology rather than in the base group. $\square$

*Solution.* For a rank-five oriented bundle,

$$H^k(B) \xrightarrow{\cong} H^{k+5}(D(E), S(E)).$$

For example a class in $H^2(B)$ corresponds to a relative class in degree 7. $\square$

**Exercise 31.40.** Compare Thom classes obtained from two bundle metrics.

*Hint.* Metrics are homotopic through positive-definite metrics.

Interpolate the two positive-definite metrics and use fiberwise radial rescaling; verify that it preserves fiber orientations and the boundary sphere pairs. $\square$

*Solution.* Two bundle metrics are joined by a continuous path of positive-definite metrics. Fiberwise radial rescaling identifies their disk/sphere pairs through an isotopy, and the normalized fiber generator is preserved. Thus the induced Thom classes correspond. $\square$

**Exercise 31.41.** Show that an orientation-preserving bundle isomorphism preserves Thom classes.

*Hint.* Use the fiber orientation generator.

Pull back the target class and restrict to each source fiber; orientation preservation gives the same generator, which determines the class uniquely. $\square$

*Solution.* An orientation-preserving bundle isomorphism sends each oriented fiber generator to the oriented fiber generator. Pullback of the target Thom class therefore satisfies the defining normalization, so by uniqueness it equals the source Thom class. $\square$

**Exercise 31.42.** Analyze an orientation-reversing bundle isomorphism.

*Hint.* Orientation reversal changes its sign.

Track the sign of the induced map on $H^r(D^r, S^{r-1}; \mathbb{Z})$; in mod-two coefficients the two possibilities become equal. $\square$

*Solution.* An orientation-reversing bundle isomorphism sends the fiber orientation generator to its negative. Consequently the pulled-back integral Thom class is $-u_E$; modulo two there is no sign change. $\square$

**Exercise 31.43.** Describe the first step in the Gysin construction for an embedding.

*Hint.* Start inside a tubular neighborhood.

For a closed oriented embedding of codimension $r$, first multiply the pulled-back base class by the normal Thom class, obtaining relative degree increased by $r$. $\square$

*Solution.* For an oriented embedding $i : N \hookrightarrow M$, first choose a tubular neighborhood $D(\nu) \subset M$ of the normal bundle. Cup with $u_\nu$ on that neighborhood, then use excision and extension to ambient cohomology. This is the geometric core of the Gysin map. $\square$

**Exercise 31.44.** Explain how the zero section meets every fiber disk.

*Hint.* The zero section selects the zero vector.

The zero vector has norm zero, so it lies in the disk interior and never on the unit sphere; this makes the zero section a map of pairs from $(B, \varnothing)$. □

*Solution.* The zero section is $s(b) = 0_b$. Since $0_b$ lies in the interior of every fiber disk $D(E_b)$, the image $s(B)$ meets every disk fiber at its center and never lies in the sphere boundary. □

**Exercise 31.45.** Compute $s^*u_E$ for a trivial positive-rank bundle.

*Hint.* A trivial positive-rank bundle admits a nonzero constant section.

Choose a nonzero constant fiber vector and normalize it. Homotope this section to the zero section and use the vanishing of the Thom class on the sphere boundary. □

*Solution.* A trivial positive-rank bundle has a nowhere-zero constant section, so its Euler class vanishes. Since $e(E) = s^*u_E$,

$$s^*u_E = 0.$$

□

**Exercise 31.46.** Preview why a nowhere-zero section constrains $s^*u_E$.

*Hint.* Compare zero and nonzero sections.

Normalize the nowhere-zero section into $S(E)$; its pullback of the relative Thom class vanishes, while linear interpolation compares it with the zero section in $D(E)$. □

*Solution.* A nowhere-zero section can be normalized to a section of $S(E)$. The Euler class is the obstruction represented by the zero section's pullback of the Thom class; a section avoiding zero makes that obstruction vanish. Hence a nowhere-zero section forces $s^*u_E = e(E) = 0$. □

**Exercise 31.47.** Relate a normal Thom class to the cohomology class of a submanifold.

*Hint.* Extend the normal Thom class into ambient cohomology.

Use the oriented normal bundle of a closed submanifold; pass from its Thom class through the ambient relative group before forgetting supports. □

*Solution.* For an oriented embedded submanifold $N \subset M$, take the Thom class of its oriented normal bundle, identify the disk bundle with a tubular neighborhood, and extend by excision into $M$. The resulting ambient cohomology class is $\mathrm{PD}[N]$. □

**Exercise 31.48.** Summarize the chain Thom class → Euler class → Poincaré duality.

*Hint.* Track the degree change at every arrow.

Assume the base is a closed oriented $n$-manifold: the rank-$r$ Thom class gives a degree-$r$ Euler class and then a degree-$(n-r)$ homology class. □

*Solution.* The chain is

$$u_E \in H^r(D(E), S(E)) \xrightarrow{s^*} e(E) \in H^r(B) \xrightarrow{\frown [B]} H_{\dim B - r}(B).$$

Thus the Thom class produces the Euler class, and cap product with the fundamental class converts it into its geometric dual cycle. □

## 31.17 Supplementary worked examples

**Example 31.49** (Thom class of a trivial oriented bundle)**.** For the oriented trivial rank-$r$ bundle $\pi : B \times \mathbb{R}^r \to B$, the Thom class is the fiberwise orientation class

$$u \in H^r(B \times D^r,\ B \times S^{r-1}).$$

Cup product with $u$ gives the Thom isomorphism $H^k(B) \cong H^{k+r}(D(E), S(E))$.

**Example 31.50** (The zero section under the Thom isomorphism)**.** The zero section $s : B \to E$ pulls the Thom class back to the Euler class:

$$e(E) = s^*u.$$

Thus the same class that generates relative cohomology in every fiber also measures the obstruction to moving the zero section off itself.

**Example 31.51** (Why orientation is required)**.** A Thom class over $\mathbb{Z}$ must restrict to a chosen generator of $H^r(D^r, S^{r-1}; \mathbb{Z})$ in every fiber. If orientations reverse around a loop, these local generators cannot be chosen coherently; twisted coefficients or mod-2 coefficients repair this failure.

## 31.18 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 31.52** (Degree of Thom class)**.** What is the degree of the Thom class of an oriented rank-$r$ real bundle?

*Hint.* Look at the top relative cohomology of the fiber disk pair. □

*Solution.* It lies in degree $r$. □

**Exercise 31.53** (Fiber restriction)**.** To what class must a Thom class restrict on each fiber pair $(D^r, S^{r-1})$?

*Hint.* Use the orientation generator. □

*Solution.* It restricts to a generator of $H^r(D^r, S^{r-1}) \cong \mathbb{Z}$. □

**Exercise 31.54** (Thom shift)**.** What degree shift appears in the Thom isomorphism?

*Hint.* Cup with a degree-$r$ class. □

*Solution.* $H^k(B) \cong H^{k+r}(D(E), S(E))$. □

**Exercise 31.55** (Euler pullback)**.** Express the Euler class in terms of the Thom class and zero section.

*Hint.* Pull back along $s : B \to E$. □

*Solution.* $e(E) = s^*u$. □

**Exercise 31.56** (Trivial bundle Euler)**.** What is the Euler class of a trivial oriented positive-rank bundle?

*Hint.* It has a nowhere-zero section. □

*Solution.* It vanishes. □

## Proofs

**Exercise 31.57** (Thom map is module map)**.** Show the Thom isomorphism is given by $\alpha \mapsto \pi^*\alpha \smile u$.

*Hint.* Check degrees and fiber restriction. ☐

*Solution.* The pullback carries $\alpha$ to the total space and cup product with the Thom class supplies the fiber orientation, giving the canonical degree-shifted isomorphism. ☐

**Exercise 31.58** (Naturality)**.** Explain the naturality of Thom classes under orientation-preserving pullback.

*Hint.* Pull back the bundle and its disk/sphere pair. ☐

*Solution.* For $f : B' \to B$, the Thom class of $f^*E$ is the pullback of $u_E$ under the induced map of Thom pairs, provided orientations are pulled back compatibly. ☐

**Exercise 31.59** (Uniqueness)**.** Why is an integral Thom class unique once the orientation is fixed?

*Hint.* Compare two candidates fiberwise and use the Thom isomorphism in degree zero. ☐

*Solution.* Their difference restricts trivially to every fiber; the relative cohomology group is a free rank-one $H^0(B)$-module on each connected component, fixing the class by the chosen orientation. ☐

**Exercise 31.60** (Orientation obstruction)**.** Explain why a nonorientable real bundle need not have an integral Thom class.

*Hint.* Transport a fiber generator around an orientation-reversing loop. ☐

*Solution.* The generator returns with opposite sign, so no globally coherent integral fiber orientation class exists. ☐

## Counterexamples and hypothesis tests

**Exercise 31.61** (Every bundle has integral Thom class)**.** Test: every real vector bundle has a Thom class over $\mathbb{Z}$.

*Hint.* Orientability is the obstruction. ☐

*Solution.* False. Nonorientable bundles require twisted coefficients or a coefficient system such as $\mathbb{F}_2$. ☐

**Exercise 31.62** (Thom class lives in ordinary cohomology of E)**.** Test: the Thom class is naturally an element of $H^r(E)$.

*Hint.* Boundary behavior matters. ☐

*Solution.* False in the disk-bundle model; it is naturally a relative class in $H^r(D(E), S(E))$, equivalently compactly supported in the total space. ☐

**Exercise 31.63** (No rank shift)**.** Test: the Thom isomorphism preserves cohomological degree.

*Hint.* Cup with $u$. ☐

*Solution.* False. It shifts degree upward by the bundle rank. ☐

## Applications and investigations

**Exercise 31.64** (Intersection localization)**.** How does the Thom class turn a submanifold into a cohomology class of the ambient manifold?

*Hint.* Use a tubular neighborhood and extend the Thom class. □

*Solution.* The normal bundle Thom class is transferred through a tubular neighborhood to a class supported near the submanifold, yielding its Poincare dual. □

**Exercise 31.65** (Zero-locus counting)**.** How can the Thom/Euler formalism count zeros of a transverse section?

*Hint.* Intersect the section with the zero section. □

*Solution.* A transverse section pulls the Thom class back to the Euler class; evaluating it on the base fundamental class gives the signed number of zeros. □

## Challenge problems

**Exercise 31.66** (Trivial rank-r bundle)**.** Write the Thom isomorphism for $B \times \mathbb{R}^r$ explicitly.

*Hint.* Use the generator $u$ of the fiber pair. □

*Solution.* $\alpha \mapsto \mathrm{pr}_B^* \alpha \smile u$ identifies $H^k(B)$ with $H^{k+r}(B \times D^r, B \times S^{r-1})$. □

**Exercise 31.67** (Thom synthesis)**.** Explain how one class links orientation, relative cohomology, Euler classes, and Poincare duality.

*Hint.* Follow the Thom class through restriction, pullback, and tubular neighborhoods. □

*Solution.* The Thom class is the coherent fiber orientation generator; cup product with it gives the Thom isomorphism, zero-section pullback gives Euler class, and transfer from normal bundles produces Poincare-dual classes of submanifolds. □

# Chapter 32

# Sphere Bundles and Euler Classes

The unit sphere bundle converts a nowhere-zero vector field into a genuine section problem. The Euler class measures the primary obstruction to solving that problem globally.

## 32.1 Unit sphere bundles

For a rank-$r$ real vector bundle $E \to B$ with metric,

$$S(E) = \{v \in E : \|v\| = 1\} \longrightarrow B$$

has fiber $S^{r-1}$.

## 32.2 Nowhere-zero sections

A nowhere-zero section of $E$ normalizes to a section of $S(E)$, and every section of $S(E)$ is a nowhere-zero section of $E$.

## 32.3 Euler class

**Definition 32.1** (Euler class)**.** For an oriented rank-$r$ bundle with Thom class $u_E$ and zero section $s : B \to E$,

$$e(E) = s^* u_E \in H^r(B; R).$$

## 32.4 The obstruction theorem

**Theorem 32.2** (Nowhere-zero section obstruction)**.** *If an oriented vector bundle admits a nowhere-zero section, then*

$$e(E) = 0.$$

The converse is an obstruction-theoretic statement that requires appropriate hypotheses; the Euler class is the primary obstruction, not in every setting a complete classifier of sections.

## 32.5 Naturality

For $f : Y \to B$,

$$e(f^* E) = f^* e(E).$$

## 32.6 Whitney sums

For oriented bundles,

$$e(E \oplus F) = e(E) \smile e(F).$$

In particular, adding a positive-rank trivial summand forces the Euler class to vanish.

## 32.7 Sphere bundles and the Gysin sequence

For an oriented rank-$r$ bundle, the sphere bundle $S(E) \to B$ has a long exact Gysin sequence. The characteristic map in the base is cup product with $e(E)$:

$$\cdots \to H^{k-r}(B) \xrightarrow{\smile e(E)} H^k(B) \to H^k(S(E)) \to H^{k-r+1}(B) \to \cdots .$$

## 32.8 Tangent bundles

For a closed oriented $n$-manifold,

$$e(TM) \in H^n(M).$$

Its evaluation on the fundamental class is the Euler number.

**Theorem 32.3** (Euler class and Euler characteristic)**.** *For a closed oriented manifold $M$,*

$$\langle e(TM), [M] \rangle = \chi(M).$$

## 32.9 The hairy-ball theorem

Since $\chi(S^{2m}) = 2$, the tangent Euler class of an even-dimensional sphere is nonzero. Thus $S^{2m}$ admits no nowhere-zero tangent vector field.

## 32.10 Odd-dimensional manifolds

A closed oriented odd-dimensional manifold has Euler characteristic zero. In particular, the tangent Euler number vanishes.

## 32.11 Zeros of transverse sections

For an oriented rank-$r$ bundle over an oriented $r$-manifold, a generic section has isolated transverse zeros. Each zero has a local sign, and the signed sum equals the Euler number.

## 32.12 Poincaré–Hopf viewpoint

For $E = TM$, the preceding statement becomes the Poincaré–Hopf theorem: the sum of indices of isolated zeros of a vector field equals $\chi(M)$.

## 32.13 Zero loci as dual cycles

More generally, a transverse section of an oriented rank-$r$ bundle has a codimension-$r$ zero set whose fundamental class is Poincaré dual to $e(E)$. This is the direct bridge to VIII/33.

## 32.14 Solved dossiers

**Problem 32.4.** Show that a nowhere-zero section of $E$ is equivalent to a section of $S(E)$ after choosing a metric.

*Solution.* Normalize the vector field fiberwise; conversely include the unit sphere bundle in $E$. ☐

**Problem 32.5.** Define the Euler class from the Thom class.

*Solution.* For an oriented rank-$r$ bundle, $e(E) = s^* u_E \in H^r(B)$. ☐

**Problem 32.6.** Why does a nowhere-zero section force $e(E) = 0$?

*Solution.* Move the zero section through a homotopy to a disjoint section; the pulled-back Thom class vanishes. ☐

**Problem 32.7.** Compute the Euler class of a trivial positive-rank oriented bundle.

*Solution.* It is zero because a constant nowhere-zero section exists. ☐

**Problem 32.8.** What is the Euler class of an oriented line bundle with a nowhere-zero section?

*Solution.* Zero. ☐

**Problem 32.9.** State multiplicativity for Whitney sums.

*Solution.* $e(E \oplus F) = e(E) \smile e(F)$ for oriented bundles. ☐

**Problem 32.10.** Relate $e(TM)$ to the Euler characteristic.

*Solution.* For a closed oriented manifold, $\langle e(TM), [M] \rangle = \chi(M)$. ☐

**Problem 32.11.** Evaluate the Euler number of $TS^2$.

*Solution.* It equals $\chi(S^2) = 2$. ☐

**Problem 32.12.** What happens for a closed oriented odd-dimensional manifold?

*Solution.* Its Euler characteristic, hence tangent Euler number, is zero. ☐

**Problem 32.13.** Why does $S^{2m}$ have no nowhere-zero tangent vector field?

*Solution.* Its Euler class pairs to $2 \neq 0$. ☐

**Problem 32.14.** Why is an Euler class an obstruction rather than always a complete classifier?

*Solution.* Vanishing is necessary for a nonzero section, but further obstructions can exist outside the standard obstruction-theoretic range. ☐

**Problem 32.15.** Where does the Euler class appear in the Gysin sequence?

*Solution.* It is the class whose cup-product map links adjacent base-cohomology terms. ☐

**Problem 32.16.** Describe the sphere bundle of a rank-$r$ bundle.

*Solution.* It has fiber $S^{r-1}$ and records nonzero directions in each vector fiber. ☐

**Problem 32.17.** What is the geometric meaning of a zero of a transverse section?

*Solution.* It contributes a local index; the signed sum represents the Euler number. ☐

**Problem 32.18.** State the Poincaré–Hopf connection.

*Solution.* For an oriented closed manifold, the sum of vector-field indices equals $\chi(M)$. ☐

**Problem 32.19.** What coefficient system avoids orientability issues?

*Solution.* Mod-2 coefficients. ☐

**Problem 32.20.** How does pullback affect Euler classes?

*Solution.* $e(f^*E) = f^*e(E)$. ☐

**Problem 32.21.** What is the Euler class of $E \oplus \varepsilon^1$?

*Solution.* Zero, since the trivial summand gives a nowhere-zero section; equivalently multiplicativity includes $e(\varepsilon^1) = 0$. ☐

**Problem 32.22.** How does VIII/32 lead to duality?

*Solution.* The zero locus of a transverse section represents the Poincaré dual of the Euler class. ☐

**Problem 32.23.** Why are sphere bundles visually natural here?

*Solution.* A global section is literally a continuous choice of one point on every sphere fiber. ☐

## 32.15 Exercises, Hints, and Solutions

**Exercise 32.24.** Construct the unit sphere bundle of a trivial rank-two bundle over $S^1$.

*Hint.* Normalize fiberwise.

Use the product Euclidean metric and impose unit norm in each $\mathbb{R}^2$ fiber; identify both the total sphere-bundle space and its projection. ☐

*Solution.* For $E = S^1 \times \mathbb{R}^2$, the unit vectors in each fiber form $S^1$, so

$$S(E) \cong S^1 \times S^1 \to S^1.$$

It is the trivial circle bundle. ☐

**Exercise 32.25.** Show that a nonzero section normalizes to a sphere-bundle section.

*Hint.* Define $\widehat{s}(b) = s(b)/\|s(b)\|$. Check that the denominator is continuous and never zero, and that the normalized vector remains over $b$. ☐

*Solution.* If $s(b) \neq 0$ for all $b$, define $\widehat{s}(b) = s(b)/\|s(b)\|$. Continuity of the norm gives a continuous section $\widehat{s} : B \to S(E)$. Conversely, including $S(E) \subset E$ turns every sphere-bundle section into a nowhere-zero section. ☐

**Exercise 32.26.** Prove naturality $e(f^*E) = f^*e(E)$.

*Hint.* Apply Thom naturality.

Draw the square formed by the bundle map and the two zero sections, then combine its commuting pullbacks with naturality of the oriented Thom class. □

*Solution.* Thom naturality gives $u_{f^*E} = \widetilde{f}^* u_E$. Pulling back along the zero sections yields

$$e(f^*E) = s_Y^* u_{f^*E} = s_Y^* \widetilde{f}^* u_E = f^* s_B^* u_E = f^* e(E).$$

□

**Exercise 32.27.** Compute the Euler class of a trivial rank-three bundle.

*Hint.* Use a constant vector.

A fixed nonzero vector gives a section in every product fiber; apply the implication from existence of such a section to vanishing Euler class. □

*Solution.* A trivial rank-three bundle admits a constant nowhere-zero section. Therefore its Euler class is zero:

$$e(B \times \mathbb{R}^3) = 0 \in H^3(B).$$

□

**Exercise 32.28.** Use multiplicativity to compute $e(E \oplus \varepsilon^1)$.

*Hint.* Use multiplicativity.

Use compatible orientations on the direct sum; one factor in the Euler cup product is the Euler class of the trivial line. □

*Solution.* By multiplicativity,

$$e(E \oplus \varepsilon^1) = e(E) \smile e(\varepsilon^1).$$

The trivial line has a nowhere-zero section, so $e(\varepsilon^1) = 0$, and the product vanishes. □

**Exercise 32.29.** Show that $e(TS^2)$ pairs to 2.

*Hint.* Pair with the fundamental class.

Normalize the positive generator $u$ by $\langle u, [S^2] \rangle = 1$; Poincare–Hopf then determines the coefficient of $u$ in the tangent Euler class. □

*Solution.* For the oriented two-sphere,

$$\langle e(TS^2), [S^2] \rangle = \chi(S^2) = 2.$$

Thus the Euler class is twice the preferred generator of $H^2(S^2; \mathbb{Z})$. □

**Exercise 32.30.** Show that $TS^2$ has no nowhere-zero section.

*Hint.* Use nonvanishing of the pairing.

Use the contrapositive of the necessary Euler-class condition; a nonzero evaluation proves the cohomology class itself cannot vanish. □

*Solution.* If $TS^2$ had a nowhere-zero section, the Euler class would vanish. But its evaluation on $[S^2]$ is 2, so $e(TS^2) \neq 0$. Hence no nowhere-zero tangent vector field exists. □

**Exercise 32.31.** Compute the Euler number of $TS^{2m}$.

*Hint.* Use $\chi(S^{2m}) = 2$.

Apply the tangent Euler-number theorem on the closed oriented even sphere and compute its Euler characteristic from its two-cell CW structure. □

*Solution.* For every $m \geq 1$,

$$\langle e(TS^{2m}), [S^{2m}] \rangle = \chi(S^{2m}) = 2.$$

So the tangent Euler number of every even-dimensional sphere is 2. □

**Exercise 32.32.** Explain why the Euler number of $TS^{2m+1}$ vanishes.

*Hint.* Use $\chi(S^{2m+1}) = 0$.

Besides the Euler calculation, view the odd sphere inside $\mathbb{C}^{m+1}$; multiplication by $i$ gives a tangent vector perpendicular to the radial vector and never zero. □

*Solution.* For $S^{2m+1}$,

$$\langle e(TS^{2m+1}), [S^{2m+1}] \rangle = \chi(S^{2m+1}) = 0.$$

Indeed odd spheres admit explicit nowhere-zero tangent vector fields. □

**Exercise 32.33.** Relate the hairy-ball theorem to the Euler class.

*Hint.* Translate vector fields into sections.

A tangent vector field is a section of $TS^{2m}$; its normalization would produce a section of the unit tangent sphere bundle, contradicting the nonzero Euler class. □

*Solution.* A tangent vector field is a section of $TM$, and after normalization a nowhere-zero vector field is a section of $S(TM)$. Since $e(TS^{2m}) \neq 0$, the Euler obstruction rules out such a section. This is the hairy-ball theorem. □

**Exercise 32.34.** Write the relevant fragment of the Gysin sequence for an oriented circle bundle.

*Hint.* Take fiber $S^1$.

The sphere fiber has dimension one, but the vector-bundle rank is two; use rank two for the cup-with-Euler degree shift. □

*Solution.* For an oriented rank-two bundle $E \to B$, the sphere fiber is $S^1$, and the Gysin sequence contains

$$\cdots \to H^{k-2}(B) \xrightarrow{\smile e(E)} H^k(B) \to H^k(S(E)) \to H^{k-1}(B) \xrightarrow{\smile e(E)} H^{k+1}(B) \to \cdots .$$

□

**Exercise 32.35.** Identify the cup-with-Euler-class map in that sequence.

*Hint.* Look for cup product by $e(E)$.

Identify the map as $H^{k-r}(B) \to H^k(B)$ and check that multiplication by the degree-$r$ class lands in the stated target. □

*Solution.* In the Gysin sequence the characteristic map from base cohomology to base cohomology is precisely

$$\alpha \longmapsto \alpha \smile e(E).$$

Its degree shift equals the rank of the oriented vector bundle. □

**Exercise 32.36.** Describe zeros of a generic section of an oriented rank-$r$ bundle over an $r$-manifold.

*Hint.* Use transversality.

Transversality subtracts the fiber rank from the base dimension. Compactness turns the resulting closed zero-dimensional zero set into a finite set. □

*Solution.* If $\operatorname{rank} E = r = \dim B$, transversality to the zero section makes the zero set of a generic section a submanifold of dimension $r - r = 0$. On a compact base it is therefore a finite set of isolated zeros. □

**Exercise 32.37.** Assign signs to isolated transverse zeros.

*Hint.* Compare orientations of base and fiber.

At a zero the vertical derivative is intrinsically $T_xB \to E_x$; compare its determinant sign using the specified orientations. □

*Solution.* At an isolated transverse zero $x$, trivialize $E$ and orient both base and fiber. The derivative of the local section is an isomorphism $T_xB \to E_x$. Assign sign $+1$ if it preserves orientation and $-1$ if it reverses orientation. □

**Exercise 32.38.** Relate their signed sum to the Euler number.

*Hint.* Sum local degrees.

For a closed oriented base of the same dimension as the rank, sum the local degrees and compare them with evaluation on its fundamental class. □

*Solution.* The local signs are local degrees. Their sum equals the Euler number:

$$\sum_{x:s(x)=0} \operatorname{ind}_x(s) = \langle e(E), [B] \rangle.$$

For $E = TB$ this is Poincaré–Hopf. □

**Exercise 32.39.** Explain the mod-2 version without orientation.

*Hint.* Ignore signs modulo two.

Count transverse zeros modulo two and use the mod-two fundamental class; no sign comparison of the base and fiber is then needed. □

*Solution.* With $\mathbb{Z}/2$ coefficients, the signs $+1$ and $-1$ coincide. Thus the zero locus is counted modulo two, and the mod-two Thom/Euler obstruction is defined without orienting the bundle. □

**Exercise 32.40.** Compare Euler classes before and after pullback to a point.

*Hint.* Pullback to the fiber over the chosen point.

Apply naturality to the inclusion of the chosen point and inspect its positive-degree cohomology; retaining the positive-rank assumption is essential. □

*Solution.* Pulling a positive-rank bundle back to a point gives a vector space over a point. Since $H^r(\{*\}) = 0$ for $r > 0$,

$$e(i^*E) = i^*e(E) = 0.$$

This is consistent with every vector space admitting nonzero vectors. □

**Exercise 32.41.** Analyze a product bundle over $S^2 \times S^2$.

*Hint.* Use Künneth and naturality.

Use the bundle chosen in the solution, $p_1^*TS^2 \oplus p_2^*TS^2$, not a trivial product bundle. Apply naturality and multiply the two pulled-back Euler classes. ☐

*Solution.* For example, on $S^2 \times S^2$ let

$$E = p_1^*TS^2 \oplus p_2^*TS^2.$$

Then

$$e(E) = p_1^*e(TS^2) \smile p_2^*e(TS^2),$$

and its evaluation on $[S^2 \times S^2]$ is $2 \cdot 2 = 4$. ☐

**Exercise 32.42.** Use Whitney-sum multiplicativity in a tangent-bundle example.

*Hint.* Split the tangent bundle if possible.

Use $T(M \times N) \cong p_M^*TM \oplus p_N^*TN$ with the product orientation; for Euler numbers assume both manifolds are closed and oriented. ☐

*Solution.* Since $T(M \times N) \cong p_M^*TM \oplus p_N^*TN$,

$$e(T(M \times N)) = p_M^*e(TM) \smile p_N^*e(TN).$$

Evaluating gives $\chi(M \times N) = \chi(M)\chi(N)$, illustrating Whitney-sum multiplicativity. ☐

**Exercise 32.43.** Explain why vanishing Euler class need not classify a bundle.

*Hint.* Remember that characteristic classes are generally not complete invariants.

Separate bundle classification from section existence: Euler data alone need not distinguish bundles, and a vanishing primary obstruction may leave higher section obstructions. ☐

*Solution.* The Euler class is only one characteristic class. In higher rank or on complicated bases, two nonisomorphic bundles can share the same Euler class but differ in other characteristic data such as Stiefel–Whitney or Pontryagin classes. Therefore $e(E) = 0$ alone is not a complete classification. ☐

**Exercise 32.44.** Describe the first obstruction to a section of $S(E)$.

*Hint.* Use obstruction theory on the sphere fiber.

For oriented rank $r \geq 2$, start with the $(r-2)$-connected sphere fiber and locate the first extension obstruction in degree $r$; higher-dimensional bases may have further obstructions. ☐

*Solution.* For an oriented rank-$r$ bundle, the fiber of $S(E)$ is $S^{r-1}$. The primary obstruction to extending a section lies in

$$H^r(B; \pi_{r-1}(S^{r-1})) \cong H^r(B; \mathbb{Z})$$

(with the appropriate orientation local system), and this obstruction is the Euler class. ☐

**Exercise 32.45.** Relate a transverse zero set to a Thom class.

*Hint.* Pull back the Thom class by the section.

Pull back the Thom class as a relative class supported near the section's zero set, then pass to ordinary base cohomology to obtain its Euler class. ☐

*Solution.* A transverse section $s$ meets the zero section cleanly. Pulling the normal Thom class through this intersection identifies the cohomology class detecting the zero set with $s^*u_E = e(E)$. Thus the Thom class localizes the zero locus. ☐

**Exercise 32.46.** Explain how the zero set becomes a dual homology class.

*Hint.* Apply Poincaré duality in the base.

On a closed oriented $n$-manifold, transversality identifies the normal bundle of the zero locus with $E|_Z$; cap the Euler class with $[B]$ to obtain its $(n-r)$-class. □

*Solution.* If $Z = s^{-1}(0)$ is transverse, then $Z$ has codimension $r$ and its normal bundle is naturally $E|_Z$. The Thom construction therefore yields

$$\mathrm{PD}[Z] = e(E).$$

□

**Exercise 32.47.** Summarize sphere bundle → section obstruction → Euler class → duality.

*Hint.* Track geometry and cohomology simultaneously.

Keep the logical arrows separate: a sphere-bundle section is equivalent to a nowhere-zero section and forces $e(E) = 0$; the reverse implication needs additional obstruction-theoretic hypotheses. □

*Solution.* The geometry and algebra line up as

$$\text{section of } S(E) \iff \text{nowhere-zero section of } E,$$

while failure of such a section is detected first by $e(E)$. For a transverse section, its zero set represents $\mathrm{PD}(e(E))$, leading directly to Poincaré duality. □

## 32.16 Supplementary worked examples

**Example 32.48** (Euler class of the tangent bundle of S2)**.** For the oriented tangent bundle $TS^2$,

$$\langle e(TS^2), [S^2] \rangle = \chi(S^2) = 2.$$

Therefore $e(TS^2) \neq 0$, so $TS^2$ admits no nowhere-zero section: the hairy-ball theorem follows immediately.

**Example 32.49** (Trivial bundle and a nowhere-zero section)**.** If $E = B \times \mathbb{R}^r$ has positive rank, the constant section $s(x) = (x, e_1)$ never vanishes. Consequently $e(E) = 0$. The vanishing is necessary for a nowhere-zero section, though in general it need not be sufficient without dimension or rank hypotheses.

**Example 32.50** (Sphere bundle as the boundary of the disk bundle)**.** Choosing a bundle metric gives the disk bundle $D(E)$ and sphere bundle $S(E) = \partial D(E)$. The pair $(D(E), S(E))$ carries the Thom class, while the sphere bundle records directions of nonzero vectors; a nowhere-zero section of $E$ is equivalent after normalization to a section of $S(E)$.

## 32.17 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 32.51** (Sphere fiber)**.** What is the fiber of the sphere bundle of a rank-$r$ real vector bundle?

*Hint.* Normalize nonzero vectors. □

*Solution.* The fiber is $S^{r-1}$. □

**Exercise 32.52** (Normalize section)**.** How does a nowhere-zero section of $E$ produce a section of $S(E)$ after choosing a metric?

*Hint.* Divide by its norm. □

*Solution.* $x \mapsto s(x)/\|s(x)\|$ is a sphere-bundle section. □

**Exercise 32.53** (Euler degree)**.** In what degree does $e(E)$ lie for an oriented rank-$r$ real bundle?

*Hint.* It is the zero-section pullback of the degree-$r$ Thom class. □

*Solution.* $e(E) \in H^r(B; \mathbb{Z})$. □

**Exercise 32.54** (Trivial Euler class)**.** Compute $e(B \times \mathbb{R}^r)$ for $r > 0$.

*Hint.* Use a constant nonzero section. □

*Solution.* It is zero. □

**Exercise 32.55** (TS2 Euler number)**.** Compute $\langle e(TS^2), [S^2] \rangle$.

*Hint.* Use the Euler characteristic. □

*Solution.* It equals 2. □

## Proofs

**Exercise 32.56** (Section forces vanishing)**.** Prove a nowhere-zero section forces $e(E) = 0$ for an oriented bundle.

*Hint.* Move the zero section along the nonzero section or use disjointness. □

*Solution.* The section is homotopic through bundle sections to a section disjoint from the zero section, so its intersection with the zero section vanishes; equivalently the pullback of the Thom class along a nonzero section is zero, hence $e(E) = 0$. □

**Exercise 32.57** (Euler naturality)**.** Show $e(f^*E) = f^*e(E)$ for orientation-preserving pullback.

*Hint.* Use naturality of the Thom class and zero section. □

*Solution.* Pullback commutes with the Thom class and with the zero-section square, giving the stated equality. □

**Exercise 32.58** (Euler number and zeros)**.** Explain why a transverse section of an oriented rank-$n$ bundle over a closed oriented $n$-manifold has signed zero count equal to $\langle e(E), [M]\rangle$.

*Hint.* Each zero is an oriented local intersection with the zero section. □

*Solution.* The pullback Thom class localizes at transverse zeros; summing local intersection signs evaluates the Euler class on the fundamental class. □

**Exercise 32.59** (Hairy ball)**.** Deduce that $S^2$ has no nowhere-zero tangent vector field.

*Hint.* Use the nonzero Euler number. □

*Solution.* A nowhere-zero tangent field would force $e(TS^2) = 0$, contradicting evaluation 2. □

## Counterexamples and hypothesis tests

**Exercise 32.60** (Euler zero implies section always)**.** Test: $e(E) = 0$ always guarantees a nowhere-zero section.

*Hint.* Euler class is only the primary obstruction in general. □

*Solution.* False without further hypotheses; higher obstructions may remain. □

**Exercise 32.61** (Sphere bundle always trivial)**.** Test: the unit sphere bundle $S(E)$ is a product $B \times S^{r-1}$.

*Hint.* Nontrivial vector bundles have nontrivial sphere bundles. □

*Solution.* False. For example, sphere bundles associated to nontrivial tangent bundles need not be products. □

**Exercise 32.62** (Odd sphere tangent Euler)**.** Test: $\langle e(TS^{2k+1}), [S^{2k+1}]\rangle = 2$.

*Hint.* Use $\chi(S^{2k+1}) = 0$. □

*Solution.* False; the Euler number is zero for odd-dimensional spheres. □

### Applications and investigations

**Exercise 32.63** (Vector-field diagnostics)**.** How can computing an Euler class guide a search for global nonvanishing fields?

*Hint.* Nonzero Euler class is a certificate of impossibility. □

*Solution.* If the Euler class is nonzero, no algorithm can produce a global nowhere-zero section without violating continuity; vanishing removes this primary obstruction. □

**Exercise 32.64** (Sphere-bundle obstruction)**.** Why can the existence of a section of $S(E)$ be formulated as an obstruction problem?

*Hint.* Build the section over skeleta. □

*Solution.* One extends a partial sphere-bundle section skeleton by skeleton; obstruction classes measure failures to extend, with the Euler class appearing as the primary obstruction in the relevant rank. □

### Challenge problems

**Exercise 32.65** (Tangent surface)**.** For a closed oriented surface $\Sigma_g$, compute the Euler number of $T\Sigma_g$.

*Hint.* Use Gauss–Bonnet/Poincare–Hopf. □

*Solution.* $\langle e(T\Sigma_g), [\Sigma_g]\rangle = \chi(\Sigma_g) = 2 - 2g$. □

**Exercise 32.66** (Euler synthesis)**.** Connect sphere-bundle sections, Thom classes, zeros of sections, and Euler characteristic.

*Hint.* Normalize nonzero sections and compare with the zero section. □

*Solution.* A nowhere-zero section is a sphere-bundle section; the Thom class detects intersection with the zero section; its pullback is the Euler class; for tangent bundles its evaluation equals the Euler characteristic, converting topology into a zero-count obstruction. □

# Chapter 33

# Poincaré Duality

Poincaré duality is the organizing theorem that turns cohomology classes into geometric homology classes on an oriented manifold. The Thom-class viewpoint from VIII/31 supplies the local normal model; the Euler-class viewpoint from VIII/32 supplies a first family of dual cycles.

## 33.1 Orientation and local homology

For an $n$-manifold $M$ and $x \in M$,

$$H_n(M, M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}.$$

An orientation chooses compatible generators of these local groups.

## 33.2 The fundamental class

**Definition 33.1** (Fundamental class)**.** For a connected closed oriented $n$-manifold, the fundamental class

$$[M] \in H_n(M; \mathbb{Z})$$

is the unique class restricting to the chosen local orientation generator at every point.

## 33.3 Cap product

Cap product combines cohomology and homology:

$$\frown : H^k(M; R) \times H_n(M; R) \longrightarrow H_{n-k}(M; R).$$

## 33.4 Poincaré duality

**Theorem 33.2** (Poincaré duality)**.** *If $M$ is a closed oriented $n$-manifold, then*

$$D_M : H^k(M; R) \longrightarrow H_{n-k}(M; R), \qquad D_M(\alpha) = \alpha \frown [M]$$

*is an isomorphism.*

## 33.5 Betti-number symmetry

Over a field,

$$b_k(M) = b_{n-k}(M).$$

This gives immediate global restrictions, including vanishing Euler characteristic in odd dimension.

## 33.6 The complementary-degree pairing

Duality can be written as the nondegenerate pairing

$$H^k(M; R) \times H^{n-k}(M; R) \longrightarrow R, \qquad (\alpha, \beta) \longmapsto \langle \alpha \smile \beta, [M] \rangle.$$

## 33.7 Spheres and surfaces

For $S^n$, degree zero and degree $n$ are dual. For a closed oriented surface of genus $g$, the $2g$ degree-one generators pair through the symplectic intersection form.

## 33.8 The torus

Choose $a, b \in H^1(T^2; \mathbb{Z})$ so that

$$a \smile b = [T^2]^*, \qquad b \smile a = -[T^2]^*.$$

The algebra records the signed intersection of the two standard generating circles.

## 33.9 Complex projective space

For $\mathbb{CP}^n$,

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[h]/(h^{n+1}), \qquad |h| = 2,$$

and the top class $h^n$ evaluates to 1 after choosing the complex orientation.

## 33.10 Geometric Poincaré duals

If $N^{n-r} \hookrightarrow M^n$ is a closed oriented submanifold with oriented normal bundle, its normal Thom class extends to

$$\mathrm{PD}[N] \in H^r(M).$$

## 33.11 Transverse intersection and cup product

If dual submanifolds meet transversely, then cup product corresponds to intersection:

$$\mathrm{PD}[A] \smile \mathrm{PD}[B] = \mathrm{PD}[A \pitchfork B]$$

with the usual orientation conventions.

## 33.12 Intersection numbers

For complementary dimensions, the signed number of transverse intersection points is

$$A \cdot B = \langle \mathrm{PD}[A] \smile \mathrm{PD}[B], [M] \rangle.$$

## 33.13 Euler classes revisited

If a transverse section of an oriented rank-$r$ bundle has zero set $Z$, then

$$\mathrm{PD}[Z] = e(E).$$

Thus Euler classes are geometric duals of zero loci.

## 33.14 Manifolds with boundary

**Theorem 33.3** (Poincaré–Lefschetz duality)**.** *For a compact oriented $n$-manifold with boundary,*

$$H^k(M;R) \cong H_{n-k}(M,\partial M;R), \qquad H^k(M,\partial M;R) \cong H_{n-k}(M;R).$$

## 33.15 Nonorientable manifolds

Integral duality requires the orientation local system. With $\mathbb{Z}/2$ coefficients the sign obstruction disappears and a mod-2 duality statement holds without orientability.

## 33.16 Bridge to intersection forms

In dimension $4m$, the middle-degree pairing

$$H^{2m}(M) \times H^{2m}(M) \to R$$

is a bilinear form obtained by cup product and evaluation on $[M]$. VIII/34 will study this intersection form as an invariant in its own right.

## 33.17 Solved dossiers

**Problem 33.4.** State Poincaré duality for a closed oriented $n$-manifold.

*Solution.* Cap product with $[M]$ gives $H^k(M;R) \cong H_{n-k}(M;R)$. □

**Problem 33.5.** Define the duality map.

*Solution.* $D_M(\alpha) = \alpha \frown [M]$. □

**Problem 33.6.** What is the fundamental class?

*Solution.* The homology class whose local image is the chosen orientation generator at every point. □

**Problem 33.7.** What symmetry follows for Betti numbers over a field?

*Solution.* $b_k = b_{n-k}$. □

**Problem 33.8.** Write the degree-pairing form.

*Solution.* $(\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle$. □

**Problem 33.9.** Why is the pairing nondegenerate?

*Solution.* Poincaré duality identifies one factor with the dual of complementary homology/cohomology. □

**Problem 33.10.** Compute the dual of the point class on a connected oriented surface.

*Solution.* It corresponds to the top-degree cohomology generator. □

**Problem 33.11.** Describe duality on $S^n$.

*Solution.* $H^0$ pairs with $H^n$ and vice versa; all intermediate groups vanish. □

**Problem 33.12.** Describe the first-cohomology pairing on $T^2$.

*Solution.* Two degree-one generators have nonzero cup product equal to the orientation class. □

**Problem 33.13.** How does a submanifold define a dual class?

*Solution.* Use the Thom class of its oriented normal bundle and extend it to ambient cohomology. □

**Problem 33.14.** What does cup product represent geometrically for transverse dual submanifolds?

*Solution.* Their intersection. □

**Problem 33.15.** What is the intersection number of complementary oriented transverse submanifolds?

*Solution.* The evaluation of the cup product of their dual classes on $[M]$. □

**Problem 33.16.** How does boundary change the theorem?

*Solution.* Poincaré–Lefschetz duality relates absolute cohomology to relative homology and relative cohomology to absolute homology. □

**Problem 33.17.** State one Poincaré–Lefschetz isomorphism.

*Solution.* $H^k(M) \cong H_{n-k}(M, \partial M)$ for compact oriented $M$. □

**Problem 33.18.** What changes for nonorientable manifolds?

*Solution.* Use the orientation local system, or mod-2 coefficients when appropriate. □

**Problem 33.19.** How is the Euler characteristic constrained in odd dimension?

*Solution.* Betti numbers pair in complementary degrees, giving zero Euler characteristic for closed oriented odd-dimensional manifolds. □

**Problem 33.20.** What is the dual of a codimension-$r$ transverse zero locus?

*Solution.* The Euler class of the rank-$r$ bundle whose section cuts it out. □

**Problem 33.21.** How is the Thom class used in the proof philosophy?

*Solution.* Local duality in normal disks is transported globally through tubular neighborhoods and excision. □

**Problem 33.22.** Why does VIII/33 lead directly to intersection forms?

*Solution.* Middle-degree cup products evaluated on $[M]$ become bilinear forms. □

**Problem 33.23.** What is the key synthesis of VIII/31–VIII/33?

*Solution.* Thom classes create dual cohomology classes, Euler classes arise from zero sections, and cap product with the fundamental class identifies cohomology with geometric homology. □

## 33.18 Exercises, Hints, and Solutions

**Exercise 33.24.** Compute Poincaré duality for $S^n$.

*Hint.* Only degrees 0 and $n$ survive.

For $n \geq 1$, cap the unit with the fundamental class and cap the normalized top class with it; distinguish the resulting top cycle and point class. ☐

*Solution.* For a closed oriented $n$-sphere,

$$H^0(S^n) \cong H_n(S^n) \cong \mathbb{Z}, \qquad H^n(S^n) \cong H_0(S^n) \cong \mathbb{Z},$$

and all other groups vanish. Cap product with $[S^n]$ gives these two isomorphisms. ☐

**Exercise 33.25.** Compute the Betti-number symmetry for a genus-$g$ surface.

*Hint.* Use $b_0 = b_2 = 1$ and $b_1 = 2g$.

Pair degree $k$ with degree $2 - k$; the middle degree is paired with itself rather than with a different Betti number. ☐

*Solution.* For a genus-$g$ closed oriented surface,

$$(b_0, b_1, b_2) = (1, 2g, 1).$$

Poincaré duality gives the symmetry $b_0 = b_2$, while the middle group is self-dual. ☐

**Exercise 33.26.** Write the cup-product pairing matrix for $T^2$ in a standard basis.

*Hint.* Choose generators $a, b$ with $a \smile b = [T^2]^*$.

Normalize $\langle a \smile b, [T^2]\rangle = 1$, then use graded commutativity and vanishing degree-one squares to fill the other three matrix entries. ☐

*Solution.* Choose $a, b \in H^1(T^2; \mathbb{Z})$ with $\langle a \smile b, [T^2]\rangle = 1$. Since degree-one cup product is skew-commutative,

$$(\langle x_i \smile x_j, [T^2]\rangle) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

☐

**Exercise 33.27.** Identify the Poincaré dual of an embedded oriented circle in $T^2$.

*Hint.* Use a transverse complementary circle.

Specify the orientation and intersection-order convention. Determine the dual functional by its intersections with both basis circles, rather than identifying it with the same-direction generator by name. ☐

*Solution.* Let $A = S^1 \times \{*\} \subset T^2$. Its Poincaré dual is the degree-one class that evaluates by intersection with $A$; in the standard basis it is the cohomology class dual to the transverse factor circle. ☐

**Exercise 33.28.** Intersect two standard generating circles on the torus.

*Hint.* There is one signed intersection.

At the common point, compare the ordered tangent vectors of the two circles with the product orientation of the torus. ☐

*Solution.* The two standard circles $S^1 \times \{*\}$ and $\{*\} \times S^1$ meet in exactly one point. With the product orientation and compatible orientations of the circles, the local intersection sign is $+1$. ☐

**Exercise 33.29.** Relate their signed intersection to cup product.

*Hint.* Evaluate the cup product on $[T^2]$.

Use the same order and orientation convention for the two dual classes as for geometric intersection; reversing the order changes the sign in degree one. ☐

*Solution.* If $\alpha, \beta$ are the dual cohomology classes of the two standard circles, then

$$A \cdot B = \langle \alpha \smile \beta, [T^2] \rangle = 1.$$

Thus the signed geometric intersection is exactly the cup-product pairing. ☐

**Exercise 33.30.** Compute the duality groups of $\mathbb{CP}^2$.

*Hint.* Use one generator in degrees $0, 2, 4$.

Use the real dimension four of $\mathbb{CP}^2$; the degree-two class is middle-dimensional, while degrees zero and four are complementary. ☐

*Solution.* For $\mathbb{CP}^2$,

$$H_k(\mathbb{CP}^2; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \\ 0, & \text{otherwise}, \end{cases}$$

and Poincaré duality pairs degrees $0 \leftrightarrow 4$ and $2 \leftrightarrow 2$. ☐

**Exercise 33.31.** Evaluate the square of the degree-two generator on $[\mathbb{CP}^2]$.

*Hint.* The square evaluates to 1 in the complex orientation.

Represent the positive generator by a complex projective line and intersect two generic such lines; the complex orientation fixes the local sign. ☐

*Solution.* If $h \in H^2(\mathbb{CP}^2; \mathbb{Z})$ is the positive generator, then $h^2$ is the orientation class and

$$\langle h^2, [\mathbb{CP}^2] \rangle = 1.$$

☐

**Exercise 33.32.** Show that closed oriented odd-dimensional manifolds have Euler characteristic zero.

*Hint.* Pair complementary Betti numbers.

For odd $n$, the paired exponents $k$ and $n-k$ have opposite parity, and there is no unpaired middle degree. ☐

*Solution.* For a closed oriented $n = 2m+1$ manifold, $b_k = b_{n-k}$. The paired terms in

$$\chi(M) = \sum_k (-1)^k b_k$$

have opposite signs because $n$ is odd, so they cancel and $\chi(M) = 0$. ☐

**Exercise 33.33.** Describe the local orientation classes defining $[M]$.

*Hint.* Use $H_n(M, M \setminus \{x\})$.

Use a coordinate ball and excision to identify each local group with $H_n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$; the global ordinary class requires the closed oriented setting. □

*Solution.* An orientation is a compatible family of generators of

$$H_n(M, M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}.$$

Compactness permits these compatible local generators to assemble into the unique global class $[M] \in H_n(M; \mathbb{Z})$ restricting to each chosen generator. □

**Exercise 33.34.** Explain the cap-product degree shift.

*Hint.* Cap lowers degree by $k$.

A degree-$k$ cochain consumes $k$ dimensions of a chain; check that capping with the degree-$n$ fundamental class lands in degree $n - k$. □

*Solution.* If $\alpha \in H^k(M)$ and $c \in H_n(M)$, then

$$\alpha \frown c \in H_{n-k}(M).$$

Thus cap product lowers homological degree by exactly the cohomological degree. □

**Exercise 33.35.** Verify naturality of evaluation under an orientation-preserving homeomorphism.

*Hint.* Track the fundamental class.

Use the evaluation identity $\langle f^*\alpha, c\rangle = \langle \alpha, f_*c\rangle$ and then insert the orientation-preserving image of the fundamental class. □

*Solution.* If $f : M \to N$ is orientation preserving, then $f_*[M] = [N]$. Naturality of cup/cap evaluation gives

$$\langle f^*\alpha, [M]\rangle = \langle \alpha, f_*[M]\rangle = \langle \alpha, [N]\rangle.$$

□

**Exercise 33.36.** Use a tubular neighborhood to build the dual class of a submanifold.

*Hint.* Use the normal Thom class.

For a closed oriented codimension-$r$ submanifold, orient the normal bundle compatibly and pass from its Thom pair to $H^r(M, M \setminus N)$. □

*Solution.* For $i : N^{n-r} \hookrightarrow M^n$, choose an oriented tubular neighborhood $D(\nu)$. The Thom class $u_\nu \in H^r(D(\nu), S(\nu))$ extends by excision to a class in $H^r(M, M \setminus N)$ and then to $H^r(M)$. This is $\mathrm{PD}[N]$. □

**Exercise 33.37.** Relate that construction to the Thom collapse.

*Hint.* Collapse the complement of the tubular neighborhood.

View the Thom class as reduced cohomology of the Thom space; pull it back through the collapse map to compare with the tubular-neighborhood construction. □

*Solution.* The Thom collapse

$$M \to \mathrm{Th}(\nu)$$

pulls the Thom class back to the ambient dual class of $N$. Thus the collapse-map construction and the tubular-neighborhood construction give the same Poincaré dual. □

**Exercise 33.38.** Show that a transverse zero set represents the dual of an Euler class.

*Hint.* Use the section's zero locus.

The derivative of a transverse section identifies normal directions to its zero set with the bundle fibers; transfer the corresponding Thom generator and forget the relative support. ☐

*Solution.* For a transverse section of an oriented rank-$r$ bundle, the zero set $Z$ has normal bundle $E|_Z$. Its normal Thom class is the restriction of the bundle Thom class, so

$$\mathrm{PD}[Z] = e(E).$$

☐

**Exercise 33.39.** State Poincaré–Lefschetz duality for a compact manifold with boundary.

*Hint.* Replace absolute homology by relative homology.

Use the relative fundamental class $[M, \partial M]$; write both absolute-to-relative and relative-to-absolute duality maps to avoid dropping the boundary term. ☐

*Solution.* For a compact oriented $n$-manifold with boundary,

$$H^k(M; R) \cong H_{n-k}(M, \partial M; R), \qquad H^k(M, \partial M; R) \cong H_{n-k}(M; R).$$

The fundamental class is relative: $[M, \partial M] \in H_n(M, \partial M)$. ☐

**Exercise 33.40.** Compute it for an interval.

*Hint.* Check endpoints as boundary.

The oriented interval has boundary equal to terminal minus initial endpoint, so it defines a relative one-cycle but not an absolute one-cycle. ☐

*Solution.* For $I = [0, 1]$,

$$H^0(I) \cong \mathbb{Z} \cong H_1(I, \partial I),$$

while $H^1(I) = 0 \cong H_0(I, \partial I)$. The relative fundamental class is the oriented interval. ☐

**Exercise 33.41.** Compute it for a disk.

*Hint.* Use $(D^n, S^{n-1})$.

The relative top class of $(D^n, S^{n-1})$ pairs with the ordinary degree-zero unit; absolute top homology of the disk is not the correct target. ☐

*Solution.* For $D^n$,

$$H^0(D^n) \cong \mathbb{Z} \cong H_n(D^n, S^{n-1}),$$

and all other corresponding groups vanish. This is the basic local model for Poincaré–Lefschetz duality. ☐

**Exercise 33.42.** Explain why ordinary integral duality fails without orientation.

*Hint.* The local signs cannot be chosen coherently.

Transport a local orientation generator around an orientation-reversing loop; its sign change prevents a globally compatible ordinary integral fundamental class. ☐

*Solution.* On a nonorientable manifold the local orientation generators cannot be chosen coherently, so there is no global integral fundamental class in ordinary coefficients. Consequently the ordinary integral cap-product isomorphism fails. ☐

**Exercise 33.43.** Repair the statement using mod-2 coefficients.

*Hint.* Signs disappear modulo two.

Use mod-two coefficients in the closed-manifold statement so that all local orientation generators agree without sign choices. □

*Solution.* With $\mathbb{Z}/2$ coefficients, the sign ambiguity disappears. Every manifold has a mod-two fundamental class, and

$$H^k(M;\mathbb{Z}/2) \cong H_{n-k}(M;\mathbb{Z}/2)$$

for closed manifolds. □

**Exercise 33.44.** Describe the complementary-degree pairing in dimension four.

*Hint.* Use degree two against degree two.

Use degree two against degree two and evaluate in degree four; over a field the pairing is perfect, while integral torsion requires passing to the free lattice. □

*Solution.* For a closed oriented four-manifold, middle-degree duality is

$$H^2(M;R) \times H^2(M;R) \to R, \qquad (\alpha,\beta) \mapsto \langle \alpha \smile \beta, [M] \rangle.$$

Because degree two is even, this pairing is symmetric. □

**Exercise 33.45.** Identify the middle-dimensional bilinear form that will appear in VIII/34.

*Hint.* Evaluate cup products on $[M]$.

Choose an integral basis for middle cohomology modulo torsion and evaluate pairwise cup products on the fundamental class to obtain the matrix. □

*Solution.* The bilinear form in the previous exercise is the intersection form

$$Q_M(\alpha,\beta) = \langle \alpha \smile \beta, [M] \rangle$$

on middle cohomology (or, integrally, on its free lattice). This is the invariant studied in VIII/34. □

**Exercise 33.46.** Explain how transverse intersections produce its coefficients.

*Hint.* Represent classes by transverse surfaces.

For smooth oriented four-manifolds, use transverse representatives of the dual two-dimensional classes and determine each coefficient by signed local intersections. □

*Solution.* Represent $\mathrm{PD}^{-1}(\alpha)$ and $\mathrm{PD}^{-1}(\beta)$ by transverse oriented surfaces. Each intersection point contributes a sign, and the signed total equals

$$\langle \alpha \smile \beta, [M] \rangle.$$

□

**Exercise 33.47.** Summarize fundamental class → cap product → duality → intersection form.

*Hint.* Track the middle degree.

Start with $[M]$, apply cap product to convert degree-$k$ classes into dimension-$(n-k)$ cycles, and specialize complementary cup evaluation to the middle dimension. □

*Solution.* The fundamental class turns cohomology into complementary-dimensional homology by cap product. Geometric representatives of those dual homology classes intersect according to cup product. In the middle dimension this complementary-degree pairing becomes the intersection form. □

## 33.19 Supplementary worked examples

**Example 33.48** (Poincare duality on a closed oriented surface)**.** For a closed oriented surface $\Sigma_g$,

$$H^1(\Sigma_g;\mathbb{Z}) \xrightarrow{\cap[\Sigma_g]} H_1(\Sigma_g;\mathbb{Z})$$

is an isomorphism. The cup-product pairing on $H^1$ therefore corresponds to the algebraic intersection pairing on one-cycles.

**Example 33.49** (Duality on a sphere)**.** For $S^n$, cap product with the fundamental class identifies $H^0(S^n)$ with $H_n(S^n)$ and $H^n(S^n)$ with $H_0(S^n)$. The two nonzero cohomology groups are exchanged with complementary homological degrees.

**Example 33.50** (Submanifold dual classes)**.** An oriented codimension-$k$ submanifold $N \subset M^n$ determines a class $\mathrm{PD}[N] \in H^k(M)$. For a complementary cycle $Z$, evaluating $\mathrm{PD}[N]$ on $[Z]$ gives the algebraic intersection number $N \cdot Z$ when transversality holds.

## 33.20 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 33.51** (Degree reversal)**.** For a closed oriented $n$-manifold, which homology group is dual to $H^k(M)$?

*Hint.* Cap with $[M]$ lowers degree by $k$. ☐

*Solution.* $H_{n-k}(M)$. ☐

**Exercise 33.52** (Fundamental class)**.** Where does the integral fundamental class of a connected closed oriented $n$-manifold live?

*Hint.* Top homology. ☐

*Solution.* $[M] \in H_n(M;\mathbb{Z}) \cong \mathbb{Z}$. ☐

**Exercise 33.53** (Sphere duality)**.** What is the Poincare dual of the point class in $H_0(S^n)$?

*Hint.* Complementary degree is $n$. ☐

*Solution.* It is a generator of $H^n(S^n)$, up to orientation sign. ☐

**Exercise 33.54** (Dual of whole manifold)**.** What cohomology class is dual to $[M]$ itself?

*Hint.* Complementary degree is zero. ☐

*Solution.* The unit $1 \in H^0(M)$. ☐

**Exercise 33.55** (Betti symmetry)**.** What equality of Betti numbers follows over a field?

*Hint.* Dual groups have complementary degrees. ☐

*Solution.* $b_k(M) = b_{n-k}(M)$. ☐

### Proofs

**Exercise 33.56** (Cap-product isomorphism)**.** State Poincare duality using cap product.

*Hint.* Use the fundamental class. □

*Solution.* $\alpha \mapsto \alpha \cap [M]$ is an isomorphism $H^k(M; R) \to H_{n-k}(M; R)$ under the standard orientation/coefficient hypotheses. □

**Exercise 33.57** (Intersection from cup product)**.** Prove the intersection pairing of complementary cycles can be computed by cup product of their dual classes.

*Hint.* Evaluate the product on $[M]$. □

*Solution.* If $a, b$ are complementary, then $a \cdot b = \langle \mathrm{PD}(a) \smile \mathrm{PD}(b), [M] \rangle$, with the standard sign conventions. □

**Exercise 33.58** (Duality implies top homology)**.** Show a connected closed oriented manifold has $H_n(M; \mathbb{Z}) \cong \mathbb{Z}$.

*Hint.* Apply duality to $H^0(M) \cong \mathbb{Z}$. □

*Solution.* Poincare duality identifies $H^0(M)$ with $H_n(M)$, giving $\mathbb{Z}$. □

**Exercise 33.59** (Cohomology-homology rank symmetry)**.** Explain why duality plus field coefficients forces the Poincare polynomial to be palindromic.

*Hint.* Use $b_k = b_{n-k}$. □

*Solution.* The coefficient of $t^k$ equals that of $t^{n-k}$, so $P_M(t) = t^n P_M(t^{-1})$. □

### Counterexamples and hypothesis tests

**Exercise 33.60** (Boundary harmless)**.** Test: ordinary Poincare duality in the same form holds for compact manifolds with boundary.

*Hint.* Relative groups enter. □

*Solution.* False. One uses Poincare–Lefschetz duality, relating absolute cohomology to relative homology or vice versa. □

**Exercise 33.61** (Orientation irrelevant over Z)**.** Test: every closed manifold has integral Poincare duality with untwisted coefficients.

*Hint.* Nonorientable manifolds lack an integral fundamental class. □

*Solution.* False. One needs orientability or appropriate local coefficients; mod-2 duality remains available. □

**Exercise 33.62** (Duality says groups equal same degree)**.** Test: Poincare duality identifies $H^k(M)$ with $H_k(M)$.

*Hint.* Degrees are complementary. □

*Solution.* False in general; the homology degree is $n - k$. □

## Applications and investigations

**Exercise 33.63** (Geometric query)**.** How can a codimension-$k$ feature be stored cohomologically for repeated intersection queries?

*Hint.* Use its Poincare-dual class. □

*Solution.* Store $\mathrm{PD}[N] \in H^k(M)$; pairing it with varying complementary cycles returns their algebraic intersections with $N$. □

**Exercise 33.64** (Sanity check for manifolds)**.** How can Betti-number symmetry be used to detect an error in a computed homology table of a closed oriented manifold?

*Hint.* Compare complementary degrees. □

*Solution.* If $b_k \neq b_{n-k}$, either the computation, manifold assumptions, or orientation/coefficient assumptions are wrong. □

## Challenge problems

**Exercise 33.65** (Surface pairing)**.** For $T^2$, choose dual degree-one classes $a, b$. Compute $\langle a \smile b, [T^2]\rangle$ after compatible orientation choices.

*Hint.* Use the standard meridian-longitude basis. □

*Solution.* It is 1; reversing the order gives $-1$ over $\mathbb{Z}$. □

**Exercise 33.66** (Duality synthesis)**.** Explain how Thom classes lead to Poincare duality for embedded submanifolds.

*Hint.* Use the normal bundle and a tubular neighborhood. □

*Solution.* The normal Thom class extends to the ambient manifold and becomes the cohomological dual of the submanifold; cap and cup products then convert intersections of geometric cycles into algebraic pairings. □

# Chapter 34

# Intersection Forms

Poincaré duality becomes especially rigid in the middle dimension. On a closed oriented manifold, cup product followed by evaluation on the fundamental class produces a bilinear pairing whose algebra records geometric intersection. In dimension four this pairing is one of the basic topological invariants.

## 34.1 The middle-dimensional pairing

Let $M$ be a closed oriented $2n$-manifold. Define

$$Q_M(\alpha,\beta) = \langle \alpha \smile \beta, [M]\rangle, \qquad \alpha,\beta \in H^n(M;\mathbb{Z})/\mathrm{tors}.$$

**Definition 34.1** (Intersection form)**.** The integral middle-dimensional intersection form is the bilinear form

$$Q_M : \left(H^n(M;\mathbb{Z})/\mathrm{tors}\right)^2 \longrightarrow \mathbb{Z}$$

given by cup product and evaluation on the fundamental class.

## 34.2 Symmetry and graded symmetry

Graded commutativity gives

$$Q_M(\beta,\alpha) = (-1)^n Q_M(\alpha,\beta).$$

Thus the middle pairing is symmetric when $n$ is even and skew-symmetric when $n$ is odd.

## 34.3 Unimodularity

**Theorem 34.2** (Unimodularity)**.** *For a closed oriented manifold, the integral intersection form on the free middle cohomology is unimodular.*

This is the lattice form of integral Poincaré duality.

## 34.4 Geometric intersections

If $\alpha$ and $\beta$ are Poincaré dual to complementary transverse cycles, then

$$Q_M(\alpha,\beta)$$

is their signed intersection number.

## 34.5 Self-intersection

A middle-dimensional submanifold can be pushed slightly in a normal direction. Its signed intersection with the push-off is its self-intersection number. For an oriented normal bundle, this number is the Euler number of the normal bundle.

## 34.6 Four-manifolds

For a closed oriented 4-manifold,

$$Q_M : H^2(M;\mathbb{Z})_{\text{free}} \times H^2(M;\mathbb{Z})_{\text{free}} \to \mathbb{Z}$$

is symmetric.

## 34.7 Positive and negative directions

After tensoring with $\mathbb{R}$, a symmetric form decomposes into positive and negative directions. Write

$$b_2^+(M), \qquad b_2^-(M)$$

for their dimensions.

**Definition 34.3** (Signature)**.**

$$\sigma(M) = b_2^+(M) - b_2^-(M).$$

## 34.8 Definite and indefinite forms

The form is positive definite if every nonzero vector has positive square, negative definite if every nonzero vector has negative square, and indefinite if both signs occur.

## 34.9 Even and odd forms

An integral symmetric form is even when

$$Q(x,x) \in 2\mathbb{Z}$$

for every integral vector $x$; otherwise it is odd.

## 34.10 The four-sphere

Since

$$H^2(S^4;\mathbb{Z}) = 0,$$

the intersection form of $S^4$ has rank zero.

## 34.11 Complex projective plane

Let $h \in H^2(\mathbb{CP}^2;\mathbb{Z})$ be the standard generator with

$$\langle h^2, [\mathbb{CP}^2] \rangle = 1.$$

Then

$$Q_{\mathbb{CP}^2} = [1].$$

It is positive definite, odd, unimodular, and has signature 1.

## 34.12 Orientation reversal

Changing the orientation changes the fundamental class by a sign:

$$Q_{-M} = -Q_M.$$

Thus

$$Q_{-\mathbb{CP}^2} = [-1].$$

## 34.13 The product $S^2 \times S^2$

Let $a, b \in H^2(S^2 \times S^2; \mathbb{Z})$ be pulled back from the two factors. Then

$$a^2 = b^2 = 0, \qquad \langle a \smile b, [S^2 \times S^2] \rangle = 1.$$

Hence

$$Q_{S^2 \times S^2} = H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The hyperbolic form $H$ is even, indefinite, unimodular, and has signature zero.

## 34.14 Connected sums

For oriented closed 4-manifolds,

$$Q_{M \# N} \cong Q_M \oplus Q_N.$$

Consequently

$$Q_{\mathbb{CP}^2 \# (-\mathbb{CP}^2)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

## 34.15 Integral change of basis

If $A$ represents $Q_M$ and $P \in GL_r(\mathbb{Z})$ changes the integral basis, then

$$A \longmapsto P^T A P.$$

Thus rank, unimodularity, parity, definiteness, and signature are invariants of the integral congruence class.

## 34.16 Intersection matrices from geometry

Choosing geometric representatives for a basis of middle homology turns the form into an intersection matrix. Diagonal entries are self-intersections; off-diagonal entries count pairwise signed intersections.

## 34.17 Bridge to Lefschetz theory

For a self-map $f : M \to M$, fixed points are intersections of the graph $\Gamma_f \subset M \times M$ with the diagonal $\Delta \subset M \times M$. Lefschetz theory converts that global intersection number into a trace on homology.

## 34.18 Solved dossiers

**Problem 34.4.** Define the middle-dimensional intersection form of a closed oriented $2n$-manifold.

*Solution.* On the free part, $Q_M(\alpha,\beta) = \langle \alpha \smile \beta, [M]\rangle$ for $\alpha,\beta \in H^n(M;\mathbb{Z})/\text{tors}$. □

**Problem 34.5.** How does graded commutativity control the symmetry of $Q_M$?

*Solution.* $Q_M(\beta,\alpha) = (-1)^n Q_M(\alpha,\beta)$; it is symmetric for even $n$ and skew-symmetric for odd $n$. □

**Problem 34.6.** Why is the integral form unimodular on the free part?

*Solution.* Integral Poincaré duality identifies the free middle cohomology lattice with its integral dual, so the representing matrix has determinant $\pm 1$. □

**Problem 34.7.** What is the intersection form of $S^4$?

*Solution.* $H^2(S^4) = 0$, so the form has rank zero. □

**Problem 34.8.** Compute the form of $\mathbb{CP}^2$ with its complex orientation.

*Solution.* If $h \in H^2(\mathbb{CP}^2)$ and $\langle h^2, [\mathbb{CP}^2]\rangle = 1$, then the matrix is $[1]$. □

**Problem 34.9.** What does orientation reversal do to the form?

*Solution.* $[-M] = -[M]$, hence $Q_{-M} = -Q_M$. □

**Problem 34.10.** Compute the form of $S^2 \times S^2$.

*Solution.* With $a,b$ dual to the two factors, $a^2 = b^2 = 0$ and $ab$ evaluates to 1, so the matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. □

**Problem 34.11.** What is the signature of the hyperbolic form?

*Solution.* Its eigenvalues are 1 and $-1$, so the signature is 0. □

**Problem 34.12.** Define positive and negative indices in dimension four.

*Solution.* After extending $Q_M$ to $\mathbb{R}$, $b_2^+$ and $b_2^-$ are the dimensions of maximal positive- and negative-definite subspaces. □

**Problem 34.13.** Define the signature.

*Solution.* $\sigma(M) = b_2^+ - b_2^-$. □

**Problem 34.14.** How does connected sum affect the form for oriented closed 4-manifolds?

*Solution.* On the middle cohomology free parts, $Q_{M\#N} \cong Q_M \oplus Q_N$. □

**Problem 34.15.** Compute $Q_{\mathbb{CP}^2\#(-\mathbb{CP}^2)}$.

*Solution.* It is $\operatorname{diag}(1,-1)$, hence indefinite of signature zero. □

**Problem 34.16.** What does even mean for an integral symmetric form?

*Solution.* $Q(x,x)$ is even for every integral lattice vector $x$. □

**Problem 34.17.** Is the $\mathbb{CP}^2$ form even or odd?

*Solution.* Odd, because the generator has self-intersection 1. □

**Problem 34.18.** Is the hyperbolic form even?

*Solution.* Yes: for $(x, y)$ its square is $2xy$, always even. □

**Problem 34.19.** Give the geometric meaning of $Q_M(\alpha, \beta)$ in dimension four.

*Solution.* Represent the dual classes by oriented embedded or immersed surfaces in general position; the form is their signed intersection number. □

**Problem 34.20.** What is self-intersection geometrically?

*Solution.* It is the intersection of a surface with a transverse push-off, equivalently the Euler number of its oriented normal bundle. □

**Problem 34.21.** Why does torsion not enter the ordinary integral matrix directly?

*Solution.* The cup-evaluation pairing to $\mathbb{Z}$ vanishes on torsion against the free fundamental class, so the lattice form is defined on the free quotient. □

**Problem 34.22.** How does a change of integral basis affect the matrix?

*Solution.* $A$ changes by integral congruence $A \mapsto P^T AP$ with $P \in GL_r(\mathbb{Z})$. □

**Problem 34.23.** Why is the chapter a direct continuation of Poincaré duality?

*Solution.* The intersection form is precisely the middle-degree Poincaré pairing promoted from a duality statement to a structured bilinear invariant. □

## 34.19 Exercises, Hints, and Solutions

**Exercise 34.24.** Compute the intersection form of $S^4$.

*Hint.* Use $H^2(S^4) = 0$.

Compute the middle cohomology lattice before writing a matrix; the answer is a rank-zero form, not a one-by-one zero matrix. □

*Solution.* Since $H^2(S^4; \mathbb{Z}) = 0$, the free middle cohomology lattice is zero. Hence $Q_{S^4}$ is the unique rank-zero bilinear form. □

**Exercise 34.25.** Compute the form of $\mathbb{CP}^2$.

*Hint.* Use the generator $h$ with $h^2[M] = 1$.

Use the complex orientation and the positive degree-two generator; evaluate its square to determine the single entry. □

*Solution.* With $h$ the positive generator of $H^2(\mathbb{CP}^2; \mathbb{Z})$,

$$\langle h^2, [\mathbb{CP}^2] \rangle = 1.$$

Thus the matrix is $[1]$. □

**Exercise 34.26.** Compute the form of $-\mathbb{CP}^2$.

*Hint.* Reverse the fundamental class.

Keep the underlying cohomology generator fixed while replacing the fundamental class by its negative; the form's value changes sign. □

*Solution.* Orientation reversal replaces the fundamental class by its negative. Therefore

$$Q_{-\mathbb{CP}^2} = -Q_{\mathbb{CP}^2} = [-1].$$

□

**Exercise 34.27.** Compute the form of $S^2 \times S^2$.

*Hint.* Use the two factor classes.

Pull back a degree-two sphere generator along each projection. Squares vanish by dimension in the individual factors, while the mixed product evaluates on the product class. □

*Solution.* For the factor classes $a, b$,

$$a^2 = b^2 = 0, \qquad \langle ab, [S^2 \times S^2] \rangle = 1.$$

Hence

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = H.$$

□

**Exercise 34.28.** Diagonalize the hyperbolic form over $\mathbb{R}$.

*Hint.* Find eigenvectors $(1,1)$ and $(1,-1)$.

Normalize the two eigenvectors to length one for a real congruence to $\operatorname{diag}(1,-1)$; the unnormalized vectors give diagonal entries $2, -2$. □

*Solution.* Over $\mathbb{R}$, the vectors $(1,1)$ and $(1,-1)$ are eigenvectors of $H$ with eigenvalues 1 and $-1$, respectively. So $H$ is congruent over $\mathbb{R}$ to $\operatorname{diag}(1,-1)$. □

**Exercise 34.29.** Compute its signature.

*Hint.* Count positive minus negative eigenvalues.

Count the positive and negative directions after real diagonalization; a zero signature does not mean the form itself is zero or degenerate. □

*Solution.* The hyperbolic form has one positive and one negative eigenvalue, so

$$b_2^+ = 1, \qquad b_2^- = 1, \qquad \sigma(H) = 0.$$

□

**Exercise 34.30.** Show that the hyperbolic form is even.

*Hint.* Evaluate $Q((x,y),(x,y))$.

Evaluate on a general integral vector $(x,y)$, not only on basis vectors; the cross terms combine to an even integer. □

*Solution.* For $v = (x,y) \in \mathbb{Z}^2$,

$$Q(v,v) = 2xy,$$

which is always even. Thus $H$ is an even integral form. □

**Exercise 34.31.** Show that [1] is odd.

*Hint.* Test the generator.

One lattice vector with an odd square suffices to disprove evenness; use the rank-one basis generator. □

*Solution.* For the generator $1 \in \mathbb{Z}$, the form [1] has $Q(1,1) = 1$, which is odd. Hence it is an odd form. □

**Exercise 34.32.** Compute the form of $\mathbb{CP}^2 \# \mathbb{CP}^2$.

*Hint.* Use orthogonal direct sum under connected sum.

Use classes supported away from the connected-sum neck so that cross terms between the two summands vanish. □

*Solution.* Connected sum gives orthogonal direct sum:

$$Q_{\mathbb{CP}^2 \# \mathbb{CP}^2} \cong [1] \oplus [1] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is positive definite with signature 2. □

**Exercise 34.33.** Compute the form of $\mathbb{CP}^2 \# (-\mathbb{CP}^2)$.

*Hint.* Combine [1] and $[-1]$.

Reverse the orientation on exactly one summand before taking the orthogonal sum; this changes only its block's sign. □

*Solution.* Similarly,

$$Q_{\mathbb{CP}^2 \# (-\mathbb{CP}^2)} \cong [1] \oplus [-1] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

It is indefinite and has signature 0. □

**Exercise 34.34.** Determine $b_2^+$ and $b_2^-$ in the preceding two examples.

*Hint.* Read signs from the diagonal.

Read the numbers of positive and negative diagonal entries in each example, and check that their sum equals the lattice rank. □

*Solution.* For $\mathbb{CP}^2 \# \mathbb{CP}^2$, $(b_2^+, b_2^-) = (2, 0)$. For $\mathbb{CP}^2 \# (-\mathbb{CP}^2)$, $(b_2^+, b_2^-) = (1, 1)$. □

**Exercise 34.35.** Verify that orientation reversal negates the matrix.

*Hint.* Replace $[M]$ by $-[M]$.

The cup product is unchanged by orientation reversal, while evaluation changes through $[-M] = -[M]$; this isolates the source of the sign. □

*Solution.* Since $Q_M(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$ and $[-M] = -[M]$,

$$Q_{-M}(\alpha, \beta) = -Q_M(\alpha, \beta).$$

Thus every representing matrix is negated. □

**Exercise 34.36.** Show that integral change of basis acts by congruence.

*Hint.* Write the new basis matrix $P$.

If columns of $P$ are the new basis in old coordinates, substitute $x = Py$ into the bilinear expression $x^T A x'$. □

*Solution.* If the old coordinate vector is $x = Py$, then

$$Q(x, x') = y^T (P^T A P) y'.$$

Hence an integral basis change $P \in GL_r(\mathbb{Z})$ sends $A$ to $P^T A P$. □

**Exercise 34.37.** Show that unimodularity is basis independent.

*Hint.* Use $\det(P) = \pm 1$.

Take determinants in $P^T A P$ and use that an integral basis change has determinant $\pm 1$; arbitrary integer matrices are not valid basis changes. □

*Solution.* If $A$ is unimodular, $\det A = \pm 1$. Under $A \mapsto P^T A P$,

$$\det(P^T A P) = \det(P)^2 \det(A) = \det(A)$$

because $\det P = \pm 1$. So unimodularity is basis independent. □

**Exercise 34.38.** Derive symmetry in dimension four from graded commutativity.

*Hint.* Degree two has even parity.

Middle cohomology has degree two, so the exchange sign is $(-1)^{2\cdot 2}$; evaluate the resulting cup equality on the fundamental class. □

*Solution.* In dimension four the middle degree is 2. Graded commutativity gives

$$\alpha \smile \beta = (-1)^{2\cdot 2} \beta \smile \alpha = \beta \smile \alpha,$$

so $Q_M$ is symmetric. □

**Exercise 34.39.** Derive skew-symmetry in middle degree on a closed oriented surface.

*Hint.* Degree one has odd parity.

For degree one, exchanging factors contributes a minus sign. Over $\mathbb{Z}$, the integer-valued self-pairing then vanishes because there is no two-torsion in the target. □

*Solution.* For a closed oriented surface the middle degree is 1, so

$$\alpha \smile \beta = -\beta \smile \alpha.$$

Therefore the degree-one intersection pairing is skew-symmetric. □

**Exercise 34.40.** Represent the standard generators of $H_2(S^2 \times S^2)$ geometrically.

*Hint.* Use the two factor spheres.

Use the factor spheres with product-compatible orientations and check their homology classes against the two Künneth summands in degree two. □

*Solution.* Take

$$A = S^2 \times \{q\}, \qquad B = \{p\} \times S^2.$$

Their homology classes generate $H_2(S^2 \times S^2) \cong \mathbb{Z}^2$. □

**Exercise 34.41.** Compute their mutual intersections.

*Hint.* Perturb to transverse representatives.

Their tangent spaces occupy the two complementary product factors at their unique common point; compare their ordered orientations with the ambient product orientation. ☐

*Solution.* The factor spheres $A$ and $B$ meet transversely at the single point $(p, q)$. With product orientations the sign is $+1$, so $A \cdot B = 1$. ☐

**Exercise 34.42.** Compute their self-intersections using push-offs.

*Hint.* Push a factor sphere in its normal direction.

Move the fixed point in the other sphere factor to construct a disjoint homologous push-off; this directly calculates the self-intersection. ☐

*Solution.* A small push-off of $A = S^2 \times \{q\}$ is $S^2 \times \{q'\}$, disjoint from $A$; hence $A^2 = 0$. The same argument gives $B^2 = 0$. ☐

**Exercise 34.43.** Relate self-intersection to the normal Euler number.

*Hint.* Use the normal-bundle Euler class.

A transverse normal section determines the push-off, and its zeros are exactly the intersections with the original submanifold; sum their normal Euler indices. ☐

*Solution.* For an oriented middle-dimensional submanifold $A$, a transverse push-off is determined by a section of its normal bundle. Its zeros count the self-intersection, so

$$A \cdot A = \langle e(\nu_A), [A] \rangle.$$

☐

**Exercise 34.44.** Explain why torsion is removed from the lattice form.

*Hint.* Pair torsion with an integer-valued functional.

If $mx = 0$, bilinearity makes $mQ(x, y) = 0$ in the torsion-free target $\mathbb{Z}$; hence the pairing factors through the free quotient. ☐

*Solution.* If $x$ is torsion, say $mx = 0$, then for any $y$,

$$mQ(x, y) = Q(mx, y) = 0$$

in $\mathbb{Z}$, hence $Q(x, y) = 0$. Therefore the integer-valued lattice form factors through the free quotient. ☐

**Exercise 34.45.** Compare rank, parity, definiteness and signature as invariants.

*Hint.* Keep the invariants logically distinct.

Compare the hyperbolic plane with $\mathrm{diag}(1, -1)$: equal rank and signature do not imply integral congruence, because their parities differ. ☐

*Solution.* Rank records lattice dimension; parity records whether all squares are even; definiteness records the sign pattern over $\mathbb{R}$; signature is $b^+ - b^-$. These are preserved under integral congruence but capture different information. ☐

**Exercise 34.46.** Write the intersection matrix for an orthogonal sum $H \oplus [1]$.

*Hint.* Write a block diagonal matrix.

Keep the hyperbolic two-by-two block and the rank-one positive block separate; compute determinant and inertia from the blocks. ☐

*Solution.* For

$$H \oplus [1],$$

a natural basis gives

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is unimodular and indefinite, with signature 1. □

**Exercise 34.47.** Explain how VIII/34 prepares the graph–diagonal intersection picture in Lefschetz theory.

*Hint.* Think of fixed points as intersections of graph and diagonal.

In $M \times M$, both the graph and diagonal have dimension $\dim M$; their transverse intersections are zero-dimensional and carry local fixed-point signs with a fixed convention. □

*Solution.* For $f : M \to M$, fixed points satisfy $(x, f(x)) = (x, x)$, so they are intersections of the graph $\Gamma_f$ with the diagonal $\Delta$ in $M \times M$. VIII/35 turns their signed intersection number into the Lefschetz trace. □

## 34.20 Supplementary worked examples

**Example 34.48** (The intersection form of CP2)**.** For the complex projective plane, $H_2(\mathbb{CP}^2;\mathbb{Z}) \cong \mathbb{Z}$ is generated by the class of a projective line. Two generic projective lines meet once positively, so the intersection form is the $1 \times 1$ matrix

$$[1].$$

**Example 34.49** (The hyperbolic form on S2 x S2)**.** Let $a = [S^2 \times \{*\}]$ and $b = [\{*\} \times S^2]$. Self-intersections are zero while $a \cdot b = 1$, giving

$$Q_{S^2\times S^2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

**Example 34.50** (Orientation reversal)**.** Reversing the orientation of a closed oriented 4-manifold changes every oriented intersection number by a sign. Hence

$$Q_{-M} = -Q_M,$$

so the signature changes from $\sigma(M)$ to $-\sigma(M)$.

## 34.21 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 34.51** (Definition in dimension four)**.** Define the intersection form of a closed oriented 4-manifold on the free part of $H_2(M;\mathbb{Z})$.

*Hint.* Intersect representative surfaces or use cup product. □

*Solution.* $Q_M(x,y) = \langle \mathrm{PD}(x) \smile \mathrm{PD}(y), [M]\rangle$. □

**Exercise 34.52** (CP2 matrix)**.** Write the intersection matrix of $\mathbb{CP}^2$ in the projective-line basis.

*Hint.* A line has self-intersection $+1$. □

*Solution.* The matrix is $[1]$. □

**Exercise 34.53** (S2xS2 matrix)**.** Write the matrix in the product-sphere basis.

*Hint.* Each factor has zero self-intersection and the factors meet once. □

*Solution.* $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. □

**Exercise 34.54** (Signature of CP2)**.** Compute $\sigma(\mathbb{CP}^2)$.

*Hint.* Count positive minus negative eigenvalues of $[1]$. □

*Solution.* $\sigma = 1$. □

**Exercise 34.55** (Signature of S2xS2)**.** Compute the signature of the hyperbolic form.

*Hint.* Its eigenvalues are 1 and $-1$. □

*Solution.* $\sigma = 0$. □

## Proofs

**Exercise 34.56** (Symmetry)**.** Prove the intersection form on a closed oriented 4-manifold is symmetric.

*Hint.* Degree-two cup products commute because $(-1)^{2\cdot 2} = 1$. □

*Solution.* Graded commutativity gives $\alpha \smile \beta = \beta \smile \alpha$ in degree two, so $Q(x, y) = Q(y, x)$. □

**Exercise 34.57** (Unimodularity)**.** Explain why the intersection form on the free part is unimodular.

*Hint.* Use Poincare duality modulo torsion. □

*Solution.* Poincare duality identifies the free lattice $H_2/\mathrm{tors}$ with its integral dual, so the pairing has determinant $\pm 1$ in any integral basis. □

**Exercise 34.58** (Basis change)**.** Show changing an integral basis by $A \in GL_n(\mathbb{Z})$ transforms the matrix by $Q \mapsto A^T QA$.

*Hint.* Expand the bilinear pairing on new basis vectors. □

*Solution.* The coordinates of both arguments transform by $A$, yielding the congruence transformation $A^T QA$. □

**Exercise 34.59** (Orientation reversal)**.** Prove $Q_{-M} = -Q_M$.

*Hint.* The fundamental class changes sign. □

*Solution.* Evaluation on $[-M] = -[M]$ multiplies every pairing value by $-1$. □

## Counterexamples and hypothesis tests

**Exercise 34.60** (Form determined by Betti number)**.** Test: $b_2(M)$ determines the intersection form.

*Hint.* Many noncongruent forms have the same rank. □

*Solution.* False. Rank alone does not determine parity, signature, or lattice structure. □

**Exercise 34.61** (Intersection matrix basis invariant entrywise)**.** Test: the matrix entries are independent of basis.

*Hint.* Only the equivalence class of the bilinear form is invariant. □

*Solution.* False. Matrices change by integral congruence under basis changes. □

**Exercise 34.62** (Self-intersection always zero)**.** Test: every oriented surface class in a 4-manifold has zero self-intersection.

*Hint.* Consider a projective line in $\mathbb{CP}^2$. □

*Solution.* False; the generator of $H_2(\mathbb{CP}^2)$ has self-intersection $+1$. □

## Applications and investigations

**Exercise 34.63** (Geometric collision matrix)**.** How can a basis of embedded surfaces be used to construct an intersection matrix computationally?

*Hint.* Perturb to transverse position and count signed intersections. ☐

*Solution.* Each matrix entry is obtained by making two representatives transverse and summing local signs; diagonal entries use a push-off determined by the normal bundle. ☐

**Exercise 34.64** (Classification diagnostic)**.** Why is signature more informative than $b_2$ alone?

*Hint.* It remembers the positive/negative decomposition of the form. ☐

*Solution.* Two manifolds with the same $b_2$ can have different signatures, so the signature distinguishes intersection lattices that rank cannot. ☐

## Challenge problems

**Exercise 34.65** (Connected sum form)**.** Under standard orientations, what happens to intersection forms under connected sum of closed oriented 4-manifolds?

*Hint.* Middle homology splits. ☐

*Solution.* $Q_{M\#N} \cong Q_M \oplus Q_N$. ☐

**Exercise 34.66** (Intersection synthesis)**.** Explain how cup product, Poincare duality, and embedded representatives produce the same intersection form.

*Hint.* Translate homology classes to degree-two cohomology and evaluate. ☐

*Solution.* Poincare duality turns surfaces into degree-two classes; their cup product evaluated on $[M]$ equals the signed geometric intersection count, linking the algebraic matrix directly to geometry. ☐

# Chapter 35

# Lefschetz Theory

Lefschetz theory is a synthesis theorem. A self-map acts linearly on homology, fixed points appear geometrically as intersections of a graph with a diagonal, and Poincaré duality proves that the two counts agree.

## 35.1 The Lefschetz number

Let $X$ be a finite CW complex and $f : X \to X$ a continuous map.

**Definition 35.1** (Lefschetz number)**.** With coefficients in a field such as $\mathbb{Q}$,

$$L(f) = \sum_{k\geq 0}(-1)^k \operatorname{tr}\big(f_* : H_k(X;\mathbb{Q}) \to H_k(X;\mathbb{Q})\big).$$

## 35.2 Homotopy invariance

If $f \simeq g$, then $f_* = g_*$ on homology, hence

$$L(f) = L(g).$$

## 35.3 The fixed-point theorem

**Theorem 35.2** (Lefschetz fixed-point theorem)**.** *For a self-map of a suitable compact triangulable space, if*

$$L(f) \neq 0,$$

*then $f$ has a fixed point.*

The converse is false: fixed points can occur with total index zero.

## 35.4 The identity map

For the identity,

$$L(\mathrm{id}_X) = \sum_k (-1)^k \dim H_k(X;\mathbb{Q}) = \chi(X).$$

Thus Euler characteristic is the Lefschetz number of the identity.

## 35.5 Contractible spaces and Brouwer

If $X$ is connected and contractible, only $H_0$ contributes:

$$L(f) = 1.$$

Applied to $D^n$, this gives Brouwer's fixed-point theorem.

## 35.6 Sphere maps

If $f : S^n \to S^n$ has degree $d$, then

$$L(f) = 1 + (-1)^n d.$$

Whenever this number is nonzero, $f$ must have a fixed point.

## 35.7 Constant maps

A constant self-map of a connected finite CW complex induces the identity on $H_0$ and zero on positive-dimensional reduced homology, so

$$L(f) = 1.$$

## 35.8 The graph and diagonal

For a self-map $f : M \to M$ of a closed manifold, define

$$\Gamma_f = \{(x, f(x)) : x \in M\}, \qquad \Delta = \{(x, x) : x \in M\}.$$

Then

$$x \in \mathrm{Fix}(f) \quad \Longleftrightarrow \quad (x, f(x)) \in \Gamma_f \cap \Delta.$$

## 35.9 Intersection-theoretic meaning

Under the usual transversality and orientation hypotheses,

$$L(f) = \Gamma_f \cdot \Delta.$$

This is the geometric bridge from VIII/34 to the homological trace formula.

## 35.10 Local fixed-point index

If a smooth fixed point $x$ is nondegenerate, so

$$\det(I - Df_x) \neq 0,$$

its local index is

$$\mathrm{ind}_x(f) = \operatorname{sign}\det(I - Df_x).$$

**Theorem 35.3** (Lefschetz–Hopf)**.** *When the fixed points are isolated and the standard hypotheses hold,*

$$L(f) = \sum_{x \in \mathrm{Fix}(f)} \mathrm{ind}_x(f).$$

## 35.11 Cancellation

The theorem only says that a nonzero Lefschetz number forces a fixed point. If positive and negative local indices cancel, fixed points can exist while

$$L(f) = 0.$$

## 35.12 Torus endomorphisms

A linear torus map induced by

$$A \in M_2(\mathbb{Z})$$

acts by 1 on $H_0$, by $A$ on $H_1$, and by $\det A$ on $H_2$. Therefore

$$L(f) = 1 - \operatorname{tr}(A) + \det(A) = \det(I - A).$$

## 35.13 A fixed-point criterion on the torus

If

$$\det(I - A) \neq 0,$$

then the induced torus map has nonzero Lefschetz number and therefore a fixed point.

## 35.14 Poincaré-duality proof philosophy

The diagonal class in $M \times M$ is expressed using dual bases in homology and cohomology. Pulling it against the graph of $f$ converts geometric intersection into the alternating trace of $f_*$.

## 35.15 Supertrace viewpoint

The Lefschetz number is the supertrace of the graded homology endomorphism:

$$L(f) = \operatorname{str}(f_*).$$

This compactly packages the alternating signs.

## 35.16 Endpoint of the canonical arc

The chapter closes the Volume VIII progression

homology → cohomology → cup/cap products → Thom and Euler classes → Poincaré duality → intersection

## 35.17 Solved dossiers

**Problem 35.4.** Define the Lefschetz number of a self-map of a finite CW complex.

*Solution.* $L(f) = \sum_k (-1)^k \operatorname{tr}(f_* : H_k(X;\mathbb{Q}) \to H_k(X;\mathbb{Q}))$. □

**Problem 35.5.** Why may rational homology be used in the basic trace formula?

*Solution.* It removes torsion from the finite-dimensional trace calculation while preserving the classical Lefschetz number. □

**Problem 35.6.** State the Lefschetz fixed-point theorem.

*Solution.* If $L(f) \neq 0$ for a self-map of a suitable compact triangulable space, then $f$ has a fixed point. □

**Problem 35.7.** What is $L(\mathrm{id}_X)$?

*Solution.* $\chi(X)$. □

**Problem 35.8.** How does Lefschetz imply Brouwer for a map $D^n \to D^n$?

*Solution.* $D^n$ is contractible, so only $H_0$ contributes and $L(f) = 1 \neq 0$. □

**Problem 35.9.** Compute $L(f)$ for $f : S^n \to S^n$ of degree $d$.

*Solution.* $L(f) = 1 + (-1)^n d$. □

**Problem 35.10.** What condition on the degree of a sphere map guarantees a fixed point?

*Solution.* $1 + (-1)^n d \neq 0$. □

**Problem 35.11.** What is the Lefschetz number of a constant self-map of a connected finite complex?

*Solution.* 1. □

**Problem 35.12.** Why is the converse of the Lefschetz theorem false?

*Solution.* A map can have fixed points whose local indices cancel, so $L(f) = 0$ does not imply fixed-point-free. □

**Problem 35.13.** Why is $L(f)$ homotopy invariant?

*Solution.* Homotopic maps induce the same homomorphisms on homology and therefore the same traces. □

**Problem 35.14.** Describe fixed points using graph and diagonal.

*Solution.* $x$ is fixed exactly when $(x, f(x))$ lies in both $\Gamma_f$ and $\Delta$. □

**Problem 35.15.** What does the global Lefschetz number measure geometrically?

*Solution.* Under transversality/nondegeneracy hypotheses it is the signed intersection number $\Gamma_f \cdot \Delta$. □

**Problem 35.16.** What is a local fixed-point index in the nondegenerate smooth case?

*Solution.* It is the local intersection sign; equivalently the sign of $\det(I - Df_x)$ when that determinant is nonzero. □

**Problem 35.17.** State the Lefschetz–Hopf refinement.

*Solution.* For isolated fixed points satisfying the usual hypotheses, $L(f)$ is the sum of their local fixed-point indices. □

**Problem 35.18.** Compute the Lefschetz number of a torus map induced by $A \in M_2(\mathbb{Z})$.

*Solution.* $L(f) = 1 - \operatorname{tr}(A) + \det(A) = \det(I - A)$. □

**Problem 35.19.** What does $\det(I - A) \neq 0$ imply for the induced torus map?

*Solution.* Its Lefschetz number is nonzero, so it has a fixed point. □

**Problem 35.20.** What happens for a fixed-point-free map?

*Solution.* Necessarily $L(f) = 0$. □

**Problem 35.21.** How does the identity map connect Lefschetz theory to Euler characteristic?

*Solution.* Its local/global fixed-point count is encoded by $L(\mathrm{id}) = \chi(X)$. □

**Problem 35.22.** How does Lefschetz theory use Poincaré duality?

*Solution.* On manifolds, duality converts the graph–diagonal intersection class into the alternating trace on homology. □

**Problem 35.23.** Why is VIII/35 a natural endpoint for Volume VIII?

*Solution.* It synthesizes homology, cohomology, cup/cap products, Poincaré duality and intersection theory into a computable fixed-point invariant. □

## 35.18 Exercises, Hints, and Solutions

**Exercise 35.24.** Compute $L(\mathrm{id}_{S^n})$.

*Hint.* Use $L(\mathrm{id}) = \chi$.

The identity has trace equal to the dimension in each homology degree; use the sphere's Betti numbers, including the separate two-component case for $S^0$. □

*Solution.* For the identity, $L(\mathrm{id}_X) = \chi(X)$. Hence

$$L(\mathrm{id}_{S^n}) = \chi(S^n) = 1 + (-1)^n,$$

equal to 2 for even $n$ and 0 for odd $n$. □

**Exercise 35.25.** Compute $L(f)$ for a degree-$d$ self-map of $S^n$.

*Hint.* Only $H_0$ and $H_n$ contribute.

For $n \geq 1$, connectedness gives trace one on rational $H_0$, and the chosen degree gives the trace on top homology. □

*Solution.* Only $H_0(S^n)$ and $H_n(S^n)$ are nonzero. The induced maps have traces 1 and $d$, so

$$L(f) = 1 + (-1)^n d.$$

□

**Exercise 35.26.** Determine for which $d$ the Lefschetz theorem forces a fixed point on $S^2$.

*Hint.* Set $1 + d \neq 0$.

Insert the even dimension into the sphere trace formula, then distinguish nonzero Lefschetz number from the inconclusive zero case. □

*Solution.* For $S^2$,

$$L(f) = 1 + d.$$

Lefschetz forces a fixed point whenever $d \neq -1$. The theorem is silent when $d = -1$. □

**Exercise 35.27.** Do the same on $S^1$.

*Hint.* Set $1 - d \neq 0$.

The degree-one contribution carries a minus sign; do not interpret the exceptional degree as proving there are no fixed points. □

*Solution.* For $S^1$,

$$L(f) = 1 - d.$$

Thus a fixed point is forced whenever $d \neq 1$. □

**Exercise 35.28.** Compute the Lefschetz number of a constant map on a connected finite CW complex.

*Hint.* Only $H_0$ contributes nontrivially.

A constant map factors through a point, giving zero on positive reduced homology; for a connected finite CW complex, its ordinary $H_0$ map is the identity. □

*Solution.* A constant map on a connected space induces the identity on $H_0$ and zero on positive reduced homology. Therefore

$$L(f) = 1.$$

□

**Exercise 35.29.** Deduce Brouwer's fixed-point theorem for $D^n$.

*Hint.* Use contractibility.

For any continuous disk self-map, only rational degree-zero homology contributes. Apply the fixed-point theorem in its compact polyhedron setting. □

*Solution.* Every self-map $f : D^n \to D^n$ has $L(f) = 1$, since the disk is contractible and connected. The Lefschetz fixed-point theorem therefore guarantees a fixed point. This is Brouwer's theorem. □

**Exercise 35.30.** Show that homotopic self-maps have the same Lefschetz number.

*Hint.* Use homotopy invariance of induced homology maps.

Use equality of induced homology maps, not just equality of their ranks; their traces then agree before taking the alternating sum. □

*Solution.* Homotopic maps induce identical maps on every homology group. Their traces agree degree by degree, hence their alternating trace sums agree:

$$f \simeq g \implies L(f) = L(g).$$

□

**Exercise 35.31.** Show that a fixed-point-free map must have Lefschetz number zero.

*Hint.* Use the contrapositive of Lefschetz.

Negate both conclusion and hypothesis of the implication correctly; the converse statement about maps with zero Lefschetz number is not available. □

*Solution.* The Lefschetz theorem says $L(f) \neq 0 \Rightarrow \mathrm{Fix}(f) \neq \varnothing$. Taking the contrapositive gives

$$\mathrm{Fix}(f) = \varnothing \Rightarrow L(f) = 0.$$

□

**Exercise 35.32.** Give an example where $L(f) = 0$ but fixed points may still occur.

*Hint.* Cancellation of local indices is possible.

Test the identity of the circle: its fixed-point set is large, but its alternating homology trace cancels. □

*Solution.* The identity on $S^1$ has every point fixed but

$$L(\mathrm{id}_{S^1}) = \chi(S^1) = 0.$$

Thus $L(f) = 0$ does not imply that the fixed-point set is empty. □

**Exercise 35.33.** Write the graph $\Gamma_f$ and diagonal $\Delta$ for a self-map of a manifold.

*Hint.* Use $x \mapsto (x, f(x))$.

Specify the ambient product and the two embeddings $x \mapsto (x, f(x))$ and $x \mapsto (x, x)$; each has the dimension of the original manifold. □

*Solution.* The graph and diagonal are

$$\Gamma_f = \{(x, f(x)) : x \in M\}, \qquad \Delta = \{(x, x) : x \in M\} \subset M \times M.$$

□

**Exercise 35.34.** Explain why their intersections are fixed points.

*Hint.* Solve $(x, f(x)) = (x, x)$.

Equality of the first coordinates identifies the two parameters; equality of the second coordinates then gives the fixed-point equation. □

*Solution.* A point lies in both submanifolds exactly when

$$(x, f(x)) = (x, x),$$

which is equivalent to $f(x) = x$. Hence $\Gamma_f \cap \Delta$ is naturally the fixed-point set. □

**Exercise 35.35.** Relate a transverse intersection sign to local index.

*Hint.* Orient the graph and diagonal.

For a smooth nondegenerate fixed point, linearize $x - f(x)$; require $I - Df_x$ invertible and use the graph–diagonal orientation convention yielding its determinant sign. □

*Solution.* At a transverse intersection of $\Gamma_f$ and $\Delta$, the orientations of their tangent spaces determine a local intersection sign. For a nondegenerate smooth fixed point this sign is

$$\operatorname{sign}\det(I - Df_x),$$

the local fixed-point index. □

**Exercise 35.36.** Compute $L(f)$ for the torus map induced by $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

*Hint.* Use $1 - \operatorname{tr} A + \det A$.

Compute the traces on $H_0, H_1, H_2$ separately; the induced top-degree map is multiplication by $\det A$. □

*Solution.* For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$,

$$L(f) = 1 - \operatorname{tr} A + \det A = 1 - 4 + 4 = 1.$$

Equivalently $\det(I - A) = \det(-I) = 1$, so a fixed point is forced. □

**Exercise 35.37.** Compute it for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

*Hint.* Compute trace and determinant.

The zero trace sum is inconclusive. For this particular linear torus endomorphism, the origin is nevertheless fixed; do not confuse absence of a guarantee with absence of fixed points. □

*Solution.* For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$,

$$L(f) = 1 - 2 + 1 = 0.$$

Indeed $I - A$ is singular, so Lefschetz gives no forced fixed point. □

**Exercise 35.38.** Verify $L(f) = \det(I - A)$ for a general $2 \times 2$ integer matrix.

*Hint.* Expand $\det(I - A)$.

Expand both the determinant and the alternating trace using the four matrix entries; this checks the identity without assuming diagonalizability. □

*Solution.* For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\det(I - A) = (1 - a)(1 - d) - bc = 1 - (a + d) + (ad - bc) = 1 - \operatorname{tr} A + \det A.$$

This is exactly the torus Lefschetz number. □

**Exercise 35.39.** Determine when the torus trace formula guarantees a fixed point.

*Hint.* Require $\det(I - A) \neq 0$.

A nonzero determinant of $I - A$ excludes eigenvalue one and makes the Lefschetz number nonzero; this is sufficient, not a necessary condition for having a fixed point. □

*Solution.* The trace formula gives $L(f) = \det(I - A)$. Therefore

$$\det(I - A) \neq 0$$

is sufficient for the induced torus map to have a fixed point. □

**Exercise 35.40.** Compute $L$ for an orientation-reversing degree $-1$ map of $S^2$.

*Hint.* Use $1 + d$ in even dimension.

The degree $-1$ cancels the degree-zero trace on the even sphere; a reflection and the antipodal map illustrate different fixed-point behavior with this same Lefschetz number. □

*Solution.* An orientation-reversing map $S^2 \to S^2$ of degree $-1$ has

$$L(f) = 1 + (-1) = 0.$$

Thus Lefschetz alone does not force a fixed point. □

**Exercise 35.41.** Compute $L$ for the antipodal map on $S^n$.

*Hint.* The antipodal degree is $(-1)^{n+1}$.

Substitute the antipodal degree into the sphere formula and simplify the parity exponents; for $S^0$, compute the trace of its component swap directly. □

*Solution.* The antipodal map on $S^n$ has degree $(-1)^{n+1}$. Hence

$$L(a) = 1 + (-1)^n(-1)^{n+1} = 1 - 1 = 0.$$

□

**Exercise 35.42.** Compare the conclusion with the actual fixed-point set.

*Hint.* Check that antipodal maps have no fixed points.

Solve $x = -x$ in the ambient real vector space and compare with the unit-norm condition defining the sphere. □

*Solution.* The antipodal map has no fixed points because $x = -x$ on a sphere would imply $x = 0$, which is impossible. Its Lefschetz number is correspondingly zero, consistent with the theorem. □

**Exercise 35.43.** Use the identity map to recover Euler characteristic.

*Hint.* Sum alternating Betti numbers.

Take rational homology so each identity trace is the corresponding Betti number, then use Euler–Poincare for a finite CW complex. □

*Solution.* For the identity,

$$L(\mathrm{id}_X) = \sum_k (-1)^k \operatorname{tr}(\mathrm{id}|H_k) = \sum_k (-1)^k b_k = \chi(X).$$

□

**Exercise 35.44.** Express the Lefschetz number as an alternating supertrace.

*Hint.* Regard the total homology as a graded vector space.

Group the even-degree traces and odd-degree traces separately; the supertrace is their difference, not the ordinary trace on total homology. □

*Solution.* If $H_*(X;\mathbb{Q}) = H_{\mathrm{even}} \oplus H_{\mathrm{odd}}$ as a graded vector space, then

$$L(f) = \operatorname{tr}(f_*|H_{\mathrm{even}}) - \operatorname{tr}(f_*|H_{\mathrm{odd}}),$$

which is the supertrace. □

**Exercise 35.45.** Explain the role of finite-dimensional homology in defining the trace.

*Hint.* Use a field such as $\mathbb{Q}$.

Require finite-dimensional groups in every degree and only finitely many nonzero degrees for this ordinary finite trace sum; arbitrary infinite-dimensional traces are not defined here. □

*Solution.* The trace is defined for finite-dimensional vector spaces. A finite CW complex has finite-dimensional $H_k(X;\mathbb{Q})$, and only finitely many degrees are nonzero, so the alternating trace sum is finite and well-defined. $\square$

**Exercise 35.46.** Describe how Poincaré duality enters the graph–diagonal proof.

*Hint.* Dualize the diagonal class.

Choose homogeneous cohomology bases and their Poincare dual bases; the graded signs in the diagonal's class are what produce the alternating, rather than ordinary, trace. $\square$

*Solution.* Poincaré duality identifies the homology class of the diagonal with a sum built from dual homology/cohomology bases. Pairing the graph class with this diagonal class converts the graph–diagonal intersection number into the alternating traces of $f_*$. $\square$

**Exercise 35.47.** Summarize homology trace $\leftrightarrow$ intersection number $\leftrightarrow$ fixed point.

*Hint.* Track the three descriptions of the same number.

State the extra hypotheses for each description: finite homology for the trace, the oriented manifold setting for intersection, and isolated fixed points with defined local indices for the finite index sum. $\square$

*Solution.* The same integer has three forms:

$$L(f) = \sum_k (-1)^k \operatorname{tr}(f_*|H_k) = \Gamma_f \cdot \Delta = \sum_{x\in \operatorname{Fix}(f)} \operatorname{ind}_x(f)$$

when the local-index hypotheses hold. This is the algebraic, geometric, and local fixed-point synthesis of Lefschetz theory. $\square$

## 35.19 Supplementary worked examples

**Example 35.48** (A degree-d map of the circle)**.** For $f : S^1 \to S^1$ of degree $d$,

$$L(f) = \operatorname{tr}(f_*|H_0) - \operatorname{tr}(f_*|H_1) = 1 - d.$$

If $d \neq 1$, then $L(f) \neq 0$, so the Lefschetz fixed-point theorem guarantees a fixed point.

**Example 35.49** (A degree-d map of a sphere)**.** For $f : S^n \to S^n$ of degree $d$, only $H_0$ and $H_n$ contribute:

$$L(f) = 1 + (-1)^n d.$$

Thus the antipodal map has Lefschetz number 0, consistent with its lack of fixed points.

**Example 35.50** (Linear torus maps)**.** A torus map induced by an integer matrix $A \in M_n(\mathbb{Z})$ acts on $H_1(T^n)$ by $A$ and on higher homology by exterior powers. The alternating trace identity gives

$$L(f_A) = \det(I - A).$$

A nonzero determinant therefore forces a fixed point.

## 35.20 Graded supplementary exercises

The following exercises progress from direct computation and construction to proof, hypothesis testing, application, and synthesis.

### Standard computations and constructions

**Exercise 35.51** (Definition)**.** Define the Lefschetz number of a self-map $f : X \to X$ of a finite-type space.

*Hint.* Alternate traces on homology. □

*Solution.* $L(f) = \sum_i (-1)^i \operatorname{tr}(f_* : H_i(X; F) \to H_i(X; F))$ over a suitable field or finite free setting. □

**Exercise 35.52** (Identity map)**.** What is $L(\mathrm{id}_X)$ for a finite CW complex?

*Hint.* The trace in each degree is the Betti number. □

*Solution.* $L(\mathrm{id}_X) = \chi(X)$. □

**Exercise 35.53** (Circle degree d)**.** Compute $L(f)$ for a degree-$d$ map $S^1 \to S^1$.

*Hint.* Use traces 1 and $d$. □

*Solution.* $L(f) = 1 - d$. □

**Exercise 35.54** (Sphere degree d)**.** Compute $L(f)$ for a degree-$d$ map $S^n \to S^n$.

*Hint.* Only degrees 0 and $n$ contribute. □

*Solution.* $L(f) = 1 + (-1)^n d$. □

**Exercise 35.55** (Constant map)**.** Compute the Lefschetz number of a constant self-map of a connected finite CW complex.

*Hint.* It is identity on $H_0$ and zero in positive degrees. □

*Solution.* $L(f) = 1$. □

## Proofs

**Exercise 35.56** (Homotopy invariance)**.** Prove homotopic self-maps have the same Lefschetz number.

*Hint.* Homotopic maps induce the same homology maps. ☐

*Solution.* The induced linear maps agree in every degree, so all traces and hence their alternating sum agree. ☐

**Exercise 35.57** (No-fixed-point contrapositive)**.** What can be concluded if a map has no fixed points?

*Hint.* Use the contrapositive of the Lefschetz theorem. ☐

*Solution.* Under the theorem's hypotheses, it must have $L(f) = 0$. ☐

**Exercise 35.58** (Torus determinant)**.** Explain why $L(f_A) = \det(I - A)$ for a linear map of $T^n$.

*Hint.* Use traces on exterior powers. ☐

*Solution.* $H_k(T^n) \cong \Lambda^k \mathbb{Z}^n$, so the alternating sum $\sum(-1)^k \operatorname{tr}(\Lambda^k A)$ equals $\det(I - A)$. ☐

**Exercise 35.59** (Local index sum)**.** State the relation between isolated fixed-point indices and the Lefschetz number.

*Hint.* Sum the local contributions. ☐

*Solution.* For isolated nondegenerate or suitably indexed fixed points, the sum of local fixed-point indices equals $L(f)$. ☐

## Counterexamples and hypothesis tests

**Exercise 35.60** (L=0 means no fixed point)**.** Test: if $L(f) = 0$, then $f$ has no fixed points.

*Hint.* The theorem gives only a sufficient condition when nonzero. ☐

*Solution.* False. Maps can have fixed points whose local indices cancel, or have zero Lefschetz number for other reasons. ☐

**Exercise 35.61** (Nonzero L counts fixed points exactly)**.** Test: $|L(f)|$ always equals the number of fixed points.

*Hint.* Local indices may have signs or magnitudes. ☐

*Solution.* False. The Lefschetz number is an algebraic index sum, not generally a raw count. ☐

**Exercise 35.62** (L depends on chosen representative in homotopy class)**.** Test: homotopic maps can have different Lefschetz numbers.

*Hint.* Induced homology maps are homotopy invariant. ☐

*Solution.* False. Lefschetz number depends only on the homotopy class. ☐

### Applications and investigations

**Exercise 35.63** (Fixed-point certificate)**.** How can $L(f)$ provide a cheap certificate of a fixed point before solving $f(x) = x$ geometrically?

*Hint.* Compute finite-dimensional traces on homology. ☐

*Solution.* If the alternating trace is nonzero, the theorem guarantees at least one fixed point without locating it. ☐

**Exercise 35.64** (Dynamics on a torus)**.** For a torus automorphism $A$, why is $\det(I - A)$ a useful dynamical diagnostic?

*Hint.* Nonzero determinant is the Lefschetz number. ☐

*Solution.* If $\det(I - A) \neq 0$, a fixed point is forced; the determinant can be computed directly from the integer matrix defining the map. ☐

### Challenge problems

**Exercise 35.65** (Antipodal sphere)**.** Check the Lefschetz number of the antipodal map on $S^n$.

*Hint.* Its degree is $(-1)^{n+1}$. ☐

*Solution.* $L = 1 + (-1)^n(-1)^{n+1} = 0$, consistent with the fact that the antipodal map has no fixed points. ☐

**Exercise 35.66** (Lefschetz synthesis)**.** Explain the chain of ideas from a geometric self-map to a fixed-point conclusion.

*Hint.* Pass to homology, take traces, then invoke the theorem. ☐

*Solution.* A self-map induces finite-dimensional endomorphisms on homology; their alternating trace is the homotopy-invariant Lefschetz number; if it is nonzero, the fixed-point theorem forces an intersection of the graph of $f$ with the diagonal, hence a fixed point. ☐